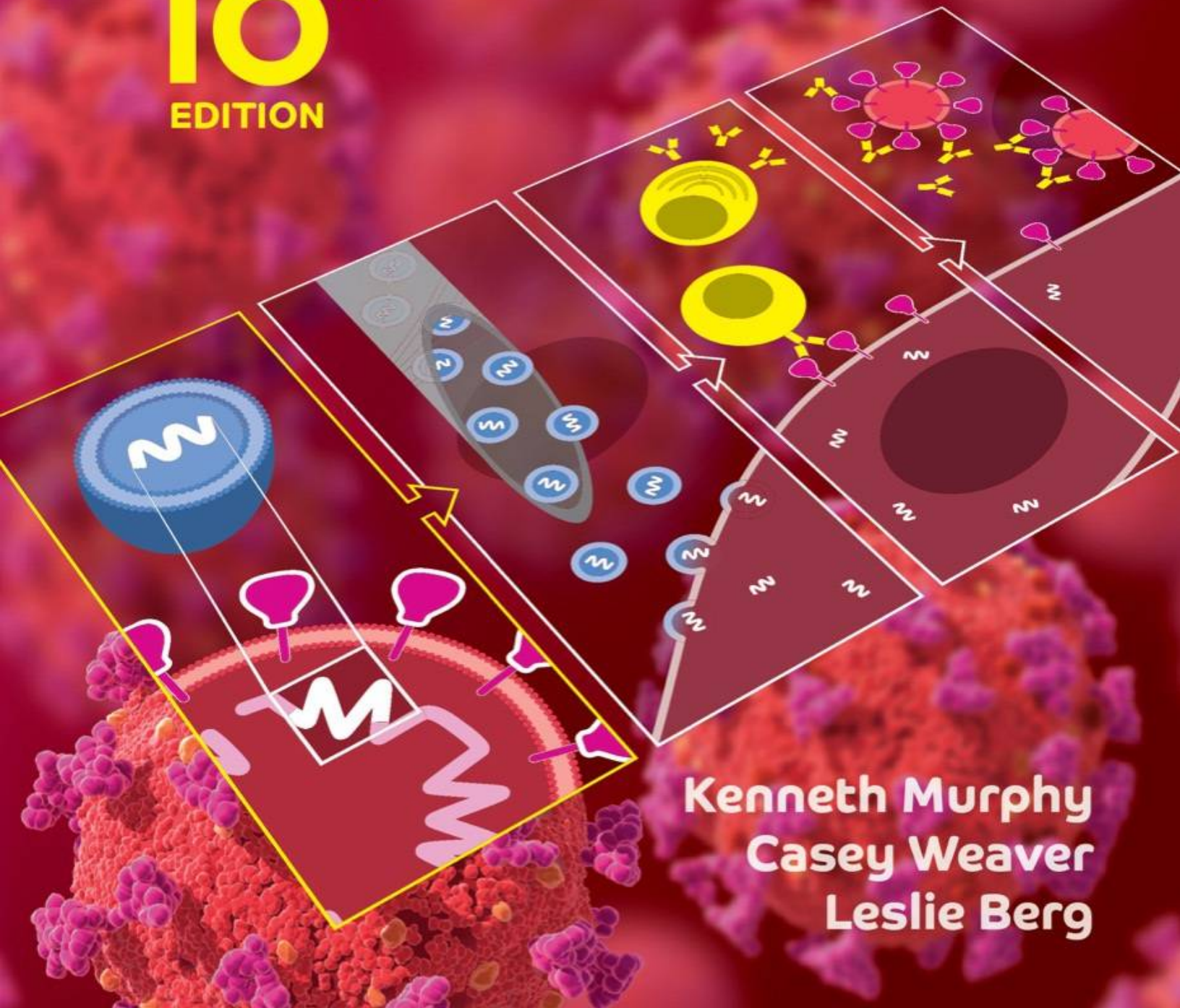


Janeway's

Immunobiology

10TH
EDITION



Kenneth Murphy
Casey Weaver
Leslie Berg

Janeway's **Immunobiology**

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Preface

Since early 2020, the field of immunology has moved to an unexpectedly prominent place in daily life. As SARS-CoV-2 spread around the globe, basic questions about how the immune system combats viruses moved into the forefront of public attention. Why were individual responses to the virus so variable? Why were outcomes in older patients so much more dire than in younger infected individuals? And, most important, how quickly would a vaccine be available, would it provide adequate protection, and for how long? While some questions have been answered, notably observing the most rapid vaccine development in human history, many other questions remain outstanding. Clearly, the need for continued research into basic mechanisms controlling human immunity is undeniable.

Though somewhat overshadowed during the COVID-19 pandemic, there has been a dramatic rise in the use of various other immunotherapies in the years since the last edition of *Janeway's Immunobiology*. Multiple methods for manipulating immune responses have significantly altered the practice of clinical medicine, benefiting patients who present with a diagnosis of cancer, autoimmune disease, or chronic inflammatory disorder. In large part, these clinical advances arose from decades of basic research aimed at uncovering the molecules and pathways that regulate immune cells at steady-state conditions or during infection, inflammation, or autoimmunity.

This Tenth Edition of *Janeway's Immunobiology* strives to carry on Charlie Janeway's legacy of providing a comprehensive, up-to-date textbook focused on the mechanisms underlying the responses to infection. In keeping with this mission, this edition's text and figures have been updated throughout. The immune response to SARS-CoV-2 and the multiple vaccine strategies utilized against this infection are introduced in [Chapter 1](#) and further discussed in [Chapters 12](#), [16](#), and [Appendix I](#). [Chapter 3](#) now includes a detailed explanation of the mechanism by which mutations in innate sensors of infection can lead to autoimmunity. New developments in the identification of ligands for $\gamma\delta$ T cells are highlighted in [Chapter 4](#), and the new cryo-electron microscopy (cryo-EM) structure of the RAG-1:RAG-2 complex bound to DNA is described in [Chapter 5](#). Recently discovered structural features of the peptide-loading machinery for antigen presentation to T cells are included in [Chapter 6](#). The treatment of antigen receptor signaling in [Chapter 7](#) now includes discussion of lymphocyte metabolism, a topic also revisited in the revised [Chapters 9](#) and [11](#). Additionally, new information in [Chapter 7](#) highlights the pathways targeted by checkpoint blockade immunotherapy and describes the signaling modules used to engineer chimeric antigen receptors (CARs) for cell-based therapies, a topic also elaborated in [Chapter 16](#). The latest findings on T-cell localization and dynamics during priming interactions with dendritic cells are covered in [Chapter 9](#), while [Chapter 11](#) describes new insights into the development and functions of tissue-resident memory T cells. [Chapter 12](#), which focuses on immunity at the barrier surfaces, has undergone a major overhaul and includes an expanded discussion of immune interactions with commensal microorganisms and a detailed discussion of the roles

of diverse epithelial cell types in the intestine, respiratory tract, and the skin. [Chapter 12](#) also includes a new section on respiratory virus infections, including influenza and SARS-CoV-2. [Chapter 16](#) brings vaccines to the forefront and includes a detailed discussion of the new vaccine approaches used for SARS-CoV-2 and poliovirus vaccines. The Immunologist's Toolbox ([Appendix I](#)) has been updated with new sections on single-cell technologies such as single-cell RNA-seq and lineage-tracing approaches, antigen tetramers for identifying antigen-specific B cells, and parabiosis used to determine lymphocyte tissue residency. Also included in [Appendix I](#) is a new description of an analytic technique used to measure antibody responses to SARS-CoV-2 in infected or vaccinated individuals. End-of-chapter study questions have been reworked in the Tenth Edition, and an online learning platform for student self-assessment, InQuizitive, has been created.

We have benefited greatly from the major contribution of Gregory Barton, an expert in innate immune signaling, who has revised and updated [Chapter 3](#). Many additional people deserve special thanks, including our thoughtful colleagues who reviewed the Ninth Edition and provided insights that helped direct revisions in this new edition. We benefited from the eagle-eyed accuracy reviewers who provided feedback on Tenth Edition chapters. All of these individuals are cited in the Acknowledgments section, and we express our gratitude for their efforts.

We have the good fortune to work with an outstanding editorial group at W. W. Norton. We thank Denise Schanck, our editorial advisor, who coordinated the entire project, guiding us gently but firmly back on track throughout the process. We thank Betsy Twitchell, our editor, who contributed her guidance, support, and wisdom. Sincere thanks also go to Judith Kromm, our developmental editor, and Christopher Curioli, our copyeditor, for their insights and attention to detail. As in all previous editions, Matt McClements has contributed his genius—and patience—reinterpreting the authors' sketches into elegant illustrations. Our editing, production, and permissions teams provided their expertise throughout; they are Maggie Stephens, David Bradley, Ben Reynolds, Stephanie Romeo, and Patricia Wong. Our media experts, Kate Brayton, Jasmine Ribeaux, Alexandra Malakhoff, and Kara Zaborowsky, created our excellent supplements package. The authors wish to thank their most important partners—Theresa, Cindy Lou, and Charles—colleagues in life who have supported this effort with their generosity of time, their own editorial insights, and their infinite patience.

We hope this Tenth Edition will continue to inspire students to appreciate immunology's beautiful subtlety—just as Charlie did. We encourage all readers to share with us their views on where we have come up short, so the next edition will further approach the asymptote. Happy reading and learning!

Kenneth Murphy

Casey Weaver

Leslie Berg

What's New in the Tenth Edition

New Coauthor

Janeway's Immunobiology is the premier text for immunology, giving students at all levels a current and authoritative understanding of immunology research. Leslie Berg joins Ken Murphy and Casey Weaver in comprehensively updating the textbook.

New Online Assessment in InQuizitive

InQuizitive for *Janeway's Immunobiology* is an award-winning, adaptive learning tool that gives students personalized learning pathways as they work through the material. Through a variety of question types and answer-specific feedback, students learn the necessary vocabulary to succeed in the immunology course and practice applying that vocabulary to core concepts.

New Norton Teaching Tools Support Classroom Instruction

The Norton Teaching Tools site for *Janeway's Immunobiology* gives instructors ready access to primary literature suggestions for each chapter, active learning activities, lecture PowerPoint files, descriptions of all the animations and videos, and much more.

New Coverage of the COVID-19 Pandemic

COVID-19 coverage has been incorporated in [Chapters 1](#), [12](#), [16](#), and [Appendix I](#) and is highlighted on the front cover with a new illustration of the novel mRNA vaccination strategy.

Highlights of New Research

[Chapter 1](#):

- Brief introduction to coronavirus, which is then elaborated in later chapters.

[Chapter 3](#):

- Latest research on the Toll-like receptors and cGAS–STING pathway.
- Discussion of mutations in RIG-I, MDA-5, and cGAS–STING pathways that lead to autoimmune disorders.
- Latest information on inflammasome activation and gasdermin pore formation in secretion of inflammatory cytokines.

Chapter 5:

- New cryo-EM structure of the RAG-1:RAG-2 tetrameric complex bound to recombination signal sequences in the DNA.
- Discussion of the latest research on the evolutionary origins of RAG-dependent V(D)J recombination.

Chapter 6:

- Newly discovered structural features of the peptide-loading machinery for TAP and the peptide-loading complex (PLC).

Chapter 7:

- Latest findings on the metabolic impact of antigen receptor stimulation.
- Up-to-date research on immune checkpoint therapy with CTLA-4 and PD-1 blockade.
- New section on chimeric antigen receptors for CAR T-cell cancer immunotherapy (CAR-T therapy).

Chapter 8:

- New discussion of development of ILCs and nonconventional T cells, including regulatory T cells, iNKT cells, and CD8 α intraepithelial lymphocytes.

Chapter 9:

- Addition of new information on dendritic-cell subsets.
- New section on metabolic control of T-cell activation and differentiation.
- Latest developments in spatiotemporal control of T-cell priming by dendritic cells.
- New insights into how IL-2 signaling regulates T-cell differentiation.

Chapter 11:

- New emphasis on the relationship between antibody isotypes and immune effector modules.
- Updates on T_{FH} cells and tissue-resident memory T cells.
- New research on T-cell metabolic reprogramming in formation and maintenance of T-cell memory.

Chapter 12:

- Major revision that includes more than 25 new or revised illustrations.
- Detailed discussion of the emergence of the COVID-19 pandemic and basis for viral tropism for host cells.
- Major expansion of discussion of mucosal immunity in the intestines, lungs, and skin.
- Expanded consideration of the interplay of barrier tissues with the commensal microbiota.
- Enhanced treatment of specialized epithelial cell types in barrier tissues, including tuft cells, goblet cells, Paneth cells, enteroendocrine cells, and others.
- Extended treatment of intraepithelial lymphocytes (IELs).
- New section introducing the latest findings in neuroimmune regulation.

Chapter 13:

- Expanded treatment and new figures on HIV/AIDS, including the latest findings on CCR5- and CXCR4-tropic HIV variants and use of combination antiretroviral therapy (cART).
- Latest structural information on broadly neutralizing antibodies against HIV envelope protein.
- Expanded and updated coverage of immune evasion by pathogens, including targeting of complement and interferon pathways.
- New section on viral mechanisms promoting exhaustion of T cells and NK cells.
- Updated research on mechanisms used by viral pathogens to subvert host cytokine and chemokine responses.

Chapter 16:

- Vaccines are brought to the forefront, illustrating new vaccine approaches to COVID-19 and modern vaccine technologies for poliovirus eradication.
- New discussion of checkpoint blockade and CAR T-cell technology for cancer immunotherapy.

Appendix I:

- Multiplex bead assay for flow cytometric analysis of antibody responses and its use to measure serum antibodies to SARS-CoV-2.
- Description of antigen tetramers for identification of antigen-specific B cells.
- New description of parabiosis for analysis of lymphocyte tissue residency.
- New discussion of cell lineage tracing techniques.
- Description of techniques for single-cell molecular analyses.

Resources for Instructors and Students

Instructor Resources

Instructor resources for *Janeway's Immunobiology* are available at www.norton.com

Art of *Janeway's Immunobiology*, Tenth Edition

All the images from the book are available in both labeled and unlabeled versions in PowerPoint and JPEG formats.

PowerPoint Lecture Slides

The section headings, concept headings, and figures from the text have been integrated into customizable PowerPoint presentations.

InQuizitive

InQuizitive is Norton's award-winning, easy-to-use adaptive learning tool that personalizes the learning experience for students, helping them to master—and retain—key learning objectives. The InQuizitive course for *Janeway's Immunobiology*, Tenth Edition, includes dynamic question types focused on vocabulary and critical thinking, with clickable glossary definitions throughout. Through a variety of question types, answer-specific feedback, and pedagogical elements that promote metacognition, students are motivated to keep working until they've mastered both the concepts and the tough terminology critical to success in the immunology course. As a result, students arrive better prepared for class, giving instructors more time for discussion and activities. Because InQuizitive motivates students to keep working on a chapter until they've mastered the content, assigning it consistently can help improve course performance. A robust activity report makes it easy to identify challenging concepts, and the convenience of learning management system (LMS) integration saves time by allowing InQuizitive scores to report directly to an LMS gradebook.

Norton Teaching Tools

Searchable by chapter, phrase, or topic, the Norton Teaching Tools site is focused on resources for active learning. Each chapter offers primary literature papers with questions for classroom discussion, concept map activities to help students connect and understand

complex immunological processes, relevant animations with suggestions on how to integrate them into a course, and more. For instructors looking to seamlessly weave case studies into their course, there are case study companion PowerPoints with integrated clicker questions, which will also be included in *Squarecap*, a next-generation classroom response system that promotes a high level of student engagement in the classroom through quality assessment.

Test Bank

Written by Debby Walser-Kuntz, Samira Aghlara-Fotovvat, Omid Veisheh, and Cynthia Leifer, the revised and expanded Test Bank includes more than 900 questions in a variety of question formats. Questions are classified by section and difficulty, making it easy to construct tests and quizzes that are meaningful and diagnostic. Norton Testmaker brings Norton's high-quality testing materials online. Create assessments for a course from anywhere with an Internet connection, without downloading files or installing specialized software. Search and filter Test Bank questions by chapter, question type, difficulty, learning objective, and Bloom's Taxonomy. Instructors can also edit questions or create their own. Then, easily export tests to Microsoft Word or Common Cartridge files for import into an LMS.

Resources for an LMS

Easily add high-quality Norton digital resources to an online, hybrid, or lecture course. Get started building a course with our easy-to-use integrated resources; all activities can be accessed right within an existing learning management system.

Student Resources

Student resources for *Janeway's Immunobiology* are available at digital.wwnorton.com/janeway10

Ebook for *Janeway's Immunobiology*

Norton Ebooks offer an enhanced reading experience at a fraction of the cost of a print textbook. They provide an active reading experience, enabling students to take notes, bookmark, search, highlight, and even read offline. Students can access the animations directly from the ebook, and clickable key terms are linked to their definitions. Instructors can even add notes that students see as they read the text. Norton Ebooks can be viewed on—and synced across—all computers and mobile devices and can be made available for offline reading.



Case Studies in Immunology, Seventh Edition Ebook

by Raif Geha and Luigi Notarangelo

Case Studies in Immunology: A Clinical Companion presents major topics of immunology through a selection of clinical cases that reinforce and extend the basic science. Each case history is preceded by essential scientific facts about the immunological mechanisms of that specific disorder. The cases themselves demonstrate how immunological problems are deconstructed in the clinic, and each one is followed by a concise summary of the clinical finding with questions that serve as discussion points. Cases are color coded to correspond to *Janeway's Immunobiology*, allowing instructors to easily find cases that complement the material in the main text. In *Janeway's Immunobiology*, diseases covered in *Case Studies* are indicated by an icon in the margin. Access to the *Case Studies in Immunology* ebook is free with every purchase of *Janeway's Immunobiology*, Tenth Edition. It is also available for purchase separately.

Animations

A set of more than three dozen animations covers some of the most important immunology concepts, such as innate recognition of pathogens, V(D)J recombination, TCR signaling, and many more. Animation-based questions in InQuizitive allow students to practice their understanding of the information presented in the animations, while links to animations in the ebook help bring these topics to life as the student reads.

Acknowledgments

We thank our spouses, families, friends, and colleagues for their continuing patience and support, without which the writing of this new edition of our textbook would not be possible. As always, we are also indebted to a large number of scientists whose generous help has been essential for making the text as clear, up to date, and accurate as possible.

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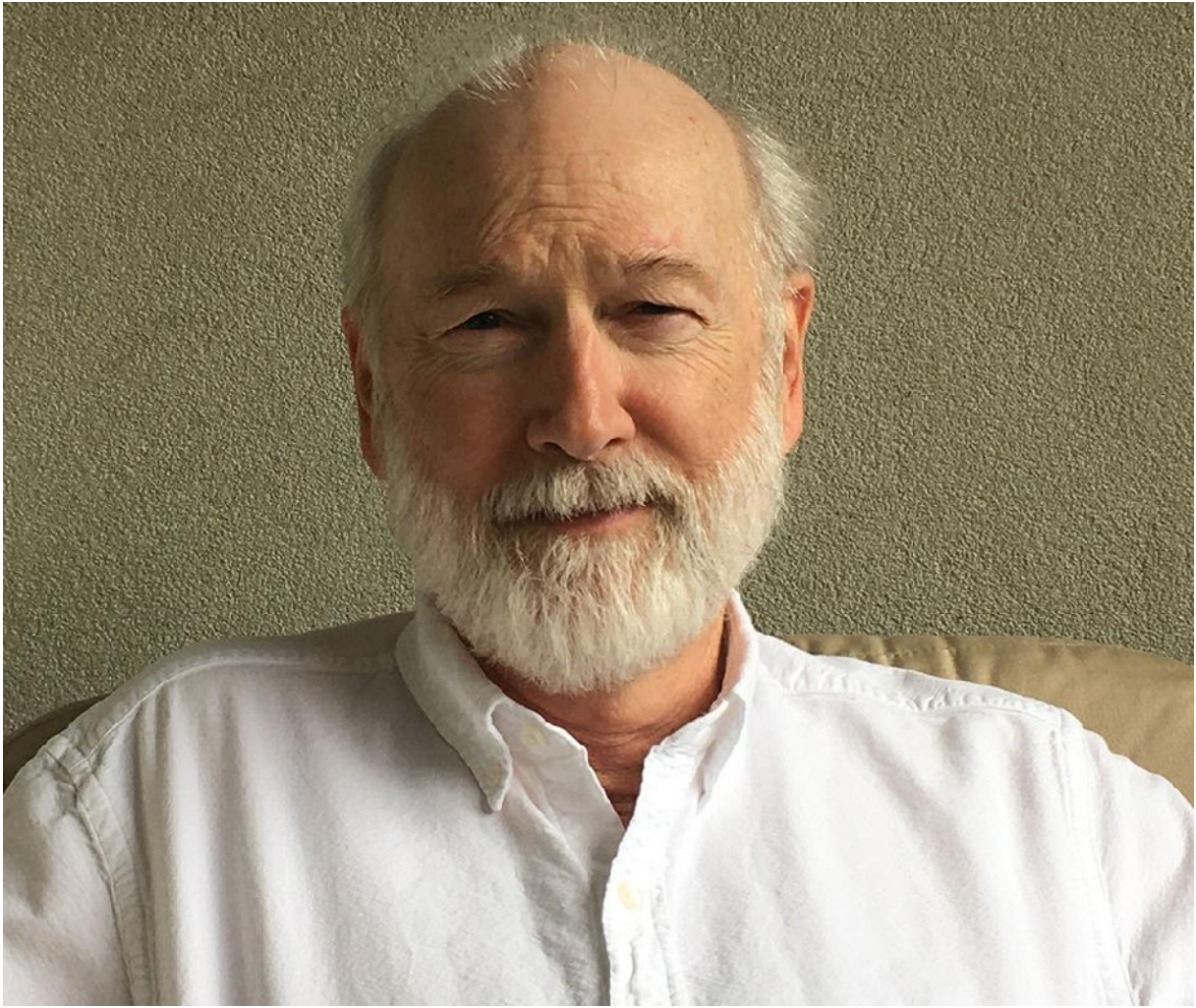
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Leslie Berg is Professor and Chair of the Department of Immunology & Microbiology at the University of Colorado Anschutz School of Medicine. Her research focuses on understanding T-cell receptor signaling and how variations in T-cell receptor signal strength regulate T-cell development, activation, and effector-cell differentiation.

Contents in Brief

[PREFACE](#) [iii](#)

1. [What's New in the Tenth Edition](#) [iv](#)
2. [Resources for Instructors and Students](#) [vi](#)
3. [Acknowledgments](#) [vii](#)
4. [About the Authors](#) [ix](#)

[PART I AN INTRODUCTION TO IMMUNOBIOLOGY AND INNATE IMMUNITY](#)

1. [Chapter 1](#)[Basic Concepts in Immunology](#) [1](#)
2. [Chapter 2](#)[Innate Immunity: The First Lines of Defense](#) [37](#)
3. [Chapter 3](#)[Cellular Mechanisms of Innate Immunity](#) [79](#)

[PART II THE RECOGNITION OF ANTIGEN](#)

1. [Chapter 4](#)[Antigen Recognition by B-cell and T-cell Receptors](#) [141](#)
2. [Chapter 5](#)[The Generation of Lymphocyte Antigen Receptors](#) [179](#)
3. [Chapter 6](#)[Antigen Presentation to T Lymphocytes](#) [217](#)

[PART III THE DEVELOPMENT OF MATURE LYMPHOCYTE RECEPTOR REPERTOIRES](#)

1. [Chapter 7](#)[Lymphocyte Receptor Signaling](#) [261](#)
2. [Chapter 8](#)[The Development of B and T Lymphocytes](#) [301](#)

[PART IV THE ADAPTIVE IMMUNE RESPONSE](#)

1. [Chapter 9](#)[T Cell–Mediated Immunity](#) [347](#)
2. [Chapter 10](#)[The Humoral Immune Response](#) [405](#)
3. [Chapter 11](#)[Integrated Dynamics of Innate and Adaptive Immunity](#) [453](#)
4. [Chapter 12](#)[The Barrier Immune System](#) [505](#)

[PART V THE IMMUNE SYSTEM IN HEALTH AND DISEASE](#)

1. [Chapter 13](#)[Failures of Host Defense Mechanisms](#) [587](#)
2. [Chapter 14](#)[Allergic Diseases and Hypersensitivity Reactions](#) [663](#)
3. [Chapter 15](#)[Autoimmunity and Transplantation](#) [699](#)
4. [Chapter 16](#)[Manipulation of the Immune Response](#) [749](#)

[APPENDICES](#)

Note: Appendices II-IV are available on digital.wwnorton.com/janeway10 and in the ebook.

[Appendix I](#)[The Immunologist's Toolbox](#) [A-1](#)

[Appendix II](#)[CD Antigens](#) [A-49](#)

[Appendix III](#)[Cytokines and Their Receptors](#) [A-69](#)

[Appendix IV](#)[Chemokines and Their Receptors](#) [A-73](#)

[Glossary](#) [G-1](#)

[Credits](#) [C-1](#)

[Index](#) [I-1](#)

Detailed Contents

[Preface](#) [iii](#)

1. [What's New in the Tenth Edition](#) [iv](#)
2. [Resources for Instructors and Students](#) [vi](#)
3. [Acknowledgments](#) [vii](#)
4. [About the Authors](#) [ix](#)

[PART I AN INTRODUCTION TO IMMUNOBIOLOGY AND INNATE IMMUNITY](#)

1. [Chapter 1 Basic Concepts in Immunology](#) [1](#)
2. [The origins of vertebrate immune cells.](#) [2](#)
3. [Principles of innate immunity.](#) [3](#)
4. [1-1 Commensal organisms cause little host damage while pathogens damage host tissues by a variety of mechanisms.](#) [3](#)
5. [1-2 Anatomic and chemical barriers are the first defense against pathogens.](#) [5](#)
6. [1-3 The immune system is activated by inflammatory inducers that indicate the presence of pathogens or tissue damage.](#) [6](#)
7. [1-4 The myeloid lineage comprises most of the cells of the innate immune system.](#) [7](#)
8. [1-5 Sensor cells express pattern-recognition receptors that provide an initial discrimination between self and nonself.](#) [8](#)
9. [1-6 Sensor cells induce an inflammatory response by producing mediators such as chemokines and cytokines.](#) [9](#)
10. [1-7 Innate lymphoid cells and natural killer cells are effector cells that share similarities with lymphoid lineages of the adaptive immune system.](#) [11](#)
11. [Summary.](#) [11](#)
12. [Principles of adaptive immunity.](#) [11](#)
13. [1-8 The interaction of antigens with antigen receptors induces lymphocytes to acquire effector and memory activity.](#) [12](#)
14. [1-9 Antibodies and T-cell receptors are composed of constant and variable regions that provide distinct functions.](#) [13](#)
15. [1-10 Antibodies and T-cell receptors recognize antigens by fundamentally different mechanisms.](#) [13](#)
16. [1-11 Antigen-receptor genes are assembled by somatic gene rearrangements of incomplete receptor gene segments.](#) [14](#)
17. [1-12 Lymphocytes activated by antigen give rise to clones of antigen-specific effector cells that mediate adaptive immunity.](#) [15](#)
18. [1-13 Lymphocytes with self-reactive receptors are normally eliminated during development or are functionally inactivated.](#) [16](#)
19. [1-14 Lymphocytes mature in the bone marrow or the thymus and then congregate in lymphoid tissues throughout the body.](#) [16](#)

20. [1-15 Adaptive immune responses are initiated by antigen and antigen-presenting cells in peripheral lymphoid tissues. 17](#)
21. [1-16 Lymphocytes encounter and respond to antigen in the peripheral lymphoid organs. 18](#)
22. [1-17 Mucosal surfaces have specialized immune structures that orchestrate responses to environmental microbial encounters. 20](#)
23. [1-18 Lymphocytes activated by antigen proliferate in the peripheral lymphoid organs, generating effector cells and immunological memory. 22](#)
24. [Summary. 24](#)
25. [The effector mechanisms of immunity. 25](#)
26. [1-19 Innate immune responses can select from several effector modules to protect against different types of pathogens. 26](#)
27. [1-20 Antibodies protect against extracellular pathogens and their toxic products. 27](#)
28. [1-21 T cells orchestrate cell-mediated immunity and regulate B-cell responses to most antigens. 29](#)
29. [1-22 Inherited and acquired defects in the immune system result in increased susceptibility to infection. 31](#)
30. [1-23 Understanding adaptive immune responses is important for the control of allergies, autoimmune disease, and the rejection of transplanted organs. 32](#)
31. [1-24 Vaccination is the most effective means of controlling infectious diseases. 33](#)
32. [1-25 Emerging pathogens present new challenges for the immune system. 33](#)
33. [Summary. 34](#)
34. [Summary to Chapter 1. 35](#)
35. [References. 35](#)
36. [Chapter 2 Innate Immunity: The First Lines of Defense 37](#)
37. [Anatomic barriers and initial chemical defenses. 38](#)
38. [2-1 Infectious diseases are caused by diverse living agents that replicate in their hosts. 38](#)
39. [2-2 Epithelial surfaces of the body provide the first barrier against infection. 42](#)
40. [2-3 Infectious agents must overcome innate host defenses to establish a focus of infection. 44](#)
41. [2-4 Epithelial cells and phagocytes produce several kinds of antimicrobial molecules. 45](#)
42. [Summary. 49](#)
43. [The complement system and innate immunity. 49](#)
44. [2-5 The complement system recognizes features of microbial surfaces and marks them for destruction by coating them with C3b. 51](#)
45. [2-6 The lectin pathway uses soluble receptors that recognize microbial surfaces to activate the complement cascade. 54](#)
46. [2-7 The classical pathway is initiated by activation of the C1 complex and is homologous to the lectin pathway. 57](#)

47. [2-8 Complement activation is largely confined to the surface on which it is initiated. 58](#)
48. [2-9 The alternative pathway is an amplification loop for C3b formation that is accelerated by properdin in the presence of pathogens. 59](#)
49. [2-10 Membrane and plasma proteins that regulate the formation and stability of C3 convertases determine the extent of complement activation. 61](#)
50. [2-11 Complement developed early in the evolution of multicellular organisms. 62](#)
51. [2-12 Surface-bound C3 convertase deposits large numbers of C3b fragments on pathogen surfaces and generates C5 convertase activity. 63](#)
52. [2-13 Ingestion of complement-tagged pathogens by phagocytes is mediated by receptors for the bound complement proteins. 64](#)
53. [2-14 The small fragments of some complement proteins initiate a local inflammatory response. 66](#)
54. [2-15 The terminal complement proteins polymerize to form pores in membranes that can kill certain pathogens. 66](#)
55. [2-16 Complement-control proteins regulate all three pathways of complement activation and protect the host from their destructive effects. 68](#)
56. [2-17 Genetic and acquired disorders in complement regulation can produce various inflammatory conditions. 71](#)
57. [2-18 Pathogens produce several types of proteins that can inhibit complement activation. 73](#)
58. [Summary. 74](#)
59. [Summary to Chapter 2. 74](#)
60. [Discussion questions. 75](#)
61. [References. 75](#)
62. [Chapter 3 Cellular Mechanisms of Innate Immunity 79](#)
63. [Pathogen recognition by cells of the innate immune system. 79](#)
64. [3-1 After entering tissues, many microbes are recognized, ingested, and killed by phagocytes. 80](#)
65. [3-2 G protein-coupled receptors on phagocytes link microbe recognition with increased efficiency of intracellular killing. 83](#)
66. [3-3 Microbial recognition and tissue damage initiate an inflammatory response. 87](#)
67. [3-4 Toll-like receptors represent an ancient pathogen-recognition system. 90](#)
68. [3-5 Mammalian Toll-like receptors are activated by many different pathogen-associated molecular patterns. 90](#)
69. [3-6 TLR-4 recognizes bacterial lipopolysaccharide in association with the host accessory proteins MD-2 and CD14. 94](#)
70. [3-7 TLRs activate NF \$\kappa\$ B, AP-1, and IRF transcription factors to induce the expression of inflammatory cytokines and type I interferons. 95](#)
71. [3-8 The RIG-I-like receptors detect cytoplasmic viral RNAs and activate MAVS to induce type I interferon production and pro-inflammatory cytokines. 98](#)

72. [3-9 Cytosolic DNA activates the cGAS–STING pathway to induce production of type I interferons. 100](#)
73. [3-10 NLRs comprise a large family of intracellular sensors with diverse functions. 102](#)
74. [3-11 Certain NLR proteins react to infection or cellular damage by forming an inflammasome that induces cell death and secretion of inflammatory cytokines. 103](#)
75. [3-12 Activation of innate sensors in macrophages and dendritic cells triggers changes in gene expression that have far-reaching effects on the immune response. 106](#)
76. [3-13 Toll signaling in *Drosophila* is downstream of a distinct set of pathogen-recognition molecules. 107](#)
77. [3-14 TLR and NOD genes have undergone extensive diversification in both invertebrates and some primitive chordates. 108](#)
78. [Summary. 109](#)
79. [Consequences of innate immune activation. 109](#)
80. [3-15 Cytokines and their receptors fall into distinct families of structurally related proteins. 110](#)
81. [3-16 Cytokine receptors of the hematopoietin superfamily are associated with the JAK family of tyrosine kinases, which activate STAT transcription factors. 112](#)
82. [3-17 Chemokines released by macrophages and dendritic cells recruit effector cells to sites of infection. 114](#)
83. [3-18 Cell-adhesion molecules control interactions between leukocytes and endothelial cells during an inflammatory response. 116](#)
84. [3-19 Neutrophils make up the first wave of cells that cross the blood vessel wall to enter an inflamed tissue. 118](#)
85. [3-20 TNF- \$\alpha\$ is an important cytokine that triggers local containment of infection but induces shock when released systemically. 120](#)
86. [3-21 Cytokines made by macrophages and dendritic cells induce a systemic reaction known as the acute-phase response. 122](#)
87. [3-22 Interferons induced by viral infection make several contributions to host defense. 124](#)
88. [3-23 Several types of innate lymphoid cells provide protection in early infection. 126](#)
89. [3-24 NK cells are activated by type I interferon and macrophage-derived cytokines. 128](#)
90. [3-25 NK cells express activating and inhibitory receptors to distinguish between healthy and infected cells. 129](#)
91. [3-26 NK-cell receptors belong to several structural families. 130](#)
92. [3-27 NK cells express activating receptors that recognize ligands induced on infected cells or tumor cells. 133](#)
93. [Summary. 134](#)
94. [Summary to Chapter 3. 134](#)

95. [Discussion questions. 135](#)

96. [References. 136](#)

[PART II THE RECOGNITION OF ANTIGEN](#)

1. [Chapter 4 Antigen Recognition by B-cell and T-cell Receptors 141](#)
2. [The structure of a typical antibody molecule. 142](#)
3. [4-1 IgG antibodies consist of four polypeptide chains. 143](#)
4. [4-2 Immunoglobulin heavy and light chains are composed of constant and variable regions. 144](#)
5. [4-3 The domains of an immunoglobulin molecule have similar structures. 144](#)
6. [4-4 The antibody molecule can readily be cleaved into functionally distinct fragments. 146](#)
7. [4-5 The hinge region of the immunoglobulin molecule allows flexibility in binding to multiple antigens. 147](#)
8. [Summary. 147](#)
9. [Structural variation in immunoglobulin constant regions. 148](#)
10. [4-6 Different classes of immunoglobulins are distinguished by the structure of their heavy-chain constant regions. 148](#)
11. [4-7 The constant region confers functional specialization on the antibody. 150](#)
12. [4-8 IgM and IgA can form polymers by interacting with the J chain. 151](#)
13. [Summary. 152](#)
14. [The interaction of the antibody molecule with specific antigen. 152](#)
15. [4-9 Localized regions of hypervariable sequence form the antigen-binding site. 152](#)
16. [4-10 Antibodies bind antigens via contacts in CDRs that are complementary to the size and shape of the antigen. 153](#)
17. [4-11 Antibodies bind to conformational shapes on the surfaces of antigens using a variety of noncovalent forces. 155](#)
18. [4-12 Antibody interaction with intact antigens is influenced by steric constraints. 156](#)
19. [4-13 Some species generate antibodies with alternative structures. 157](#)
20. [Summary. 158](#)
21. [Antigen recognition by T cells. 159](#)
22. [4-14 The TCR \$\alpha\beta\$ heterodimer is very similar to a Fab fragment of immunoglobulin. 159](#)
23. [4-15 A T-cell receptor recognizes antigen in the form of a complex of a foreign peptide bound to an MHC molecule. 161](#)
24. [4-16 There are two classes of MHC molecules with distinct subunit compositions but similar three-dimensional structures. 162](#)
25. [4-17 Peptides are stably bound to MHC molecules and also serve to stabilize the MHC molecule on the cell surface. 164](#)
26. [4-18 MHC class I molecules bind short peptides of 8–10 amino acids by both ends. 165](#)

27. [4-19 The length of the peptides bound by MHC class II molecules is not constrained. 166](#)
28. [4-20 The crystal structures of several peptide:MHC:T-cell receptor complexes show a similar orientation of the T-cell receptor over the peptide:MHC complex. 168](#)
29. [4-21 The CD4 and CD8 cell-surface proteins of T cells directly contact MHC molecules and are required to make an effective response to antigen. 169](#)
30. [4-22 The two classes of MHC molecules are expressed differentially on cells. 172](#)
31. [4-23 A distinct subset of T cells bears an alternative receptor made up of \$\gamma\$ and \$\delta\$ chains. 173](#)
32. [Summary. 174](#)
33. [Summary to Chapter 4. 174](#)
34. [Discussion questions. 175](#)
35. [References. 176](#)
36. [Chapter 5 The Generation of Lymphocyte Antigen Receptors 179](#)
37. [Primary immunoglobulin gene rearrangements and expression. 180](#)
38. [5-1 Complete immunoglobulin genes are generated by the somatic recombination of separate gene segments. 180](#)
39. [5-2 Multiple contiguous V gene segments are present at each immunoglobulin locus. 182](#)
40. [5-3 Rearrangement of V, D, and J gene segments is guided by flanking DNA sequences. 184](#)
41. [5-4 The reaction that recombines V, D, and J gene segments involves both lymphocyte-specific and ubiquitous DNA-modifying enzymes. 187](#)
42. [5-5 The diversity of the immunoglobulin repertoire is generated by four main processes. 190](#)
43. [5-6 The multiple inherited gene segments are used in different combinations. 191](#)
44. [5-7 Variable addition and subtraction of nucleotides at the junctions between gene segments contributes to the diversity of the third hypervariable region. 191](#)
45. [5-8 IgM and IgD are derived from the same pre-mRNA transcript and are both expressed on the surface of mature B cells. 193](#)
46. [5-9 Transmembrane and secreted forms of immunoglobulin are generated from different heavy-chain mRNA transcripts. 194](#)
47. [Summary. 196](#)
48. [T-cell receptor gene rearrangement. 196](#)
49. [5-10 The T-cell receptor gene segments are arranged in a pattern similar to that of immunoglobulin gene segments and are rearranged by the same enzymes. 196](#)
50. [5-11 T-cell receptors concentrate diversity in the third hypervariable region. 199](#)
51. [5-12 \$\gamma\delta\$ T-cell receptors are also generated by gene rearrangement. 200](#)
52. [Summary. 201](#)

53. [Evolution of the adaptive immune response. 201](#)
54. [5-13 Some invertebrates generate extensive diversity in a repertoire of immunoglobulin-like genes. 202](#)
55. [5-14 Agnathans possess an adaptive immune system that uses somatic gene rearrangement to diversify receptors built from LRR domains. 203](#)
56. [5-15 RAG-dependent adaptive immunity based on a diversified repertoire of immunoglobulin-like genes appeared abruptly in the cartilaginous fishes. 205](#)
57. [5-16 Different species generate immunoglobulin diversity in different ways. 207](#)
58. [5-17 Both \$\alpha\beta\$ and \$\gamma\delta\$ T-cell receptors are present in cartilaginous fishes. 210](#)
59. [5-18 MHC class I and class II molecules are also first found in the cartilaginous fishes. 210](#)
60. [Summary. 211](#)
61. [Summary to Chapter 5. 211](#)
62. [Discussion questions. 212](#)
63. [References. 213](#)
64. [Chapter 6 Antigen Presentation to T Lymphocytes 217](#)
65. [The generation of \$\alpha\beta\$ T-cell receptor ligands. 218](#)
66. [6-1 Antigen presentation functions both in arming effector T cells and in triggering their effector functions to attack pathogen-infected cells. 218](#)
67. [6-2 Peptides are generated from ubiquitinated proteins in the cytosol by the proteasome. 220](#)
68. [6-3 Peptides from the cytosol are transported by TAP into the endoplasmic reticulum and further processed before binding to MHC class I molecules. 222](#)
69. [6-4 Newly synthesized MHC class I molecules are retained in the endoplasmic reticulum until they bind a peptide. 224](#)
70. [6-5 Dendritic cells use cross-presentation to present exogenous proteins on MHC class I molecules so as to prime CD8 T cells. 226](#)
71. [6-6 Peptide:MHC class II complexes are generated in acidified endocytic vesicles from proteins obtained through endocytosis, phagocytosis, and autophagy. 227](#)
72. [6-7 The invariant chain directs newly synthesized MHC class II molecules to acidified intracellular vesicles. 229](#)
73. [6-8 The MHC class II-like molecules HLA-DM and HLA-DO regulate exchange of CLIP for other peptides. 230](#)
74. [6-9 Cessation of MHC class II antigen processing occurs after dendritic-cell activation through reduced expression of the MARCH-1 E3 ligase. 233](#)
75. [Summary. 234](#)
76. [The major histocompatibility complex and its function. 235](#)
77. [6-10 Many proteins involved in antigen processing and presentation are encoded by genes within the MHC. 235](#)
78. [6-11 The protein products of MHC class I and class II genes are highly polymorphic. 238](#)
79. [6-12 MHC polymorphism affects antigen recognition by T cells by](#)

[influencing both peptide binding and the contacts between the T-cell receptor and MHC molecule. 240](#)

80. [6-13 Alloreactive T cells recognizing nonself MHC molecules are very abundant. 243](#)
81. [6-14 Many T cells respond to superantigens. 245](#)
82. [6-15 MHC polymorphism extends the range of antigens to which the immune system can respond. 246](#)
83. [Summary. 246](#)
84. [Recognition of nonpeptide ligands by unconventional T-cell subsets. 247](#)
85. [6-16 A variety of genes with specialized functions in immunity are also encoded in the MHC. 247](#)
86. [6-17 Specialized MHC class I molecules act as ligands for the activation and inhibition of NK cells and unconventional T-cell subsets. 249](#)
87. [6-18 Members of the CD1 family of MHC class I-like molecules present lipid-based antigens to NKT cells. 251](#)
88. [6-19 The nonclassical MHC class I molecule MR1 presents microbial riboflavin metabolites to MAIT cells. 253](#)
89. [6-20 \$\gamma\delta\$ T cells can recognize a variety of diverse ligands. 254](#)
90. [Summary. 255](#)
91. [Summary to Chapter 6. 255](#)
92. [Discussion questions. 256](#)
93. [References. 257](#)

[PART III THE DEVELOPMENT OF MATURE LYMPHOCYTE RECEPTOR REPERTOIRES](#)

1. [Chapter 7 Lymphocyte Receptor Signaling 261](#)
2. [General principles of signal transduction and propagation. 261](#)
3. [7-1 Transmembrane receptors convert extracellular signals into intracellular biochemical events. 262](#)
4. [7-2 Intracellular signal propagation is mediated by large multiprotein signaling complexes. 263](#)
5. [7-3 Small G proteins act as molecular switches in many different signaling pathways. 264](#)
6. [7-4 Signaling proteins are recruited to the membrane by a variety of mechanisms. 265](#)
7. [7-5 Post-translational modifications of proteins can both activate and inhibit signaling responses. 266](#)
8. [7-6 The activation of many receptors generates small-molecule second messengers. 267](#)
9. [Summary. 268](#)
10. [Antigen receptor signaling and lymphocyte activation. 268](#)
11. [7-7 Antigen receptors consist of variable antigen-binding chains associated with invariant chains that carry out the signaling function of the receptor. 268](#)
12. [7-8 Antigen recognition by the T-cell receptor and its co-receptors transduces a signal across the plasma membrane to initiate signaling. 270](#)

13. [7-9 Antigen recognition by the T-cell receptor and its co-receptors leads to phosphorylation of ITAMs by Src family kinases, generating the first intracellular signal in a signaling cascade. 272](#)
14. [7-10 Phosphorylated ITAMs recruit and activate the tyrosine kinase ZAP-70. 273](#)
15. [7-11 ITAMs are also found in other receptors on leukocytes that signal for cell activation. 274](#)
16. [7-12 Activated ZAP-70 phosphorylates adaptor proteins and promotes PI 3-kinase activation. 275](#)
17. [7-13 Activated PLC- \$\gamma\$ generates the second messengers diacylglycerol and inositol trisphosphate that lead to transcription factor activation. 276](#)
18. [7-14 \$\text{Ca}^{2+}\$ entry activates the transcription factor NFAT. 277](#)
19. [7-15 Ras activation stimulates the mitogen-activated protein kinase \(MAPK\) relay and induces expression of the transcription factor AP-1. 278](#)
20. [7-16 Protein kinase C activates the transcription factors NF \$\kappa\$ B and AP-1. 280](#)
21. [7-17 PI 3-kinase activation up-regulates cellular metabolic pathways via the serine/threonine kinase Akt. 281](#)
22. [7-18 T-cell receptor signaling leads to enhanced integrin-mediated cell adhesion. 283](#)
23. [7-19 T-cell receptor signaling induces cytoskeletal reorganization by activating the small GTPase Cdc42. 283](#)
24. [7-20 The logic of B-cell receptor signaling is similar to that of T-cell receptor signaling, but some of the signaling components are specific to B cells. 285](#)
25. [7-21 Antigen receptor signaling is a dynamic process that evolves over time. 287](#)
26. [Summary. 288](#)
27. [Co-stimulatory and inhibitory receptors modulate antigen receptor signaling in T and B lymphocytes. 288](#)
28. [7-22 The cell-surface protein CD28 is a required co-stimulatory signaling receptor for naive T-cell activation. 289](#)
29. [7-23 Maximal activation of PLC- \$\gamma\$, which is important for transcription factor activation, requires a co-stimulatory signal induced by CD28. 290](#)
30. [7-24 TNF receptor superfamily members augment T-cell and B-cell activation. 291](#)
31. [7-25 Inhibitory receptors on lymphocytes down-regulate immune responses by interfering with co-stimulatory signaling pathways. 292](#)
32. [7-26 Inhibitory receptors on lymphocytes down-regulate immune responses by recruiting protein or lipid phosphatases. 293](#)
33. [7-27 Knowledge of lymphocyte receptor signaling pathways can be exploited to engineer chimeric antigen receptors. 295](#)
34. [Summary. 296](#)
35. [Summary to Chapter 7. 297](#)
36. [Discussion questions. 297](#)
37. [References. 298](#)

38. [Chapter 8 The Development of B and T Lymphocytes 301](#)
39. [Development of B lymphocytes. 303](#)
40. [8-1 Lymphocytes derive from hematopoietic stem cells in the bone marrow. 303](#)
41. [8-2 Commitment to B cell, T cell, or innate lymphoid cell lineages is regulated by distinct networks of transcription factors. 305](#)
42. [8-3 B-cell development begins by rearrangement of the heavy-chain locus. 305](#)
43. [8-4 The pre-B-cell receptor tests for successful production of a complete heavy chain and signals for the transition from the pro-B cell to the pre-B cell stage. 308](#)
44. [8-5 Pre-B-cell receptor signaling inhibits further heavy-chain locus rearrangement and enforces allelic exclusion. 310](#)
45. [8-6 Pre-B cells rearrange the light-chain locus and express cell-surface immunoglobulin. 310](#)
46. [8-7 Immature B cells are tested for autoreactivity before they leave the bone marrow. 312](#)
47. [8-8 Lymphocytes that encounter sufficient quantities of self antigens in the periphery are eliminated. 313](#)
48. [8-9 Newly formed B cells arriving in the spleen turn over rapidly and require cytokines and positive signals through the B-cell receptor for maturation and long-term survival. 314](#)
49. [8-10 B-1 B cells are an innate lymphocyte subset that arises early in development. 316](#)
50. [Summary. 317](#)
51. [Development of T lymphocytes. 319](#)
52. [8-11 T-cell progenitors originate in the bone marrow, but all the important events in T-cell development occur in the thymus. 320](#)
53. [8-12 T-cell precursors proliferate extensively in the thymus, but most die there. 321](#)
54. [8-13 Successive stages in the development of thymocytes are marked by changes in cell-surface molecules. 321](#)
55. [8-14 Commitment to the T-cell lineage occurs in the thymus after Notch signaling. 322](#)
56. [8-15 After T-cell commitment, thymocyte development is regulated by T-cell receptor rearrangement, expression, and signaling. 325](#)
57. [8-16 Thymocytes at different developmental stages are found in distinct parts of the thymus. 326](#)
58. [8-17 T cells with \$\alpha\beta\$ or \$\gamma\delta\$ receptors arise from a common progenitor. 326](#)
59. [8-18 T cells expressing \$\gamma\delta\$ T-cell receptors arise in two distinct phases during development. 327](#)
60. [8-19 Successful synthesis of a rearranged \$\beta\$ chain allows the production of a pre-T-cell receptor that triggers cell proliferation and blocks further \$\beta\$ -chain gene rearrangement. 329](#)

61. [8-20 T-cell \$\alpha\$ -chain genes undergo successive rearrangements until positive selection or cell death intervenes. 331](#)
62. [Summary. 332](#)
63. [Positive and negative selection of T cells. 333](#)
64. [8-21 Overview of positive and negative selection of \$\alpha\beta\$ lineage T cells. 333](#)
65. [8-22 Only thymocytes whose receptors interact with self peptide:self MHC complexes can survive and mature. 334](#)
66. [8-23 Positive selection coordinates the expression of CD4 or CD8 with the specificity of the T-cell receptor and the potential effector functions of the T cell. 335](#)
67. [8-24 Thymic cortical epithelial cells mediate positive selection of developing thymocytes. 337](#)
68. [8-25 T cells that react strongly with self antigens are deleted in the thymus. 338](#)
69. [8-26 The specificity and/or the strength of signals for negative and positive selection must differ. 339](#)
70. [8-27 \$\alpha\beta\$ T cells diverge into multiple lineages at the CD4⁺ and CD8⁺ single-positive stage in the thymic medulla. 339](#)
71. [8-28 Thymic emigration is controlled by signaling through a G protein-coupled receptor. 340](#)
72. [8-29 T cells that encounter sufficient quantities of self antigens for the first time in the periphery are eliminated or inactivated. 341](#)
73. [Summary. 341](#)
74. [Summary to Chapter 8. 341](#)
75. [Discussion questions. 342](#)
76. [References. 343](#)

[PART IV THE ADAPTIVE IMMUNE RESPONSE](#)

1. [Chapter 9 T Cell-Mediated Immunity 347](#)
2. [Development and function of secondary lymphoid organs—sites for the initiation of adaptive immune responses. 349](#)
3. [9-1 Secondary lymphoid structures are specialized to facilitate the interaction of circulating T and B lymphocytes with antigen. 349](#)
4. [9-2 The development of secondary lymphoid tissues is initiated by lymphoid tissue inducer cells and cytokines of the tumor necrosis factor family. 351](#)
5. [9-3 T-cell entry into lymphoid tissues occurs across specialized small blood vessels and is mediated by the sequential actions of adhesion molecules and chemokine receptors. 353](#)
6. [9-4 T cells are highly motile within T-cell zones as they survey dendritic cells for cognate antigen. 356](#)
7. [9-5 The exit of T cells from T-cell zones of lymphoid tissues is controlled by a chemotactic lipid. 357](#)
8. [9-6 Activation of conventional dendritic cells stimulates their maturation and migration to T-cell zones to induce naive T-cell activation. 359](#)
9. [9-7 Subsets of conventional dendritic cells are specialized for different](#)

- [modes of antigen presentation. 364](#)
10. [Summary. 366](#)
 11. [Naive T-cell activation, clonal expansion, and differentiation into effector and memory T cells. 367](#)
 12. [9-8 Conventional dendritic cells deliver multiple signals for the activation of naive T cells. 368](#)
 13. [9-9 T-cell activation induces metabolic changes that prepare the T cell for rapid clonal expansion and differentiation. 369](#)
 14. [9-10 The rapid expression of interleukin-2 and the high-affinity component of its receptor allow activated T cells to respond to this cytokine coincident with entry into the cell cycle. 372](#)
 15. [9-11 Additional co-stimulatory and inhibitory receptors are engaged after initial T-cell activation to control clonal expansion and contraction. 373](#)
 16. [9-12 Proliferating T cells differentiate into effector T cells that are programmed for altered tissue homing and loss of a requirement for co-stimulation to act. 375](#)
 17. [9-13 CD8 T cells can be activated in different ways to become cytotoxic effector cells. 376](#)
 18. [9-14 CD4 T cells differentiate into several functionally distinct subsets. 378](#)
 19. [9-15 Cytokines play a central role in guiding the differentiation of CD4 T cells down distinct pathways. 381](#)
 20. [9-16 CD4 T-cell subsets can cross-regulate each other's differentiation through the cytokines they produce. 385](#)
 21. [9-17 Regulatory T cells are a subset of CD4 T cells that restrains immune responses to maintain self-tolerance. 386](#)
 22. [Summary. 388](#)
 23. [General properties of effector T cells and their cytokines. 389](#)
 24. [9-18 An immunological synapse can form between effector T cells and their targets to direct the release of effector molecules. 389](#)
 25. [9-19 The effector functions of T cells are determined by the array of effector molecules they produce, which can act locally or at a distance. 391](#)
 26. [9-20 Effector T cells express several TNF family cytokines. 392](#)
 27. [Summary. 394](#)
 28. [T cell-mediated cytotoxicity. 394](#)
 29. [9-21 Cytotoxic T cells induce target cells to undergo programmed cell death via extrinsic or intrinsic pathways of apoptosis. 394](#)
 30. [9-22 Cytotoxic effector proteins that trigger apoptosis are contained in the granules of cytotoxic CD8 T cells. 398](#)
 31. [9-23 Cytotoxic T cells are selective serial killers of targets expressing a specific antigen. 399](#)
 32. [9-24 Cytotoxic T cells also act by releasing cytokines. 400](#)
 33. [Summary. 400](#)
 34. [Summary to Chapter 9. 400](#)
 35. [Discussion questions. 401](#)

36. [References. 402](#)
37. [Chapter 10 The Humoral Immune Response 405](#)
38. [B-cell activation by antigen and helper T cells. 406](#)
39. [10-1 Activation of B cells by antigen involves signals from the B-cell receptor and either helper T cells or microbial antigens. 406](#)
40. [10-2 Linked recognition of antigen by T cells and B cells promotes robust antibody responses. 408](#)
41. [10-3 B cells that encounter their antigens migrate toward the boundaries between B-cell and T-cell areas in secondary lymphoid tissues. 409](#)
42. [10-4 T cells express surface molecules and cytokines that activate B cells, which in turn promote T_{FH} cell development. 412](#)
43. [10-5 Some activated B cells differentiate into antibody-secreting plasmablasts and plasma cells. 413](#)
44. [10-6 The second phase of a primary B-cell immune response occurs when activated B cells migrate into follicles and proliferate to form germinal centers. 414](#)
45. [10-7 Germinal center B cells undergo V-region somatic hypermutation, and cells with mutations that improve affinity for antigen are selected. 417](#)
46. [10-8 Positive selection of germinal center B cells involves contact with T_{FH} cells and CD40 signaling. 419](#)
47. [10-9 Activation-induced cytidine deaminase introduces mutations into genes transcribed in B cells. 420](#)
48. [10-10 Mismatch and base-excision repair pathways contribute to somatic hypermutation after initiation by AID. 421](#)
49. [10-11 AID initiates class switching to allow the same assembled V_H exon to be associated with different C_H genes in the course of an immune response. 423](#)
50. [10-12 Cytokines made by T_H cells direct the choice of isotype for class switching in T-dependent antibody responses. 425](#)
51. [10-13 B cells that survive the germinal center reaction eventually differentiate into either plasma cells or memory cells. 426](#)
52. [10-14 Some antigens do not require T-cell help to induce antibody responses. 427](#)
53. [Summary. 429](#)
54. [The distributions and functions of immunoglobulin classes. 430](#)
55. [10-15 Antibodies of different classes operate in distinct places and have distinct effector functions. 431](#)
56. [10-16 Polymeric immunoglobulin receptor binds to the Fc regions of IgA and IgM and transports them across epithelial barriers. 432](#)
57. [10-17 The neonatal Fc receptor carries IgG across the placenta and prevents IgG excretion from the body. 433](#)
58. [10-18 High-affinity IgG and IgA antibodies can neutralize toxins and block the infectivity of viruses and bacteria. 434](#)
59. [10-19 Antigen:antibody complexes activate the classical pathway of](#)

- [complement by binding to C1q. 436](#)
60. [10-20 Complement receptors and Fc receptors both contribute to removal of immune complexes from the circulation. 438](#)
 61. [Summary. 439](#)
 62. [The destruction of antibody-coated pathogens via Fc receptors. 440](#)
 63. [10-21 The Fc receptors of accessory cells are signaling receptors specific for immunoglobulins of different classes. 440](#)
 64. [10-22 Fc receptors on phagocytes are activated by antibodies bound to the surface of pathogens and enable the phagocytes to ingest and destroy pathogens. 441](#)
 65. [10-23 Fc receptors activate NK cells to destroy antibody-coated targets. 443](#)
 66. [10-24 Mast cells and basophils bind IgE antibody via the high-affinity Fcε receptor. 444](#)
 67. [10-25 IgE-mediated activation of accessory cells has an important role in resistance to parasite infection. 445](#)
 68. [Summary. 446](#)
 69. [Summary to Chapter 10. 446](#)
 70. [Discussion questions. 447](#)
 71. [References. 448](#)
 72. [Chapter 11 Integrated Dynamics of Innate and Adaptive Immunity 453](#)
 73. [Integration of innate and adaptive immunity in response to specific types of pathogens. 454](#)
 74. [11-1 The course of an infection can be divided into several distinct phases. 454](#)
 75. [11-2 The effector mechanisms that are recruited to clear an infection can be organized into immune modules. 458](#)
 76. [11-3 Subsets of ILCs are early responders of the immune response. 460](#)
 77. [Summary. 462](#)
 78. [Effector CD4 T cells augment the effector functions of innate immune cells. 462](#)
 79. [11-4 Effector T cells are guided to specific tissues and sites of infection by changes in their expression of adhesion molecules and chemokine receptors. 463](#)
 80. [11-5 Pathogen-specific effector T cells are progressively enriched at sites of infection where they may undergo further maturation to acquire full effector function. 467](#)
 81. [11-6 T_H1 cells coordinate and amplify the host response to intracellular pathogens through classical activation of macrophages. 467](#)
 82. [11-7 Activation of macrophages by T_H1 cells must be tightly regulated to avoid tissue damage. 469](#)
 83. [11-8 Chronic activation of macrophages by T_H1 cells mediates the formation of granulomas to contain intracellular pathogens that cannot be cleared. 471](#)
 84. [11-9 Defects in type 1 immunity reveal its important role in the elimination of intracellular pathogens. 471](#)
 85. [11-10 T_H2 cells coordinate type 2 responses to expel intestinal helminths and repair tissue injury. 472](#)

86. [11-11 T_H17 cells coordinate type 3 responses to enhance the clearance of extracellular bacteria and fungi. 475](#)
87. [11-12 Effector T cells continue to respond to maturational and maintenance signals as they carry out their effector functions in target tissues. 476](#)
88. [11-13 Effector T cells can be activated to release cytokines independently of antigen recognition. 478](#)
89. [11-14 Effector CD4 T cells demonstrate plasticity and cooperativity that enable adaptation during antipathogen responses. 479](#)
90. [11-15 Integration of cell- and antibody-mediated immunity is critical for protection against many types of pathogens. 480](#)
91. [11-16 Resolution of an infection is accompanied by the death of most effector T cells and persistence of a small number of long-lived memory T cells. 482](#)
92. [Summary. 483](#)
93. [Immunological memory. 484](#)
94. [11-17 Immunological memory is long lived after infection or vaccination. 484](#)
95. [11-18 MHC tetramers and adoptive transfers of clonal T-cell populations have enabled the study of T-cell memory. 485](#)
96. [11-19 The developmental origins of memory T cells parallel those of effector T cells. 486](#)
97. [11-20 Memory T cells are heterogeneous and include central memory, effector memory, and tissue-resident subsets. 488](#)
98. [11-21 Circulating memory T cells acquire heightened sensitivity to IL-7 and/or IL-15 and undergo metabolic reprogramming to survive long-term. 491](#)
99. [11-22 CD4 T-cell help is required for CD8 T-cell memory and involves CD40 and IL-2 signaling. 492](#)
100. [11-23 Memory B-cell responses are more rapid and have higher affinity for antigen compared with responses of naive B cells. 494](#)
101. [11-24 Memory B cells can reenter germinal centers and undergo additional somatic hypermutation and affinity maturation during secondary immune responses. 496](#)
102. [11-25 In immune individuals, secondary and subsequent responses are mainly attributable to memory lymphocytes. 496](#)
103. [Summary. 498](#)
104. [Summary to Chapter 11. 498](#)
105. [Discussion questions. 500](#)
106. [References. 500](#)
107. [Chapter 12 The Barrier Immune System 505](#)
108. [Organization of the mucosal immune system. 505](#)
109. [12-1 The mucosal immune system protects the internal surfaces of the body. 506](#)
110. [12-2 Immune cells of the mucosal immune system are located both within and outside of specialized lymphoid tissues in proximity to the epithelium. 510](#)
111. [12-3 Maturation of the gut-associated lymphoid tissue is driven by acquisition of the commensal microbiota. 516](#)

- 112. [Summary. 519](#)
- 113. [Innate immune defenses of the intestinal immune system. 520](#)
- 114. [12-4 The intestines are lined by a diversity of epithelial cell types that develop from a common progenitor and play distinct roles in mucosal immunity. 520](#)
- 115. [12-5 The intestinal epithelium contains conventional and unconventional T cells that are focused on barrier maintenance and defense. 528](#)
- 116. [12-6 Innate lymphoid cells and unconventional lymphocytes are present in GALT and in the lamina propria and are rapid responders to microbes that breach the epithelium. 534](#)
- 117. [Summary. 535](#)
- 118. [The role of adaptive immunity in regulating the intestinal mucosal immune system at homeostasis. 536](#)
- 119. [12-7 The mucosal immune system must establish and maintain tolerance to harmless foreign antigens. 536](#)
- 120. [12-8 Macrophages and dendritic cells have complementary roles in the maintenance of immune tolerance in the intestines. 537](#)
- 121. [12-9 The intestines have multiple routes for uptake and delivery of antigen to antigen-presenting cells. 540](#)
- 122. [12-10 Intestinal dendritic cells favor the induction of antigen-specific T_{reg} cells that are critical for the maintenance of mucosal immune homeostasis. 543](#)
- 123. [12-11 Lymphocytes primed within the mucosal immune system are directed to return to the mucosal tissue by tissue-specific adhesion molecules and chemokine receptors. 546](#)
- 124. [12-12 Secretory IgA is the dominant class of antibody associated with the mucosal immune system at homeostasis. 547](#)
- 125. [12-13 T cell-independent processes can contribute to IgA production in some species. 550](#)
- 126. [12-14 IgA deficiency is relatively common in humans but may be compensated for by secretory IgM. 551](#)
- 127. [12-15 Large numbers of antigen-experienced T cells are present in the intestinal lamina propria even in the absence of disease. 551](#)
- 128. [12-16 Priming of lymphocytes in one mucosal tissue may induce protective immunity at other mucosal surfaces. 552](#)
- 129. [Summary. 553](#)
- 130. [The intestinal immune response in host defense and immune-mediated disease. 553](#)
- 131. [12-17 Enteric pathogens elicit 'danger signals' by activating pattern-recognition receptors that are sequestered in the intestinal epithelium. 554](#)
- 132. [12-18 Pathogens induce inflammatory adaptive immune responses when innate defenses have been breached. 556](#)
- 133. [12-19 Effector T-cell responses in the intestine protect the function of the epithelium. 557](#)
- 134. [12-20 Noninvasive and invasive enteric bacterial pathogens use different strategies to colonize the intestines. 557](#)

- 135. [12-21 Dysregulated immune responses to commensal bacteria provoke intestinal disease. 563](#)
- 136. [Summary. 564](#)
- 137. [Immunity at other barrier tissues. 564](#)
- 138. [12-22 Anatomy and mucosal immunity of the airways. 564](#)
- 139. [12-23 Respiratory immunity to inhaled pathogens; respiratory viruses. 567](#)
- 140. [12-24 Cutaneous immunity. 574](#)
- 141. [12-25 The sensory nervous system communicates with the immune system to enhance defense against infections in the skin. 577](#)
- 142. [Summary. 579](#)
- 143. [Summary to Chapter 12. 580](#)
- 144. [Discussion questions. 581](#)
- 145. [References. 581](#)

[PART V THE IMMUNE SYSTEM IN HEALTH AND DISEASE](#)

- 1. [Chapter 13 Failures of Host Defense Mechanisms 587](#)
- 2. [Immunodeficiency diseases. 588](#)
- 3. [13-1 Primary immunodeficiency diseases are caused by inherited gene variants that typically cause recurrent infections early in life. 588](#)
- 4. [13-2 Defects in T-cell development can result in severe combined immune deficiencies. 589](#)
- 5. [13-3 SCID can also be due to defects in the purine salvage pathway. 592](#)
- 6. [13-4 Defects in antigen receptor gene rearrangement can result in SCID. 593](#)
- 7. [13-5 Defects in signaling from T-cell antigen receptors can cause severe immunodeficiency. 594](#)
- 8. [13-6 Genetic defects in thymic development or function that block T-cell development result in combined immunodeficiencies. 594](#)
- 9. [13-7 Defects in B-cell development result in deficiencies in antibody production that cause an inability to clear extracellular bacteria. 596](#)
- 10. [13-8 Combined immunodeficiencies \(CIDs\) can be caused by defects in T-cell differentiation and function that impair activation of B cells, dendritic cells and macrophages, and innate effector cells. 598](#)
- 11. [13-9 B cell-specific activation defects cause hyper IgM syndromes that are limited to defects in the antibody response. 600](#)
- 12. [13-10 Common variable immunodeficiency results from a variety of inherited defects that cause more limited defects in the antibody response. 600](#)
- 13. [13-11 Unidentified genetic defects cause isotype-specific antibody deficiencies. 601](#)
- 14. [13-12 Normal pathways for host defense against different infectious agents are pinpointed by genetic deficiencies of cytokine pathways central to type 1/T_H1 and type 3/T_H17 responses. 601](#)
- 15. [13-13 Inherited defects in the cytolytic pathway of lymphocytes can cause uncontrolled lymphoproliferation and inflammatory responses to viral infections. 603](#)
- 16. [13-14 X-linked lymphoproliferative syndrome is associated with fatal infection](#)

- by Epstein–Barr virus and the development of lymphomas. 605
17. [13-15 Immunodeficiency is caused by inherited defects in the development of dendritic cells. 606](#)
 18. [13-16 Defects in complement components and complement-regulatory proteins cause defective humoral immune function and tissue damage. 607](#)
 19. [13-17 Defects in phagocytic cells result in widespread bacterial infections. 608](#)
 20. [13-18 Mutations in the molecular regulators of inflammation can cause uncontrolled inflammatory responses that result in ‘autoinflammatory disease.’ 612](#)
 21. [13-19 Factor-replacement therapies, hematopoietic stem-cell transplantation, and gene therapy can be used successfully to correct genetic defects. 613](#)
 22. [13-20 Noninherited, secondary immunodeficiencies are major predisposing causes of infection and death. 615](#)
 23. [Summary. 616](#)
 24. [Evasion and subversion of immune defenses. 616](#)
 25. [13-21 Extracellular bacterial pathogens have evolved different strategies to avoid detection by pattern-recognition receptors and destruction by antibody, complement, and antimicrobial peptides. 617](#)
 26. [13-22 Intracellular bacterial pathogens can evade the immune system by seeking shelter within phagocytes. 622](#)
 27. [13-23 Immune evasion is also practiced by protozoan parasites. 623](#)
 28. [13-24 Many viruses target type I and type III interferon pathways to impair host antiviral defense. 625](#)
 29. [13-25 RNA viruses use different mechanisms of antigenic variation to keep a step ahead of the adaptive immune system. 626](#)
 30. [13-26 DNA viruses use multiple mechanisms to subvert CTL and NK-cell responses. 628](#)
 31. [13-27 Viruses that establish latency persist *in vivo* by ceasing to replicate until immunity wanes. 631](#)
 32. [13-28 DNA and RNA viruses can induce exhaustion of CTL and NK cell responses, thereby impairing the antiviral response. 633](#)
 33. [Summary. 633](#)
 34. [Acquired immune deficiency syndrome. 634](#)
 35. [13-29 HIV is a retrovirus that establishes a chronic infection that slowly progresses to AIDS. 636](#)
 36. [13-30 HIV infects and replicates within cells of the immune system. 636](#)
 37. [13-31 Activated CD4 T cells are the major source of HIV replication. 638](#)
 38. [13-32 There are several routes by which HIV is transmitted and establishes infection. 641](#)
 39. [13-33 HIV variants with tropism for different co-receptors play different roles in transmission and progression of disease. 641](#)
 40. [13-34 A genetic deficiency of the co-receptor CCR5 confers resistance to HIV infection. 644](#)
 41. [13-35 An immune response controls but does not eliminate HIV. 644](#)

42. [13-36 Lymphoid tissue is the major reservoir of HIV infection. 646](#)
43. [13-37 Genetic variation in the host can alter the rate of HIV disease progression. 647](#)
44. [13-38 The destruction of immune function as a result of HIV infection leads to increased susceptibility to opportunistic infection and eventually to death. 649](#)
45. [13-39 Drugs that block HIV replication lead to a rapid decrease in titer of infectious virus and an increase in CD4 T cells. 649](#)
46. [13-40 In the course of infection, HIV accumulates many mutations, which can result in the outgrowth of drug-resistant variants. 652](#)
47. [13-41 HIV treatment and prophylaxis are leading to a decline in the spread of HIV and AIDS. 652](#)
48. [13-42 Vaccination against HIV remains an attractive solution but poses many difficulties. 653](#)
49. [Summary. 655](#)
50. [Summary to Chapter 13. 656](#)
51. [Discussion questions. 656](#)
52. [References. 657](#)
53. [Chapter 14 Allergic Diseases and Hypersensitivity Reactions 663](#)
54. [IgE and IgE-mediated allergic diseases. 664](#)
55. [14-1 Sensitization involves class switching to IgE production on first contact with an allergen. 665](#)
56. [14-2 Although many types of antigens can cause allergic sensitization, proteases are common sensitizing agents. 666](#)
57. [14-3 Genetic factors contribute to the development of IgE-mediated allergic disease. 667](#)
58. [14-4 Environmental factors may interact with genetic susceptibility to cause allergic disease. 669](#)
59. [Summary. 671](#)
60. [Effector mechanisms in IgE-mediated allergic reactions. 671](#)
61. [14-5 Most IgE is cell-bound and engages effector mechanisms of the immune system by pathways different from those of other antibody isotypes. 671](#)
62. [14-6 Mast cells reside in tissues and orchestrate allergic reactions. 673](#)
63. [14-7 Eosinophils and basophils cause inflammation and tissue damage in allergic reactions. 675](#)
64. [14-8 IgE-mediated allergic reactions have a rapid onset but can also lead to chronic responses. 677](#)
65. [14-9 Allergen introduced into the bloodstream can cause anaphylaxis and urticaria. 679](#)
66. [14-10 Allergen inhalation is associated with the development of rhinitis and asthma. 681](#)
67. [14-11 Allergy to particular foods causes systemic reactions as well as symptoms limited to the gut. 682](#)
68. [14-12 IgE-mediated allergic disease can be treated by inhibiting effector pathways or by desensitization to restore biological tolerance. 684](#)

69. [Summary. 686](#)
70. [Non-IgE-mediated allergic diseases. 686](#)
71. [14-13 Non-IgE-dependent drug-induced hypersensitivity reactions in susceptible individuals occur by binding of the drug to the surface of circulating blood cells. 687](#)
72. [14-14 Systemic disease caused by immune-complex formation can follow the administration of large quantities of poorly catabolized antigens. 687](#)
73. [14-15 Hypersensitivity reactions can be mediated by T_H1 cells and cytotoxic CD8 T cells. 689](#)
74. [14-16 Celiac disease has features of both an allergic response and autoimmunity. 692](#)
75. [Summary. 694](#)
76. [Summary to Chapter 14. 694](#)
77. [Discussion questions. 695](#)
78. [References. 695](#)
79. [Chapter 15 Autoimmunity and Transplantation 699](#)
80. [The making and breaking of self-tolerance. 699](#)
81. [15-1 Multiple mechanisms normally contribute to prevent autoimmunity. 699](#)
82. [15-2 Lymphocytes that bind self antigens with relatively low affinity usually ignore them but in some circumstances become activated. 702](#)
83. [15-3 Antigens in immunologically privileged sites do not induce immune attack but can serve as targets. 703](#)
84. [15-4 Autoimmune responses can be controlled at various stages by regulatory T cells. 704](#)
85. [Summary. 705](#)
86. [Autoimmune diseases and pathogenic mechanisms. 706](#)
87. [15-5 Autoimmunity can be classified into either organ-specific or systemic disease. 706](#)
88. [15-6 Multiple components of the immune system are typically recruited in autoimmune disease. 707](#)
89. [15-7 Chronic autoimmune disease develops through positive feedback from inflammation, inability to clear the self antigen, and a broadening of the autoimmune response. 709](#)
90. [15-8 Both antibody and effector T cells can cause tissue damage in autoimmune disease. 711](#)
91. [15-9 Autoantibodies against blood cells promote their destruction. 713](#)
92. [15-10 The fixation of sublytic doses of complement to cells in tissues stimulates a powerful inflammatory response. 713](#)
93. [15-11 Autoantibodies against receptors cause disease by stimulating or blocking receptor function. 714](#)
94. [15-12 Autoantibodies against extracellular antigens cause inflammatory injury. 715](#)
95. [15-13 T cells specific for self antigens can cause direct tissue injury and sustain autoantibody responses. 717](#)

96. [Summary. 721](#)
97. [The genetic and environmental basis of autoimmunity. 721](#)
98. [15-14 Autoimmune diseases have a strong genetic component. 722](#)
99. [15-15 Monogenic defects of immune tolerance. 722](#)
100. [15-16 MHC genes have an important role in controlling susceptibility to autoimmune disease. 726](#)
101. [15-17 Genetic variants that impair innate immune responses can predispose to T cell-mediated chronic inflammatory disease. 727](#)
102. [15-18 Cross-reactivity between foreign molecules on pathogens and self molecules can promote autoimmune disease. 730](#)
103. [Summary. 731](#)
104. [Responses to alloantigens and transplant rejection. 732](#)
105. [15-19 Transplant rejection is mediated primarily by T-cell responses to MHC molecules. 732](#)
106. [15-20 Rejection of MHC-identical grafts is caused by peptides from other alloantigens bound to graft MHC molecules. 734](#)
107. [15-21 Alloreactive T cells reject transplanted organs by direct and indirect allorecognition. 735](#)
108. [15-22 Antibodies that react with endothelium cause hyperacute graft rejection. 737](#)
109. [15-23 Late failure of transplanted organs is caused by chronic injury to the graft. 737](#)
110. [15-24 A variety of organs are transplanted routinely in clinical medicine. 738](#)
111. [15-25 The converse of graft rejection is graft-versus-host disease. 740](#)
112. [15-26 The fetus is an allograft that is tolerated repeatedly. 741](#)
113. [Summary. 742](#)
114. [Summary to Chapter 15. 743](#)
115. [Discussion questions. 744](#)
116. [References. 744](#)
117. [Chapter 16 Manipulation of the Immune Response 749](#)
118. [Fighting infectious diseases with vaccination. 749](#)
119. [16-1 Vaccines can be based on attenuated pathogens or material from killed organisms. 750](#)
120. [16-2 Most effective vaccines generate antibodies that prevent damage caused by toxins or that neutralize pathogens and stop infection. 752](#)
121. [16-3 Effective vaccines must induce long-lasting protection while being safe and inexpensive. 753](#)
122. [16-4 Live-attenuated viral vaccines are usually more potent than 'killed' vaccines. 753](#)
123. [16-5 Live-attenuated viral vaccines can be made safer by the use of recombinant DNA technology. 754](#)
124. [16-6 Vaccines against bacteria or parasites can be developed using nonpathogenic or disabled bacteria or genetically attenuated parasites \(GAPs\). 755](#)

125. [16-7 Conjugate vaccines rely on linked recognition between T and B cells. 756](#)
126. [16-8 Adjuvants enhance the immunogenicity of vaccines, but few are approved for use in humans. 758](#)
127. [16-9 Protective immunity can be induced by RNA- and DNA-based vaccination. 759](#)
128. [16-10 The route of vaccination is an important determinant of success. 760](#)
129. [16-11 *Bordetella pertussis* vaccination illustrates the importance of the perceived safety of a vaccine. 761](#)
130. [16-12 Peptide-based vaccines can elicit protective immunity, but they require adjuvants and must be targeted to the appropriate cells and cell compartment to be effective. 762](#)
131. [16-13 Vaccination and checkpoint blockade may be useful in controlling existing chronic infections. 763](#)
132. [Summary. 763](#)
133. [Using the immune response to attack tumors. 764](#)
134. [16-14 The development of transplantable tumors in mice led to the discovery of protective immune responses to tumors. 764](#)
135. [16-15 Tumors are 'edited' by the immune system as they evolve and can escape rejection in many ways. 765](#)
136. [16-16 Tumor-rejection antigens can be recognized by T cells and form the basis of immunotherapies. 769](#)
137. [16-17 Checkpoint blockade can augment immune responses to existing tumors. 771](#)
138. [16-18 T cells expressing chimeric antigen receptors are an effective treatment in some leukemias. 773](#)
139. [16-19 Monoclonal antibodies against tumor antigens, alone or linked to toxins, can control tumor growth. 774](#)
140. [16-20 Enhancing the immune response to tumors by vaccination holds promise for cancer prevention and therapy. 776](#)
141. [Summary. 777](#)
142. [Treatment of unwanted immune responses. 778](#)
143. [16-21 Corticosteroids are powerful anti-inflammatory drugs that alter the transcription of many genes. 779](#)
144. [16-22 Cytotoxic drugs cause immunosuppression by killing dividing cells and have serious side effects. 780](#)
145. [16-23 Cyclosporin A, tacrolimus, and rapamycin are effective immunosuppressive agents that interfere with various T-cell signaling pathways. 781](#)
146. [16-24 JAK inhibitors can be used to treat autoimmune and inflammatory diseases. 783](#)
147. [16-25 Antibodies against cell-surface molecules can be used to eliminate lymphocyte subsets or to inhibit lymphocyte function. 783](#)
148. [16-26 Antibodies can be engineered to reduce their immunogenicity in](#)

- [humans. 784](#)
149. [16-27 Monoclonal antibodies can be used to prevent allograft rejection. 785](#)
150. [16-28 Depletion of autoreactive lymphocytes can treat autoimmune disease. 787](#)
151. [16-29 Biologics that block TNF- \$\alpha\$, IL-1, or IL-6 can alleviate autoimmune diseases. 788](#)
152. [16-30 Biologics can block cell migration to sites of inflammation and reduce immune responses. 789](#)
153. [16-31 Blockade of co-stimulatory pathways that activate lymphocytes can be used to treat autoimmune disease. 790](#)
154. [16-32 Some commonly used drugs have immunomodulatory properties. 791](#)
155. [16-33 Controlled administration of antigen can be used to manipulate the nature of an antigen-specific response. 791](#)
156. [Summary. 792](#)
157. [Summary to Chapter 16. 792](#)
158. [Discussion questions. 794](#)
159. [References. 794](#)
160. [APPENDICES](#)
161. Note: Appendices II–IV are available on digital.wwnorton.com/janeway10 and in the ebook.
162. [Appendix I The Immunologist's Toolbox A-1](#)
163. [A-1 Immunization. A-1](#)
164. [A-2 Antibody responses. A-4](#)
165. [A-3 Affinity chromatography. A-5](#)
166. [A-4 Radioimmunoassay \(RIA\), enzyme-linked immunosorbent assay \(ELISA\), and competitive inhibition assay. A-5](#)
167. [A-5 Multiplex bead assay for detection of cytokines, antibody responses, or antigens. A-7](#)
168. [A-6 Hemagglutination and blood typing. A-8](#)
169. [A-7 Coombs tests and the detection of rhesus incompatibility. A-9](#)
170. [A-8 Monoclonal antibodies. A-11](#)
171. [A-9 Phage display libraries for antibody V-region production. A-12](#)
172. [A-10 Generation of human monoclonal antibodies from vaccinated individuals. A-13](#)
173. [A-11 Microscopy and imaging using fluorescent dyes. A-13](#)
174. [A-12 Immunoelectron microscopy. A-15](#)
175. [A-13 Immunohistochemistry. A-15](#)
176. [A-14 Immunoprecipitation and co-immunoprecipitation. A-16](#)
177. [A-15 Immunoblotting \(western blotting\). A-17](#)
178. [A-16 Use of antibodies in the isolation and characterization of multiprotein complexes by mass spectrometry. A-18](#)
179. [A-17 Isolation of peripheral blood lymphocytes by density-gradient fractionation. A-20](#)
180. [A-18 Isolation of lymphocytes from tissues other than blood. A-20](#)

181. [A-19 Flow cytometry and FACS analysis. A-21](#)
182. [A-20 Lymphocyte isolation using antibody-coated magnetic beads. A-23](#)
183. [A-21 Isolation of homogeneous T-cell lines. A-24](#)
184. [A-22 ELISPOT assay. A-24](#)
185. [A-23 Identification of functional subsets of T cells based on cytokine production or transcription factor expression. A-25](#)
186. [A-24 Identification of T or B lymphocytes on the basis of antigen receptor specificity. A-28](#)
187. [A-25 Biosensor assays for measuring the rates of association and dissociation of antigen receptors for their ligands. A-28](#)
188. [A-26 Assays of lymphocyte proliferation. A-30](#)
189. [A-27 Measurements of apoptosis. A-32](#)
190. [A-28 Assays for cytotoxic T cells. A-33](#)
191. [A-29 Assays for CD4 T cells. A-34](#)
192. [A-30 Transfer of protective immunity. A-35](#)
193. [A-31 Adoptive transfer of lymphocytes. A-36](#)
194. [A-32 Assessment of lymphocyte tissue residency by parabiosis. A-37](#)
195. [A-33 Hematopoietic stem-cell transfers. A-38](#)
196. [A-34 In vivo administration of antibodies. A-38](#)
197. [A-35 Transgenic mice. A-39](#)
198. [A-36 Gene knockout by targeted disruption. A-40](#)
199. [A-37 Knockdown of gene expression by RNA interference \(RNAi\). A-43](#)
200. [A-38 Cell-lineage tracing techniques. A-44](#)
201. [A-39 Single-cell analysis A-45](#)
202. [eAppendix II CD Antigens A-49](#)
203. [eAppendix III Cytokines and Their Receptors A-69](#)
204. [eAppendix IV Chemokines and Their Receptors A-73](#)
205. [Glossary G-1](#)
206. [Credits C-1](#)
207. [Index I-1](#)

PART I

AN INTRODUCTION TO IMMUNOBIOLOGY AND INNATE IMMUNITY

- [1 Basic Concepts in Immunology](#)
- [2 Innate Immunity: The First Lines of Defense](#)
- [3 Cellular Mechanisms of Innate Immunity](#)

1

Basic Concepts in Immunology

IN THIS CHAPTER

- [The origins of vertebrate immune cells.](#)
 - [Principles of innate immunity.](#)
 - [Principles of adaptive immunity.](#)
 - [The effector mechanisms of immunity.](#)
-

[Immunology](#) is the study of the body's defense against infection. We are continually exposed to microorganisms, many of which cause disease, and yet become ill only rarely. How does the body defend itself? When infection does occur, how does the body eliminate the invader and cure itself? And why do we develop long-lasting immunity to many infectious diseases encountered once and overcome? These are the questions addressed by immunology, which we study to understand our body's defenses against infection at the cellular and molecular levels.



Fig. 1.1 Edward Jenner. Portrait by John Raphael Smith.

The beginning of immunology as a science is usually attributed to Edward Jenner for his work in the late 18th century (**Fig. 1.1**). The notion of immunity—that surviving a disease confers greater protection against it later—was known since ancient Greece. [Variolation](#)—the inhalation or transfer into superficial skin wounds of material from smallpox pustules—had been practiced since at least the 1400s in the Middle East and China as a form of protection against that disease and was known to Jenner. Jenner had observed that the relatively mild disease of cowpox seemed to confer protection against the often fatal disease of smallpox, and in 1796, he demonstrated that inoculation with cowpox protected the

recipient against smallpox. His scientific proof relied on the deliberate exposure of the inoculated individual to infectious smallpox material 2 months after inoculation. This scientific test was his original contribution.

Jenner called the procedure [vaccination](#). This term is still used to describe the inoculation of healthy individuals with weakened or attenuated strains of disease-causing agents in order to provide protection from disease. Although Jenner's bold experiment was successful, it took almost two centuries for smallpox vaccination to become universal. This advance enabled the World Health Organization to announce in 1980 that smallpox had been eradicated (**Fig. 1.2**), arguably the greatest triumph of modern medicine.

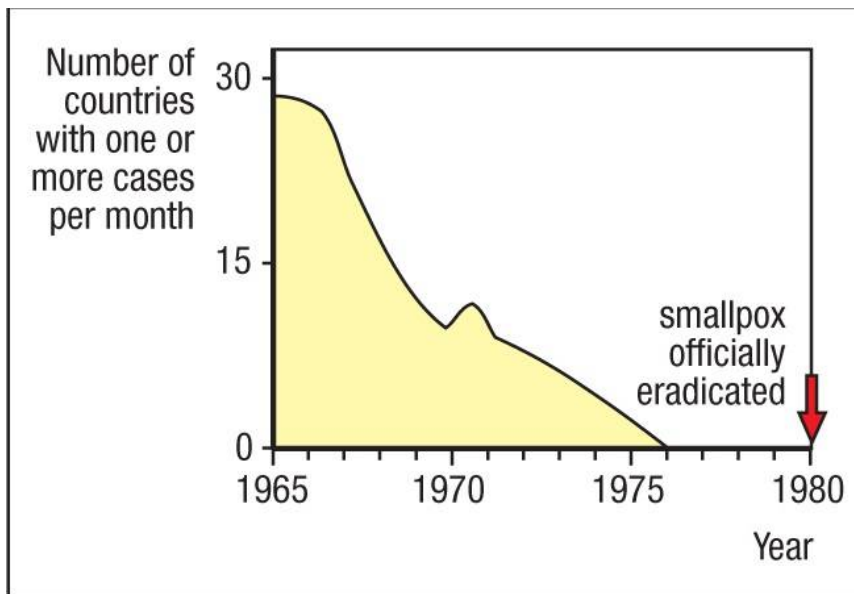


Fig. 1.2 The eradication of smallpox by vaccination. After a period of 3 years in which no cases of smallpox were recorded, the World Health Organization was able to announce in 1980 that smallpox had been eradicated, and vaccination stopped (upper panel). A few laboratory stocks have been retained, however, and some fear that these are a source from which the virus might reemerge. Ali Maow Maalin (lower panel) contracted and survived the last case of smallpox in Somalia in 1977.

Jenner's strategy of vaccination was extended in the late 19th century by the discoveries of many great microbiologists. Robert Koch proved that infectious diseases are caused by specific microorganisms. In the 1880s, Louis Pasteur devised a vaccine against cholera in chickens and also developed a rabies vaccine that proved to be a spectacular success upon its first trial in a boy bitten by a rabid dog.

These practical triumphs led to a search for vaccination's mechanism of protection and to the development of the science of immunology. In the early 1890s, Emil von Behring and Shibasaburo Kitasato discovered that the serum of animals immune to diphtheria or tetanus contained a specific 'antitoxic activity' that could confer short-lived protection against the effects of diphtheria or tetanus toxins in unimmunized animals and people. This activity was later determined to be due to the proteins we now call [antibodies](#), which bind specifically to the toxins and neutralize their activity. That these antibodies might have a crucial role in immunity was reinforced by Jules Bordet's discovery in 1899 of [complement](#), a component of serum that acts in conjunction with antibodies to destroy pathogenic bacteria.

A specific response against infection by potential pathogens, such as the production of antibodies against a particular pathogen, is known as [adaptive immunity](#), because it develops during the lifetime of an individual as an adaptation to infection with that pathogen. Adaptive immunity is distinguished from [innate immunity](#), which—at the time von Behring was developing serum therapy for diphtheria—was already known chiefly through the work of the great Russian immunologist Élie Metchnikoff, who discovered that many microorganisms could be engulfed and digested by phagocytic cells, which thus provide defenses against infection that are nonspecific. Whereas these cells—which Metchnikoff called 'macrophages'—are always present and ready to act, adaptive immunity requires time to develop but is highly specific.

It was soon clear that specific antibodies could be induced against a vast range of substances called [antigens](#) because they could stimulate *antibody generation*. Paul Ehrlich advanced the development of an [antisera](#) as a treatment for diphtheria and developed methods to standardize therapeutic serums. Today the term 'antigen' refers to any substance recognized by the adaptive immune system. Typically antigens are common proteins, glycoproteins, and polysaccharides of pathogens, but they can include a much wider range of chemical structures; for example, metals such as nickel, drugs such as penicillin, and organic chemicals such as the urushiol (a mix of pentadecylcatechols) in the leaves of poison ivy. Metchnikoff and Ehrlich shared the 1908 Nobel Prize for their respective work on immunity.

This chapter introduces the principles of innate immunity and adaptive immunity, the cells of the immune system, the tissues in which they develop, and the tissues through which they circulate. We then outline the specialized functions of the different types of cells by which they eliminate infection.

Glossary

[immunology](#)

The study of all aspects of host defense against infection and also of the adverse consequences of immune responses.

[variolation](#)

The intentional inhalation of or skin infection with material taken from smallpox pustules of an infected person for the purpose of deriving protective immunity.

[vaccination](#)

The deliberate induction of adaptive immunity to a pathogen by injecting a dead or attenuated (nonpathogenic) live form of the pathogen or its antigens (a vaccine).

[antibody](#)

A protein that binds specifically to a particular substance—called its antigen. Each antibody molecule has a unique structure that enables it to bind specifically to its corresponding antigen, but all antibodies have the same overall structure and are known collectively as immunoglobulins. Antibodies are produced by differentiated B cells (plasma cells) in response to infection or immunization and bind to and neutralize pathogens or prepare them for uptake and destruction by phagocytes.

[complement](#)

A set of plasma proteins that act together as a defense against pathogens in extracellular spaces. The pathogen becomes coated with complement proteins that facilitate its removal by phagocytes and that can also kill certain pathogens directly. Activation of the complement system can be initiated in several different ways. *See* **classical pathway**, **alternative pathway**, **lectin pathway**.

[adaptive immunity](#)

Immunity to infection conferred by an adaptive immune response.

[innate immunity](#)

The various innate resistance mechanisms that are encountered first by a pathogen, before adaptive immunity is induced, such as anatomical barriers, antimicrobial peptides, the complement system, and macrophages and neutrophils carrying nonspecific pathogen-recognition receptors. Innate immunity is present in all individuals at all times, does not increase with repeated exposure to a given pathogen, and discriminates between groups of similar pathogens, rather than responding to a particular pathogen. Cf. **adaptive immunity**.

[antigen](#)

Any molecule that can bind specifically to an antibody or generate peptide fragments that are recognized by a T-cell receptor.

[antiserum \(pl. antisera\)](#)

The fluid component of clotted blood from an immune individual that contains antibodies against the antigen that was used for immunization. An antiserum contains a mixture of different antibodies that all bind the antigen, each of which has a different structure, its own epitope on the antigen, and its own set of cross-reactions. This heterogeneity makes each antiserum unique.

The origins of vertebrate immune cells.

The body is protected from infectious agents, their toxins, and the damage they cause by a variety of effector cells and molecules that together make up the [immune system](#). Both innate and adaptive immune responses depend on the activities of white blood cells, or [leukocytes](#). Most cells of the immune system arise from the [bone marrow](#), where many of them develop and mature. But some, particularly certain tissue-resident macrophage and lymphocyte populations (for example, the microglia of the central nervous system), originate from the yolk sac or fetal liver during embryonic development. They seed tissues before birth and are maintained throughout life as independent, self-renewing populations.

After birth, all the cellular elements of blood, including the red blood cells that transport oxygen, the platelets that trigger blood clotting in damaged tissues, and the white blood cells of the immune system, ultimately derive from the [hematopoietic stem cells \(HSCs\)](#) of the bone marrow. Because these can give rise to all the different types of blood cells, they are often known as [pluripotent](#) (or multipotent) hematopoietic stem cells. The hematopoietic stem cells give immediate rise to cells of more limited developmental potential, which are the progenitors of red blood cells, platelets, and the two main categories of white blood cells, the [lymphoid](#) and [myeloid](#) lineages. Once mature, immune cells reside within peripheral tissues, circulate in the bloodstream, or circulate in a specialized system of vessels called the [lymphatic system](#). The lymphatic system drains extracellular fluid and immune cells from tissues and transports them as [lymph](#), which is eventually emptied back into the blood system. The different types of blood cells and their lineage relationships are summarized in **Fig. 1.3**.

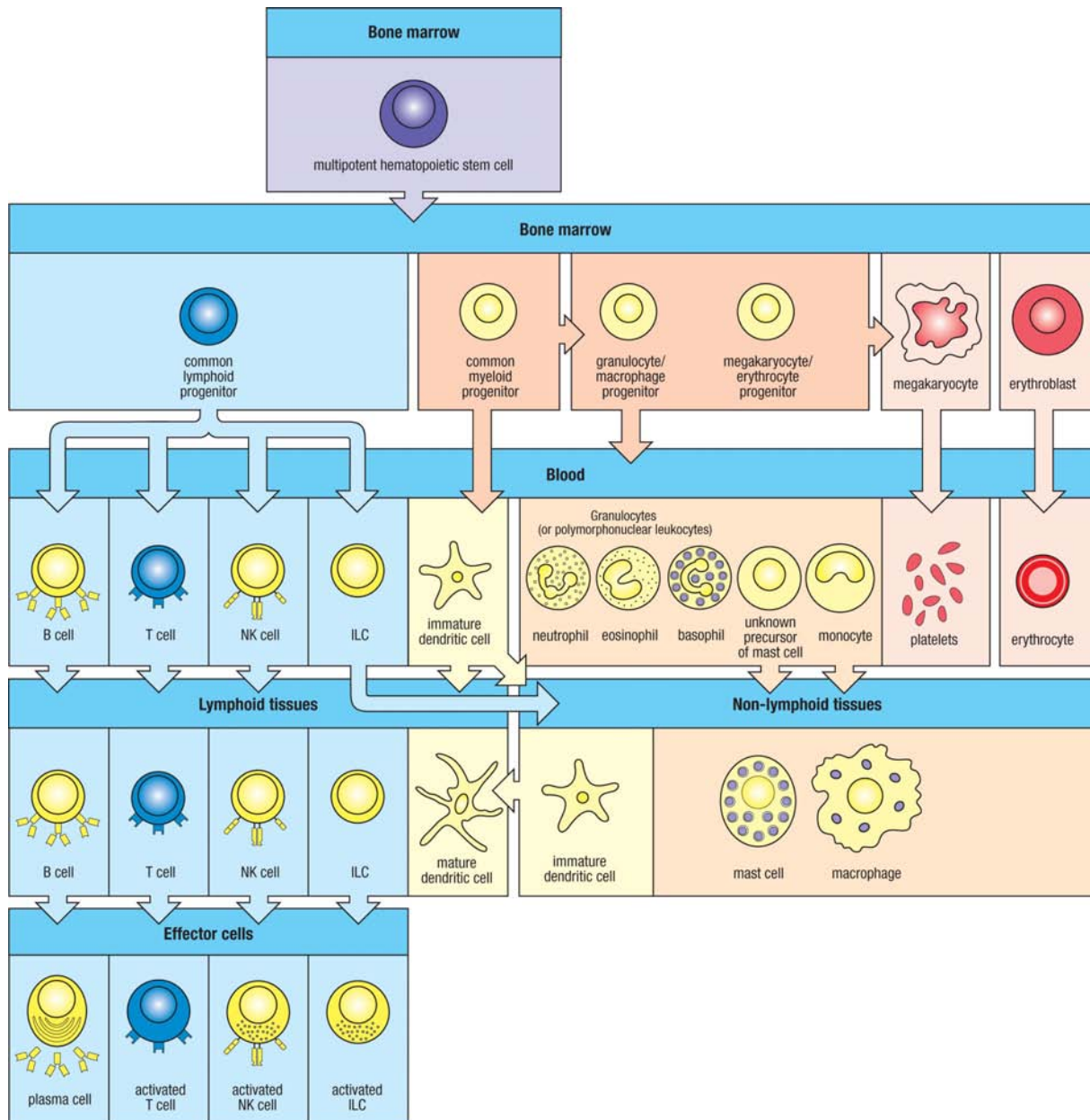


Fig. 1.3 All the cellular elements of the blood, including the cells of the immune system, arise from multipotent hematopoietic stem cells in the bone marrow. These multipotent cells divide to produce two types of stem cells. A common lymphoid progenitor gives rise to the lymphoid lineage (blue background) of white blood cells, or leukocytes—the innate lymphoid cells (ILCs) and natural killer (NK) cells and the T and B lymphocytes. A common myeloid progenitor gives rise to the myeloid lineage (pink and yellow backgrounds), which comprises the rest of the leukocytes, the erythrocytes (red blood cells), and the megakaryocytes that produce platelets important in blood clotting. T and B lymphocytes are distinguished from the other leukocytes by having antigen receptors and from each other by their sites of differentiation—the thymus and bone marrow, respectively. After encounter with antigen, B cells differentiate into antibody-secreting plasma cells, while T cells differentiate into effector T cells with a variety of functions. Unlike T and B cells, ILCs and NK cells lack antigen specificity. The remaining leukocytes are the monocytes, the dendritic

cells, and the neutrophils, eosinophils, and basophils. The last three of these circulate in the blood and are termed granulocytes, because of the cytoplasmic granules whose staining gives these cells a distinctive appearance in blood smears, or polymorphonuclear leukocytes, because of their irregularly shaped nuclei. Immature dendritic cells (yellow background) are phagocytic cells that enter the tissues; they mature after they have encountered a potential pathogen. The majority of dendritic cells are derived from the common myeloid progenitor cells, but some may also arise from the common lymphoid progenitor. Monocytes enter tissues, where they differentiate into phagocytic macrophages or dendritic cells. Mast cells also enter tissues and complete their maturation there.

Glossary

[immune system](#)

The tissues, cells, and molecules involved in innate immunity and adaptive immunity.

[leukocyte](#)

A white blood cell. Leukocytes include lymphocytes, polymorphonuclear leukocytes, and monocytes.

[bone marrow](#)

The tissue where all the cellular elements of the blood—red blood cells, white blood cells, and platelets—are initially generated from hematopoietic stem cells. The bone marrow is also the site of further B-cell development in mammals and the source of stem cells that give rise to T cells on migration to the thymus. Thus, bone marrow transplantation can restore all the cellular elements of the blood, including the cells required for adaptive immune responses.

[hematopoietic stem cells \(HSCs\)](#)

Type of pluripotent (or multipotent) cell in the bone marrow that can give rise to all the different blood cell types.

[pluripotent](#)

Typically referring to the capacity of a progenitor cell to generate all possible lineages of an organ system.

[lymphoid](#)

Describes tissues composed mainly of lymphocytes.

[myeloid](#)

Refers to the lineage of blood cells that includes all leukocytes except lymphocytes.

[lymphatic system](#)

The system of lymph-carrying vessels and peripheral lymphoid tissues through which extracellular fluid from tissues passes before it is returned to the blood via the thoracic duct.

[lymph](#)

The extracellular fluid that accumulates in tissues and is drained by lymphatic vessels that carry it through the lymphatic system to the thoracic duct, which returns it to the

blood.

Principles of innate immunity.

In this part of the chapter we will outline the principles of innate immunity and describe the molecules and cells that provide continuous defense against invasion by pathogens.

Although a subset of white blood cells, known as [lymphocytes](#), possesses the most powerful ability to recognize and target pathogenic microorganisms, these cells need the participation of the innate immune system to initiate and mount their offensive. Indeed, the adaptive immune response and the innate immune response use many of the same destructive mechanisms to eliminate invading microorganisms.

1-1 Commensal organisms cause little host damage while pathogens damage host tissues by a variety of mechanisms.

We recognize four broad categories of disease-causing microorganisms, or [pathogens](#): [viruses](#), [bacteria](#) and archaea, [fungi](#), and the unicellular and multicellular eukaryotic organisms collectively termed [parasites](#) (Fig. 1.4). These microorganisms vary tremendously in size and in how they damage host tissues. The smallest are viruses, which range from five to a few hundred nanometers in size and are obligate intracellular pathogens. Viruses can kill cells directly by inducing lysis during viral replication. Somewhat larger are intracellular bacteria and mycobacteria. These can kill cells directly or damage cells by producing toxins. Many single-celled intracellular parasites, such as members of the *Plasmodium* genus that cause malaria, also directly kill infected cells. Pathogenic bacteria and fungi growing in extracellular spaces can induce shock and sepsis by releasing toxins into the blood or tissues. The largest pathogens—parasitic worms, or helminths—are too large to infect host cells but can injure tissues by forming cysts that induce damaging cellular responses in the tissues into which the worms migrate.

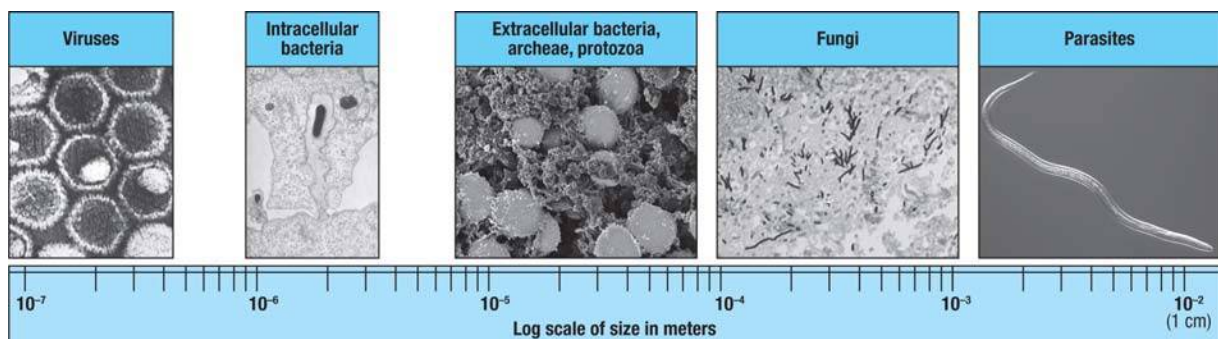


Fig. 1.4 Pathogens vary greatly in size and lifestyle. Intracellular pathogens include viruses, such as herpes simplex (first panel), and various bacteria, such as *Listeria monocytogenes* (second panel). Many bacteria, such as *Staphylococcus aureus* (third

panel), or fungi, such as *Aspergillus fumigatus* (fourth panel), can grow in the extracellular spaces and directly invade tissues, as do some archaea and protozoa (third panel). Many parasites, such as the nematode *Strongyloides stercoralis* (fifth panel), are large multicellular organisms that can move throughout the body in a complex life cycle.

Not all microbes are pathogens. Many tissues, especially those at the barriers of the body, such as the skin, oral mucosa, conjunctiva, and gastrointestinal tract, are constantly colonized by microbial communities—called the [microbiome](#)—that consist of archaea, bacteria, and fungi but cause no damage to the host. These are also called [commensal microorganisms](#), because they can have a [symbiotic](#) relationship with the host. Indeed, some commensal organisms perform important functions, as in the case of the bacteria that aid in cellulose digestion in the stomachs of ruminants. The difference between commensal organisms and pathogens lies in whether they induce damage. Even enormous numbers of microbes in the intestinal microbiome normally cause no damage and are confined within the intestinal lumen by a protective layer of mucus, whereas pathogenic bacteria can penetrate this barrier, injure intestinal epithelial cells, and spread into the underlying tissues.

1-2 Anatomic and chemical barriers are the first defense against pathogens.

The host can adopt three strategies to deal with the threat posed by microbes: [avoidance](#), [resistance](#), and [tolerance](#). Avoidance mechanisms prevent exposure to microbes and include both anatomic barriers and behavior modifications. If an infection is established, resistance is aimed at reducing or eliminating pathogens. To defend against the great variety of microbes, the immune system has numerous molecular and cellular functions, collectively called mediators, or [effector mechanisms](#), suited to resist different categories of pathogens. Their description is a major aspect of this book. Finally, tolerance involves responses that enhance a tissue's capacity to resist damage induced by microbes. This meaning of the term 'tolerance' has been used extensively in the context of disease susceptibility in plants rather than animal immunity. For example, increasing growth by activating dormant meristems, the undifferentiated cells that generate new parts of the plant, is a common tolerance mechanism in response to damage. This should be distinguished from the term [immunological tolerance](#), which refers to mechanisms that prevent an immune response from being mounted against the host's own tissues.

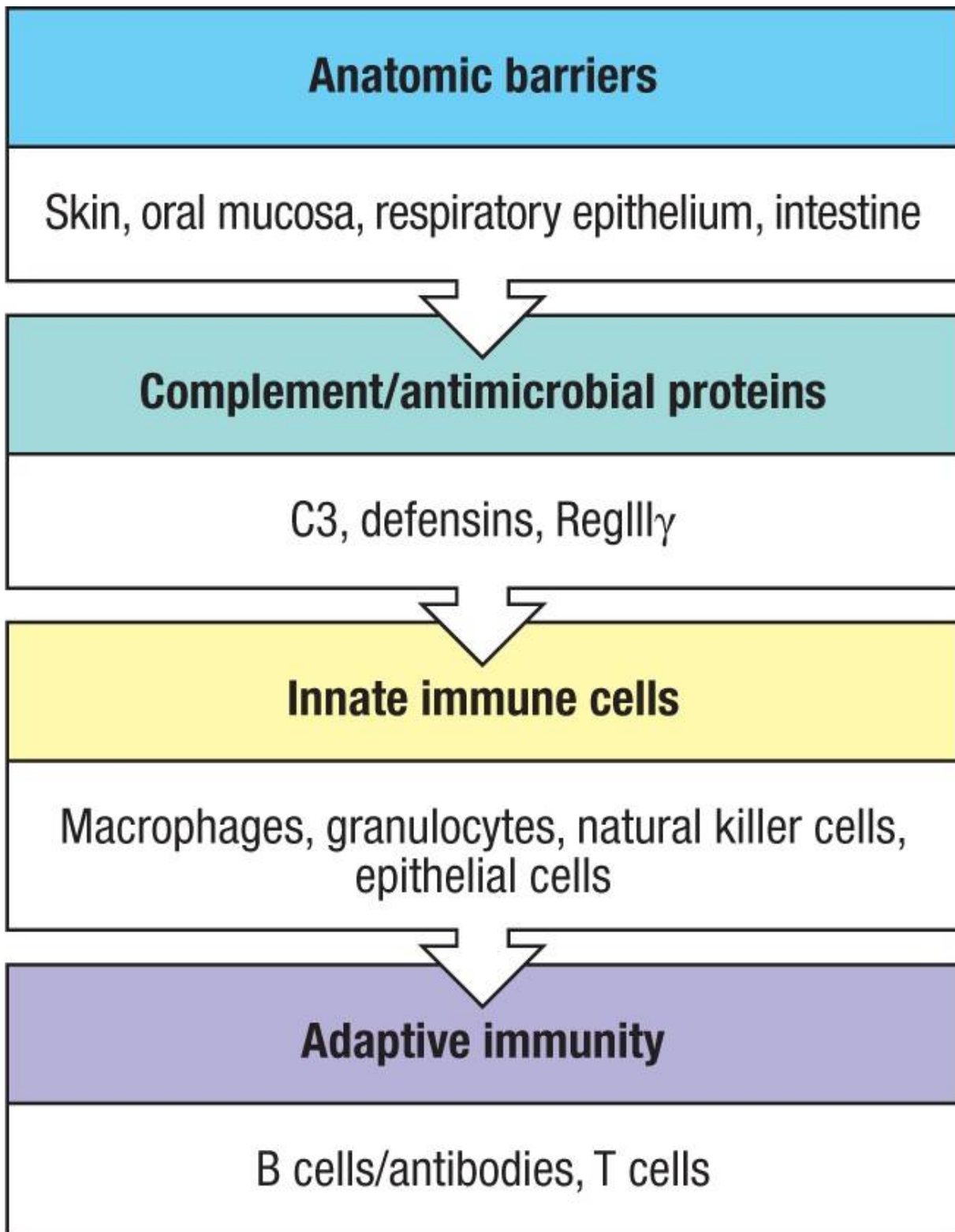


Fig. 1.5 Protection against pathogens relies on several levels of defense. The first is the anatomic barrier provided by the body's epithelial surfaces. Second, various chemical and enzymatic systems, including complement, act as an immediate antimicrobial barrier near these epithelia. If epithelia are breached, various nearby innate lymphoid and myeloid cells can coordinate a rapid cell-mediated defense. If the pathogen overcomes these barriers, the slower-acting defenses of the adaptive immune system are brought to bear.

Anatomic and chemical barriers are the initial defenses against infection (**Fig. 1.5**). The skin and mucosal surfaces represent a kind of avoidance strategy that prevents exposure of internal tissues to microbes. At most anatomic barriers, additional resistance mechanisms further strengthen host defenses. For example, mucosal surfaces produce a variety of [antimicrobial proteins](#) that act as natural antibiotics to prevent microbes from entering the body.

If these barriers are breached or evaded, other components of the innate immune system can immediately come into play. We mentioned earlier the discovery by Jules Bordet of complement, which acts with antibodies to lyse bacteria. Complement is a group of around 30 different plasma proteins that act together and are one of the most important effector mechanisms in serum and interstitial tissues. Complement not only acts in conjunction with antibodies but can also target foreign organisms in the absence of a specific antibody; thus it contributes to both innate and adaptive responses. We will examine anatomic barriers, the antimicrobial proteins, and complement in greater detail in [Chapter 2](#).

1-3 The immune system is activated by inflammatory inducers that indicate the presence of pathogens or tissue damage.

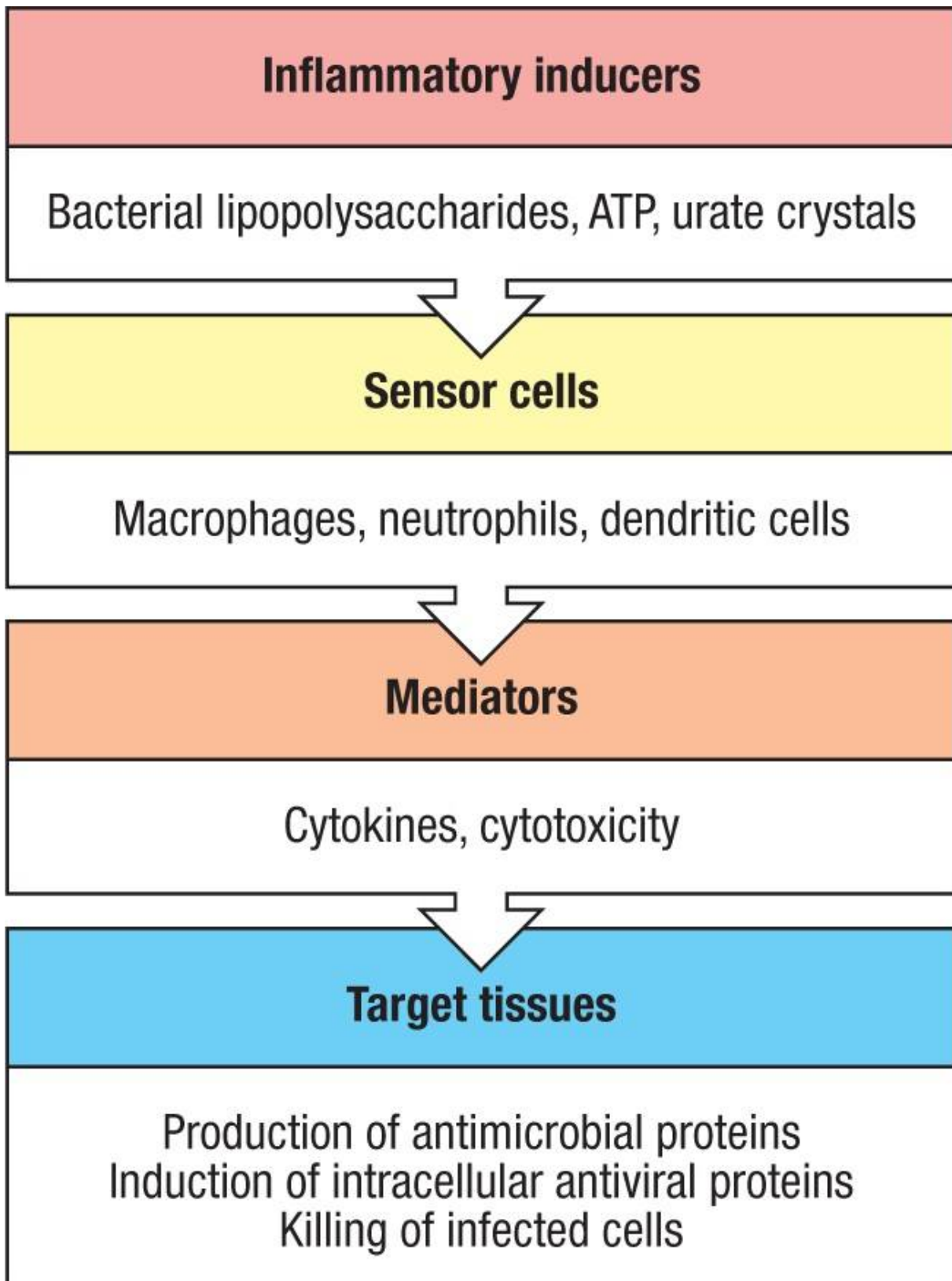


Fig. 1.6 Cell-mediated immunity proceeds in a series of steps. Inflammatory inducers are chemical structures that indicate the presence of invading microbes or the cellular damage produced by them. Sensor cells detect these inducers by their various innate recognition receptors and in response produce a variety of inflammatory mediators that act directly in defense or that further propagate the immune response. The mediators

include many cytokines, and they act on various target tissues (such as epithelial cells) to induce antimicrobial proteins and resist intracellular viral growth or on other immune cells (such as ILCs), which produce other cytokines that amplify the immune response.

A pathogen that breaches the host's anatomic and chemical barriers will encounter the cellular defenses of innate immunity. Cellular immune responses are initiated when various types of [sensor cells](#) detect [inflammatory inducers](#) (**Fig. 1.6**) through their many [innate recognition receptors](#), which are encoded by a relatively small number of genes that remain constant over an individual's lifetime. Inflammatory inducers that trigger these receptors include molecular components unique to bacteria or viruses, such as bacterial lipopolysaccharides, or molecules such as ATP, which is not normally found in the extracellular space. Triggering these receptors can activate the innate immune sensor cells to produce various [inflammatory mediators](#) that either act directly to destroy invading microbes or act on other cells to propagate the immune response. For example, macrophages can ingest microbes and produce toxic chemical mediators, such as degradative enzymes or reactive oxygen intermediates, to kill them. Dendritic cells may produce cytokine mediators, including many cytokines that activate target tissues, such as epithelial or other immune cells, to resist or kill invading microbes more efficiently. We will discuss these receptors and mediators briefly below and in much greater detail in [Chapter 3](#).

Innate immune responses occur rapidly on exposure to an infectious organism (**Fig. 1.7**). In contrast, responses by the adaptive immune system take days rather than hours to develop. However, the adaptive immune system is capable of eliminating infections more efficiently because of the exquisite specificity of antigen recognition by its lymphocytes. In contrast to a limited repertoire of receptors expressed by innate immune cells, lymphocytes express highly specialized [antigen receptors](#) that collectively possess a vast repertoire of specificity. This enables the adaptive immune system to respond to virtually any pathogen and effectively focus resources to eliminate pathogens that have evaded or overwhelmed innate immunity. But the adaptive immune system interacts with, and relies on, cells of the innate immune system for many of its functions. The next several sections will introduce the major components of the innate immune system and prepare us to consider adaptive immunity later in the chapter.

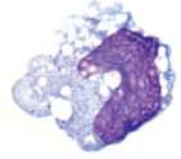
Phases of the immune response			
Response		Typical time after infection to start of response	Duration of response
Innate immune response	Inflammation, complement activation, phagocytosis, and destruction of pathogen	Minutes	Days
	Interaction between antigen-presenting dendritic cells and antigen-specific T cells: recognition of antigen, adhesion, costimulation, T-cell proliferation and differentiation	Hours	Days

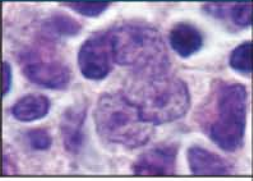
Adaptive immune response	Activation of antigen-specific B cells	Hours	Days
	Formation of effector and memory T cells	Days	Weeks
	Interaction of T cells with B cells, formation of germinal centers. Formation of effector B cells (plasma cells) and memory B cells. Production of antibody	Days	Weeks
	Emigration of effector lymphocytes from peripheral lymphoid organs	A few days	Weeks
	Elimination of pathogen by effector cells and antibody	A few days	Weeks
Immunological memory	Maintenance of memory B cells and T cells and high serum or mucosal antibody levels. Protection against reinfection	Days to weeks	Can be lifelong

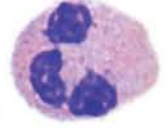
Fig. 1.7 Phases of the immune response.

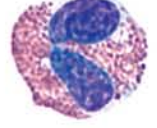
1-4 The myeloid lineage comprises most of the cells of the innate immune system.

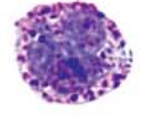
The [common myeloid progenitor \(CMP\)](#) is the precursor of the macrophages, granulocytes (the collective term for the white blood cells called neutrophils, eosinophils, and basophils), mast cells, and dendritic cells of the innate immune system. Macrophages, granulocytes, and dendritic cells make up the three classes of phagocytes in the immune system. The CMP also generates megakaryocytes and red blood cells, which will not concern us here. The cells of the myeloid lineage are shown in **Fig. 1.8**.

Macrophage	
	
Phagocytosis and activation of bactericidal mechanisms Antigen presentation and cytokine production	

Dendritic cell	
	
Antigen uptake in peripheral sites Antigen presentation and cytokine production	

Neutrophil	
	
Phagocytosis and activation of bactericidal mechanisms	

Eosinophil	
	
Killing of antibody-coated parasites	

Basophil	
	
Promotion of allergic responses and augmentation of anti-parasitic immunity	

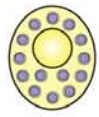
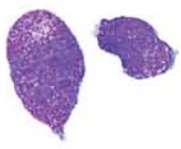
Mast cell	
	
Release of granules containing histamine and active agents	

Fig. 1.8 Myeloid cells in innate and adaptive immunity. In the rest of the book, these cells will be represented in the schematic form shown on the left. A photomicrograph of each cell type is shown on the right. Macrophages and neutrophils are primarily phagocytic cells that engulf pathogens and destroy them in intracellular vesicles, a function they perform in both innate and adaptive immune responses. Dendritic cells are phagocytic when they are immature and can take up pathogens; after maturing, they function as specialized cells that present pathogen antigens to T lymphocytes in a form they can recognize, thus activating T lymphocytes and initiating adaptive immune responses. Macrophages can also present antigens to T lymphocytes and can activate them. The other myeloid cells are primarily secretory cells that release the contents of their prominent granules upon activation via antibody during an adaptive immune response. Eosinophils are thought to be involved in attacking large antibody-coated parasites such as worms; basophils are also thought to be involved in anti-parasite immunity. Mast cells are tissue cells that trigger a local inflammatory response to antigen by releasing substances that act on local blood vessels. Mast cells, eosinophils, and basophils are also important in allergic responses. Original magnification $\times 15,000$.

[Macrophages](#) are resident in almost all tissues. Many tissue-resident macrophages arise during embryonic development, but some macrophages that arise from the bone marrow in the adult animal are the mature form of [monocytes](#), which circulate in the blood and continually migrate into tissues, where they differentiate. Macrophages are relatively long-lived cells and perform several different functions throughout the innate immune response and the subsequent adaptive immune response. One is to engulf and kill invading microorganisms. This phagocytic function provides a first defense in innate immunity. Macrophages also dispose of pathogens and infected cells targeted by an adaptive immune response. Both monocytes and macrophages are phagocytic, but most infections occur in the tissues, and so it is primarily macrophages that perform this important protective function. An additional and crucial role of macrophages is to orchestrate immune responses: they help induce inflammation, which, as we shall see, is a prerequisite to a successful immune response, and they produce many inflammatory mediators that activate other immune-system cells and recruit them into an immune response.

Local inflammation and the phagocytosis of invading bacteria can also be triggered by the activation of complement. Bacterial surfaces can activate the complement system, inducing a cascade of proteolytic reactions that coat the microbes with fragments of specific proteins of the complement system. Microbes coated in this way are recognized by specific [complement receptors](#) on macrophages and neutrophils, taken up by phagocytosis, and destroyed. In addition to their specialized role in the immune system, macrophages act as general scavenger cells in the body, clearing it of dead cells and cell debris.

The [granulocytes](#) are named for the densely staining granules in their cytoplasm; they are also called [polymorphonuclear leukocytes](#) because of their oddly shaped nuclei. The three types of granulocytes—neutrophils, eosinophils, and basophils—are distinguished by the

different staining properties of their granules, which serve distinct functions. Granulocytes are generally relatively short-lived, surviving for only a few days. They mature in the bone marrow, and their production increases during immune responses, when they migrate to sites of infection or inflammation. The phagocytic [neutrophils](#) are the most numerous and important cells in innate immune responses: they take up a variety of microorganisms by phagocytosis and efficiently destroy them in intracellular vesicles by using degradative enzymes and other antimicrobial substances stored in their cytoplasmic granules. Hereditary deficiencies in neutrophil function open the way to overwhelming bacterial infection, which is fatal if untreated. Their role is discussed further in [Chapter 3](#).

[Eosinophils](#) and [basophils](#) are less abundant than neutrophils, but like neutrophils, they have granules containing a variety of enzymes and toxic proteins, which are released when these cells are activated. Eosinophils and basophils are thought to be important chiefly in defense against parasites, which are too large to be ingested by macrophages or neutrophils. They can also contribute to allergic inflammatory reactions, in which their effects are damaging rather than protective.

[Mast cells](#) begin development in the bone marrow but migrate as immature precursors that mature in peripheral tissues, especially skin, intestines, and airway mucosa. Their granules contain many inflammatory mediators, such as histamine and various proteases, which play a role in protecting the internal surfaces from pathogens, including parasitic worms. Because of their localization in barrier tissues, mast cells also function as early sensors of infection or injury and, in this role, contribute to the initiation of inflammation. We cover eosinophils, basophils, and mast cells and their role in allergic inflammation further in [Chapters 10](#) and [14](#).

[Dendritic cells](#) were discovered in the 1970s by Ralph Steinman, a finding for which he shared the 2011 Nobel Prize. These cells form the third class of phagocytic cells of the immune system and include several related lineages whose distinct functions are still being clarified. Most dendritic cells have elaborate membranous processes, like the dendrites of nerve cells. Immature dendritic cells migrate through the bloodstream from the bone marrow to enter tissues. They take up particulate matter by phagocytosis and also continually ingest large amounts of the extracellular fluid and its contents by a process known as [macropinocytosis](#). They degrade the pathogens that they take up, but their main role in the immune system is not the clearance of microorganisms. Instead, dendritic cells are a major class of sensor cells whose encounter with pathogens triggers them to produce mediators that activate other immune cells. Dendritic cells were discovered because of their role in activating a particular class of lymphocytes—T lymphocytes—of the adaptive immune system, and we will return to this activity when we discuss T-cell activation in [Section 1-15](#). But dendritic cells and the mediators they produce also play a critical role in controlling responses of cells of the innate immune system.

1-5 Sensor cells express pattern-recognition

receptors that provide an initial discrimination between self and nonself.

Long before the mechanisms of innate recognition were discovered, it was recognized that purified antigens such as proteins often did not evoke an immune response in an experimental immunization; that is, they were not [immunogenic](#). Rather, the induction of strong immune responses against purified proteins required the inclusion of microbial constituents, such as killed bacteria or bacterial extracts, famously called the immunologist's 'dirty little secret' by Charles Janeway (see Appendix I, [Section A-1](#)). This additional material was termed an [adjuvant](#), because it helped intensify the response to the immunizing antigen (*adjuvare* is Latin for 'to help'). We know now that adjuvants are needed, at least in part, to activate innate receptors on various types of sensor cells to help activate T cells in the absence of an infection.

Macrophages, neutrophils, and dendritic cells are important classes of sensor cells that detect infection and initiate immune responses by producing inflammatory mediators, although other cells, even cells of the adaptive immune system, can serve this function. As mentioned in [Section 1-3](#), these cells express a limited number of invariant innate recognition receptors as a means of detecting pathogens or the damage induced by them. Also called [pattern-recognition receptors \(PRRs\)](#), they recognize simple molecules and regular patterns of molecular structure known as [pathogen-associated molecular patterns \(PAMPs\)](#) that are part of many microorganisms but not of the host body's own cells. Such structures include mannose-rich oligosaccharides, peptidoglycans, and lipopolysaccharides of the bacterial cell wall, as well as unmethylated CpG DNA (unmethylated cytosine–guanine dinucleotide in DNA), which is common to many pathogens. All of these microbial elements have been conserved during evolution, making them excellent targets for recognition because they do not change (**Fig. 1.9**).

Macrophages express receptors for many microbial constituents

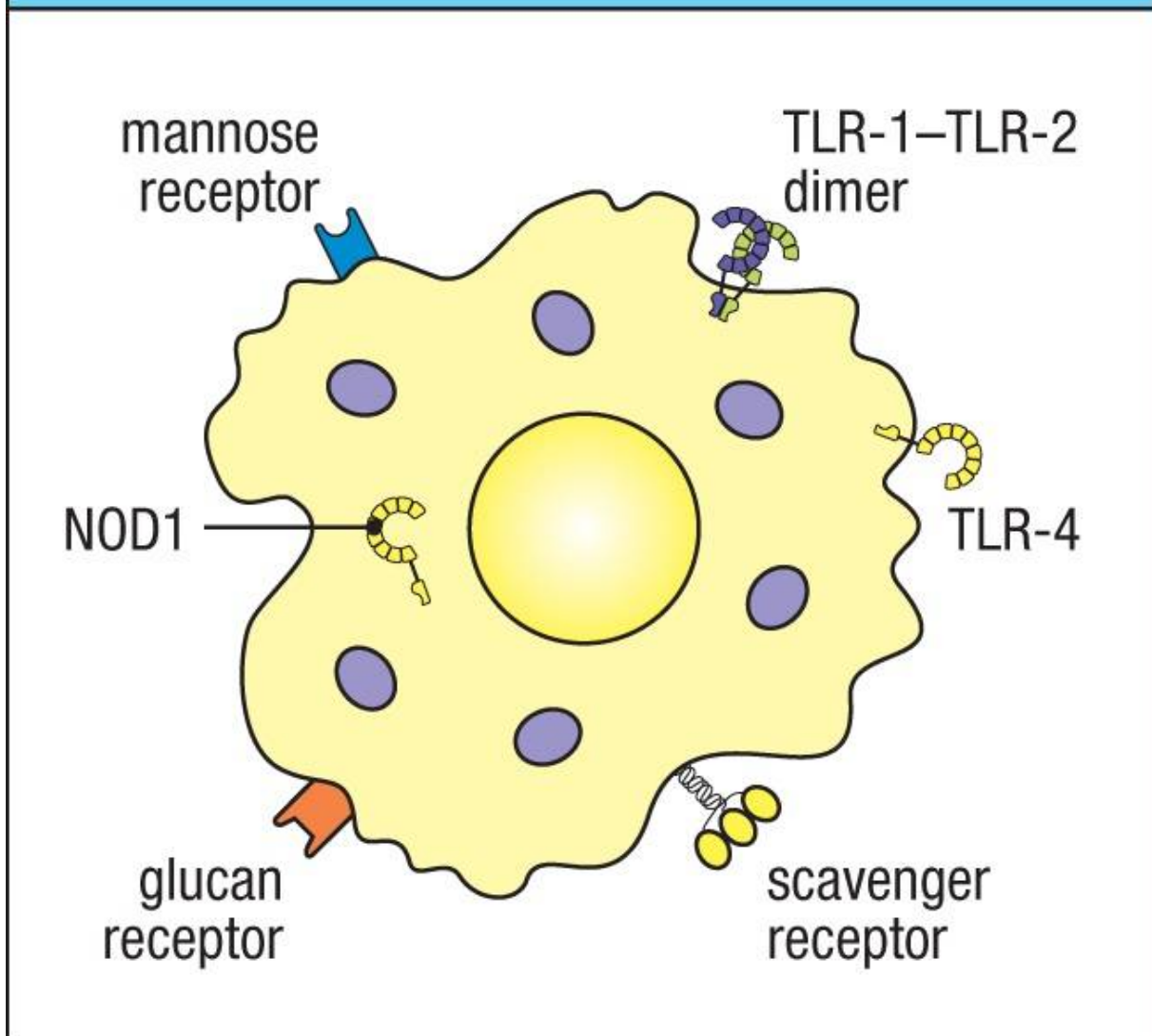


Fig. 1.9 Macrophages express a number of receptors that allow them to recognize different pathogens. Macrophages express a variety of receptors, each of which is able to recognize specific components of microbes. Some, like the mannose and glucan receptors and the scavenger receptor, bind cell-wall carbohydrates of bacteria, yeast, and fungi. The Toll-like receptors (TLRs) are an important family of pattern-recognition receptors present on macrophages, dendritic cells, and other immune cells. TLRs recognize different microbial components; for example, a heterodimer of TLR-1 and TLR-2 binds certain lipopeptides from pathogens such as Gram-positive bacteria, while TLR-4 binds both lipopolysaccharides from Gram-negative and lipoteichoic acids from Gram-positive bacteria. Other pattern-recognition receptors are intracellular, such as the nucleotide-binding and oligomerization domain (NOD)-like receptors. NOD1, for example, binds to diaminopimelic acid, a component of the cell wall of most Gram-negative bacteria.

Some PRRs are transmembrane proteins, such as the [Toll-like receptors \(TLRs\)](#) that detect PAMPs derived from extracellular bacteria or bacteria taken into vesicular pathways by phagocytosis. The role of the Toll receptor in immunity was discovered first in *Drosophila melanogaster* by Jules Hoffmann and later extended to homologous TLRs in mice by Charles Janeway and Bruce Beutler. (Hoffmann and Beutler shared the 2011 Nobel Prize along with Ralph Steinman—see [Section 1-4](#)—for their work in the activation of innate immunity.) Other PRRs are cytoplasmic proteins, such as the [NOD-like receptors \(NLRs\)](#) that sense intracellular bacterial invasion. Yet other cytoplasmic receptors detect viral infection on the basis of differences in the structures and locations of the host mRNA and virally derived RNA species and the differences between host and microbial DNA. Some receptors expressed by sensor cells detect cellular damage induced by pathogens, rather than the pathogens themselves. Much of our knowledge of innate recognition has emerged only within the past two decades and is still an active area of discovery. We describe these innate recognition systems further in [Chapter 3](#) and discuss how adjuvants are used as a component of vaccines in [Chapter 16](#).

1-6 Sensor cells induce an inflammatory response by producing mediators such as chemokines and cytokines.

Activation of PRRs on sensor cells such as macrophages and neutrophils can directly induce effector functions in these cells, such as the phagocytosis and degradation of bacteria they encounter. But sensor cells serve to amplify the immune response by the production of inflammatory mediators. Two important categories of inflammatory mediators are the secreted proteins called [cytokines](#) and [chemokines](#), which act in a manner similar to hormones to convey important signals to other immune cells.

‘Cytokine’ is a term for any protein secreted by immune cells that affects the behavior of nearby cells bearing appropriate receptors. There are more than 60 different cytokines; some are produced by many different cell types, and others are produced by only a few specific cell types. Some cytokines influence many types of cells, while others influence only a few, this through the expression pattern of each cytokine’s specific receptor. The response that a cytokine induces in a target cell is typically related to amplification of an effector mechanism of the target cell, as illustrated in the remaining sections of this chapter.

Instead of presenting all the cytokines together all at once, we introduce each cytokine as it arises during our description of cellular and functional responses. We list the cytokines, their producer and target cells, and their general functions in [eAppendix III](#), found in the ebook and on the [Digital Landing Page](#) for this book.

Chemokines are a specialized subgroup of secreted proteins that act as chemoattractants,

attracting cells bearing chemokine receptors, such as neutrophils and monocytes, out of the bloodstream and into infected tissue (**Fig. 1.10**). Beyond this role, chemokines also help organize the various cells in lymphoid tissues into discrete regions where specialized responses can take place. There are about 50 different chemokines. [eAppendix IV](#), found in the ebook and on the [Digital Landing Page](#) for this book, lists the chemokines, their target cells, and their general functions. We will discuss chemokines as the need arises during our descriptions of particular cellular immune processes.

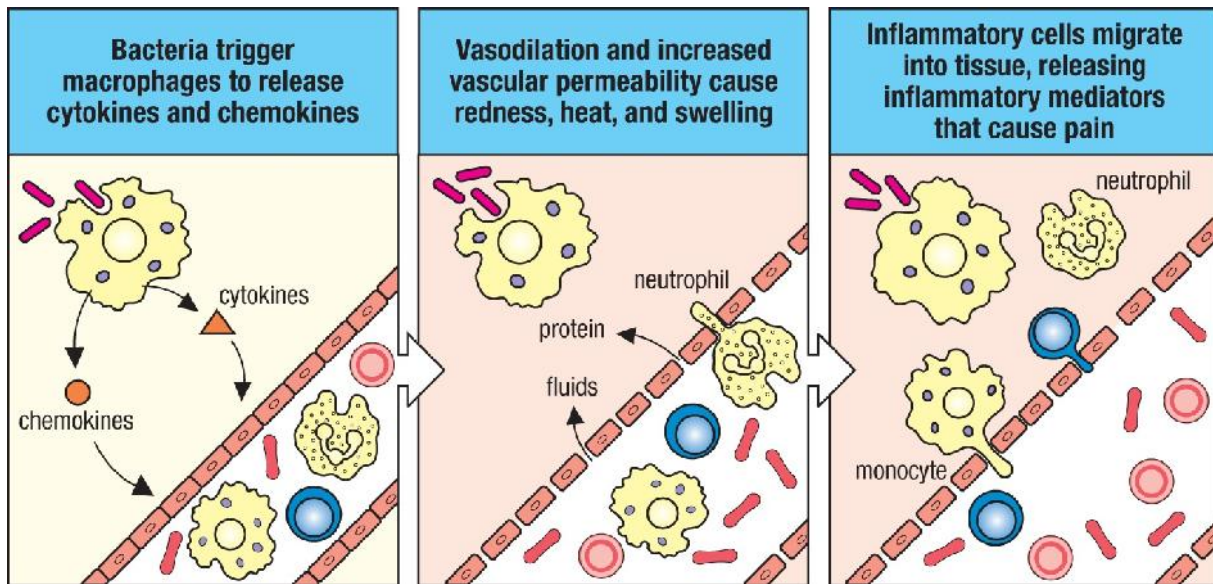


Fig. 1.10 Infection triggers an inflammatory response. Macrophages encountering bacteria or other types of microorganisms in tissues are triggered to release cytokines (left panel) that increase the permeability of blood vessels, allowing fluid and proteins to pass into the tissues (center panel). Macrophages also produce chemokines, which direct the migration of neutrophils to the site of infection. The stickiness of the endothelial cells of the blood vessel wall is also changed, so that circulating cells of the immune system adhere to the wall and are able to crawl through it; first neutrophils and then monocytes are shown entering the tissue from a blood vessel (right panel). The accumulation of fluid and cells at the site of infection causes the redness, swelling, heat, and pain known collectively as inflammation. Neutrophils and macrophages are the principal inflammatory cells. Later in an immune response, activated lymphocytes can also contribute to inflammation.

The cytokines and chemokines released by activated macrophages act to recruit cells from the blood into infected tissues, a process known as [inflammation](#), which helps to destroy the pathogen. Inflammation increases the flow of lymph, which carries microbes or cells bearing their antigens from the infected tissue to nearby lymphoid tissues, where the adaptive immune response is initiated. Once adaptive immunity has been generated, inflammation also recruits these effector components to the site of infection.

Inflammation is described clinically by the Latin words *calor*, *dolor*, *rubor*, and *tumor*, meaning heat, pain, redness, and swelling. Each of these features reflects an effect of

cytokines or other inflammatory mediators on the local blood vessels. Heat, redness, and swelling result from the dilation and increased permeability of blood vessels during inflammation, leading to increased local blood flow and leakage of fluid and blood proteins into the tissues. Cytokines and complement fragments have important effects on the [endothelium](#) that lines blood vessels; the [endothelial cells](#) themselves also produce cytokines in response to infection. These alter the adhesive properties of the endothelial cells and cause circulating leukocytes to stick to the endothelial cells and migrate between them into the site of infection, to which they are attracted by chemokines. The migration of cells into the tissue and their local actions account for the pain.

The main cell types seen in the initial phase of an inflammatory response are macrophages and neutrophils, the latter being recruited into the inflamed, infected tissue in large numbers. Macrophages and neutrophils are thus also known as [inflammatory cells](#). The influx of neutrophils is followed a short time later by the increased entry of monocytes, which rapidly differentiate into macrophages, thus reinforcing and sustaining the innate immune response. Later, if the inflammation continues, eosinophils also migrate into inflamed tissues and contribute to the destruction of the invading microorganisms.

1-7 Innate lymphoid cells and natural killer cells are effector cells that share similarities with lymphoid lineages of the adaptive immune system.

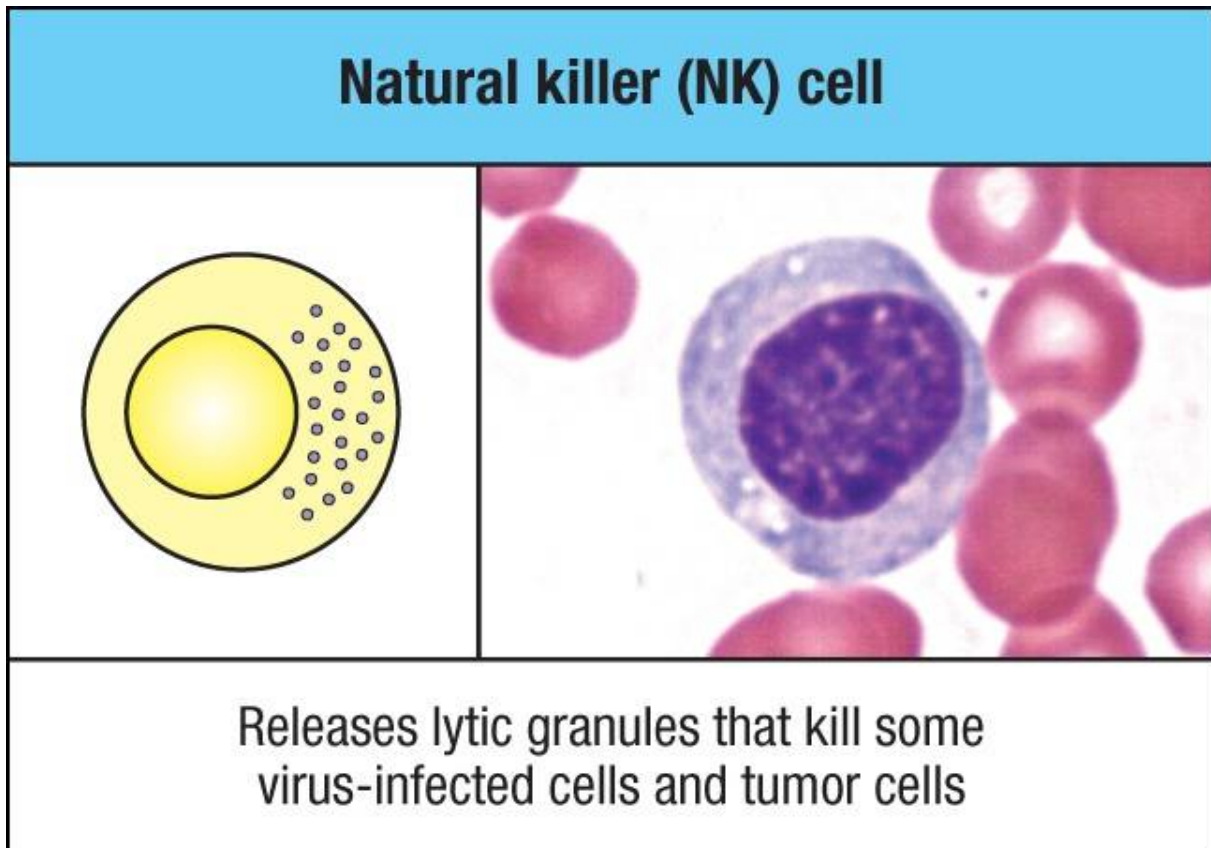


Fig. 1.11 Natural killer (NK) cells. These are large, granular, lymphoid-like cells with important functions in innate immunity (especially against intracellular infections), being able to kill other cells. Unlike lymphocytes, they lack antigen-specific receptors.

The [common lymphoid progenitor \(CLP\)](#) in the bone marrow gives rise both to antigen-specific lymphocytes of the adaptive immune system and to several innate lineages that lack antigen-specific receptors. Although the B and T lymphocytes of the adaptive immune system were recognized in the 1960s, the [natural killer \(NK\) cells](#) (**Fig. 1.11**) of the innate immune system were not discovered until the 1970s. NK cells are large lymphocyte-like cells with a distinctive granular cytoplasm that were identified because of their ability to recognize and kill certain tumor cells and cells infected with herpesviruses. Initially, the distinction between these cells and T lymphocytes was unclear, but we now recognize that NK cells are a distinct lineage of cells that arise from the CLP in the bone marrow. They lack the antigen-specific receptors of the adaptive immune system cells but express members of various families of innate receptors that can respond to cellular stress and to infections by very specific viruses. NK cells play an important role in the early innate response to viral infections, before the adaptive immune response has developed.

More recently, additional lineages of cells related to NK cells have been identified. Collectively, these cells are called [innate lymphoid cells \(ILCs\)](#). Arising from the CLP, ILCs reside in peripheral tissues, such as the intestine, where they function as the sources of mediators of inflammatory responses. The functions of NK cells and ILCs are discussed in [Chapter 3](#).

Summary.

Strategies of avoidance, resistance, and tolerance represent different ways to deal with pathogens. Anatomic barriers and various chemical barriers, such as complement and antimicrobial proteins, may be considered a primitive form of avoidance, and they are the first line of defense against entry of both commensal organisms and pathogens into host tissues. If these barriers are breached, the vertebrate immune response becomes largely focused on resistance. Inflammatory inducers, which may be either chemical structures unique to microbes (PAMPs) or the chemical signals of tissue damage, act on receptors expressed by sensor cells to inform the immune system of infection. Sensor cells are typically innate immune cells such as macrophages or dendritic cells. Sensor cells can either directly respond with effector activity or produce inflammatory mediators, typically cytokines and chemokines that act on other immune cells, such as the innate NK cells and ILCs. These cells then are recruited into target tissues to provide specific types of immune-response effector activities, such as cell killing or production of cytokines that have direct antiviral activity, all aimed to reduce or eliminate infection by pathogens. Inflammatory mediators released in target tissues can recruit and activate several types of inflammatory cells that are specially suited for eliminating viruses, intracellular bacteria, extracellular pathogens, or parasites.

Glossary

[lymphocyte](#)

A class of white blood cells that bear variable cell-surface receptors for antigen and are responsible for adaptive immune responses. There are two main types—B lymphocytes (B cells) and T lymphocytes (T cells)—which mediate humoral and cell-mediated immunity, respectively. On antigen recognition, a lymphocyte enlarges to form a lymphoblast and then proliferates and differentiates into an antigen-specific effector cell.

[pathogen](#)

Microorganism that typically causes disease when it infects a host.

[virus](#)

Pathogen composed of a nucleic acid genome enclosed in a protein coat. Viruses can replicate only in a living cell, because they do not possess the metabolic machinery for independent life.

[bacteria](#)

A vast kingdom of unicellular prokaryotic microorganisms, some species of which cause infectious diseases in humans and animals, while others make up most of the body's commensal microbiota. Disease-causing bacteria may live in the extracellular spaces or inside cells in vesicles or in the cytosol.

[fungi](#)

A kingdom of single-celled and multicellular eukaryotic organisms, including the

yeasts and molds, that can cause a variety of diseases. Immunity to fungi is complex and involves both humoral and cell-mediated responses.

[parasites](#)

Organisms that obtain sustenance from a live host. In immunology, it refers to worms and protozoa, the subject matter of parasitology.

[microbiome](#)

The collection of genomes from all the microorganisms in an environment, such as the commensal microorganisms of the gut or skin. The term is to be contrasted with 'microbiota,' which refers to the collection of microorganisms themselves. *See* **commensal microorganisms**.

[symbiotic](#)

Relationship between two agents, typically diverse species, that confers benefits to both.

[avoidance](#)

Mechanisms that prevent a host's exposure to microbes, such as anatomic barriers or particular behaviors.

[resistance](#)

A general immune strategy aimed at reducing or eliminating pathogens. Cf. **avoidance** and **tolerance**.

[tolerance](#)

The failure to respond to an antigen. Tolerance to self antigens is an essential feature of the immune system; when tolerance is lost, the immune system can destroy self tissues, as happens in autoimmune disease.

[effector mechanisms](#)

Those processes by which pathogens are destroyed and cleared from the body. Innate and adaptive immune responses use most of the same effector mechanisms to eliminate pathogens.

[immunological tolerance](#)

The failure to respond to an antigen. Tolerance to self antigens is an essential feature of the immune system; when tolerance is lost, the immune system can destroy self tissues, as happens in autoimmune disease.

[sensor cells](#)

Cells of the innate immune system that detect the presence of pathogens using their innate recognition receptors. In response to triggering of their innate recognition receptors, sensor cells produce inflammatory mediators that promote pathogen clearance and recruit additional cells into the immune response.

[inflammatory inducers](#)

Chemical structures that indicate the presence of invading microbes or cellular damage, such as bacterial lipopolysaccharides, extracellular ATP, or urate crystals.

[innate recognition receptors](#)

General term for a large group of proteins that recognize many different inflammatory inducers and that are encoded in the germline and do not need gene rearrangement in somatic cells to be expressed.

[inflammatory mediators](#)

Chemicals such as cytokines produced by immune cells that act on target cells to promote defense against microbes.

[antigen receptor](#)

The cell-surface receptor by which lymphocytes recognize antigen. Each individual lymphocyte bears receptors of a single antigen specificity.

[common myeloid progenitor \(CMP\)](#)

Stem cell that can give rise to the myeloid cells of the immune system (the macrophages, granulocytes, mast cells, and dendritic cells of the innate immune system). This stem cell also gives rise to megakaryocytes and red blood cells.

[macrophages](#)

Large mononuclear phagocytic cells present in most tissues that have many functions; for example, scavenger cells, pathogen-recognition cells, and production of pro-inflammatory cytokines. Macrophages arise both embryonically and from bone marrow precursors throughout life.

[monocyte](#)

Type of white blood cell with a bean-shaped nucleus; it is a precursor of tissue macrophages.

[complement receptors \(CRs\)](#)

Cell-surface proteins of various types that recognize and bind complement proteins that have become bound to an antigen such as a pathogen. Complement receptors on phagocytes enable them to identify and bind pathogens coated with complement proteins and to ingest and destroy them. *See* **CR1, CR2, CR3, CR4, CR1g**, and the **C1 complex**.

[granulocytes](#)

White blood cells with multilobed nuclei and cytoplasmic granules. They comprise the neutrophils, eosinophils, and basophils. Also known as polymorphonuclear leukocytes.

[polymorphonuclear leukocytes](#)

White blood cells with multilobed nuclei and cytoplasmic granules. They comprise the neutrophils, eosinophils, and basophils. Also known as granulocytes.

[neutrophil](#)

The most numerous type of white blood cell in human peripheral blood. Neutrophils are phagocytic cells with a multilobed nucleus and granules that stain with neutral stains. They enter infected tissues and engulf and kill extracellular pathogens.

[eosinophil](#)

A type of white blood cell containing granules that stain with eosin. It is thought to be important chiefly in defense against parasitic infections but is also medically important as an effector cell in allergic reactions.

[basophil](#)

Type of white blood cell containing granules that stain with basic dyes. It is thought to have a function similar to that of mast cells.

[mast cell](#)

A large granule-rich cell found in connective tissues throughout the body, most abundantly in the submucosal tissues and the dermis. The granules store bioactive molecules including the vasoactive amine histamine, which are released on mast-cell

activation. Mast cells are thought to be involved in defenses against parasites, and they have a crucial role in allergic reactions.

[dendritic cells](#)

Bone marrow–derived cells found in most tissues, including lymphoid tissues. There are two main functional subsets. Conventional dendritic cells take up antigen in peripheral tissues, are activated by contact with pathogens, and travel to the peripheral lymphoid organs, where they are the most potent stimulators of T-cell responses. Plasmacytoid dendritic cells can also take up and present antigen, but their main function in an infection is to produce large amounts of the antiviral interferons as a result of pathogen recognition through receptors such as TLRs. Both these types of dendritic cells are distinct from the follicular dendritic cell that presents antigen to B cells in lymphoid follicles.

[macropinocytosis](#)

A process in which large amounts of extracellular fluid are taken up into an intracellular vesicle. This is one way in which dendritic cells can take up a wide variety of antigens from their surroundings.

[immunogenic](#)

Elicitation of an adaptive immune response.

[adjuvant](#)

Any substance that enhances the immune response to an antigen with which it is mixed.

[pattern-recognition receptors \(PRRs\)](#)

Receptors of the innate immune system that recognize common molecular patterns on pathogen surfaces.

[pathogen-associated molecular patterns \(PAMPs\)](#)

Molecules specifically associated with groups of pathogens that are recognized by cells of the innate immune system.

[Toll-like receptors \(TLRs\)](#)

Innate receptors on macrophages, dendritic cells, and some other cells that recognize pathogens and their products, such as bacterial lipopolysaccharide. Recognition stimulates the receptor-bearing cells to produce cytokines that help initiate immune responses.

[NOD-like receptors \(NLRs\)](#)

Large family of proteins containing a nucleotide-binding and oligomerization domain (NOD) associated with various other domains, and whose general function is the detection of microbes and of cellular stress.

[cytokines](#)

Proteins made by a cell that affect the behavior of other cells, particularly immune cells. Cytokines made by lymphocytes are often called interleukins (abbreviated ILs). Cytokines and their receptors are listed in eAppendix III. Cf. **chemokines**.

[chemokines](#)

Small chemoattractant proteins that stimulate the migration and activation of cells, especially phagocytic cells and lymphocytes. Chemokines have a central role in inflammatory responses. Properties of individual chemokines are listed in eAppendix

IV.

inflammation

General term for the local accumulation of fluid, plasma proteins, and white blood cells that is initiated by physical injury, infection, or a local immune response.

endothelium

The epithelium that forms the walls of blood capillaries and the lining of larger blood vessels.

endothelial cell

Cell type that forms the endothelium, the epithelium of a blood vessel wall.

inflammatory cells

Cells such as macrophages, neutrophils, and effector T_H1 lymphocytes that invade inflamed tissues and contribute to the inflammation.

common lymphoid progenitor (CLP)

Stem cell that can give rise to all the types of lymphocytes but is biased toward producing B cells.

natural killer (NK) cell

A type of ILC that is important in innate immunity to viruses and other intracellular pathogens and in antibody-dependent cell-mediated cytotoxicity (ADCC). NK cells express activating and inhibitory receptors but not the antigen-specific receptors of T or B cells.

innate lymphoid cells (ILCs)

These are a class of innate immune cells having overlapping characteristics with T cells but lacking an antigen receptor. They arise in several groups, ILC1, ILC2, ILC3, and NK cells, which exhibit properties roughly similar to T_H1, T_H2, T_H17, and CD8 T cells.

commensal microbiota, commensal microorganisms

Microorganisms (predominantly bacteria) that normally live harmlessly in symbiosis with their host (e.g., the gut bacteria in humans and other animals). Many commensals confer a positive benefit on their host in some way.

antimicrobial peptides, antimicrobial proteins

Amphipathic peptides or proteins secreted by epithelial cells and phagocytes that kill a variety of microbes nonspecifically, mainly by disrupting cell membranes. Antimicrobial peptides in humans include the defensins, the cathelicidins, the histatins, and RegIIIγ.

Principles of adaptive immunity.

We come now to the components of adaptive immunity, the antigen-specific lymphocytes. Unless indicated otherwise, we shall use the term 'lymphocyte' in the remainder of this chapter to refer only to the antigen-specific lymphocytes. Lymphocytes allow responses against a vast array of antigens from various pathogens encountered during a person's lifetime and confer the important feature of immunological memory. Lymphocytes make this possible through the highly variable antigen receptors on their surface, by which they recognize and bind antigens. Each lymphocyte matures bearing a unique variant of a prototype antigen receptor, so that the population of lymphocytes expresses a huge repertoire of receptors that are highly diverse in their antigen-binding sites. Among the billion or so lymphocytes circulating in the body at any one time there will always be some that can recognize a given foreign antigen.

A unique feature of the adaptive immune system is that it is capable of generating [immunological memory](#), so that having been exposed once to an infectious agent, a person will make an immediate and stronger response against any subsequent exposure to it; that is, the individual will have protective immunity against it. Finding ways of generating long-lasting immunity to pathogens that do not naturally provoke it is one of the greatest challenges facing immunologists today.

1-8 The interaction of antigens with antigen receptors induces lymphocytes to acquire effector and memory activity.

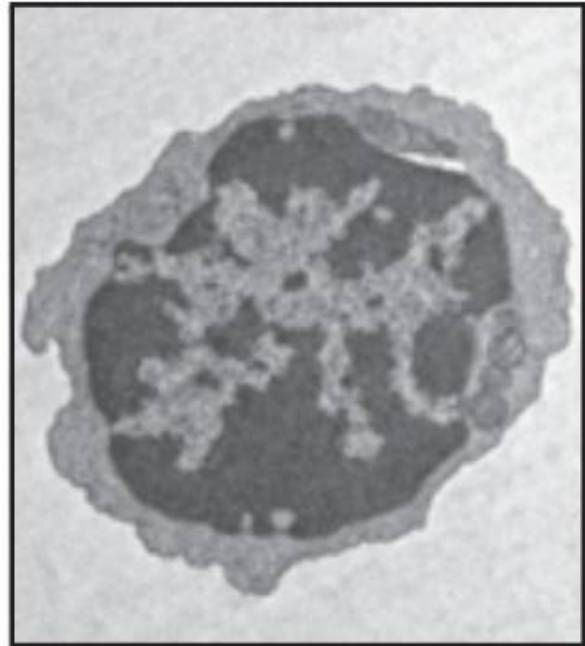
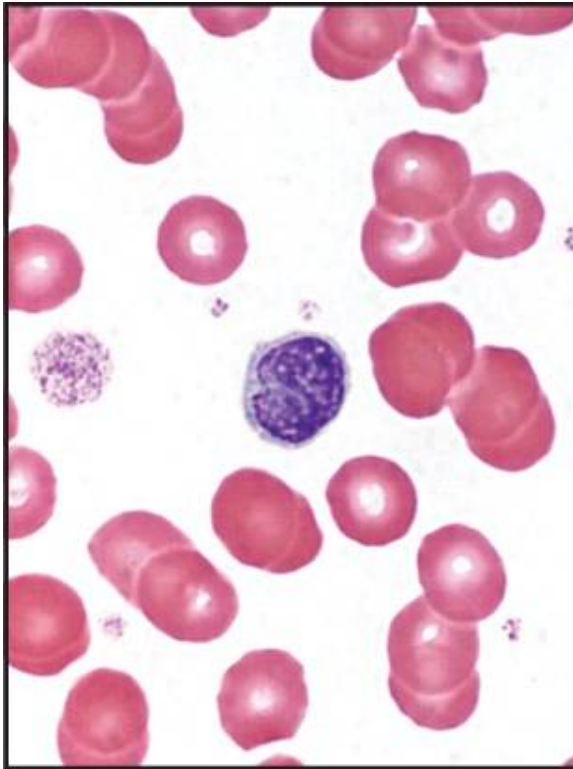


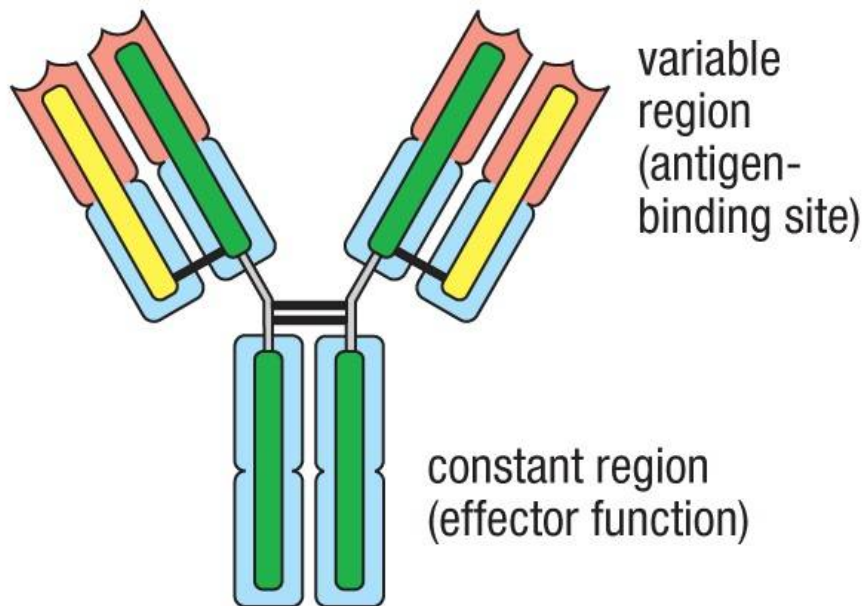
Fig. 1.12 Lymphocytes are mostly small and inactive cells. The left panel shows a light micrograph of a small lymphocyte—in which the nucleus has been stained purple by hematoxylin and eosin dye—surrounded by red blood cells (which have no nuclei). Note the darker purple patches of condensed chromatin of the lymphocyte nucleus, indicating little transcriptional activity and the relative absence of cytoplasm. The right panel shows a transmission electron micrograph of a small lymphocyte. Again, note the evidence of functional inactivity: the condensed chromatin, the scanty cytoplasm, and the absence of rough endoplasmic reticulum.

There are two major types of lymphocytes in the vertebrate immune system, the [B lymphocytes \(B cells\)](#) and [T lymphocytes \(T cells\)](#). These express distinct types of antigen receptors and have quite different roles in the immune system, as was discovered in the 1960s. Most lymphocytes circulating in the body appear as rather unimpressive small cells with few cytoplasmic organelles and a condensed, inactive-appearing nuclear chromatin (**Fig. 1.12**). Lymphocytes manifest little functional activity until they encounter a specific antigen that interacts with an antigen receptor on their cell surface. Lymphocytes that have not yet been activated by antigen are known as [naive lymphocytes](#); those that have met their antigen, become activated, and have differentiated further into fully functional lymphocytes are known as [effector lymphocytes](#).

B cells and T cells are distinguished by the structure of the antigen receptor that they express. The [B-cell antigen receptor](#), or [B-cell receptor \(BCR\)](#), is formed by the same genes that encode antibodies, therefore B-cell receptors and antibodies compose the class of proteins known as [immunoglobulin \(Ig\)](#) (**Fig. 1.13**). Thus, the antigen receptor of B lymphocytes is also known as [membrane immunoglobulin \(mIg\)](#) or [surface immunoglobulin](#)

(sIg). The [T-cell antigen receptor](#), or [T-cell receptor \(TCR\)](#), is related to the immunoglobulins but is quite distinct in its structure and recognition properties.

Schematic structure of an antibody molecule



Schematic structure of the T-cell receptor

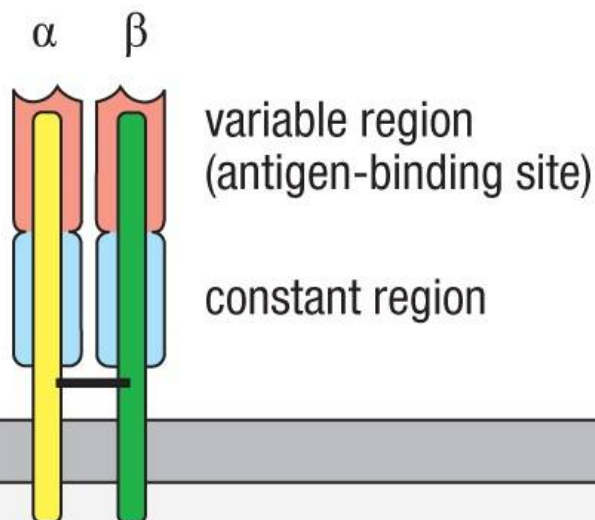


Fig. 1.13 Schematic structure of antigen receptors. Upper panel: an antibody molecule, which is secreted by activated B cells as an antigen-binding effector molecule. A membrane-bound version of this molecule acts as the B-cell antigen receptor (not shown). An antibody is composed of two identical heavy chains (green) and two identical light chains (yellow). Each chain has a constant part (shaded blue) and a variable part (shaded red). Each arm of the antibody molecule is formed by a light chain and a heavy chain, with the variable parts of the two chains coming together to create a variable region that contains the antigen-binding site. The stem is formed from the constant parts of the heavy chains and takes a limited number of forms. This constant region is involved in the elimination of the bound antigen. Lower panel: a T-cell antigen receptor. This is also composed of two chains, an α chain (yellow) and a β chain (green), each of which has a variable and a constant part. As with the antibody molecule, the variable parts of the two chains create a variable region, which forms the antigen-binding site. The T-cell receptor is not produced in a secreted form.

After antigen binds to a B-cell receptor, the B cell will proliferate and differentiate into [plasma cells](#). These are the effector form of B lymphocytes, and they secrete antibodies that have the same antigen specificity as the plasma cell's B-cell receptor. Thus the antigen that activates a given B cell becomes the target of the antibodies produced by that B cell's progeny.

When a T cell first encounters an antigen that its receptor can bind, it proliferates and differentiates into one of several different functional types of [effector T lymphocytes](#). When effector T cells subsequently detect antigen, they can manifest three broad classes of activity. [Cytotoxic T cells](#) kill other cells that are infected with viruses or other intracellular pathogens bearing the antigen. [Helper T cells](#) provide signals, often in the form of specific cytokines that activate the functions of other cells, such as B-cell production of antibody and macrophage killing of engulfed pathogens. [Regulatory T cells](#) suppress the activity of other lymphocytes and help to limit the possible damage of immune responses. We discuss the detailed functions of cytotoxic, helper, and regulatory T cells in [Chapters 9, 11, 12, and 15](#).

Some of the B cells and T cells activated by antigen will differentiate into [memory cells](#), the lymphocytes that are responsible for the long-lasting immunity that can follow exposure to disease or vaccination. Memory cells will readily differentiate into effector cells on a second exposure to their specific antigen. Immunological memory is described in [Chapter 11](#).

1-9 Antibodies and T-cell receptors are composed of constant and variable regions that provide distinct functions.

Antibodies were studied by traditional biochemical techniques long before recombinant

DNA technology allowed the study of the membrane-bound forms of the antigen receptors on B and T cells. These early studies found that antibody molecules are composed of two distinct regions. One is a [constant region](#), also called the [fragment crystallizable region](#), or [Fc region](#), which takes one of only five biochemically distinguishable forms (see Fig. 1.13). The [variable region](#) of different antibody molecules, by contrast, can be composed of a vast number of different amino acid sequences that allow antibodies to recognize an equally vast variety of antigens. The basic structure of antibody molecules was determined by Gerald Edelman and Rodney Porter, who shared the 1972 Nobel Prize for their work. These biochemical studies led to subsequent X-ray crystallographic determination of the structure of the antibody Fc region.

The antibody molecule is composed of two identical [heavy chain](#) and two identical [light chain](#) polypeptides. Heavy and light chains each have variable and constant regions. The variable regions of a heavy chain and a light chain combine to form an [antigen-binding site](#) that determines the antigen-binding specificity of the antibody. Thus, both heavy and light chains contribute to the antigen-binding specificity of the antibody molecule. Also, each antibody has two identical variable regions, and so has two identical antigen-binding sites. The constant region determines the effector function of the antibody; that is, how the antibody will interact with various immune cells to dispose of antigen once it is bound.

The T-cell receptor shows many similarities to the B-cell receptor and antibody (see Fig. 1.13). It is composed of two chains, the TCR α and β chains, that are roughly equal in size and which span the T-cell membrane. Like antibody, each TCR chain has a variable region and a constant region, and the combination of the α - and β -chain variable regions creates a single site for binding antigen. The structures of both antibodies and T-cell receptors are described in detail in [Chapter 4](#), and functional properties of antibody constant regions are discussed in [Chapters 4](#) and [10](#).

1-10 Antibodies and T-cell receptors recognize antigens by fundamentally different mechanisms.

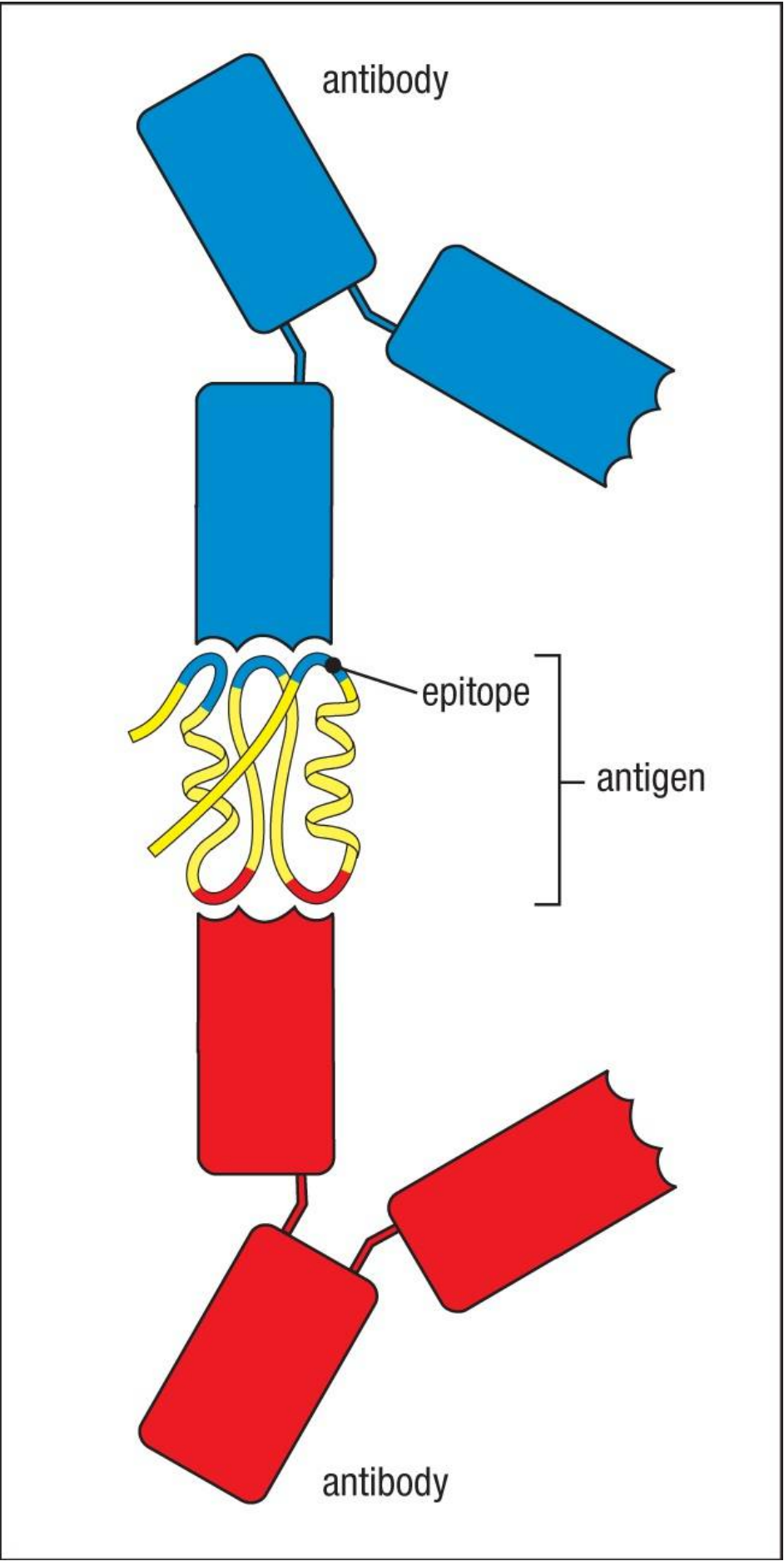


Fig. 1.14 Antigens are the molecules recognized by the immune response, while epitopes are sites within antigens to which antigen receptors bind. Antigens can be complex macromolecules such as proteins, as shown in yellow. Most antigens are larger than the sites on the antibody or antigen receptor to which they bind, and the actual portion of the antigen that is bound is known as the antigenic determinant, or epitope, for that receptor. Large antigens such as proteins can contain more than one epitope (indicated in red and blue) and thus may bind different antibodies (shown here in the same color as the epitopes they bind). Antibodies generally recognize epitopes on the surface of the antigen.

In principle, almost any chemical structure can be recognized as an antigen by the adaptive immune system, but the usual antigens encountered in an infection are the proteins, glycoproteins, and polysaccharides of pathogens. An individual antigen receptor or antibody recognizes a small portion of the antigen's molecular structure, and the part recognized is known as an [antigenic determinant](#), or [epitope](#) (**Fig. 1.14**). Typically, proteins and glycoproteins have many different epitopes that can be recognized by different antigen receptors.

Antibodies and B-cell receptors directly recognize the epitopes of native antigen in the serum or the extracellular spaces. It is possible for different antibodies to simultaneously recognize an antigen by its different epitopes; such simultaneous recognition increases the efficiency of clearing or neutralizing the antigen.

Whereas antibodies can recognize nearly any type of chemical structure, T-cell receptors usually recognize protein antigens and do so very differently from antibodies. The T-cell receptor recognizes a peptide epitope derived from a partially degraded protein, but only if the peptide is bound to specialized cell-surface glycoproteins called [MHC molecules](#) (**Fig. 1.15**). The members of this large family of cell-surface glycoproteins are encoded in a cluster of genes called the [major histocompatibility complex \(MHC\)](#). The antigens recognized by T cells can be derived from proteins arising from intracellular pathogens, such as a virus, or from extracellular pathogens. A further difference from the antibody molecule is that there is no secreted form of the T-cell receptor; the T-cell receptor functions solely to signal to the T cell that it has bound its antigen, and the subsequent immunological effects depend on the actions of the T cells themselves. We will further describe how epitopes from antigens are placed on MHC proteins in [Chapter 6](#) and how T cells carry out their subsequent functions in [Chapter 9](#).

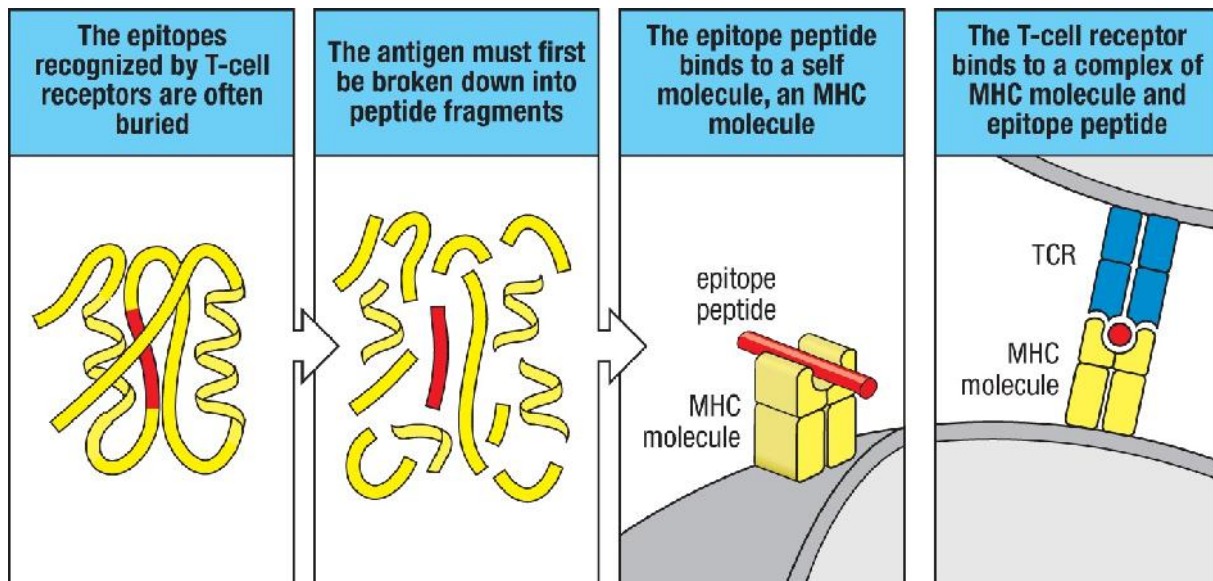


Fig. 1.15 T-cell receptors bind a complex of an antigen fragment and a self molecule. Unlike most antibodies, T-cell receptors can recognize epitopes that are buried within antigens (first panel). These antigens must first be degraded by proteases (second panel) and the epitope peptide delivered to a self molecule, called an MHC molecule (third panel). It is in this form, as a complex of peptide and MHC molecule, that antigens are recognized by T-cell receptors (TCRs; fourth panel).

1-11 Antigen-receptor genes are assembled by somatic gene rearrangements of incomplete receptor gene segments.

The innate immune system detects inflammatory stimuli by means of a relatively limited number of sensors, such as the TLR and NOD proteins, numbering fewer than 100 different types of proteins. Antigen-specific receptors of adaptive immunity provide a seemingly infinite range of specificities and yet are encoded by a finite number of genes. The basis for this extraordinary range of specificity was discovered in 1976 by Susumu Tonegawa, for which he was awarded the 1987 Nobel Prize. Immunoglobulin variable regions are inherited as sets of [gene segments](#), each encoding a part of the variable region of one of the immunoglobulin polypeptide chains. During B-cell development in the bone marrow, these gene segments are irreversibly joined by a process of DNA recombination to form a stretch of DNA encoding a complete variable region. A similar process of antigen-receptor gene rearrangement takes place for the T-cell receptor genes during development of T cells in the thymus.

Just a few hundred different gene segments can combine in different ways to generate thousands of different receptor chains. This [combinatorial diversity](#) allows a small amount of genetic material to encode a truly staggering diversity of receptors. During this

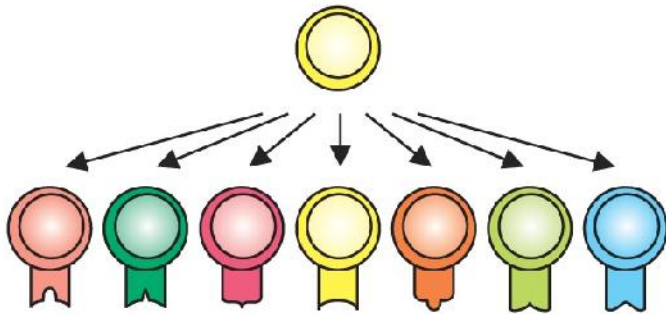
recombination process, the random addition or subtraction of nucleotides at the junctions of the gene segments creates additional diversity known as [junctional diversity](#). Diversity is amplified further by the fact that each antigen receptor has two different variable chains, each encoded by distinct sets of gene segments. In [Chapter 5](#) we will describe the gene-rearrangement process that assembles complete antigen receptors from gene segments.

1-12 Lymphocytes activated by antigen give rise to clones of antigen-specific effector cells that mediate adaptive immunity.

There are two critical features of lymphocyte development that distinguish adaptive immunity from innate immunity. First, the process that assembles antigen receptors from incomplete gene segments, described earlier, is carried out in a manner ensuring that each developing lymphocyte expresses only one receptor specificity. Whereas the cells of the innate immune system express many different pattern-recognition receptors and recognize features shared by many pathogens, the antigen-receptor expression of lymphocytes is 'clonal,' so that each mature lymphocyte differs from others in the specificity of its antigen receptor. Second, because the gene-rearrangement process irreversibly changes the lymphocyte's DNA, all its progeny inherit the same receptor specificity. Because this specificity is inherited by a cell's progeny, the proliferation of an individual lymphocyte forms a [clone](#) of cells with identical antigen receptors.

There are lymphocytes of at least 10^8 different specificities in an individual human at any one time, composing the [lymphocyte receptor repertoire](#) of the individual. These lymphocytes are continually undergoing a process similar to natural selection: only those lymphocytes that encounter an antigen to which their receptor binds will be activated to proliferate and differentiate into effector cells. This selective mechanism was first proposed in the 1950s by Frank Macfarlane Burnet, who postulated the preexistence in the body of many different potential antibody-producing cells, each displaying on its surface a membrane-bound version of the antibody that served as a receptor for the antigen. On binding antigen, the cell is activated to divide and to produce many identical progeny, a process known as [clonal expansion](#); this clone of identical cells can now secrete [clonotypic](#) antibodies with a specificity identical to that of the surface receptor that first triggered activation and clonal expansion (**Fig. 1.16**). Burnet called this the [clonal selection theory](#) of antibody production; its four basic postulates are listed in **Fig. 1.17**. Clonal selection of lymphocytes is the single most important principle in adaptive immunity.

A single progenitor cell gives rise to a large number of lymphocytes, each with a different specificity



Removal of potentially self-reactive immature lymphocytes by clonal deletion



Pool of mature naive lymphocytes



Proliferation and differentiation of activated specific lymphocytes to form a clone of effector cells

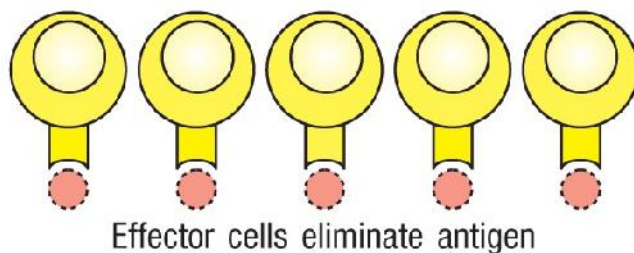


Fig. 1.16 Clonal selection. Each lymphoid progenitor gives rise to a large number of lymphocytes, each bearing a distinct antigen receptor. Lymphocytes with receptors that bind ubiquitous self antigens are eliminated before they become fully mature, ensuring tolerance to such self antigens. When a foreign antigen (red dot) interacts with the receptor on a mature naive lymphocyte, that cell is activated and starts to divide. It gives rise to a clone of identical progeny, all of whose receptors bind the same antigen. Antigen specificity is thus maintained as the progeny proliferate and differentiate into effector cells. Once antigen has been eliminated by these effector cells, the immune response ceases, although some lymphocytes are retained to mediate immunological memory.

Postulates of the clonal selection hypothesis
Each lymphocyte bears a single type of receptor with a unique specificity
Interaction between a foreign molecule and a lymphocyte receptor capable of binding that molecule with high affinity leads to lymphocyte activation
The differentiated effector cells derived from an activated lymphocyte will bear receptors of identical specificity to those of the parental cell from which that lymphocyte was derived
Lymphocytes bearing receptors specific for ubiquitous self molecules are deleted at an early stage in lymphoid cell development and are therefore absent from the repertoire of mature lymphocytes

Fig. 1.17 The four basic principles of clonal selection.

1-13 Lymphocytes with self-reactive receptors are normally eliminated during development or are functionally inactivated.

When Burnet formulated his theory, nothing was known of the antigen receptors or indeed the function of lymphocytes themselves. In the early 1960s, James Gowans discovered that removal of the small lymphocytes from rats resulted in the loss of all known adaptive immune responses, which were restored when the small lymphocytes were replaced. This led to the realization that lymphocytes must be the units of clonal selection, and their biology

became the focus of the new field of [cellular immunology](#).

Clonal selection of lymphocytes with diverse receptors elegantly explained adaptive immunity, but it raised one significant conceptual problem. With so many different antigen receptors being generated randomly during the lifetime of an individual, there is a possibility that some receptors might react against an individual's own [self antigens](#). How are lymphocytes prevented from recognizing native antigens on the tissues of the body and attacking them? Ray Owen had shown in the late 1940s that genetically different twin calves with a common placenta, and thus a shared placental blood circulation, were immunologically unresponsive, or [tolerant](#), to one another's tissues. Peter Medawar then showed in 1953 that exposure to foreign tissues during embryonic development caused mice to become immunologically tolerant to these tissues. Burnet proposed that developing lymphocytes that are potentially self-reactive are removed before they can mature, a process known as [clonal deletion](#). Medawar and Burnet shared the 1960 Nobel Prize for their work on tolerance. This process was demonstrated experimentally in the late 1980s. Some lymphocytes that receive either too much or too little signal through their antigen receptor during development are eliminated by a form of cell suicide called [apoptosis](#)—derived from a Greek word meaning the falling of leaves from trees—or [programmed cell death](#). Other types of mechanisms of immunological tolerance have been identified since then that rely on the induction of an inactive state, called [anergy](#), as well as mechanisms of active suppression of self-reactive lymphocytes. [Chapter 8](#) will describe lymphocyte development and tolerance mechanisms that shape the lymphocyte receptor repertoire. [Chapters 14](#) and [15](#) will discuss how immune-tolerance mechanisms can sometimes fail.

1-14 Lymphocytes mature in the bone marrow or the thymus and then congregate in lymphoid tissues throughout the body.

Lymphocytes circulate in the blood and the lymph and are also found in large numbers in [lymphoid tissues](#) or [lymphoid organs](#), the latter being organized aggregates of lymphocytes in a framework of nonlymphoid cells. Lymphoid organs can be divided broadly into the [central \(or primary\) lymphoid organs](#), where lymphocytes are generated, and the [peripheral \(or secondary\) lymphoid organs](#), where mature naive lymphocytes are maintained and adaptive immune responses are initiated. The central lymphoid organs are the bone marrow and the [thymus](#), an organ in the upper chest. The peripheral lymphoid organs comprise the [lymph nodes](#), the [spleen](#), and the mucosal lymphoid tissues of the gut, the nasal and respiratory tract, the urogenital tract, and other mucosa. The locations of the main lymphoid tissues are shown schematically in **Fig. 1.18**; we describe the individual peripheral lymphoid organs in more detail later in the chapter. Lymph nodes are interconnected by a system of lymphatic vessels, which drain extracellular fluid from tissues, carry it through the lymph nodes, and deposit it back into the blood.

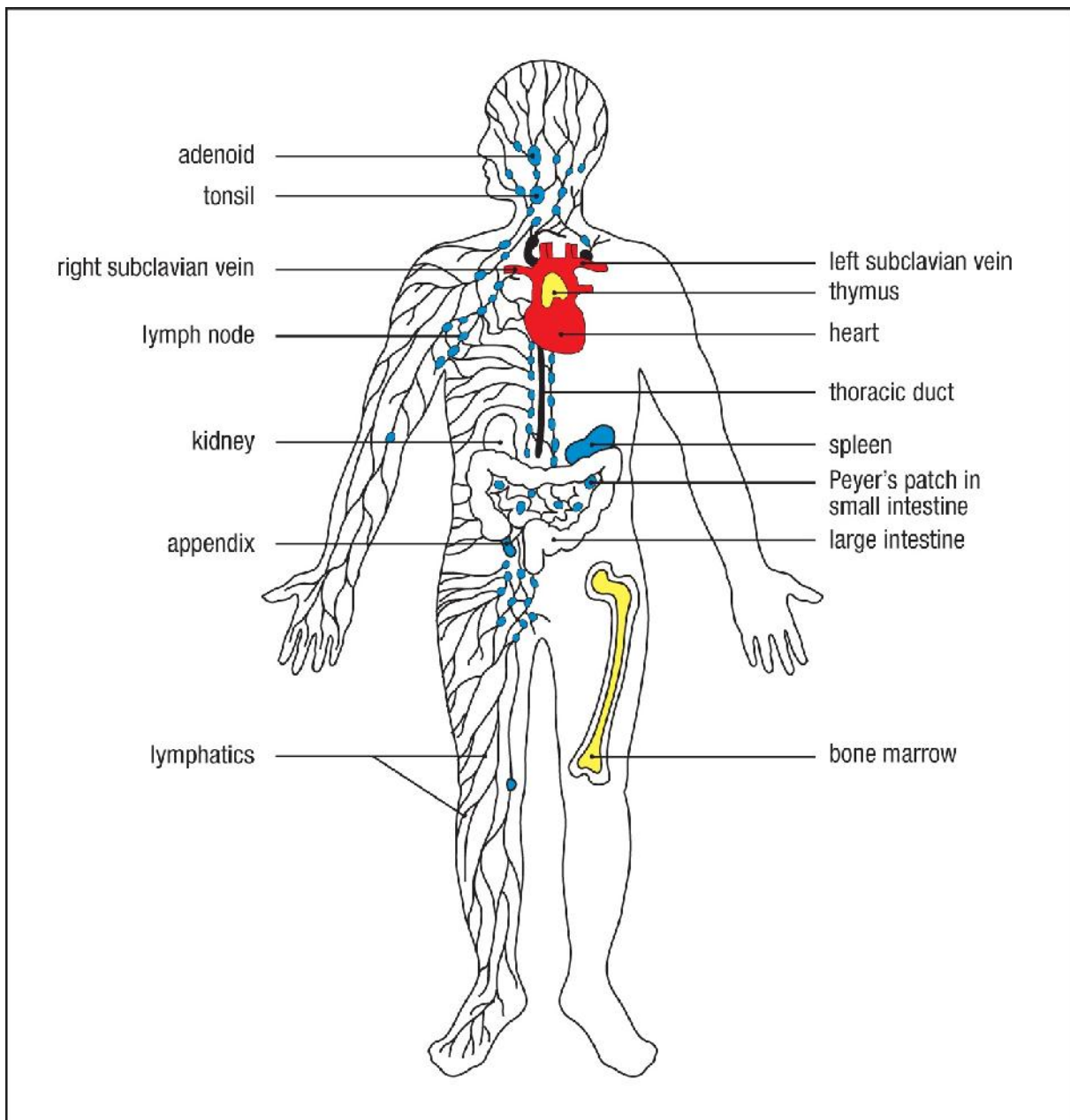


Fig. 1.18 The distribution of lymphoid tissues in the body. Lymphocytes arise from stem cells in bone marrow and differentiate in the central lymphoid organs (yellow)—B cells in the bone marrow and T cells in the thymus. They migrate from these tissues and are carried in the bloodstream to the peripheral lymphoid organs (blue). These include lymph nodes, spleen, and lymphoid tissues associated with mucosa, such as the gut-associated tonsils, Peyer's patches, and appendix. The peripheral lymphoid organs are the sites of lymphocyte activation by antigen, and lymphocytes recirculate between the blood and these organs until they encounter their specific antigen. Lymphatics drain extracellular fluid from the peripheral tissues, through the lymph nodes, and into the thoracic duct, which empties into the left subclavian vein. This fluid, known as lymph, carries antigen taken up by dendritic cells and macrophages to the lymph nodes, and it recirculates lymphocytes from the lymph nodes back into the blood. Lymphoid tissue is also associated with other mucosa such as the bronchial linings (not shown).

The progenitors that give rise to B and T lymphocytes originate in the bone marrow. B cells complete their development within the bone marrow. Although the 'B' in B lymphocytes originally stood for the [bursa of Fabricius](#), a lymphoid organ in young chicks in which lymphocytes mature, it is a useful mnemonic for bone marrow. The immature precursors of T lymphocytes migrate to the thymus, from which they get their name, and complete their development there. Once they have completed maturation, both types of lymphocytes enter the bloodstream as mature naive lymphocytes and continually circulate through the peripheral lymphoid tissues.

1-15 Adaptive immune responses are initiated by antigen and antigen-presenting cells in peripheral lymphoid tissues.

Adaptive immune responses are initiated when B or T lymphocytes encounter antigens for which their receptors have specific reactivity, provided that there are appropriate inflammatory signals to support activation. For T cells, this activation occurs via encounters with dendritic cells that have picked up antigens at sites of infection and migrated to peripheral (also called secondary) lymphoid organs. Activation of the dendritic cells' PRRs by PAMPs at the site of infection stimulates the dendritic cells in the tissues to engulf the pathogen and degrade it intracellularly. They also take up extracellular material, including virus particles and bacteria, by receptor-independent macropinocytosis. These processes lead to the display of peptide antigens on the MHC molecules of the dendritic cells, a display that activates the antigen receptors of lymphocytes. Activation of PRRs also triggers the dendritic cells to express cell-surface proteins called [co-stimulatory molecules](#), which support the ability of the T lymphocyte to proliferate and differentiate into its final, fully functional form (**Fig. 1.19**). For these reasons dendritic cells are also called [antigen-presenting cells \(APCs\)](#), and as such, they form a crucial link between the innate immune response and the adaptive immune response (**Fig. 1.20**). In certain situations, macrophages and B cells can also act as antigen-presenting cells, but dendritic cells are the cells that are specialized in initiating the adaptive immune response. Free antigens can also stimulate the antigen receptors of B cells, but most B cells respond better to antigens on the surface of cells, such as pathogens, and in addition require 'help' from activated helper T cells for optimal antibody responses. The activation of naive T lymphocytes is therefore an essential first stage in virtually all adaptive immune responses. [Chapter 6](#) returns to dendritic cells to discuss how antigens are processed for presentation to T cells. [Chapters 7](#) and [9](#) discuss co-stimulation and lymphocyte activation. [Chapter 10](#) describes how T cells help in activating B cells.

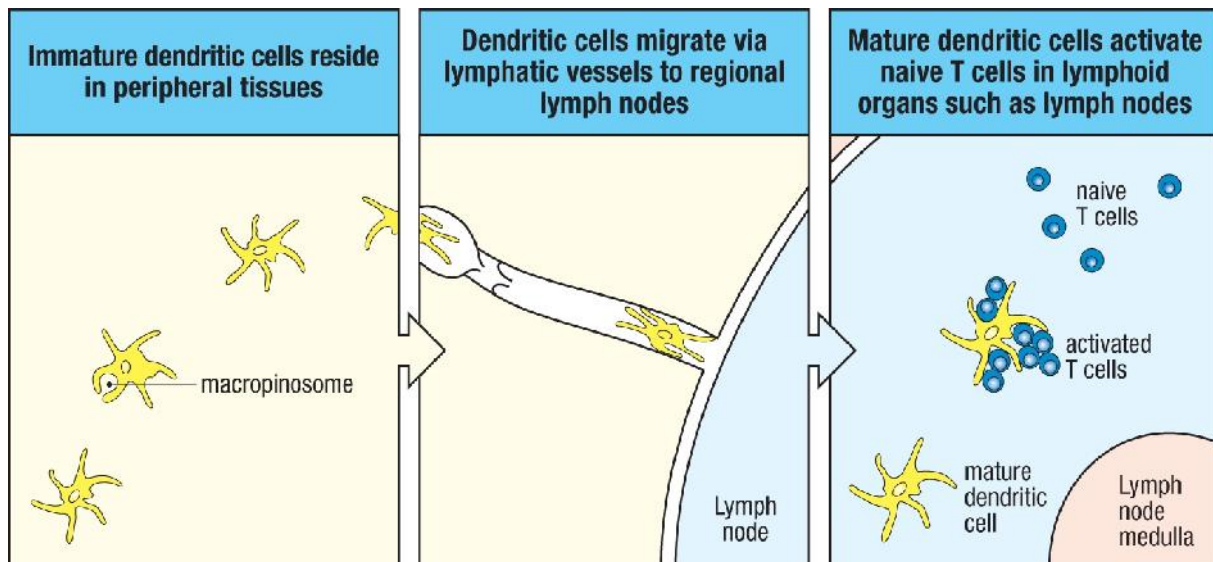


Fig. 1.19 Dendritic cells initiate adaptive immune responses. Immature dendritic cells residing in a tissue take up pathogens and their antigens by macropinocytosis and by receptor-mediated endocytosis. They are stimulated by recognition of the presence of pathogens to migrate through the lymphatics to regional lymph nodes, where they arrive as fully mature nonphagocytic dendritic cells that express both antigen and the co-stimulatory molecules necessary to activate a naive T cell that recognizes the antigen. Thus the dendritic cells stimulate lymphocyte proliferation and differentiation.

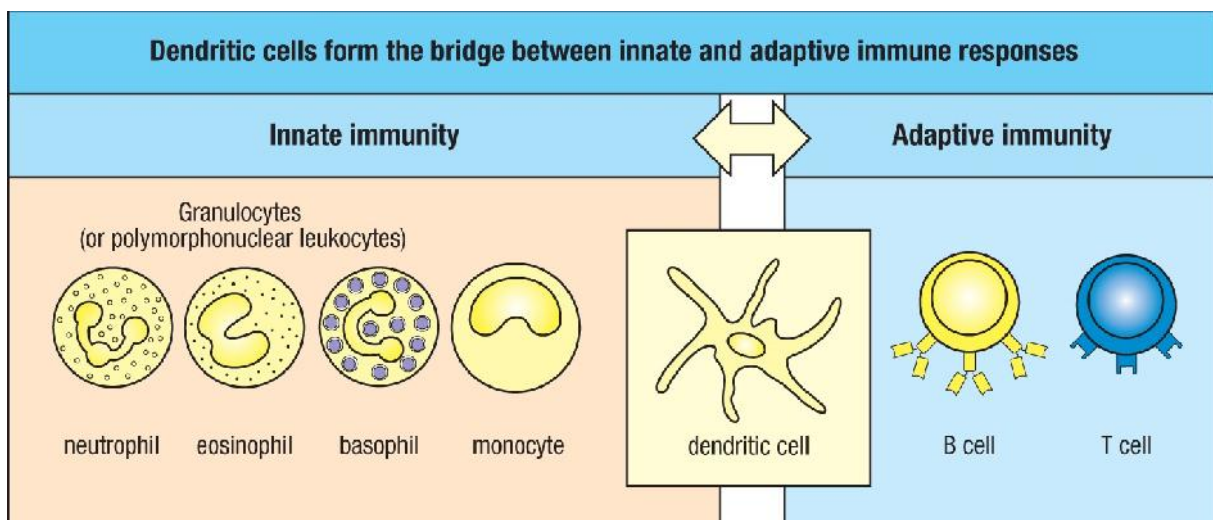


Fig. 1.20 Dendritic cells form a key link between the innate immune system and the adaptive immune system. Like the other cells of innate immunity, dendritic cells recognize pathogens via invariant cell-surface receptors for pathogen molecules and are activated by these stimuli early in an infection. Dendritic cells in tissues are phagocytic; they are specialized to ingest a wide range of pathogens and to display their antigens at the dendritic cell surface in a form that can be recognized by T cells.

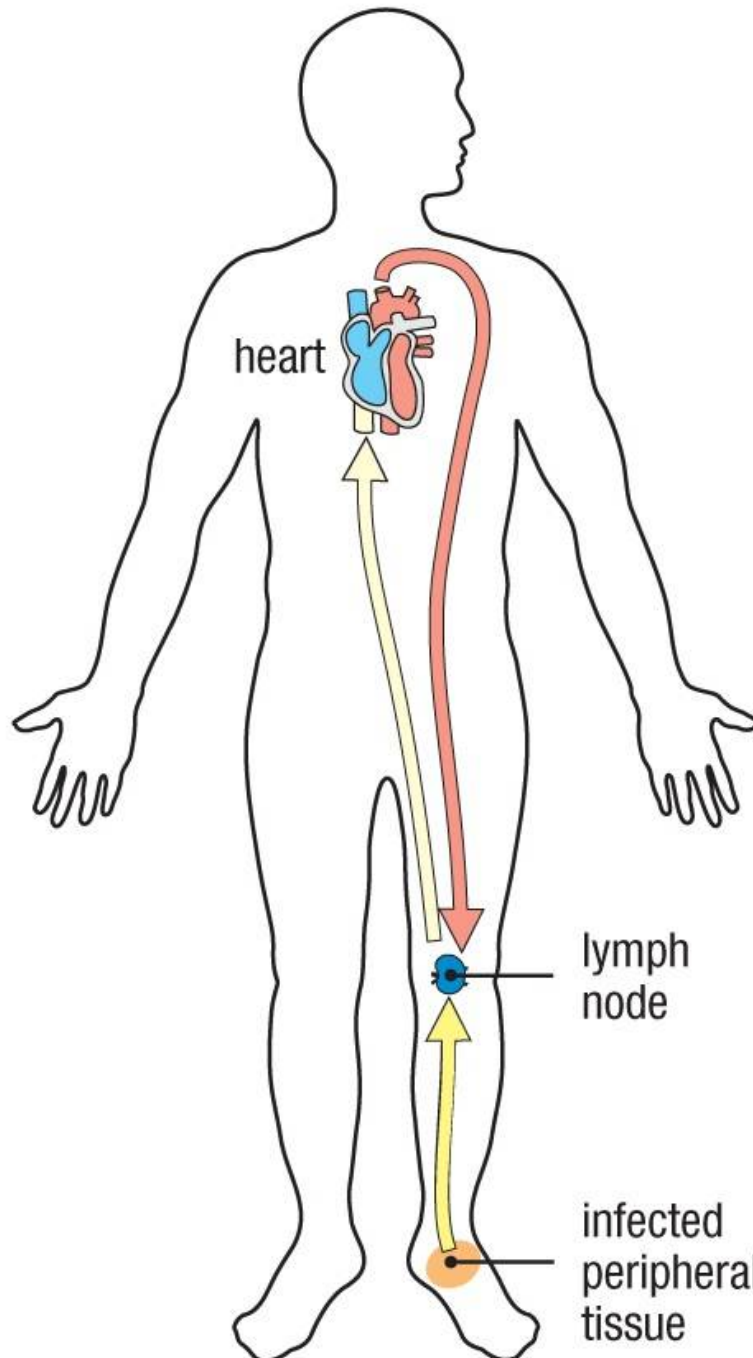
1-16 Lymphocytes encounter and respond to

antigen in the peripheral lymphoid organs.

Antigen and lymphocytes eventually encounter each other in the peripheral lymphoid organs—the lymph nodes, spleen, and mucosal lymphoid tissues (see Fig. 1.18). Mature naive lymphocytes are continually recirculating through these tissues, to which pathogen antigens are carried from sites of infection, primarily by dendritic cells. The peripheral lymphoid organs are specialized to trap antigen-bearing dendritic cells and to facilitate the initiation of adaptive immune responses. Peripheral lymphoid organs are composed of aggregations of lymphocytes in a framework of nonleukocyte stromal cells. The stromal cells provide both the basic structural organization of the tissue and survival signals to help sustain the life of the lymphocytes. Besides lymphocytes, peripheral lymphoid organs also contain resident macrophages and dendritic cells.

Lymphocytes and lymph return to blood via the thoracic duct

Naive lymphocytes enter lymph nodes from blood



Antigens from sites of infection reach lymph nodes via lymphatics

Fig. 1.21 Circulating lymphocytes encounter antigen in peripheral lymphoid organs. Naive lymphocytes recirculate constantly through peripheral lymphoid organs, here illustrated as a popliteal lymph node—a lymph node situated behind the knee. In the case of an infection in the foot, this will be the draining lymph node, where lymphocytes may encounter their specific antigens and become activated. Both activated and nonactivated lymphocytes are returned to the bloodstream via the lymphatic system.

When an infection occurs in a tissue such as the skin, free antigen and antigen-bearing dendritic cells travel from the site of infection through the afferent lymphatic vessels into the [draining lymph nodes](#) (**Fig. 1.21**)—peripheral lymphoid organs in which the free antigen and antigen-bearing dendritic cells activate antigen-specific lymphocytes. The activated lymphocytes then undergo a period of proliferation and differentiation, after which most leave the lymph nodes as effector cells via the efferent lymphatic vessel. This eventually returns them to the bloodstream (see Fig. 1.18), which then carries them to the tissues where they will act. This whole process takes about 4–6 days from the time that the antigen is recognized, which means that an adaptive immune response to an antigen that has not been encountered before does not become effective until about a week after infection (see Fig. 1.7). Naive lymphocytes that do not recognize their antigen also leave through the efferent lymphatic vessel and are returned to the blood, from which they continue to recirculate through lymphoid tissues until they recognize antigen or die.

The lymph nodes are highly organized lymphoid organs located at the points of convergence of vessels of the lymphatic system, which is the extensive system that collects extracellular fluid from the tissues and returns it to the blood (see Fig. 1.18). This extracellular fluid is produced continually by filtration from the blood and is called lymph. Lymph flows away from the peripheral tissues under the pressure exerted by its continual production and is carried by [lymphatic vessels](#), or [lymphatics](#). One-way valves in the lymphatic vessels prevent a reverse flow, and the movements of one part of the body in relation to another are important in driving the lymph along.

As noted above, [afferent lymphatic vessels](#) drain fluid from the tissues and carry pathogens and antigen-bearing cells from infected tissues to the lymph nodes (**Fig. 1.22**). Free antigens simply diffuse through the extracellular fluid to the lymph node, while the dendritic cells actively migrate into the lymph node, attracted by chemokines. The same chemokines also attract lymphocytes from the blood, and these enter lymph nodes by squeezing through the walls of specialized blood vessels called [high endothelial venules \(HEVs\)](#), named for their thicker, more rounded appearance relative to flatter endothelial cells in other locations. In the lymph nodes, B lymphocytes are localized in [follicles](#), which make up the outer [cortex](#) of the lymph node, with T cells more diffusely distributed in the surrounding [paracortical areas](#), also referred to as the deep cortex or [T-cell zones](#) (see Fig. 1.22). Lymphocytes migrating from the blood into lymph nodes enter the paracortical areas first, and because they are attracted by the same chemokines, antigen-presenting dendritic cells and macrophages also

become localized there. Free antigen diffusing through the lymph node can become trapped on these dendritic cells and macrophages. This juxtaposition of antigen, antigen-presenting cells, and naive T cells in the T-cell zone creates an ideal environment in which naive T cells can bind their specific antigen and thus become activated.

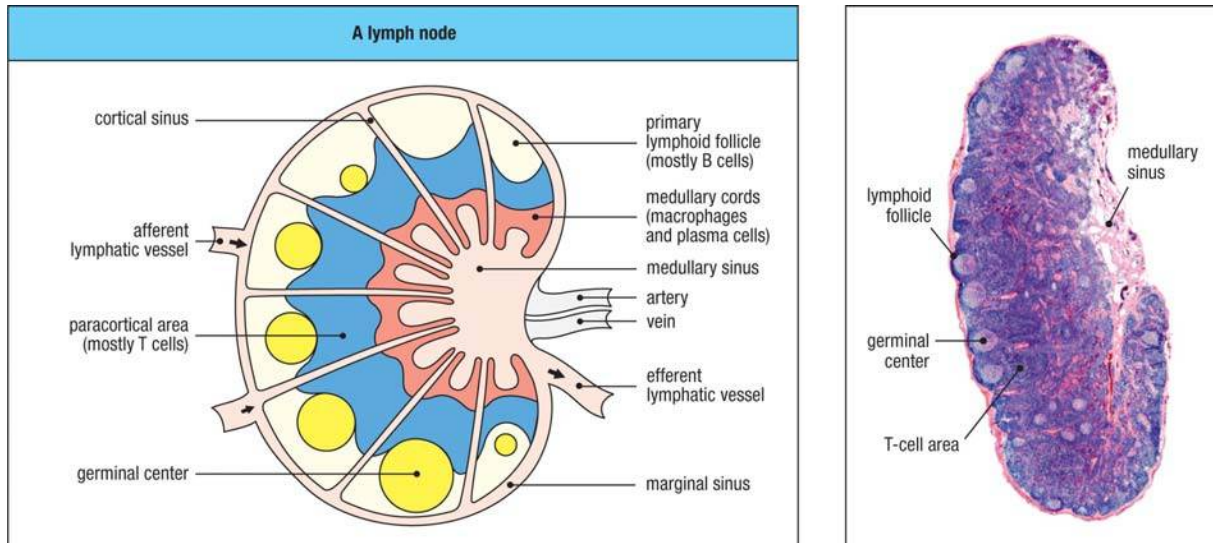


Fig. 1.22 Organization of a lymph node. As shown at left in the diagram of a lymph node in longitudinal section, a lymph node consists of an outermost cortex (the yellow and blue areas) and an inner medulla (dark pink). The cortex is composed of an outer cortex of B cells organized into lymphoid follicles and of adjacent, or paracortical, areas made up mainly of T cells and dendritic cells. When an immune response is under way, some of the follicles contain central areas of intense B-cell proliferation called germinal centers. Lymph draining from the extracellular spaces of the body carries antigens in phagocytic dendritic cells and phagocytic macrophages from the tissues to the lymph node via the afferent lymphatics. These migrate directly from the sinuses into the cellular parts of the node. Lymph leaves via the efferent lymphatic in the medulla. The medulla consists of strings of macrophages and antibody-secreting plasma cells known as the medullary cords. Naive lymphocytes enter the node from the bloodstream through specialized postcapillary venules (not shown) and leave with the lymph through the efferent lymphatic. The light micrograph at right shows a longitudinal section through a lymph node that has follicles containing germinal centers. Original magnification $\times 40$.

As noted earlier, activation of B cells usually requires not only antigen, which binds to the B-cell receptor, but also the cooperation of activated helper T cells, a type of effector T cell. The location of B cells and T cells within the lymph node is dynamically regulated by their state of activation. When they become activated, T cells and B cells both move to the border of the follicle and T-cell zone, where T cells can first provide their helper function to B cells. Some of the B-cell follicles include [germinal centers](#), where activated B cells are undergoing intense proliferation and differentiation into plasma cells. These mechanisms are described in detail in [Chapter 10](#).

In humans, the spleen is a fist-sized organ situated just behind the stomach (see Fig. 1.18). It

has no direct connection with the lymphatic system; instead, it collects antigen from the blood and is involved in immune responses to blood-borne pathogens. Lymphocytes enter and leave the spleen via blood vessels. The spleen also collects and disposes of senescent red blood cells. Its organization is shown schematically in **Fig. 1.23**. The bulk of the spleen is composed of [red pulp](#), which is the site of red blood cell disposal. The lymphocytes surround the arterioles running through the spleen, forming isolated areas of [white pulp](#). The sheath of lymphocytes around an arteriole is called the [periarteriolar lymphoid sheath \(PALS\)](#) and contains mainly T cells. Lymphoid follicles occur at intervals along it, and these contain mainly B cells. An area called the [marginal zone](#) surrounds the follicle; it has few T cells, is rich in macrophages, and has a resident, noncirculating population of B cells known as [marginal zone B cells](#). These B cells are poised to rapidly produce antibodies that have low affinity to bacterial capsular polysaccharides. These antibodies, which are discussed in [Chapter 8](#), provide some degree of protection before the adaptive immune response is fully activated. Blood-borne microbes, soluble antigens, and antigen:antibody complexes are filtered from the blood by macrophages and immature dendritic cells within the marginal zone. Like the migration of immature dendritic cells from peripheral tissues to the T-cell areas of lymph nodes, dendritic cells in the marginal zones in the spleen migrate to the T-cell areas after taking up antigen and becoming activated; here they are able to present the antigens they carry to T cells.

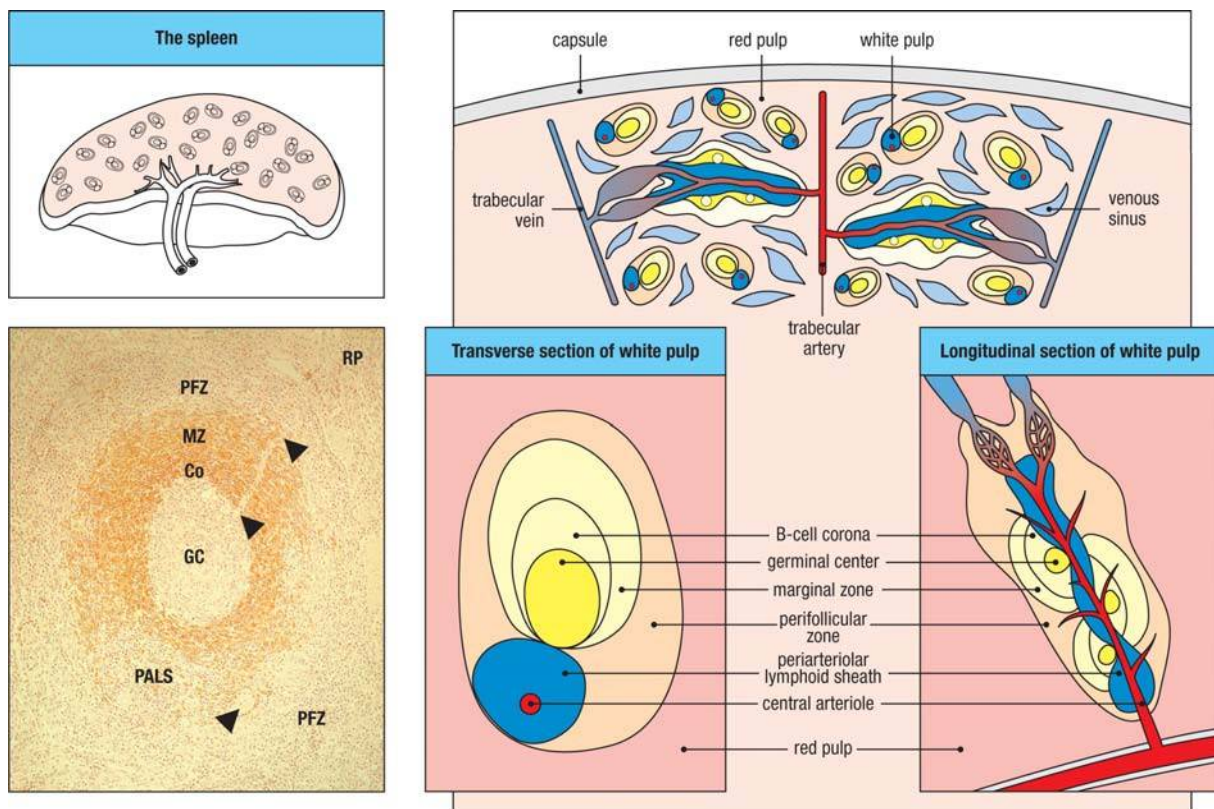


Fig. 1.23 Organization of the lymphoid tissues of the spleen. The schematic at top left shows that the spleen consists of red pulp (pink areas), which is a site of red blood cell destruction, interspersed with the lymphoid white pulp. An enlargement of a small section of a human spleen (top right) shows the arrangement of discrete areas of white pulp (yellow and blue) around central arterioles. Most of the white pulp is shown in

transverse section, with two portions in longitudinal section. The two schematics below this diagram show enlargements of a transverse section (bottom center) and longitudinal section (bottom right) of white pulp. Surrounding the central arteriole is the periarteriolar lymphoid sheath (PALS), made up of T cells. Lymphocytes and antigen-loaded dendritic cells come together here. The follicles consist mainly of B cells; in secondary follicles, a germinal center is surrounded by a B-cell corona. The follicles are surrounded by a so-called marginal zone of lymphocytes. In each area of white pulp, blood carrying both lymphocytes and antigen flows from a trabecular artery into a central arteriole. From this arteriole smaller blood vessels fan out, eventually terminating in a specialized zone in the human spleen called the perifollicular zone (PFZ), which surrounds each marginal zone. Cells and antigen then pass into the white pulp through open blood-filled spaces in the perifollicular zone. The light micrograph at bottom left shows a transverse section of white pulp of human spleen immunostained for mature B cells. Both follicle and PALS are surrounded by the perifollicular zone. The follicular arteriole emerges in the PALS (arrowhead at bottom), traverses the follicle, goes through the marginal zone, and opens into the perifollicular zone (upper arrowheads). Co, follicular B-cell corona; GC, germinal center; MZ, marginal zone; RP, red pulp; arrowheads, central arteriole.

1-17 Mucosal surfaces have specialized immune structures that orchestrate responses to environmental microbial encounters.

Most pathogens enter the body through mucosal surfaces, which are also exposed to a vast load of other potential antigens from the air, food, and the natural microbial flora of the body. Mucosal surfaces are protected by an extensive system of lymphoid tissues known generally as the [mucosal immune system](#) or [mucosa-associated lymphoid tissue \(MALT\)](#). Collectively, the mucosal immune system is estimated to contain as many lymphocytes as all the rest of the body, and these lymphocytes form a specialized set of cells obeying somewhat different rules of recirculation from those of lymphocytes in the other peripheral lymphoid organs. The [gut-associated lymphoid tissue \(GALT\)](#) includes the [tonsils](#), [adenoids](#), [appendix](#), and specialized structures in the small intestine called [Peyer's patches](#), and they collect antigen from the epithelial surfaces of the gastrointestinal tract. In Peyer's patches, which are the most important and highly organized of these tissues, the antigen is collected by specialized epithelial cells called [microfold cells](#), or [M cells](#) (**Fig. 1.24**). The lymphocytes form a follicle consisting of a large central dome of B lymphocytes surrounded by smaller numbers of T lymphocytes. Dendritic cells resident within the Peyer's patch present the antigen to T lymphocytes. Lymphocytes enter Peyer's patches from the blood and leave through efferent lymphatics. Effector lymphocytes generated in Peyer's patches travel through the lymphatic system and into the bloodstream, from where they are disseminated back into mucosal tissues to carry out their effector actions.

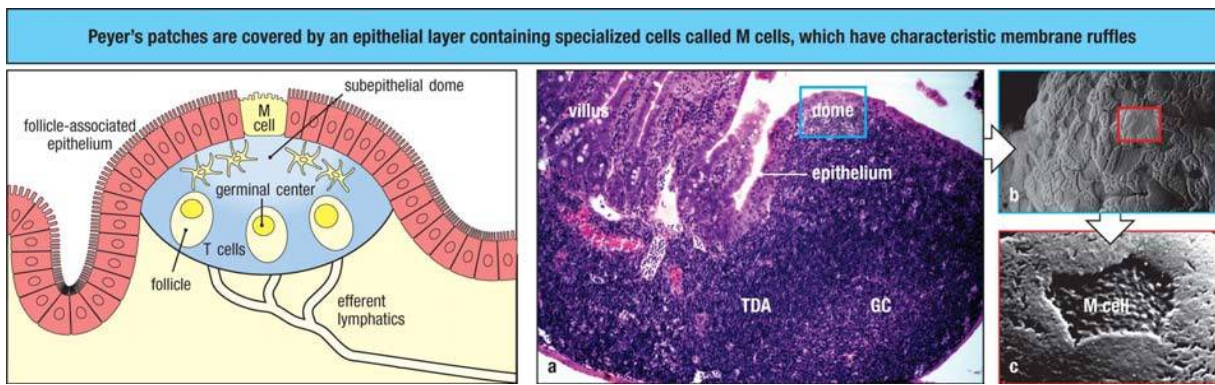


Fig. 1.24 Organization of a Peyer's patch in the gut mucosa. As the diagram on the left shows, a Peyer's patch contains numerous B-cell follicles with germinal centers. The areas between follicles are occupied by T cells and are therefore called the T cell–dependent areas. The layer between the surface epithelium and the follicles is known as the subepithelial dome and is rich in dendritic cells, T cells, and B cells. Peyer's patches have no afferent lymphatics, and the antigen enters directly from the gut across a specialized epithelium made up of so-called microfold (M) cells. Although this tissue looks very different from other lymphoid organs, the basic divisions are maintained. As in the lymph nodes, lymphocytes enter Peyer's patches from the blood across the walls of high endothelial venules (not shown), and leave via the efferent lymphatic. Panel a: The light micrograph shows a section through a Peyer's patch in the gut wall of a mouse. The Peyer's patch can be seen lying beneath the epithelial tissues. GC, germinal center; TDA, T cell–dependent area. Panel b: A scanning electron micrograph of the follicle-associated epithelium boxed in panel a shows the M cells, which lack the microvilli and the mucus layer present on normal epithelial cells. Each M cell appears as a sunken area on the epithelial surface. Panel c: A higher-magnification view of the boxed area in panel b shows the characteristic ruffled surface of an M cell. M cells are the portal of entry for many pathogens and other particles. Panel a, hematoxylin and eosin stain, magnification $\times 100$; panel b, magnification $\times 5000$; panel c, magnification $\times 23,000$.

Similar but more diffuse aggregates of lymphocytes are present in the respiratory tract and other mucosa: [nasal-associated lymphoid tissue \(NALT\)](#) and [bronchus-associated lymphoid tissue \(BALT\)](#) are present in the respiratory tract. Like the Peyer's patches, these mucosal lymphoid tissues are also overlaid by M cells, through which inhaled microbes and antigens that become trapped in the mucous covering of the respiratory tract can pass. The mucosal immune system is discussed in [Chapter 12](#).

Although very different in appearance, the lymph nodes, spleen, and mucosa-associated lymphoid tissue all share the same basic architecture. They all operate on the same principle, trapping antigens and antigen-presenting cells from sites of infection in order to present antigen to migratory small lymphocytes, thus inducing adaptive immune responses. The peripheral lymphoid organs also provide sustaining signals to lymphocytes that do not encounter their specific antigen immediately, so that they survive and continue to recirculate.

Because they are involved in initiating adaptive immune responses, the peripheral lymphoid organs are not static structures but vary quite markedly, depending on whether or not infection is present. The diffuse mucosal lymphoid tissues may appear in response to infection and then disappear, whereas the architecture of the organized tissues changes in a more defined way during an infection. For example, the B-cell follicles of the lymph nodes expand as B lymphocytes proliferate to form germinal centers (see Fig. 1.22), and the entire lymph node enlarges, a phenomenon familiarly known as swollen glands.

Finally, specialized populations of lymphocytes and innate lymphoid cells can be found distributed throughout particular sites in the body rather than being found in organized lymphoid tissues. Such sites include the liver and the [lamina propria](#) of the gut, as well as the base of the epithelial lining of the gut, reproductive epithelia, and, in mice but not in humans, the epidermis. These lymphocyte populations seem to have an important role in protecting these tissues from infection and are described further in [Chapters 8](#) and [12](#).

1-18 Lymphocytes activated by antigen proliferate in the peripheral lymphoid organs, generating effector cells and immunological memory.

The great diversity of the lymphocyte receptor repertoire means that there will usually be some lymphocytes bearing a receptor for any given foreign antigen. Recent experiments suggest this number to be perhaps a few hundred per mouse, certainly not enough to mount a response against a pathogen. To generate sufficient antigen-specific effector lymphocytes to fight an infection, a lymphocyte with an appropriate receptor specificity is activated first to proliferate. Only when a large clone of identical cells has been produced do these finally differentiate into effector cells, a process that requires 4–6 days. This means the adaptive immune response to a pathogen occurs nearly a week after the initial infection has occurred and been detected by the innate immune system.

On recognizing its specific antigen on an activated antigen-presenting cell, a naive lymphocyte stops migrating, the volume of its nucleus and cytoplasm increases, and new mRNA molecules and new proteins are synthesized. Within a few hours, the cell looks completely different and is known as a [lymphoblast](#). Dividing lymphoblasts are able to duplicate themselves two to four times every 24 hours over the course of 3–5 days, so that a single naive lymphocyte can produce a clone of around 1000 daughter cells of identical specificity. These then differentiate into effector cells. In the case of B cells, the differentiated effector cells are the plasma cells, which secrete antibody. In the case of T cells, the effector cells are either cytotoxic T cells, which are able to destroy infected cells, or helper T cells, which activate other cells of the immune system (see [Section 1-8](#)).

Effector lymphocytes do not recirculate like naive lymphocytes. Some effector T cells detect sites of infection and migrate into them from the blood; others stay in the lymphoid tissues to activate B cells. Some antibody-secreting plasma cells remain in the peripheral lymphoid organs, but most plasma cells generated in the lymph nodes and spleen will migrate to the bone marrow and take up residence there, secreting large amounts of antibodies into the blood system. Effector cells generated in the mucosal immune system generally stay within the mucosal tissues. Most lymphocytes generated by clonal expansion in an immune response will eventually die. However, a significant number of activated antigen-specific B cells and T cells persist after antigen has been eliminated. These cells are known as [memory cells](#) and form the basis of immunological memory. They can be reactivated much more quickly than naive lymphocytes, which ensures a more rapid and effective response on a second encounter with a pathogen and thereby usually provides lasting protective immunity.

The characteristics of immunological memory are readily observed by comparing an individual's antibody response to a first or [primary immunization](#) with the response to a [secondary \(or booster\) immunization](#) with the same antigen. As shown in **Fig. 1.25**, the secondary antibody response occurs after a shorter lag phase and achieves a markedly higher level than in the primary response. During the secondary response, antibodies can also acquire higher affinity, or strength of binding, for the antigen because of a process called [affinity maturation](#), which takes place in the specialized germinal centers within B-cell follicles (see [Section 1-16](#)). Importantly, helper T cells are required for the process of affinity maturation, but T-cell receptors do not undergo affinity maturation. Compared with naive T cells, memory T cells show a lower threshold for activation, but as a result of changes in the responsiveness of the cell and not because of changes in the receptor. We describe the mechanisms of antibody affinity maturation in [Chapter 10](#).

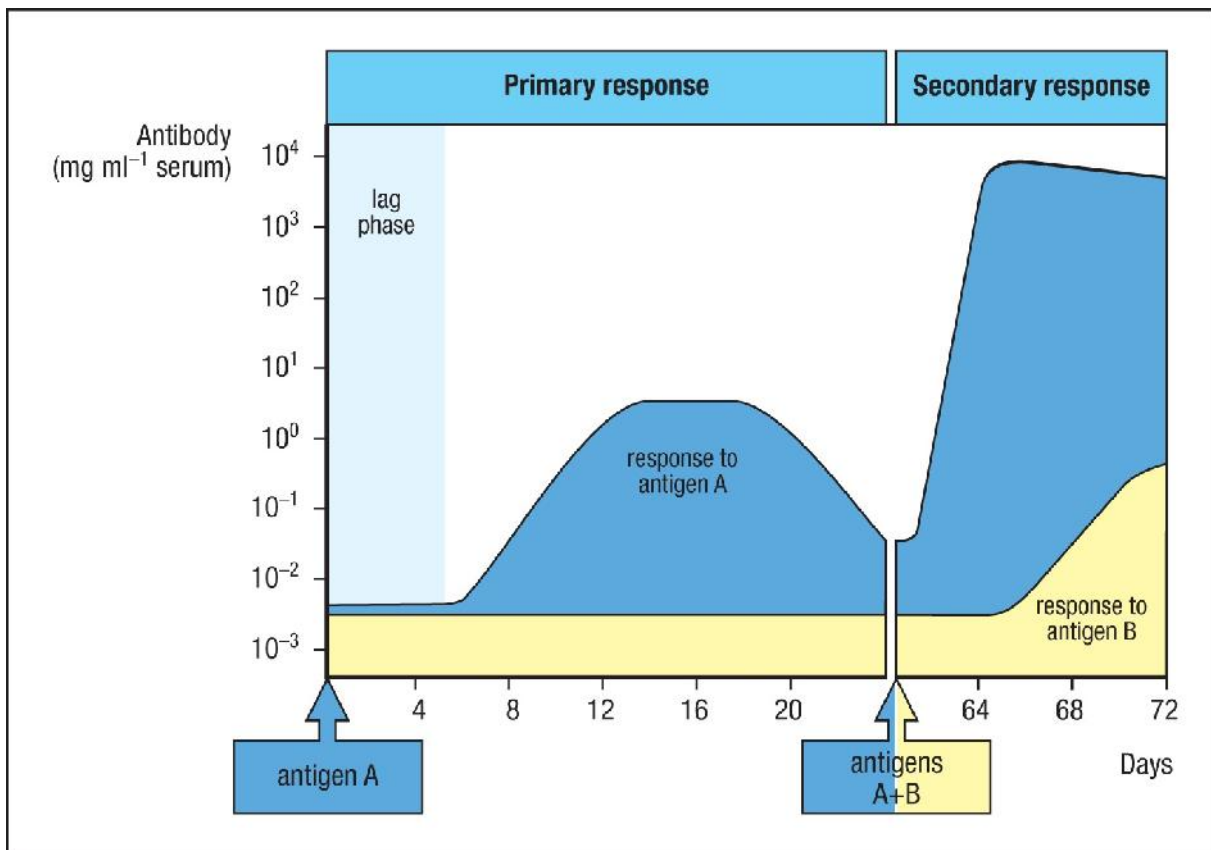


Fig. 1.25 The course of a typical antibody response. The first encounter with an antigen produces a primary response. Antigen A introduced at time zero encounters little specific antibody in the serum. After a lag phase (light blue), antibody against antigen A (dark blue) appears; its concentration rises to a plateau and then gradually declines, as is typical of a primary response. When the serum is tested for antibody against another antigen, B (yellow), there is little preset. When the animal is later challenged with a mixture of antigens A and B, a very rapid and intense antibody secondary response to A occurs, illustrating immunological memory. This is the main reason for giving booster injections after an initial vaccination. Note that the response to B resembles the primary response to A, as this is the first encounter with antigen B.

The cellular basis of immunological memory is the clonal expansion and clonal differentiation of cells that have a specific attraction for the eliciting antigen, and the memory is therefore entirely antigen-specific. It is immunological memory that enables successful vaccination and prevents reinfection with pathogens that have been repelled successfully by an adaptive immune response. In [Chapter 11](#), we will return to immunological memory, which is perhaps the most important biological consequence of adaptive immunity.

Summary.

While the innate immune system relies on invariant pattern-recognition receptors to detect

common microbial structures or the damage they cause, the adaptive immune system relies on a repertoire of antigen receptors to recognize structures that are specific to individual pathogens. This feature provides adaptive immunity with greater sensitivity and specificity. The clonal expansion of antigen-reactive lymphocytes also confers the property of immunological memory, which enhances protection against reinfection by the same pathogen.

Adaptive immunity relies on two major types of lymphocytes. B cells mature in the bone marrow and are the source of circulating antibodies. T cells mature in the thymus and recognize peptides from pathogens presented by MHC molecules on infected cells or antigen-presenting cells. An adaptive response involves the selection and amplification of clones of lymphocytes bearing receptors that recognize the foreign antigen. This clonal selection provides the theoretical framework for understanding all the key features of an adaptive immune response.

Each lymphocyte carries cell-surface receptors of a single antigen specificity. These receptors are generated by the random recombination of variable receptor gene segments and the pairing of distinct variable protein chains: heavy and light chains in immunoglobulins or the two chains of T-cell receptors. The large antigen-receptor repertoire of lymphocytes can recognize virtually any antigen. Adaptive immunity is initiated when an innate immune response fails to eliminate a new infection, and activated antigen-presenting cells—typically dendritic cells that bear antigens from pathogens and co-stimulatory receptors—migrate to the draining lymphoid tissues.

Immune responses are initiated in several peripheral lymphoid organs. The spleen serves as a filter for blood-borne infections. Lymph nodes draining various tissues and the mucosa-associated and gut-associated lymphoid tissues (MALT and GALT) are organized into specific zones where T and B cells can be activated efficiently by antigen-presenting cells or helper T cells. When a recirculating lymphocyte encounters its corresponding antigen in these peripheral lymphoid organs, it proliferates, and its clonal progeny differentiate into effector T and B lymphocytes that can eliminate the infectious agent. A subset of these proliferating lymphocytes differentiates into memory cells, ready to respond rapidly to the same pathogen if it is encountered again. The details of these processes of recognition, development, and differentiation form the main material of the central three parts of this book.

Glossary

[immunological memory](#)

The ability of the immune system to respond more rapidly and more effectively on a second encounter with an antigen. Immunological memory is specific for a particular antigen and is long-lived.

[B lymphocytes \(B cells\)](#)

One of the two types of antigen-specific lymphocytes responsible for adaptive immune responses, the other being the T cells. The function of B cells is to produce antibodies. B cells are divided into two classes. Conventional B cells have highly diverse antigen receptors and are generated in the bone marrow throughout life, emerging to populate the blood and lymphoid tissues. B-1 cells have much less diverse antigen receptors and form a population of self-renewing B cells in the peritoneal and pleural cavities.

T lymphocytes (T cells)

One of the two types of antigen-specific lymphocytes responsible for adaptive immune responses, the other being the B cells. T cells are responsible for the cell-mediated adaptive immune reactions. They originate in the bone marrow but undergo most of their development in the thymus. The highly variable antigen receptor on T cells is called the T-cell receptor and recognizes a complex of peptide antigen bound to MHC molecules on cell surfaces. There are two main lineages of T cells: those carrying $\alpha\beta$ receptors and those carrying $\gamma\delta$ receptors. Effector T cells perform a variety of functions in an immune response, acting always by interacting with another cell in an antigen-specific manner. Some T cells activate macrophages, some help B cells produce antibody, and some kill cells infected with viruses and other intracellular pathogens.

naïve lymphocytes

T cells or B cells that have undergone normal development in the thymus or the bone marrow but have not yet been activated by foreign (or self) antigens.

effector lymphocytes

The cells that differentiate from naïve lymphocytes after initial activation by antigen and can then mediate the removal of pathogens from the body without further differentiation. They are distinct from memory lymphocytes, which must undergo further differentiation to become effector lymphocytes.

immunoglobulin (Ig)

The protein family to which antibodies and B-cell receptors belong.

membrane immunoglobulin (mIg)

Transmembrane immunoglobulin present on B cells; it is the B-cell receptor for antigen.

surface immunoglobulin (sIg)

The membrane-bound immunoglobulin that acts as the antigen receptor on B cells.

T-cell antigen receptor, T-cell receptor (TCR)

The cell-surface receptor for antigen on T lymphocytes. It consists of a disulfide-linked heterodimer of the highly variable α and β chains in a complex with the invariant CD3 and ζ proteins, which have a signaling function. T cells carrying this type of receptor are often called $\alpha\beta$ T cells. An alternative receptor made up of variable γ and δ chains is expressed with CD3 and ζ on a subset of T cells.

T-cell antigen receptor, T-cell receptor (TCR)

The cell-surface receptor for antigen on T lymphocytes. It consists of a disulfide-linked heterodimer of the highly variable α and β chains in a complex with the invariant CD3 and ζ proteins, which have a signaling function. T cells carrying this type of receptor are often called $\alpha\beta$ T cells. An alternative receptor made up of variable γ and δ chains

is expressed with CD3 and ζ on a subset of T cells.

[plasma cells](#)

Terminally differentiated activated B lymphocytes. Plasma cells are the main antibody-secreting cells of the body. They are found in the medulla of the lymph nodes, in splenic red pulp, in bone marrow, and in mucosal tissues.

[regulatory T cells \(T_{reg} cells\)](#)

Effector CD4 T cells that inhibit T-cell responses and are involved in controlling immune reactions and preventing autoimmunity. Several different subsets have been distinguished, notably the natural regulatory T-cell lineage that is produced in the thymus, and the induced regulatory T cells that differentiate from naive CD4 T cells in the periphery in certain cytokine environments.

[memory cells](#)

B and T lymphocytes that mediate immunological memory. They are more sensitive to antigen than are naive lymphocytes and respond rapidly on reexposure to the antigen that originally induced them.

[memory cells](#)

B and T lymphocytes that mediate immunological memory. They are more sensitive to antigen than are naive lymphocytes and respond rapidly on reexposure to the antigen that originally induced them.

[constant region \(C region\)](#)

That part of an immunoglobulin or a T-cell receptor that is relatively constant in amino acid sequence between different molecules. Also known as the Fc region in antibodies. The constant region of an antibody determines its particular effector function. Cf.

variable region.

[fragment crystallizable region](#)

The carboxy-terminal halves of the two heavy chains of an IgG molecule disulfide-bonded to each other by the residual hinge region. It is produced by cleavage of IgG by papain. In the complete antibody this portion is often called the Fc region.

[variable region \(V region\)](#)

The region of an immunoglobulin or T-cell receptor that is formed of the amino-terminal domains of its component polypeptide chains. These are the most variable parts of the molecule and contain the antigen-binding sites. Cf. **constant region**.

[antigen-binding site](#)

The site at the tip of each arm of an antibody that makes physical contact with the antigen and binds it noncovalently. The antigen specificity of the site is determined by its shape and the amino acids present.

[antigenic determinant](#)

That portion of an antigenic molecule that is bound by the antigen-binding site of a given antibody or antigen receptor; it is also known as an epitope.

[epitope](#)

A site on an antigen recognized by an antibody or an antigen receptor. T-cell epitopes are short peptides bound to MHC molecules. B-cell epitopes are typically structural motifs on the surface of the antigen. Also called an antigenic determinant.

[MHC molecules](#)

Highly polymorphic cell-surface proteins encoded by MHC class I and MHC class II genes involved in presentation of peptide antigens to T cells. They are also known as histocompatibility antigens.

major histocompatibility complex (MHC)

A cluster of genes on human chromosome 6 that encodes a set of membrane glycoproteins called the MHC molecules. The MHC also encodes proteins involved in antigen processing and other aspects of host defense. The genes for the MHC molecules are the most polymorphic in the human genome, having large numbers of alleles at the various loci.

gene segments

Sets of short DNA sequences at the immunoglobulin and T-cell receptor loci that encode different regions of the variable domains of antigen receptors. Gene segments of each type are joined together by somatic recombination to form a complete variable-domain exon. There are three types of gene segments: V gene segments encode the first 95 amino acids, D gene segments (in heavy-chain and TCR α chain loci only) encode about 5 amino acids, and J gene segments encode the last 10–15 amino acids of the variable domain. There are multiple copies of each type of gene segment in the germline DNA, but only one of each type is joined together to form the variable domain.

combinatorial diversity

The diversity among antigen receptors generated by combining separate units of genetic information, comprising two types. First, receptor gene segments are joined in many different combinations to generate diverse receptor chains; second, two different receptor chains (heavy and light in immunoglobulins; α and β , or γ and δ , in T-cell receptors) are combined to make the antigen-recognition site.

junctional diversity

The variability in sequence present in antigen-specific receptors that is created during the process of joining V, D, and J gene segments and which is due to imprecise joining and insertion of non-templated nucleotides at the joins between gene segments.

clone

A population of cells all derived from the same progenitor cell.

lymphocyte receptor repertoire

All the highly variable antigen receptors carried by B and T lymphocytes.

clonal expansion

The proliferation of antigen-specific lymphocytes in response to antigenic stimulation that precedes their differentiation into effector cells. It is an essential step in adaptive immunity, allowing rare antigen-specific cells to increase in number so that they can effectively combat the pathogen that elicited the response.

clonotypic

Describes a feature unique to members of a clone. For example, the distribution of antigen receptors in the lymphocyte population is said to be clonotypic, as the cells of a given clone all have identical antigen receptors.

clonal selection theory

The central paradigm of adaptive immunity. It states that adaptive immune responses

derive from individual antigen-specific lymphocytes that are self-tolerant. These specific lymphocytes proliferate in response to antigen and differentiate into antigen-specific effector cells that eliminate the eliciting pathogen and into memory cells to sustain immunity. The theory was formulated by Frank Macfarlane Burnet and in earlier forms by Niels K. Jerne and David Talmage.

cellular immunology

The study of the cellular basis of immunity.

self antigens

The potential antigens on the tissues of an individual, against which an immune response is not usually made except in the case of autoimmunity.

tolerant

Describes the state of immunological tolerance, in which the individual does not respond to a particular antigen.

clonal deletion

The elimination of immature lymphocytes when they bind to self antigens, which produces tolerance to self as required by the clonal selection theory of adaptive immunity. Clonal deletion is the main mechanism of central tolerance and can also occur in peripheral tolerance.

apoptosis

A form of cell death common in the immune system, in which the cell activates an internal death program. It is characterized by nuclear DNA degradation, nuclear degeneration and condensation, and the rapid phagocytosis of cell remains.

Proliferating lymphocytes experience high rates of apoptosis during their development and during immune responses.

programmed cell death

A form of cell death common in the immune system, in which the cell activates an internal death program. It is characterized by nuclear DNA degradation, nuclear degeneration and condensation, and the rapid phagocytosis of cell remains.

Proliferating lymphocytes experience high rates of apoptosis during their development and during immune responses.

anergy

A state of nonresponsiveness to antigen. People are said to be anergic when they cannot mount delayed-type hypersensitivity reactions to a test antigen, whereas T cells and B cells are said to be anergic when they cannot respond to their specific antigen under optimal conditions of stimulation.

lymphoid tissue

Tissue composed of large numbers of lymphocytes.

lymphoid organs

Organized tissues characterized by very large numbers of lymphocytes interacting with a nonlymphoid stroma. The central, or primary, lymphoid organs, where lymphocytes are generated, are the thymus and bone marrow. The main peripheral, or secondary, lymphoid organs, in which adaptive immune responses are initiated, are the lymph nodes, spleen, and mucosa-associated lymphoid organs such as tonsils and Peyer's patches.

thymus

A central lymphoid organ, in which T cells develop, situated in the upper part of the middle of the chest, just behind the breastbone.

lymph nodes

A type of peripheral lymphoid organ present in many locations throughout the body where lymphatic vessels converge.

spleen

An organ in the upper left side of the peritoneal cavity containing a red pulp, involved in removing senescent blood cells, and a white pulp of lymphoid cells that respond to antigens delivered to the spleen by the blood.

bursa of Fabricius

Lymphoid organ associated with the gut that is the site of B-cell development in chickens.

co-stimulatory molecules

Cell-surface proteins on antigen-presenting cells that deliver co-stimulatory signals to naive T cells. Examples are the B7 molecules on dendritic cells, which are ligands for CD28 on naive T cells.

antigen-presenting cells (APCs)

Highly specialized cells that can process antigens and display their peptide fragments on the cell surface together with other, co-stimulatory, proteins required for activating naive T cells. The main antigen-presenting cells for naive T cells are dendritic cells, macrophages, and B cells.

draining lymph node

A lymph node downstream of a site of infection that receives antigens and microbes from the site via the lymphatic system. Draining lymph nodes often enlarge enormously during an immune response and can be palpated; they were originally called swollen glands.

afferent lymphatic vessels

Vessels of the lymphatic system that drain extracellular fluid from the tissues and carry antigen, macrophages, and dendritic cells from sites of infection to lymph nodes or other peripheral lymphoid organs.

follicles

An area of predominantly B cells in a peripheral lymphoid organ, such as a lymph node, which also contains follicular dendritic cells.

cortex

The outer part of a tissue or organ; in lymph nodes it refers to the follicles, which are mainly populated by B cells.

paracortical areas

The T-cell areas of lymph nodes.

T-cell zones

Regions of peripheral lymphoid organs that are enriched in naive T cells and are distinct from the follicles. They are the sites at which adaptive immune responses are initiated.

germinal centers

Sites of intense B-cell proliferation and differentiation that develop in lymphoid follicles during an adaptive immune response. Somatic hypermutation and class switching occur in germinal centers.

[red pulp](#)

The nonlymphoid area of the spleen in which red blood cells are broken down.

[white pulp](#)

The discrete areas of lymphoid tissue in the spleen.

[periarteriolar lymphoid sheath \(PALS\)](#)

Part of the inner region of the white pulp of the spleen; it contains mainly T cells.

[marginal zone](#)

Area of lymphoid tissue lying at the border of the white pulp in the spleen.

[marginal zone B cells](#)

A unique population of B cells found in the spleen marginal zones; they do not circulate and are distinguished from conventional B cells by a distinct set of surface proteins.

[mucosal immune system](#)

The immune system that protects internal mucosal surfaces (such as the linings of the gut, respiratory tract, and urogenital tracts), which are the site of entry for virtually all pathogens and other antigens. *See also* **mucosa-associated lymphoid tissue**.

[mucosa-associated lymphoid tissue \(MALT\)](#)

Generic term for all organized lymphoid tissue found at mucosal surfaces, in which an adaptive immune response can be initiated. It comprises gut-associated lymphoid tissue (GALT), nasal-associated lymphoid tissue (NALT), and bronchus-associated lymphoid tissue (BALT) (when present).

[gut-associated lymphoid tissue \(GALT\)](#)

Lymphoid tissues associated with the gastrointestinal tract, comprising Peyer's patches, the appendix, and isolated lymphoid follicles found in the intestinal wall, where adaptive immune responses are initiated.

[tonsils](#)

Paired masses of organized peripheral lymphoid tissue situated at the base of the tongue, in which adaptive immune responses can be initiated. They are part of the mucosal immune system. *See also* **palatine tonsils**.

[adenoids](#)

Paired mucosa-associated lymphoid tissue located in the nasal cavity.

[appendix](#)

A gut-associated lymphoid tissue located at the beginning of the colon.

[Peyer's patches](#)

Organized peripheral lymphoid organs under the epithelium in the small intestine, especially the ileum, and in which an adaptive immune response can be initiated. They contain lymphoid follicles and T-cell areas. They are part of the gut-associated lymphoid tissue (GALT).

[microfold cells](#)

Specialized epithelial cell type in the intestinal epithelium over Peyer's patches, through which antigens and pathogens enter from the gut.

M cells

Specialized epithelial cell type in the intestinal epithelium over Peyer's patches, through which antigens and pathogens enter from the gut.

bronchus-associated lymphoid tissue (BALT)

Organized lymphoid tissue found in the bronchi in some animals. Adult humans do not normally have such organized lymphoid tissue in the respiratory tract, but it may be present in some infants and children.

lamina propria

A layer of connective tissue underlying a mucosal epithelium. It contains lymphocytes and other immune-system cells.

lymphoblast

A lymphocyte that has enlarged after activation and has increased its rate of RNA and protein synthesis but is not yet fully differentiated.

primary immunization

The first encounter with a given antigen, which generates the primary adaptive immune response.

affinity maturation

The increase in affinity of antibodies for their specific antigen as an adaptive immune response progresses. This phenomenon is particularly prominent in secondary and subsequent immunizations.

B-cell antigen receptor, B-cell receptor (BCR)

The cell-surface receptor on B cells for specific antigen. It is composed of a transmembrane immunoglobulin molecule (which recognizes antigen) associated with the invariant Ig α and Ig β chains (which have a signaling function). On activation by antigen, B cells differentiate into plasma cells producing antibody molecules of the same antigen specificity as this receptor.

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effector T lymphocytes, effector T cells

The T cells that perform the functions of an immune response (such as cell killing and cell activation) that clear the infectious agent from the body. There are several different subsets, each with a specific role in an immune response.

effector T lymphocytes, effector T cells

The T cells that perform the functions of an immune response (such as cell killing and cell activation) that clear the infectious agent from the body. There are several different subsets, each with a specific role in an immune response.

helper CD4 T cells, helper T cells

Effector CD4 T cells that stimulate or 'help' B cells to make antibody in response to antigenic challenge. T_{H2}, T_{H1}, and the T_{FH} subsets of effector CD4 T cells can perform this function.

Fc fragment, Fc region

The carboxy-terminal halves of the two heavy chains of an IgG molecule disulfide-bonded to each other by the residual hinge region. It is produced by cleavage of IgG by papain. In the complete antibody this portion is often called the Fc region.

heavy chain, H chain

One of the two types of protein chains in an immunoglobulin molecule, the other being called the light chain. There are several different classes, or isotypes, of heavy chain (α , δ , ϵ , γ , and μ), each of which confers a distinctive functional activity on the antibody molecule. Each immunoglobulin molecule contains two identical heavy chains.

light chain, L chain

The smaller of the two types of polypeptide chains that make up an immunoglobulin molecule. It consists of one V and one C domain and is disulfide-bonded to the heavy chain. There are two classes, or isotypes, of light chain, known as κ and λ , which are produced from separate genetic loci.

primary lymphoid organs

The sites of lymphocyte development; in humans, these are the bone marrow and thymus. B lymphocytes develop in bone marrow, whereas T lymphocytes develop within the thymus from bone marrow-derived progenitors. Also called the primary lymphoid organs and tissues.

secondary lymphoid organs

The lymph nodes, spleen, and mucosa-associated lymphoid tissues, in which adaptive immune responses are induced, as opposed to the central lymphoid organs, in which lymphocytes develop. They are also called secondary lymphoid organs and tissues.

lymphatic vessels, lymphatics

Thin-walled vessels that carry lymph.

lymphatic vessels, lymphatics

Thin-walled vessels that carry lymph.

high endothelial cells, high endothelial venules (HEVs)

Specialized small venous blood vessels in lymphoid tissues. Lymphocytes migrate from the blood into lymphoid tissues by attaching to the high endothelial cells in the walls of the venules and squeezing between them.

secondary immunization

A second or booster injection of an antigen, given some time after the initial immunization. It stimulates a secondary immune response.

nasal- (or nasopharynx-) associated lymphoid tissue (NALT)

Organized lymphoid tissues found in the upper respiratory tract. In humans, NALT consists of Waldeyer's ring, which includes the adenoids and palatine and lingual tonsils, plus other similarly organized lymphoid tissue located around the pharynx. It is part of the mucosal immune system.

The effector mechanisms of immunity.

For activated innate and adaptive immune cells to destroy pathogens, they must employ an appropriate effector mechanism suited to each infecting agent. The different types of pathogens noted in **Fig. 1.26** have different lifestyles and require different responses for both their recognition and their destruction. Perhaps it is not surprising, then, that defenses against different pathogen types are organized into [effector modules](#) suited for these different lifestyles. For example, defense against extracellular pathogens can involve both phagocytic cells and B cells, which recognize extracellular antigens and become plasma cells that secrete antibody into the extracellular environment. Defense against intracellular pathogens involves T cells that can detect peptides generated inside the infected cell. Some effector T cells directly kill cells infected with intracellular pathogens such as viruses. These examples illustrate components of two distinct effector modules, one specialized for elimination of extracellular pathogens, and a second specialized for elimination of intracellular pathogens. In general, an effector module is a collection of cell-mediated and humoral mechanisms that involve components of both the innate and adaptive immune systems; these mechanisms act together to achieve elimination of a particular category of pathogen. One important component of each effector module is the effector T cell response. Depending on the nature of the pathogen, activated T cells differentiate into three major subsets of helper T cells, each of which produces different patterns of cytokines. These three subsets, discussed later, generally specialize in promoting defenses against pathogens having three major lifestyles: they can defend against intracellular infection, destroy extracellular bacteria and fungi, or provide barrier immunity directed at parasites. T cells also promote defense against extracellular pathogens by helping B cells make antibody.

The immune system protects against four classes of pathogens		
Type of pathogen	Examples	Diseases
Viruses (intracellular)	Variola Influenza Varicella	Smallpox Flu Chickenpox
Intracellular bacteria, protozoa, parasites	<i>Mycobacterium leprae</i> <i>Leishmania donovani</i> <i>Plasmodium falciparum</i> <i>Toxoplasma gondii</i>	Leprosy Leishmaniasis Malaria Toxoplasmosis
Extracellular bacteria, parasites, fungi	<i>Streptococcus pneumoniae</i> <i>Clostridium tetani</i>	Pneumonia Tetanus

	<i>Trypanosoma brucei</i>	Sleeping sickness
	<i>Pneumocystis jirovecii</i>	<i>Pneumocystis pneumonia</i>
Parasitic worms (extracellular)	<i>Ascaris</i>	Ascariasis
	<i>Schistosoma</i>	Schistosomiasis

Fig. 1.26 The major types of pathogens confronting the immune system, and some of the diseases they cause.

Most of the other effector mechanisms used by an adaptive immune response to dispose of pathogens are the same as those of innate immunity and involve cells such as macrophages and neutrophils and proteins such as complement. Indeed, it seems likely that the vertebrate adaptive immune response evolved by the addition of specific recognition properties to innate defense mechanisms already existing in invertebrates. This is supported by recent findings that the innate lymphoid cells—the ILCs—show similar patterns of differentiation into different cytokine-producing subsets as those of T cells.

We begin this section by outlining the effector actions of antibodies, which depend almost entirely on recruiting cells and molecules of the innate immune system.

1-19 Innate immune responses can select from several effector modules to protect against different types of pathogens.

As we mentioned in Section [1-7](#), the innate immune system contains several types of cells—NK cells and ILCs—that have similarities to lymphocytes, particularly T cells. NK cells lack the antigen-specific receptors of T cells but can exhibit the cytotoxic capacity of T cells and produce some of the cytokines that effector T cells produce. ILCs develop from the same progenitor cells in the bone marrow as NK cells, and they also lack antigen-specific receptors. Very recent discoveries indicate that ILCs actually comprise several closely related lineages that differ in the specific cytokines that they will produce when activated. Remarkably, there is a striking similarity between the patterns of cytokines produced by ILC subsets and by helper T-cell subsets, as mentioned above. It appears that subsets of ILCs are the innate homologs of their helper T-cell counterparts, and NK cells are the innate homolog of cytotoxic T cells.

As mentioned in [Section 1-6](#), there are a large number of cytokines with different functions (see [eAppendix III](#), found in the ebook and on the [Digital Landing Page](#) for this book). A convenient way to organize the effects of cytokines is by the effector module that each cytokine promotes. Some cytokines tend to promote immunity to intracellular pathogens. One such cytokine is [interferon- \$\gamma\$ \(IFN- \$\gamma\$ \)](#), which acts both by activating phagocytes to more efficiently kill intracellular pathogens and by inducing target tissues to resist intracellular

pathogens. This effector module is called [type 1 immunity](#). IFN- γ is produced by some but not all subsets of innate and adaptive lymphocytes, and the ILC subset making IFN- γ is called [ILC1](#). Other ILC subsets produce cytokines favoring effector modules called [type 2 immunity](#) and [type 3 immunity](#), which coordinate defense against parasitic and extracellular pathogens, respectively. The modular nature of immune effector functions will be encountered frequently throughout this book. One principle seems to be that activated sensor cells from either the innate or the adaptive immune system can activate different subsets of innate or adaptive lymphocytes that are specialized for amplifying particular effector modules that are directed against different categories of pathogens (**Fig. 1.27**).

Effector module	Cell types, functions, and mechanisms
Cytotoxicity	NK cells, CD8 T cells
	Elimination of virally infected and metabolically stressed cells
Intracellular immunity (Type 1)	ILC1, T _H 1 cells
	Elimination of intracellular pathogens; activation of macrophages
Mucosal and barrier immunity (Type 2)	ILC2, T _H 2 cells
	Elimination and expulsion of parasites; recruitment of eosinophils, basophils, and mast cells
Extracellular immunity (Type 3)	ILC3, T _H 17 cells
	Elimination of extracellular bacteria and fungi; recruitment and activation of neutrophils

Fig. 1.27 Innate and adaptive lymphocytes share a variety of functions. The different effector modules are served by both innate and adaptive immune mechanisms. For each of the four major types of innate lymphocytes, there is a corresponding type of T cell with generally similar functional characteristics. Each set of innate lymphocyte and T cell exerts an effector activity that is broadly directed at a distinct category of pathogen.

1-20 Antibodies protect against extracellular pathogens and their toxic products.

Antibodies are found in plasma—the fluid component of blood—and in extracellular fluids. Because body fluids were once known as humors, immunity mediated by antibodies and other proteins in the plasma is known as [humoral immunity](#).

Antibodies are Y-shaped molecules with two identical antigen-binding sites and one

constant, or Fc, region. As mentioned in [Section 1-9](#), there are five forms of the constant region of an antibody, known as the antibody [classes](#) or [isotypes](#). The constant region determines an antibody's functional properties—how it will engage with the effector mechanisms that dispose of antigen once it is recognized. Each class carries out its particular function by engaging a distinct set of effector mechanisms. We describe the antibody classes and their actions in [Chapters 4](#) and [10](#).

The first and most direct way in which antibodies can protect against pathogens or their products is by binding to them and thereby blocking their access to cells that they might infect or destroy (**Fig. 1.28**, left panels). This function is known as [neutralization](#) and is important for protection against viruses, which are prevented from entering cells and replicating, and against bacterial toxins. Neutralization is the form of immunity elicited by most vaccines against bacterial toxins.

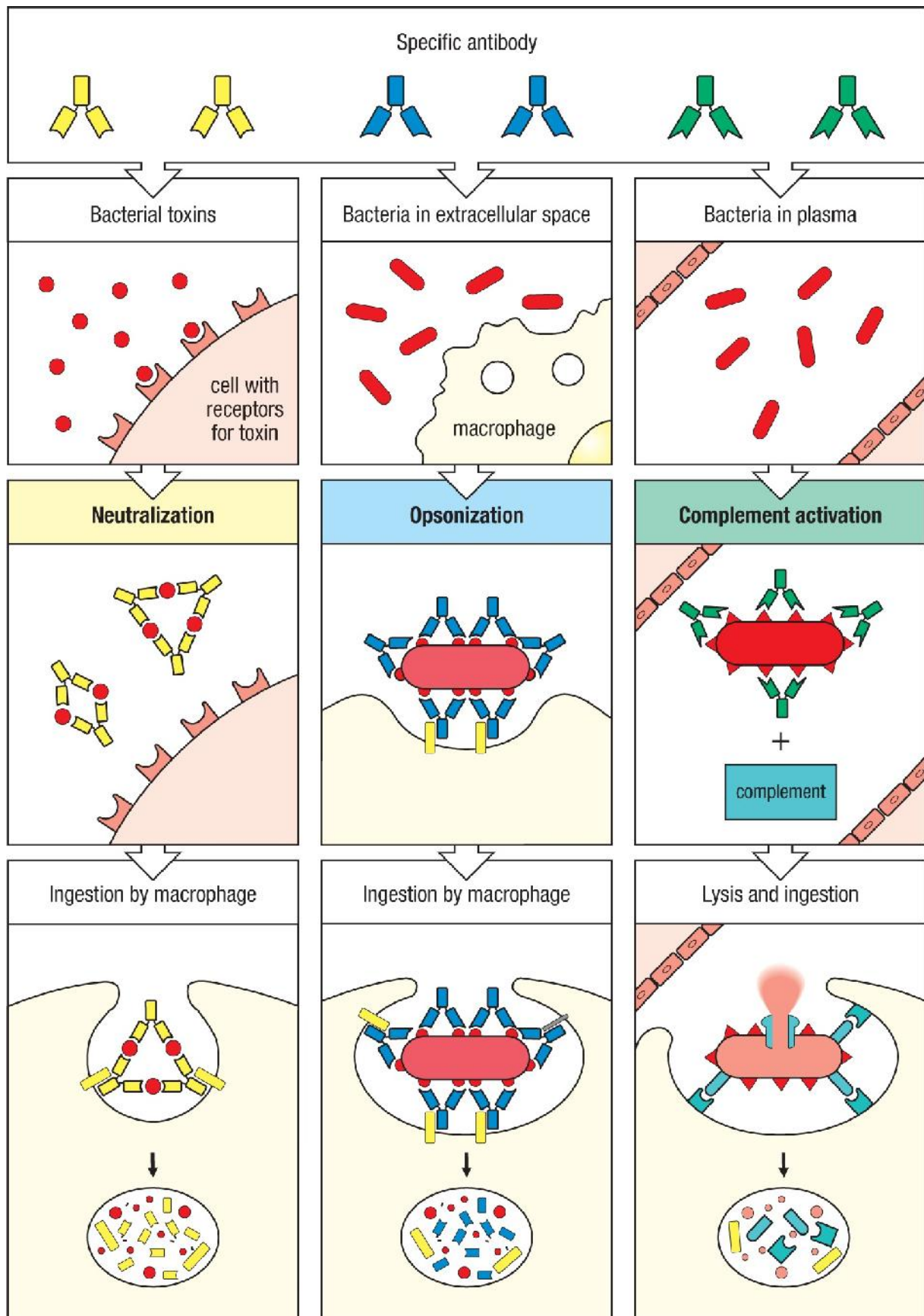


Fig. 1.28 Antibodies can participate in host defense in three main ways. The left panels show antibodies binding to and neutralizing a bacterial toxin, thus preventing it from interacting with host cells and causing pathology. Unbound toxin can react with

receptors on the host cell, whereas the toxin:antibody complex cannot. Antibodies also neutralize complete virus particles and bacterial cells by binding to them and, as for toxins, preventing these microbes from interacting with host cells. For these intact microbes, neutralization by bound antibodies prevents host-cell infection and/or damage by the microbes. The antigen:antibody complex is eventually scavenged and degraded by macrophages.

Antibodies coating an antigen render it recognizable as foreign by phagocytes (macrophages and neutrophils), which then ingest and destroy it; this is called opsonization. The center panels show opsonization and phagocytosis of a bacterial cell. Antibody first binds to antigens (red) on the bacterial cell through the variable regions. Then the antibody's Fc region binds to Fc receptors (yellow) expressed by macrophages and other phagocytes, facilitating phagocytosis.

The right panels show activation of the complement system by antibodies coating a bacterial cell. Bound antibodies form a platform that activates the first protein in the complement system, which deposits complement proteins (blue) on the surface of the bacterium. This can lead in some cases to formation of a pore that lyses the bacterium directly. More generally, complement proteins on the bacterium can be recognized by complement receptors on phagocytes; this stimulates the phagocytes to ingest and destroy the bacterium. Thus, antibodies target pathogens and their toxic products for disposal by phagocytes.

For bacteria, however, binding by antibodies is not sufficient to stop their replication. In this case, a second function of antibodies is to enable a phagocytic cell such as a macrophage or a neutrophil to ingest and destroy the bacterium. Many bacteria evade the innate immune system because they have an outer coat that is not recognized by the pattern-recognition receptors of phagocytes. However, antigens in the coat can be recognized by antibodies, and phagocytes have receptors, called [Fc receptors](#), that bind the constant region and facilitate phagocytosis of the bacterium (see Fig. 1.28, center panels). The coating of pathogens and foreign particles in this way is known as [opsonization](#).

The third function of antibodies is [complement activation](#). In [Section 1-2](#) we briefly mentioned Bordet's discovery of complement as a serum factor that 'complements' the activities of antibodies. Complement can be activated by microbial surfaces even without the help of antibodies, which leads to the covalent deposition of certain complement proteins onto the bacterial surface. But when an antibody binds first to the bacterial surface, its constant region provides a platform that is much more efficient in complement activation than microbial activation of complement alone. Thus, once antibodies are produced, complement activation against a pathogen can be substantially increased.

Certain complement components that are deposited on the bacterial surface can directly lyse the membranes of some bacteria, and this is important in a few bacterial infections (see Fig. 1.28, right panels). The major function of complement, however, is to enable phagocytes to

engulf and destroy bacteria that the phagocytes would not otherwise recognize. Most phagocytes express receptors that bind certain complement proteins; called complement receptors, these receptors bind to the complement proteins deposited onto the bacterial surface and thus facilitate bacterial phagocytosis. Certain other complement proteins also enhance the phagocytes' bactericidal capacity. The end result is that all pathogens and free molecules bound by antibody are eventually delivered to phagocytes for ingestion, degradation, and removal from the body (see Fig. 1.28, bottom panels). The complement system and the phagocytes that antibodies recruit are not themselves antigen-specific; they depend on antibody molecules to mark the particles as foreign.

1-21 T cells orchestrate cell-mediated immunity and regulate B-cell responses to most antigens.

Some bacteria and parasites, and all viruses, replicate inside cells, where they cannot be detected by antibodies, which access only the blood and extracellular space. The destruction of intracellular invaders is the function of the T lymphocytes, which are responsible for the [cell-mediated immune responses](#) of adaptive immunity. But T lymphocytes participate in responses to a wide variety of pathogens, including extracellular organisms, and so must exert a wide variety of effector activities.

T lymphocytes, of which there are several types, develop in the thymus. They are characterized by the type of T-cell receptors they express and by the expression of certain markers. The two main classes of T cells express either a cell-surface protein called [CD8](#) or another called [CD4](#). These are not just random markers, but are important for a T cell's function, because they help to determine the interactions between the T cell and other cells. Recall from [Section 1-10](#) that T cells detect peptides derived from foreign antigens that are displayed by MHC molecules on a cell's surface. CD8 and CD4 function in antigen recognition by recognizing different regions of MHC molecules and by being involved in the signaling of the T-cell receptor that is engaged with its antigen. Thus, CD4 and CD8 are known as [co-receptors](#) and they provide a functional difference between CD8 and CD4 T cells.

Importantly, there are two main types of MHC molecules, called [MHC class I](#) and [MHC class II](#). These have slightly different structures, but both have an elongated groove on the outer surface that can bind a peptide (**Fig. 1.29**). The peptide becomes trapped in this groove during the synthesis and assembly of the MHC molecule inside the cell, and the peptide:MHC complex is then transported to the cell surface and displayed to T cells (**Fig. 1.30**). Because CD8 recognizes a region of the MHC class I protein while CD4 recognizes a region of the MHC class II protein, the two co-receptors functionally distinguish T cells. Therefore, CD8 T cells selectively recognize peptides that are presented by MHC class I molecules, while CD4 T cells selectively recognize peptides that are presented by MHC

class II molecules.

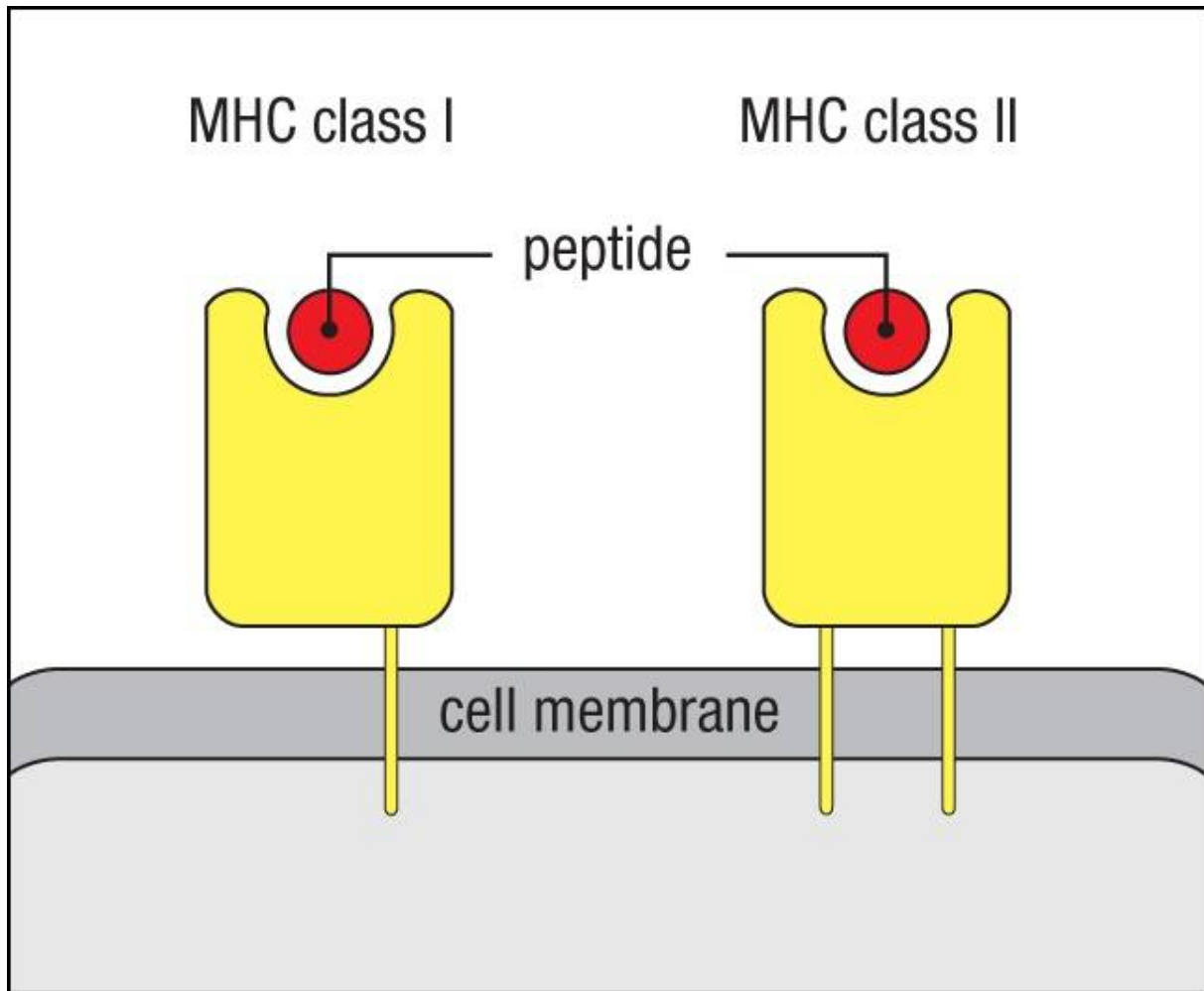


Fig. 1.29 MHC molecules on the cell surface display peptide fragments of antigens. MHC molecules are membrane proteins whose outer extracellular domains form a cleft in which a peptide fragment is bound. These fragments are derived from proteins degraded inside the cell and include both self and foreign protein antigens. The peptides are bound by the newly synthesized MHC molecule before it reaches the cell surface. There are two kinds of MHC molecules, MHC class I and MHC class II; they have related but distinct structures and functions. Although not shown here for simplicity, both MHC class I and MHC class II molecules are trimers of two protein chains and the bound self or nonself peptide.

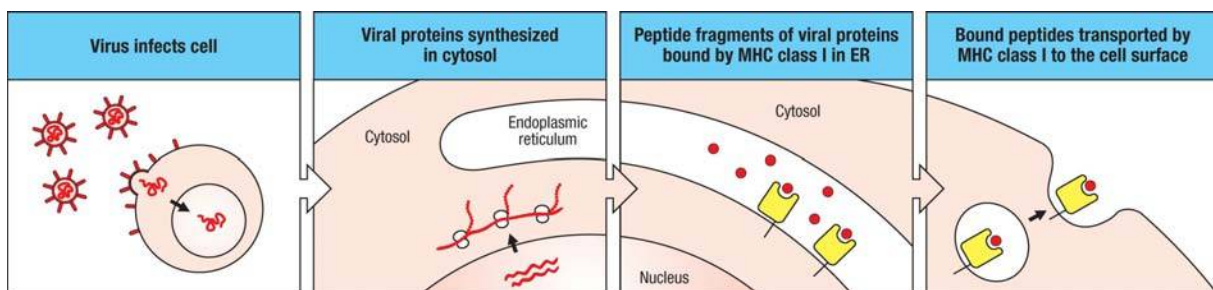


Fig. 1.30 MHC class I molecules present antigen derived from proteins in the

cytosol. In cells infected with viruses, viral proteins are synthesized in the cytosol. Peptide fragments of viral proteins are transported into the endoplasmic reticulum (ER), where they are bound by MHC class I molecules, which then deliver the peptides to the cell surface.

The most direct action of T cells is cytotoxicity. Cytotoxic T cells are effector T cells that act against cells infected with viruses. Antigens derived from the virus multiplying inside the infected cell are displayed on the cell's surface, where they are recognized by the antigen receptors of cytotoxic T cells. These T cells can then control the infection by directly killing the infected cell before viral replication is complete and new viruses are released (**Fig. 1.31**). Cytotoxic T cells carry CD8, and so recognize antigen presented by MHC class I molecules. Because MHC class I molecules are expressed on most cells of the body, they serve as an important mechanism to defend against viral infections. MHC class I molecules bearing viral peptides are recognized by CD8-bearing cytotoxic T cells, which then kill the infected cell (**Fig. 1.32**).

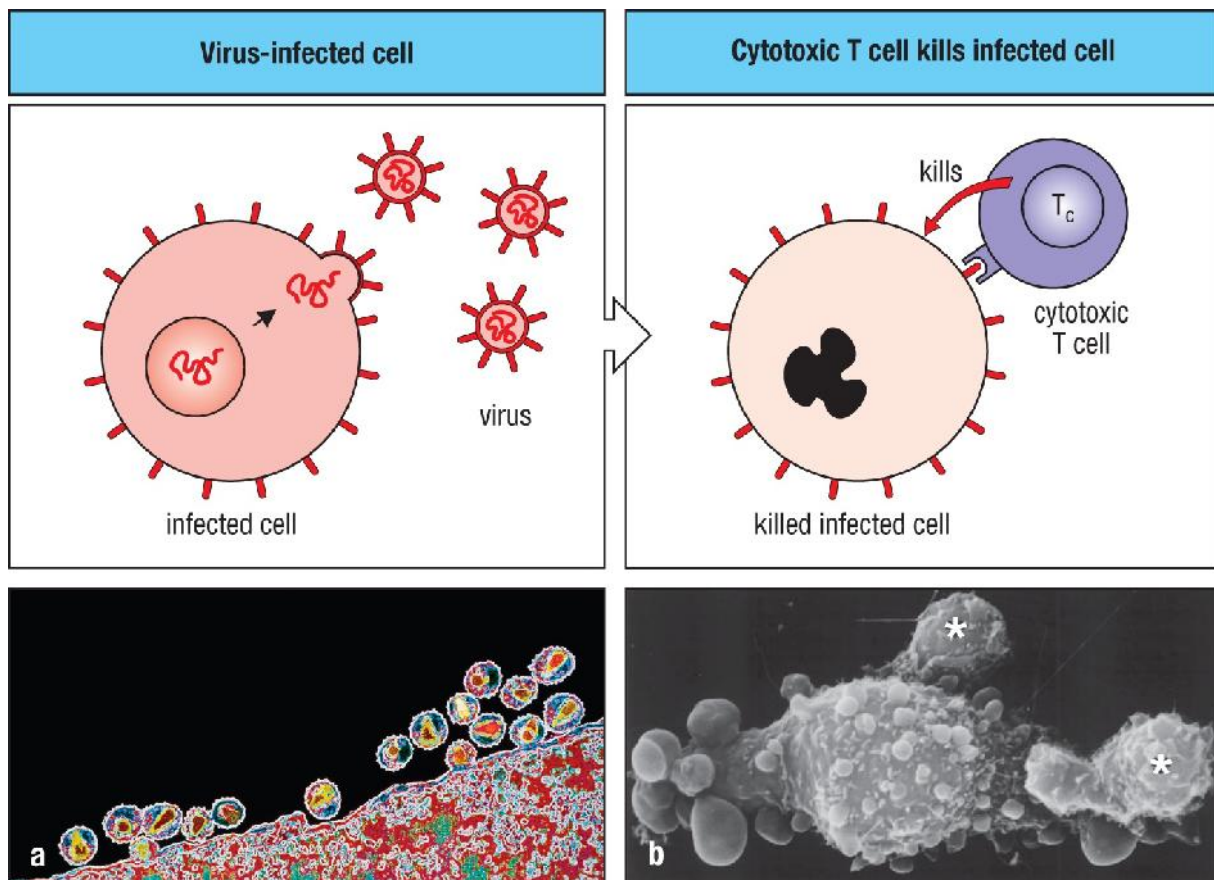


Fig. 1.31 Mechanism of host defense against intracellular infection by viruses. Cells infected by viruses are recognized by specialized T cells called cytotoxic T cells, which kill the infected cells directly. The killing mechanism involves the activation of enzymes known as caspases, which contain cysteine in their active site and cleave target proteins at aspartic acid. The caspases in turn activate a cytosolic nuclease that cleaves host and viral DNA in the infected cell. Panel a is a colored transmission electron micrograph of human immunodeficiency viruses (HIV) budding from the surface of an

infected T-cell. Panel b is a scanning electron micrograph of a target cell being killed by two cytotoxic T lymphocytes (asterisks).

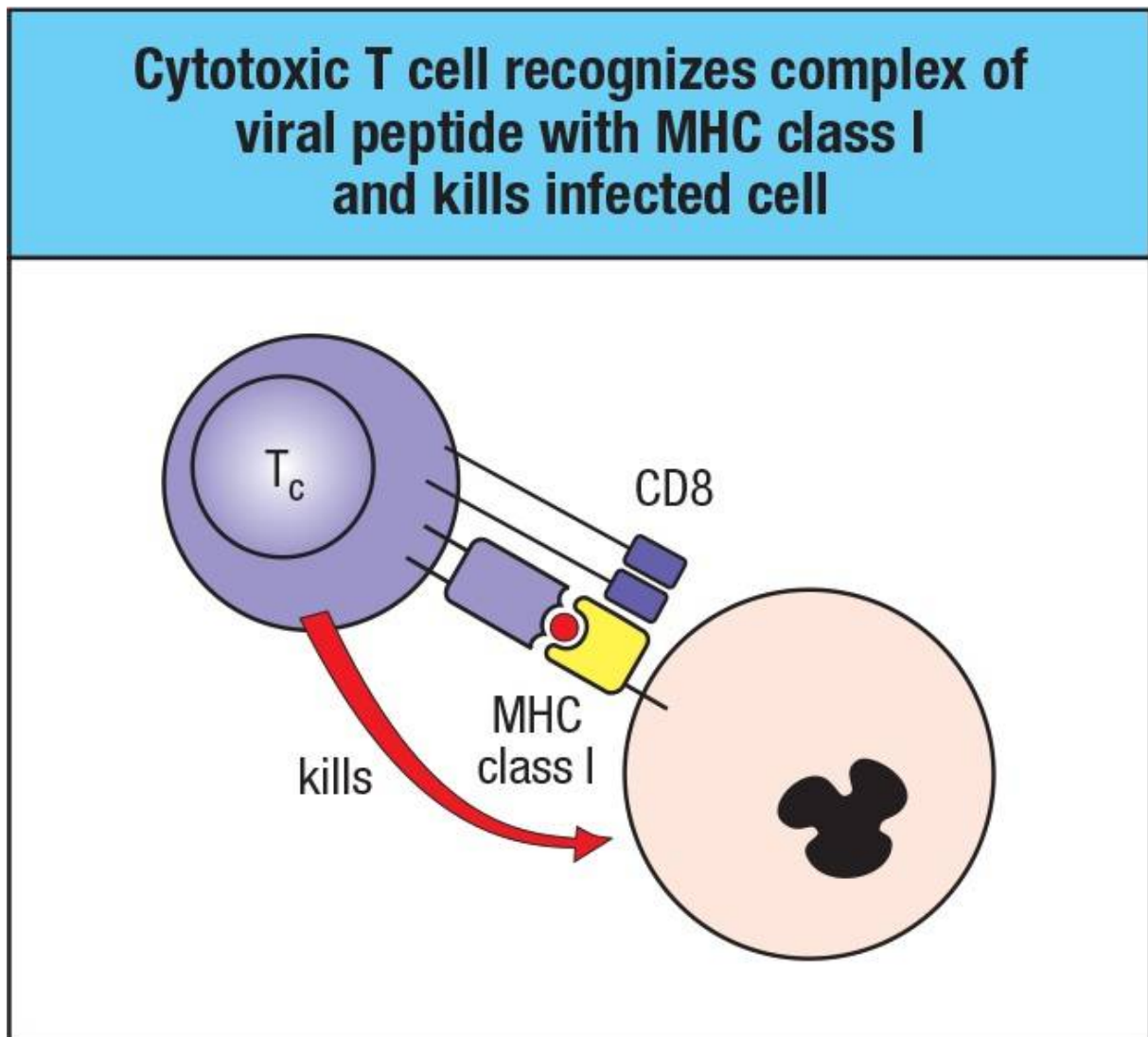
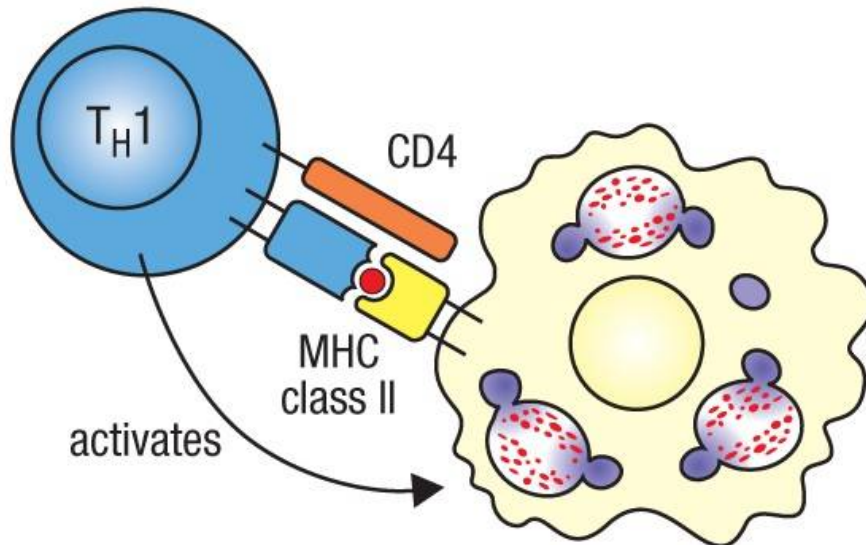


Fig. 1.32 Cytotoxic CD8 T cells recognize antigen presented by MHC class I molecules and kill the cell. The peptide:MHC class I complex on virus-infected cells is detected by antigen-specific cytotoxic T cells. Cytotoxic T cells are preprogrammed to kill the cells they recognize.

CD4 T cells recognize antigen presented by MHC class II molecules, which are expressed by the predominant antigen-presenting cells of the immune system: dendritic cells, macrophages, and B cells (**Fig. 1.33**). Thus CD4 T cells tend to recognize antigens taken up by phagocytosis from the extracellular environment. CD4 T cells are the helper T cells mentioned earlier in the chapter. They develop into a variety of different effector subsets, called [T_H1](#) (for T helper type 1), [T_H2](#), [T_H17](#), and so on, and they produce cytokines in patterns similar to those of the subsets of ILCs mentioned earlier that activate effector modules protective against different pathogens. These subsets act primarily at sites of

infection or injury in peripheral tissues. In the lymphoid tissues, a subset of CD4 T cells, called the [T follicular helper \(T_{FH}\)](#) cell, interacts with B cells to regulate antibody production during the immune response. The various T helper subsets are described in [Chapter 9](#).

T_H1 cell recognizes complex of bacterial peptide with MHC class II and activates macrophage



T follicular helper cell recognizes complex of antigenic peptide with MHC class II and activates B cell

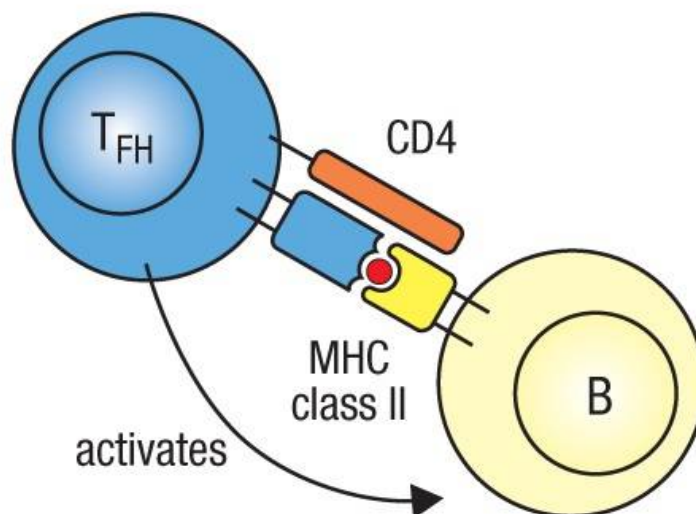


Fig. 1.33 CD4 T cells recognize antigen presented by MHC class II molecules. On recognition of their specific antigen on infected macrophages, T_H1 cells activate the macrophage, leading to the destruction of the intracellular bacteria (top panel). When T follicular helper (T_{FH}) cells recognize antigen on B cells (bottom panel), they activate these cells to proliferate and differentiate into antibody-producing plasma cells (not shown).

For example, the T_H1 subset of CD4 T cells helps to control certain bacteria that take up residence in membrane-enclosed vesicles inside macrophages. They produce the same cytokine as ILC1 cells, $IFN-\gamma$, which activates macrophages to increase their intracellular killing power and destroy these bacteria. Important infections that are controlled by this function are tuberculosis and leprosy, which are caused by the bacteria *Mycobacterium tuberculosis* and *M. leprae*, respectively. Mycobacteria survive intracellularly because they prevent the vesicles they occupy from fusing with lysosomes, which contain a variety of degradative enzymes and antimicrobial substances (**Fig. 1.34**). However, on its surface, the infected macrophage presents mycobacteria-derived antigens that can be recognized by activated antigen-specific T_H1 cells, which in turn secrete particular cytokines that induce the macrophage to overcome the block on vesicle fusion. T_H2 and T_H17 subsets produce cytokines that are specialized for promoting responses against parasites or extracellular bacteria and fungi, respectively. CD4 T cells, and their specialized subsets, play a pervasive role in adaptive immunity, and we will be returning to them many times in this book, including in [Chapters 8, 9, 11, and 12](#).

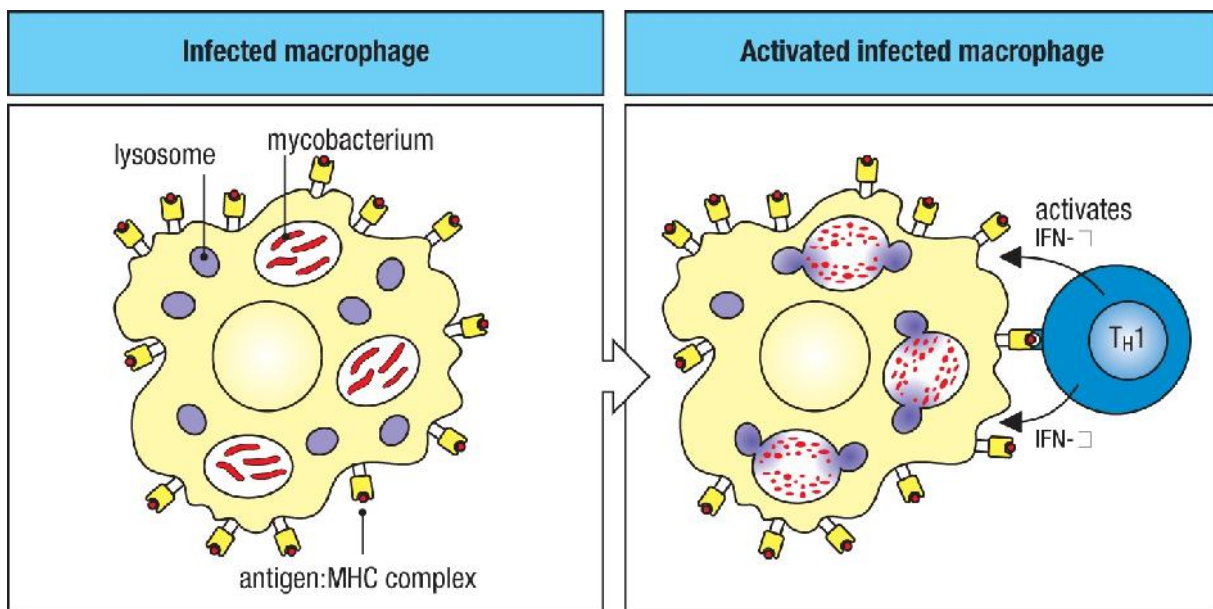


Fig. 1.34 Mechanism of host defense against intracellular infection by mycobacteria. Mycobacteria are engulfed by macrophages but resist being destroyed by preventing the intracellular vesicles in which they reside from fusing with lysosomes containing bactericidal agents. Thus the bacteria are protected from being killed. In resting macrophages, mycobacteria persist and replicate in these vesicles. When the

phagocyte is recognized and activated by a T_H1 cell, however, the phagocytic vesicles fuse with lysosomes, and the bacteria can be killed. Macrophage activation is controlled by T_H1 cells, both to avoid tissue damage and to save energy.

1-22 Inherited and acquired defects in the immune system result in increased susceptibility to infection.

We tend to take for granted the ability of our immune systems to free our bodies of infection and prevent its recurrence. In some people, however, parts of the immune system fail. In the most severe of these [immunodeficiency diseases](#), adaptive immunity is completely absent, and death occurs in infancy from overwhelming infection unless heroic measures are taken. Other less catastrophic failures lead to recurrent infections with particular types of pathogens, depending on the particular deficiency. Much has been learned about the functions of the different components of the human immune system through the study of these immunodeficiencies, many of which are caused by inherited genetic defects. Because understanding the features of immunodeficiencies requires a detailed knowledge of normal immune mechanisms, we have postponed discussion of most of these diseases until [Chapter 13](#), where they can be considered together.

In the 1980s, a devastating form of immunodeficiency appeared, the [acquired immune deficiency syndrome](#), or [AIDS](#), which is caused by an infectious agent, the human immunodeficiency viruses HIV-1 and HIV-2. This disease destroys T cells, dendritic cells, and macrophages bearing CD4, leading to infections caused by intracellular bacteria and other pathogens normally controlled by such cells. These infections are the major cause of death from this increasingly prevalent immunodeficiency disease, which is discussed fully in [Chapter 13](#) together with the inherited immunodeficiencies.



Case Study: Acquired Immune Deficiency Syndrome (AIDS)

1-23 Understanding adaptive immune responses is important for the control of allergies, autoimmune disease, and the

rejection of transplanted organs.

The main function of our immune system is to protect the human host from infectious agents. However, many medically important diseases are associated with a normal immune response directed against an inappropriate antigen, often in the absence of infectious disease. Immune responses directed at noninfectious antigens occur in [allergy](#), in which the antigen is an innocuous foreign substance; in [autoimmune disease](#), in which the response is to a self antigen; and in [graft rejection](#), in which the antigen is borne by a transplanted foreign cell (discussed in [Chapter 15](#)). The major antigens provoking graft rejection are, in fact, the MHC molecules, as each of these is present in many different versions in the human population—that is, they are highly [polymorphic](#)—and most unrelated people differ in the set of MHC molecules they express, a property commonly known as their ‘tissue type.’ The MHC was originally recognized by the work of Peter Gorer in the 1930s as a gene locus in mice, the [H-2 locus](#), that controlled the acceptance or rejection of transplanted tumors. Genes in this locus were later studied by George Snell, who examined their role in tissue transplantation by developing mouse strains differing only at these histocompatibility loci. The human MHC molecules were first discovered during the Second World War, when attempts were made to use skin grafts from donors to repair badly burned pilots and bomb victims. The patients rejected the grafts, which were recognized by their immune systems as being ‘foreign.’ What we call a successful immune response or a failure, and whether the response is considered harmful or beneficial to the host, depends not on the response itself but rather on the nature of the antigen and the circumstances in which the response occurs (**Fig. 1.35**). Snell was awarded the 1980 Nobel Prize (together with Baruj Benacerraf and Jean Dausset) for his work on the MHC.

Antigen	Effect of response to antigen	
	Normal response	Deficient response
Infectious agent	Protective immunity	Recurrent infection
Innocuous substance	Allergy	No response
Grafted organ	Rejection	Acceptance
Self organ	Autoimmunity	Self tolerance
Tumor	Tumor immunity	Cancer

Fig. 1.35 Immune responses can be beneficial or harmful, depending on the nature of the antigen. Beneficial responses are shown in white boxes, harmful responses in red-shaded boxes. Where the response is beneficial, its absence is harmful.

Allergic diseases, which include asthma, are an increasingly common cause of disability in the developed world. Autoimmunity is also now recognized as the cause of many important diseases. An autoimmune response directed against pancreatic β cells is the leading cause of diabetes in the young. In allergies and autoimmune diseases, the powerful protective mechanisms of the adaptive immune response cause serious damage to the individual.

Immune responses to harmless antigens, to body tissues, or to organ grafts are, like all other immune responses, highly specific. At present, the usual way to treat these responses is with immunosuppressive drugs, which inhibit all immune responses, desirable and undesirable alike. If it were possible to suppress only those lymphocyte clones responsible for the unwanted response, the disease could be cured or the grafted organ protected without impeding protective immune responses. At present, antigen-specific immunoregulation is outside the reach of clinical treatment. But as we shall see in [Chapter 16](#), many new drugs have been developed recently that offer more selective immune suppression to control autoimmune and other unwanted immune responses. Among these, therapies using highly specific [monoclonal antibodies](#) were made possible by Georges Köhler and César Milstein, who shared the 1984 Nobel Prize along with Niels K. Jerne. Köhler and Milstein received the prize for their “discovery of the principle for production of monoclonal antibodies.” We shall discuss the current state of understanding of allergies, autoimmune disease, graft rejection, and immunosuppressive drugs and monoclonal antibodies in [Chapters 14–16](#), and we shall see in [Chapter 15](#) how the mechanisms of immune regulation are beginning to emerge from a better understanding of the functional subsets of lymphocytes and the cytokines that control them.

1-24 Vaccination is the most effective means of controlling infectious diseases.

The deliberate stimulation of an immune response by immunization, or vaccination, has achieved many successes in the two centuries since Jenner’s pioneering experiment. Mass immunization programs have led to the virtual eradication of several diseases that used to be associated with significant morbidity (illness) and mortality (**Fig. 1.36**). Immunization is considered so safe and so important that most states in the United States require children to be immunized against up to seven common childhood diseases. Impressive as these accomplishments are, there are still many diseases for which we lack effective vaccines. And even though vaccines for diseases such as measles can be used effectively in developed countries, technical and economic problems can prevent their widespread use in developing countries, where mortality from these diseases is still high.

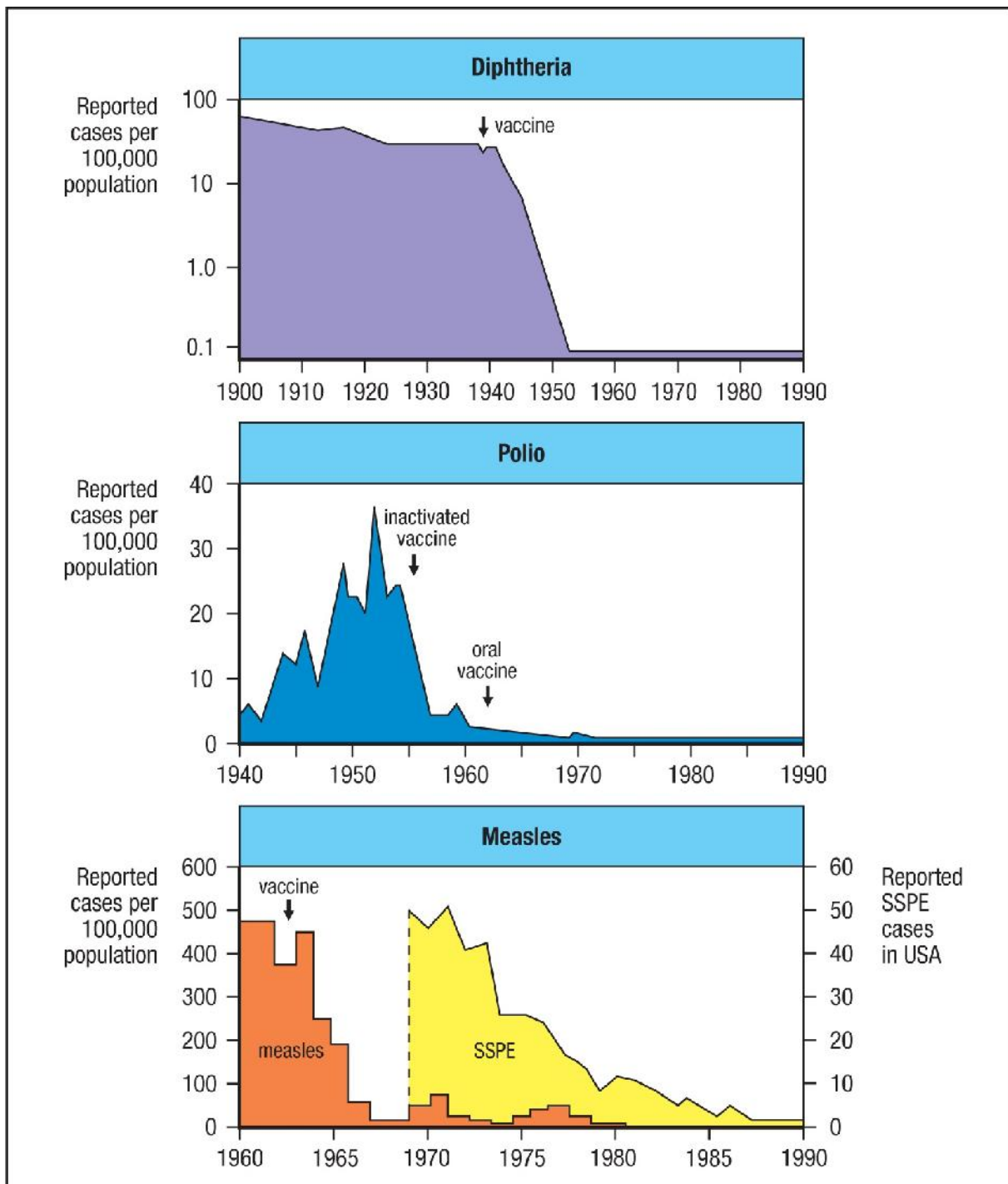


Fig. 1.36 Successful vaccination campaigns. Diphtheria, polio, and measles and their consequences have been virtually eliminated in the United States, as shown in these three graphs. SSPE stands for subacute sclerosing panencephalitis, a brain disease that is a late consequence of measles infection in few individuals. When measles was prevented, SSPE disappeared 15–20 years later. However, because these diseases have not been eradicated worldwide, immunization must be maintained in a very high percentage of the population to prevent their reappearance.

The tools of modern immunology and molecular biology are being applied to develop new

vaccines and improve old ones, and we discuss these advances in [Chapter 16](#). The prospect of controlling these important diseases is tremendously exciting. The guarantee of good health is a critical step toward population control and economic development. At a cost of pennies per person, great hardship and suffering can be alleviated.

Many serious pathogens have resisted efforts to develop vaccines against them, often because they can evade or subvert the protective mechanisms of an adaptive immune response. We examine some of the evasive strategies used by successful pathogens in [Chapter 13](#). The conquest of many of the world's leading diseases, including malaria and diarrheal diseases (the leading killers of children) as well as the more recent threat from AIDS, depends on a better understanding of the pathogens that cause them and their interactions with the cells of the immune system.

1-25 Emerging pathogens present new challenges for the immune system.

Pathogenic microbes have been besieging humans for centuries, if not millennia. Reports of the bubonic plague, caused by the bacterium *Yersinia pestis*, date back to the Philistines in 1320 BCE. Influenza virus has been causing human disease for centuries, with the earliest report of an influenza-like illness by Hippocrates, the Greek physician, in ~410 BCE. Current estimates indicate that *Mycobacterium tuberculosis*, the causative agent of tuberculosis, has existed for more than 70,000 years. However, as recent history has shown us, new pathogenic organisms occasionally arise and wreak devastation on the human population. Frequently, these emerging pathogens are viruses with relatively small genomes and high mutation rates, ideal conditions for the development of human-to-human transmission capabilities.

The early 21st century witnessed the emergence of several novel coronaviruses that cause respiratory infections with a high rate of transmission. These viruses are believed to have crossed into humans from bats, their natural reservoir. Two of these viruses caused epidemics with a high fatality rate, SARS-CoV in 2002–2003 and MERS-CoV in 2012, but ultimately were responsible for fewer than 1000 deaths each. A third coronavirus, SARS-CoV-2, emerged in late 2019 and rapidly spread across the globe, causing a worldwide pandemic that disrupted societies and economies and by June 2020 was responsible for more than 4000 deaths a day. Studies of the immune response to SARS-CoV-2 are proceeding at a remarkable pace, and by early 2021 multiple effective vaccines had been developed and introduced to the public.

Summary.

The responses to infection can be organized into several effector modules that target the

various types of pathogen lifestyles. Innate sensor cells that detect infection generate mediators that activate innate lymphoid cells (ILCs) and T cells, which amplify the immune response and also activate various effector modules. ILCs include subsets that produce different cytokines and activate distinct effector modules. T cells fall into two major classes that are based on the expression of the co-receptors CD8 and CD4. CD8 and CD4 T cells recognize antigen presented by MHC class I or MHC class II proteins, respectively. These subsets of T cells, like their ILC counterparts, also promote the actions of distinct effector modules. NK cells and CD8 T cells can exert cytotoxic activity to target intracellular infections such as viruses. Other subsets of innate lymphoid and helper T cells can secrete mediators that activate other effector functions, ones that target intracellular bacteria, extracellular bacteria and fungi, and parasites. T cells also provide signals that help regulate B cells and stimulate them to produce antibodies. Specific antibodies mediate the clearance and elimination of soluble toxins and extracellular pathogens. They interact not only with the toxins or the antigens on microbes but also with the Fc region of specific receptors that are expressed by many types of phagocytes. Phagocytes also express receptors for complement proteins that are deposited on microbial surfaces, particularly in the presence of antibody.

Failures of immunity can be caused by genetic defects or by infections that target important components of the immune system. Misdirected immune responses can damage host tissues, as in autoimmune diseases or allergy, or lead to the failure of transplanted organs. While vaccination is still the greatest tool of immunology to fight diseases, modern approaches have added new tools, such as monoclonal antibodies, that have become progressively more important in the clinic since their initial use in the mid-1980s.

Summary to [Chapter 1](#).

The immune system defends the host against infection. Innate immunity serves as a first line of defense but lacks the ability to recognize certain pathogens and to provide the specific protective immunity that prevents reinfection. Adaptive immunity is based on clonal selection from a repertoire of lymphocytes bearing highly diverse antigen-specific receptors that enable the immune system to recognize any foreign antigen. In the adaptive immune response, antigen-specific lymphocytes proliferate and differentiate into clones of effector lymphocytes that eliminate the pathogen. Figure 1.7 summarizes the phases of the immune response and their approximate timings. Host defense requires different recognition systems and a wide variety of effector mechanisms to seek out and destroy the wide variety of pathogens in their various habitats within the body and at its external and internal surfaces. Not only can the adaptive immune response eliminate a pathogen, but, in the process, it also generates increased numbers of differentiated memory lymphocytes through clonal selection, and this allows a more rapid and effective response upon reinfection. The regulation of immune responses, whether to suppress them when unwanted or to stimulate them in the prevention of infectious disease, is the major medical goal of research in immunology.

Glossary

[effector module](#)

A collection of cell-mediated and humoral mechanisms that involve components of both the innate and adaptive immune systems. These mechanisms act together to achieve elimination of a particular category of pathogen.

[type 1 immunity](#)

Class of effector activities aimed at elimination of intracellular pathogens.

[ILC1](#)

A subset of innate lymphoid cells characterized by production of IFN- γ .

[type 2 immunity](#)

Class of effector activities aimed at elimination of parasites and promoting barrier and mucosal immunity.

[type 3 immunity](#)

Class of effector activities aimed at elimination of extracellular pathogens such as bacteria and fungi.

[classes](#)

The class of an antibody is defined by the type of heavy chain it contains. There are five main antibody classes: IgA, IgD, IgM, IgG, and IgE, containing heavy chains α , δ , μ , γ , and ϵ , respectively. The IgG class has several subclasses. *See also* **isotypes**.

[isotype](#)

The designation of an immunoglobulin chain with respect to the type of constant region it has. Light chains can be of either κ or λ isotype. Heavy chains can be of μ , δ , γ , α , or

ϵ isotype. The different heavy-chain isotypes have different effector functions and determine the class and functional properties of antibodies (IgM, IgD, IgG, IgA, and IgE, respectively).

[neutralization](#)

Inhibition of the infectivity of a virus or the toxicity of a toxin molecule by the binding of antibodies.

[Fc receptors](#)

Family of cell-surface receptors that bind the Fc portions of different immunoglobulins: Fc γ receptors bind IgG, for example, and Fc ϵ receptors bind IgE.

[opsonization](#)

The coating of the surface of a pathogen by antibody and/or complement that makes it more easily ingested by phagocytes.

[complement activation](#)

The activation of the normally inactive proteins of the complement system that occurs on infection. See **classical pathway**, **alternative pathway**, **lectin pathway**.

[cell-mediated immune responses](#)

An adaptive immune response in which antigen-specific effector T cells have the main role. The immunity to infection conferred by such a response is called cell-mediated immunity. A primary cell-mediated immune response is the T-cell response that occurs the first time a particular antigen is encountered.

[CD8](#)

The co-receptor for T-cell receptors that recognize peptide antigens bound to MHC class I molecules. It binds to the lateral face of the MHC molecule.

[CD4](#)

The co-receptor for T-cell receptors that recognize peptide antigens bound to MHC class II molecules. It binds to the lateral face of the MHC molecule.

[co-receptor](#)

Cell-surface protein that increases the sensitivity of a receptor to its ligand by binding to associated ligands and participating in signaling. The antigen receptors on T cells and B cells act in conjunction with co-receptors, which are either CD4 or CD8 on T cells, and a co-receptor complex of three proteins, one of which is the complement receptor CR2, on B cells.

[MHC class I](#)

Polymorphic cell-surface proteins encoded in the MHC locus and expressed on most cells. They present antigenic peptides generated in the cytosol to CD8 T cells and also bind the co-receptor CD8.

[MHC class II](#)

Polymorphic cell-surface proteins encoded in the MHC locus and expressed primarily on specialized antigen-presenting cells. They present antigenic peptides derived from internalized extracellular pathogens to CD4 T cells and also bind the co-receptor CD4.

[T follicular helper \(T_{FH}\)](#)

An effector T cell found in lymphoid follicles that provides help to B cells for antibody production and class switching.

[immunodeficiency diseases](#)

Any inherited or acquired disorder in which some aspect or aspects of host defense are absent or functionally defective.

acquired immune deficiency syndrome (AIDS)

A disease caused by infection with the human immunodeficiency virus (HIV-1 or HIV-2). AIDS occurs when an infected individual has lost most of his or her CD4 T cells, so that infections with opportunistic pathogens occur.

AIDS

A disease caused by infection with the human immunodeficiency virus (HIV-1 or HIV-2). AIDS occurs when an infected individual has lost most of his or her CD4 T cells, so that infections with opportunistic pathogens occur.

allergy

The state in which a symptomatic immune reaction is made to a normally innocuous environmental antigen. It involves the interaction between the antigen and antibody or primed T cells produced by earlier exposure to the same antigen.

autoimmune disease

Disease in which the pathology is caused by adaptive immune responses to self antigens.

graft rejection

The immunologically mediated rejection of grafted tissues or organs from a genetically nonidentical donor. It is due chiefly to recognition of nonself MHC molecules on the graft.

polymorphic

Existing in a variety of different forms; applied to a gene, occurring in a variety of different alleles.

monoclonal antibodies

Antibodies produced by a single clone of B lymphocytes, so that they are all identical.

IFN- γ

A cytokine of the interferon structural family produced by effector CD4 T_H1 cells, CD8 T cells, and NK cells. Its primary function is the activation of macrophages, and it acts through a different receptor from that of the type I interferons.

humoral immunity, humoral immune response

Immunity due to proteins circulating in the blood, such as antibodies (in adaptive immunity) or complement (in innate immunity). Adaptive humoral immunity can be transferred to unimmunized recipients by the transfer of serum containing specific antibody.

T_H1

A subset of effector CD4 T cells characterized by the cytokines they produce, notably IFN- γ and TNF- α . They are mainly involved in activating macrophages and promoting inflammation.

T_H2

A subset of effector CD4 T cells that are characterized by the cytokines they produce, notably IL-4, IL-5, and IL-13. They are involved in recruiting eosinophils and activating mast cells and basophils, and in promoting enhanced barrier protection.

T_H17

A subset of CD4 T cells that are characterized by production of the cytokine IL-17.
They help recruit neutrophils to sites of infection.

[H-2 locus, H-2 genes](#)

The major histocompatibility complex of the mouse. Haplotypes are designated by a lowercase superscript, as in H-2^b.

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2

Innate Immunity: The First Lines of Defense

IN THIS CHAPTER

- [Anatomic barriers and initial chemical defenses.](#)
 - [The complement system and innate immunity.](#)
-

As stated in [Chapter 1](#), most microbial invaders can be detected and destroyed within minutes or hours by the body's defense mechanisms of [innate immunity](#), which do not rely on expansion of antigen-specific lymphocytes. The innate immune system uses a limited number of secreted proteins and cell-associated receptors to detect infection and to distinguish between pathogens and host tissues. These are called innate receptors because they are inborn; they are encoded by genes directly inherited from an individual's parents and do not need to be generated by the gene rearrangements used to assemble antigen receptors of lymphocytes described in [Section 1-11](#). The importance of innate immunity is illustrated by several immunodeficiencies that result when it is impaired, increasing susceptibility to infection even in the presence of an intact adaptive immune system, as we will discuss in [Chapter 13](#).

As we saw in Fig. 1.5, an infection starts when a pathogen breaches one of the host's anatomic barriers. Some innate immune mechanisms start acting immediately (**Fig. 2.1**). These immediate defenses include several classes of preformed soluble molecules that are present in extracellular fluid, blood, and epithelial secretions and which can either kill the pathogen or weaken its effect. [Antimicrobial enzymes](#) such as lysozyme begin to digest bacterial cell walls; [antimicrobial peptides](#) such as the defensins lyse bacterial cell membranes directly; and a system of plasma proteins known as the [complement system](#) targets pathogens both for lysis and for phagocytosis by cells of the innate immune system, such as macrophages. If these fail, innate immune cells become activated by pattern-recognition receptors (PRRs) that detect molecules called pathogen-associated molecular patterns (PAMPs) that are typical of microbes (see [Section 1-5](#)). The activated innate cells can engage various effector mechanisms to eliminate the infection. By themselves, neither the soluble nor the cellular components of innate immunity generate long-term protective immunological memory. If an infectious organism breaches these first two lines of defense, mechanisms will be engaged to induce an adaptive immune response—the third phase of the response to a pathogen. This leads to the expansion of antigen-specific lymphocytes that

target the pathogen specifically and to the formation of memory cells that provide long-lasting specific immunity.

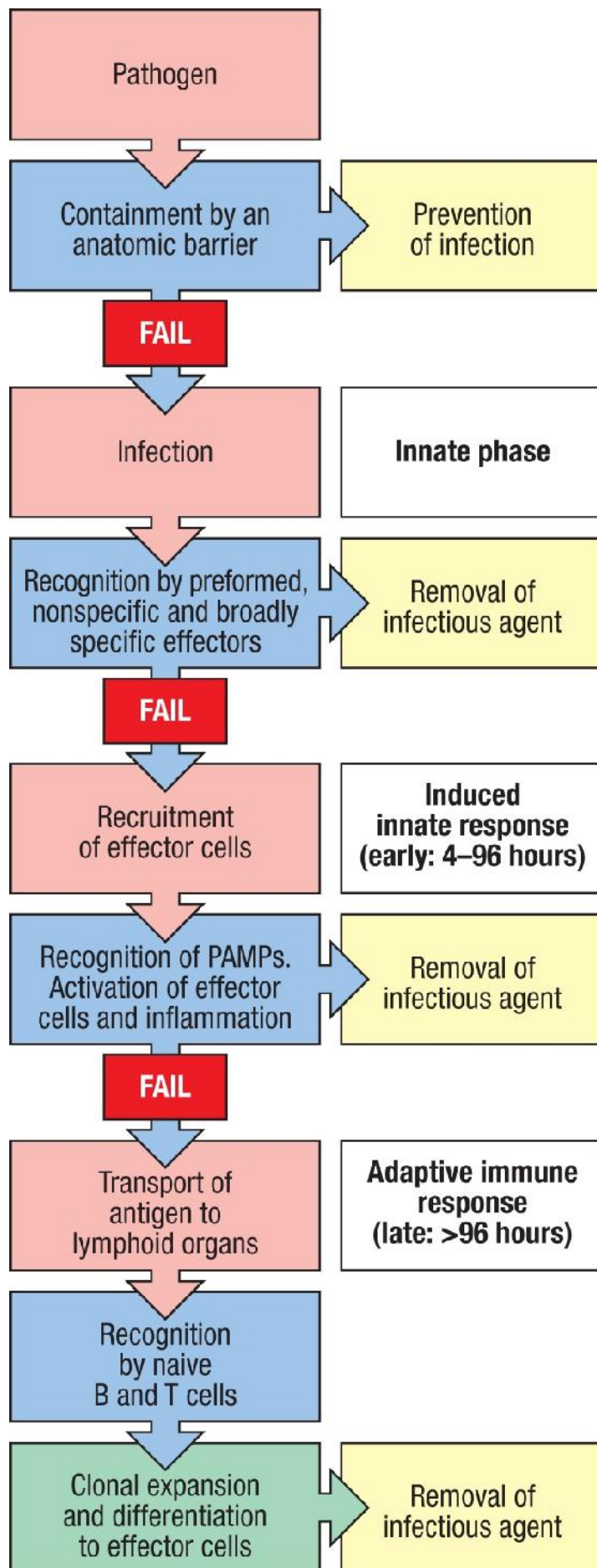


Fig. 2.1 The response to an initial infection occurs in three phases. These are the innate phase, the induced innate response, and the adaptive immune response. The first two phases rely on the recognition of pathogens by germline-encoded receptors of the innate immune system, whereas adaptive immunity uses variable antigen-specific receptors that are produced as a result of gene segment rearrangements. Adaptive immunity occurs late, because the rare B cells and T cells specific for the invading pathogen must first undergo clonal expansion before they differentiate into effector cells that migrate to the site of infection and clear the infection. The effector mechanisms that remove the infectious agent are similar or identical in each phase.

This chapter considers the first phase of the innate immune response. We first describe the anatomic barriers that protect the host against infection and examine the immediate innate defenses provided by various secreted soluble proteins. The anatomic barriers are fixed defenses against infection and consist of the epithelia that line the internal and external surfaces of the body, along with the phagocytes residing beneath all epithelial surfaces. These phagocytes act directly by engulfing and digesting invading microorganisms. Epithelia are also protected by many kinds of chemical defenses, including antimicrobial enzymes and peptides. Next we describe the complement system, which directly kills some microorganisms and interacts with others to promote their removal by phagocytic cells. The complement system together with other soluble circulating defensive proteins is sometimes referred to as [humoral](#) innate immunity, from the old word ‘humor’ for body fluids. If these early defenses fail, the phagocytes at the site of infection help recruit new cells and circulating effector molecules, a process called inflammation, which we will discuss in [Chapter 3](#).

Glossary

[innate immunity](#)

The various innate resistance mechanisms that are encountered first by a pathogen, before adaptive immunity is induced, such as anatomical barriers, antimicrobial peptides, the complement system, and macrophages and neutrophils carrying nonspecific pathogen-recognition receptors. Innate immunity is present in all individuals at all times, does not increase with repeated exposure to a given pathogen, and discriminates between groups of similar pathogens, rather than responding to a particular pathogen. Cf. **adaptive immunity**.

[antimicrobial enzymes](#)

Enzymes that kill microorganisms by their actions. An example is lysozyme, which digests bacterial cell walls.

[complement system](#)

A set of plasma proteins that act together as a defense against pathogens in extracellular spaces. The pathogen becomes coated with complement proteins that facilitate its removal by phagocytes and which can also kill certain pathogens directly.

Activation of the complement system can be initiated in several different ways. *See* **classical pathway, alternative pathway, lectin pathway.**

[humoral](#)

Referring to effector proteins in the blood or body fluids, such as antibodies in adaptive immunity or complement proteins in innate immunity.

[antimicrobial peptides, antimicrobial proteins](#)

Amphipathic peptides or proteins secreted by epithelial cells and phagocytes that kill a variety of microbes nonspecifically, mainly by disrupting cell membranes.

Antimicrobial peptides in humans include the defensins, the cathelicidins, the histatins, and RegIIIγ.

Anatomic barriers and initial chemical defenses.

Microorganisms that cause disease in humans and animals enter the body at different sites and produce disease symptoms by a variety of mechanisms. Microorganisms that cause disease and produce damage, or pathology, to tissues are referred to as [pathogenic microorganisms](#), or simply [pathogens](#). As innate immunity eliminates most microorganisms that may occasionally cross an anatomic barrier, pathogens are microorganisms that have evolved ways of overcoming the body's innate defenses more effectively than have other microorganisms. Once infection is established, both innate and adaptive immune responses are typically required to eliminate pathogens from the body. Even in these cases, the innate immune system performs a valuable function by reducing pathogen numbers during the time needed for the adaptive immune system to gear up for action. In the first part of this chapter, we briefly describe the different types of pathogens and their invasive strategies and then examine the immediate innate defenses that, in most cases, prevent microorganisms from establishing an infection.

2-1 Infectious diseases are caused by diverse living agents that replicate in their hosts.

The agents that cause disease fall into five groups: viruses, bacteria, fungi, protozoa, and helminths (worms). Protozoa and worms are usually grouped together as parasites and are the subject of the discipline of parasitology, whereas viruses, bacteria, and fungi are the subject of microbiology. **Figure 2.2** lists some examples of the different classes of microorganisms and parasites, and the diseases they cause. The characteristic features of each pathogen are its mode of transmission, its mechanism of replication, its mechanism of [pathogenesis](#)—the means by which it causes disease—and the response it elicits from the host. The distinct pathogen habitats and life cycles mean that a range of different innate and adaptive immune mechanisms have to be deployed for pathogen destruction.

Routes of infection for pathogens				
Route of entry	Mode of transmission	Pathogen	Disease	Type of pathogen
Mucosal surfaces				

Mouth and respiratory tract	Inhalation or ingestion of infective material (e.g., saliva droplets)	Measles virus Influenza virus Varicella-zoster Epstein–Barr virus <i>Streptococcus pyogenes</i> <i>Haemophilus influenzae</i> <i>Neisseria meningitidis</i>	Measles Influenza Chickenpox Mononucleosis Tonsillitis Pneumonia, meningitis Meningococcal meningitis	Paramyxovirus Orthomyxovirus Herpesvirus Herpesvirus Gram-positive bacterium Gram-negative bacterium Gram-negative bacterium
	Spores	<i>Bacillus anthracis</i>	Inhalation anthrax	Gram-positive bacterium
Gastrointestinal tract	Contaminated water or food	Rotavirus Hepatitis A <i>Salmonella enterica</i> ssp. <i>enteritidis</i> , <i>Salmonella enterica</i> ssp. <i>typhimurium</i> <i>Vibrio cholerae</i> <i>Salmonella enterica</i> ssp. <i>typhi</i> <i>Trichuris trichiura</i>	Diarrhea Jaundice Food poisoning Cholera Typhoid fever Trichuriasis	Rotavirus Picornavirus Gram-negative bacterium Gram-negative bacterium Gram-negative bacterium Helminth
	Sexual transmission/ infected blood	Hepatitis B virus Human immunodeficiency	Hepatitis B Acquired immune deficiency	Hepadnavirus Retrovirus

Reproductive tract and other routes		virus (HIV)	syndrome (AIDS)	
	Sexual transmission	<i>Neisseria gonorrhoeae</i> <i>Treponema pallidum</i>	Gonorrhea Syphilis	Gram-negative bacterium Bacterium (spirochete)
Opportunistic infections	Resident microbiota	<i>Candida albicans</i>	Candidiasis, thrush	Fungus
	Resident lung microbiota	<i>Pneumocystis jirovecii</i>	Pneumonia	Fungus
External epithelia				
External surface	Physical contact	<i>Trichophyton</i>	Athlete's foot	Fungus
Wounds and abrasions	Minor skin abrasions	<i>Bacillus anthracis</i>	Cutaneous anthrax	Gram-positive bacterium
	Puncture wounds	<i>Clostridium tetani</i>	Tetanus	Gram-positive bacterium
	Handling infected animals	<i>Francisella tularensis</i>	Tularemia	Gram-negative bacterium
Insect bites	Mosquito bites (<i>Aedes aegypti</i>)	Flavivirus	Yellow fever	Virus
	Deer tick bites	<i>Borrelia burgdorferi</i>	Lyme disease	Bacterium (spirochete)
	Mosquito bites (<i>Anopheles</i>)	<i>Plasmodium</i> spp.	Malaria	Protozoan

Fig. 2.2 A variety of microorganisms can cause disease. Pathogenic organisms are of five main types: viruses, bacteria, fungi, protozoa, and helminths (worms). Some well-known pathogens are listed.

Infectious agents can grow in all body compartments, as shown schematically in **Fig. 2.3**. We saw in [Chapter 1](#) that two major compartments can be defined: extracellular and intracellular. Both innate and adaptive immune responses have different ways of dealing with pathogens found in these two compartments. Many bacterial pathogens live and replicate in extracellular spaces, either within tissues or on the surface of the epithelia that line body cavities. Extracellular bacteria are usually susceptible to killing by phagocytes, an important arm of the innate immune system, but some pathogens, such as *Staphylococcus* and *Streptococcus* species, achieve a very basic form of [immune evasion](#) by having a protective polysaccharide capsule that resists engulfment. This can be overcome to some extent by the help of another component of innate immunity (complement), which deposits specific molecules on the surface of the bacteria (opsonization), rendering them more susceptible to phagocytosis. In the adaptive immune response, bacteria are rendered more susceptible to phagocytosis by a combination of antibodies and complement.

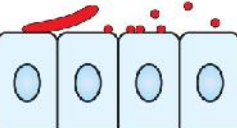
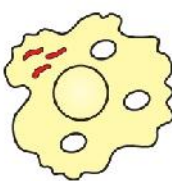
Site of infection	Extracellular		Intracellular	
	Interstitial spaces, blood, lymph	Epithelial surfaces	Cytoplasmic	Vesicular
				
Organisms	Viruses Bacteria Protozoa Fungi Worms	<i>Neisseria gonorrhoeae</i> <i>Streptococcus pneumoniae</i> <i>Vibrio cholerae</i> <i>Helicobacter pylori</i> <i>Candida albicans</i> Worms	Viruses <i>Chlamydia</i> spp. <i>Rickettsia</i> spp. Protozoa	<i>Mycobacterium</i> spp. <i>Yersinia pestis</i> <i>Legionella pneumophila</i> <i>Cryptococcus neoformans</i> <i>Leishmania</i> spp.
Protective immunity	Complement Phagocytosis Antibodies	Antimicrobial peptides Antibodies, especially IgA	NK cells Cytotoxic T cells	T cell–dependent and NK cell–dependent macrophage activation

Fig. 2.3 Pathogens can be found in various compartments of the body, where they must be combated by different host-defense mechanisms. Virtually all pathogens have an extracellular phase in which they are vulnerable to the circulating molecules and cells of innate immunity and to the antibodies of the adaptive immune response. All

these clear the microorganism mainly by promoting its uptake and destruction by the phagocytes of the immune system. Intracellular phases of pathogens such as viruses are not accessible to these mechanisms; instead, the infected cell is attacked by the NK cell of innate immunity or by the cytotoxic T cells of adaptive immunity. Activation of macrophages as a result of NK-cell or T-cell activity can induce the macrophage to kill pathogens that are living inside macrophage vesicles.

Infectious diseases differ in their symptoms and outcome depending on where the causal pathogen replicates within the body—the intracellular or the extracellular compartment—and what damage it does to the tissues (**Fig. 2.4**). Pathogens that live intracellularly frequently cause disease by damaging or killing the cells they infect. Obligate intracellular pathogens, such as viruses, must invade host cells to replicate. Facultative intracellular pathogens, such as mycobacteria, can replicate either intracellularly or outside the cell. Two strategies of innate immunity defend against intracellular pathogens. One strategy aims to destroy pathogens before they infect cells and includes soluble defenses such as antimicrobial peptides, as well as phagocytic cells that can engulf and destroy pathogens before they become intracellular. Alternatively, the innate immune system can recognize and kill cells infected by some pathogens. This is the role of the natural killer cells (NK cells), which act to limit certain viral infections before cytotoxic T cells of the adaptive immune system become functional. Intracellular pathogens can be subdivided further into those that replicate freely in the cell, such as viruses and certain bacteria (for example, *Chlamydia*, *Rickettsia*, and *Listeria*), and those that replicate inside intracellular vesicles, such as mycobacteria. Pathogens that live inside macrophage vesicles may become more susceptible to being killed after activation of the macrophage as a result of NK-cell or T-cell actions (see Fig. 2.3).

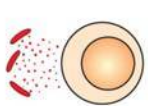
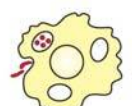
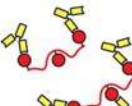
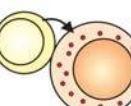
	Direct mechanisms of tissue damage by pathogens			Indirect mechanisms of tissue damage by pathogens		
	Exotoxin production	Endotoxin	Direct cytopathic effect	Immune complexes	Anti-host antibody	Cell-mediated immunity
Pathogenic mechanism						
Infectious agent	<i>Streptococcus pyogenes</i> <i>Staphylococcus aureus</i> <i>Corynebacterium diphtheriae</i> <i>Clostridium tetani</i> <i>Vibrio cholerae</i>	<i>Escherichia coli</i> <i>Haemophilus influenzae</i> <i>Salmonella enterica</i> ssp. <i>typhi</i> <i>Shigella</i> <i>Pseudomonas aeruginosa</i> <i>Yersinia pestis</i>	Varicella-zoster Hepatitis B virus Polio virus Measles virus Influenza virus Herpes simplex virus Human herpes virus 8 (HHV8)	Hepatitis B virus <i>Plasmodium</i> <i>Streptococcus pyogenes</i> <i>Treponema pallidum</i> Most acute infections	<i>Streptococcus pyogenes</i> <i>Mycoplasma pneumoniae</i>	Lymphocytic choriomeningitis virus Herpes simplex virus <i>Mycobacterium tuberculosis</i> <i>Mycobacterium leprae</i> <i>Borrelia burgdorferi</i> <i>Schistosoma mansoni</i>
Disease	Tonsillitis, scarlet fever Boils, toxic shock syndrome, food poisoning Diphtheria Tetanus Cholera	Gram-negative sepsis Meningitis, pneumonia Typhoid fever Bacillary dysentery Wound infection Plague	Smallpox Chickenpox, shingles Hepatitis Poliomyelitis Measles, subacute sclerosing panencephalitis Influenza Cold sores Kaposi's sarcoma	Kidney disease Vascular deposits Glomerulonephritis Kidney damage in secondary syphilis Transient renal deposits	Rheumatic fever Hemolytic anemia	Aseptic meningitis Herpes stromal keratitis Tuberculosis Tuberculous leprosy Lyme arthritis Schistosomiasis

Fig. 2.4 Pathogens can damage tissues in a variety of different ways. The mechanisms of damage, representative infectious agents, and the common names of the diseases associated with each are shown. Exotoxins are released by microorganisms and act at the surface of host cells, for example, by binding to receptors. Endotoxins, which are intrinsic components of microbial structure, trigger phagocytes to release cytokines that produce local or systemic symptoms. Many pathogens are cytopathic, directly damaging the cells they infect. Finally, an adaptive immune response to the pathogen can generate antigen:antibody complexes that activate neutrophils and macrophages, antibodies that can cross-react with host tissues, or T cells that kill infected cells. All of these have some potential to damage the host's tissues. In addition, neutrophils, the most abundant cells early in infection, release many proteins and small-molecule inflammatory mediators that both control infection and cause tissue damage.

Many of the most dangerous extracellular bacterial pathogens cause disease by releasing protein toxins; these secreted toxins are called [exotoxins](#) (see Fig. 2.4). The innate immune system has little defense against such toxins, and highly specific antibodies produced by the adaptive immune system are required to neutralize their action (see Fig. 1.28). The damage caused by a particular infectious agent also depends on where it grows; *Streptococcus pneumoniae* in the lung causes pneumonia, for example, whereas in the blood it causes a potentially fatal systemic illness, pneumococcal sepsis. In contrast, nonsecreted constituents of bacterial structure that trigger phagocytes to release cytokines with local and systemic effects are called [endotoxins](#). An endotoxin of major medical importance is the [lipopolysaccharide \(LPS\)](#) of the outer cell membrane of [Gram-negative bacteria](#), such as *Salmonella*. Many of the clinical symptoms of infection by such bacteria—including fever, pain, rash, hemorrhage, septic shock—are due largely to LPS.

Most pathogenic microorganisms can overcome innate immune responses and continue to grow, making us ill. An adaptive immune response is required to eliminate them and to prevent subsequent reinfections. Certain pathogens are never entirely eliminated by the immune system and persist in the body for years. But most pathogens are not universally lethal. Those that have lived for thousands of years in the human population are highly evolved to exploit their human hosts; they cannot alter their pathogenicity without upsetting the compromise they have achieved with the human immune system. Rapidly killing every host it infects is no better for the long-term survival of a pathogen than being wiped out by the immune response before the microbe has had time to infect someone else. In short, we have adapted to live with many microbes, and they with us. Nevertheless, highly pathogenic strains of avian influenza, the episode in 2002–2003 of SARS (severe acute respiratory syndrome), a severe pneumonia in humans that is caused by the coronavirus SARS-CoV from bats, and the SARS-CoV-2 pandemic of 2020 all remind us that new and deadly infections can transfer from animal reservoirs to humans. Even though infection with SARS-CoV-2 has a case fatality rate substantially lower than that seen in SARS, the 2020 pandemic has had an enormous negative impact on the global economy. Such transmission appears responsible for the Ebola virus epidemic in West Africa in 2014–2015. These are known as [zoonotic](#) infections—and we must be on the alert at all times for the emergence of new pathogens and new threats to health. The human immunodeficiency virus that causes AIDS (discussed in [Chapter 13](#)) serves as a warning that we remain constantly vulnerable.



Case Study: Acquired Immune Deficiency Syndrome (AIDS)

2-2 Epithelial surfaces of the body provide the first barrier against infection.

Our body surfaces are composed of epithelia with diverse functions. They can provide a protective physical barrier between the internal milieu and the external world that contains pathogens. Epithelia compose the skin and the linings of the body's tubular structures—the respiratory, urogenital, and gastrointestinal tracts. Epithelia in these locations are specialized for their particular functions and possess unique innate defense strategies against the microbes they typically encounter (**Fig. 2.5** and **Fig. 2.6**). The necessity to carry out various functions in these locations, such as absorption of nutrients or oxygen for example, means that these epithelia are not perfect barriers to infection.

	Skin	Gut	Lungs	Eyes/nose/oral cavity

	Stratified epithelium	Single cell layer of columnar epithelium	Upper airway: pseudostratified columnar epithelium Lower airway: single cell layer of columnar epithelium	Pseudostratified columnar epithelium
Mechanical	Epithelial cells joined by tight junctions			
	Longitudinal flow of air or fluid	Longitudinal flow of air or fluid	Movement of mucus by cilia	Tears Nasal cilia
Chemical	Fatty acids	Low pH	Pulmonary surfactant	Enzymes in tears and saliva (lysozyme)
		Enzymes (pepsin)		
	β -defensins Lamellar bodies Cathelicidin	α -defensins (cryptidins) RegIII (lecticidins) Cathelicidin	α -defensins Cathelicidin	Histatins β -defensins
Microbiological	Normal microbiota			

Fig. 2.5 Many barriers prevent pathogens from crossing epithelia and colonizing tissues. Surface epithelia provide mechanical, chemical, and microbiological barriers to infection. The epithelium of the skin is stratified, having multiple layers. Epithelia in the various locations in the intestine have a single cell layer. The epithelia of the upper airway have a single cell layer, but the cell nuclei are staggered in a way that suggests multiple layers.

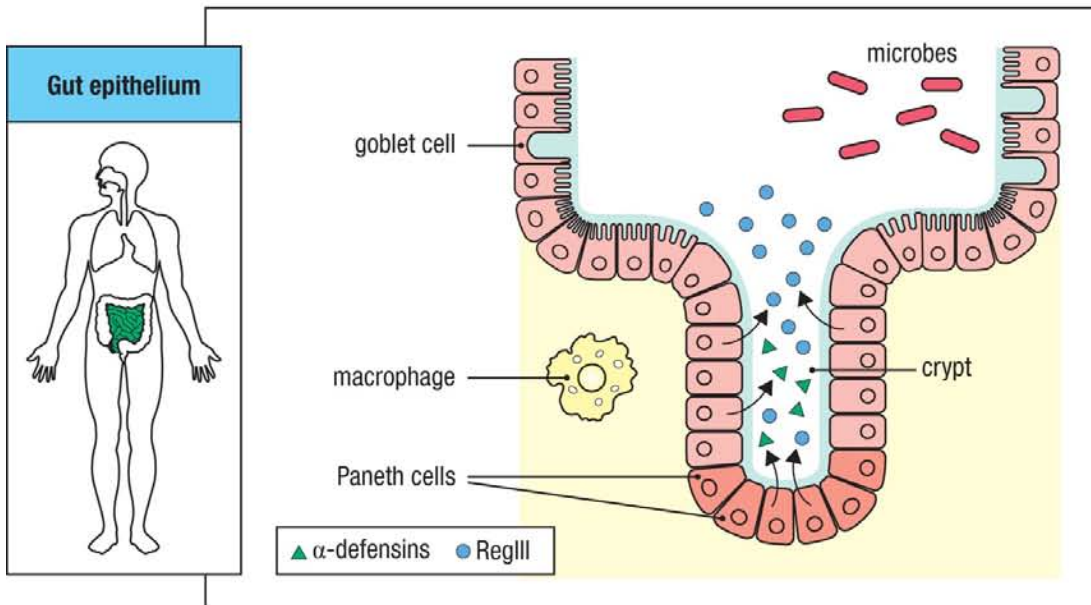
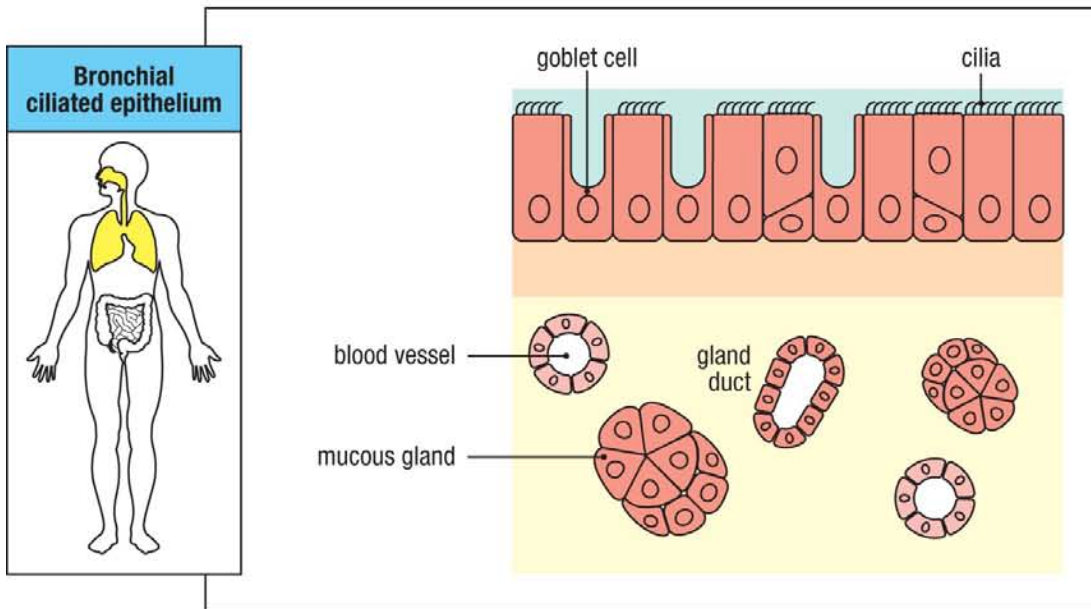
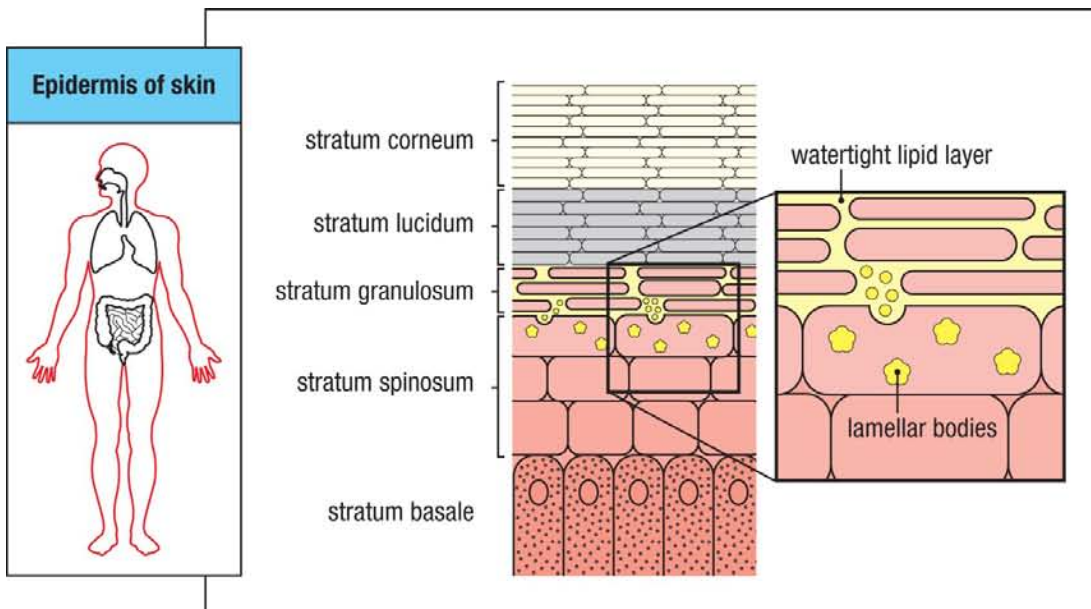


Fig. 2.6 Epithelia form specialized physical and chemical barriers that provide innate defenses in different locations. Top panel: The epidermis has multiple layers of keratinocytes in different stages of differentiation arising from the basal layer of stem cells. Differentiated keratinocytes in the stratum spinosum produce β -defensins and cathelicidins, which are incorporated into secretory organelles called lamellar bodies (yellow) and secreted into the intercellular space to form a waterproof lipid layer (the stratum corneum) containing antimicrobial activity. Center panel: In the lung, the airways are lined with ciliated epithelium. Beating of the cilia moves a continuous stream of mucus (green) secreted by goblet cells outward, trapping and ejecting potential pathogens. Type II pneumocytes in the lung alveoli (not shown) also produce and secrete antimicrobial defensins. Bottom panel: In the intestine, epithelial cells and Paneth cells—specialized cells deep in the epithelial crypts—produce several kinds of antimicrobial proteins: α -defensins (cryptidins) and the antimicrobial RegIII proteins.

Epithelia can be a layer of single cells or multiple layers of cells with varying geometry. Epithelial cells are held together by tight junctions, which effectively form a seal against the external environment. The body's internal epithelia are known as [mucosal epithelia](#) because they secrete a viscous fluid called [mucus](#), which contains many glycoproteins called [mucins](#). Mucus has a number of protective functions. The mucus layer establishes a viscous physical barrier that impedes microbial access to various epithelial surfaces. In the respiratory tract, microorganisms can be expelled in the outward flow of mucus driven by the beating of cilia on the mucosal epithelium (**Fig. 2.7**). The importance of mucus flow in clearing infection is illustrated by people with the inherited disease [cystic fibrosis](#), in which the mucus becomes abnormally thick and dehydrated because of defects in a gene, *CFTR*, encoding a chloride channel in the epithelium. Such individuals frequently develop lung infections caused by bacteria that colonize the epithelial surface but do not cross it (see Fig. 2.7). Mucus also plays an important role in the gut. Peristalsis is an important mechanism for keeping both food and infectious agents moving through the body. Failure of peristalsis is typically accompanied by the overgrowth of pathogenic bacteria within the lumen of the gut. The colon of the large intestine has two layers of mucus. The inner layer, nearest the epithelium, has a denser, less porous composition that excludes the passage of bacteria from the colonic lumen to the epithelial surface. An outer mucus layer, farther from the epithelium, is more porous, forming the habitat for colonic bacteria.

Ciliated respiratory epithelium is covered by a layer of mucus



Lung of a person with cystic fibrosis

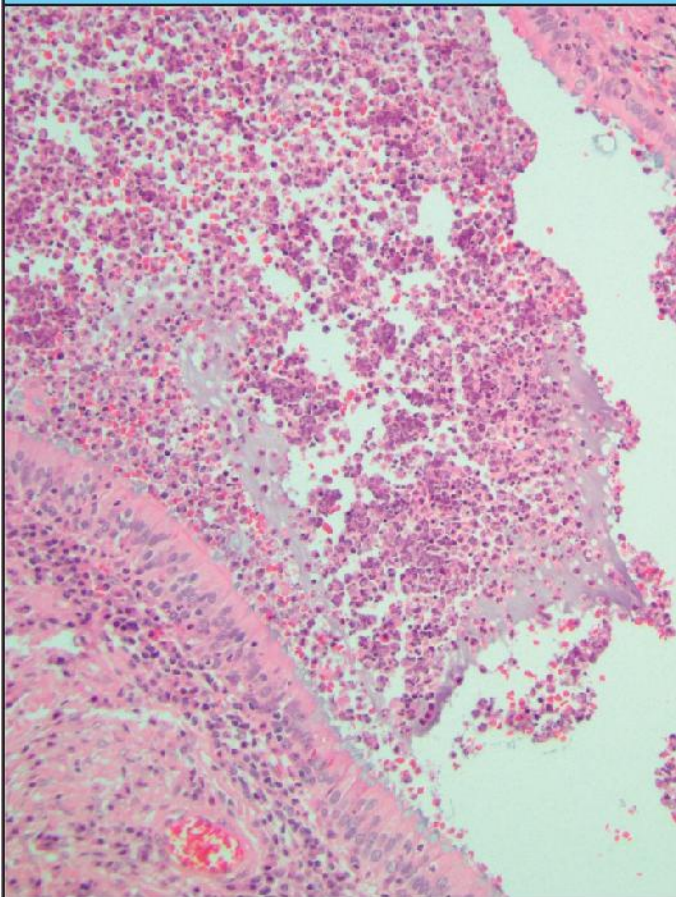


Fig. 2.7 Ciliated respiratory epithelium propels the overlying mucus layer for clearance of environmental microbes. Top panel: The ciliated respiratory epithelium in the airways of the lung is covered by a layer of mucus. The cilia propel the mucus outward and help prevent colonization of the airways by bacteria. Bottom panel: Section of a lung from an individual with cystic fibrosis. The dehydrated mucus layer impairs the ability of cilia to propel it, leading to frequent bacterial colonization and resulting inflammation of the airway.

Most healthy epithelial surfaces are associated with populations of normally nonpathogenic bacteria, known as [commensal bacteria](#) or the [microbiota](#), which help keep pathogens at bay. The microbiota can also make antimicrobial substances, such as the lactic acid produced by vaginal lactobacilli, some strains of which also produce antimicrobial peptides (bacteriocins). Commensal microorganisms in the outer mucus layer generate short-chain fatty acids (SCFAs) and other metabolites that can strengthen the barrier functions of epithelia by stimulating production of antimicrobial peptides (see [Section 2-4](#)). When commensal microorganisms are killed by antibiotic treatment, pathogens frequently replace them and cause disease (see Fig. 12.13). Under some circumstances commensal microbes themselves can cause disease if their growth is not kept in check or if the immune system is compromised. In [Chapter 12](#), we will discuss how commensal microorganisms play an important role in the setting of normal immunity, particularly in the intestine; and in [Chapter 15](#), we will see how these normally nonpathogenic organisms can cause disease in the context of inherited immunodeficiencies.

2-3 Infectious agents must overcome innate host defenses to establish a focus of infection.

Our bodies are constantly exposed to microorganisms present in our environment, including infectious agents that have been shed by other individuals. Contact with these microorganisms may occur through external or internal epithelial surfaces. To establish an infection, a microorganism must first invade the body by binding to or crossing an epithelium (**Fig. 2.8**). Infection is a major cause of mortality and morbidity in cases of epithelial damage caused by wounds, burns, or loss of the integrity of the body's internal epithelia. The body rapidly repairs damaged epithelial surfaces, but even without epithelial damage, pathogens may establish infection by specifically adhering to and colonizing epithelial surfaces, using the attachment to avoid being dislodged by the flow of air or fluid across the surface.

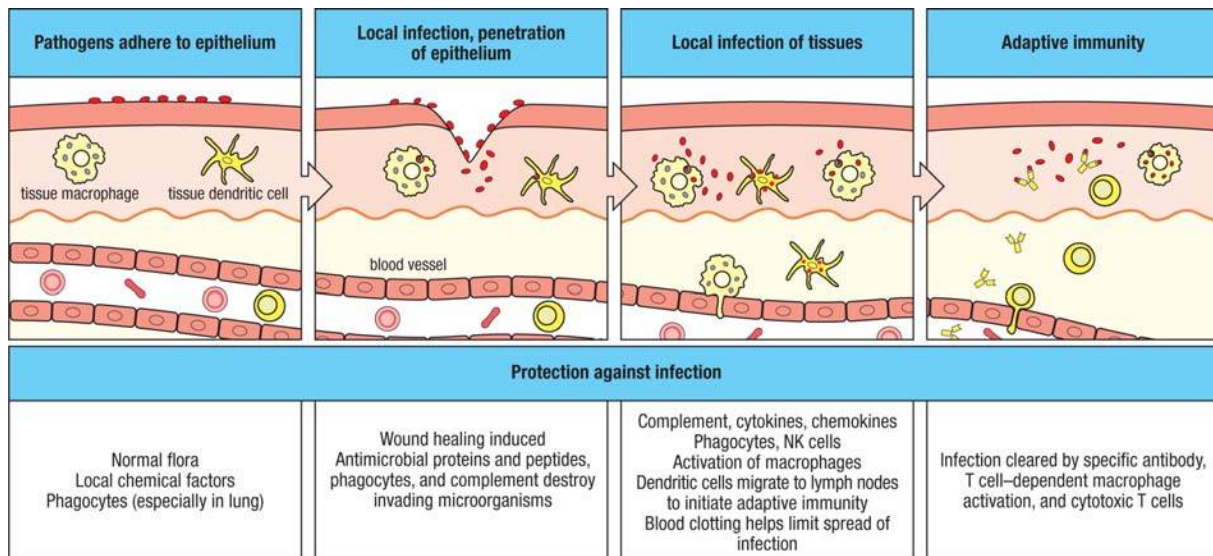


Fig. 2.8 An infection and the response to it can be divided into a series of stages.

These are illustrated here for an infectious microorganism entering through a wound in the skin. The infectious agent must first adhere to the epithelial cells and then cross the epithelium. A local immune response may prevent the infection from becoming established. If not, it helps to contain the infection and also delivers the infectious agent carried in lymph and inside dendritic cells, to local lymph nodes. This initiates the adaptive immune response and eventual clearance of the infection.

Disease occurs when a microorganism succeeds in evading or overwhelming innate host defenses to establish a local site of infection, and then replicates there to allow its further transmission within our bodies. The epithelium lining the respiratory tract provides a route of entry into tissues for airborne microorganisms, and the lining of the gastrointestinal tract does the same for microorganisms ingested in food and water. The intestinal pathogens *Salmonella enterica* ssp. *typhi*, which causes typhoid fever, and *Vibrio cholerae*, which causes cholera, are spread through fecally contaminated food and water, respectively. Insect bites and wounds allow microorganisms to penetrate the skin, and direct contact between individuals offers opportunities for infection through the skin, the gut, and the reproductive tract (see Fig. 2.2).

In spite of this exposure, infectious disease is fortunately quite infrequent. Most of the microorganisms that succeed in crossing an epithelial surface are efficiently removed by innate immune mechanisms that function in the underlying tissues, preventing infection from becoming established. It is difficult to know how many infections are repelled in this way, because they cause no symptoms and pass undetected.

In general, pathogenic microorganisms are distinguished from the mass of microorganisms in the environment by having special adaptations that evade the immune system. In some cases, such as the fungal disease athlete's foot, the initial infection remains local and does not cause significant pathology. In other cases, such as tetanus, the bacterium (*Clostridium tetani* in this case) secretes a powerful neurotoxin, and the infection causes serious illness as

it spreads through the lymphatics or the bloodstream, invades and destroys tissues, and disrupts the body's workings.

The spread of a pathogen is often initially countered by an [inflammatory response](#) that recruits more effector cells and molecules of the innate immune system out of the blood and into the tissues, while inducing clotting in small blood vessels further downstream so that the microbe cannot spread through the circulation (see Fig. 2.8). The cellular responses of innate immunity act over several days. During this time, the adaptive immune response may also begin if antigens derived from the pathogen are delivered to local lymphoid tissues by dendritic cells (see [Section 1-15](#)). While an innate immune response may eliminate some infections, an adaptive immune response can target particular strains and variants of pathogens and protect the host against reinfection by using either effector T cells or antibodies to generate immunological memory.

2-4 Epithelial cells and phagocytes produce several kinds of antimicrobial molecules.

Our surface epithelia are more than mere physical barriers to infection; they also produce a wide variety of chemical substances that are microbicidal or that inhibit microbial growth. For example, the acid pH of the stomach and the digestive enzymes, bile salts, fatty acids, and lysolipids present in the upper gastrointestinal tract create a substantial chemical barrier to infection (see Fig. 2.5). One important group of antimicrobial proteins comprises enzymes that attack chemical features specific to bacterial cell walls. Such antimicrobial enzymes include [lysozyme](#) and [secretory phospholipase A₂](#), which are secreted in tears and saliva and by phagocytes. Lysozyme is a glycosidase that breaks a specific chemical bond in the [peptidoglycan](#) layer of the bacterial cell wall. Peptidoglycan is composed of a network of polymers that consist of alternating residues of *N*-acetylglucosamine (GlcNAc) and *N*-acetylmuramic acid (MurNAc); the peptidoglycan layer is formed by peptide bridges that cross-link these polysaccharide polymers (**Fig. 2.9**). Lysozyme selectively cleaves the β -(1,4) linkage between the two sugars of the polymer and is more effective in acting against Gram-positive bacteria, in which the peptidoglycan cell wall is exposed, than against Gram-negative bacteria, which have an outer layer of LPS covering the peptidoglycan layer. Lysozyme is also produced by [Paneth cells](#), specialized epithelial cells in the base of the crypts in the small intestine that secrete many antimicrobial proteins into the gut (see Fig. 2.6). Paneth cells also produce secretory phospholipase A₂, a highly basic enzyme that can enter the bacterial cell wall to access and hydrolyze phospholipids in the cell membrane, killing the bacteria.

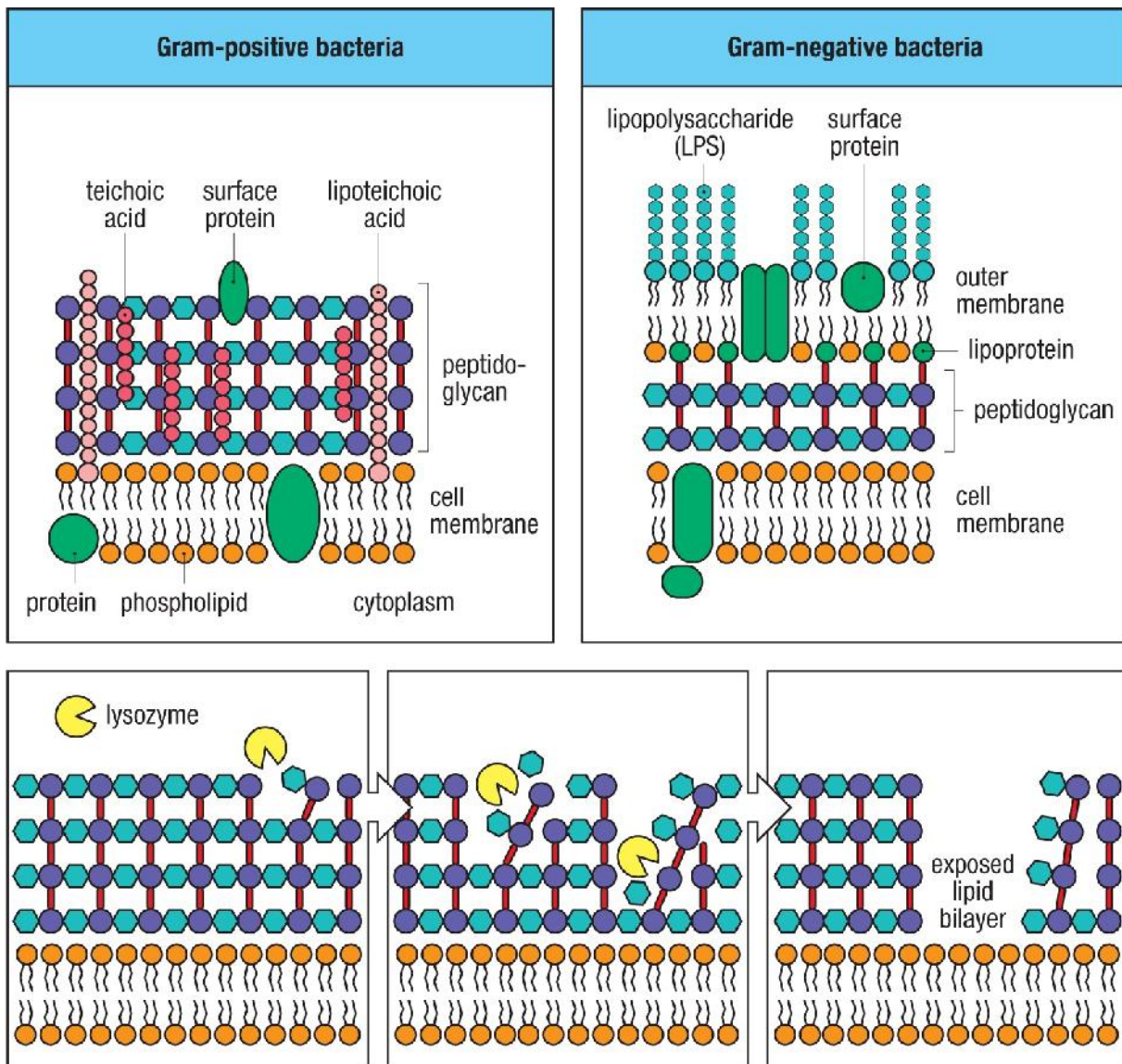
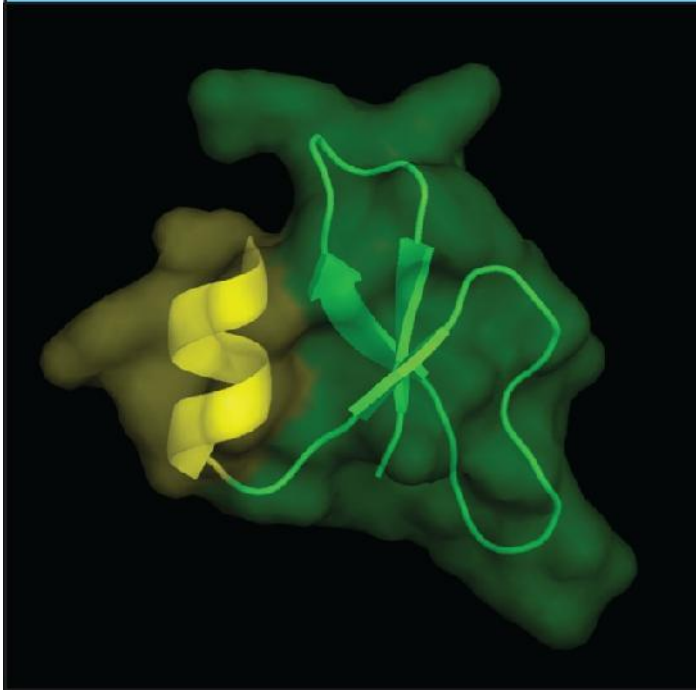


Fig. 2.9 Lysozyme digests the cell walls of Gram-positive and Gram-negative bacteria. Upper panels: The peptidoglycan layer of bacterial cell walls is a dense, three-dimensional network of polymers of alternating residues of β -(1,4)-linked *N*-acetylglucosamine (GlcNAc; large turquoise hexagons) and *N*-acetylmuramic acid (MurNAc; purple circles) that are cross-linked by peptide bridges (red bars). In Gram-positive bacteria (upper left panel), peptidoglycan forms the outer layer in which other molecules are embedded such as teichoic acid and the lipoteichoic acids that link the peptidoglycan layer to the bacterial cell membrane itself. In Gram-negative bacteria (upper right panel), a thin inner wall of peptidoglycan is covered by an outer lipid membrane that contains proteins and lipopolysaccharide (LPS). Lipopolysaccharide is composed of a lipid, lipid A (turquoise circles), to which is attached a polysaccharide core (small turquoise hexagons). Lysozyme (lower panels) cleaves β -(1,4) linkages between GlcNAc and MurNAc, creating a defect in the peptidoglycan layer and exposing the underlying cell membrane to other antimicrobial agents. Lysozyme is more effective against Gram-positive bacteria because of the relatively greater accessibility of the peptidoglycan.

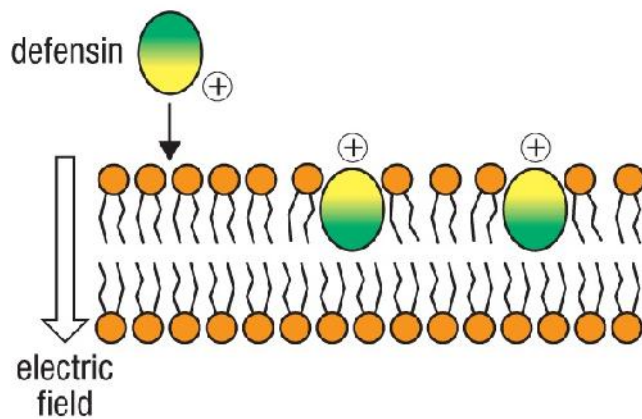
A second group of antimicrobial agents secreted by epithelial cells and phagocytes is the antimicrobial peptides. These represent one of the most ancient forms of defense against infection. These peptides can be produced by phagocytic immune cells, such as neutrophils, where they may be retained within phagocytic granules. They are also produced by epithelial cells, such as in the lung, where they can be secreted onto mucosal surfaces. Three important classes of antimicrobial peptides in mammals are [defensins](#), [cathelicidins](#), and [histatins](#).

Defensins are an ancient, evolutionarily conserved class of antimicrobial peptides made by many eukaryotic organisms, including mammals, insects, and plants (**Fig. 2.10**). They are short cationic peptides of around 30–40 amino acids that usually have three disulfide bonds stabilizing a common [amphipathic](#) structure—a positively charged region separated from a hydrophobic region. Defensins act within minutes to disrupt the cell membranes of bacteria and fungi, as well as the membrane envelopes of some viruses. The mechanism is thought to involve insertion of the hydrophobic region into the membrane bilayer and the formation of a pore that makes the membrane leaky (see Fig. 2.10). Most multicellular organisms make many different defensins—the plant *Arabidopsis thaliana* produces 13 and the fruit fly *Drosophila melanogaster* at least 15. Human Paneth cells make as many as 21 different defensins, many of which are encoded by a cluster of genes on chromosome 8.

Human β_1 -defensin



Electrostatic attraction and the transmembrane electric field bring the defensin into the lipid bilayer



Defensin peptides form a pore

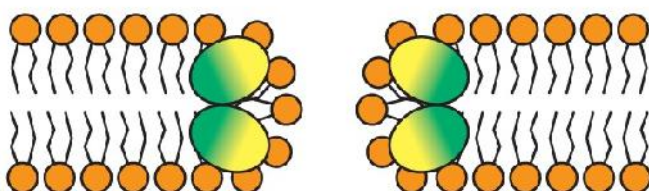
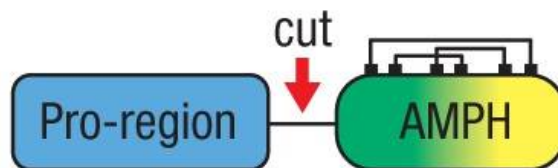


Fig. 2.10 Defensins are amphipathic peptides that disrupt the cell membranes of microbes. The structure of human β_1 -defensin is shown in the top panel. It is composed of a short segment of α helix (yellow) resting against three strands of antiparallel β sheet (green), generating an amphipathic peptide with charged and hydrophobic residues residing in separate regions. This general feature is shared by defensins from plants and insects and allows the defensins to interact with the charged surface of the cell membrane and become inserted in the lipid bilayer (center panel). Although the details are still unclear, a transition in the arrangement of the defensins in the membrane leads to the formation of pores and a loss of membrane integrity (bottom panel).

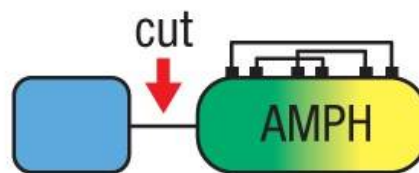
Three subfamilies of defensins— α -, β -, and θ -defensins—are distinguished on the basis of amino acid sequence. Each family has members with distinct activities, some being active against Gram-positive bacteria and some against Gram-negative bacteria, while others are specific for fungal pathogens. All the antimicrobial peptides, including the defensins, are generated by proteolytic processing from inactive [propeptides](#) (**Fig. 2.11**). In humans, developing neutrophils produce [\$\alpha\$ -defensins](#) through the processing of an initial propeptide of about 90 amino acids by cellular proteases to remove an anionic propiece, generating a mature cationic defensin that is stored in so-called [primary granules](#). The primary granules of neutrophils are specialized membrane-enclosed vesicles, rather similar to lysosomes, that contain a number of other antimicrobial agents as well as defensins. We will explain in [Chapter 3](#) how these primary granules are induced to fuse with phagocytic vesicles (phagosomes) after the cell has engulfed a pathogen, helping to kill the microbe. The Paneth cells of the gut constitutively produce α -defensins, called [cryptdins](#), which are processed by proteases such as the metalloprotease matrilysin in mice or trypsin in humans, before being secreted into the gut lumen. The [\$\beta\$ -defensins](#) lack the long propiece of α -defensins and are generally produced specifically in response to the presence of microbial products. β -Defensins (and some α -defensins) are made by epithelia outside the gut, primarily in the respiratory and urogenital tracts, skin, and tongue. β -Defensins made by keratinocytes in the epidermis and by type II pneumocytes in the lungs are packaged into [lamellar bodies](#) (see [Fig. 2.6](#)), lipid-rich secretory organelles that release their contents into the extracellular space to form a watertight lipid sheet in the epidermis and the pulmonary surfactant layer in the lung. The θ -defensins arose in the primates, but the single human θ -defensin gene has been inactivated by a mutation.

Defensins, cathelicidins, and histatins are activated by proteolysis to release an amphipathic antimicrobial peptide

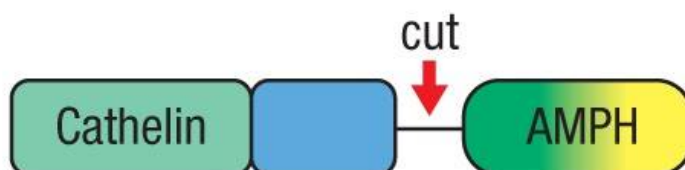
α -Defensins



β -Defensins



Cathelicidins



RegIII 'lecticidins'

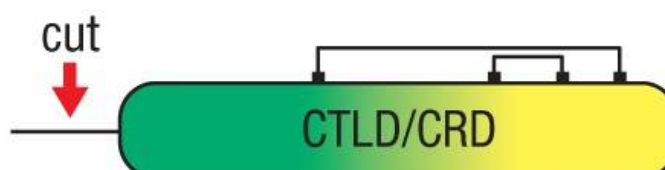


Fig. 2.11 Defensins, cathelicidins, and RegIII proteins are activated by proteolysis.

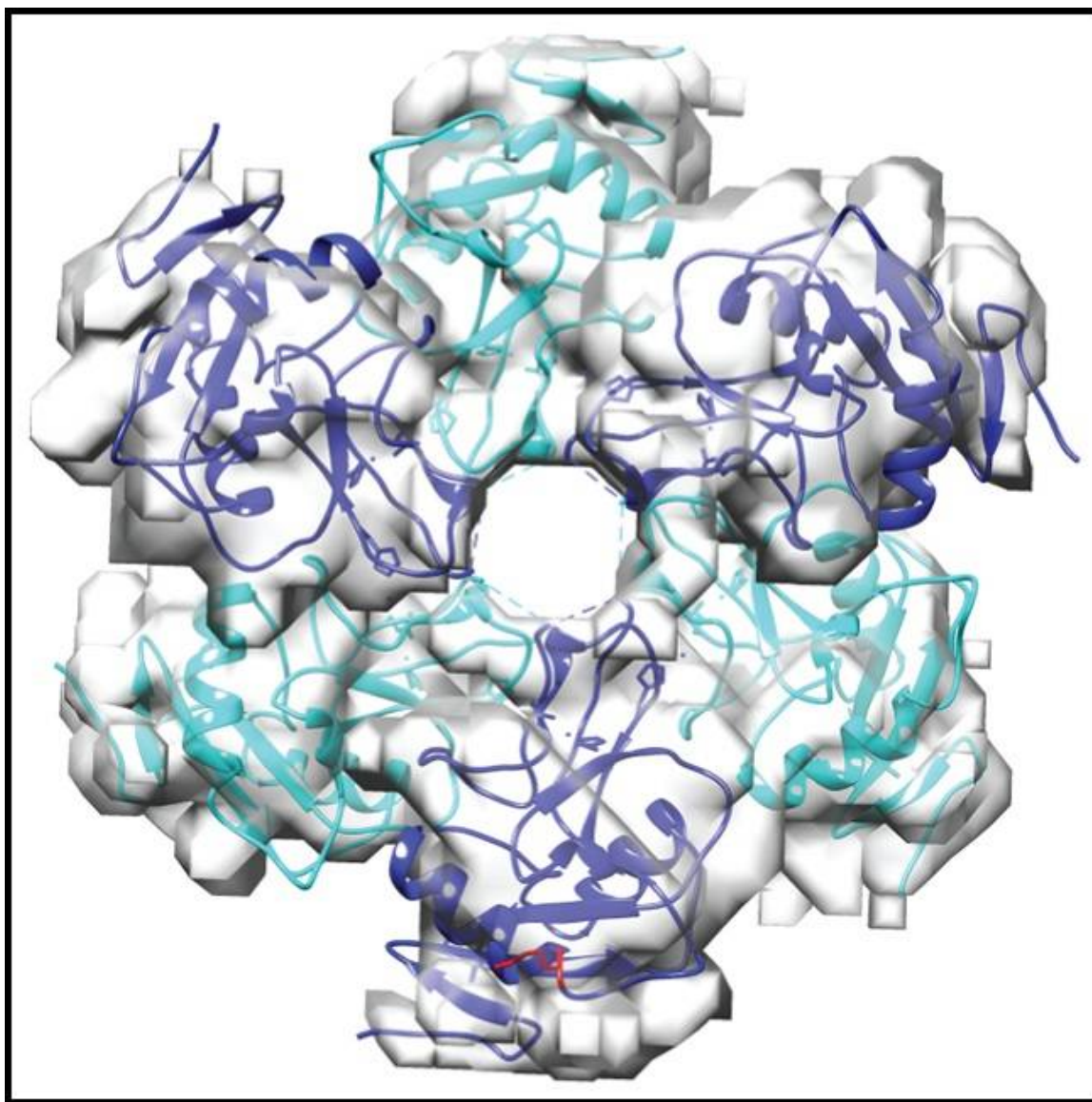
When α - and β -defensins are first synthesized, they contain a signal peptide (not shown); a pro-region (blue), which is shorter in the β -defensins; and an amphipathic domain (AMPH; green–yellow); the pro-region represses the membrane-inserting properties of the amphipathic domain. After defensins are released from the cell or into phagosomes, they undergo cleavage by proteases, which releases the amphipathic domain in active form. Newly synthesized cathelicidins contain a signal peptide (not shown), a cathelin domain, a short pro-region, and an amphipathic domain; they, too, are activated by proteolytic cleavage. RegIII proteins contain a C-type lectin domain (CTLN), also known as a carbohydrate-recognition domain (CRD). After release of the signal peptide, further proteolytic cleavage of a RegIII protein also regulates its antimicrobial activity.

The antimicrobial peptides belonging to the cathelicidin family lack the disulfide bonds that stabilize the defensins. Humans and mice have one cathelicidin gene, but some other mammals, including cattle and sheep, have several. Cathelicidins are made constitutively by neutrophils and macrophages, and in response to infection by keratinocytes in the skin and epithelial cells in the lungs and intestine. They are made as inactive propeptides composed of two linked domains and are processed before secretion (see Fig. 2.11). In neutrophils, the inactive cathelicidin propeptides are stored in another type of specialized cytoplasmic granules called [secondary granules](#). Cathelicidin is activated by proteolytic cleavage only when primary and secondary granules are induced to fuse with phagosomes, where it is cleaved by [neutrophil elastase](#) that has been stored in primary granules. Cleavage separates the two domains, and the cleavage products either remain in the phagosome or are released from the neutrophil by exocytosis. The carboxyl-terminal peptide is a cationic amphipathic peptide that disrupts membranes and is toxic to a wide range of microorganisms. The amino-terminal peptide is similar in structure to a protein called [cathelin](#), an inhibitor of cathepsin L (a lysosomal enzyme involved in antigen processing and protein degradation), but its role in immune defense is unclear. In keratinocytes, cathelicidins, like β -defensins, are stored and processed in the lamellar bodies.

A class of antimicrobial peptides called histatins are constitutively produced in the oral cavity by the parotid, sublingual, and submandibular glands. These short, histidine-rich, cationic peptides are active against pathogenic fungi such as *Cryptococcus neoformans* and *Candida albicans*. More recently, histatins were found to promote the rapid wound healing that is typical in the oral cavity, but the mechanism of this effect is unclear.

Another type of bactericidal proteins made by epithelia is carbohydrate-binding proteins, or [lectins](#). Most [C-type lectins](#) require calcium for the binding activity of their carbohydrate-recognition domain (CRD), which provides a variable interface for binding carbohydrate structures. C-type lectins of the RegIII family include several bactericidal proteins expressed by intestinal epithelium in humans and mice. However, carbohydrate recognition by RegIII proteins is not calcium dependent, but instead is inhibited by calcium. In mice, [RegIII \$\gamma\$](#) is

produced by Paneth cells and secreted into the gut, where it binds to peptidoglycans in bacterial cell walls and exerts direct bactericidal activity. Like other bactericidal peptides, RegIII γ is produced in inactive form but is cleaved by the protease trypsin, which removes a short amino-terminal fragment to activate the bactericidal potential of RegIII γ within the intestinal lumen (see Fig. 2.11). Human RegIII α (also called [HIP/PAP](#) for hepatocarcinoma intestine pancreas/pancreatitis-associated protein) kills bacteria directly by forming a hexameric pore in the bacterial membrane (**Fig. 2.12**). RegIII family proteins preferentially kill Gram-positive bacteria, in which the peptidoglycan is exposed on the outer surface (see Fig. 2.9). In fact, the LPS of Gram-negative bacteria inhibits the pore-forming ability of RegIII α , further enforcing the selectivity of RegIII proteins for Gram-positive bacteria.



a

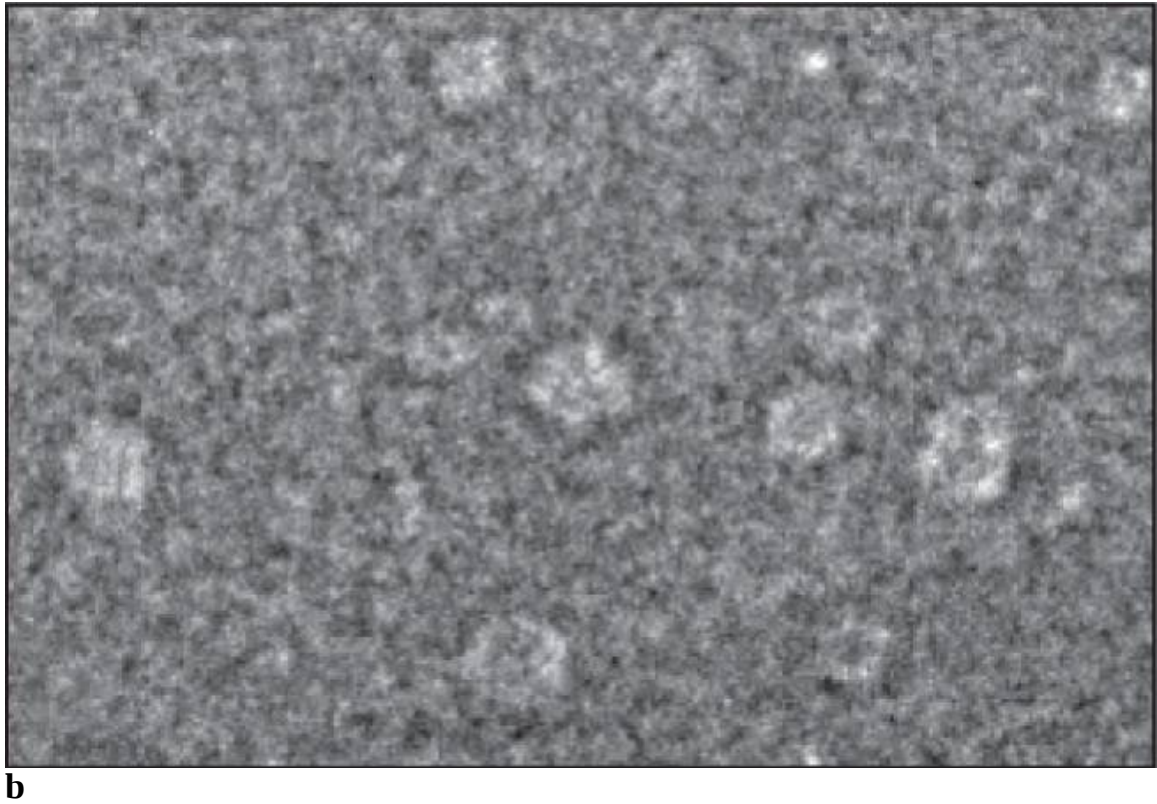


Fig. 2.12 Pore formation by human RegIII α . Panel a: A model of the RegIII α pore was generated by docking the human pro-RegIII α structure (PDB ID: 1UV0), shown as individual purple and turquoise ribbon diagrams, into the cryo-electron microscopic map of the RegIII α filament. LPS blocks the pore-forming activity of RegIII α , explaining its selective bactericidal activity against Gram-positive but not Gram-negative bacteria. Panel b: Electron microscopic images of RegIII α pores assembled in the presence of lipid bilayers.

Finally, we briefly mention the peptides S100A8 and S100A9, which heterodimerize to form [calprotectin](#), which we will discuss in [Section 11-11](#) in the context of production by T cells. Calprotectin acts to sequester magnesium and iron required by microorganisms and exerts an antimicrobial effect at sites of its production. This activity requires binding of calcium to one of calprotectin's two metal-binding sites, which allows its second site to bind other metal ions. Calprotectin is produced in high amounts by neutrophils, and by T cells, but it is also made by intestinal epithelia, and so is part of the innate barrier defense system.

Summary.

The mammalian immune response to invading organisms proceeds in three phases, beginning with immediate innate defenses, then the induced innate defenses, and finally adaptive immunity. The first phase of host defense consists of those mechanisms that are present and ready to resist an invader at any time. Epithelial surfaces provide a physical

barrier against pathogen entry, but they also have other, more specialized strategies. Mucosal surfaces have a protective barrier of mucus. Through particular cell-surface interactions, highly differentiated epithelia protect against both microbial colonization and invasion. Defense mechanisms of epithelia include the prevention of pathogen adherence, secretion of antimicrobial enzymes and bactericidal peptides, and the flow caused by the actions of cilia. Antimicrobial peptides and the bactericidal lectins of the RegIII family are made as inactive proproteins that require a proteolytic step to complete their activation, whereupon they become capable of killing microbes by forming pores in the microbial cell membranes. The actions of antimicrobial enzymes and peptides described in this part of the chapter often involve binding to unique glycan/carbohydrate structures on the microbe. Thus, these soluble molecular defenses are both pattern-recognition receptors and effector molecules at the same time, representing the simplest form of innate immunity.

Glossary

[pathogenic microorganism](#)

Microorganism that typically causes disease when it infects a host.

[pathogen](#)

Microorganism that typically causes disease when it infects a host.

[pathogenesis](#)

The origin or cause of the pathology of a disease.

[immune evasion](#)

Mechanisms used by pathogens to avoid detection and/or elimination by host immune defenses.

[exotoxin](#)

A protein toxin produced and secreted by a bacterium.

[endotoxins](#)

Toxins derived from bacterial cell walls released by damaged cells. They can potentially induce cytokine synthesis and in large amounts can cause a systemic reaction called septic shock or endotoxic shock.

[lipopolysaccharide \(LPS\)](#)

The surface lipopolysaccharide of Gram-negative bacteria, which stimulates TLR-4 on macrophages and dendritic cells.

[Gram-negative bacteria](#)

Bacteria that fail to retain crystal violet stain after alcohol wash because of a thin peptidoglycan layer.

[zoonotic](#)

Describes a disease of animals that can be transmitted to humans.

[mucosal epithelia](#)

Mucus-coated epithelia lining the body's internal cavities that connect with the outside (such as the gut, airways, and vaginal tract).

[mucus](#)

Sticky solution of proteins (mucins) secreted by goblet cells of internal epithelia,

forming a protective layer on the epithelial surface.

[mucins](#)

Highly glycosylated cell-surface proteins. Mucin-like molecules are bound by L-selectin in lymphocyte homing.

[cystic fibrosis](#)

Disease caused by defect in the CFTR gene, leading to abnormally thick mucus and causing serious recurrent infections of the lung.

[commensal bacteria](#)

Microorganisms (predominantly bacteria) that normally live harmlessly in symbiosis with their host (e.g., the gut bacteria in humans and other animals). Many commensals confer a positive benefit on their host in some way.

[microbiota](#)

Microorganisms (predominantly bacteria) that normally live harmlessly in symbiosis with their host (e.g., the gut bacteria in humans and other animals). Many commensals confer a positive benefit on their host in some way.

[inflammatory response](#)

General term for the local accumulation of fluid, plasma proteins, and white blood cells that is initiated by physical injury, infection, or a local immune response.

[lysozyme](#)

Antimicrobial enzyme that degrades bacterial cell walls.

[peptidoglycan](#)

A component of bacterial cell walls that is recognized by certain receptors of the innate immune system.

[Paneth cells](#)

Specialized epithelial cells at the base of the crypts in the small intestine that secrete antimicrobial peptides.

[defensins](#)

See **α -defensins**, **β -defensins**.

[cathelicidins](#)

Family of antimicrobial peptides that in humans has one member.

[histatins](#)

Antimicrobial peptides constitutively produced by the parotid, sublingual, and submandibular glands in the oral cavity. Active against pathogenic fungi such as *Cryptococcus neoformans* and *Candida albicans*.

[amphipathic](#)

Describes molecules that have a positively charged (or hydrophilic) region separated from a hydrophobic region.

[propeptide](#)

Inactive precursor form of a polypeptide or peptide, which requires proteolytic processing to produce the active peptide.

[\$\alpha\$ -defensins](#)

A class of antimicrobial peptides produced by neutrophils and the Paneth cells of the intestine.

[primary granules](#)

Granules in neutrophils that correspond to lysosomes and contain antimicrobial peptides such as defensins and other antimicrobial agents.

[cryptdins](#)

α -Defensins (antimicrobial peptides) made by the Paneth cells of the small intestine.

[\$\beta\$ -defensins](#)

Antimicrobial peptides made by virtually all multicellular organisms. In mammals they are produced by the epithelia of the respiratory and urogenital tracts, skin, and tongue.

[lamellar bodies](#)

Lipid-rich secretory organelles in keratinocytes and lung pneumocytes that release β -defensins into the extracellular space.

[secondary granules](#)

Type of granule in neutrophils that stores certain antimicrobial peptides.

[neutrophil elastase](#)

Proteolytic enzyme stored in the granules of neutrophils that is involved in the processing of antimicrobial peptides.

[cathelin](#)

A cathepsin L inhibitor.

[lectin](#)

A carbohydrate-binding protein.

[C-type lectins](#)

Large class of carbohydrate-binding proteins that require Ca^{2+} for binding, including many that function in innate immunity.

[HIP/PAP](#)

An antimicrobial C-type lectin secreted by intestinal cells in humans. Also known as RegIII α .

[calprotectin](#)

A complex of heterodimers of the antimicrobial peptides S100A8 and S100A9, which sequester zinc and manganese from microbes. Produced in abundance by neutrophils and in lesser amounts by macrophages and epithelial cells.

[secretory phospholipase A₂](#)

Antimicrobial enzyme present in tears and saliva and also secreted by the Paneth cells of the gut.

[RegIII \$\gamma\$](#)

An antimicrobial protein of the C-type lectin family, produced by Paneth cells in the gut in mice.

The complement system and innate immunity.

When a pathogen breaches the host's epithelial barriers and initial antimicrobial defenses, it next encounters a major component of innate immunity known as the complement system, or [complement](#). Complement is a collection of soluble proteins present in blood and other body fluids. It was discovered in the 1890s by Jules Bordet as a heat-labile substance in normal plasma whose activity could 'complement' the bactericidal activity of immune sera. Part of the process is [opsonization](#), which refers to coating a pathogen with antibodies and/or complement proteins so that it can be more readily taken up and destroyed by phagocytic cells. Although complement was first discovered as an effector arm of the antibody response, we now understand that it originally evolved as part of the innate immune system and that it still provides protection early in infection, in the absence of antibodies, through more ancient pathways of complement activation.

The complement system is composed of more than 30 different plasma proteins, which are produced mainly by the liver. In the absence of infection, these proteins circulate in an inactive form. In the presence of pathogens or of antibody bound to pathogens, the complement system becomes 'activated.' Particular [complement proteins](#) interact with each other to form several different pathways of complement activation, all of which have the final outcome of killing the pathogen, either directly or by facilitating its phagocytosis, and inducing inflammatory responses that help to fight infection. There are three pathways of [complement activation](#). As the antibody-triggered pathway of complement activation was discovered first, this became known as the classical pathway of complement activation. The next to be discovered, the so-called alternative pathway, can be activated by the presence of the pathogen alone; and the most recently discovered is the lectin pathway, which is activated by lectin-type proteins that recognize and bind to carbohydrates on pathogen surfaces.

We learned in [Section 2-4](#) that proteolysis can be used as a means of activating antimicrobial proteins. In the complement system, activation by proteolysis is inherent, with many of the complement proteins being proteases that successively cleave and activate one another. The proteases of the complement system are synthesized as inactive pro-enzymes, or [zymogens](#), which become enzymatically active only after proteolytic cleavage, usually by another complement protein. The complement pathways are triggered by proteins that act as pattern-recognition receptors to detect the presence of pathogens. This detection activates an initial zymogen, triggering a cascade of proteolysis in which complement zymogens are activated sequentially, each becoming an active protease that cleaves and activates many molecules of the next zymogen in the pathway, amplifying the signal as the cascade proceeds. This results in activation of three distinct effector pathways—[inflammation](#), [phagocytosis](#), and [membrane attack](#)—that help eliminate the pathogen. In this way, the detection of even a

small number of pathogens produces a rapid response that is greatly amplified at each step. This overall scheme for complement is shown in **Fig. 2.13**.

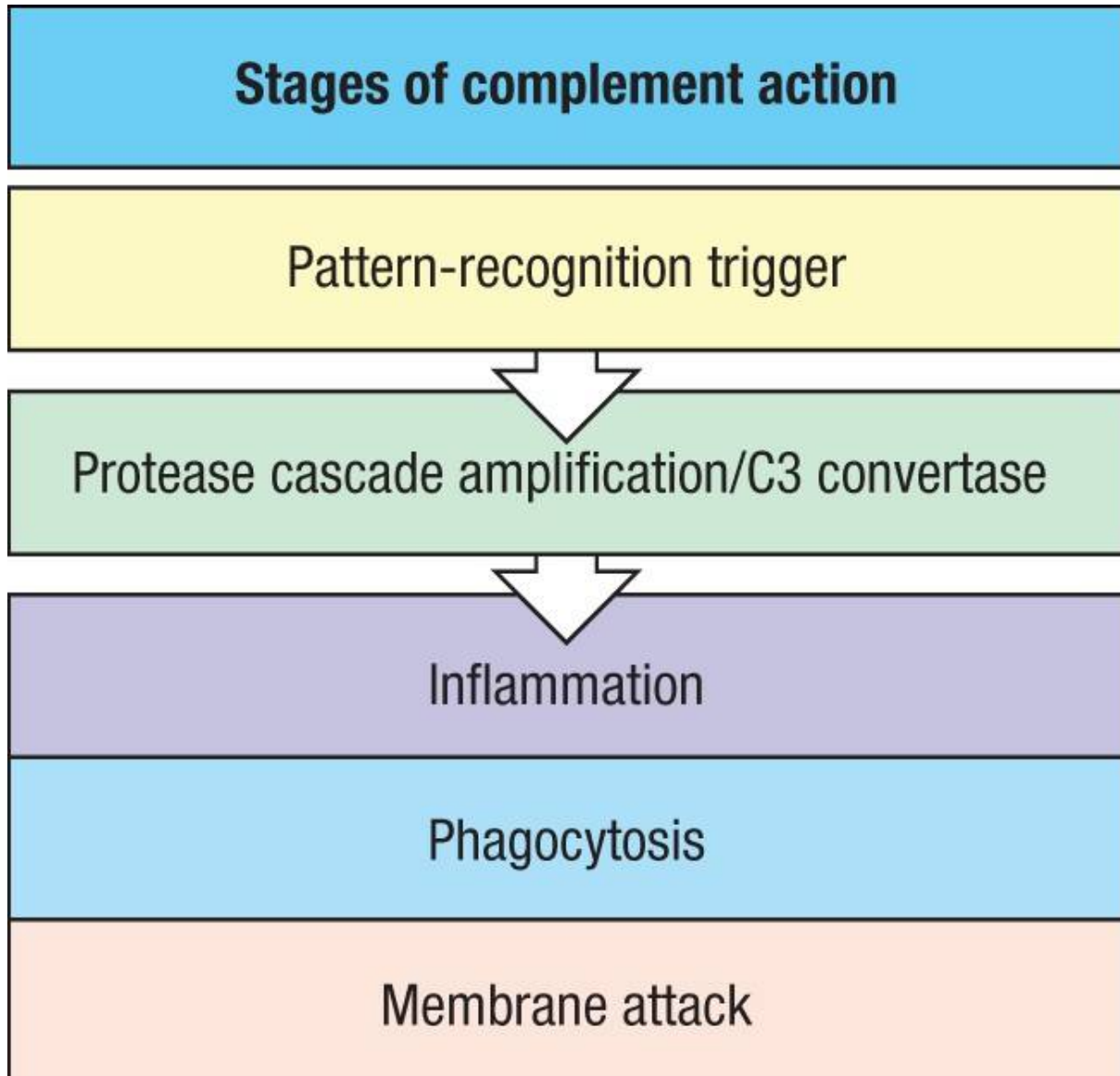


Fig. 2.13 The complement system proceeds in distinct phases in the elimination of microbes. Proteins that can distinguish self from microbial surfaces (yellow) activate a proteolytic amplification cascade that ends in the formation of the critical enzymatic activity (green) of C3 convertase, a family of proteases. This activity is the gateway to the three effector arms of complement, which produce inflammation (purple), enhance phagocytosis of the microbe (blue), and lyse microbial membranes (pink). We will use this color scheme in the figures throughout this chapter to illustrate which activity each complement protein serves.

Nomenclature for complement proteins can seem confusing, so we will start by explaining their names. The first proteins discovered belong to the classical pathway, and they are designated by the letter C followed by a number. The native complement proteins—such as the inactive zymogens—have a simple number designation; for example, C1 and C2.

Unfortunately, they were named in the order of their discovery rather than the sequence of reactions. The reaction sequence in the classical pathway, for example, is C1, C4, C2, C3, C5, C6, C7, C8, and C9 (note that not all of these are proteases). Products of cleavage reactions are designated by adding a lowercase letter as a suffix. For example, cleavage of C3 produces a small protein fragment called C3a and the remaining larger fragment, C3b. By convention, the larger fragment for other factors is designated by the suffix b. There is an exception to this convention. For [C2](#), the larger fragment was originally named [C2a](#) in the literature by its discoverers. Although there have been discussions regarding changes to this nomenclature, there has not been uniform agreement and so we will preserve the original usage of C2a as the large active fragment of C2. Another exception is the naming of C1q, C1r, and C1s: these are not cleavage products of C1 but are distinct proteins that together compose C1. The proteins of the alternative pathway were discovered later and are designated by different capital letters; for example, factor B and factor D. The factor B cleavage products are also designated by the addition of lowercase a and b; thus, the large fragment of B is called Bb and the small fragment is called Ba. Activated complement components were often designated in the past by a horizontal line, for example, C2a, but we will not use this convention. All the components of the complement system are listed in **Fig. 2.14**.

Functional protein classes in the complement system	
Binding to antigen:antibody complexes and pathogen surfaces	C1q
Binding to carbohydrate structures such as mannose or GlcNAc on microbial surfaces	MBL Ficolins Properdin (P)
Activating enzymes*	C1r C1s C2a Bb Factor D MASP-1 MASP-2

	MASP-3
Surface-binding proteins and opsonins	C4b C3b
Peptide mediators of inflammation	C5a C3a C4a
Membrane-attack proteins	C5b C6 C7 C8 C9
Complement receptors	CR1 CR2 CR3 CR4 CRIg
Complement-regulatory proteins	C1INH C4BP CR1/CD35 MCP/CD46 DAF/CD55

	Factor H
	Factor I
	Properdin (P)
	CD59

Fig. 2.14 Functional protein classes in the complement system. *In this book, C2a is used to denote the larger, active fragment of C2.

Besides acting in innate immunity, complement also influences adaptive immunity. Opsonization of pathogens by complement facilitates their uptake by phagocytic antigen-presenting cells that express complement receptors; this enhances the presentation of pathogen antigens to T cells, which we discuss in more detail in [Chapter 6](#). B cells express receptors for complement proteins that enhance their responses to complement-coated antigens, as we describe later in [Chapter 10](#). In addition, several of the complement fragments can act to influence cytokine production by antigen-presenting cells, thereby influencing the direction and extent of the subsequent adaptive immune response, as we describe in [Chapter 11](#).

2-5 The complement system recognizes features of microbial surfaces and marks them for destruction by coating them with C3b.

Figure 2.15 gives a highly simplified preview of the initiation mechanisms and outcomes of complement activation. The three pathways of complement activation are initiated in different ways. The [lectin pathway](#) is initiated by soluble carbohydrate-binding proteins—mannose-binding lectin (MBL) and ficolins—that bind to particular carbohydrate structures on microbial surfaces. Specific proteases, called MBL-associated serine proteases (MASPs), that associate with these recognition proteins trigger the cleavage of complement proteins and activation of the pathway. The [classical pathway](#) is initiated when complement component 1 (C1), which comprises a recognition protein (C1q) associated with proteases (C1r and C1s), either recognizes a microbial surface directly or binds to antibodies already bound to a pathogen. Finally, the [alternative pathway](#) can be initiated by spontaneous hydrolysis and activation of complement component 3 (C3), which can then bind directly to microbial surfaces.

These three pathways converge at the central and most important step in complement activation. When any of the pathways interacts with a pathogen surface, the enzymatic activity of a [C3 convertase](#) is generated. There are various types of C3 convertase, depending

on the complement pathway activated, but each is a multi-subunit protein with protease activity that cleaves C3. The C3 convertase is bound covalently to the pathogen surface, where it cleaves C3 to generate large amounts of [C3b](#), the main effector molecule of the complement system, and [C3a](#), a small peptide that binds to specific receptors and helps induce inflammation. Cleavage of C3 is the critical step in complement activation and leads directly or indirectly to all the effector activities of the complement system (see Fig. 2.15). C3b binds covalently to the microbial surface and acts as an opsonin, enabling phagocytes that carry receptors for complement to take up and destroy the C3b-coated microbe. Later in the chapter, we will describe the different complement receptors that bind C3b and are involved in this function of complement and explain how C3b is degraded by a serum protease into inactive smaller fragments called [C3f](#) and [C3dg](#) and C3d. C3b can also bind to the C3 convertases produced by the classical and lectin pathways and form another multi-subunit enzyme, the [C5 convertase](#), which cleaves C5, liberating the highly inflammatory peptide [C5a](#) and generating [C5b](#). C5b initiates the 'late' events of complement activation, also called the terminal pathway, in which additional complement proteins interact with C5b to form a [membrane-attack complex \(MAC\)](#) on the pathogen surface, creating a pore in the cell membrane that leads to cell lysis (see Fig. 2.15, bottom right).

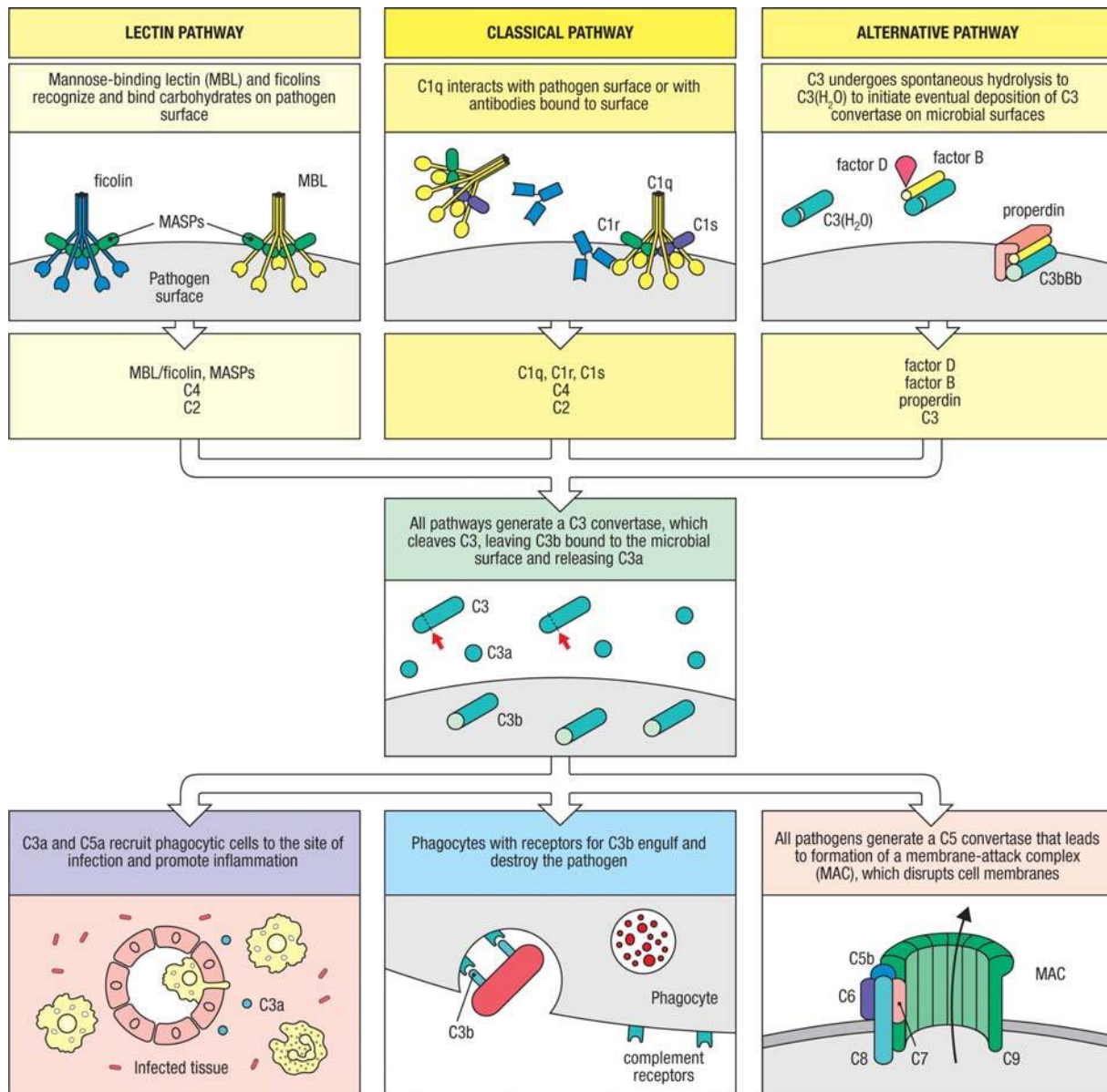


Fig. 2.15 Complement is a system of soluble pattern-recognition receptors and effector molecules that detect and destroy microorganisms. The pathogen-recognition mechanisms of the three complement-activation pathways are shown in the top row, along with the complement components used in the proteolytic cascades leading to formation of a C3 convertase. This enzyme activity cleaves complement component 3 (C3) into a small soluble protein, C3a, and a larger component, C3b, which becomes covalently bound to the pathogen surface (middle row). The components are listed by biochemical function in Fig. 2.14 and are described in detail in later figures. The lectin pathway of complement activation (top left) is triggered by the binding of mannose-binding lectin (MBL) or ficolins to carbohydrate residues in microbial cell walls and capsules. The classical pathway (top center) is triggered by binding of C1 either to the pathogen surface or to antibody bound to the pathogen. In the alternative pathway (top right), soluble C3 undergoes spontaneous hydrolysis in the fluid phase, generating C3(H₂O), which, in concert with factors B, D, and properdin, propagates the cascade. All pathways thus converge on the formation of C3b bound to a

pathogen and lead to all of the effector activities of complement, which are shown in the bottom row. C3b bound to a pathogen acts as an opsonin, enabling phagocytes that express receptors for C3b to ingest the complement-coated microbe more easily (bottom center). C3b can also bind to C3 convertases to produce another enzyme, a C5 convertase (detail not shown here), which cleaves C5 to C5a and C5b. C5b triggers the late events of the complement pathway in which the terminal components of complement—C6 to C9—assemble into a membrane-attack complex (MAC) that can damage the membrane of certain pathogens (bottom right). C3a and C5a act as chemoattractants that recruit immune-system cells to the site of infection and cause inflammation (bottom left).

The key feature of C3b is its ability to form a covalent bond with microbial surfaces, which allows the innate recognition of microbes to be translated into effector responses. Covalent bond formation is due to a highly reactive thioester bond that is hidden inside the folded C3 protein and cannot react until C3 is cleaved. When C3 convertase cleaves C3 and releases the C3a fragment, large conformational changes occur in C3b that allow the thioester bond to react with a hydroxyl or amino group on the nearby microbial surface (**Fig. 2.16**). If no bond is made, the thioester is rapidly hydrolyzed, inactivating C3b, which is one way that alternative pathway activity is confined to a target area in healthy individuals. As we will see below, some of the individual components of C3 and C5 convertases differ between the various complement pathways; the components that are different are listed in **Fig. 2.17**.

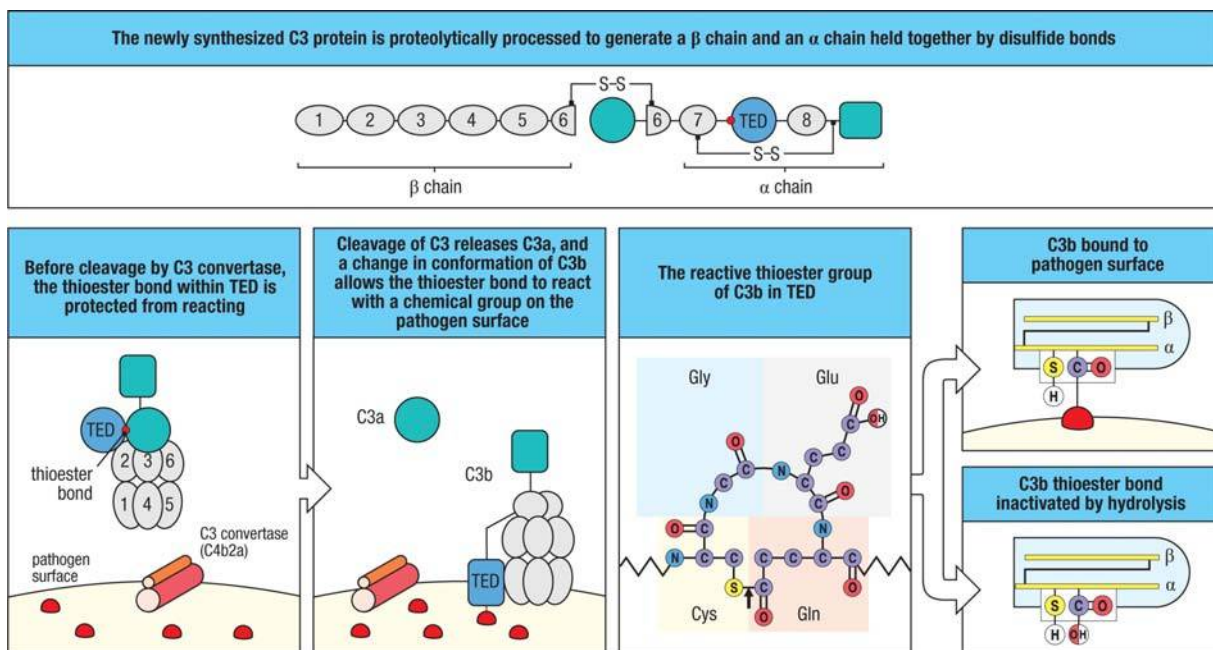


Fig. 2.16 C3 convertase activates C3 for covalent bonding to microbial surfaces by cleaving it into C3a and C3b and exposing a highly reactive thioester bond in C3b. Top panel: C3 in blood plasma consists of an α chain and a β chain (formed by proteolytic processing from the native C3 polypeptide) held together by a disulfide bond. The thioester-containing domain (TED) of the α chain contains a potentially highly reactive thioester bond (red spot). Bottom left panels: Cleavage by C3 convertase

(the lectin and classical pathway convertase C4b2a is shown here) and release of C3a from the amino terminus of the α chain causes a conformational change in C3b that exposes the thioester bond. This can now react with hydroxyl or amino groups on molecules on microbial surfaces, covalently bonding C3b to the surface. Bottom right panels: Schematic view of the thioester reaction. If a bond is not made with a microbial surface, the thioester is rapidly hydrolyzed (that is, cleaved by water), rendering C3b inactive.

C3 convertase	
Lectin pathway	C4b2a
Classical pathway	C4b2a
Alternative pathway	C3bBb
Fluid phase	C3(H ₂ O)Bb
C5 convertase	
Lectin pathway	C4b2a3b
Classical pathway	C4b2a3b
Alternative pathway	C3b ₂ Bb

Fig. 2.17 C3 and C5 convertases of the complement pathways. Note the C5 convertase of the alternative pathway consists of two C3b subunits and one Bb subunit.

Pathways leading to such potent inflammatory and destructive effects—and which have a series of built-in amplification steps—are potentially dangerous and must be tightly regulated. One important safeguard is that the key activated complement components are rapidly inactivated unless they bind to the pathogen surface on which their activation was initiated. There are also several points in the pathway at which regulatory proteins act to prevent the activation of complement on the surfaces of healthy host cells, thereby protecting them from accidental damage, as we shall see later in the chapter. Complement can, however, be activated by dying cells, such as those at sites of ischemic injury, and by cells undergoing apoptosis, or programmed cell death. In these cases, the complement coating helps phagocytes dispose of the dead and dying cells neatly, thus limiting the risk of cell contents being released and triggering an autoimmune response (discussed in [Chapter 15](#)).

Having introduced some of the main complement components, we are ready for a more detailed account of the three pathways. To help indicate the types of functions carried out by each of the complement components in the tables throughout the rest of the chapter, we will

use the color code introduced in Fig. 2.13 and Fig. 2.14: yellow for recognition and activation, green for amplification, purple for inflammation, blue for phagocytosis, and pink for membrane attack.

2-6 The lectin pathway uses soluble receptors that recognize microbial surfaces to activate the complement cascade.

Microorganisms typically bear on their surface repeating patterns of molecular structures, known generally as pathogen-associated molecular patterns (PAMPs). The cell walls of Gram-positive and Gram-negative bacteria, for example, are composed of a matrix of proteins, carbohydrates, and lipids in a repetitive array (see Fig. 2.9). The lipoteichoic acids of Gram-positive bacterial cell walls and the lipopolysaccharide of the outer membrane of Gram-negative bacteria are not present on animal cells and are important in the recognition of bacteria by the innate immune system. Similarly, the glycans of yeast surface proteins commonly terminate in mannose residues rather than the sialic acid residues (*N*-acetylneuraminic acid) that terminate the glycans of vertebrate cells (**Fig. 2.18**). The lectin pathway uses these features of microbial surfaces to detect and respond to pathogens.

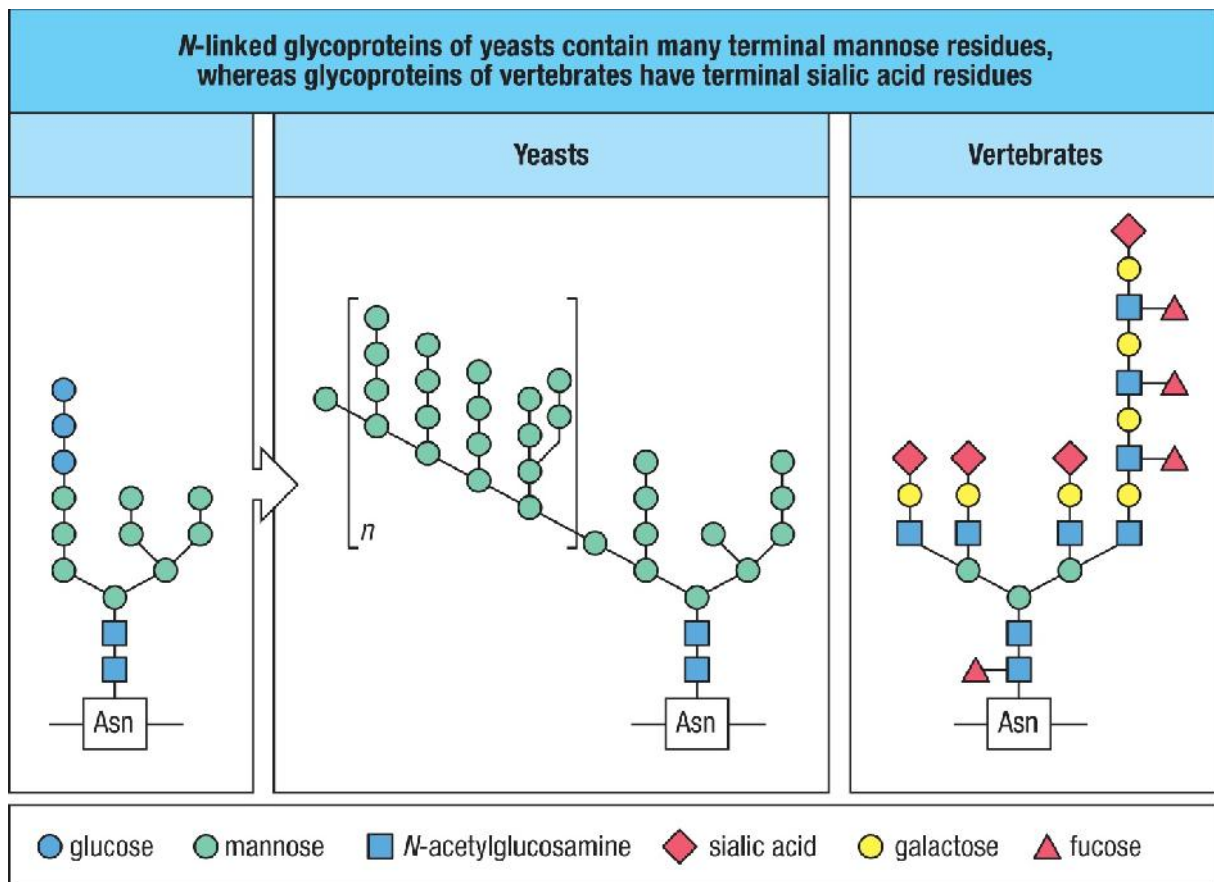


Fig. 2.18 The carbohydrate side chains on yeast and vertebrate glycoproteins are terminated with different patterns of sugars. *N*-linked glycosylation in fungi and

animals is initiated by the addition of the same precursor oligosaccharide, Glc₃-Man₉-GlcNAc₂ (left panel), to an asparagine residue. In many yeasts this is processed to high-mannose glycans (middle panel). In contrast, in vertebrates, the initial glycan is trimmed and processed, and the *N*-linked glycoproteins have terminal sialic acid residues (right panel).

The lectin pathway can be triggered by any of four different pattern-recognition receptors that circulate in blood and extracellular fluids and recognize carbohydrates on microbial surfaces. The first such receptor to be discovered was [mannose-binding lectin \(MBL\)](#), which is shown in **Fig. 2.19**, and which is synthesized in the liver. MBL is an oligomeric protein built up from a monomer that contains an amino-terminal collagen-like domain and a carboxyl-terminal C-type lectin domain (see [Section 2-4](#)). Proteins of this type are called [collectins](#). MBL monomers assemble into trimers through the formation of a triple helix by their collagen-like domains. Trimers then assemble into oligomers by disulfide bonding between the cysteine-rich collagen domains. The MBL present in the blood is composed of two to six trimers, with the major forms of human MBL being trimers and tetramers. A single carbohydrate-recognition domain of MBL has a low affinity for mannose, fucose, and *N*-acetylglucosamine (GlcNAc) residues, which are common on microbial glycans, but does not bind sialic acid residues, which terminate vertebrate glycans. Thus, multimeric MBL has high total binding strength, or [avidity](#), for repetitive carbohydrate structures on a wide variety of microbial surfaces, including Gram-positive and Gram-negative bacteria, mycobacteria, yeasts, and some viruses and parasites, while not interacting with host cells. MBL is present at low concentrations in the plasma of most individuals, but in the presence of infection, its production is increased during the [acute-phase response](#). This is part of the induced phase of the innate immune response and is discussed in [Chapter 3](#).

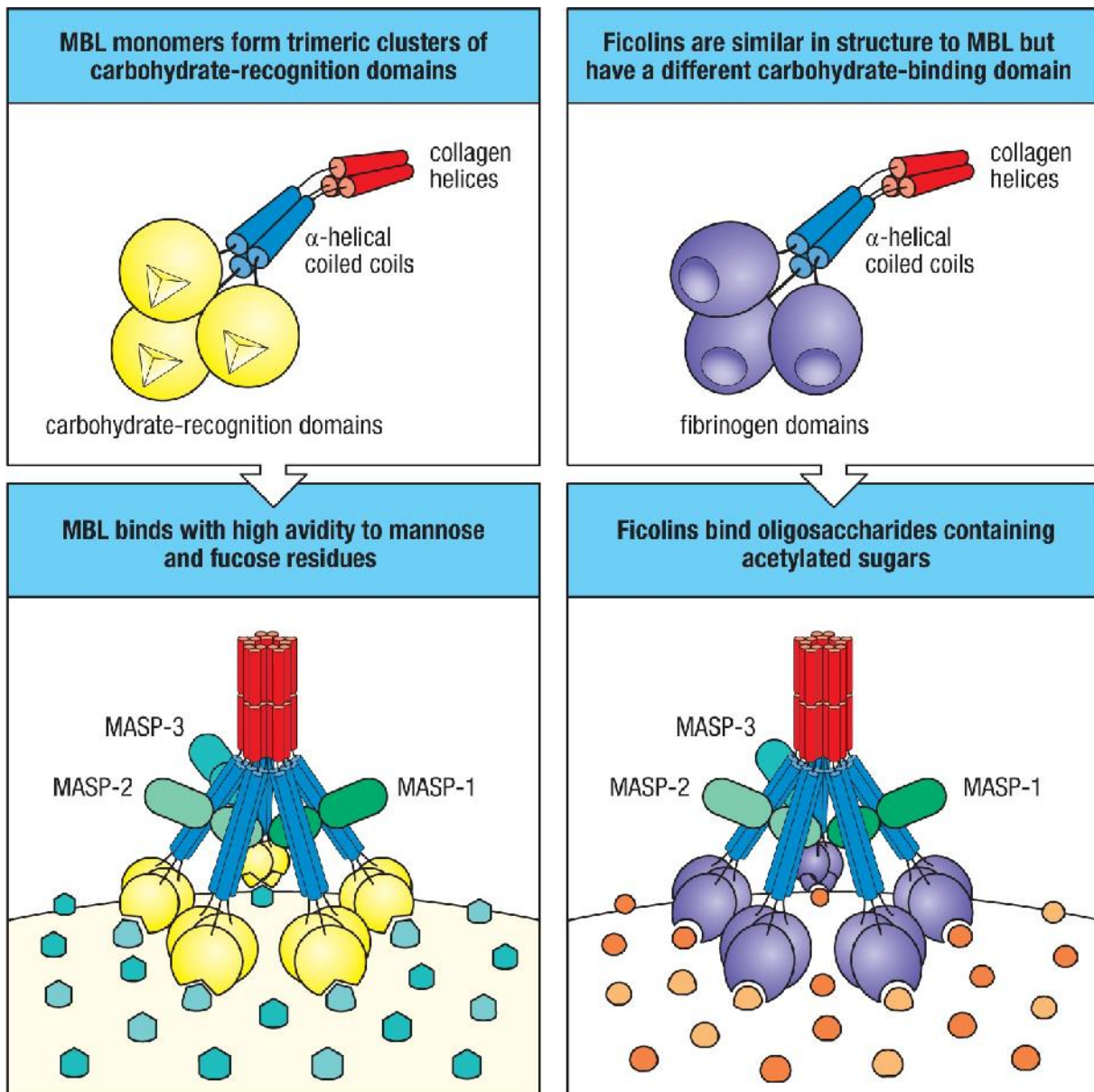


Fig. 2.19 Mannose-binding lectin and ficolins form complexes with serine proteases and recognize particular carbohydrates on microbial surfaces. Mannose-binding lectin (MBL; left panels) is an oligomeric protein in which two to six clusters of carbohydrate-binding heads arise from a central stalk that is formed from the collagen-like tails of the MBL monomers. An MBL monomer is composed of a collagen region (red), an α -helical neck region (blue), and a carbohydrate-recognition domain (yellow). Three MBL monomers associate to form a trimer (top left panel), and between two and six trimers assemble to form a mature MBL molecule (bottom left panel). MBL binds to bacterial surfaces that display a particular spatial arrangement of mannose or fucose residues. An MBL molecule associates with the serine proteases MASP-1, MASP-2, and MASP-3, and with the nonenzymatic proteins MAp19 and MAp44 (not shown), and can activate C4 and C2 after binding to carbohydrate molecules present on microbial surfaces. The ficolins (right panels) resemble MBL in their overall structure and activation mechanism. The carbohydrate-binding domain of ficolins is a fibrinogen-like domain, rather than the lectin domain present in MBL.

The other pathogen-recognition molecules used by the lectin pathway are known as [ficolins](#). Although related in overall shape and function to MBL, they have a fibrinogen-like domain, rather than a lectin domain, attached to the collagen-like stalk (see Fig. 2.19). The fibrinogen-like domain gives ficolins a general specificity for oligosaccharides containing acetylated sugars, but it does not bind mannose-containing carbohydrates. Humans have three ficolins: ficolin-1 (M-ficolin), ficolin-2 (L-ficolin), and ficolin-3 (H-ficolin). Each binds to a spectrum of bacteria. Ficolins are secreted into the blood and tissues by various cells, including cells of the liver and lung, and by neutrophils.

MBL in plasma can bind to five proteins: three [MBL-associated serine proteases](#), [MASP-1](#), [MASP-2](#), and [MASP-3](#), and two nonenzymatic proteins, [MAp19](#) and [MAp44](#). These five proteins are encoded by alternative splice variants of two genes: *MASP1*, which encodes MASP-1, MASP-3, and MAp44, and *MASP2*, which encodes MASP-2 and MAp19. MASP-1 and MASP-3 use different serine protease domains and have different substrate specificity. When MBL bound to MASP-1 and MASP-2 interacts with a pathogen surface, a conformational change occurs in MASP-1 that enables it to cleave and activate a MASP-2 molecule in the same MBL complex. Activated MASP-2 can then cleave complement components 4 and 2 (C4 and C2) to initiate the lectin and classical complement pathways (**Fig. 2.20**). When MBL bound to MASP-3 and MAp44 binds a pathogen surface, MASP-3 cleaves a pro-form of factor D, producing active factor D, which is a key event in the initiation of the alternative complement pathway. Like MBL, ficolins form oligomers that make a complex with MASP-1 and MASP-2, which similarly activate complement upon recognition of a microbial surface by the ficolin. [C4](#), like C3, contains a buried thioester bond. When MASP-2 cleaves C4, it releases C4a, allowing a conformational change in C4b that exposes the reactive thioester as described for C3b (see Fig. 2.16). C4b bonds covalently via this thioester to the microbial surface nearby, where it then binds one molecule of C2 (see Fig. 2.20). C2 is cleaved by MASP-2, producing C2a, an active serine protease that remains bound to C4b to form [C4b2a](#), which is the C3 convertase of the lectin and classical pathways. (Remember, we use the original convention by which C2a is the active large C2 fragment.) C4b2a now cleaves many molecules of C3 into C3a and C3b. The C3b fragments bond covalently to the nearby pathogen surface, and the released C3a initiates a local inflammatory response. The complement-activation pathway initiated by ficolins proceeds like the MBL lectin pathway (see Fig. 2.20).

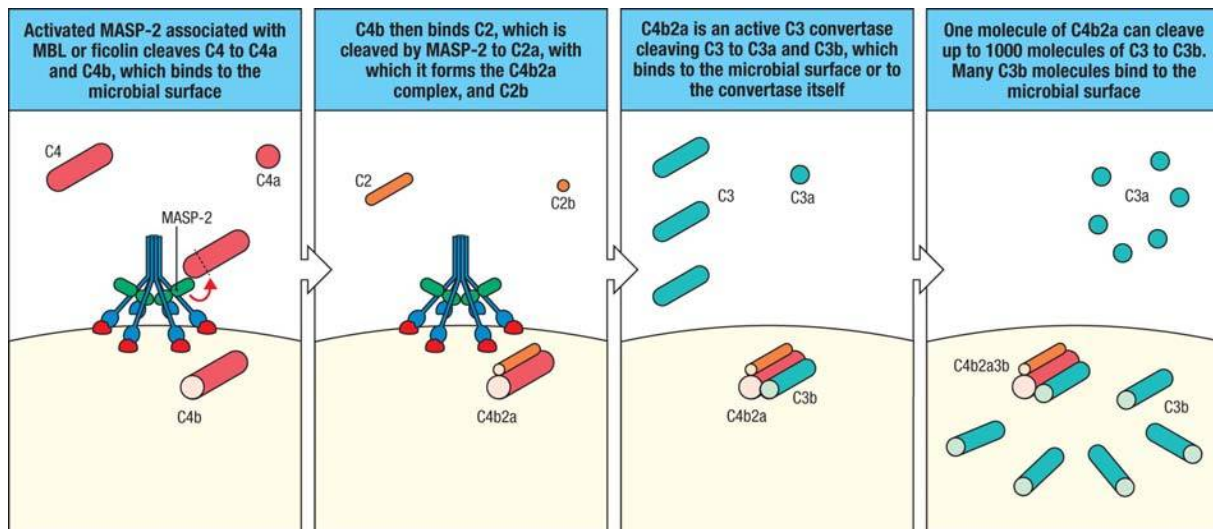


Fig. 2.20 The actions of the C3 convertase result in the binding of large numbers of C3b molecules to the pathogen surface. Binding of mannose-binding lectin or ficolins to their carbohydrate ligands on microbial surfaces induces MASP-1 to cleave and activate the serine protease MASP-2. MASP-2 then cleaves C4, exposing the thioester bond in C4b that allows it to react covalently with the pathogen surface. C4b then binds C2, making C2 susceptible to cleavage by MASP-2 and thus generating the C3 convertase C4b2a. C2a is the active protease component of the C3 convertase and cleaves many molecules of C3 to produce C3b, which binds to the pathogen surface, and C3a, an inflammatory mediator. The covalent attachment of C3b and C4b to the pathogen surface is important in confining subsequent complement activity to pathogen surfaces.

Individuals deficient in MBL or MASP-2 experience substantially more respiratory infections by common extracellular bacteria during early childhood, indicating the importance of the lectin pathway for host defense. This susceptibility illustrates the particular importance of innate defense mechanisms in early childhood, when adaptive immune responses are not yet fully developed but the maternal antibodies transferred across the placenta and present in the mother's milk are gone. Other members of the collectin family are the [surfactant proteins A and D \(SP-A and SP-D\)](#), which are present in the fluid that bathes the epithelial surfaces of the lung. There they coat the surfaces of pathogens, making them more susceptible to phagocytosis by macrophages that have left the subepithelial tissues to enter the alveoli. Because SP-A and SP-D do not associate with MASPs, they do not activate complement.

We have used MBL here as our prototype activator of the lectin pathway, but the ficolins are more abundant than MBL in plasma and so may actually be more important. Ficolin-2 recognizes acetylated sugars, such as GlcNAc and *N*-acetylgalactosamine (GalNAc), and particularly recognizes lipoteichoic acid, a component of the cell walls of Gram-positive bacteria that contains GalNAc. It can also activate complement after binding to a variety of capsulated bacteria. Ficolin-1 also recognizes acetylated sugar residues; ficolin-3 shows a more restricted binding specificity, for d-fucose and d-galactose, and has been linked to

activity against the Gram-positive bacterium *Aerococcus viridans*, a cause of bacterial endocarditis.

2-7 The classical pathway is initiated by activation of the C1 complex and is homologous to the lectin pathway.

In its overall scheme, the classical pathway is similar to the lectin pathway, except that it uses a pathogen sensor known as the [C1 complex](#), or [C1](#). Because C1 interacts directly with some pathogens but can also interact with antibodies, C1 allows the classical pathway to function both in innate immunity, which we describe now, and in adaptive immunity, which we examine in more detail in [Chapter 10](#).

Like the MBL:MASP complex, the C1 complex is composed of a large subunit ([C1q](#)), which acts as the pathogen sensor, and two serine proteases ([C1r](#) and [C1s](#)), which are initially in their inactive form (**Fig. 2.21**). C1q is a hexamer of trimers, composed of monomers that contain an amino-terminal globular domain and a carboxyl-terminal collagen-like domain. The trimers assemble through interactions of the collagen-like domains, bringing the globular domains together to form a globular head. Six of these trimers assemble to form a complete C1q molecule, which has six globular heads held together by their collagen-like tails. C1r and C1s are closely related to MASP-2 and somewhat more distantly related to MASP-1 and MASP-3; all five enzymes are likely to have evolved from the duplication of a gene for a common precursor. C1r and C1s interact noncovalently and form tetramers that fold into the arms of C1q, with at least part of the C1r:C1s complex being external to C1q, as illustrated in Fig. 2.21.

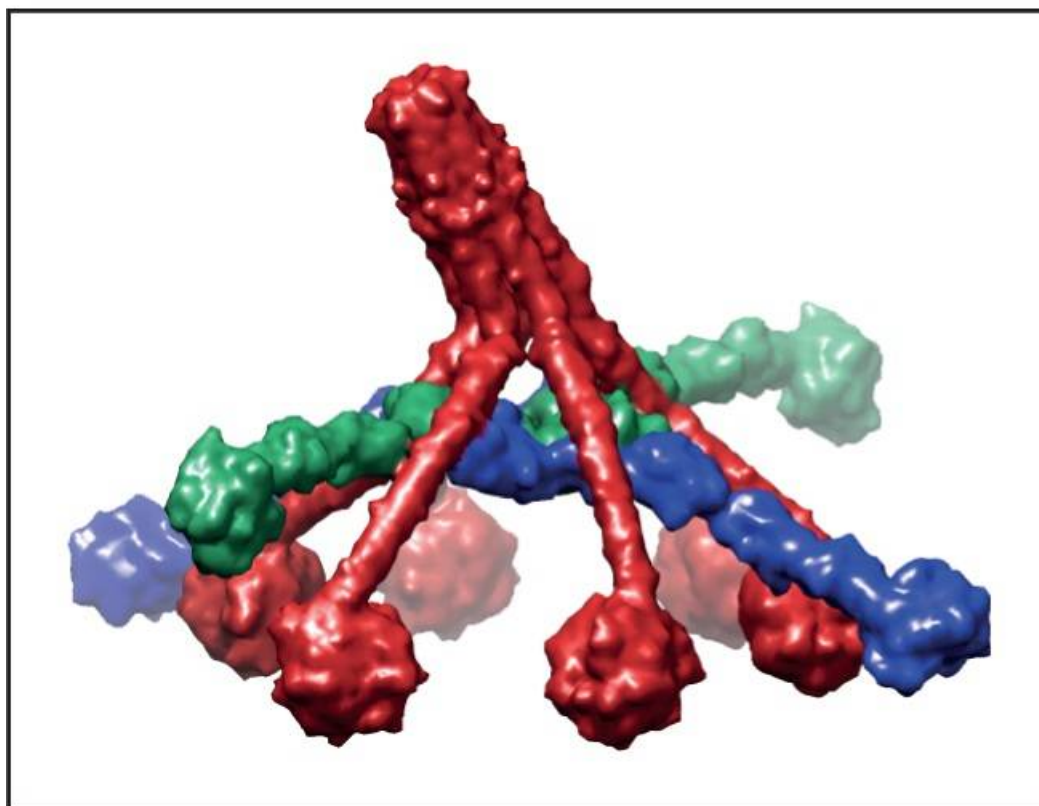
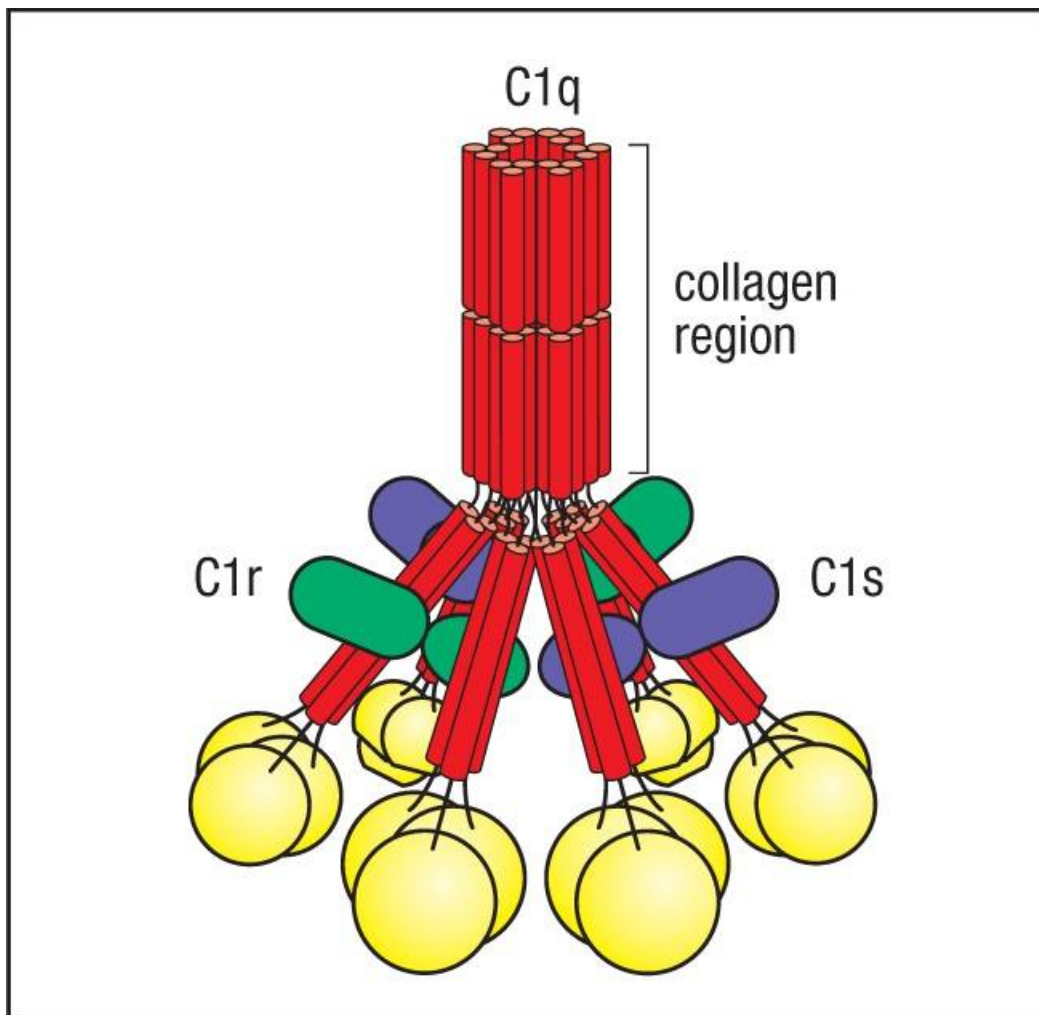


Fig. 2.21 The first protein in the classical pathway of complement activation is C1, which is a complex of C1q, C1r, and C1s. Upper panel: C1q is composed of six identical subunits with globular heads (yellow) and long collagen-like tails (red); it has been described as looking like “a bunch of tulips.” The tails combine to bind to two molecules each of C1r and C1s, forming the C1 complex C1q:C1r₂:C1s₂. The heads can bind to the constant regions of immunoglobulin molecules or directly to the pathogen surface, causing a conformational change in C1r, which then cleaves and activates the C1s zymogen (proenzyme). The C1 complex is similar in overall structure to the MBL:MASP complex, and it has an identical function, cleaving C4 and C2 to form the C3 convertase C4b2a (see Fig. 2.20). Lower panel: Structure of the C1 complex solved by cryo-electron microscopy (cryo-EM): C1q (red), C1r (green), C1s (blue). Modified from Sharp, T.H., et al.: *Proc. Natl. Acad. Sci. USA* 2019, 116:11900–11905.

The recognition function of C1 resides in the six globular heads of C1q. When two or more of these heads interact with a ligand, this causes a conformational change in the C1r:C1s complex, which leads to the activation of an autocatalytic enzymatic activity in C1r; the active form of C1r then cleaves its associated C1s to generate an active serine protease. The activated C1s acts on the next two components of the classical pathway, C4 and C2. C1s cleaves C4 to produce C4b, which binds covalently to the pathogen surface as described earlier for the lectin pathway (see Fig. 2.20). C4b then also binds one molecule of C2, which is cleaved by C1s to produce the serine protease C2a. This produces the active C3 convertase C4b2a, which is the C3 convertase of both the lectin and the classical pathways. However, because it was first discovered as part of the classical pathway, C4b2a is often known as the [classical C3 convertase](#) (see Fig. 2.17). The proteins involved in the classical pathway, along with their active forms, are listed in **Fig. 2.22**.

Proteins of the classical pathway of complement activation		
Native component	Active form	Function of the active form
C1 (C1q:C1r ₂ :C1s ₂)	C1q	Binds directly to pathogen surfaces or indirectly to antibody bound to pathogens, thus allowing autoactivation of C1r
	C1r	Cleaves C1s to active protease
	C1s	Cleaves C4 and C2
	C4b	Covalently binds to pathogen and opsonizes it. Binds C2 for cleavage by C1s

C4	C4a	Peptide mediator of inflammation (weak activity)
C2	C2a	Active enzyme of classical pathway C3/C5 convertase: cleaves C3 and C5
	C2b	Inactive small fragment
C3	C3b	Binds to pathogen surface and acts as opsonin. Initiates amplification via the alternative pathway. Binds C5 for cleavage by C2a
	C3a	Peptide mediator of inflammation (intermediate activity)

Fig. 2.22 The proteins of the classical pathway of complement activation.

C1q can attach itself to the surface of pathogens in several different ways. One is by binding directly to surface components on some bacteria, including certain proteins of bacterial cell walls and polyanionic structures, such as the lipoteichoic acid on Gram-positive bacteria. A second is through binding to C-reactive protein, an acute-phase protein in human plasma that binds to phosphocholine residues in bacterial surface molecules such as pneumococcal C polysaccharide—hence the name C-reactive protein. We discuss the acute-phase proteins in detail in [Chapter 3](#). However, a main function of C1q in an immune response is to bind to the constant, or Fc, regions of antibodies (see [Section 1-9](#)) that have bound pathogens via their antigen-binding sites. C1q thus links the effector functions of complement to recognition provided by adaptive immunity. This might seem to limit the usefulness of C1q in fighting the first stages of an infection, before the adaptive immune response has generated pathogen-specific antibodies. However, some antibodies, called [natural antibodies](#), are produced by the immune system in the apparent absence of infection. These antibodies have a low affinity for many microbial pathogens and are highly cross-reactive, recognizing common membrane constituents such as phosphocholine and even recognizing some antigens of the body's own cells (that is, self antigens). Natural antibodies may be produced in response to commensal microbiota or to self antigens, but do not seem to be the consequence of an adaptive immune response to infection by pathogens. Most natural antibody is of the isotype, or class, known as IgM (see [Chapter 5](#)) and represents a considerable amount of the total IgM circulating in humans. IgM is the class of antibody most efficient at binding C1q, making natural antibodies an effective means of activating complement on microbial surfaces immediately after infection and leading to the clearance of bacteria such as *Streptococcus pneumoniae* (the pneumococcus) before they become dangerous.

2-8 Complement activation is largely confined to the surface on which it is initiated.

We have seen that both the lectin and the classical pathways of complement activation are initiated by proteins that bind to pathogen surfaces. During the triggered enzyme cascade that follows, it is important that activating events are confined to this same site, so that C3 activation also occurs on the surface of the pathogen and not in the plasma or on host-cell surfaces. This is achieved principally by the covalent binding of C4b to the pathogen surface. In innate immunity, C4 cleavage is catalyzed by a ficolin or MBL complex that is bound to the pathogen surface, and so the C4b cleavage product can bind adjacent proteins or carbohydrates on the pathogen surface. If C4b does not rapidly form this bond, the thioester bond is cleaved by reaction with water, and C4b is irreversibly inactivated. This helps to prevent C4b from diffusing from its site of activation on the microbial surface and becoming attached to healthy host cells.

C2 becomes susceptible to cleavage by C1s only when it is bound by C4b, and the active C2a serine protease is thereby also confined to the pathogen surface, where it remains associated with C4b, forming the C3 convertase C4b2a. Cleavage of C3 to C3a and C3b is thus also confined to the surface of the pathogen. Like C4b, C3b is inactivated by hydrolysis unless its exposed thioester rapidly makes a covalent bond (see Fig. 2.16), and it therefore opsonizes only the surface on which complement activation has taken place. Opsonization by C3b is more effective when antibodies are also bound to the pathogen surface, as phagocytes have receptors for both complement and Fc receptors that bind the Fc region of antibody (see [Sections 1-20](#) and [10-20](#)). Because the reactive forms of C3b and C4b are able to form a covalent bond with any adjacent protein or carbohydrate, when complement is activated by bound antibody a proportion of the reactive C3b or C4b will become linked to the antibody molecules themselves. Antibody that is chemically cross-linked to complement is likely the most efficient trigger for phagocytosis.

2-9 The alternative pathway is an amplification loop for C3b formation that is accelerated by properdin in the presence of pathogens.

Although probably the most ancient of the complement pathways, the alternative pathway is so named because it was discovered as a second, or 'alternative,' pathway for complement activation after the classical pathway had been defined. Its key feature is its ability to be spontaneously activated. It has a unique C3 convertase, the [alternative pathway C3 convertase](#), which differs from the C4b2a convertase of the lectin or classical pathways (see Fig. 2.17). The alternative pathway C3 convertase is composed of C3b itself bound to Bb, which is a cleavage fragment of the plasma protein [factor B](#). This C3 convertase, designated

[C3bBb](#), has a special place in complement activation because, by producing C3b, it can generate more of itself. This means that once some C3b has been formed, by whichever pathway, the alternative pathway can act as an amplification loop to increase C3b production rapidly.

The alternative pathway can be activated in two different ways. The first is by the action of the lectin or classical pathway. C3b generated by either of these pathways and covalently linked to a microbial surface can bind factor B (**Fig. 2.23**). This alters the conformation of factor B, enabling a plasma protease called [factor D](#) to cleave it into Ba and Bb. Bb remains stably associated with C3b, forming the C3bBb C3 convertase. The second way of activating the alternative pathway involves the spontaneous hydrolysis (known as ‘[tickover](#)’) of the thioester bond in C3 to form C3(H₂O), as shown in **Fig. 2.24**. C3 is abundant in plasma, and tickover causes a steady, low-level production of C3(H₂O). This C3(H₂O) can bind factor B, which is then cleaved by factor D, producing a short-lived [fluid-phase C3 convertase](#), [C3\(H₂O\)Bb](#). Although formed in only small amounts by C3 tickover, fluid-phase C3(H₂O)Bb can cleave many molecules of C3 to C3a and C3b. Much of this C3b is inactivated by hydrolysis, but some attaches covalently via its thioester bond to the surfaces of any microbes present. C3b formed in this way is no different from C3b produced by the lectin or classical pathways and binds factor B, leading to the formation of C3 convertase and a stepping up of C3b production (see Fig. 2.23).

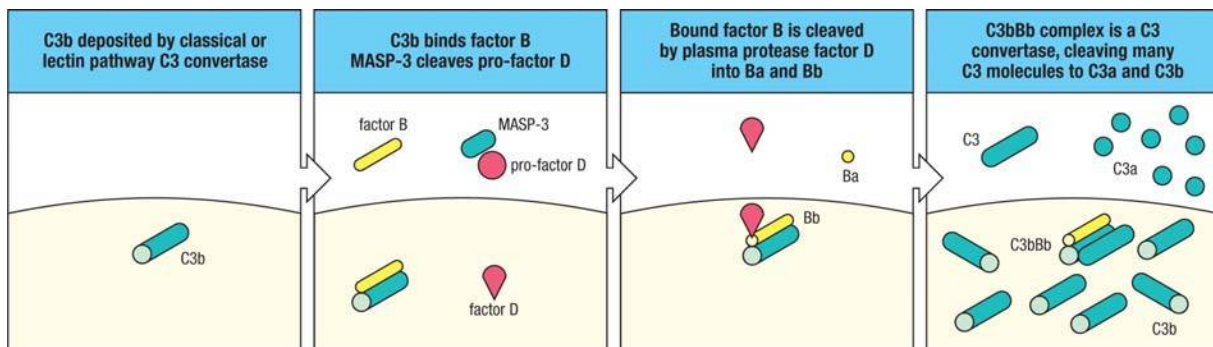


Fig. 2.23 The alternative pathway of complement activation can amplify the classical or the lectin pathway by forming an alternative C3 convertase and depositing more C3b molecules on the pathogen. C3b deposited by the classical or lectin pathway can bind factor B, making it susceptible to cleavage by factor D. MASP-3 cleaves pro-factor D (red circle) to form the active form of factor D (red fan). Cleavage of factor B produces the C3bBb complex, which is the C3 convertase of the alternative pathway of complement activation. This action of this C3 convertase, like that of C4b2a, results in the deposition of many molecules of C3b on the pathogen surface.

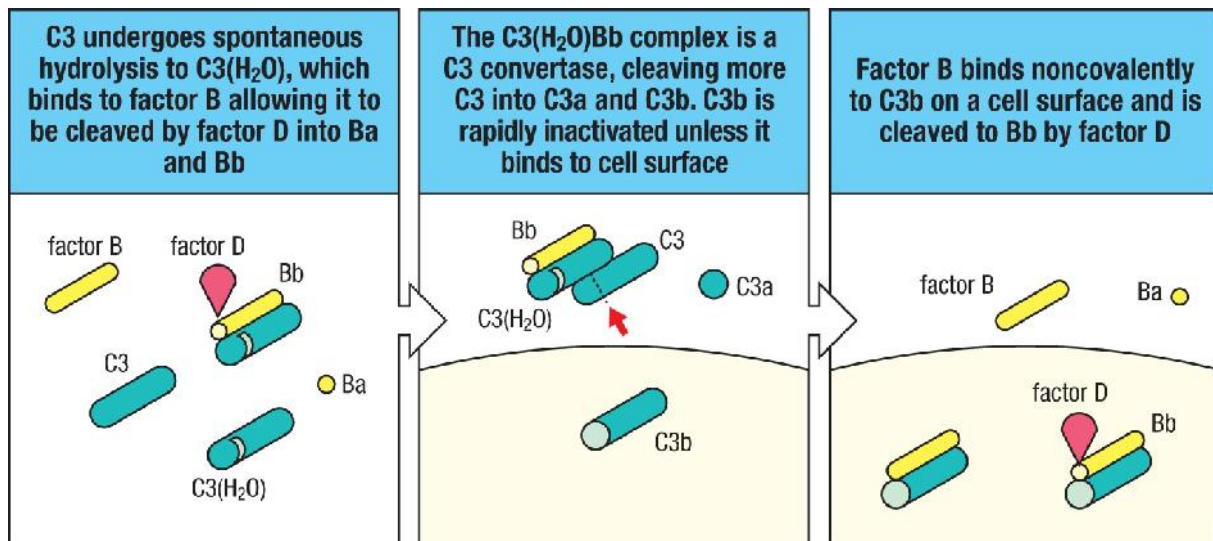
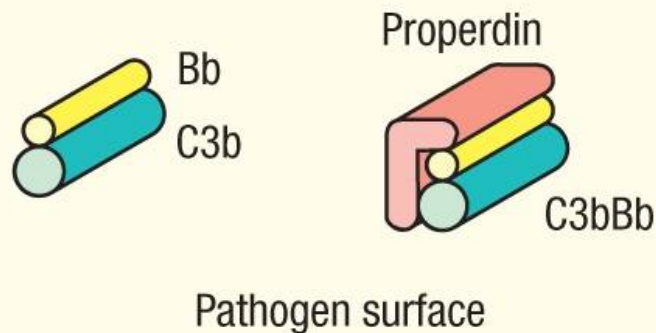


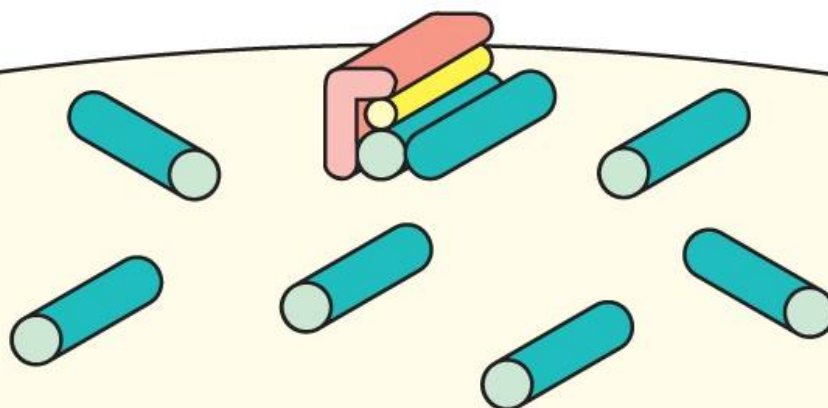
Fig. 2.24 The alternative pathway can be activated by spontaneous activation of C3. Complement component C3 hydrolyzes spontaneously in plasma to give C3(H₂O), which binds factor B and enables the bound factor B to be cleaved by factor D (first panel). The resulting ‘soluble C3 convertase’ cleaves C3 to give C3a and C3b, which can attach to host cells or pathogen surfaces (second panel). Covalently bound to the cell surface, C3b binds factor B; in turn, factor B is rapidly cleaved by factor D to Bb, which remains bound to C3b to form a C3 convertase (C3bBb), and Ba, which is released (third panel). This convertase functions in the alternative pathway as the C3 convertase C4b2a does in the lectin and classical pathways (see Fig. 2.17).

On their own, the alternative pathway C3 convertases C3bBb and C3(H₂O)Bb are very short-lived (~90 seconds). They are stabilized, however, by around 5- to 10-fold by binding the plasma protein [properdin](#) (**Fig. 2.25**). Properdin is made by neutrophils and stored in secondary granules. It is released when neutrophils are activated by the presence of pathogens. Properdin may have some properties of a pattern-recognition receptor, as it can bind to some microbial surfaces. Properdin-deficient individuals are particularly susceptible to infections with *Neisseria meningitidis*, the main agent of bacterial meningitis. Properdin’s ability to bind to bacterial surfaces may direct the activity of the alternative complement pathway to these pathogens, thus aiding their removal by phagocytosis. Properdin can also bind to mammalian cells that are undergoing apoptosis or have been damaged or modified by ischemia, viral infection, or antibody binding, leading to the deposition of C3b on these cells and facilitating their removal by phagocytosis. The distinctive components of the alternative pathway are listed in **Fig. 2.26**.

Pathogens lack complement-regulatory proteins. Binding of properdin stabilizes the C3bBb complex



C3bBb complex is a C3 convertase and deposits many molecules of C3b on the pathogen surface



Opsonization, activation of terminal complement components

Fig. 2.25 Properdin stabilizes the alternative pathway C3 convertase on pathogen surfaces. Bacterial surfaces do not express complement-regulatory proteins and favor the binding of properdin, which stabilizes the C3bBb convertase. This convertase activity is the equivalent of C4b2a of the lectin and classical pathways. C3bBb then cleaves many more molecules of C3, coating the pathogen surface with bound C3b.

Proteins of the alternative pathway of complement activation		
Native component	Active fragments	Function
C3	C3b	Binds to pathogen surface; binds B for cleavage by D; C3bBb is a C3 convertase and C3b ₂ Bb is a C5 convertase
Factor B (B)	Ba	Small fragment of B, unknown function
	Bb	Bb is the active enzyme of the C3 convertase C3bBb and the C5 convertase C3b ₂ Bb
Factor D (D)	D	Plasma serine protease, cleaves B when it is bound to C3b to Ba and Bb
Properdin (P)	P	Plasma protein that binds to bacterial surfaces and stabilizes the C3bBb convertase

Fig. 2.26 The proteins of the alternative pathway of complement activation.

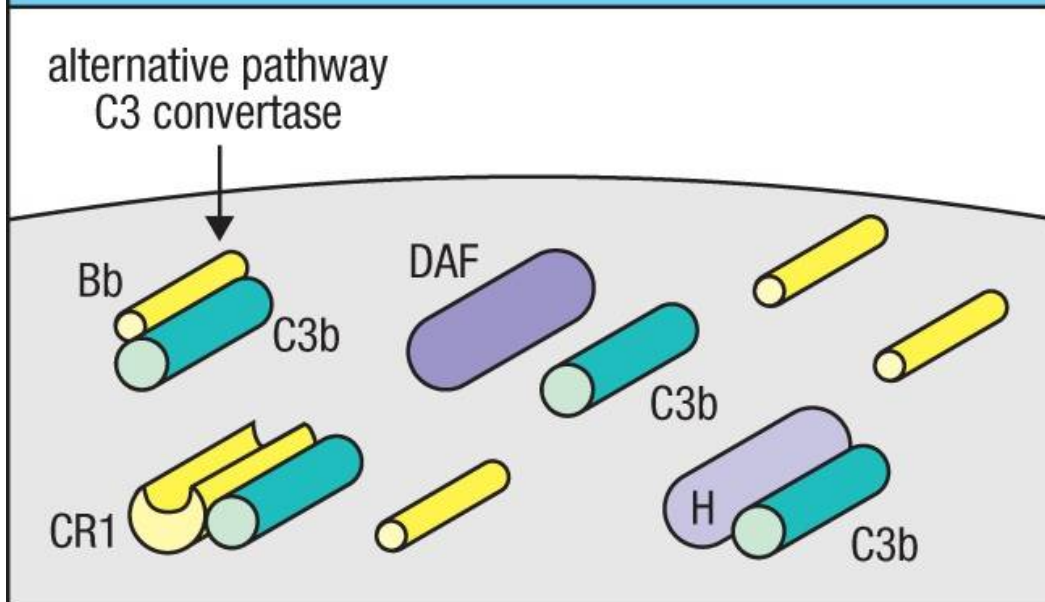
2-10 Membrane and plasma proteins that regulate the formation and stability of C3 convertases determine the extent of complement activation.

Several mechanisms ensure that complement activation will proceed only on the surface of a pathogen or on damaged host cells, and not on normal host cells and tissues. After initial complement activation by any pathway, the extent of amplification via the alternative pathway is critically dependent on the stability of the C3 convertase C3bBb. This stability is controlled by both positive and negative regulatory proteins. We have already described how

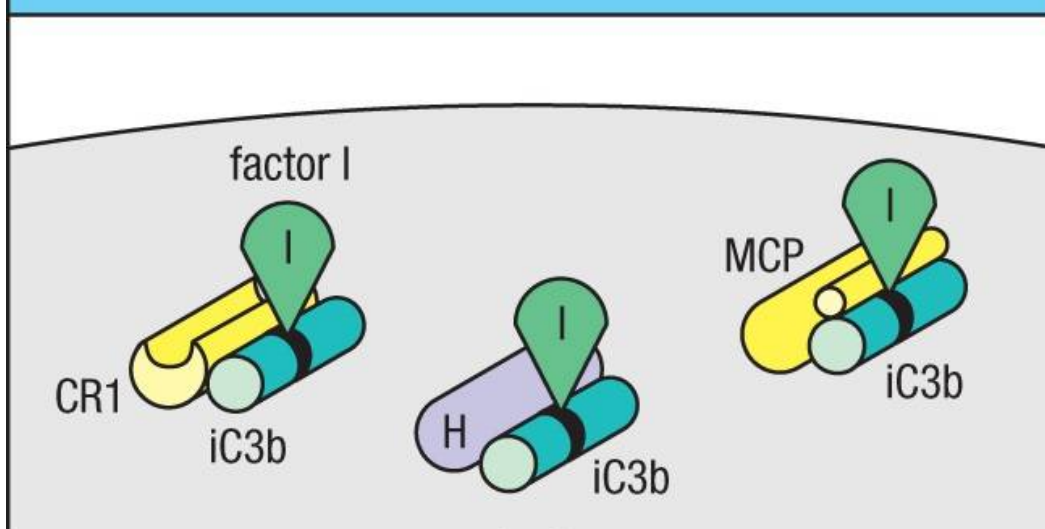
properdin acts as a positive regulatory protein on foreign surfaces, such as those of bacteria or damaged host cells, by stabilizing C3bBb.

Several negative regulatory proteins, present in plasma and in host-cell membranes, protect healthy host cells from the injurious effects of inappropriate complement activation on their surfaces. These [complement-regulatory proteins](#) interact with C3b and either prevent the convertase from forming or promote its rapid dissociation (**Fig. 2.27**). For example, a membrane-attached protein known as [decay-accelerating factor \(DAF or CD55\)](#) competes with factor B for binding to C3b on the cell surface and can displace Bb from a convertase that has already formed. Convertase formation can also be prevented by cleaving C3b to an inactive derivative, [iC3b](#). This is achieved by a plasma protease, [factor I](#), in conjunction with C3b-binding proteins that act as cofactors, such as [membrane cofactor of proteolysis \(MCP or CD46\)](#), another host-cell membrane protein (see Fig. 2.27). Cell-surface complement receptor type 1 ([CR1](#), also known as [CD35](#)) behaves similarly to DAF and MCP in that it inhibits C3 convertase formation and promotes the catabolism of C3b to inactive products, respectively, but it has a more limited tissue distribution. [Factor H](#) is another complement-regulatory protein in plasma that binds C3b, and like CR1, it is able to compete with factor B to displace Bb from the convertase; in addition, it acts as a cofactor for factor I. Factor H binds preferentially to C3b bound to vertebrate cells because it has an affinity for the sialic acid residues present on their cell surfaces (see Fig. 2.18). Thus, the amplification loop of the alternative pathway is allowed to proceed on the surface of a pathogen or on damaged host cells, but not on normal host cells or on tissues that express these negative regulatory proteins.

On host cells, complement-regulatory proteins CR1, DAF, and H cause a displacement of Bb from C3b, eliminating C3 convertase activity



C3b bound to CR1, H, and MCP is cleaved by factor I to yield inactive C3b (iC3b)



No activation of
complement on host-cell surfaces

Fig. 2.27 Complement activation spares host cells, which are protected by complement-regulatory proteins. If C3bBb forms on the surface of host cells, it is rapidly inactivated by complement-regulatory proteins expressed by the host cell: complement receptor 1 (CR1), decay-accelerating factor (DAF), and membrane cofactor of proteolysis (MCP). Host-cell surfaces also favor the binding of factor H from plasma. CR1, DAF, and factor H displace Bb from C3b, and CR1, MCP, and factor H catalyze the cleavage of bound C3b by the plasma protease factor I to produce inactive C3b (known as iC3b).

The C3 convertase of the classical and lectin pathways (C4b2a) is molecularly distinct from that of the alternative pathway (C3bBb). However, understanding of the complement system is simplified somewhat by recognition of the close evolutionary relationships between the different complement proteins (**Fig. 2.28**). Thus the complement zymogens factor B and C2 are closely related proteins encoded by homologous genes located in tandem within the major histocompatibility complex (MHC) on human chromosome 6. Furthermore, their respective binding partners, C3 and C4, both contain thioester bonds that provide the means of covalently attaching the C3 convertases to a pathogen surface.

Step in pathway	Protein serving function in pathway			Relationship
	Alternative	Lectin	Classical	
Initiating serine protease	D	MASP	C1s	Homologous (C1s and MASP)
Covalent binding to cell surface	C3b	C4b		Homologous
C3/C5 convertase	Bb	C2a		Homologous
Control of activation	CR1	CR1		Identical Homologous
	H	C4BP		
Opsonization	C3b			Identical
Initiation of effector pathway	C5b			Identical

Local inflammation	C5a, C3a		Identical
Stabilization	P	None	Unique

Fig. 2.28 There is a close evolutionary relationship among the factors of the alternative, lectin, and classical pathways of complement activation. Most of the factors are either identical or the homologous products of genes that have duplicated and then diverged in sequence. The proteins C4 and C3 are homologous and contain the unstable thioester bond by which their large fragments, C4b and C3b, bind covalently to membranes. The genes encoding proteins C2 and factor B are adjacent in the MHC region of the genome and arose by gene duplication. The regulatory proteins factor H, CR1, and C4BP share a repeat sequence common to many complement-regulatory proteins. The greatest divergence between the pathways is in their initiation: in the classical pathway the C1 complex binds either to certain pathogens or to bound antibody, and in the latter circumstance it serves to convert antibody binding into enzyme activity on a specific surface; in the lectin pathway, mannose-binding lectin (MBL) associates with a serine protease, activating MBL-associated serine protease (MASP), to serve the same function as C1r:C1s; in the alternative pathway this enzyme activity is provided by factor D.

Only one component of the alternative pathway seems entirely unrelated to its functional equivalents in the classical and lectin pathways: the initiating serine protease, factor D. Factor D can also be singled out as the only activating protease of the complement system to circulate as an active enzyme rather than a zymogen. This is both necessary for the initiation of the alternative pathway (through the cleavage of factor B bound to spontaneously activated C3) and safe for the host, because factor D has no other substrate than factor B bound to C3b. This means factor D finds its substrate only at pathogen surfaces and at a very low level in plasma, where the alternative pathway of complement activation can be allowed to proceed.

2-11 Complement developed early in the evolution of multicellular organisms.

The complement system was originally known only from vertebrates, but homologs of C3 and factor B and a prototypical ‘alternative pathway’ have been discovered in nonchordate invertebrates. This is not altogether surprising as C3, which is cleaved and activated by serine proteases, is evolutionarily related to the serine protease inhibitor α_2 -macroglobulin, whose first appearance likely was in an ancestor to all modern vertebrates. The amplification loop of the alternative pathway also has an ancestral origin, as it is present in echinoderms (which include sea urchins and sea stars) and is based on a C3 convertase formed by the echinoderm homologs of C3 and factor B. These factors are expressed by phagocytic cells called [amoeboid coelomocytes](#) present in the coelomic fluid. Expression of C3 by these cells

increases when bacteria are present. This simple system seems to function to opsonize bacterial cells and other foreign particles and facilitate their uptake by coelomocytes. C3 homologs in invertebrates are clearly related to each other. They all contain the distinctive thioester linkage and form a family of proteins, the [thioester-containing proteins](#), or [TEPs](#). In the mosquito *Anopheles*, the production of protein TEP1 is induced in response to infection, and the protein may directly bind to bacterial surfaces to mediate phagocytosis of Gram-negative bacteria. Some form of C3 activity may even predate the evolution of the Bilateria—animals with bilateral symmetry, flatworms being the most primitive modern representatives—because genomic evidence of C3, factor B, and some later-acting complement components has been found in the Anthozoa (corals and sea anemones).

After its initial appearance, the complement system seems to have evolved by the acquisition of new activation pathways that allow specific targeting of microbial surfaces. The first to evolve was likely the ficolin pathway, which is present both in vertebrates and in some closely related invertebrates, such as the urochordates. Evolutionarily, the ficolins may predate the collectins, which are also first seen in the urochordates. Homologs of MBL and of the classical pathway complement component 1q (C1q), both collectins, have been identified in the genome of the ascidian urochordate *Ciona* (sea squirt). Two invertebrate homologs of mammalian MASPs also have been identified in *Ciona*, and it seems likely that they may be able to cleave and activate C3. Thus, the minimal complement system of the echinoderms appears to have been expanded in the urochordates by the recruitment of a specific activation system that may target C3 deposition onto microbial surfaces. This also suggests that when adaptive immunity evolved, much later, the ancestral antibody molecule used an already diversified C1q-like collectin member to activate the complement pathway, and that the complement activation system evolved further by use of this collectin and its associated MASPs to become the initiating components of the classical complement pathway; namely, C1q, C1r, and C1s.

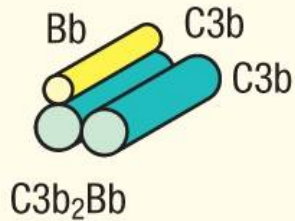
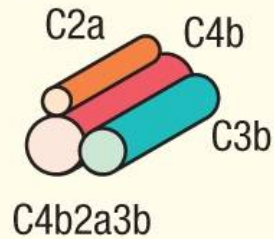
2-12 Surface-bound C3 convertase deposits large numbers of C3b fragments on pathogen surfaces and generates C5 convertase activity.

We now return to the present-day complement system. The formation of C3 convertases is the point at which the three pathways of complement activation converge. The convertase of the lectin and classical pathways, C4b2a, and the convertase of the alternative pathway, C3bBb, initiate the same subsequent events—they cleave C3 to C3b and C3a. C3b binds covalently through its thioester bond to adjacent molecules on the pathogen surface; otherwise it is inactivated by hydrolysis. C3 is the most abundant complement protein in plasma, occurring at a concentration of 1.2 mg/ml, and up to 1000 molecules of C3b can bind in the vicinity of a single active C3 convertase (see Fig. 2.23). Thus the main effect of complement activation is to deposit large quantities of C3b on the surface of the infecting pathogen, where the C3b forms a covalently bonded coat that can signal the ultimate

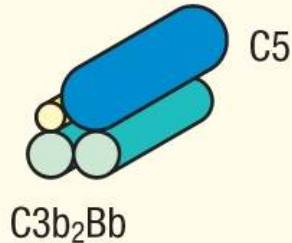
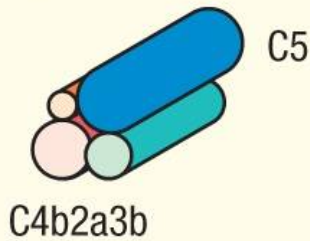
destruction of the pathogen by phagocytes.

The next step in the complement cascade is the generation of the C5 convertases. C5 is a member of the same family of proteins as C3, C4, α_2 -macroglobulin, and the thioester-containing proteins of invertebrates. C5 does not form an active thioester bond during its synthesis, but, like C3 and C4, it is cleaved by a specific protease into C5a and C5b fragments, each of which exerts specific downstream actions that are important in propagating the complement cascade. In the classical and the lectin pathways, a C5 convertase is formed by the binding of C3b to C4b2a to yield [C4b2a3b](#). The C5 convertase of the alternative pathway is formed by the binding of C3b to the C3bBb convertase to form [C3b₂Bb](#). A C5 is captured by these C5 convertase complexes through binding to an acceptor site on C3b and is thus rendered susceptible to cleavage by the serine protease activity of C2a or Bb. This reaction, which generates C5b and C5a, is much more limited than cleavage of C3, because C5 can be cleaved only when it binds to C3b that is in turn bound to C4b2a or C3bBb to form the active C5 convertase complex. Thus, complement activated by all three pathways leads to the binding of large numbers of C3b molecules on the surface of the pathogen, the generation of a more limited number of C5b molecules, and the release of C3a and a smaller amount of C5a (**Fig. 2.29**).

**C3b binds to both C4b2a and C3bBb,
forming the active C5 convertases
C4b2a3b and C3b₂Bb**



**C5 binds to the C3b component of the
C5 convertase enzyme**



**C5 is cleaved by C2a or Bb to form
C5b and C5a**

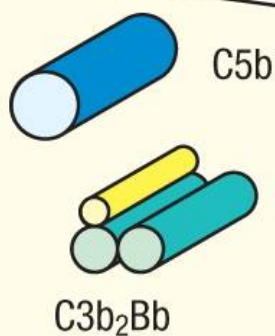
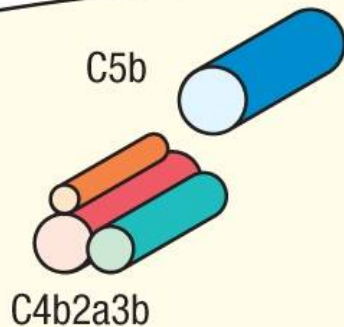


Fig. 2.29 Complement component 5 (C5) is cleaved when captured by a C3b molecule that is part of a C5 convertase complex. As shown in the top panel, C5 convertases are formed when C3b binds either the classical or lectin pathway C3 convertase C4b2a to form C4b2a3b or the alternative pathway C3 convertase C3bBb to form C3b₂Bb. C5 binds to C3b in these complexes (center panel). The bottom panel shows that C5 is cleaved by the active enzyme C2a or Bb to form C5b and the inflammatory mediator C5a. Unlike C3b and C4b, C5b is not covalently bound to the cell surface. The production of C5b initiates the assembly of the terminal complement components.

2-13 Ingestion of complement-tagged pathogens by phagocytes is mediated by receptors for the bound complement proteins.

The most important action of complement is to facilitate the uptake and destruction of pathogens by phagocytic cells. This occurs by the specific recognition of bound complement components by [complement receptors \(CRs\)](#) on phagocytes. These complement receptors bind pathogens opsonized with complement components: opsonization of pathogens is a major function of C3b and its proteolytic derivatives. C4b also acts as an opsonin but has a relatively minor role, largely because so much more C3b than C4b is generated.

The known receptors for bound complement components and for C5a and C3a are listed, with their functions and distributions, in **Fig. 2.30**. The C3b receptor CR1, described in [Section 2-10](#), is a negative regulator of complement activation (see Fig. 2.27). CR1 is expressed on many types of immune cells, including macrophages and neutrophils. Binding of C3b to CR1 cannot by itself stimulate phagocytosis, but it can lead to phagocytosis in the presence of other immune mediators that activate macrophages. For example, the small complement fragment C5a can activate macrophages to ingest bacteria that are bound to their CR1 receptors (**Fig. 2.31**). C5a accomplishes this by binding to another receptor expressed by macrophages, the [C5a receptor 1](#) (C5aR1, CD88), which has seven membrane-spanning domains. Receptors of this type transduce their signals via intracellular guanine nucleotide-binding proteins called G proteins and are known generally as [G protein-coupled receptors \(GPCRs\)](#); they are discussed in [Section 3-2](#). [C5a receptor 2](#) (C5L2, GPR77) is expressed by neutrophils and macrophages, but its function is not firmly established. It is a nonsignaling receptor that may act as a decoy receptor for C5a and regulate activity of the C5a receptor. Proteins associated with the extracellular matrix, such as fibronectin, can also contribute to phagocyte activation; these are encountered when phagocytes are recruited to connective tissue and activated there.

Receptor	Specificity	Functions	Cell types

CR1 (CD35)	C3b, C4b	Promotes C3b and C4b decay Stimulates phagocytosis (requires C5a) Erythrocyte transport of immune complexes	Erythrocytes, macrophages, monocytes, neutrophils, B cells, FDCs
CR2 (CD21)	C3d, iC3b, C3dg	Enhances B-cell response to antigens bearing C3d, iC3b, or C3dg, part of B-cell co-receptor Epstein–Barr virus receptor	B cells, FDCs
CR3 (Mac-1) (CD11b: CD18)	iC3b	Stimulates phagocytosis	Macrophages, monocytes, polymorphonuclear leukocytes, FDCs
CR4 (gp150, 95) (CD11c: CD18)	iC3b	Stimulates phagocytosis	Macrophages, monocytes, polymorphonuclear leukocytes, dendritic cells
CR1g	C3b, iC3b	Phagocytosis of circulating pathogens	Tissue-resident macrophages Hepatic sinusoid macrophages
C5aR1 (CD88)	C5a	Binding of C5a, activates G protein	Neutrophils, macrophages, endothelial cells, mast cells
C5aR2 (C5L2, GPR77)	C5a	Decoy receptor, regulates C5a receptor	Neutrophils, macrophages
C3a receptor	C3a	Binding of C3a, activates G protein	Macrophages, endothelial cells, mast cells
PAR1 PAR4	C4a	Increases vascular permeability Coagulation	Epithelium, platelets

Fig. 2.30 Distribution and function of cell-surface receptors for complement proteins. A variety of complement receptors are specific for bound C3b and its cleavage products (iC3b and C3dg). CR1 and CR3 are important in inducing the phagocytosis of bacteria with complement components bound to their surface. CR2 is found mainly on B cells, where it is part of the B-cell co-receptor complex. CR1 and CR2 share structural features with the complement-regulatory proteins that bind C3b and C4b. CR3 and CR4 are integrins composed of integrin $\beta 2$ (also known as CD18, and sometimes gp95) paired with either integrin αM (CD11b) or integrin αX (CD11c), respectively (see [eAppendix II](#), found in the ebook and on the [Digital Landing Page](#) for this book); CR3, also called Mac-1, is also important for leukocyte adhesion and migration, as we shall see in [Chapter 3](#), whereas CR4 is only known to function in phagocytosis. The receptors for C5a and C3a are seven-span G protein–coupled receptors. FDCs (follicular dendritic cells) are not involved in innate immunity and are discussed in later chapters.

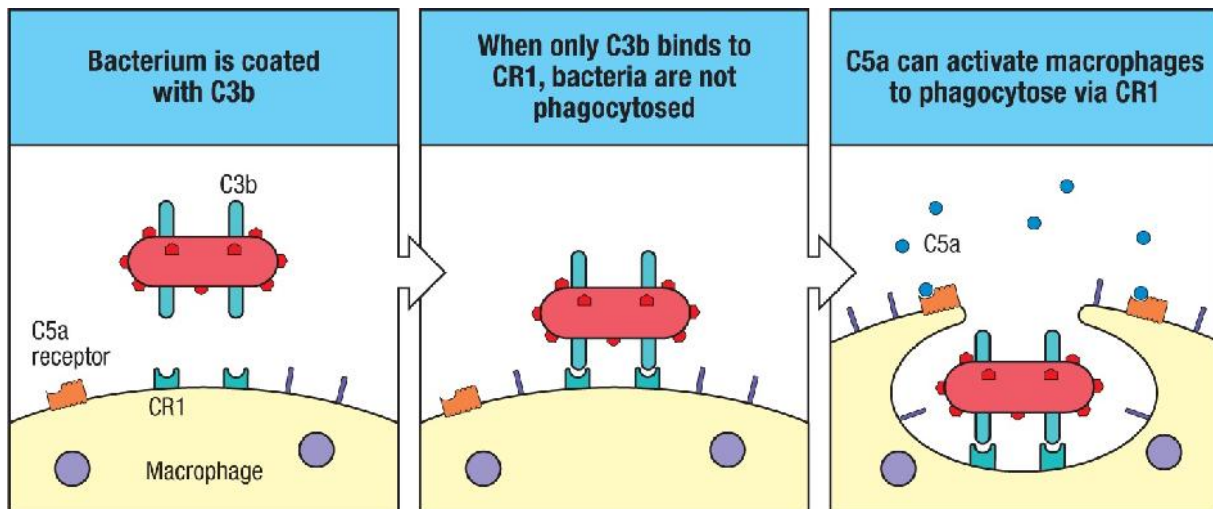


Fig. 2.31 The anaphylatoxin C5a can enhance the phagocytosis of microorganisms opsonized in an innate immune response. Activation of complement leads to the deposition of C3b on the surface of microorganisms (left panel). C3b can be bound by complement receptor 1 (CR1) on the surface of phagocytes, but this on its own is insufficient to induce phagocytosis (center panel). Phagocytes also express receptors for the anaphylatoxin C5a, and binding of C5a will activate the cell to phagocytose microorganisms bound through CR1 (right panel).

Four other complement receptors—[CR2](#) (also known as [CD21](#)), [CR3 \(CD11b:CD18\)](#), [CR4 \(CD11c:CD18\)](#), and [CRIg](#) (complement receptor of the immunoglobulin superfamily)—bind to forms of C3b that have been cleaved by factor I but that remain attached to the pathogen surface. Like several other key components of complement, C3b is subject to regulatory mechanisms that cleave it into derivatives, such as iC3b, that cannot form an active convertase. C3b bound to the microbial surface can be cleaved by factor I and MCP to remove the small fragment C3f, leaving the inactive iC3b form bound to the surface (**Fig. 2.32**). iC3b is recognized by several complement receptors—CR2, CR3, CR4, and CRIg.

Unlike the binding of C3b to CR1, the binding of iC3b to the receptor CR3 is sufficient on its own to stimulate phagocytosis. Factor I and CR1 cleave iC3b to release C3c, leaving C3dg bound to the pathogen. C3dg is recognized only by CR2. CR2 is found on B cells as part of a co-receptor complex that can augment the signal received through the antigen-specific immunoglobulin receptor. Thus, a B cell whose antigen receptor is specific for an antigen of a pathogen will receive a strong signal on binding this antigen if it or the pathogen is also coated with C3dg. The activation of complement can therefore contribute to producing a strong antibody response.

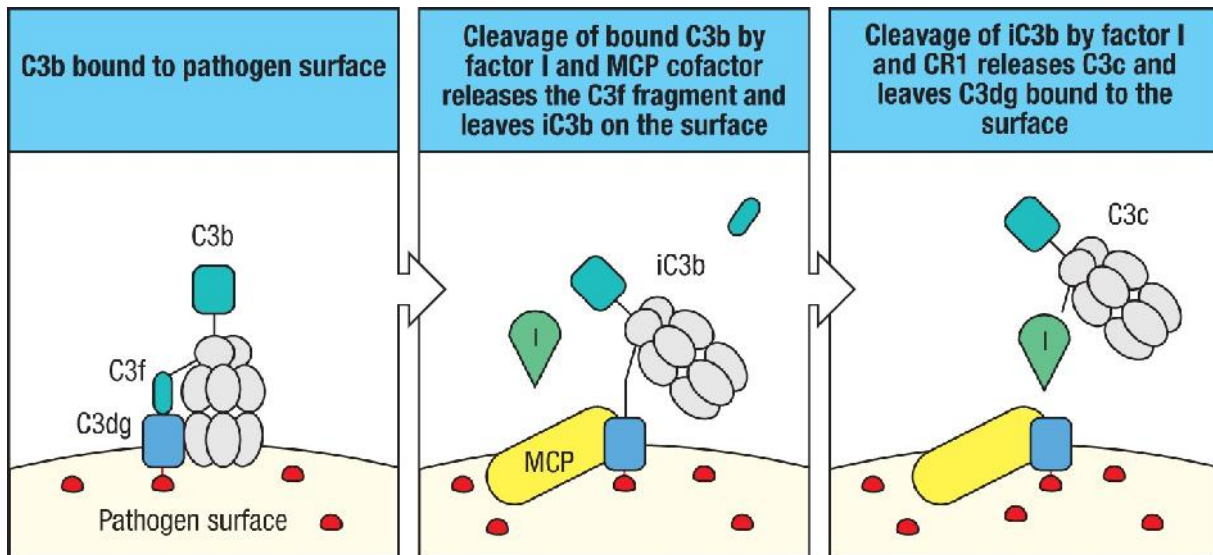


Fig. 2.32 The cleavage products of C3b are recognized by different complement receptors. After C3b is deposited on the surface of pathogens, it can undergo several conformational changes that alter its interaction with complement receptors. Factor I and MCP can cleave the C3f fragment from C3b, producing iC3b, which is a ligand for the complement receptors CR3, CR4, CR1g, and weakly for CR2, but not CR1. Factor I and CR1 cleave iC3b to release C3c, leaving C3dg bound. C3dg is then recognized by CR2.

The importance of opsonization by C3b and its inactive fragments in destroying extracellular pathogens can be seen in the effects of various complement deficiencies. For example, individuals deficient in C3 or in molecules that catalyze C3b deposition show an increased susceptibility to infection by a wide range of extracellular bacteria, including *Streptococcus pneumoniae*. We describe the effects of various defects in complement and the diseases they cause in [Chapter 13](#).

2-14 The small fragments of some complement proteins initiate a local inflammatory response.

The small complement fragments C3a and C5a act on specific receptors on endothelial cells

and mast cells (see Fig. 2.30) to produce local inflammatory responses. Like C5a, C3a also signals through a GPCR family member. When produced in large amounts or injected systemically, C3a and C5a induce a generalized circulatory collapse, producing a shocklike syndrome similar to that seen in a systemic allergic reaction involving antibodies of the IgE class, discussed in [Chapter 14](#). Such a reaction is termed [anaphylactic shock](#), and these small fragments of complement are therefore often referred to as [anaphylatoxins](#). C5a has the highest specific biological activity, but both C3a and C5a induce the contraction of smooth muscle, increase vascular permeability, and act on the endothelial cells lining blood vessels to induce the synthesis of adhesion molecules. In addition, C3a and C5a can activate the mast cells that populate submucosal tissues to release inflammatory molecules such as histamine and the cytokine tumor necrosis factor- α (TNF- α), which cause similar effects. C4a is also generated during C4 cleavage and is inactive at C3a and C5a receptors. Instead, C4a acts at two other GPCRs, protease-activated receptor 1 (PAR1) and protease-activated receptor 4 (PAR4), which are expressed on endothelial cells and platelets and are also activated by thrombin, a protease activated in the clotting cascade. C4a activates signaling in these cells to increase endothelial-cell permeability and by actions on platelets may link complement activation with the clotting cascade.

The changes induced by C5a, C3a, and C4a combine to recruit antibody, complement, and phagocytic cells to the site of an infection (**Fig. 2.33**). The increased fluid in the tissues hastens the movement of pathogen-bearing antigen-presenting cells to the local lymph nodes, contributing to the prompt initiation of the adaptive immune response. C5a also acts directly on neutrophils and monocytes to increase their adherence to vessel walls, their migration toward sites of antigen deposition, and their ability to ingest particles; it also increases the expression of CR1 and CR3 on the surfaces of these cells. In this way, C5a, and to a lesser extent C3a and C4a, act in concert with other complement components to hasten the destruction of pathogens by phagocytes.

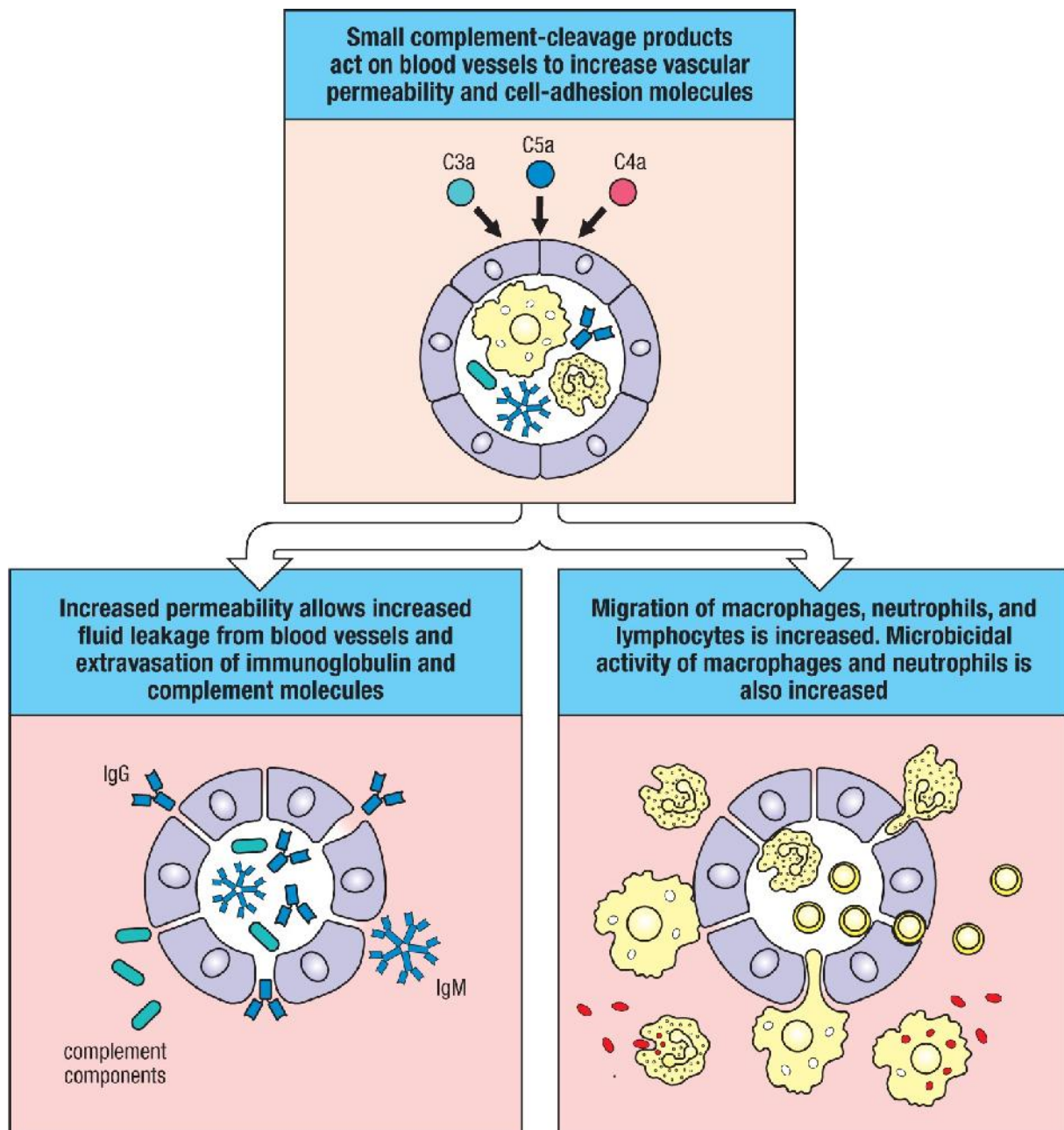


Fig. 2.33 Local inflammatory responses can be induced by small complement fragments, especially C5a. The small complement fragments are differentially active: C5a is more active than C3a. C5a and C3a cause local inflammatory responses by acting directly on local blood vessels, stimulating an increase in blood flow, increased vascular permeability, and increased binding of phagocytes to endothelial cells. C3a and C5a also activate mast cells (not shown) to release mediators, such as histamine and $\text{TNF-}\alpha$, which contribute to the inflammatory response. C4a binds to receptors expressed on endothelium and platelets that are normally activated by thrombin, PAR1, and PAR4, and so also promotes vascular permeability and local coagulation. In response to all these signals, an increase in vessel diameter and permeability leads to the accumulation of fluid and protein in the surrounding tissue. Fluid accumulation increases lymphatic drainage, bringing pathogens and their antigenic components to nearby lymph nodes. The antibodies, complement, and cells thus recruited participate in pathogen clearance by enhancing phagocytosis. The small complement fragments can also directly increase

the activity of the phagocytes.

2-15 The terminal complement proteins polymerize to form pores in membranes that can kill certain pathogens.



Case Study: Deficiency of the C8 Complement Component

One of the important effects of complement activation is the assembly of the terminal components of complement (**Fig. 2.34**) to form a membrane-attack complex. The reactions leading to the formation of this complex are shown schematically in **Fig. 2.35**. The end result is a pore in the lipid bilayer membrane that destroys membrane integrity. This is thought to kill the pathogen by destroying the proton gradient across the pathogen's cell membrane.

The terminal complement components that form the membrane-attack complex		
Native protein	Active component	Function
C5	C5a	Small peptide mediator of inflammation (high activity)
	C5b	Initiates assembly of the membrane-attack system
C6	C6	Binds C5b; forms acceptor for C7
C7	C7	Binds C5b6; amphiphilic complex inserts into lipid bilayer
C8	C8	Binds C5b67; initiates C9 polymerization
C9	C9 _n	Polymerizes to C5b678 to form a membrane-spanning channel, lysing the cell

Fig. 2.34 The terminal complement components.

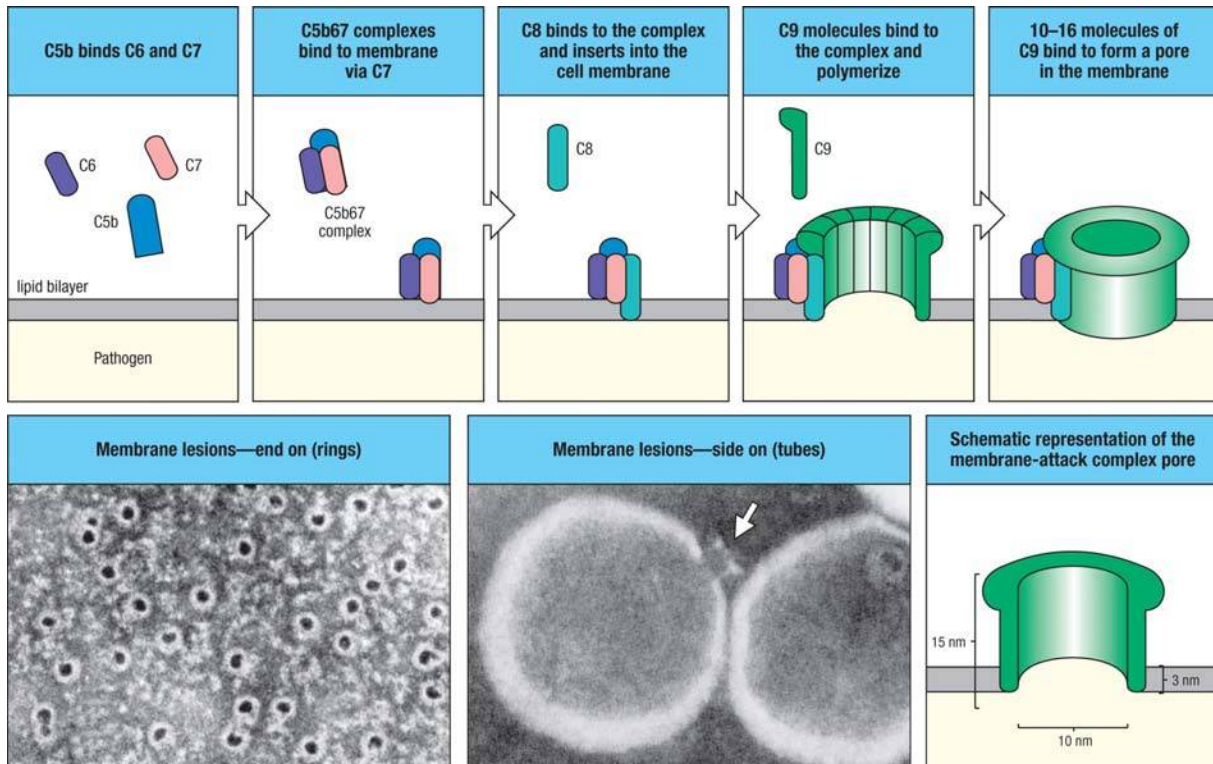


Fig. 2.35 Assembly of the membrane-attack complex generates a pore in the lipid bilayer membrane. The sequence of steps and their approximate appearance are shown here in schematic form. C5b triggers the assembly of a complex of one molecule each of C6, C7, and C8, in that order. C7 and C8 undergo conformational changes, exposing hydrophobic domains that insert into the membrane. This complex causes moderate membrane damage in its own right, and also serves to induce the polymerization of C9, again with the exposure of a hydrophobic site. Up to 16 molecules of C9 are then added to the assembly to generate a channel 10 nm in diameter in the membrane. This channel disrupts the bacterial cell membrane, killing the bacterium. The electron micrographs show erythrocyte membranes with membrane-attack complexes in two orientations, end on and side on.

The first step in the formation of the membrane-attack complex is the cleavage of C5 by a C5 convertase to release C5b (see Fig. 2.29). In the next stages (see Fig. 2.35), C5b initiates the assembly of the terminal complement components and their insertion into the cell membrane. The process begins when one molecule of C5b binds one molecule of C6, and the C5b6 complex then binds one molecule of C7. This reaction leads to a conformational change that exposes a hydrophobic site on C7, which inserts into the lipid bilayer. Similar hydrophobic sites are exposed on the later components C8 and C9 when they are bound to the complex, allowing these proteins also to insert into the lipid bilayer. C8 is a complex of two proteins, C8 β and C8 α - γ . The C8 β protein binds to C5b, and this binding of C8 β to the membrane-associated C5b67 complex allows the hydrophobic domain of C8 α - γ to insert into the lipid bilayer. Finally, C8 α - γ induces the polymerization of 10–16 molecules of C9 into a

pore-forming structure called the membrane-attack complex. The membrane-attack complex has a hydrophobic external face, allowing it to associate with the lipid bilayer, but a hydrophilic internal channel. The diameter of this channel is about 10 nm, allowing the free passage of solutes and water across the lipid bilayer. The pore damage to the lipid bilayer leads to the loss of cellular homeostasis, the disruption of the proton gradient across the membrane, the penetration of enzymes such as lysozyme into the cell, and the eventual destruction of the pathogen.

Although the effect of the membrane-attack complex is very dramatic, particularly in experimental demonstrations in which antibodies against red blood cell membranes are used to trigger the complement cascade, the significance of these components in host defense seems to be quite limited. So far, deficiencies in C5–C9 have been associated with susceptibility only to *Neisseria* species, the bacteria that cause the sexually transmitted disease gonorrhea and a common form of bacterial meningitis. Thus, the opsonizing and inflammatory actions of the earlier components of the complement cascade are clearly more important for host defense against infection. Formation of the membrane-attack complex seems to be important only for the killing of a few pathogens, although, as we will see in [Chapter 15](#), this complex might well have a major role in immunopathology.

2-16 Complement-control proteins regulate all three pathways of complement activation and protect the host from their destructive effects.

Complement activation usually is initiated on a pathogen surface, and the activated complement fragments that are produced usually bind nearby on the pathogen surface or are rapidly inactivated by hydrolysis. Even so, all complement components are activated spontaneously at a low rate in plasma, and these activated complement components will sometimes bind proteins on host cells. [Section 2-10](#) introduced the soluble host proteins factor I and factor H and the membrane-bound proteins MCP and DAF that regulate the alternative pathway of complement activation. In addition, several other soluble and membrane-bound complement-control proteins can regulate the complement cascade at various steps to protect normal host cells while allowing complement activation to proceed on pathogen surfaces (**Fig. 2.36**).

Regulatory proteins of the classical and alternative pathways			
Soluble factors regulating complement			
Name	Ligand/Binding factor	Action	Pathology if defective

C1 inhibitor (C1INH)	C1r, C1s (C1); MASP-2 (MBL)	Binds and deactivates C1r, C1s, and MASP-2	Hereditary angiodema
C4-binding protein (C4BP)	C4b	Displaces C2a from C4b cofactor for C4b cleavage by factor I	
CPN1 (Carboxypeptidase N)	C3a, C5a	Inactivates C3a and C5a	Recurrent angioedema
Factor H	C3b	Displaces Bb from C3b, cofactor for C3b cleavage by factor I	Age-related macular degeneration, atypical hemolytic uremic syndrome
Factor I	C3b, C4b	Serine protease, cleaves C3b and C4b	Low C3 levels, atypical hemolytic uremic syndrome
Membrane-bound factors regulating complement			
Name	Ligand/Binding factor	Action	Pathology if defective
CR1g	C3b, iC3b	Inhibits activation of alternative pathway	Increased susceptibility to blood-borne infections
Complement receptor 1 (CR1, CD35)	C3b, C4b	Cofactor for factor I; displaces Bb from C3b, and C2a from C4b	
Decay-accelerating factor (DAF, CD55)	C3 convertase	Displaces Bb and C2a from C3b and C4b respectively	Paroxysmal nocturnal hemoglobinuria
Membrane cofactor of proteolysis (MCP, CD46)	C3b, C4b	Cofactor for factor I	Atypical hemolytic uremic syndrome
Protectin (CD59)	C8	Inhibits MAC formation	Paroxysmal nocturnal hemoglobinuria

Fig. 2.36 The soluble and membranebound proteins that regulate the activity of complement.



Case Study: Hereditary Angioedema

The activation of C1 is controlled by the [C1 inhibitor \(C1INH\)](#), which is a plasma [serine protease inhibitor](#), or [serpin](#). C1INH binds to the catalytic sites of C1r and C1s and

irreversibly inactivates them (**Fig. 2.37**). In this way, C1INH limits cleavage and activation of C4 and C2 by C1. By the same means, C1INH limits the spontaneous activation of C1 in plasma. Its importance can be seen in the C1INH deficiency disease [hereditary angioedema \(HAE\)](#), in which chronic spontaneous complement activation leads to the production of excess cleaved fragments of C4 and C2. The large activated fragments from this cleavage (C2a and C4b), which normally combine to form the C3 convertase, do not damage host cells in such individuals because C4b is rapidly inactivated by hydrolysis in plasma, and the convertase does not form. C1INH also regulates the plasma protease kallikrein, a component of the kinin system (discussed in [Section 3-3](#)). Kallikrein is normally activated by tissue damage, but in HAE the lack of C1INH leads to increased kallikrein activity and thus to the uncontrolled production of bradykinin. Hereditary angioedema is fully corrected by replacing C1INH. A similar, extremely rare human disease stems from a partial deficiency of [carboxypeptidase N \(CPN\)](#), a metalloproteinase that inactivates the anaphylatoxins C3a and C5a as well as bradykinin and kallikrein. Humans with partial CPN deficiency exhibit recurrent angioedema due to delayed inactivation of serum C3a and bradykinin.

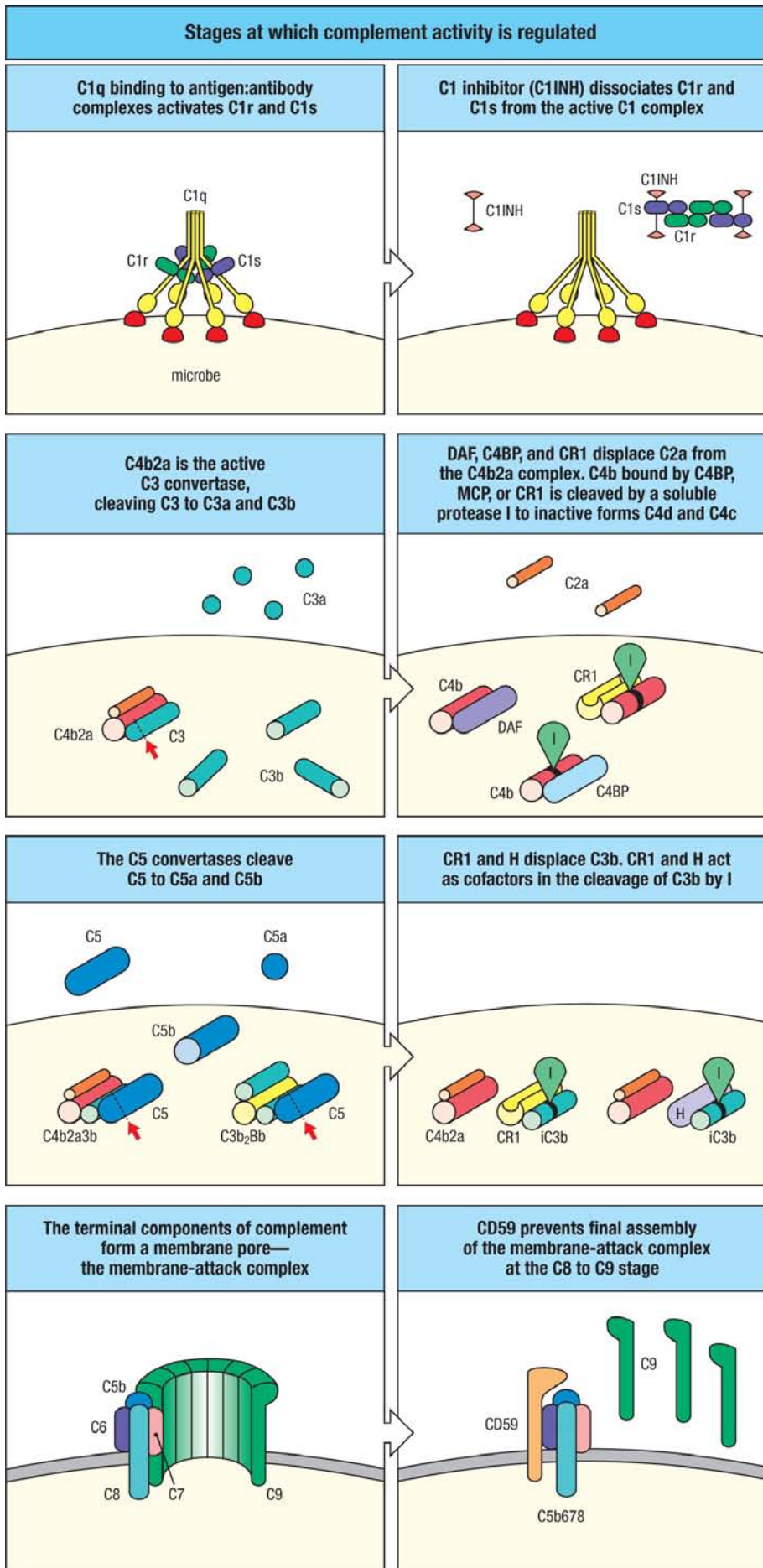


Fig. 2.37 Complement activation is regulated by a series of proteins that serve to protect host cells from accidental damage. These act on different stages of the complement cascade, dissociating complexes or catalyzing the enzymatic degradation of covalently bound complement proteins. Stages in the complement cascade are shown schematically down the left side of the figure, with the regulatory reactions on the right. The alternative pathway C3 convertase is similarly regulated by DAF, CR1, MCP, and factor H.

Because the highly reactive thioester bond of activated C3 and C4 cannot distinguish acceptor groups on a host cell from those on the surface of a pathogen, mechanisms have evolved to prevent the small amounts of C3 or C4 molecules deposited on host cells from fully triggering complement activation. We introduced these mechanisms in the context of control of the alternative pathway (see Fig. 2.27), but they are also important regulators of the classical pathway convertase (see Fig. 2.37, second and third rows). [Section 2-10](#) described the proteins that inactivate any C3b or C4b that has bound to host cells. These are the plasma factor I and its cofactors MCP and CR1, which are membrane proteins. Circulating factor I is an active serine protease, but it can cleave C3b and C4b only when they are bound to MCP, factor H, or CR1 (see Fig. 2.27). In these circumstances, factor I cleaves C3b, first into iC3b and then further to C3dg, thus permanently inactivating it. C4b is similarly inactivated by cleavage into C4c and C4d. Microbial cell walls lack MCP and CR1 and thus cannot promote the breakdown of C3b and C4b, which instead act as binding sites for factor B and C2, promoting complement activation. The importance of factor I can be seen in people with genetically determined [factor I deficiency](#). Because of uncontrolled complement activation, complement proteins rapidly become depleted, and such people suffer repeated bacterial infections, especially with ubiquitous pyogenic bacteria.

There are also plasma proteins with cofactor activity for factor I, most notably [C4b-binding protein \(C4BP\)](#) (see Fig. 2.36). It binds C4b and acts mainly as a regulator of the classical pathway in the fluid phase. Factor H binds C3b in the fluid phase, as well as at cell membranes, and helps to distinguish the C3b that is bound to host cells from that bound to microbial surfaces. The higher affinity of factor H for sialic acid residues on host-membrane glycoproteins allows it to displace factor B in binding to C3b on host cells. Also, C3b at host-cell membranes is bound by cofactor protein MCP. Factor H competes with factor B for binding to C3b bound to host cells, so that the bound C3b is catabolized by factor I into iC3b and C3dg, and complement activation is inhibited. In contrast, microbial membranes do not express MCP and lack the sialic acid modifications that attract factor H, and so factor B is favored for binding C3b on these surfaces. The greater amount of factor B on a microbial surface stimulates formation of more C3bBb, the alternative pathway C3 convertase, and thus amplifies complement activation.

2-17 Genetic and acquired disorders in complement regulation can produce various

inflammatory conditions.

The critical balance between the inhibition and the activation of complement on cell surfaces is illustrated in individuals heterozygous for mutations in various complement-regulatory proteins. Several renal diseases are caused by complement dysregulation. A condition known as [atypical hemolytic uremic syndrome \(aHUS\)](#) can be caused by relatively rare mutations in factor H, factor I, and MCP. The reduced action of these regulatory factors can increase the alternative pathway activation on the surface of platelets and red blood cells. The damage that this produces can lead to low levels of platelets (thrombocytopenia) and red blood cells (anemia) and can damage the kidneys. aHUS can also be caused by an acquired autoantibody that binds to factor H and reduces its activity, again leading to increased alternative pathway activity. A related condition, called [C3 glomerulopathy](#), can be caused by similar mutations in these factors. It is characterized by deposition of C3b in the renal glomerulus but is not associated with thrombocytopenia or anemia. Like aHUS, C3 glomerulopathy is more commonly caused by autoantibodies that inhibit factor H. Such autoantibodies are called [nephritic factors](#), as they lead to inflammation in the kidney, and they can be directed at other components of complement. One type of nephritic factor is targeted to C3, and stabilizes the alternative pathway C3 convertase C3bBb, rather than inhibiting a complement regulatory protein.

Another serious health problem related to complement malfunction is a significantly increased risk of [age-related macular degeneration \(AMD\)](#). AMD is the leading cause of blindness in the elderly in developed countries. It has been linked to single-nucleotide polymorphisms in the factor H gene, where tyrosine 402 is changed to histidine (Y402H). This polymorphism in factor H is not the primary cause of AMD, but it influences the response to retinal injury. Cellular debris produced during injury or renewal of photoreceptors, such as oxidized lipids, can activate complement and induce local inflammation. Normal factor H can bind these oxidized lipids and inactivate C3b that would otherwise trigger further complement activation. The Y402H variant of factor H interacts more weakly with oxidized lipids than does normal factor H, and consequently allows a chronic background of complement activation and inflammation to take place, which damages retinal epithelial cells and activates macrophages. Polymorphisms in other complement genes have also been found to be either detrimental or protective for this disease. Thus, even small alterations in the efficiency of either the activation or the regulation of this powerful effector system can contribute to the progression of degenerative or inflammatory disorders.

DAF and MCP also mediate protection against complement through a third mechanism, which augments the dissociation of C4b2a and C3bBb convertases that have already formed. CR1 is among the host-cell membrane molecules that regulate complement through both these mechanisms; that is, by promoting the dissociation of convertase and exhibiting cofactor activity. All the proteins that bind the homologous C4b and C3b molecules contain three or more tandem copies of a structural element called the short consensus repeat (SCR),

the complement-control protein (CCP) repeat, or (especially in Japan) the sushi domain.

In addition to the mechanisms for preventing C3 convertase formation and C4 and C3 deposition on cell membranes, there are also inhibitory mechanisms that prevent the inappropriate insertion of the membrane-attack complex (MAC) into membranes. We saw in [Section 2-15](#) that the membrane-attack complex polymerizes onto C5b molecules created by the action of C5 convertase. The membrane-attack complex mainly inserts into cell membranes adjacent to the site of the C5 convertase; that is, close to the site of complement activation on a pathogen. However, some newly formed membrane-attack complexes may diffuse from the site of complement activation and insert into adjacent host-cell membranes. Several plasma proteins, including, notably, [vitronectin](#) (also known as [S-protein](#)), bind to the C5b67, C5b678, and C5b6789 complexes and thereby inhibit their random insertion into cell membranes. Host-cell membranes also contain the intrinsic protein [CD59](#), or [protectin](#), which inhibits the binding of C9 to the C5b678 complex (see Fig. 2.37, bottom row). CD59 and DAF are both linked to the cell surface by a [glycosylphosphatidylinositol \(GPI\) tail](#), like many other peripheral membrane proteins. One of the enzymes involved in the synthesis of GPI tails is encoded by a gene, *PIGA*, on the X chromosome. In people with a somatic mutation in this gene in a clone of hematopoietic cells, both CD59 and DAF fail to function. This causes the disease [paroxysmal nocturnal hemoglobinuria](#), which is characterized by episodes of intravascular red blood cell lysis by complement. Red blood cells that lack only CD59 are also susceptible to destruction as a result of spontaneous activation of the complement cascade.

2-18 Pathogens produce several types of proteins that can inhibit complement activation.

Bacterial pathogens have evolved various strategies to avoid activation of complement, and thereby to avoid elimination by this first line of innate defense (**Fig. 2.38**). One strategy that many pathogens employ is to mimic host surfaces by attracting host complement regulators to their own surfaces. A mechanism to achieve this is for the pathogen to express surface proteins that bind to soluble complement-regulatory proteins such as C4BP and factor H; for example, the Gram-negative pathogen *Neisseria meningitidis* produces [factor H binding protein \(fHbp\)](#), which recruits factor H (see [Section 2-10](#)), and the outer membrane protein [PorA](#), which binds to C4BP. By recruiting factor H and C4BP to the pathogen membrane, the pathogen is able to inactivate C3b that is deposited on its surface and thereby avoid the consequences of complement activation. Complement is important in defense against *Neisseria* species, and several complement deficiencies are associated with increased susceptibility to this pathogen.

Pathogen	Evasion molecule	Host target	Mechanism of action
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Membrane proteins			
<i>Neisseria meningitidis</i>	Factor H binding protein (fHbp)	Factor H	Inactivates bound C3b
<i>Borrelia burgdorferi</i>	Outer surface protein E (OspE)	Factor H	Inactivates bound C3b
<i>Streptococcus pneumoniae</i>	Pneumococcal surface protein C (PspC)	Factor H	Inactivates bound C3b
Secreted proteins			
<i>Neisseria meningitidis</i>	PorA	C4BP	Inactivates bound C3b
<i>Staphylococcus aureus</i>	Clumping factor A (ClfA)	Factor I	Inactivates bound C3b
<i>Staphylococcus aureus</i>	Staphylococcal protein A (Spa)	Immunoglobulin	Binds to Fc regions and interferes with C1 activation
<i>Staphylococcus aureus</i>	Staphylokinase (SAK)	Immunoglobulin	Cleaves immunoglobulins
<i>Staphylococcus aureus</i>	Complement inhibitor (SCIN, CINB, SCINC)	C3 convertase (C4b2a, C3bBb)	Inhibition of convertase activity

Fig. 2.38 Complement evasion proteins produced by various pathogens.

Another strategy employed by pathogens is to secrete proteins that directly inhibit components of complement. The Gram-positive pathogen *Staphylococcus aureus* provides several examples of this type of strategy. [Staphylococcal protein A \(Spa\)](#) binds to the Fc regions of immunoglobulins and interferes with the recruitment and activation of C1. This binding specificity was used as an early biochemical technique in the purification of antibodies. The staphylococcal protein [staphylokinase \(SAK\)](#) acts by cleaving immunoglobulins bound to the pathogen membrane, preventing complement activation and avoiding phagocytosis. The [staphylococcal complement inhibitor \(SCIN\)](#) protein binds to the classical C3 convertase, C4b2a, and the alternative pathway C3 convertase, C3bBb, and inhibits their activity. Other stages of complement activation, including formation of the C5 convertase, are targets of inhibition by proteins produced by these and other pathogens. We will return to this topic of complement regulation in [Chapter 13](#) when we discuss how the immune system sometimes fails or is evaded by pathogens.

Summary.

The complement system is one of the major mechanisms by which pathogen recognition is converted into an effective host defense against initial infection. Complement is a system of plasma proteins that can be activated directly by pathogens or indirectly by pathogen-bound antibody, leading to a cascade of reactions that occurs on the surface of pathogens and generates active components with various effector functions. There are three pathways of complement activation: the lectin pathway, triggered by the pattern-recognition receptors MBL and the ficolins; the classical pathway, triggered directly by antibody binding to the pathogen surface; and the alternative pathway, which utilizes spontaneous C3 deposition onto microbial surfaces, is augmented by properdin, and provides an amplification loop for the other two pathways. The early events in all pathways consist of a sequence of cleavage reactions in which the larger cleavage product binds covalently to the pathogen surface and contributes to the activation of the next component. The pathways converge with the formation of a C3 convertase enzyme, which cleaves C3 to produce the active cleavage product C3b. The binding of large numbers of C3b molecules to the pathogen is the central event in complement activation. Bound complement components, especially bound C3b and its inactive fragments, are recognized by specific complement receptors on phagocytic cells, which engulf pathogens opsonized by C3b and its inactive fragments. The small cleavage fragments of C3, C4, and C5 act on specific trimeric G protein-coupled receptors to recruit phagocytes, such as neutrophils, to sites of infection or to alter local vascular permeability and coagulation. Together, these activities promote the uptake and destruction of pathogens by phagocytes. The molecules of C3b that bind the C3 convertase itself initiate the late events of complement, binding C5 to make it susceptible to cleavage by C2a or Bb. The larger C5b fragment triggers the assembly of a membrane-attack complex, which can result in the lysis of certain pathogens. A system of soluble and membrane-bound complement-regulatory proteins acts to limit complement activation on host tissues, in order to prevent tissue damage from the inadvertent binding of activated complement components to host cells or from the spontaneous activation of complement components in plasma. Many pathogens produce a variety of soluble and membrane-associated proteins that can counteract complement activation and contribute to infection of the host by the microbe.

Summary to [Chapter 2](#).

This chapter has described the preexisting, constitutive components of innate immunity. The body's epithelial surfaces are a constant but imperfect barrier to pathogen entry. In addition to their other specialized functions, epithelia have specialized adaptations, such as cilia, various antimicrobial molecules, and mucus that provide the simplest form of innate immunity. The complement system is a more specialized system that combines direct recognition of microbes with a complex effector system. The lectin pathway relies on pattern-recognition receptors that detect microbial membranes, while the alternative pathway relies on spontaneous complement activation that is down-regulated by host molecules expressed on self membranes. The main event in complement activation is accumulation of C3b on microbial membranes, which is recognized by complement receptors on phagocytic cells to promote microbial clearance by cells recruited to sites of infection by C3a and C5a. In addition, C5b initiates the membrane-attack complex that is able to lyse some microbes directly. The complement cascade is under regulation to prevent attack on host tissues, and genetic variation in regulatory pathways can result in autoimmune syndromes and age-related tissue damage.

Glossary

[complement](#)

A set of plasma proteins that act together as a defense against pathogens in extracellular spaces. The pathogen becomes coated with complement proteins that facilitate its removal by phagocytes and that can also kill certain pathogens directly. Activation of the complement system can be initiated in several different ways. *See* **classical pathway, alternative pathway, lectin pathway**.

[opsonization](#)

The coating of the surface of a pathogen by antibody and/or complement that makes it more easily ingested by phagocytes.

[complement proteins](#)

See **C1, C2, C3**, etc.

[complement activation](#)

The activation of the normally inactive proteins of the complement system that occurs on infection. *See* **classical pathway, alternative pathway, lectin pathway**.

[zymogen](#)

An inactive form of an enzyme, usually a protease, that must be modified in some way, for example by selective cleavage of the protein chain, before it can become active.

[inflammation](#)

General term for the local accumulation of fluid, plasma proteins, and white blood cells that is initiated by physical injury, infection, or a local immune response.

[phagocytosis](#)

The internalization of particulate matter by cells by a process of engulfment, in which the cell membrane surrounds the material, eventually forming an intracellular vesicle (phagosome) containing the ingested material.

[membrane attack](#)

Effector pathway of complement that is based on formation of the membrane-attack complex (MAC).

[C2](#)

Complement protein of the classical and lectin pathways that is cleaved by the C1 complex to yield C2b and C2a. C2a is an active protease that forms part of the classical C3 convertase C4b2a.

[C2a](#)

The large fragment of C2 formed when C2 is cleaved by C1s. When bound to C4b, C2a is the serine protease of the C3 convertase (C4b2a) of the classical and the lectin pathways.

[lectin pathway](#)

Complement-activation pathway that is triggered by mannose-binding lectin (MBL) or ficolins bound to bacteria.

[classical pathway](#)

The complement-activation pathway that is initiated by C1 binding either directly to bacterial surfaces or to antibody bound to the bacteria, thus flagging the bacteria as foreign. *See also* **alternative pathway**, **lectin pathway**.

[alternative pathway](#)

A form of complement activation that is initiated by spontaneous hydrolysis of C3 and which uses factor B and factor D to form the unique C3 convertase C3bBb.

[C3 convertase](#)

Enzyme complex that cleaves C3 to C3b and C3a on the surface of a pathogen. The C3 convertase of the classical and lectin pathways is formed from membrane-bound C4b complexed with the protease C2a. The alternative pathway C3 convertase is formed from membrane-bound C3b complexed with the protease Bb.

[C3b](#)

Complement protein on which all complement activation pathways converge. C3 cleavage forms C3b, which can bind covalently to microbial surfaces, where it promotes destruction by phagocytes.

[C3b₂Bb](#)

The C5 convertase of the alternative pathway of complement activation.

[C3a](#)

Pro-inflammatory complement fragments C5a and C3a released by cleavage during complement activation. They are recognized by specific receptors and recruit fluid and inflammatory cells to the site of their release.

[C3f](#)

A small fragment of C3b that is removed by factor I and MCP to leave iC3b on the microbial surface.

[C3dg](#)

Breakdown product of iC3b that remains attached to the microbial surface, where it can

bind complement receptor 2 (CR2).

[C5 convertase](#)

Enzyme complex that cleaves C5 to C5a and C5b.

[C5a](#)

Pro-inflammatory complement fragments C5a and C3a released by cleavage during complement activation. They are recognized by specific receptors and recruit fluid and inflammatory cells to the site of their release.

[C5b](#)

Fragment of C5 that initiates the formation of the membrane-attack complex (MAC).

[membrane-attack complex \(MAC\)](#)

Protein complex composed of C5b to C9 that assembles a membrane-spanning hydrophilic pore on pathogen surfaces, causing cell lysis.

[mannose-binding lectin \(MBL\)](#)

Mannose-binding protein present in the blood. It can opsonize pathogens bearing mannose on their surfaces and can activate the complement system via the lectin pathway, an important part of innate immunity.

[collectins](#)

A family of calcium-dependent sugar-binding proteins (lectins) containing collagen-like sequences. An example is mannose-binding lectin (MBL).

[avidity](#)

The sum total of the strength of binding of two molecules or cells to one another at multiple sites. It is distinct from affinity, which is the strength of binding of one site on a molecule to its ligand.

[acute-phase response](#)

A change in the proteins present in the blood that occurs during the early phases of an infection. It includes the production of acute-phase proteins, many of which are produced in the liver.

[ficolins](#)

Carbohydrate-binding proteins that can initiate the lectin pathway of complement activation. They are members of the collectin family and bind to the N-acetylglucosamine present on the surface of some pathogens.

[MBL-associated serine proteases](#)

See **MASP-1, MASP-2, MASP-3**.

[MAp19](#)

A non-enzymatically active splice variant encoded by the MASP-2 gene that can associate with mannose-binding lectin (MBL) and may regulate lectin pathway complement activation.

[MAp44](#)

A non-enzymatically active splice variant encoded by the MASP-1 gene that can associate with mannose-binding lectin (MBL) and may regulate lectin pathway complement activation.

[C4](#)

Complement protein of the classical and lectin pathways. C4 is cleaved by C1s to C4b, which forms part of the classical C3 convertase.

C4b2a

C3 convertase of the classical and lectin pathways of complement activation.

surfactant proteins A and D (SP-A and SP-D)

Acute-phase proteins that help protect the epithelial surfaces of the lung against infection.

C1 complex (C1)

Protein complex activated as the first step in the classical pathway of complement activation, composed of C1q bound to two molecules each of the proteases C1r and C1s. Binding of a pathogen or antibody to C1q activates C1r, which cleaves and activates C1s, which cleaves C4 and C2.

C1 complex (C1)

Protein complex activated as the first step in the classical pathway of complement activation, composed of C1q bound to two molecules each of the proteases C1r and C1s. Binding of a pathogen or antibody to C1q activates C1r, which cleaves and activates C1s, which cleaves C4 and C2.

C1q

A component of the C1 complex formed by a hexamer of trimers of a polypeptide with a globular domain that can bind bacterial surfaces or interact with Fc regions of antibodies and a carboxy-terminal collagen-like domain that mediates its assembly and association with C1r and C1s.

C1r

A serine protease that associates with C1q and C1s to help form the C1 complex. Its autocatalytic activity is triggered by conformational changes resulting from C1q binding to Fc or bacterial surfaces; C1r then cleaves C1s.

C1s

A serine protease that associates with C1q and C1r to help form the C1 complex. It is cleaved and activated by C1r and then functions to activate the downstream complement components 4 and 2 (C4 and C2).

classical C3 convertase

The complex of activated complement components C4b2a, which cleaves C3 to C3b on pathogen surfaces in the classical pathway of complement activation.

natural antibodies

Antibodies produced by the immune system in the apparent absence of any infection. They have a broad specificity for self and microbial antigens, can react with many pathogens, and can activate complement.

alternative pathway C3 convertase

The C3 convertase composed of C4b and C2a. This is also the C3 convertase of the classical complement pathway.

factor B

Protein in the alternative pathway of complement activation, in which it is cleaved to Ba and an active protease, Bb, the latter binding to C3b to form the alternative pathway C3 convertase, C3bBb.

C3bBb

The C3 convertase of the alternative pathway of complement activation.

[factor D](#)

A serine protease in the alternative pathway of complement activation, which cleaves factor B into Ba and Bb.

[tickover](#)

The low-level generation of C3b continually occurring in the blood in the absence of infection.

[properdin](#)

Plasma protein released by activated neutrophils that stabilizes the C3 convertase C3bBb of the alternative pathway. Also known as factor P.

[complement-regulatory proteins](#)

Proteins that control complement activity and prevent complement from being activated on the surfaces of host cells.

[iC3b](#)

Inactive complement fragment produced by cleavage of C3b.

[factor I](#)

Complement-regulatory protease in plasma that cleaves C3b to the inactive derivative iC3b, thus preventing the formation of a C3 convertase.

[CR1 \(CD35\)](#)

A receptor expressed by phagocytic cells that binds to C3b. It stimulates phagocytosis and inhibits C3 convertase formation on host-cell surfaces. It also acts in conjunction with factor I to cleave C3b to its inactive derivative iC3b and thus prevent convertase formation.

[CR1 \(CD35\)](#)

A receptor expressed by phagocytic cells that binds to C3b. It stimulates phagocytosis and inhibits C3 convertase formation on host-cell surfaces. It also acts in conjunction with factor I to cleave C3b to its inactive derivative iC3b and thus prevent convertase formation.

[factor H](#)

Complement-regulatory protein in plasma that binds C3b and competes with factor B to displace Bb from the convertase. It also acts in conjunction with factor I to cleave C3b to its inactive derivative iC3b and thus prevent convertase formation.

[amoeboid coelomocytes](#)

Phagocytic cells found in more primitive organisms (such as echinoderms and others) that carry out basic protective functions; these cells are a source of C3 homologs, thioester-containing proteins (TEPs), during infection by bacteria.

[thioester-containing proteins \(TEPs\)](#)

Homologs of complement component 3 (C3) that are found in insects and are thought to have some function in insect innate immunity.

[thioester-containing proteins \(TEPs\)](#)

Homologs of complement component 3 (C3) that are found in insects and are thought to have some function in insect innate immunity.

[C4b2a3b](#)

C5 convertase of the classical and lectin pathways of complement activation.

[complement receptors \(CRs\)](#)

Cell-surface proteins of various types that recognize and bind complement proteins that have become bound to an antigen such as a pathogen. Complement receptors on phagocytes enable them to identify and bind pathogens coated with complement proteins and to ingest and destroy them. See **CR1, CR2, CR3, CR4, CR1g**, and the **C1 complex**.

[C5a receptor 1](#)

A G protein–coupled receptor expressed by macrophages and activated by C5a that facilitates ingestion of opsonized bacteria bound to CR1.

[G protein–coupled receptors \(GPCRs\)](#)

A large class of seven-span transmembrane cell-surface receptors that associate with intracellular heterotrimeric G proteins after ligand binding and signal by activation of the G protein. Important examples are the chemokine receptors.

[C5a receptor 2](#)

A nonsignaling surface receptor expressed by neutrophils and macrophages and which may act as a decoy receptor for C5a and thus regulate activity of the C5a receptor.

[CR2 \(CD21\)](#)

Complement receptor that is part of the B-cell co-receptor complex. It binds to antigens coated with breakdown products of C3b, especially C3dg, and, by cross-linking the B-cell receptor, enhances sensitivity to antigen at least 100-fold. It is also the receptor used by the Epstein–Barr virus to infect B cells.

[CR2 \(CD21\)](#)

Complement receptor that is part of the B-cell co-receptor complex. It binds to antigens coated with breakdown products of C3b, especially C3dg, and, by cross-linking the B-cell receptor, enhances sensitivity to antigen at least 100-fold. It is also the receptor used by the Epstein–Barr virus to infect B cells.

[CR3 \(CD11b:CD18\)](#)

Complement receptor 3. A $\beta 2$ integrin that acts both as an adhesion molecule and as a complement receptor. CR3 on phagocytes binds iC3b, a breakdown product of C3b on pathogen surfaces, and stimulates phagocytosis.

[CR4 \(CD11c:CD18\)](#)

A $\beta 2$ integrin that acts both as an adhesion molecule and as a complement receptor. CR4 on phagocytes binds iC3b, a breakdown product of C3b on pathogen surfaces, and stimulates phagocytosis.

[CR1g \(complement receptor of the immunoglobulin superfamily\)](#)

A complement receptor that binds to inactivated forms of C3b.

[anaphylactic shock](#)

A rapid-onset and systemic allergic reaction to antigen; for example, to insect venom injected directly into the bloodstream or to foods such as peanuts. Severe systemic reactions can potentially be fatal due to circulatory collapse and suffocation from tracheal swelling. Anaphylaxis usually results from antigens binding to IgE bound by Fc ϵ receptors on mast cells, leading to systemic release of inflammatory mediators.

[anaphylatoxins](#)

Pro-inflammatory complement fragments C5a and C3a released by cleavage during complement activation. They are recognized by specific receptors and recruit fluid and

inflammatory cells to the site of their release.

C1 inhibitor (C1INH)

An inhibitor protein for C1 that binds and inactivates C1r:C1s enzymatic activity.

Deficiency in C1INH causes hereditary angioedema through production of vasoactive peptides that cause subcutaneous and laryngeal swelling.

serine protease inhibitor (serpin)

Class of proteins that inhibit various proteases, originally referring to those specific to serine proteases.

serine protease inhibitor (serpin)

Class of proteins that inhibit various proteases, originally referring to those specific to serine proteases.

hereditary angioedema (HAE)

A genetic deficiency of the C1 inhibitor of the complement system. In the absence of C1 inhibitor, spontaneous activation of the complement system can cause diffuse fluid leakage from blood vessels, the most serious consequence of which is swelling of the larynx, leading to suffocation.

carboxypeptidase N (CPN)

A metalloproteinase that inactivates C3a and C5a. CPN deficiency causes a condition of recurrent angioedema.

factor I deficiency

A genetically determined lack of the complement-regulatory protein factor I. This results in uncontrolled complement activation, so that complement proteins rapidly become depleted. Those with the deficiency suffer repeated bacterial infections, especially with ubiquitous pyogenic bacteria.

C4b-binding protein (C4BP)

A complement-regulatory protein that inactivates the classical pathway C3 convertase formed on host cells by displacing C2a from the C4b2a complex. C4BP binds C4b attached to host cells, but cannot bind C4b attached to pathogens.

atypical hemolytic uremic syndrome (aHUS)

A condition characterized by damage to platelets and red blood cells and inflammation of the kidneys that is caused by uncontrolled complement activation in individuals with inherited deficiencies in complement-regulatory proteins.

C3 glomerulopathy

An inflammatory condition in the kidney in which C3b is abnormally deposited in the glomerulus leading to proteinuria and hematuria. It can be associated with autoantibodies, called nephritic factors, that can inhibit factor H, other complement-regulatory components, or that stabilize C3 convertase, leading to increased C3 deposition.

nephritic factor

An autoantibody that leads to inflammation in the kidney by increasing C3 deposition. Nephritic factors may inhibit the activity of complement-regulatory proteins, such as factor H, or may stabilize the alternative pathway C3 convertase C3bBb.

age-related macular degeneration (AMD)

A leading cause of blindness in the elderly, for which some single-nucleotide

polymorphisms (SNPs) in the factor H genes confer an increased risk.

vitronectin

Plasma protein that binds incompletely formed MAC complexes, such as C5b67, preventing bystander complement damage to host membranes.

S-protein (vitronectin)

Plasma protein that binds incompletely formed MAC complexes, such as C5b67, preventing bystander complement damage to host membranes.

glycosylphosphatidylinositol (GPI) tail

A glycolipid modification of proteins that can allow attachment to host membranes without the requirement of a transmembrane protein domain.

paroxysmal nocturnal hemoglobinuria

A disease in which complement-regulatory proteins are defective, so that activation of complement binding to red blood cells leads to episodes of spontaneous hemolysis.

factor H binding protein (fHbp)

A protein produced by the pathogen *Neisseria meningitidis* that recruits factor H to its membrane, thereby inactivating C3b deposited on its surface, and evading destruction by complement.

PorA

Outer membrane protein of *Neisseria meningitidis* that binds C4BP, thereby inactivating C3b deposited on its surface.

staphylococcal protein A (Spa)

Staphylococcal protein that blocks the binding of the antibody Fc region with C1, thereby preventing complement activation.

staphylokinase (SAK)

Staphylococcal protease that cleaves immunoglobulins bound to its surface, thereby preventing complement activation.

staphylococcal complement inhibitor (SCIN)

Staphylococcal protein that inhibits the activity of the classical and alternative C3 convertases, promoting the evasion of destruction by complement.

MASP-1, MASP-2, MASP-3

Serine proteases of the classical and lectin pathways of complement activation that bind to C1q, ficolins, and mannose-binding lectin, and function in their activation to cleave C4.

MASP-1, MASP-2, MASP-3

Serine proteases of the classical and lectin pathways of complement activation that bind to C1q, ficolins, and mannose-binding lectin, and function in their activation to cleave C4.

fluid-phase C3 convertase

Short-lived alternative pathway C3 convertase, C3(H₂O)Bb, that is continually produced at a low level in the plasma that can initiate activation of the alternative pathway of complement activation.

decay-accelerating factor (DAF or CD55)

A cell-surface protein that protects cells from lysis by complement. Its absence causes the disease paroxysmal nocturnal hemoglobinuria.

membrane cofactor of proteolysis (MCP or CD46)

A complement-regulatory protein; a host-cell membrane protein that acts in conjunction with factor I to cleave C3b to its inactive derivative iC3b and thus prevent convertase formation.

C6, C7, C8, C9

Complement proteins that act with C5b to form the membrane-attack complex, producing a pore that leads to lysis of the target cell.

C6, C7, C8, C9

Complement proteins that act with C5b to form the membrane-attack complex, producing a pore that leads to lysis of the target cell.

C6, C7, C8, C9

Complement proteins that act with C5b to form the membrane-attack complex, producing a pore that leads to lysis of the target cell.

C6, C7, C8, C9

Complement proteins that act with C5b to form the membrane-attack complex, producing a pore that leads to lysis of the target cell.

CD59, protectin

Cell-surface protein that protects host cells from complement damage by blocking binding of C9 to the C5b678 complex, thus preventing MAC formation.

CD59, protectin

Cell-surface protein that protects host cells from complement damage by blocking binding of C9 to the C5b678 complex, thus preventing MAC formation.

Discussion Questions

1. What activities and structural features of the skin, lungs, and intestine are used to keep microbes from penetrating beyond the epithelial boundary?
2. Most antimicrobial peptides do not exert their actions within the host cells that produce them. What structural features of such peptides prevent their activation within neutrophils and Paneth cells?
3. The proteins of the complement cascade all circulate freely within the blood plasma. How is it that complement is normally activated only during an infection?
4. Recognition of bacteria by mannose-binding lectin and ficolins leads to their destruction by the same mechanisms as those of recognition of bacteria by antibodies. Explain how this is so. What evolutionary adaptation was necessary to allow for this?
5. Spontaneous hydrolysis of the thioester bond in C3 occurs continually at a low rate even in the absence of infection. Contrast the reactions that occur as a consequence of this C3 'tickover' in a healthy person with those that occur in a person infected with bacteria.
6. What is the central feature of the amplification loop in the alternative pathway? As this pathway is initiated spontaneously, how are host tissues protected from nonspecific damage by this pathway?
7. How do acquired antibodies directed against factor H lead to atypical hemolytic uremic syndrome? What are such antibodies called?
8. Age-related macular degeneration is associated with inherited variants of factor H. If this disease is not caused by these polymorphisms, how does this variant contribute to this disease?

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3

Cellular Mechanisms of Innate Immunity

IN THIS CHAPTER

- [Pathogen recognition by cells of the innate immune system.](#)
 - [Consequences of innate immune activation.](#)
-

In [Chapter 2](#) we introduced the innate defenses—such as epithelial barriers, secreted antimicrobial proteins, and the complement system—that act immediately upon encounter with microbes to protect the body from infection. We also introduced the phagocytic cells that lie beneath the epithelial barriers and stand ready to engulf and digest invading microorganisms that have been flagged for destruction by complement. These phagocytes also initiate the next phase, the induced innate immune response, leading to an inflammatory reaction that recruits new phagocytic cells and circulating effector molecules to the site of infection.

In this chapter we describe how phagocytic cells of the innate immune system detect microbes or the cellular damage they cause, how they destroy these pathogens, and how they orchestrate downstream inflammatory responses through production of cytokines and chemokines (chemoattractant cytokines). We also introduce other cells of the innate immune system—a diverse array of specialized innate lymphoid cells (ILCs) including the natural killer cells (NK cells)—that contribute to innate host defenses against viruses and other intracellular pathogens. Also in this stage of infection, dendritic cells initiate adaptive immune responses.

Pathogen recognition by cells of the innate immune system.

The basis of the adaptive immune system’s enormous capacity for antigen recognition has long been appreciated. In contrast, the basis of recognition of microorganisms by innate immune receptors was discovered only in the late 1990s. Overall, the diversity of features recognized by innate immune receptors is quite limited compared with the diversity of features recognized by antigen receptors of the adaptive immune system. Nevertheless, the innate immune system has the capacity to recognize essentially any microorganism because innate immune receptors recognize conserved molecular features shared by related groups of microbes. These features are often referred to as [pathogen-associated molecular patterns \(PAMPs\)](#), and the receptors that recognize them are called [pattern-recognition receptors \(PRRs\)](#).

We will discuss specific examples of PAMPs and PRRs later in this chapter, but before doing so it is useful to establish some key principles that shape our understanding of innate immunity and distinguish it from adaptive immunity. The general characteristics of PRRs are contrasted with those of the antigen-specific receptors of adaptive immunity in **Fig. 3.1**. One key feature of PRRs is that their specificities are genetically encoded and inherited through the germline, in contrast to the random generation of antigen receptors of the adaptive immune system. Therefore, the specificities of PRRs have been shaped through evolution to recognize PAMPs, which are typically molecules that cannot be mutated without a significant fitness cost to the microbe. This selection of receptors that recognize conserved, largely invariant microbial features is how the innate immune system can recognize the broad diversity of potential pathogens with a limited number of receptors. In general, ligands for PRRs do not occur in healthy host cells, which ensures that immune responses are only directed against microbes; however, we will cover some interesting exceptions to this rule later in the chapter. Another critical feature of innate immune recognition is that cells express multiple PRRs, which we refer to as nonclonal expression, to distinguish it from the clonal expression of antigen receptors of the adaptive immune system. Nonclonal expression enables a given cell to respond to a range of pathogens so that large numbers of cells can rapidly mobilize when an infection is first detected. For example, macrophages, neutrophils, and dendritic cells express a variety of PRRs and can therefore respond to many distinct pathogens.

Receptor characteristic	Innate immunity	Adaptive immunity
Specificity inherited in the genome	Yes	No

Triggers immediate response	Yes	No
Recognizes broad classes of pathogens	Yes	No
Encoded in multiple gene segments	No	Yes
Requires gene rearrangement	No	Yes
Clonal expression	No	Yes
Able to discriminate between even closely related molecular structures	Yes	Yes

Fig. 3.1 Comparison of the characteristics of recognition molecules of the innate and adaptive immune systems. The innate immune system uses germline-encoded receptors while the adaptive immune system uses antigen receptors of unique specificity assembled from incomplete gene segments during lymphocyte development. Antigen receptors of the adaptive immune system are clonally expressed on individual lymphocytes and their progeny. Typically, receptors of the innate immune system are expressed nonclonally; that is, they are expressed on many, if not all, the cells of a given cell type.

In addition to direct PAMP recognition by PRRs, a more recently appreciated concept is that certain innate immune receptors respond to the consequences of infection. Pathogens often disrupt specific cellular functions while establishing infection, and some innate receptors detect these perturbations. The actions of pathogens can also induce the release or production of self-derived host molecules, which in turn activate innate immune receptors. Such molecules have been termed [damage-associated molecular patterns \(DAMPs\)](#). While the cellular perturbations and DAMPs are often the result of infection, they can also be induced in other contexts, such as tissue damage or neoplastic transformation.

Coordination of the innate immune response relies on the information provided by many types of receptors. Innate immune receptors can be classified into four main groups on the basis of their cellular localization and their function: free receptors in the serum, such as ficolins and histatins (discussed in [Chapter 2](#)); membrane-bound phagocytic receptors; membrane-bound signaling receptors; and cytoplasmic signaling receptors. Phagocytic receptors primarily signal for phagocytosis of the microbes they recognize. A diverse group of chemotactic receptors helps to guide cells to sites of infection; other receptors, including PRRs and cytokine receptors, help to control the activity of effector molecules at those sites.

In this part of the chapter we first look at the recognition properties of phagocytic receptors

and of signaling receptors that activate phagocytic microbial killing mechanisms. Next, we describe the main PRR families that detect infection, including the Toll-like receptors (TLRs), the first of the innate sensor systems to be discovered, and several families of receptors that detect intracellular infections.

3-1 After entering tissues, many microbes are recognized, ingested, and killed by phagocytes.

If a microorganism crosses an epithelial barrier and begins to replicate in the tissues of the host, in most cases it is immediately recognized by resident phagocytic cells. The main classes of phagocytic cells in the innate immune system are macrophages and monocytes, granulocytes, and dendritic cells. [Macrophages](#) are the major phagocyte population resident in most normal tissues at homeostasis. They can arise either from progenitor cells that enter the tissues during embryonic development, and then self-renew at steady state during life, or from circulating [monocytes](#). Studies suggest that the embryonic progenitors arise from either the fetal liver, the yolk sac, or an embryonic region near the dorsal aorta called the [aorta–gonad–mesonephros \(AGM\)](#), although the relative contribution of these origins is still debated. Macrophages are found in especially large numbers in connective tissue: for example, in the submucosal layer of the gastrointestinal tract; in the submucosal layer of the bronchi and in the lung interstitium—the tissue and intercellular spaces around the air sacs (alveoli)—and in the alveoli themselves; along some blood vessels in the liver; and throughout the spleen, where they remove senescent blood cells. Macrophages in different tissues were historically given different names; for example, [microglial cells](#) in neural tissue and [Kupffer cells](#) in the liver. The self-renewal of these two types of cells is dependent on a cytokine called interleukin-34 (IL-34) that is produced in these tissues and acts on the same receptor as macrophage-colony stimulating factor (M-CSF).

During infection or inflammation, macrophages can also arise from monocytes that leave from the circulation to enter into tissues. Monocytes in both mouse and human develop in the bone marrow and circulate in the blood as two main populations. In humans, 90% of circulating monocytes are [‘classical’ monocytes](#) that express CD14, a co-receptor for a PRR described later, and function during infection by entering tissues and differentiating into activated inflammatory monocytes or macrophages. In mice, this monocyte population expresses high levels of the surface marker Ly6C. A smaller population of [‘patrolling’ monocytes](#) roll along the endothelium rather than circulating freely in the blood. In humans, they express CD14 and CD16, a type of Fc receptor (FcγRIII; see [Section 10-21](#)), and are thought to survey for injury to the endothelium, but they do not differentiate into tissue macrophages. In mice, they express low levels of Ly6C.

The second major class of phagocytes comprises the granulocytes, which include [neutrophils](#), [eosinophils](#), and [basophils](#). Of these, neutrophils have the greatest phagocytic

activity and are the cells most immediately involved in innate immunity against infectious agents. Also called polymorphonuclear neutrophilic leukocytes (PMNs, or polys), they are short-lived cells that are abundant in the blood but are not present in healthy tissues. Macrophages and granulocytes have an important role in innate immunity because they can recognize, ingest, and destroy many pathogens without the aid of an adaptive immune response. Phagocytic cells that scavenge incoming pathogens represent an ancient mechanism of innate immunity, as they are found in both invertebrates and vertebrates.

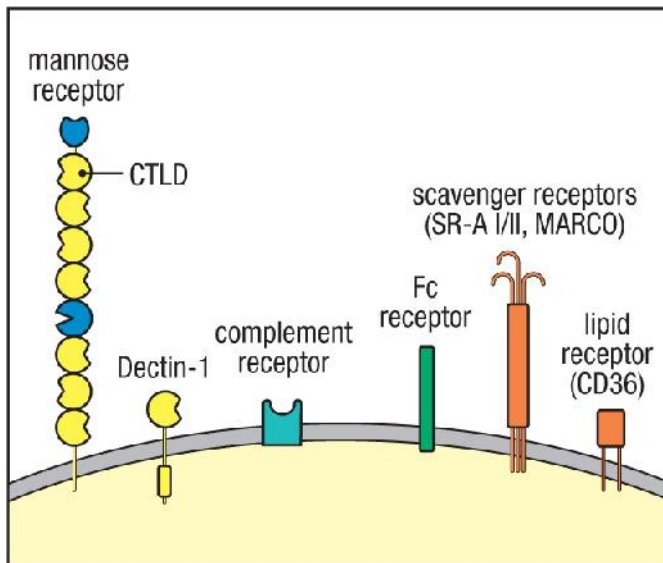
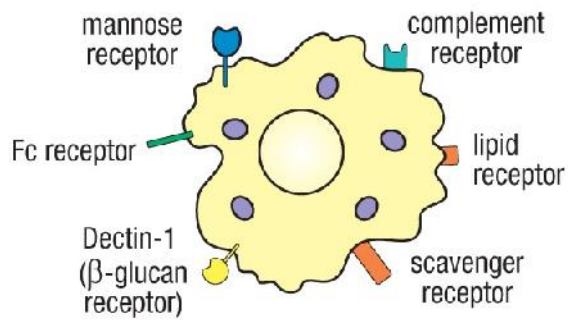
The third class of phagocytes in the immune system is the immature [dendritic cells](#) that reside in lymphoid organs and in peripheral tissues. There are two main functional types of dendritic cells: [conventional \(or classical\) dendritic cells \(cDCs\)](#) and [plasmacytoid dendritic cells \(pDCs\)](#). Both types of cells arise from progenitors within the bone marrow, and they migrate via the blood to tissues throughout the body and to peripheral lymphoid organs. Dendritic cells ingest and break down microbes, but, unlike macrophages and neutrophils, their primary role in immune defense is not the frontline, large-scale direct killing of microbes. A major role of cDCs is to process ingested microbes in order to generate peptide antigens that can activate T cells and induce an adaptive immune response. They also produce cytokines in response to microbial recognition (via PRRs) that activate other types of cells against infection. cDCs are thus considered to act as a bridge between innate and adaptive immune responses. pDCs can produce large amounts of a class of cytokines known as type I [interferons](#), or antiviral interferons, and are considered to be part of innate immunity; they are discussed in detail later in the chapter.

Because most microorganisms enter the body through the mucosa of the gut and respiratory system, the skin, or the urogenital tract, macrophages in the submucosal tissues are the first cells to encounter most pathogens, but they are soon reinforced by the recruitment of large numbers of neutrophils to sites of infection. Macrophages and neutrophils recognize pathogens by means of cell-surface PRRs that bind PAMPs on the surface of pathogens, as discussed at the beginning of this chapter. Although they are both phagocytic, macrophages and neutrophils have distinct properties and functions in innate immunity.

The process of [phagocytosis](#) is initiated when certain receptors on the surface of the cell—typically a macrophage, neutrophil, or dendritic cell—interact with the microbial surface. The bound pathogen is first surrounded by the phagocyte plasma membrane and then internalized in a large membrane-enclosed endocytic vesicle known as a [phagosome](#). The phagosome fuses with one or more lysosomes to generate a [phagolysosome](#), in which the lysosomal contents are released. The phagolysosome also becomes acidified, acquires antimicrobial peptides and enzymes, and undergoes enzymatic processes that produce highly reactive [superoxide radicals](#) and [nitric oxide radicals](#), which together kill the microbe (**Fig. 3.2**). Neutrophils are highly specialized for the intracellular killing of microbes and contain different types of cytoplasmic granules—the [primary granules](#) and [secondary granules](#) described in [Section 2-4](#). These granules fuse with phagosomes, releasing additional enzymes and antimicrobial peptides that attack the microbe. Another pathway by which extracellular material, including microbial material, can be taken up into the endosomal compartment of cells and degraded is [receptor-mediated endocytosis](#), which is not restricted

to phagocytes. Dendritic cells and other phagocytes can also take up pathogens by a nonspecific process called [macropinocytosis](#), in which large amounts of extracellular fluid and its contents are ingested.

Macrophages have phagocytic receptors that bind microbes and their components



Bound material is internalized in phagosomes and broken down in phagolysosomes

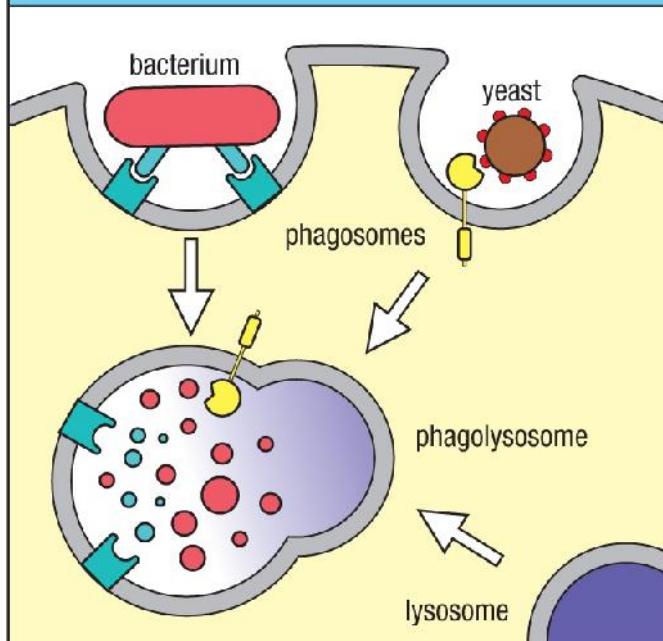


Fig. 3.2 Macrophages express receptors that enable them to take up microbes by phagocytosis. First panel: Macrophages residing in tissues throughout the body are among the first cells to encounter and respond to pathogens. They carry cell-surface receptors that bind to various molecules on microbes, in particular carbohydrates and lipids, and induce phagocytosis of the bound material. Second panel: Dectin-1 is a member of the C-type lectin-like family built around a single C-type lectin-like domain (CTLD). Lectins in general are based on a carbohydrate-recognition domain (CRD). The macrophage mannose receptor contains many CTLDs and has a fibronectin-like domain and a cysteine-rich region at its amino terminus. Class A scavenger receptors, such as MARCO, are built from collagen-like domains and form trimers. The receptor protein CD36 is a class B scavenger receptor that recognizes and internalizes lipids. Various complement receptors bind and internalize complement-coated bacteria. Third panel: In phagocytosis, receptor-bound material is taken into intracellular phagosomes, which fuse with lysosomes to form an acidified phagolysosome in which the ingested material is broken down by lysosomal hydrolases.

Macrophages and neutrophils constitutively express a number of cell-surface PRRs that stimulate the phagocytosis and intracellular killing of microbes bound to them, although some also signal through other pathways to trigger responses such as cytokine production. This set of phagocytic PRRs includes several members of the C-type lectin-like family (see Fig. 3.2). For example, [Dectin-1](#) is strongly expressed by macrophages and neutrophils and recognizes β -(1,3)-linked glucans (polymers of glucose), which are common components of fungal cell walls in particular. Dendritic cells also express Dectin-1, as well as several other C-type lectin-like phagocytic receptors. Another C-type lectin, the [mannose receptor](#) expressed by macrophages and dendritic cells, recognizes various mannosylated ligands, including some present on fungi, bacteria, and viruses; it was once suspected of having an important role in resistance to microbes. However, experiments with mice that lack this receptor do not support this idea. The macrophage mannose receptor is now thought to function mainly as a clearance receptor for host glycoproteins such as β -glucuronidase and lysosomal hydrolases, which have mannose-containing carbohydrate side chains and whose extracellular concentrations are raised during inflammation.

Another set of phagocytic PRRs on macrophages, called [scavenger receptors](#), recognize various anionic polymers and acetylated low-density lipoproteins. These receptors are structurally heterogeneous, consisting of at least six different molecular families. Class A scavenger receptors are membrane proteins composed of trimers of collagen domains (see Fig. 3.2). They include [SR-A I](#), [SR-A II](#), and [MARCO](#) (macrophage receptor with a collagenous structure), all of which bind various bacterial cell-wall components and help to internalize bacteria, although the basis of their specificity is poorly understood. Class B scavenger receptors bind high-density lipoproteins, and they internalize lipids. One of these receptors is CD36, which binds many ligands, including long-chain fatty acids.

A third set of receptors of crucial importance in macrophage and neutrophil phagocytosis is

the complement receptors and Fc receptors introduced in [Chapters 1](#) and [2](#). These receptors bind to complement-coated microbes or to antibodies that have bound to the surface of microbes and facilitate the phagocytosis of a wide range of microorganisms.

3-2 G protein–coupled receptors on phagocytes link microbe recognition with increased efficiency of intracellular killing.

Phagocytosis of microbes by macrophages and neutrophils is generally followed by the death of the microbe inside the phagocyte. In addition to the phagocytic receptors described above, macrophages and neutrophils have other receptors that signal to stimulate antimicrobial killing. These receptors belong to the evolutionarily ancient family of [G protein–coupled receptors \(GPCRs\)](#), which are characterized by seven membrane-spanning segments. Members of this family are crucial to immune-system function because they also direct responses to anaphylatoxins such as the complement fragment C5a (see [Section 2-14](#)), leukotrienes, and to many chemokines, recruiting phagocytes to sites of infection and promoting inflammation.

The [fMet-Leu-Phe \(fMLF\) receptor](#) is a G protein–coupled receptor that senses the presence of bacteria by recognizing a unique feature of bacterial polypeptides. Protein synthesis in bacteria is typically initiated with an *N*-formylmethionine (fMet) residue, an amino acid present in prokaryotes but not in eukaryotes. The fMLF receptor is named after a tripeptide, formyl-methionyl-leucyl phenylalanine, for which it has a high affinity, although it also binds other peptide motifs. Bacterial polypeptides binding to this receptor activate intracellular signaling pathways that direct the cell to move toward the most concentrated source of the ligand. Signaling through the fMLF receptor also induces the production of microbicidal [reactive oxygen species \(ROS\)](#) in the phagolysosome. The C5a receptor 1 recognizes the small fragment of C5 generated when complement is activated, usually by the presence of microbes (see [Section 2-13](#)), and signals by a similar pathway as the fMLF receptor. Thus, stimulation of these receptors both guides monocytes and neutrophils toward a site of infection and leads to increased antimicrobial activity; these cell responses can be activated by directly sensing unique bacterial products or by messengers such as C5a that indicate previous recognition of a microbe.

The G protein–coupled receptors are so named because ligand binding activates a member of a class of intracellular GTP-binding proteins called [G proteins](#), sometimes referred to as [heterotrimeric G proteins](#) to distinguish them from the family of ‘small’ GTPases typified by Ras. Heterotrimeric G proteins are composed of three subunits: $G\alpha$, $G\beta$, and $G\gamma$, of which the α subunit is similar to the small GTPases (**Fig. 3.3**). In the resting state, the G protein is inactive, not associated with the receptor, and a molecule of GDP is bound to the α subunit. Ligand binding induces conformational changes in the receptor that allow it to bind the G protein, which results in the displacement of the GDP from the G protein and its replacement

with GTP. The active G protein dissociates into two components, the $G\alpha$ subunit and a complex consisting of a $G\beta$ and a $G\gamma$ subunit. Each of these components can interact with other intracellular signaling molecules to transmit and amplify the signal. G proteins can activate a wide variety of downstream enzymatic targets, such as adenylate cyclase, which produces the second messenger cyclic AMP, and phospholipase C, whose activation gives rise to the second messenger inositol 1,4,5-trisphosphate (IP_3) and the release of Ca^{2+} from stores in the endoplasmic reticulum. The subsequent increase in cytosolic Ca^{2+} can induce several signaling pathways, including the activation of protein kinase C (PKC) (see [Chapter 7](#) for a more detailed discussion of these and other signaling pathways).

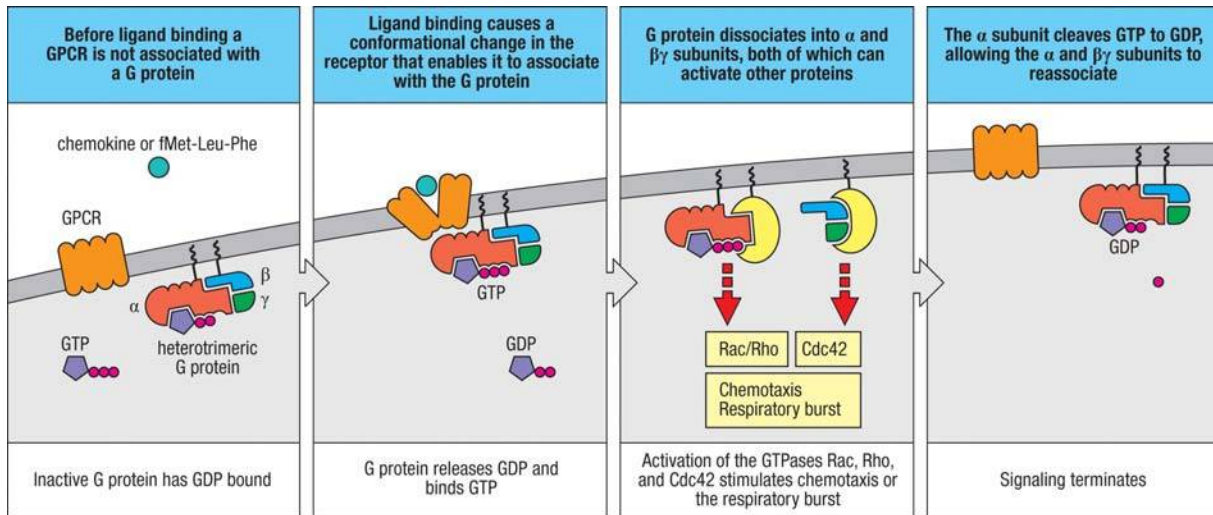


Fig. 3.3 G protein-coupled receptors signal by coupling with intracellular heterotrimeric G proteins. First panel: G protein-coupled receptors (GPCRs) such as the fMet-Leu-Phe (fMLF) and chemokine receptors signal through GTP-binding proteins known as heterotrimeric G proteins. In the inactive state, the α subunit of the G protein binds GDP and is associated with the β and γ subunits. Second panel: The binding of a ligand to the receptor induces a conformational change that allows the receptor to interact with the G protein, which results in the displacement of GDP and binding of GTP by the α subunit. Third panel: GTP binding triggers the dissociation of the G protein into the α subunit and the $\beta\gamma$ subunit, each of which can activate other proteins at the inner face of the cell membrane. In the case of fMLF signaling in macrophages and neutrophils, the α subunit of the activated G protein indirectly activates the GTPases Rac and Rho, whereas the $\beta\gamma$ subunit indirectly activates the GTPase Cdc42. The actions of these proteins result in the assembly of NADPH oxidase, resulting in a respiratory burst, a brief spurt in oxygen consumption leading to the production of reactive oxygen species. Chemokine signaling acts by a similar pathway and activates chemotaxis, which is the process whereby a cell migrates toward the source of chemokine. Fourth panel: The activated response ceases when the intrinsic GTPase activity of the α subunit hydrolyzes GTP to GDP, and the α and $\beta\gamma$ subunits reassociate. The intrinsic rate of GTP hydrolysis by α subunits is relatively slow, and signaling is regulated by additional GTPase-activating proteins (not shown), which accelerate the rate of GTP hydrolysis.

Signaling by fMLF and C5a receptors influences cell motility, metabolism, gene expression, and cell division through activation of several [Rho family small GTPase proteins](#). The α subunit of the activated G protein indirectly activates [Rac](#) and [Rho](#), while the $\beta\gamma$ subunit indirectly activates the small GTPase Cdc42 (see Fig. 3.3). Activation of these GTPases is controlled by [guanine nucleotide exchange factors \(GEFs\)](#) (see Fig. 7.5), which exchange GTP for GDP bound to the GTPase. The G proteins activated by fMLF activate the GEF protein [PREX1](#) (phosphatidylinositol 3,4,5-trisphosphate–dependent Rac exchanger 1 protein), which can directly activate Rac. Other GEFs, including members of the Vav family that are controlled by other types of receptors (see [Section 7-19](#)), can also activate Rac activity, and their activity synergizes with the actions of fMLF and C5a.

The activation of Rac and Rho helps to increase the microbicidal capacity of macrophages and neutrophils that have ingested pathogens. Upon phagocytosing microbes, macrophages and neutrophils produce a variety of toxic products that help to kill the engulfed microorganism (**Fig. 3.4**). The most important of these are the antimicrobial peptides described in [Section 2-4](#), reactive nitrogen species such as nitric oxide (NO), and ROS such as the superoxide anion (O_2^-) and hydrogen peroxide (H_2O_2). Nitric oxide is produced by a high-output form of nitric oxide synthase, inducible NOS2 (iNOS2), whose expression is induced by a variety of stimuli, including fMLF.

Antimicrobial mechanisms of phagocytes		
Class of mechanism	Macrophage products	Neutrophil products
Acidification	pH $\approx$ 3.5-4.0, bacteriostatic or bactericidal	
Toxic oxygen–derived products	Superoxide O_2^- , hydrogen peroxide H_2O_2 , singlet oxygen 1O_2 , hydroxyl radical $\cdot OH$, hypochlorite OCl^-	
Toxic nitrogen oxides	Nitric oxide NO	
Antimicrobial peptides	Cathelicidin, macrophage elastase–derived peptide	α -Defensins (HNP1–4), β -defensin HBD4, cathelicidin, azurocidin, bacterial permeability inducing protein (BPI), lactoferricin
Enzymes	Lysozyme: digests cell walls of some Gram-positive bacteria Acid hydrolases (e.g., elastase and other proteases): break down ingested microbes	
Competitors		Lactoferrin (sequesters Fe^{2+}), vitamin B_{12} –binding protein

Fig. 3.4 Bactericidal agents produced or released by phagocytes after uptake of microorganisms. Most of the agents listed are directly toxic to microbes and can act directly in the phagolysosome. They can also be secreted into the extracellular environment, and many of these substances are toxic to host cells. Other phagocyte products sequester

essential nutrients in the extracellular environment, rendering them inaccessible to microbes and hindering microbial growth. Besides being directly bacteriostatic or bactericidal, acidification of lysosomes also activates the many acid hydrolases that degrade the contents of the vacuole.

Activation of phagocytic receptors, including the fMLF and C5a receptors, can promote ROS generation by inducing assembly of a multicomponent, membrane-associated [NADPH oxidase](#), also called [phagocyte oxidase](#). In unstimulated phagocytes, this enzyme is inactive because it is not fully assembled. One set of subunits, the cytochrome b_{558} complex (composed of p22 and gp91), is localized in the plasma membranes of resting macrophages and neutrophils, and it appears in lysosomes after the maturation of phagolysosomes. The other components, p40, p47, and p67, are in the cytosol. Activation of phagocytes induces the cytosolic subunits to join with the membrane-associated cytochrome b_{558} to form a complete, functional NADPH oxidase in the phagolysosome membrane (**Fig. 3.5**). Activation of Rac promotes the movement of the cytosolic components to the membrane to assemble the active NADPH oxidase. Phosphorylation of certain residues of the NADPH oxidase complex also plays a critical role in activation and is mediated by PKC activation downstream of GPCRs. PKC activation can also occur downstream of phagocytic receptors, such as Dectin-1 and Fc receptors, which contain [immunoreceptor tyrosine-based activation motifs \(ITAMs\)](#) in their cytosolic tails. The signaling mediated by ITAMs is discussed in greater detail later in this chapter ([Section 3-26](#)) as well as in later chapters.

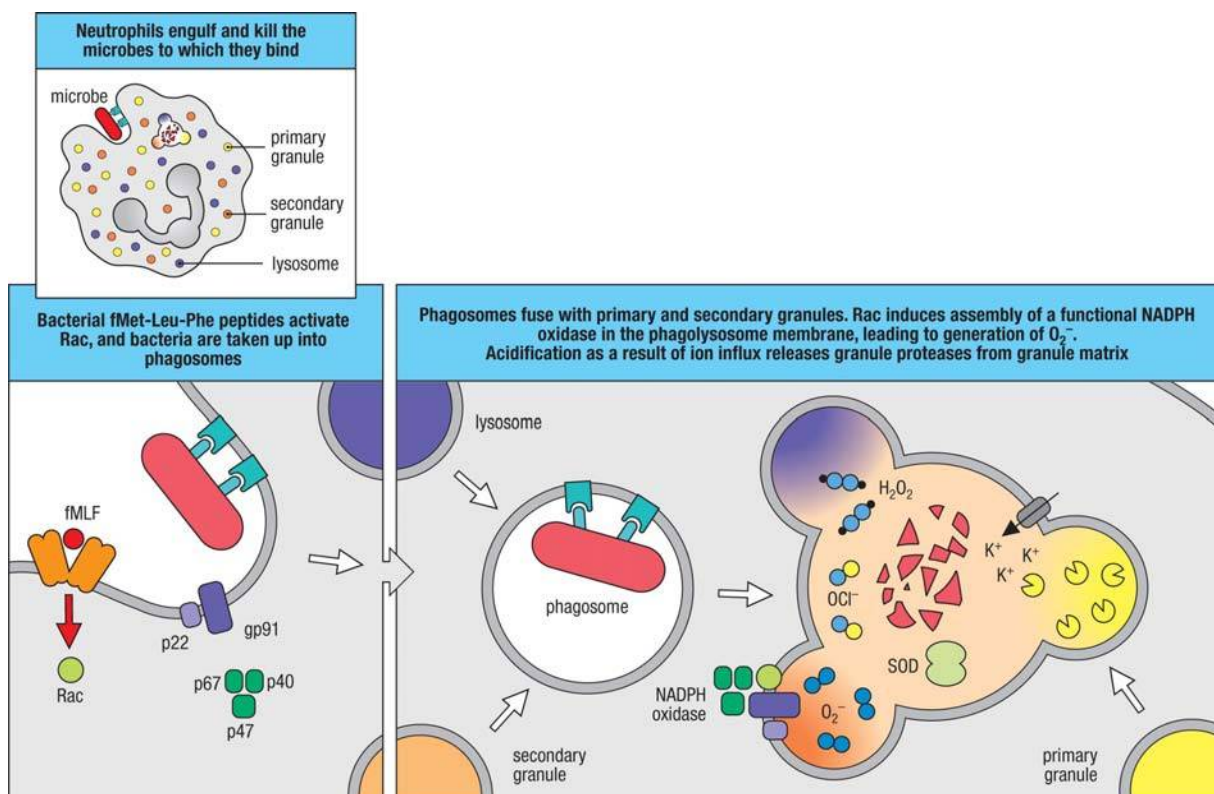


Fig. 3.5 The microbicidal respiratory burst in phagocytes is initiated by activation-induced assembly of the phagocyte NADPH oxidase. First panel: Neutrophils are highly specialized for the uptake and killing of pathogens and contain several different

kinds of cytoplasmic granules, such as the primary and secondary granules shown in the first panel. These granules contain antimicrobial peptides and enzymes. Second panel: In resting neutrophils, the cytochrome b_{558} subunits (gp91 and p22) of NADPH oxidase are localized in the plasma membrane; the other oxidase components (p40, p47, and p67) are located in the cytosol. Signaling by phagocytic receptors and by fMLF or C5a receptors synergizes to activate Rac and induce the assembly of the complete, active NADPH oxidase in the membrane of the phagolysosome, which has formed by the fusion of the phagosome with lysosomes and primary and secondary granules. Third panel: Active NADPH oxidase transfers electrons from NADPH to molecular oxygen, forming the superoxide ion O_2^- (blue) and other free oxygen radicals in the lumen of the phagolysosome. Potassium and hydrogen ions are then drawn into the phagolysosome to neutralize the charged superoxide ion, increasing acidification of the vesicle. Acidification dissociates granule enzymes such as cathepsin G and elastase (yellow) from their proteoglycan matrix, leading to their cleavage and activation by lysosomal proteases. O_2^- is converted by superoxide dismutase (SOD) to hydrogen peroxide (H_2O_2), which can kill microorganisms and can be converted by myeloperoxidase, a heme-containing enzyme, to microbicidal hypochlorite (OCl^-) and by chemical reaction with ferrous (Fe^{2+}) ions to the hydroxyl radical ($\cdot OH$).

The NADPH oxidase reaction results in a transient increase in oxygen consumption by the cell, which is known as the [respiratory burst](#). It generates superoxide anion within the lumen of the phagolysosome, and this is converted by the enzyme [superoxide dismutase \(SOD\)](#) into H_2O_2 . Further chemical and enzymatic reactions produce a range of toxic ROS from H_2O_2 , including the hydroxyl radical ($\cdot OH$), hypochlorite (OCl^-), and hypobromite (OBr^-). In this way, the direct recognition of bacterially derived polypeptides or previous pathogen recognition by the complement system activates a potent killing mechanism within macrophages and neutrophils that have ingested microbes via their phagocytic receptors. However, phagocyte activation can also cause extensive tissue damage because hydrolytic enzymes, membrane-disrupting peptides, and reactive oxygen species can be released into the extracellular environment and are toxic to host cells.

Neutrophils use the respiratory burst described above in their role as an early responder to infection. Neutrophils are not tissue-resident cells, and they need to be recruited to a site of infection from the bloodstream. Their main function is to ingest and kill microorganisms, but they also produce cytokines and lipid mediators that promote and orchestrate the inflammatory response. Although neutrophils are eventually present in much larger numbers than macrophages in some types of acute infection, they are short-lived, dying soon after they have accomplished a round of phagocytosis and used up their primary and secondary granules. Dead and dying neutrophils are a major component of the [pus](#) that forms in abscesses and in wounds infected by certain extracellular capsulated bacteria such as streptococci and staphylococci, which are thus known as [pus-forming, or pyogenic, bacteria](#). Macrophages, in contrast, are long-lived cells and continue to generate new lysosomes.



Case Study: Chronic Granulomatous Disease

Individuals with a disease called [chronic granulomatous disease \(CGD\)](#) have a genetic deficiency of NADPH oxidase, which means their phagocytes do not produce the toxic oxygen derivatives characteristic of the respiratory burst and so are less able to kill ingested microorganisms and clear an infection. The most common form of CGD is an X-linked heritable disease that arises from inactivating mutations in the gene encoding the gp91 subunit of cytochrome b_{558} . People with this defect are unusually susceptible to bacterial and fungal infections, especially in infancy, though they remain susceptible for life. One autosomal recessive form of NADPH oxidase deficiency, p47phox deficiency, has very low but detectable activity and causes a milder form of CGD.

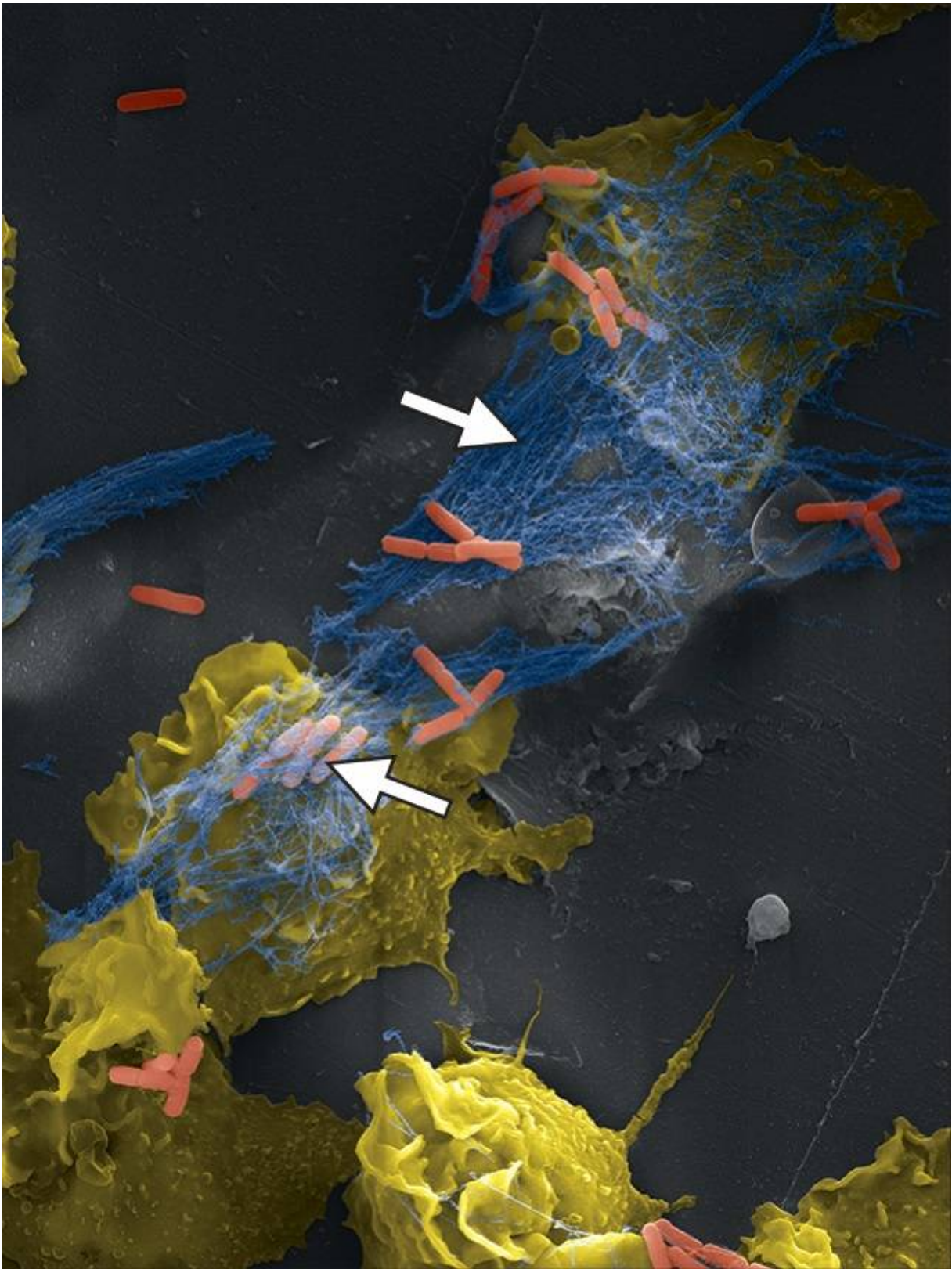


Fig. 3.6 Neutrophil extracellular traps (NETs) can trap bacteria and fungi. This scanning electron micrograph of activated human neutrophils infected with a virulent strain of *Shigella flexneri* (pink rods) shows the stimulated neutrophils forming NETs (blue, indicated by arrows). Bacteria trapped within NETs are visible (lower arrow).

In addition to killing microbes engulfed by phagocytosis, neutrophils use another rather novel mechanism of destruction that is directed at extracellular pathogens. During infection, some activated neutrophils undergo a unique form of cell death in which the nuclear chromatin, rather than being degraded as occurs during apoptosis, is released into the extracellular space and forms a fibril matrix known as [neutrophil extracellular traps, or NETs](#) (**Fig. 3.6**). It has been proposed that NETs act to capture microorganisms, which may then be more efficiently phagocytosed by other neutrophils or macrophages, but the importance of this mechanism during infection remains rather unclear.

Macrophages can phagocytose pathogens and produce the respiratory burst immediately upon encountering an infecting microorganism, and this can be sufficient to prevent an infection from becoming established. In the 19th century, the immunologist Élie Metchnikoff believed that the innate response of macrophages encompassed all host defenses; indeed, invertebrates such as the sea star that he was studying rely entirely on innate immunity to overcome infection. Although this is not the case in humans and other vertebrates, the innate response of macrophages still provides an important front line of defense that must be overcome if a microorganism is to establish an infection that can be passed on to a new host.

Pathogens have, however, developed a variety of strategies to avoid immediate destruction by macrophages and neutrophils. Many extracellular pathogenic bacteria coat themselves with a thick polysaccharide capsule that is not recognized by any phagocytic receptor. In such cases, however, the complement system can recognize microbial surfaces and coat them with C3b, thereby flagging them for phagocytosis via complement receptors, as described in [Chapter 2](#). Other pathogens, for example, mycobacteria, have evolved ways to grow inside macrophage phagosomes by inhibiting their acidification and fusion with lysosomes. Without such strategies, a microorganism must enter the body in sufficient numbers to overwhelm the immediate innate host defenses and to establish a focus of infection.

3-3 Microbial recognition and tissue damage initiate an inflammatory response.

An important effect of the interaction between microbes and tissue macrophages is the activation of macrophages and other immune cells to release small proteins called [cytokines](#) and [chemokines](#), and other chemical mediators, such as lipid mediators. Collectively, these signals induce a state of inflammation in the tissue, attract monocytes and neutrophils to the infection, and allow plasma proteins to enter the tissue from the blood. An inflammatory response is usually initiated within minutes to hours of infection or wounding. Macrophages are stimulated to secrete [pro-inflammatory](#) cytokines, such as TNF- α , and chemokines by interactions between microbes or microbial products and specific receptors expressed by macrophages. We will examine how the cytokines interact with pathogens later in the chapter, but first we describe some general aspects of inflammation and how it contributes to host defense.

Inflammation has three essential roles in combating infection. The first is to deliver additional effector molecules and cells from the blood into sites of infection, and so increase the destruction of invading microorganisms. The second is to induce local blood clotting, which provides a physical barrier to the spread of the infection in the bloodstream. The third is to promote the repair of injured tissue, which occurs during the resolution phase, after the infection has been cleared.

Inflammatory responses are characterized by heat, pain, redness, and swelling at the site of an infection, reflecting four types of change in the local blood vessels, as shown in **Fig. 3.7**. The first is an increase in vascular diameter, leading to increased local blood flow—hence the heat and redness—and a reduction in the velocity of blood flow, especially along the inner walls of small blood vessels. The second change is the activation of endothelial cells lining the blood vessel to express [cell-adhesion molecules](#) that promote the binding of circulating leukocytes. The combination of slowed blood flow and adhesion molecules allows leukocytes to attach to the endothelium and migrate into the tissues, a process known as [extravasation](#). All these changes are initiated by the pro-inflammatory cytokines and chemokines produced by activated macrophages and other cells in the tissue.

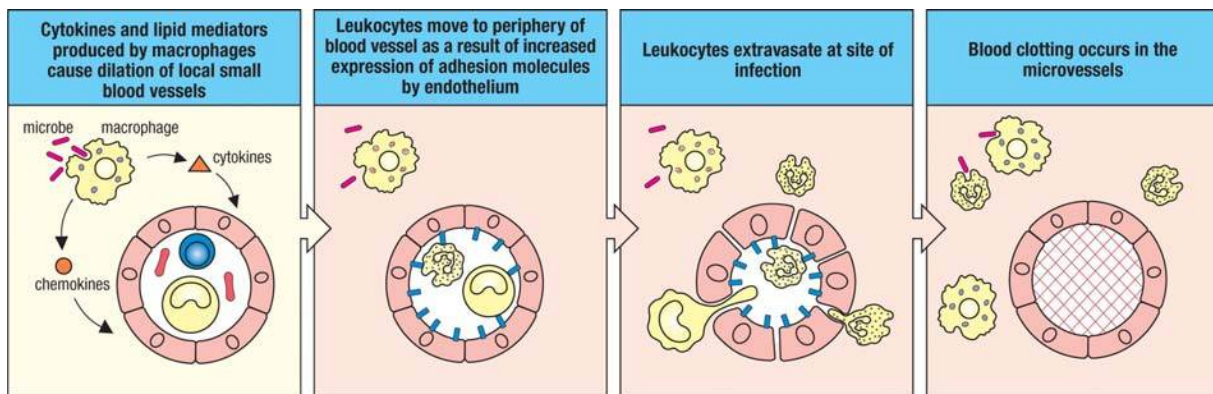


Fig. 3.7 Infection stimulates macrophages to release cytokines and chemokines that initiate an inflammatory response. Cytokines and lipid mediators produced by tissue macrophages at the site of infection cause the dilation of local small blood vessels and changes in the endothelial cells of their walls. These changes lead to the movement of leukocytes, such as neutrophils and monocytes, out of the blood vessel (extravasation) and into the infected tissue; this movement is guided by chemokines produced by the activated macrophages. The blood vessels also become more permeable, allowing plasma proteins and fluid to leak into the tissues. Together, these changes cause the characteristic inflammatory signs of heat, pain, redness, and swelling at the site of infection.

Once inflammation has begun, the first white blood cells attracted to the site are neutrophils. These are followed by monocytes (**Fig. 3.8**), which upon activation are called [inflammatory monocytes](#). Monocytes can produce various pro-inflammatory cytokines but are distinguishable from macrophages by their lack of expression of the adhesion G protein-coupled receptor E1, commonly called F4/80. Monocytes are also able to give rise to

dendritic cells in the tissues, depending on signals that they receive from their environment. In the later stages of inflammation, other leukocytes such as eosinophils and lymphocytes also enter the infected site.

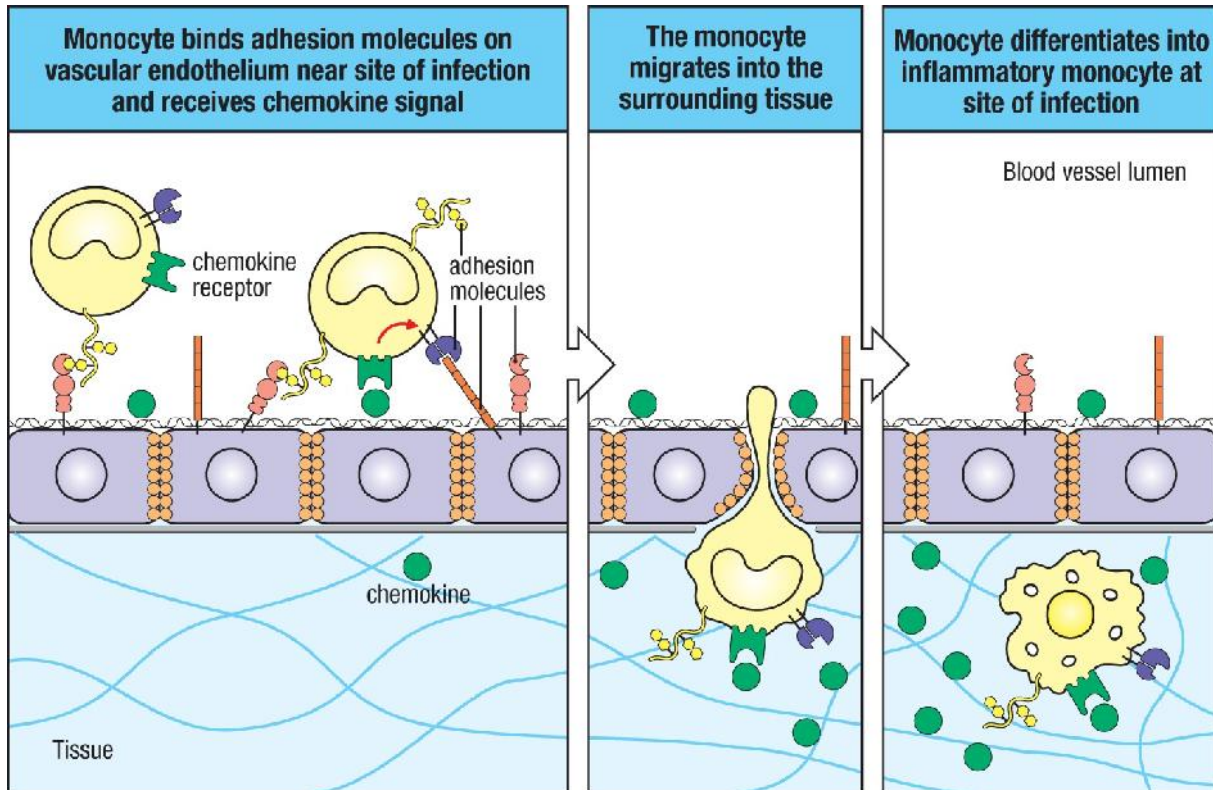


Fig. 3.8 Monocytes circulating in the blood migrate into infected and inflamed tissues. Adhesion molecules on the endothelial cells of the blood vessel wall capture the monocyte and cause it to adhere to the vascular endothelium. Chemokines bound to the vascular endothelium then signal the monocyte to migrate across the endothelium into the underlying tissue. The monocyte, now differentiating into an inflammatory monocyte, continues to migrate, under the influence of chemokines released during inflammatory responses, toward the site of infection. Monocytes leaving the blood are also able to differentiate into dendritic cells (not shown), depending on the signals that they receive from their environment.

The third major change in local blood vessels is an increase in vascular permeability. Thus, instead of being tightly joined together, the endothelial cells lining the blood vessel walls become separated, leading to an exit of fluid and proteins from the blood and their local accumulation in the tissue. This accounts for the swelling, or [edema](#), and pain—as well as the accumulation in tissues of plasma proteins such as complement and mannose-binding lectin (MBL) that aid in host defense. The changes that occur in endothelium as a result of inflammation are known generally as [endothelial activation](#). The fourth change, clotting in microvessels in the site of infection, prevents the spread of the pathogen via the blood.

These changes are induced by a variety of inflammatory mediators released as a consequence of the recognition of pathogens by macrophages, and later by neutrophils and other white

blood cells. Macrophages and neutrophils secrete lipid mediators of inflammation—[prostaglandins](#), [leukotrienes](#), and [platelet-activating factor \(PAF\)](#)—which are rapidly produced by enzymatic pathways that degrade membrane phospholipids. Their actions are followed by those of the chemokines and cytokines that are synthesized and secreted by macrophages and inflammatory monocytes in response to pathogens. The cytokine [tumor necrosis factor- \$\alpha\$ \(TNF- \$\alpha\$, also known simply as TNF\)](#), for example, is a potent activator of endothelial cells. We describe TNF- α and related cytokines in more detail in [Section 3-15](#).

Besides stimulating the respiratory burst in phagocytes and acting as a chemoattractant for neutrophils and monocytes, C5a also promotes inflammation by increasing vascular permeability and inducing the expression of certain adhesion molecules on endothelium. C5a also activates local [mast cells](#) (see [Section 1-4](#)), which are stimulated to release their granules containing the small inflammatory molecule histamine as well as TNF- α and cathelicidins.

If wounding has occurred, the injury to blood vessels immediately triggers two protective enzyme cascades. One is the [kinin system](#) of plasma proteases that is triggered by tissue damage to generate several polypeptides that regulate blood pressure, coagulation, and pain. Although we will not fully describe its components here, one inflammatory mediator produced is the vasoactive peptide [bradykinin](#), which increases vascular permeability to promote the influx of plasma proteins to the site of tissue injury. It also causes pain. Although unpleasant to the victim, pain draws attention to the problem and leads to immobilization of the affected part of the body, which helps to limit the spread of the infection.

The [coagulation system](#) is another protease cascade that is triggered in the blood after damage to blood vessels, although its full description is also outside our present scope. Its activation leads to the formation of a fibrin clot, whose normal role is to prevent blood loss. With regard to innate immunity, however, the clot physically encases the infectious microorganisms and prevents their entry into the bloodstream. The kinin and the coagulation cascades are also triggered by activated endothelial cells, and so they can have important roles in the inflammatory response to pathogens even if wounding or gross tissue injury has not occurred. Thus, within minutes of the penetration of tissues by a pathogen, the inflammatory response causes an influx of proteins and cells that may control the infection.

Damage to tissues can occur in the absence of infection by microbes; for example, as a result of physical trauma, ischemia, or metabolic or autoimmune disorders. In such [sterile injury](#), many of the changes associated with infection, such as neutrophil recruitment, can occur, in addition to activation of the kinin system and clot formation.

3-4 Toll-like receptors represent an ancient pathogen-recognition system.

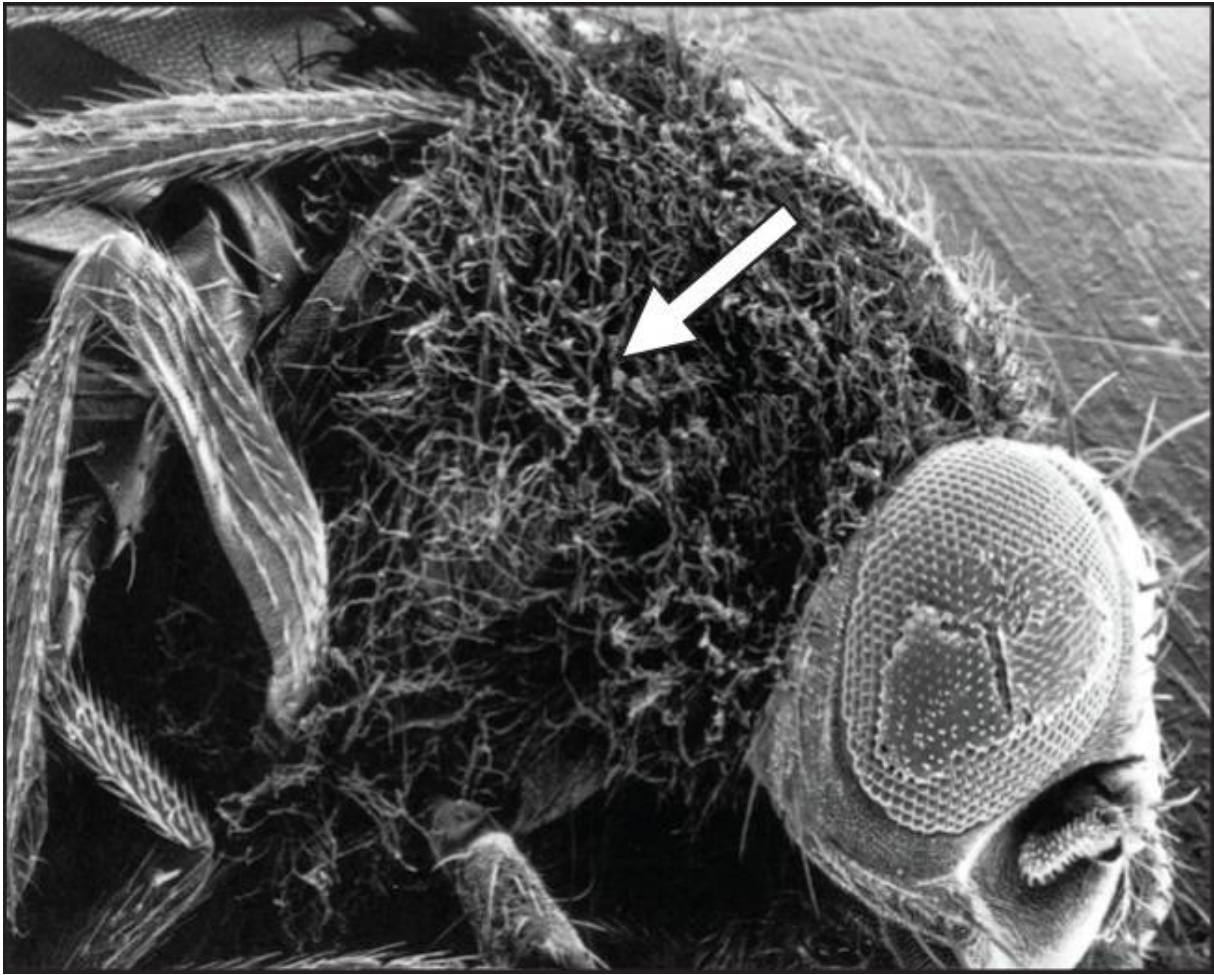


Fig. 3.9 Toll is required for antifungal responses in *Drosophila melanogaster*. Flies that are deficient in the Toll receptor are dramatically more susceptible than wild-type flies to fungal infection. This is illustrated here by the uncontrolled hyphal growth (arrow) of the normally weak pathogen *Aspergillus fumigatus* in a Toll-deficient fly.

[Section 1-5](#) introduced pattern-recognition receptors (PRRs), which function as sensors for pathogen-associated molecular patterns (PAMPs). The existence of these receptors was predicted by Charles Janeway, Jr., before mechanisms of innate recognition of PAMPs were known, to explain how complex mixtures of microbial molecules, called adjuvants, were able to stimulate adaptive immune responses to purified antigens. The first example of such a receptor was the [Toll](#) protein, discovered in the fruit fly *Drosophila melanogaster*. Although Toll was identified earlier as a gene controlling the dorso-ventral patterning during development of *Drosophila*, in 1996 Jules Hoffmann discovered that Toll signaling induces the expression of several host-defense mechanisms. The importance of this mechanism was illustrated by the decreased production of antimicrobial peptides and the enhanced susceptibility to infection of adult flies with mutations in the Toll pathway (**Fig 3.9**). For this work, Hoffmann was awarded part of the 2011 Nobel Prize in Physiology or Medicine.

In 1997, Janeway and Ruslan Medzhitov discovered a human homolog of Toll, subsequently termed [Toll-like receptor \(TLR\)](#), and showed that signaling by this receptor activates innate

and adaptive immune responses. Subsequently, numerous TLRs associated with resistance to infection by pathogens have been found in humans and other animals. In plants, proteins with domains resembling the ligand-binding regions of TLR proteins are involved in the production of antimicrobial peptides, indicating the ancient association of these domains with host defense.

3-5 Mammalian Toll-like receptors are activated by many different pathogen-associated molecular patterns.

There are 10 expressed TLR genes in humans and 12 in mice. Each TLR recognizes one or more distinct molecular ‘patterns’ present on microbes, which are referred to as pathogen-associated molecular patterns (PAMPs), as discussed at the beginning of this chapter. These molecules are general components of both pathogenic and nonpathogenic microorganisms and are sometimes called microbial-associated molecular patterns, or MAMPs, to reflect this fact. Collectively, the mammalian TLRs recognize molecules from bacteria, fungi, viruses, and protozoa. The mammalian TLRs and their known microbial ligands are listed in **Fig. 3.10**. Note that in almost all cases the ligands represent highly conserved features that are characteristic of these microbial groups. For example, [lipoteichoic acids](#) of Gram-positive bacterial cell walls and [lipopolysaccharide \(LPS\)](#) of the outer membrane of Gram-negative bacteria (see Fig. 2.9) are recognized by TLRs. In general, these microbial molecules are not found in vertebrate cells, which prevents inappropriate immune recognition of host cells. The major exceptions to this rule, however, are the DNA and RNA ligands for certain TLRs. As discussed later in this section, additional mechanisms are used to distinguish between microbial and host DNA and RNA.

Innate immune recognition by mammalian Toll-like receptors	
Toll-like receptor	Ligand
TLR-1:TLR-2 heterodimer	Lipomannans (mycobacteria)
TLR-2:TLR-6 heterodimer	Diacyl and triacyl lipopeptides (bacteria)
	Lipoteichoic acids (Gram-positive bacteria)
	Cell-wall β -glucans (fungi)
TLR-3	Double-stranded RNA (viruses), poly I:C
TLR-4	LPS (Gram-negative bacteria)
TLR-5	Flagellin (bacteria)

TLR-7	Single-stranded RNA (viruses)
TLR-8	Single-stranded RNA (viruses)
TLR-9	DNA with unmethylated CpG (bacteria and DNA viruses)
TLR-10 (human only)	Unknown
TLR-11 (mouse only)	Profilin and profilin-like proteins (<i>Toxoplasma gondii</i> , uropathogenic bacteria)
TLR-12 (mouse only)	Profilin (<i>Toxoplasma gondii</i>)
TLR-13 (mouse only)	Single-stranded RNA (bacterial ribosomal RNA)

Fig. 3.10 Innate immune recognition by Toll-like receptors. Each of the human or mouse TLRs recognizes one or more microbial molecular patterns, generally by direct interaction with molecules on the pathogen surface. Some Toll-like receptor proteins form heterodimers (for example, TLR-1:TLR-2 and TLR-2:TLR-6). LPS, lipopolysaccharide.

TLRs are sensors for microbes present in extracellular spaces. Some mammalian TLRs are cell-surface receptors, but others are located intracellularly in the membranes of endosomes, where they detect pathogens or their components that have been taken into cells by phagocytosis, receptor-mediated endocytosis, or macropinocytosis (**Fig. 3.11**). TLRs localized to the plasma membrane can also be incorporated into phagosomes, which further enables detection of phagocytosed microbes. TLRs are transmembrane proteins in which the extracellular region of the monomer is composed of 18–25 copies of a [leucine-rich repeat \(LRR\)](#). Each LRR of a TLR protein is composed of approximately 20–25 amino acids, and multiple LRRs create a horseshoe-shaped protein scaffold that is adaptable for ligand binding and recognition on both the outer (convex) and inner (concave) surfaces. Signaling by mammalian TLRs is activated when binding of a ligand induces formation of a dimer or induces conformational changes in a preformed TLR dimer. All mammalian TLR proteins have in their cytoplasmic tails a [TIR \(for Toll-IL-1 receptor\) domain](#), which is also found in the cytoplasmic tail of the receptor for the cytokines interleukin-1 α ([IL-1 \$\alpha\$](#)) and interleukin-1 β ([IL-1 \$\beta\$](#)). The TLR TIR domains mediate interaction with other TIR domain-containing signaling molecules, discussed in greater detail in [Section 3-7](#). For years after the discovery of the mammalian TLRs it was not known whether they made direct contact with microbial products or whether they sensed the presence of microbes by some indirect means. *Drosophila* Toll, for example, does not recognize pathogen products directly, but instead is activated when it binds a cleaved version of a self protein, Spätzle. *Drosophila* has other direct pathogen-recognition molecules, and these trigger the proteolytic cascade that ends in the cleavage of Spätzle. In this sense, Toll is not a classical pattern-recognition receptor. However, X-ray crystal structures of several mammalian dimeric TLRs bound to their ligands show that mammalian TLRs make direct contact with microbial ligands.

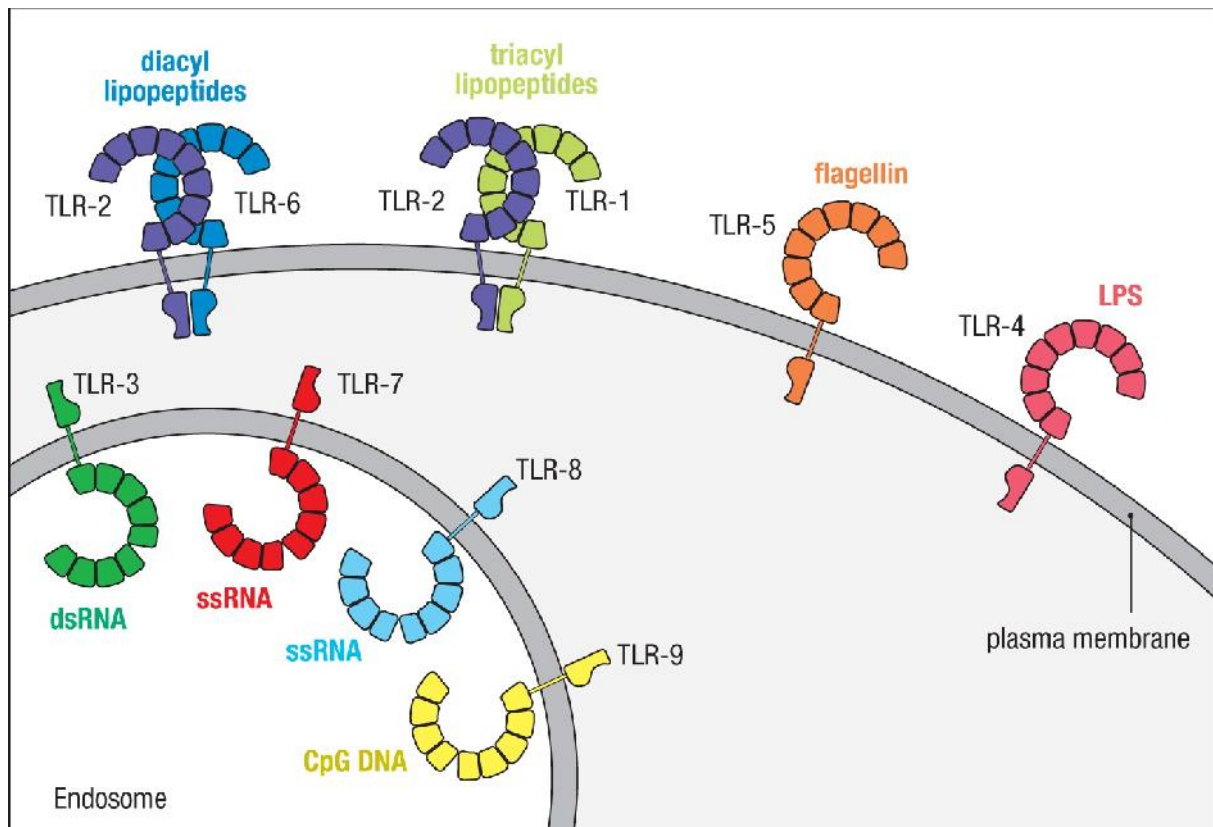


Fig. 3.11 The cellular locations of the mammalian Toll-like receptors. TLRs are transmembrane proteins where the extracellular region of the monomer contains 18–25 copies of the leucine-rich repeat (LRR), but these diagrams depict only 9 LRRs in each monomer for simplicity. Some TLRs are located on the cell surface of dendritic cells, macrophages, and other cells, where they are able to detect extracellular pathogen molecules. TLRs are thought to act as dimers. Only those that form heterodimers are shown in dimeric form here; the rest act as homodimers but are shown as monomers for simplicity. TLRs located intracellularly, in the membranes of endosomes, can recognize microbial components, such as DNA or RNA, that are accessible only after the microbe has been broken down. The ectodomains of TLR-3, TLR-7, TLR-8, and TLR-9 undergo proteolytic cleavage once they reach the endosome. This cleavage is required for their function and is thought to prevent responses to self nucleic acids in the extracellular space (see text for further discussion). TLR-11, TLR-12, and TLR-13 (also localized to endosomes) are not shown.

TLRs are expressed on many cell types, including macrophages, dendritic cells, B cells, stromal cells, and certain epithelial cells, enabling the initiation of antimicrobial responses in many tissues. Mammalian [TLR-1](#), [TLR-2](#), and [TLR-6](#) are cell-surface receptors that are activated by various ligands, including lipoteichoic acid from Gram-positive bacteria and [diacyl](#) and [triacyl lipoproteins](#) present in both Gram-positive and Gram-negative bacteria. Ligand binding induces the formation of heterodimers of TLR-2 and TLR-1 or of TLR-2 and TLR-6. The X-ray crystal structure of a synthetic triacyl lipopeptide ligand bound to TLR-1 and TLR-2 shows exactly how ligand binding induces dimerization (**Fig. 3.12**). Two of the three lipid chains bind to the convex surface of TLR-2, while the third binds to the convex

surface of TLR-1. Dimerization brings the cytoplasmic TIR domains of the TLR chains into close proximity with each other to initiate signaling. Similar interactions are presumed to occur with the diacyl lipopeptide ligands that induce the dimerization of TLR-2 and TLR-6.

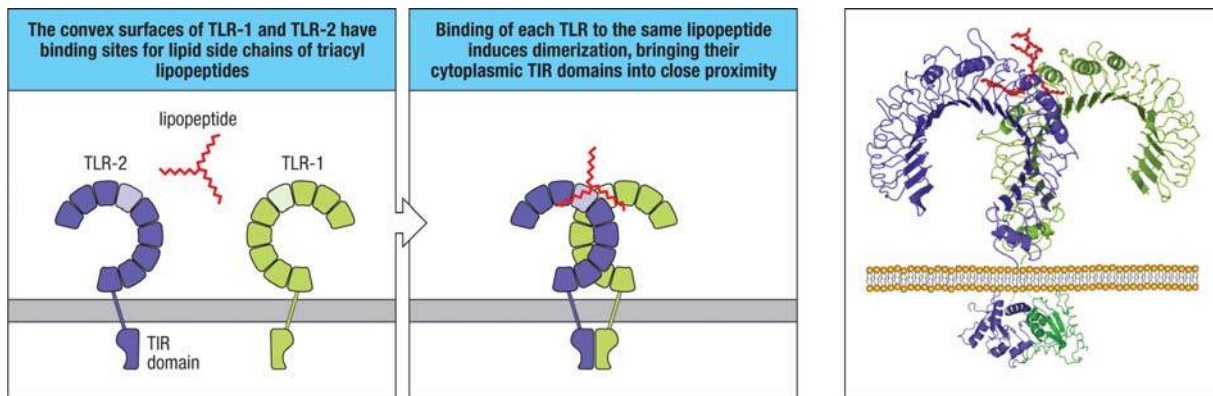


Fig. 3.12 Direct recognition of pathogen-associated molecular patterns by TLR-1 and TLR-2 induces dimerization of the TLRs and signaling. TLR-1 and TLR-2 are located on cell surfaces (left panel), where they can directly recognize bacterial triacyl lipoproteins (middle panel). The convex surfaces of their extracellular domains have binding sites for the lipid side chains of triacyl lipopeptides. In the crystal structure (right panel), the ligand is a synthetic lipid that can activate TLR-1:TLR-2 dimers; it has three fatty-acid chains bound to a polypeptide backbone. Two fatty-acid chains bind to a pocket on the convex surface of the TLR-2 ectodomain, and the third chain associates with a hydrophobic channel in the convex binding surface of TLR-1, inducing dimerization of the two TLR subunits and bringing their cytoplasmic Toll–IL-1 receptor (TIR) domains together to initiate signaling.

[TLR-5](#) is expressed on the cell surface of certain macrophages and dendritic cells and also in intestinal epithelial cells; it recognizes flagellin, a protein subunit of bacterial flagella. TLR-5 recognizes a highly conserved site on flagellin that is buried and inaccessible in the assembled flagellar filament. This means the receptor is activated only by monomeric flagellin, which can be shed during flagellar assembly or when flagella are degraded.

Mice, but not humans, express [TLR-11](#) and [TLR-12](#). TLR-11-deficient mice develop urinary infections caused by uropathogenic strains of *Escherichia coli*, although the bacterial ligand for TLR-11 has not yet been identified. TLR-11 and TLR-12 also both recognize protozoan parasites such as *Toxoplasma gondii* and *Plasmodium falciparum*. They bind to motifs that are present in the protozoan actin-binding protein [profilin](#) but absent in mammalian profilins. TLR-11 and TLR-12 are both required in macrophages and conventional dendritic cells for activation by *T. gondii* profilin, but TLR-12 plays a more dominant role. Mice lacking TLR-11 develop more severe tissue injury than normal mice on infection with *Toxoplasma*, whereas mice lacking TLR-12 die rapidly after infection.

Five of the mammalian TLRs (TLR3, TLR7, TLR8, TLR9, and TLR13) recognize nucleic acids. These receptors are all localized to endosomes. The endosome lumen is acidic and

contains many proteases, which degrade internalized microbes and release their nucleic acids for recognition by TLRs. All microbes possess some form of nucleic acid and so can be recognized by one or more of the nucleic acid-sensing TLRs; however, analysis of mice and humans with defects in one or more of these TLRs indicates that these receptors are most important for detection of viruses. In general, viruses lack other conserved features that could serve as PAMPs.

[TLR-7](#) and [TLR-8](#) share very similar amino acid sequences, and both receptors are activated by degradation products of [single-stranded RNA \(ssRNA\)](#). Many viruses (for example, orthomyxoviruses such as influenza; flaviviruses such as West Nile virus; and coronaviruses such as SARS-CoV-2) utilize ssRNA genomes, and TLR-7 and TLR-8 bind to small RNA fragments released when viruses are degraded in endosomes. Recent X-ray crystal structures of TLR-7 and TLR-8 have revealed two pairs of binding sites for distinct RNA fragments: guanosine and a short RNA fragment bind TLR-7, while uridine and a short RNA fragment bind TLR-8. Whether differences in the RNA sequences that activate TLR-7 and TLR-8 have consequences for antiviral immunity remains unclear. In mice, TLR-7 plays a dominant role over TLR-8 for detection of ssRNA viruses. For example, mice lacking TLR-7 have impaired immune responses to influenza. A viral detection function for TLR-8 in mice has not been established as clearly. The difference between mice and humans may be based, at least in part, on which cell types express the receptors in each species. In mice, TLR-7 is expressed broadly in many innate immune cell types, but in humans TLR-7 expression is limited to plasmacytoid dendritic cells and B cells, while TLR-8 is expressed more broadly. [TLR-13](#) also binds RNA but has specificity for certain bacterial ribosomal RNA sequences instead of viral RNA. TLR-13 is present in mice but not humans.

[TLR-9](#) recognizes single-stranded DNA sequences that contain [unmethylated CpG dinucleotides](#). In mammals, CpG dinucleotides in genomic DNA are often methylated on the cytosine by DNA methyltransferases. In the genomes of bacteria and many viruses, CpG dinucleotides remain unmethylated, so these sequences serve as ligands for TLR-9. The requirement for unmethylated CpG sequences is thought to limit the potential for recognition of self DNA by TLR-9. The structure of the TLR-9 dimer bound to DNA reveals a ligand-binding modality similar to what was previously described for TLR-7 and TLR-8: two pairs of binding sites within the dimer bind distinct DNA sequences. While the DNA genomes of viruses and bacteria are typically double-stranded, TLR-9 recognizes single-stranded DNA, so the DNA strands of pathogen genomes must be melted apart within endosomes prior to TLR-9 binding.

[TLR-3](#) is expressed by macrophages, conventional dendritic cells, and intestinal epithelial cells; it recognizes [double-stranded RNA \(dsRNA\)](#). Some viruses have genomes composed of dsRNA, but dsRNA is also generated when ssRNA viruses replicate or express genes. dsRNA can be recognized by TLR-3 after direct endocytosis of viruses with double-stranded RNA genomes, such as rotavirus, or by the phagocytosis of dying cells in which viruses are replicating. Crystallographic analysis shows that TLR-3 binds directly to dsRNA. TLR-3 has two contact sites for dsRNA: one on the amino terminus and a second near the membrane-proximal carboxyl terminus. Mutations in the ectodomain of human TLR-3, which produce a

dominantly acting loss-of-function mutant receptor, have been associated with encephalitis that is caused by a failure to control the herpes simplex virus. This increased susceptibility is somewhat surprising because herpes simplex virus is a DNA virus; the most likely explanation for sensing by TLR-3 is generation of dsRNA after transcription of overlapping viral genes encoded on opposing strands of DNA.



Case Study: Recurrent Herpes Simplex Encephalitis

To reach endosomes, TLR-3, TLR-7, TLR-8, TLR-9, TLR-11, TLR-12, and TLR-13 must interact with a specific protein, [UNC93B1](#). UNC93B1, which is composed of 12 transmembrane domains, binds TLRs in the endoplasmic reticulum and shuttles them to endosomes. Mice lacking this protein have defects in signaling by these endosomal TLRs because the receptors cannot exit the endoplasmic reticulum. Rare human mutations in UNC93B1 have been identified as causing susceptibility to herpes simplex encephalitis, similarly to TLR-3 deficiency, but they do not impair immunity to many other viral pathogens, presumably because of the existence of other viral sensors, which are discussed later in this chapter.

Sensing of nucleic acids enables broad recognition of diverse pathogens, but, in abnormal settings, TLRs may be activated by self-derived nucleic acids, leading to autoimmunity. In healthy cells, RNA and DNA are normally confined to the nucleus and cytoplasm and are not present in endosomes; however, nucleic acids can be released into the extracellular space during tissue damage or inflammation (for example, NETs). The localization of nucleic acid-sensing TLRs to endosomes reduces the likelihood that these receptors encounter such extracellular RNA and DNA. RNases and DNases in the extracellular space degrade RNA and DNA before they are taken up by endocytosis and delivered to TLRs. Another mechanism that limits TLR activation to endosomes is that the ectodomains of each of the nucleic acid-sensing TLRs must undergo proteolytic cleavage before the receptors can be activated; this cleavage occurs when TLRs arrive in endosomes, which prevents receptors from responding to ligands in other subcellular compartments, such as the cell surface (see Fig. 3.11). The two halves of the cleaved TLR remain associated and both contribute to ligand binding. Without this proteolytic event, the receptor cannot undergo the necessary conformational change required to initiate signaling.

In mouse models of systemic lupus erythematosus (SLE), an autoimmune disease associated with responses to self nucleic acids, TLR-7 and TLR-9 both contribute to disease. Several studies have identified polymorphisms in the human TLR-7 gene that are associated with increased risk of SLE, suggesting a potential role in this disease.

3-6 TLR-4 recognizes bacterial lipopolysaccharide in association with the host accessory proteins MD-2 and CD14.

Not all mammalian TLRs bind their ligands so directly. [TLR-4](#) is expressed by several types of immune-system cells, including dendritic cells and macrophages, and is important in sensing and responding to numerous bacterial infections. TLR-4 recognizes the LPS of Gram-negative bacteria by a mechanism that is partly direct and partly indirect. LPS varies in composition among different bacteria but essentially consists of a polysaccharide core attached to an amphipathic lipid, lipid A, with a variable number of fatty-acid chains per molecule. To recognize LPS, the ectodomain of TLR-4 uses an accessory protein, [MD-2](#). MD-2 initially binds to TLR-4 within the cell and is necessary both for the correct trafficking of TLR-4 to the cell surface and for the recognition of LPS. MD-2 associates with the central section of the curved ectodomain of TLR-4, binding off to one side as shown in **Fig. 3.13**. When the TLR-4:MD-2 complex encounters LPS, five lipid chains of LPS bind to a deep hydrophobic pocket of MD-2, but not directly to TLR-4, while a sixth lipid chain remains exposed on the surface of MD-2. This last lipid chain and parts of the LPS polysaccharide backbone can then bind to the convex side of a second TLR-4 ectodomain, inducing TLR-4 dimerization, which activates intracellular signaling pathways.

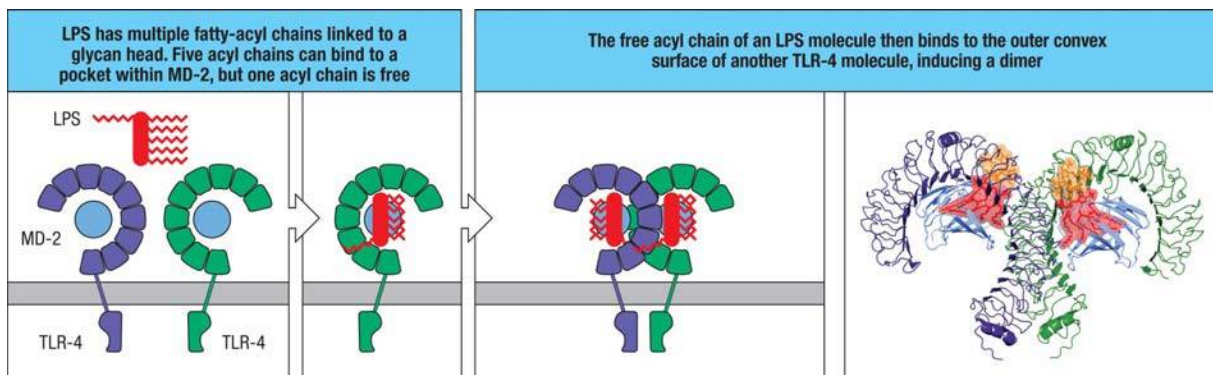


Fig. 3.13 TLR-4 recognizes LPS in association with the accessory protein MD-2. A side view of the symmetrical complex of TLR-4, MD-2, and LPS. TLR-4 polypeptide backbones are shown in green and purple. The MD-2 protein is shown in light blue. Five of the LPS acyl chains (shown in red) are inserted into a hydrophobic pocket within MD-2. The remainder of the LPS glycan (orange in the crystal structure) and one lipid chain make contact with the convex surface of a second TLR-4 monomer. The crystal structure shows the entire extracellular region of TLR-4, composed of the LRR region (shown in green and purple), but lacks the intracellular signaling domains.

TLR-4 activation by LPS involves two other accessory proteins besides MD-2. While LPS is normally an integral component of the outer membrane of Gram-negative bacteria, during infections it can become detached from the membrane and be picked up by the host [LPS-](#)

[binding protein](#) present in the blood and in extracellular fluid in tissues. LPS is transferred from LPS-binding protein to a second protein, CD14, which is present on the surface of macrophages, neutrophils, and dendritic cells. On its own, CD14 can act as a phagocytic receptor, but on macrophages and dendritic cells it also acts as an accessory protein for TLR-4.

The systemic injection of LPS causes a collapse of the circulatory and respiratory systems, a condition known as [shock](#). These dramatic effects of LPS are seen in humans as [septic shock](#), which results from an uncontrolled systemic bacterial infection, or [sepsis](#). In this case, LPS induces an overwhelming secretion of cytokines, including IL-1 β and TNF- α , causing systemic vascular permeability, an undesirable effect of its normal role in containing local infections. Mutant mice lacking TLR-4 function are resistant to LPS-induced septic shock but are highly sensitive to LPS-bearing pathogens such as *Salmonella enterica* ssp. *typhimurium*. In fact, TLR-4 (the first human homolog of Toll discovered by Medzhitov and Janeway, as discussed earlier) was identified as a receptor for LPS by positional cloning of its gene from the LPS-resistant C3H/HeJ mouse strain, which harbors a naturally occurring mutation in the cytoplasmic tail of TLR-4 that interferes with the receptor's ability to signal. For this discovery, the 2011 Nobel Prize in Physiology or Medicine was partly awarded to Bruce Beutler.

3-7 TLRs activate NF κ B, AP-1, and IRF transcription factors to induce the expression of inflammatory cytokines and type I interferons.

Signaling by mammalian TLRs in various cell types induces a diverse range of intracellular responses that together result in the production of inflammatory cytokines, chemotactic factors, antimicrobial peptides, and the antiviral cytokines interferon- α ([IFN- \$\alpha\$](#)) and interferon- β ([IFN- \$\beta\$](#)), the [type I interferons](#). TLR signaling achieves this by activating several signaling pathways that each activate different transcription factors. As mentioned earlier, ligand-induced dimerization of two TLR ectodomains brings the cytoplasmic TIR domains together, allowing them to interact with the TIR domains of cytoplasmic adaptor molecules that initiate intracellular signaling. There are five such adaptors used by mammalian TLRs: [MyD88](#), [TRIF](#), [MAL](#) (also known as TIRAP), [TRAM](#), and [BCAP](#). It is significant that the TIR domains of the different TLRs interact with different combinations of these adaptors because the choice of adaptor influences which of several downstream signals will be activated by the TLR. Recruitment of MyD88 and/or TRIF controls most of the signaling downstream of TLR activation, while MAL and TRAM facilitate the recruitment of MyD88 and TRIF, respectively, but do not directly participate in signal transduction. Most TLRs interact with MyD88 and not TRIF. TLR-3 interacts only with TRIF, and TLR-4 interacts with both MyD88 and TRIF. BCAP links TLR activation to distinct signaling pathways, but

the details of this signaling are less clear because BCAP was only recently discovered as a TLR adaptor.

Signaling by most TLRs activates the transcription factor **NFκB** (**Fig. 3.14**). Mammalian TLRs also activate several members of the **interferon regulatory factor (IRF)** transcription factor family through a second pathway, and they activate members of the **activator protein 1 (AP-1)** family, such as c-Jun, through yet another signaling pathway involving **mitogen-activated protein kinases (MAPKs)**. NFκB and AP-1 act primarily to induce the expression of pro-inflammatory cytokines and chemotactic factors. The IRF factors IRF3 and IRF7 are particularly important for inducing antiviral type I interferons, whereas a related factor, IRF5, is involved in the production of pro-inflammatory cytokines. Here we will describe how TLR signaling induces the transcription of various cytokine genes; later in the chapter, we will explain how those cytokines exert their various actions.

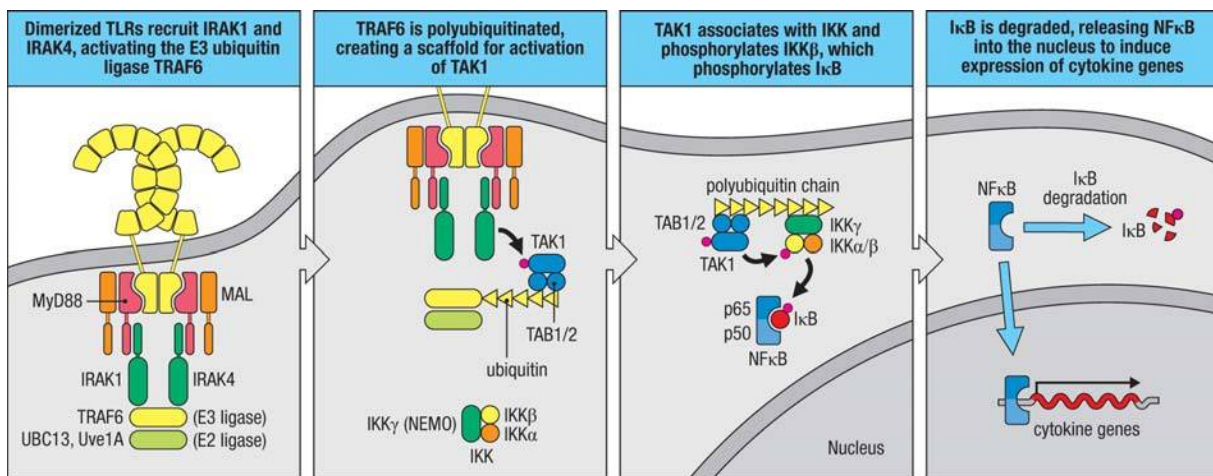


Fig. 3.14 TLR signaling can activate the transcription factor NFκB, which induces the expression of pro-inflammatory cytokines. First panel: TLRs signal via their cytoplasmic TIR domains, which are brought into proximity to each other by ligand-induced dimerization of their ectodomains. Some TLRs use the adaptor protein MyD88, and others use the MyD88/MAL pair to initiate signaling. The MyD88 death domain recruits the serine/threonine kinases IRAK1 and IRAK4, in association with the ubiquitin E3 ligase TRAF6. IRAK undergoes autoactivation and phosphorylates TRAF6, activating its E3 ligase activity. Second panel: TRAF6 cooperates with an E2 ligase (UBC13) and a cofactor (Uve1A) to generate polyubiquitin scaffolds (yellow triangles) by attachment of ubiquitin through its lysine 63 (K63) residue. This scaffold recruits a complex of proteins composed of the kinase TAK1 (transforming growth factor-β-activated kinase 1) and two adaptor proteins, TAB1 (TAK1-binding protein 1) and TAB2. TAB1 and TAB2 bind to polyubiquitin, bringing TAK1 into proximity with IRAK to become phosphorylated (red dot). Third panel: Activated TAK1 activates IKK the IκB kinase complex. First, the IKKγ subunit (NEMO) binds to the polyubiquitin scaffold and brings the IKK complex into proximity to TAK1. TAK1 then phosphorylates and activates IKKβ. IKKβ then phosphorylates IκB, the cytoplasmic inhibitor of NFκB. Fourth panel: Phosphorylated IκB is targeted by a process of ubiquitination (not shown) that leads to its degradation. This releases NFκB, which is

composed of two subunits, p50 and p65, into the nucleus, driving the transcription of many genes including those encoding inflammatory cytokines. TAK1 also stimulates activation of the mitogen-activated protein kinases (MAPKs) JNK and p38, which phosphorylate and activate AP-1 transcription factors (not shown).

We consider first the signaling pathway triggered by TLRs that use MyD88. Two protein domains of MyD88 are responsible for its function as an adaptor. MyD88 has a TIR domain at its carboxyl terminus that associates with the TIR domains in the TLR cytoplasmic tails. At its amino terminus, MyD88 has a [death domain](#), so named because it was first identified in signaling proteins involved in apoptosis, a type of programmed cell death. The MyD88 death domain associates with a similar death domain present in other intracellular signaling proteins. Both MyD88 domains are required for signaling, and rare mutations in either domain are associated with immunodeficiency characterized by recurrent bacterial infections in humans. The MyD88 death domain recruits and activates two serine/threonine protein kinases—[IRAK4](#) (IL-1 receptor–associated kinase 4) and [IRAK1](#)—via their death domains. This IRAK complex performs two functions: it recruits enzymes that produce a [signaling scaffold](#), and it uses this scaffold to recruit other molecules that are then phosphorylated by the IRAKs.



Case Study: Interleukin-1 Receptor–Associated Kinase 4 Deficiency

To form a signaling scaffold, the IRAK complex recruits the enzyme [TRAF6 \(tumor necrosis factor receptor–associated factor 6\)](#), which is an E3 [ubiquitin ligase](#) that acts in cooperation with [UBC13](#), an E2 ubiquitin ligase, and its cofactor [Uve1A](#) (together called [TRIKA1](#)) (see Fig. 3.14). The combined activity of TRAF6 and UBC13 is to ligate (unite with a chemical bond) one ubiquitin molecule to another protein, which can be another ubiquitin molecule, and thereby generate protein polymers. The polyubiquitin involved in signaling contains linkages between the lysine 63 residue on one ubiquitin and the carboxyl terminus of the next, forming [K63 linkages](#). This polyubiquitin polymer can be initiated on other proteins, including TRAF6 itself, or produced as free linear ubiquitin polymers, and can be extended to produce [polyubiquitin chains](#) that act as a platform—or scaffold—to which other signaling molecules bind. Next, the scaffold recruits a signaling complex consisting of the polyubiquitin-binding adaptor proteins [TAB1](#), [TAB2](#), and the serine/threonine kinase [TAK1](#) (see Fig. 3.14). By being brought onto the scaffold, TAK1 is phosphorylated by the IRAK complex, and activated TAK1 propagates signaling by activating certain MAPKs, such as c-Jun terminal kinase (JNK) and MAPK14 (p38 MAPK). These then activate AP-1 family transcription factors that transcribe cytokine genes.

TAK1 also phosphorylates and activates the [IKB kinase \(IKK\)](#) complex, which is composed

of three proteins: IKK α , IKK β , and IKK γ (the latter also known as [NEMO](#), for NFKB essential modifier). IKK γ functions by binding to polyubiquitin chains, which brings the IKK complex into proximity with TAK1. TAK1 phosphorylates and activates IKK β . IKK β then phosphorylates [IKB](#) (inhibitor of KB), which is a distinct molecule whose name should not be confused with IKK β . IKB is a cytoplasmic protein that constitutively binds to the transcription factor NFKB, which is a family of transcription factors. One version of NFKB is composed of the subunits [p50](#) and [p65](#), but other heterodimers and homodimers also exist. The binding of IKB traps the NFKB proteins in the cytoplasm. Phosphorylation by IKK induces the degradation of IKB, and this releases NFKB into the nucleus, where it can drive transcription of genes for pro-inflammatory cytokines such as TNF- α , IL-1 β , and [IL-6](#). The actions of these cytokines in the innate immune response are described in the second half of this chapter. The outcome of TLR activation can also vary depending on the cell type in which it occurs.

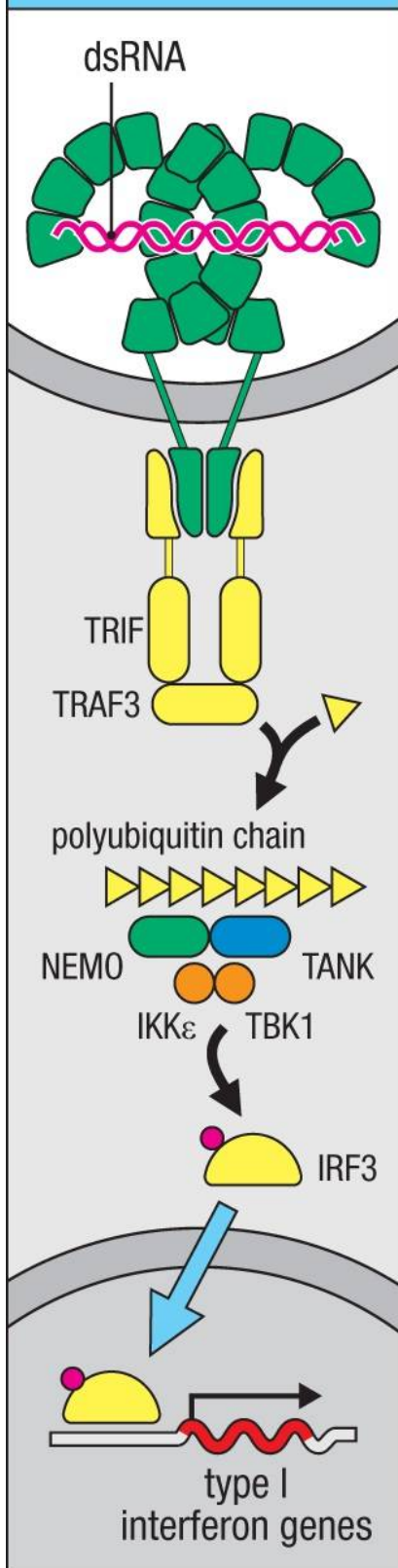
The ability of TLRs to activate NFKB is crucial to their role of alerting the immune system to the presence of bacterial pathogens. Rare instances of inactivating mutations in IRAK4 in humans cause an immunodeficiency, [IRAK4 deficiency](#), which, like MyD88 deficiency, is characterized by recurrent bacterial infections. Mutations in human IKK γ (NEMO) produce a syndrome known as [X-linked hypohidrotic ectodermal dysplasia and immunodeficiency](#) or [NEMO deficiency](#), which is characterized by both immunodeficiency and developmental defects.

TLR signaling also activates several members of the IRF family. IRF proteins reside in the cytoplasm and are inactive until they become phosphorylated on serine and threonine residues in their carboxyl termini. They then move to the nucleus as active transcription factors. Of the nine IRF family members, IRF3 and IRF7 are particularly important for TLR signaling and expression of antiviral type I interferons. For TLR-3, the cytoplasmic TIR domain interacts with the adaptor protein TRIF. TRIF interacts with the E3 ubiquitin ligase [TRAF3](#), which, like TRAF6, generates a polyubiquitin scaffold. In TLR-3 signaling, this scaffold recruits a multiprotein complex containing the kinases [IKK \$\epsilon\$](#) and [TBK1](#), which phosphorylate IRF3 (**Fig. 3.15**). TLR-4 also triggers this pathway by binding TRIF, but the IRF3 response induced by TLR-4 is relatively weak compared with that induced by TLR-3, and its functional role *in vivo* remains elusive. In contrast to TLR-3, the TLRs TLR-7, TLR-8, and TLR-9 use MyD88-dependent signaling to activate IRFs and induce expression of type I interferons. This signaling pathway is particularly relevant in plasmacytoid dendritic cells and represents a good example of how the outcome of TLR signaling can vary depending on the cell type in which it occurs. Upon TLR-7 and TLR-9 signaling in plasmacytoid dendritic cells, the MyD88 TIR domain recruits the IRAK1:IRAK4 complex as described earlier. However, the IRAK complex carries out a distinct function beyond recruiting TRAFs that generate a signaling scaffold. In these cells, IRAK1 can also physically associate with IRF7, which is highly expressed by plasmacytoid dendritic cells. This allows IRF7 to become phosphorylated by IRAK1, leading to induction of type I interferons (see Fig. 3.15). Not all IRF factors regulate type I interferon genes; IRF5, for example, plays a role in the induction of pro-inflammatory cytokines.



Case Study: X-linked Hypohidrotic Ectodermal Dysplasia and Immunodeficiency

TLR-3 in endosome binds dsRNA and signals via TRIF to induce IFN gene expression



TLR-7 in endosome binds ssRNA and signals via MyD88 to induce IFN gene expression

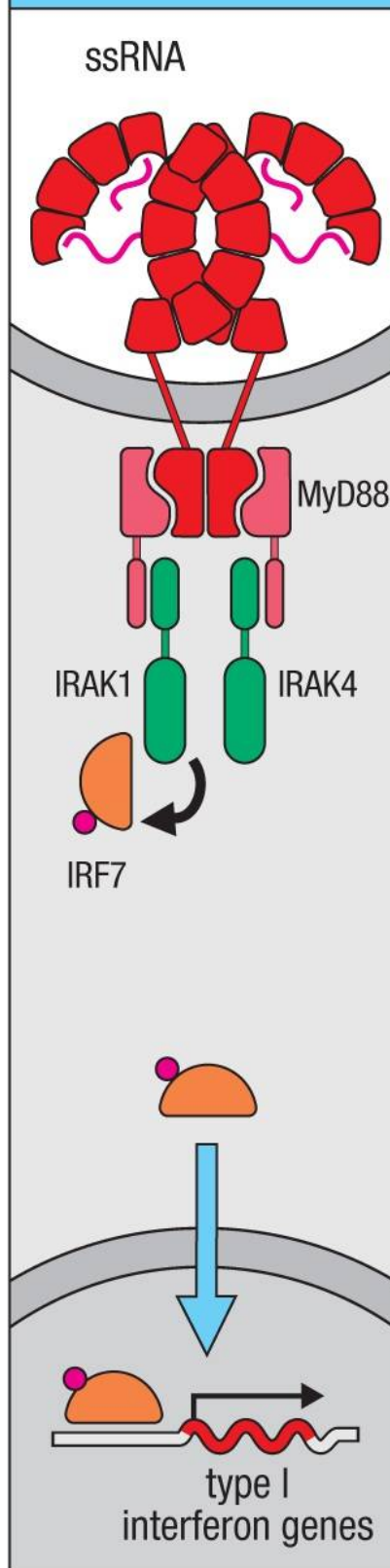


Fig. 3.15 Expression of antiviral interferons in response to viral nucleic acids can be stimulated by two different pathways from different TLRs. Left panel: TLR-3, expressed by dendritic cells and macrophages, senses double-stranded viral RNA (dsRNA). TLR-3 signaling uses the adaptor protein TRIF, which recruits the E3 ligase TRAF3 to generate K63-linked polyubiquitin chains. This scaffold recruits NEMO and TANK (TRAF family member–associated NFκB activator), which associate with the serine/threonine kinases IKKε (IκB kinase ε) and TBK1 (TANK-binding kinase 1). TBK1 phosphorylates (red dot) the transcription factor IRF3, and IRF3 then enters the nucleus and induces expression of type I interferon genes. Right panel: TLR-7, expressed by plasmacytoid dendritic cells, detects single-stranded RNA (ssRNA) and signals through MyD88. Here, IRAK1 directly recruits and phosphorylates IRF7, which is also highly expressed in plasmacytoid dendritic cells. IRF7 then enters the nucleus to induce expression of type I interferons.

The collective ability of TLRs to activate both IRFs and NFκB means that they can stimulate either antiviral or antibacterial responses as needed. TLRs are expressed by different types of cells involved in innate immunity and by some stromal and epithelial cells, and the responses generated will differ in some respects depending on what type of cell is being activated.

3-8 The RIG-I–like receptors detect cytoplasmic viral RNAs and activate MAVS to induce type I interferon production and pro-inflammatory cytokines.

The TLRs, being expressed on the cell's plasma membrane or endocytic vesicles, are primarily sensors of extracellular microbial products. Since the discovery of the mammalian TLRs, additional families of innate sensors have been identified that detect microbial products in the cytosol (**Fig. 3.16**). Certain TLR family members detect extracellular viral RNAs and DNAs that enter the cell from the endocytic pathway. By contrast, viral RNAs produced within a cell are sensed by a separate family of proteins called the [RIG-I–like receptors \(RLRs\)](#). These proteins serve as viral sensors by binding to viral dsRNAs using an RNA helicase–like domain. The first of these sensors to be discovered was [RIG-I \(retinoic acid–inducible gene I\)](#). RIG-I is widely expressed across tissues and cell types. Mice deficient in RIG-I are highly susceptible to infection by several kinds of single-stranded RNA viruses, including paramyxoviruses, rhabdoviruses, orthomyxoviruses, and flaviviruses, but not picornaviruses.

Ligand	Recognition strategies
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RIG-I	Triphosphate dsRNA
MDA-5	dsRNA
cGAS	DNA
NOD1	γ -Glutamyl diaminopimelic acid (iE-DAP)
NOD2	Muramyl dipeptide (MDP)
NLRP1	Pathogen protease activity
NLRP3	Reduced intracellular potassium, ROS, disruption of lysosomes
NAIP1 with NLRC4 (mouse)	Needle subunit of bacterial T3SS
NAIP2 with NLRC4 (mouse)	Rod subunit of bacterial T3SS
NAIP5 with NLRC4 (mouse)	Flagellin
NAIP with NLRC4 (human)	Flagellin
Pyrin	Inactivation of Rho GTPases
AIM2	DNA

Fig. 3.16 Ligands and recognition strategies for cytosolic innate immune receptors.

Some cytosolic innate immune receptors detect pathogens directly, while others are activated by the activities of pathogens. This figure lists the major receptors for which the ligands or mechanisms of activation are known. ROS, reactive oxygen species; T3SS, type 3 secretion system.

The RLR proteins also contain two amino-terminal [caspase recruitment domains \(CARDs\)](#). CARD was initially recognized in a family of proteases called [caspases](#) (for cysteine–aspartic acid proteases), which are important in many intracellular pathways, including those leading to cell death by apoptosis. CARD is structurally related to the death domain in MyD88 and can dimerize with CARDs on other proteins to activate signaling to produce type I interferons when viral RNAs are bound (**Fig. 3.17**).

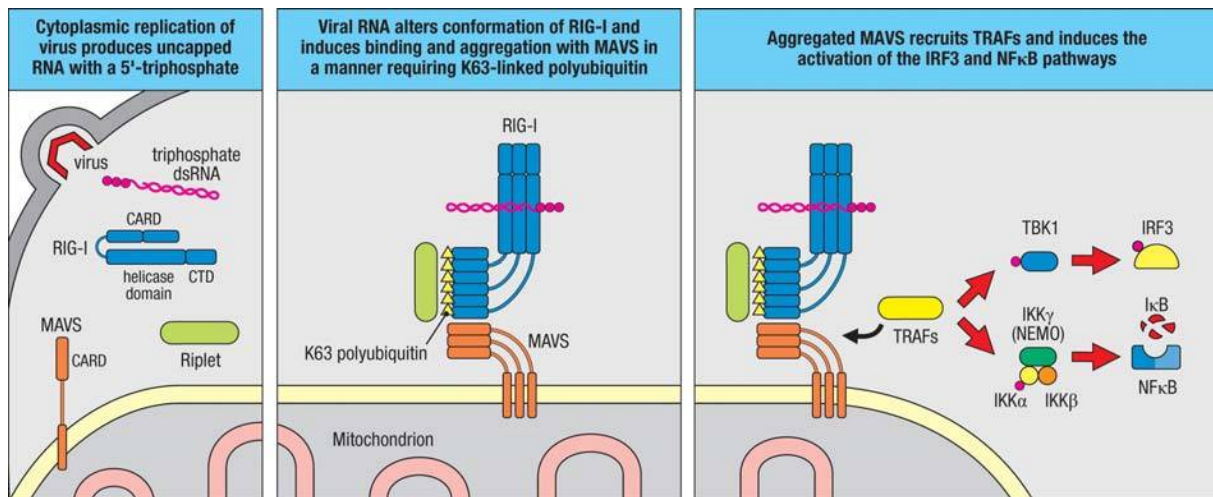


Fig. 3.17 RIG-I and other RLRs are cytoplasmic sensors of viral RNA. First panel: Before detecting viral RNA, RIG-I is cytoplasmic and in an inactive, autoinhibited conformation. The adaptor protein MAVS is attached to the mitochondrial outer membrane. Second panel: Detection of uncapped 5'-triphosphate dsRNA by RIG-I leads to assembly of RIG-I proteins along the dsRNA, enabling aggregation of the CARDS and interaction with the amino-terminal CARD of MAVS. This interaction involves the generation of K63-linked polyubiquitin from the E3 ligases Riplet and TRIM25 (the latter not shown). Third panel: The aggregation induces a proline rich region of MAVS to interact with TRAFs (see text) and leads to the generation of additional K63-linked polyubiquitin scaffold. As in TLR signaling, this scaffold recruits TBK1 and IKK complexes (see Figs. 3.14 and 3.15) to activate IRF3 and NFκB, producing type I interferons and pro-inflammatory cytokines, respectively. CTD, C-terminal domain.

RIG-I discriminates between host and viral RNA by sensing differences at the 5' end of RNA transcripts. Eukaryotic RNA is transcribed in the nucleus and contains a 5'-triphosphate group on its initial nucleotide that undergoes subsequent enzymatic modification called [capping](#) by the addition of a 7-methylguanosine to the 5'-triphosphate. Most RNA viruses do not replicate in the nucleus, where capping normally occurs, so their RNA genomes do not undergo this modification. Biochemical and structural studies have determined that RIG-I senses dsRNAs with the unmodified 5'-triphosphate. Many viruses produce dsRNAs with 5'-triphosphate that are detected by RIG-I, either during viral replication or due to hairpins that form when base pairing occurs within the viral RNA. In contrast, the picornaviruses, which include poliovirus and hepatitis A, replicate by a mechanism that involves the covalent attachment of a viral protein to the 5' end of the viral RNA, so that the 5'-triphosphate is masked, which explains why RIG-I is not involved in sensing them.

[MDA-5 \(melanoma differentiation-associated 5\)](#) is similar in structure to RIG-I, but it senses longer dsRNAs and does not require the 5'-triphosphate for recognition. In contrast to RIG-I-deficient mice, mice deficient in MDA-5 are susceptible to picornaviruses, indicating that these two sensors of viral RNAs have crucial but distinct roles in host defense. The RLR family member [LGP2](#) (encoded by *DHX58*) contains a helicase domain but lacks CARDS. LGP2 appears to cooperate with RIG-I and MDA-5 in the recognition of viral RNA, as mice

lacking LGP2 have impaired antiviral responses normally mediated by RIG-I or MDA-5. This cooperative viral recognition by LGP2 appears to depend on its helicase domain, because in mice, mutations that disrupt the helicase ATPase activity result in impaired IFN- β production in response to various RNA viruses.

Sensing of viral RNAs by RIG-I and MDA-5 activates signaling that leads to type I interferon production appropriate for defense against viral infection (see Fig. 3.17). Upon binding dsRNA, both RIG-I and MDA-5 assemble into filaments along the dsRNA, which promotes the aggregation of CARDs from multiple RIG-I or MDA-5 molecules. For RIG-I, aggregation of the CARDs is facilitated by recruitment of E3 ligases [Riplet](#) (encoded by *RNF153*) and [TRIM25](#), which generate K63-linked polyubiquitin scaffolds (see [Section 3-7](#)) that stabilize interactions between the CARDs. The initiation of MDA-5 signaling does not require Riplet or TRIM25, most likely because MDA-5 more readily assembles into filaments along dsRNA, which enables aggregation of the CARDs. The aggregated CARDs interact with a downstream adaptor protein called [MAVS \(mitochondrial antiviral signaling protein\)](#). MAVS is attached to the outer mitochondrial membrane and contains its own CARD that mediates interaction with RIG-I and MDA-5 assembled on dsRNAs. Activated MAVS also forms polymers. In this state, MAVS propagates signals by recruiting various TRAF family E3 ubiquitin ligases, including TRAF2, TRAF3, TRAF5, and TRAF6. The relative importance of each E3 ligase may differ between cell types, but their further production of K63-linked polyubiquitin leads to activation of TBK1 and IRF3 and production of type I interferons, as described for TLR-3 signaling (see Fig. 3.15), and also to activation of NF κ B.

The recognition of specific features of dsRNA (5'-triphosphate for RIG-I and longer dsRNAs for MDA-5) reduces the chance that cellular RNAs will trigger activation of RLRs. However, humans have been identified with mutations in RIG-I and MDA-5 that result in inappropriate activation of the receptors, which manifests as severe diseases such as SLE, Aicardi–Goutières syndrome, and Singleton–Merten syndrome. How these mutations facilitate RIG-I or MDA-5 activation is an area of active investigation, but in some cases they appear to enable the inappropriate activation of the receptors by endogenous RNAs.

3-9 Cytosolic DNA activates the cGAS–STING pathway to induce production of type I interferons.

Innate sensors that recognize cytosolic RNA use specific modifications, such as the 5' cap, to discriminate between host and viral origin. Host DNA is generally restricted to the nucleus, but viral, microbial, or protozoan DNA may become located in the cytosol during various stages of infection. The fact that introduction of DNA into the cytosol induces the production of type I interferons was known for a number of years before the sensor or signaling proteins that mediate the response were identified. It is now known that the main sensor of cytosolic

DNA is an enzyme called [cGAS, for cyclic GAMP synthase](#). cGAS binds directly to cytosolic DNA, which stimulates its enzymatic activity to produce a second messenger molecule called [cyclic guanosine monophosphate–adenosine monophosphate \(cyclic GMP–AMP\), or cGAMP](#). cGAMP binds to and activates [STING \(stimulator of interferon genes\)](#), which is a protein encoded by *TMEM173* and localized within the endoplasmic reticulum membrane by an amino-terminal tetraspan transmembrane domain (**Fig. 3.18**). STING exists as an inactive homodimer until cGAMP binding, which induces a conformation change, recruitment of TBK1, activation of IRF3, and production of type I interferons, similarly to signaling by TLR-3 and RLRs (see Figs. 3.15 and 3.17).

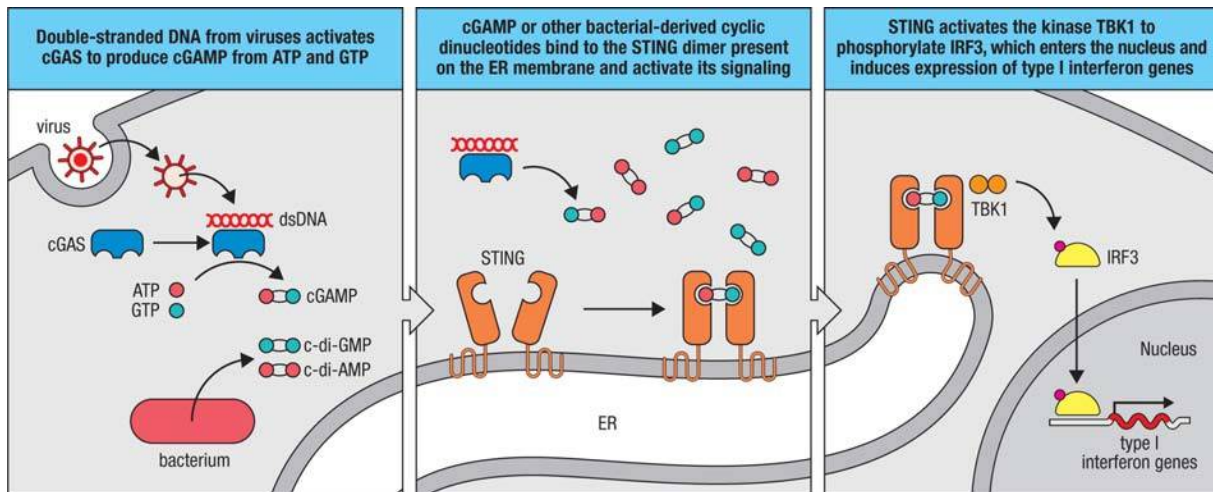


Fig. 3.18 cGAS is a cytosolic sensor of DNA that signals through STING to activate type I interferon production. First panel: cGAS resides in the cytoplasm and serves as a sensor of double-stranded DNA (dsDNA) from viruses. When cGAS binds dsDNA, its enzymatic activity is stimulated, leading to production of cyclic GMP–AMP (cGAMP). Bacteria that infect cells produce second messengers such as cyclic dinucleotides, including cyclic diguanylate monophosphate (c-di-GMP) and cyclic diadenylate monophosphate (c-di-AMP). Second panel: cGAMP and other bacterial dinucleotides can bind and activate the STING dimer present on the endoplasmic reticulum (ER) membrane. Third panel: In this state, STING activates TBK1. Active TBK1 activates IRF3, as described in Fig. 3.15.

cGAMP is a member of a larger family of [cyclic dinucleotides \(CDNs\)](#) produced by diverse organisms. In fact, the importance of CDNs was first appreciated in bacteria, where CDNs such as cyclic diguanylate monophosphate (c-di-GMP) and cyclic diadenylate monophosphate (c-di-AMP) are synthesized by bacterial enzymes and function as signaling molecules. Before the discovery of mammalian cGAS and cGAMP, these bacterial CDNs were shown to bind and activate STING directly, suggesting that STING may itself serve as a direct sensor of intracellular bacterial infection.

cGAS and STING are expressed in most immune-cell types and many nonimmune cells, and there is good evidence that detection of DNA by the cGAS–STING pathway is important for the immune response to infection, especially viral infection. Mice harboring inactivated

cGAS or STING genes show increased susceptibility to a number of DNA viruses. The cGAS–STING pathway can also be activated during infections with certain intracellular bacterial pathogens, either by bacterial DNA released during infection or by bacterial CDNs that bind STING directly, as discussed above. However, the relevance of cGAS–STING signaling for the host response to bacterial infections remains rather unclear, as the type I interferon response induced by cGAS–STING signaling is predominantly antiviral. In some contexts, the production of type I interferons can impair the response to bacteria, so it is possible that activation of cGAS–STING by some pathogens represents a pathogen-evolved mechanism of immune evasion. It is also possible that STING activation can induce other host-defense mechanisms besides production of type I interferons, such as autophagy, which may help to degrade intracellular bacteria.

As with many of the nucleic acid–sensing PRRs discussed earlier, inappropriate activation of the cGAS–STING pathway can lead to inflammatory diseases. An enzyme called [Trex1](#) limits activation of cGAS by degrading DNA present in the cytosol. Mice and humans with defective Trex1 develop autoimmune diseases similar to those described earlier for humans with gain-of-function mutations in MDA-5 and RIG-I (see [Section 3-8](#)). DNA accumulates in the cytosol of Trex1-deficient cells, where it activates cGAS. Accordingly, loss of cGAS reverses disease symptoms seen in Trex1-mutant mice. The origin of the cytosolic DNA in Trex1-mutant cells is an area of active investigation, but one source appears to be endogenous retroelements, mobile genetic elements present within mammalian genomes. Other individuals have been identified with mutations in STING that result in constitutive signaling, leading to an autoimmune disorder called STING-associated vasculopathy with onset in infancy (SAVI).

3-10 NLRs comprise a large family of intracellular sensors with diverse functions.

One large group of cytosolic innate sensors is collectively called [NOD-like receptors \(NLRs\)](#), and their family members contain a [nucleotide-binding and oligomerization domain \(NOD\)](#) and leucine-rich repeats (LRRs) in their structure. Each NLR possesses additional domains that result in diverse functions and, in some instances, enable detection of infection. Some NLRs activate NFκB to initiate the same inflammatory responses as the TLRs, while other NLRs trigger a very different response that leads to cell death and the release of different pro-inflammatory cytokines. We will discuss this second type of response in the next section. Not all NLRs are involved in pathogen detection. For example, NLRC5 and CIITA regulate the transcription of MHC class I and class II genes, respectively.

NOD1 and NOD2 are NLRs that recognize fragments of bacterial cell-wall peptidoglycans, although it is not known whether this occurs through direct binding or via accessory proteins. [NOD1](#) senses [γ-glutamyl diaminopimelic acid \(iE-DAP\)](#), a breakdown product of peptidoglycans of Gram-negative bacteria such as *Salmonella* and some Gram-positive bacteria such as *Listeria*, whereas [NOD2](#) recognizes [muramyl dipeptide \(MDP\)](#), which is

present in the peptidoglycans of most bacteria. NOD ligands may enter the cytoplasm as a result of intracellular infection, but they may also be transported from materials captured by endocytosis, as mice lacking certain oligopeptide transporters have reduced responses to NOD ligands.

Both NOD1 and NOD2 contain CARDs. When NOD1 or NOD2 is activated, it recruits the CARD-containing serine/threonine kinase [RIP2](#) (also known as RICK and RIPK2) (**Fig. 3.19**). RIP2 associates with the E3 ligases cIAP1, cIAP2, and XIAP, whose activity generates a polyubiquitin scaffold similar to the scaffold already discussed in the context of TLR signaling. This scaffold recruits TAK1 and IKK and results in activation of NF κ B as shown in Fig. 3.14. NF κ B then induces the expression of genes for inflammatory cytokines and for enzymes involved in the production of [nitric oxide \(NO\)](#), which is toxic to bacteria and intracellular parasites. In keeping with their role as sensors of bacterial components, NOD proteins are expressed in cells that are routinely exposed to bacteria. These include epithelial cells forming the barrier that bacteria must cross to establish an infection in the body and the macrophages and dendritic cells that ingest bacteria that have succeeded in entering the body. Macrophages and dendritic cells express TLRs as well as NOD1 and NOD2 and are activated by both pathways.

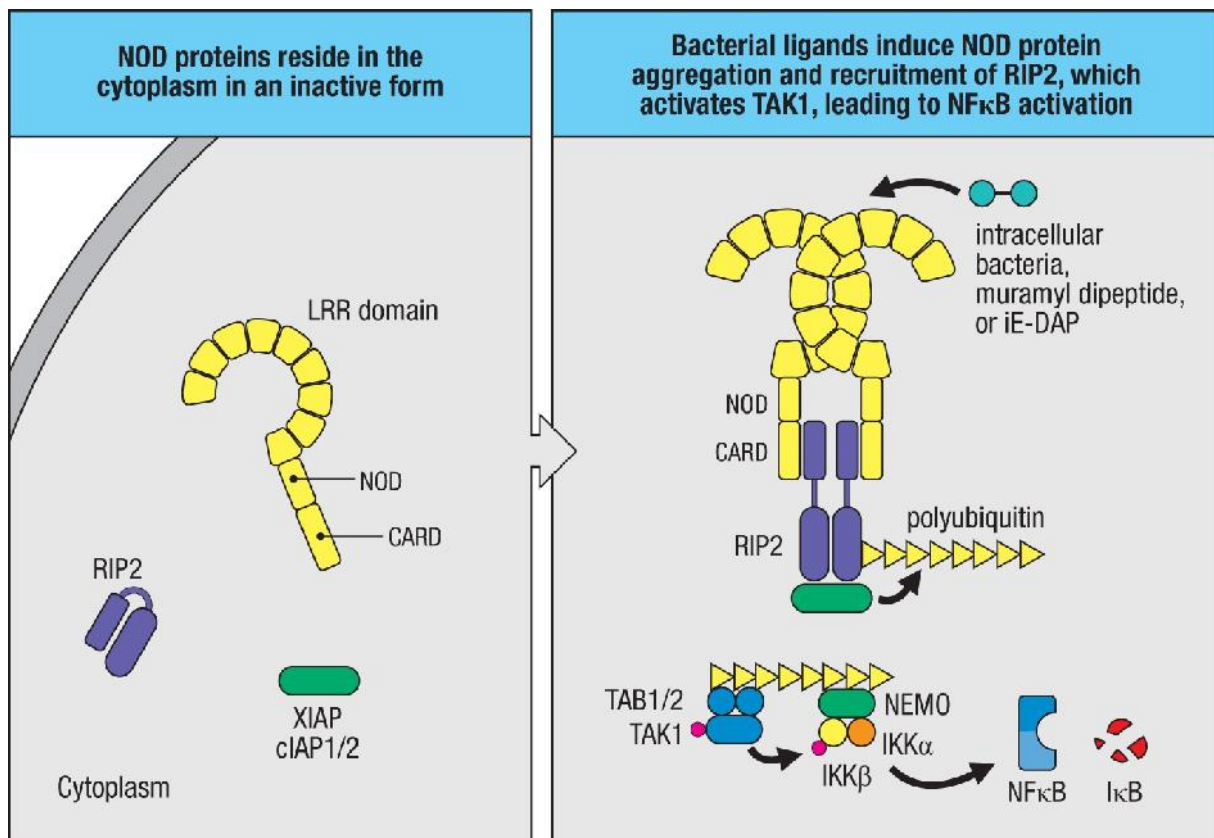


Fig. 3.19 Intracellular NOD proteins sense the presence of bacteria by recognizing bacterial peptidoglycans and activate NF κ B to induce the expression of pro-inflammatory genes. First panel: NOD proteins reside in an inactive state in the cytoplasm, where they serve as sensors for various bacterial components. Second panel: Degradation of bacterial cell-wall peptidoglycans produces muramyl dipeptide, which activates NOD2. NOD1 is activated by γ -glutamyl diaminopimelic acid (iE-DAP), a

breakdown product of Gram-negative bacterial cell walls. Whether these ligands are recognized directly by NOD1 and NOD2 remains unclear. Activation of NOD1 and NOD2 induces aggregation, allowing CARD-dependent recruitment of the serine/threonine kinase RIP2, which associates with E3 ligases, including XIAP (X-linked inhibitor of apoptosis protein), cIAP1 (cellular inhibitor of apoptosis 1), and cIAP2. This recruited E3 ligase activity produces a polyubiquitin scaffold, as in TLR signaling, and the association of TAK1 and the IKK complex with this scaffold leads to the activation of NF κ B, as shown in Fig. 3.14. In this pathway, RIP2 acts as a scaffold to recruit XIAP, and RIP2 kinase activity is not required for signaling.



Case Study: Crohn's Disease

NOD2 seems to have a more specialized role, being strongly expressed in the Paneth cells of the gut, where it regulates the expression of potent antimicrobial peptides such as the α - and β -defensins (see [Chapter 2](#)). Loss-of-function mutations in NOD2 in humans are associated with the inflammatory bowel condition known as [Crohn's disease](#) (discussed in [Chapter 15](#)). Lack of NOD2 function is thought to diminish the production of defensins and other antimicrobial peptides, thereby weakening the natural barrier function of the intestinal epithelium and leading to the inflammation characteristic of this disease. Gain-of-function mutations in human NOD2 are associated with the inflammatory disorders [early-onset sarcoidosis](#) and [Blau syndrome](#), which are characterized by spontaneous inflammation in tissues such as the liver or in the joints, eyes, and skin. Activating mutations in the NOD domain seem to promote the signaling cascade in the absence of ligand, leading to an inappropriate inflammatory response in the absence of pathogens.



Case Study: Hereditary Periodic Fever Syndromes

3-11 Certain NLR proteins react to infection or cellular damage by forming an inflammasome that induces cell death and secretion of inflammatory cytokines.

Some NLR proteins respond to infection or, in some cases, cellular damage in the absence of infection by inducing the formation of a multiprotein complex known as an [inflammasome](#). Inflammasomes serve as platforms for the activation of inflammatory caspases such as [caspase 1](#), a protease that when activated induces the proteolytic cleavage and release of pro-inflammatory cytokines IL-1 β and IL-18. Cleavage of these cytokines converts them into their active forms. Activated caspase 1 also cleaves [gasdermin D](#), a cytosolic protein that, once cleaved, inserts into the plasma membrane and assembles into a pore (**Fig. 3.20**). This pore is required for exit of active IL-1 β and IL-18 from the cell, as these cytokines lack signal peptides and are otherwise confined to the cytosol. The gasdermin D pore also induces a lytic form of cell death called [pyroptosis](#) ('fiery death'). While the induction of host-cell death in response to infection may seem counterproductive at first glance, pyroptosis exposes intracellular pathogens that otherwise hide within host cells (for example, macrophages). These pathogens are now visible and thus susceptible to killing by neutrophils and activated macrophages recruited by the inflammatory cytokines produced by inflammasome activation.

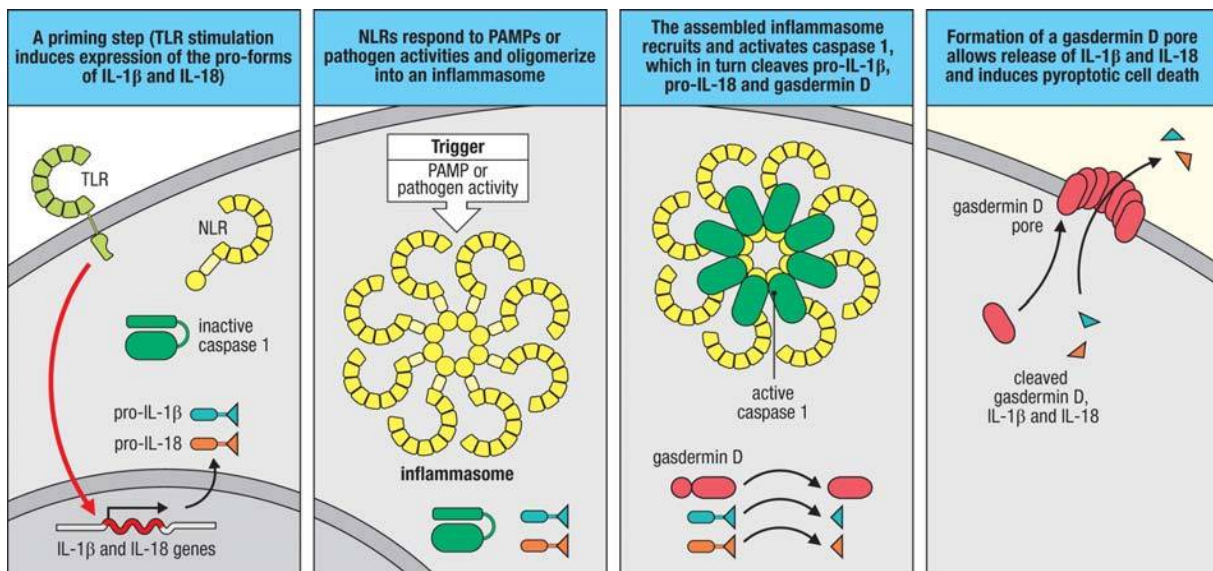


Fig. 3.20 Inflammasomes activate inflammatory caspases, leading to secretion of pro-inflammatory cytokines and pyroptosis. The general steps of inflammasome activation and the downstream events after activation are shown for a 'generic' NLR protein. First panel: The NLR is in an inactive conformation in unstimulated cells. Second and third panels: Some NLRs detect pathogens directly while others respond to pathogen activities. NLR activation leads to a conformation change that enables oligomerization of the NLR and recruitment of the inflammatory protease caspase 1. In unstimulated cells caspase 1 is inactive, but binding to the assembled inflammasome activates its protease activity. Active caspase 1 cleaves pro-IL-1 β , pro-IL-18, and gasdermin D, converting these proteins into their active forms. Fourth panel: Cleaved gasdermin D proteins form a pore in the plasma membrane, which allows release of IL-1 β and IL-18. The loss of membrane integrity also induces lysis of cells. This form of cell death is called pyroptosis. Most cells require a separate priming step to induce expression of IL-1 β or IL-18 (and in some cases inflammasome components themselves). This priming step, which can be achieved through TLR activation, may

help prevent accidental inflammasome activation.

Certain NLR proteins function as PRRs by inducing inflammasome assembly after directly binding to molecules produced by pathogens. For example, [NAIPs \(NLR family, apoptosis inhibitory proteins\)](#) assemble into inflammasomes with [NLRC4](#) after recognizing certain bacterial proteins that access the host cytosol. Mice encode a small family of NAIP proteins, each of which recognizes a distinct ligand. For example, [NAIP5](#) recognizes bacterial flagellin, whereas NAIP1 and [NAIP2](#) recognize distinct protein subunits of the [type III secretion system \(T3SS\)](#), a needle-like macromolecular machine that bacterial pathogens use to inject proteins into host cells. Humans encode a single NAIP protein that has a broad specificity for flagellin and T3SS proteins. NAIP5 recognizes the highly conserved D0 domain of flagellin, which is distinct from the region of flagellin recognized by TLR-5. As a cell-surface receptor, TLR-5 can detect flagellin from potentially any extracellular flagellated bacterium, including harmless commensal species, whereas cytosolic NAIP5 will respond only to bacteria that have the capacity to invade or access the cytosol—typically only pathogenic species.

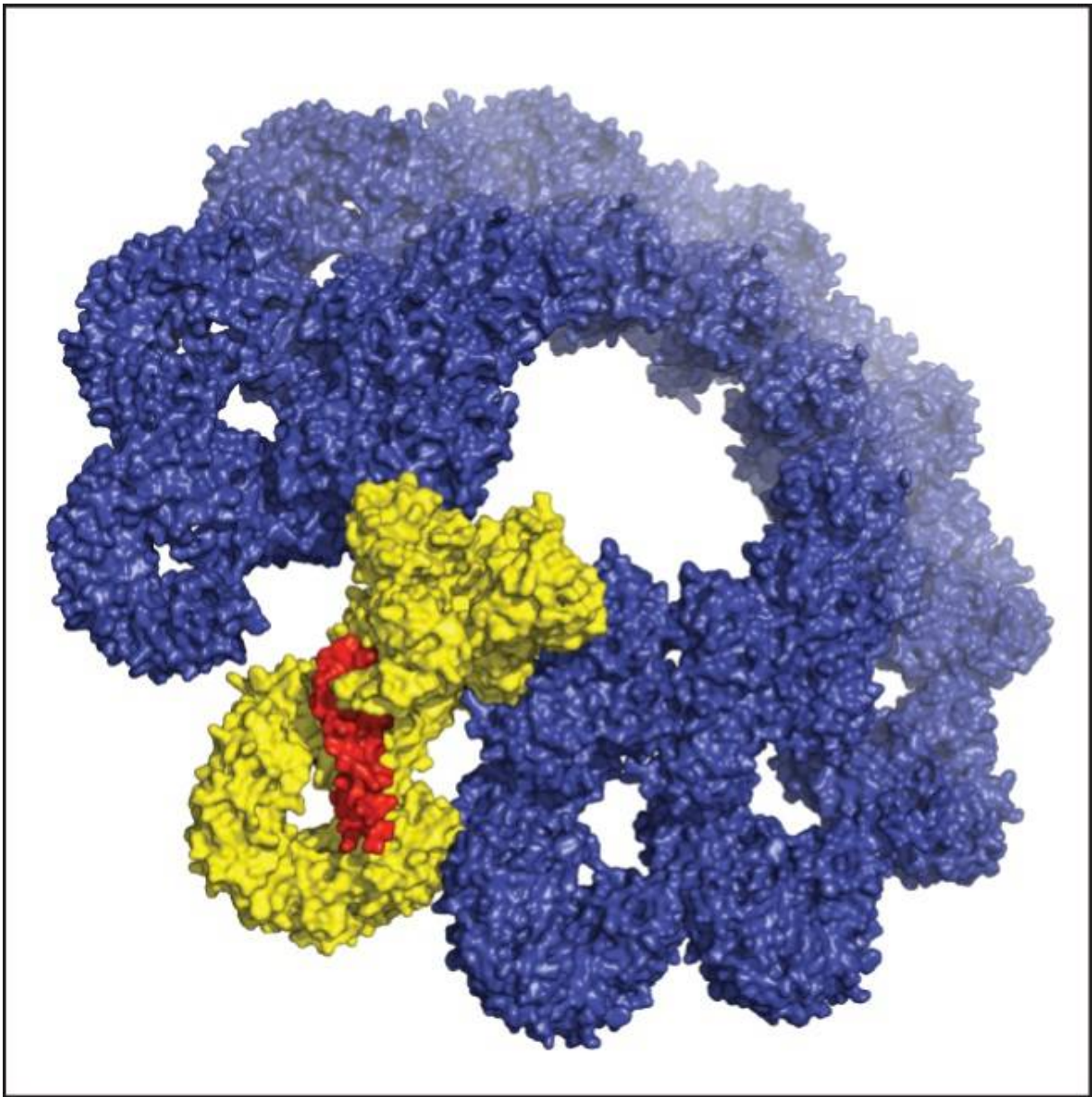


Fig. 3.21 NAIP5 and NLRC4 form an inflammasome upon flagellin recognition. A model of the structure of the NAIP5–NLRC4 inflammasome bound to flagellin. Note that NAIP5 (shown in yellow) directly contacts the D0 domain of flagellin (shown in red). This interaction induces a conformational change in NAIP5 that promotes interaction with multiple NLRC4 proteins (shown in blue) and assembly into an inflammasome. The CARDS of NLRC4 (not shown) form an inner ring and are responsible for the recruitment and activation of caspase 1.

Structural analyses of the NAIP5–NLRC4 inflammasome bound to flagellin have confirmed that NAIP5 binds flagellin directly. This binding induces a conformational change in NAIP5, promoting interaction with NLRC4 and formation of an inflammasome (**Fig. 3.21**). NLRC4 contains an N-terminal CARD that, upon inflammasome assembly, interacts with the CARD present in caspase 1 to mediate its dimerization and activation. Active caspase 1 then cleaves multiple substrates, including the pro-forms of IL-1 β , IL-18, and gasdermin D, as discussed

earlier.

The importance of NAIP–NLRC4 inflammasomes for the host response to infection has been established by studies of cells or mice with disrupted NAIP or NLRC4 genes. Mice lacking NLRC4 are more susceptible to infection by numerous intracellular bacterial pathogens, including *Salmonella enterica* ssp. *typhimurium* and *Legionella pneumophila*. NAIP and NLRC4 genes are highly expressed in intestinal epithelial cells. In these cells, inflammasome activation leads to pyroptotic cell death, which is accompanied by expulsion of the infected cell back into the intestinal lumen, which limits the ability of bacteria to establish infection.

While the NAIPs represent classic PRRs, other NLRs do not recognize pathogens directly but rather assemble inflammasomes in response to the consequences of infection, such as pathogen-induced disruptions of cellular structures. The most studied member of this subgroup of NLRs is [NLRP3](#) (also known as NALP3 or cryopyrin). In contrast to NLRC4, NLRP3 has a [pyrin](#) domain at its N terminus, hence its [NLRP](#) designation (NLRs with CARDs are designated NLRC). Pyrin domains are structurally related to CARDs and interact with other pyrin domains. Several seemingly unrelated triggers can induce NLRP3 activation: reduced intracellular potassium, the generation of reactive oxygen species (ROS), or the disruption of lysosomes (for example, via phagocytosis of crystalline material). It remains unclear how such diverse stimuli are sensed by this NLR, but each represents a potential consequence of infection. The loss of intracellular potassium through efflux can occur during infection with, for example, intracellular bacteria such as *Staphylococcus aureus* that produce pore-forming toxins. Also, the death of nearby cells during infection can release ATP into the extracellular space; this ATP can activate the [purinergic receptor P2X7](#), which itself is a potassium channel, and allow K⁺ ion efflux. As discussed earlier in this chapter, ROS are produced by phagocytes upon infection. Finally, a number of pathogens are known to disrupt or permeabilize the phagosomal membrane as a mechanism of escape or to manipulate the host cell.

While it is still not understood how NLRP3 senses such diverse changes to cellular homeostasis, structural and biochemical approaches have delineated the steps of NLRP3 inflammasome assembly. As mentioned earlier, NLRP3 lacks a CARD, so it cannot recruit caspase 1 directly. Instead, upon activation, NLRP3 oligomerizes, and this aggregation induces the pyrin domains of NLRP3 to interact with pyrin domains of another protein named [ASC](#) (also called PYCARD). ASC is an adaptor protein composed of an amino-terminal pyrin domain and a carboxyl-terminal CARD. Recruitment of ASC creates a platform for caspase 1 recruitment and activation, via CARD–CARD interactions, as described earlier for the NAIP–NLRC4 inflammasomes. Once caspase 1 is activated, the downstream events are similar for all inflammasomes.

[NLRP1](#) is another NLR that assembles into an inflammasome in response to the activity of pathogens. Specifically, NLRP1 is partially degraded by several different bacterial enzymes; however, instead of inactivating NLRP1, this partial degradation releases a fragment of NLRP1 containing a CARD that assembles into an inflammasome that recruits and activates caspase 1. The best-understood example of this mechanism comes from studies of the [lethal](#)

[factor protease](#), a toxin secreted by the anthrax bacterium, *Bacillus anthracis*. Lethal factor enters into cells and promotes infection by cleaving MAP kinase kinases, which disrupts signaling and the ensuing immune response. However, lethal factor also cleaves NLRP1, leading to its partial degradation and inflammasome formation. Because responses downstream of caspase 1 are effective in preventing anthrax, it appears unlikely that *B. anthracis* evolved to activate NLRP1. Instead, NLRP1 appears to serve as a ‘decoy’ substrate for pathogen proteases and assembles into an inflammasome when such protease activity is detected.

Inflammasome activation can also be initiated by proteins other than NLRs. As discussed earlier for NLRs, some of these proteins detect pathogen-derived ligands while others sense the consequences of infection. As an example of the former, [AIM2 \(absent in melanoma 2\)](#), a member of the [PYHIN](#) family of proteins, is activated by direct binding to cytosolic DNA. PYHIN proteins contain an N-terminal pyrin domain but lack the LRR domains present in the NLR family. In place of LRR domains, PYHIN proteins have an HIN (hematopoietic expression, interferon-inducible nature, and nuclear localization) domain. The HIN domain of AIM2 recognizes the DNA genomes of pathogens and triggers caspase 1 activation through pyrin domain interactions with ASC. The protein [Pyrin](#) (not to be confused with the pyrin domain that is present in multiple proteins and was described earlier) is another inflammasome sensor protein with an N-terminal pyrin domain but no LRRs. Pyrin senses the inactivation of Rho GTPases, enzymes that regulate the cytoskeleton and cell motility (see [Section 3-2](#)). Kinases activated by RhoA phosphorylate specific residues in the Pyrin protein, and this phosphorylation keeps Pyrin in an inactive conformation. Certain pathogens, such as *Clostridium difficile*, produce toxins that inactivate RhoA, which correspondingly reduces phosphorylation of Pyrin and enables its activation. Activated Pyrin recruits ASC, leading to caspase 1 activation in a manner analogous to the other inflammasomes discussed above.

Many cell types, including most macrophages, must undergo a priming step before inflammasome activation can lead to release of inflammatory cytokines. This priming step induces the transcription and translation of the mRNAs that encode the pro-forms of IL-1 β , IL-18, and other cytokines (see Fig. 3.20). In some cases, inflammasome components themselves (for example, NLRP3) are induced upon priming. TLR signaling can satisfy this priming step, which may help ensure that inflammasome activation proceeds primarily during infections. Priming is not always necessary, however, as some cells in tissues (for example, the intestine) express pro-IL-1 β and pro-IL-18 constitutively.

[Caspase 11](#) (in mice) and [caspase 4](#) and [caspase 5](#) (in humans) are pro-inflammatory caspases that also function as cytosolic sensors of LPS. Once activated, these caspases are able to cleave and activate gasdermin D directly, leading to pyroptosis, but they do not appear to directly cleave the pro-forms of IL-1 β and IL-18. However, once pyroptosis is initiated, K⁺ efflux occurs via gasdermin D pores, resulting in NLRP3 activation, caspase 1 activation, and IL-1 β and IL-18 processing and release. This response to LPS has been referred to as a ‘non-canonical’ inflammasome pathway. The ability of LPS to induce toxic

shock in mice is due primarily to activation of caspase 11. TLR-4 is still involved, however, because caspase 11 requires TLR-4 signaling to be transcriptionally induced.

Inappropriate inflammasome activation has been associated with various diseases. [Gout](#) has been known for many years to cause inflammation in the cartilaginous tissues by the deposition of monosodium urate crystals, but how urate crystals caused inflammation was a mystery. Although the precise mechanism is still unclear, urate crystals are known to activate the NLRP3 inflammasome, which induces the inflammatory cytokines associated with the symptoms of gout. Mutations in NLRs can activate inflammasomes inappropriately, and they are the cause of some inherited [autoinflammatory diseases](#), in which inflammation occurs in the absence of infection. Mutations in NLRP3 in humans are associated with hereditary periodic fever syndromes, such as [familial cold autoinflammatory syndrome](#) and [Muckle-Wells syndrome](#) (discussed in more detail in [Chapter 13](#)). Individuals with activating NLRP3 mutations have also been identified. Macrophages from individuals with these conditions show spontaneous production of inflammatory cytokines such as IL-1 β and IL-18. We will discuss how pathogens can interfere with formation of the inflammasome in [Chapter 13](#).



Case Study: Hereditary Periodic Fever Syndromes

3-12 Activation of innate sensors in macrophages and dendritic cells triggers changes in gene expression that have far-reaching effects on the immune response.

Besides activating effector functions and cytokine production, another outcome of the activation of innate sensing pathways is the induction of [co-stimulatory molecules](#) on tissue dendritic cells and macrophages (see [Section 1-15](#)). We will describe these in more detail later in the book but mention them now because they provide an important link between innate and adaptive immune responses. Two important co-stimulatory molecules are the cell-surface proteins CD80 and CD86, which are induced on macrophages and tissue dendritic cells by innate sensors such as TLRs in response to pathogen recognition (**Fig. 3.22**). CD80 and CD86 are recognized by specific [co-stimulatory receptors](#) expressed by cells of the adaptive immune response, particularly CD4 T cells, and their activation is an important step in activating adaptive immune responses.

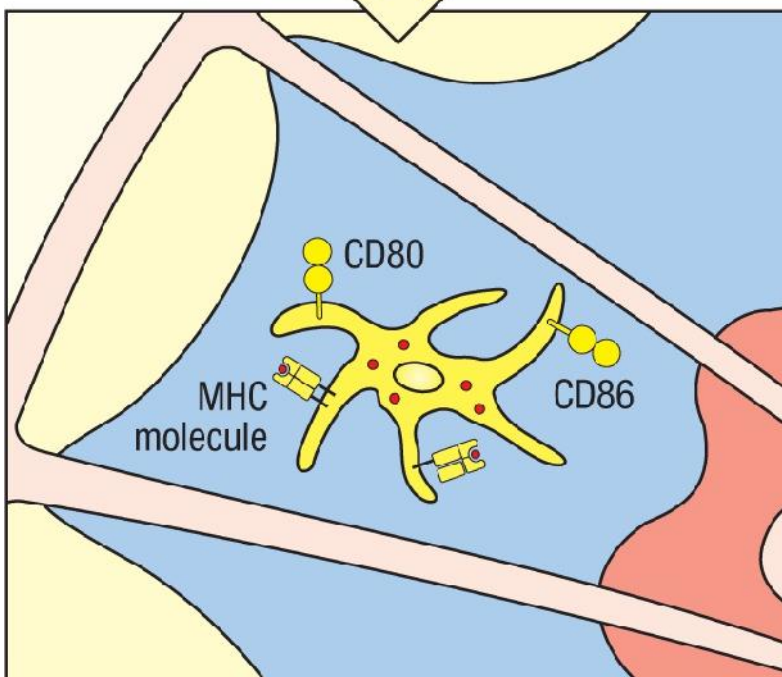
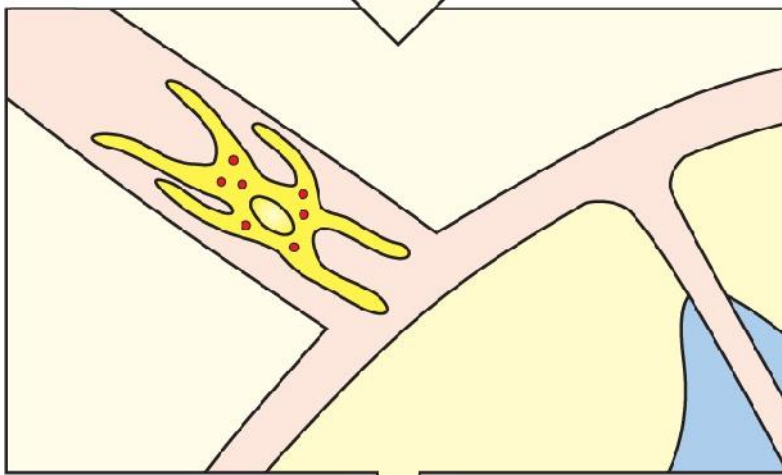
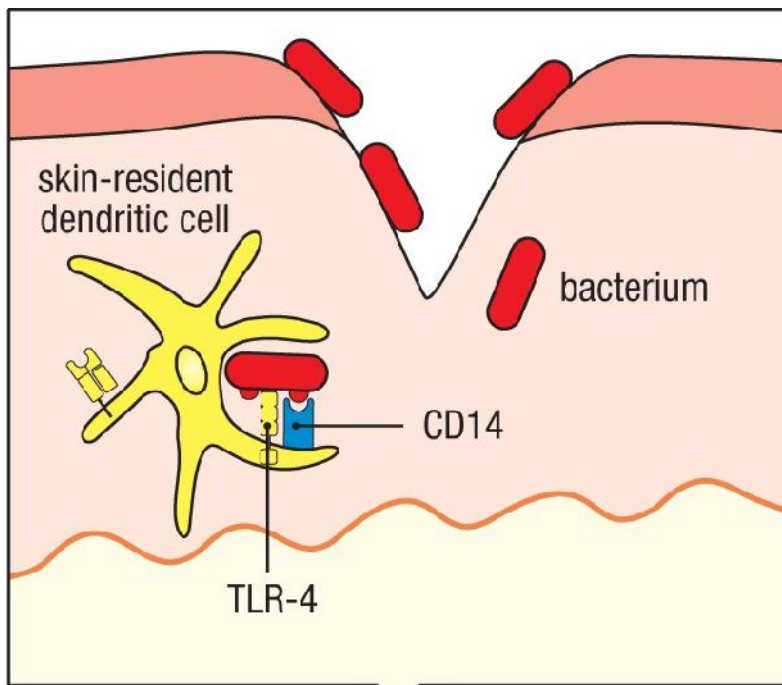


Fig. 3.22 Bacterial LPS induces changes in dendritic cells, stimulating them to migrate and to initiate adaptive immunity by activating T cells. Top panel:

Immature dendritic cells in the skin are highly phagocytic and macropinocytic but lack the ability to activate T lymphocytes. Dendritic cells residing in the skin ingest microbes and their products and degrade them. During a bacterial infection, the dendritic cells are activated by various innate sensors, and the activation induces two types of changes.

Second panel: The dendritic cells migrate out of the tissues into the lymphatic system and begin to mature. They lose the ability to ingest antigen but gain the ability to stimulate T cells. Third panel: In the regional lymph nodes, they become mature dendritic cells. They change the character of their cell-surface molecules, increasing the number of MHC molecules on their surface and the expression of the co-stimulatory molecules CD80 and CD86.

Substances such as LPS that induce co-stimulatory activity have been used for years in mixtures that are co-injected with protein antigens to enhance their immunogenicity. These substances are known as [adjuvants](#) (see [Appendix I, Section A-1](#)), and it was found empirically that the best adjuvants contain microbial components that induce macrophages and tissue dendritic cells to express co-stimulatory molecules and cytokines. We now know that the effect of these microbial components is due to activation of PRRs. As we shall see in [Chapters 9 and 11](#), the cytokines produced in response to infections influence the functional character of the adaptive immune response that develops. In this way, the ability of the innate immune system to discriminate among different types of pathogens is used by the organism to ensure an appropriate module of the adaptive immune response is induced.

3-13 Toll signaling in *Drosophila* is downstream of a distinct set of pathogen-recognition molecules.

Before leaving innate sensing, we shall look briefly at how Toll, TLRs, and NODs are used in invertebrate innate immunity. Although Toll is central to defense against both bacterial and fungal pathogens in *Drosophila*, Toll itself is not a pattern-recognition receptor but is downstream of other proteins that detect pathogens (**Fig. 3.23**). In *Drosophila*, there are 13 genes encoding [peptidoglycan-recognition proteins \(PGRPs\)](#) that bind the peptidoglycan components of bacterial cell walls. Another family, the [Gram-negative binding proteins \(GNBPs\)](#), recognizes LPS and β -(1,3)-linked glucans. GNBPs recognize Gram-negative bacteria and fungi rather than Gram-positive bacteria. The family members GGBP1 and PGRP-SA cooperate in the recognition of peptidoglycan from Gram-positive bacteria. They interact with a serine protease called [Grass](#), which initiates a proteolytic cascade that terminates in the cleavage of the protein Spätzle. One of the cleaved fragments forms a homodimer that binds to Toll and induces its dimerization, which in turn stimulates the antimicrobial response. A fungus-specific recognition protein, GGBP3, also activates the

proteolytic cascade, causing cleavage of Spätzle and activation of Toll.

Consequences of innate immune activation.

We will now examine the responses of innate immunity induced as an immediate consequence of pathogen recognition by the sensors described in the first half of this chapter. We will focus on the major phagocytes—neutrophils, macrophages, and dendritic cells—and the cytokines they produce that induce and maintain inflammation. First, we will introduce the families of cytokines and chemokines that coordinate many cellular responses, such as the recruitment of neutrophils and other immune cells to sites of infection. We will discuss the various adhesion molecules that are induced on immune cells circulating in the blood and on endothelial cells of blood vessels to coordinate movement of cells out of the blood and into infected tissues. We will consider in some detail how macrophage-derived chemokines and cytokines promote the continued destruction of infecting microbes. This is achieved both by stimulating the production and recruitment of fresh phagocytes and by inducing another phase of the innate immune response—the acute-phase response—in which the liver produces proteins that act as opsonizing molecules, helping to augment the actions of complement. We will also look at the mechanism of action of antiviral interferons, the type I interferons, and finally examine the growing class of innate lymphoid cells, or ILCs, which include the NK cells that contribute to innate immune defense against viruses, other intracellular pathogens, and even cancerous cells. ILCs exert a diverse array of effector functions that contribute to a rapid innate immune response to infection. They respond to early cytokine signals provided by innate sensor cells and amplify the response by producing various types of effector cytokines. Some time after induction of the innate response, an adaptive response will ensue that uses many of the same effector mechanisms used by the innate immune system but targets them with much greater precision. The effector mechanisms described here therefore serve as a primer for the focus on adaptive immunity in the later parts of this book.

3-15 Cytokines and their receptors fall into distinct families of structurally related proteins.

Cytokines are small proteins (about 25 kDa) that are released by various cells in the body, usually in response to an activating stimulus, and that induce responses through binding to specific receptors. Cytokines can act in an [autocrine](#) manner, affecting the behavior of the cell that releases the cytokine, or in a [paracrine](#) manner, affecting adjacent cells. Some cytokines are even stable enough to act in an [endocrine](#) manner, affecting distant cells, although this depends on their ability to enter the circulation and on their half-life in the

blood. In an attempt to develop a standardized nomenclature for molecules secreted by, and acting on, leukocytes, many cytokines are called by the name [interleukin \(IL\)](#) followed by a number (for example, IL-1 or IL-2). However, not all cytokines are included in this system; thus, students of immunology are still faced with a somewhat confusing and difficult task. The cytokines are listed alphabetically, together with their receptors, in [eAppendix III](#).

Cytokines can be grouped by structure into families—the IL-1 family, the hematopoietin superfamily, the interferons (described in [Section 3-7](#)), the TNF family, and chemokines—and their receptors can likewise be grouped (**Fig. 3.24**). The [IL-1 family](#) contains 11 members, notably IL-1 α , IL-1 β , IL-18, and IL-33. Most members of this family are produced as inactive proproteins that are cleaved (removing an amino-terminal peptide) to produce the mature cytokine. The exception to this rule is IL-1 α , for which both the proprotein and its cleaved forms are biologically active. As discussed earlier (see [Section 3-11](#)), mature IL-1 β and IL-18 are produced by macrophages through the action of caspase 1 in response to TLR signaling and inflammasome activation. The IL-1 family receptors have TIR domains in their cytoplasmic tails and signal by the NFKB pathway described earlier for TLRs. The IL-1 receptor functions in concert with a second transmembrane protein, the IL-1 receptor accessory protein (IL1RAP), which is required for IL-1 signal transduction.

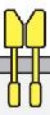
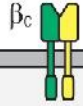
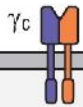
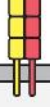
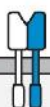
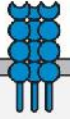
Homodimeric receptors		Receptors for erythropoietin and growth hormone
Heterodimeric receptors with a common chain		Receptors for IL-3, IL-5, GM-CSF share a common chain, CD131 or β_c (common β chain)
		Receptors for IL-2, IL-4, IL-7, IL-9, IL-15, and IL-21 share a common chain, CD132 or γ_c (common γ chain). IL-2 receptor also has a third chain, a high-affinity subunit IL-2R α (CD25)
Heterodimeric receptors (no common chain)		IL-1 family receptors
		Receptors for IL-13, IFN- α , IFN- β , IFN- γ , IL-10
TNF receptor family		Tumor necrosis factor (TNF) receptors I and II, CD40, Fas (Apo1, CD95), CD30, CD27, nerve growth factor receptor
Chemokine receptor family		CCR1–10, CXCR1–5, XCR1, CX3CR1

Fig. 3.24 Cytokine receptors belong to families of receptor proteins, each with a distinctive structure. Many cytokines signal through receptors of the hematopoietin receptor superfamily, named after its first member, the erythropoietin receptor. The hematopoietin receptor superfamily includes homodimeric and heterodimeric receptors, which are subdivided into families on the basis of protein sequence and structure. Examples of these are given in the first three rows. Heterodimeric class I cytokine receptors have an α chain that often defines the ligand specificity of the receptor; they may share with other receptors a common β or γ chain that confers the intracellular signaling function. Heterodimeric class II cytokine receptors have no common chain and include receptors for interferons or interferon-like cytokines. Many cytokine receptors signal through the JAK–STAT pathway. The IL-1 family receptors have extracellular immunoglobulin domains and signal as dimers through TIR domains in their cytoplasmic tails and through MyD88. Other families of cytokine receptors are the tumor necrosis factor receptor (TNFR) family and the chemokine receptor family, the latter belonging to the very large family of G protein–coupled receptors. The ligands of the TNFR family act as trimers and may be associated with the cell membrane rather than being secreted.

The [hematopoietin superfamily](#) of cytokines is quite large and includes non-immune-system growth and differentiation factors, such as erythropoietin (which stimulates red blood cell development), as well as interleukins with roles in innate immunity and adaptive immunity. IL-6 is a member of this superfamily, as is the cytokine GM-CSF, which stimulates the production of new monocytes and granulocytes in the bone marrow. Many of the soluble cytokines made by activated T cells are members of the hematopoietin superfamily. The receptors for the hematopoietin cytokines are tyrosine kinase–associated receptors that form dimers when their cytokine ligand binds. Dimerization initiates intracellular signaling from the tyrosine kinases associated with the cytoplasmic domains of the receptor. Some types of cytokine receptors are composed of two identical subunits, but others have two different subunits. An important feature of cytokine signaling is the large variety of different receptor subunit combinations that occur.

These cytokines and their receptors can also be further divided into subfamilies characterized by functional similarities and genetic linkage. For instance, IL-3, IL-4, IL-5, IL-13, and GM-CSF are related structurally, their genes are closely linked in the genome, and they are often produced together by the same kinds of cells. In addition, they bind to closely related receptors, which belong to the family of [class I cytokine receptors](#). The IL-3, IL-5, and GM-CSF receptors form a subgroup that shares a [common \$\beta\$ chain \(\$\beta_c\$ \)](#). Another subgroup of class I cytokine receptors is defined by a [common \$\gamma\$ chain \(\$\gamma_c\$ \)](#) of the IL-2 receptor. This chain is shared by receptors for the cytokines IL-2, IL-4, IL-7, IL-9, IL-15, and IL-21 and is encoded by a gene located on the X chromosome. Mutations that inactivate γ_c cause an [X-linked severe combined immune deficiency \(X-linked SCID\)](#) because of inactivation of the signaling pathways for several cytokines—IL-7, IL-15, and IL-2—that are required for normal lymphocyte development (see [Section 13-2](#)). More distantly related, the receptor for

IFN- γ is a member of a small family of heterodimeric cytokine receptors with some similarities to the hematopoietin receptor superfamily. These so-called [class II cytokine receptors](#) (also known as interferon receptors) include the receptors for IFN- α and IFN- β and the IL-10 receptor. The hematopoietin and interferon receptors all signal through the JAK–STAT pathway described in the next section and activate different combinations of STATs with different effects.



Case Study: X-linked Severe Combined Immune Deficiency

The [TNF family](#), of which TNF- α is the prototype, contains more than 17 cytokines with important functions in adaptive immunity and innate immunity. Unlike most of the other immunologically important cytokines, many members of the TNF family are transmembrane proteins, a characteristic that gives them distinct properties and limits their range of action. Some, however, can be released from the membrane in some circumstances. They are usually found as homotrimers of a membrane-bound subunit, although some heterotrimers consisting of different subunits also occur. TNF- α (sometimes called simply TNF) is initially expressed as a trimeric membrane-bound cytokine but can be released from the membrane. The effects of TNF- α are mediated by either of two [TNF receptors](#). TNF receptor I (TNFR-I) is expressed on a wide range of cells, including endothelial cells and macrophages, whereas TNFR-II is expressed largely by lymphocytes. The receptors for cytokines of the TNF family are structurally unrelated to the receptors described above and also have to cluster to become activated. Because TNF family cytokines are produced as trimers, the binding of these cytokines induces the clustering of three identical receptor subunits. The signaling pathway activated by these receptors is described in [Chapter 7](#), where we see that signaling uses members of the TRAF family to activate the so-called non-canonical NF κ B pathway.

Members of the chemokine receptor family are listed in [eAppendix IV](#), along with the chemokines they recognize. These receptors have a seven-transmembrane structure and signal by interacting with G proteins as described in [Section 3-2](#).

3-16 Cytokine receptors of the hematopoietin superfamily are associated with the JAK family of tyrosine kinases, which activate STAT transcription factors.

The signaling chains of the hematopoietin superfamily of cytokine receptors are

noncovalently associated with protein tyrosine kinases of the [Janus kinase \(JAK\) family](#)—so called because they have two tandem kinase-like domains and thus resemble the two-headed mythical Roman god Janus. There are four members of the JAK family: Jak1, Jak2, Jak3, and Tyk2. Because mice deficient for individual JAK family members show different phenotypes, each kinase must have a distinct function. For example, Jak3 is used by γ_c for signaling by several of the cytokines described above. Mutations that inactivate Jak3 cause a form of [SCID](#) that is not X-linked.

The dimerization or clustering of receptor signaling chains brings the JAKs into close proximity, causing phosphorylation of each JAK on a tyrosine residue that stimulates its kinase activity. The activated JAKs then phosphorylate their associated receptors on specific tyrosine residues. This phosphotyrosine, and the specific amino acid sequence surrounding it, creates a binding site that is recognized by [SH2 domains](#) found in other proteins, in particular members of a family of transcription factors known as [signal transducers and activators of transcription \(STATs\)](#) (**Fig. 3.25**).

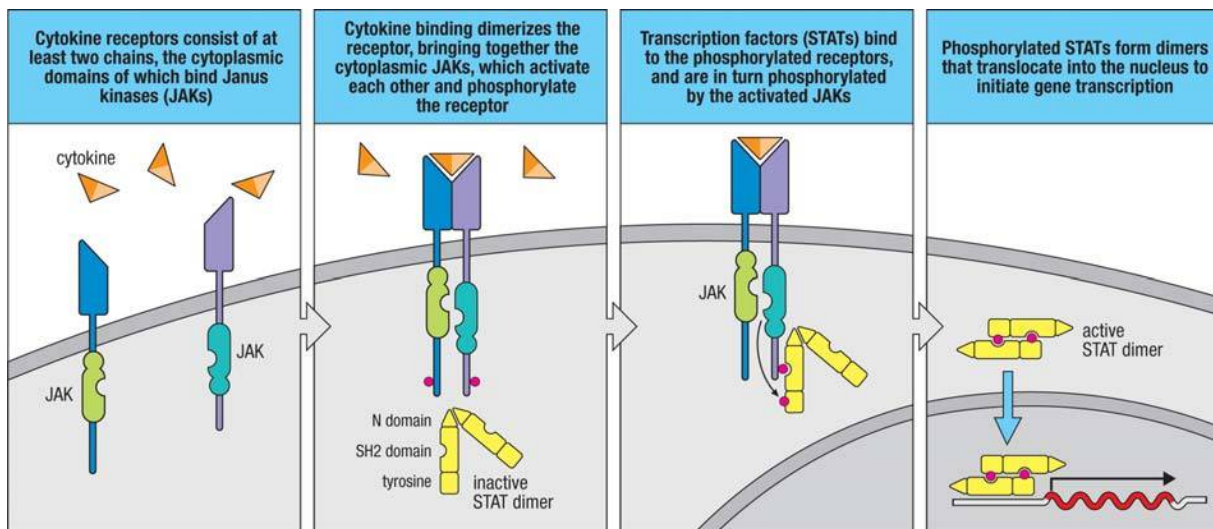


Fig. 3.25 Many cytokine receptors signal using a rapid pathway called the JAK–STAT pathway. First panel: Many cytokines act via receptors that are associated with cytoplasmic Janus kinases (JAKs). The receptor consists of at least two chains, each associated with a specific JAK. Second panel: Binding of ligand brings the two chains together, allowing the JAKs to phosphorylate and activate each other, and then to phosphorylate (red dots) specific tyrosines in the receptor tails. Each member of the STAT (signal transducer and activator of transcription) family of proteins has an N-terminal domain that homodimerizes STAT monomers in the cytosol before activation and an SH2 domain that binds to a tyrosine-phosphorylated cytokine receptor tail. Third panel: Upon binding, the STAT homodimers are phosphorylated on a conserved tyrosine residue in their carboxyl terminus by JAKs. Fourth panel: After phosphorylation, the STAT dimer reconfigures into a dimer that is stabilized by the SH2 domain of one monomer binding to a phosphotyrosine residue on the other monomer. They then translocate to the nucleus, where they bind to and activate the transcription of a variety of genes important for adaptive immunity.

There are seven STATs (1–4, 5a, 5b, and 6), which reside in the cytoplasm in an inactive form until activated by cytokine receptors. Before activation, most STATs form homodimers, due to a specific homotypic interaction between domains present at the amino termini of the individual STAT monomers. The receptor specificity of each STAT is determined by the recognition of the distinctive phosphotyrosine sequence on each activated receptor by the different SH2 domains within the various STAT proteins. Recruitment of a STAT to the activated receptor brings the STAT close to an activated JAK, which can then phosphorylate the conserved tyrosine residue in the carboxyl terminus of each of the STATs. This leads to a rearrangement of the STAT dimer, in which the phosphotyrosine of each STAT monomer binds to the SH2 domain of the other STAT monomer, forming a configuration that can bind DNA with high affinity. Activated STATs predominantly form homodimers, with a cytokine typically activating one type of STAT. For example, IFN- γ activates STAT1 and generates STAT1 homodimers, whereas IL-4 activates STAT6, generating STAT6 homodimers. Other cytokine receptors can activate several STATs, and some STAT heterodimers can be formed, such as STAT1–STAT2 heterodimers formed downstream of the receptor for IFN- α and IFN- β . The phosphorylated STAT dimer enters the nucleus, where it acts as a transcription factor to initiate the expression of selected genes that can regulate growth, differentiation, and/or survival of particular subsets of lymphocytes.

Because signaling by these receptors depends on tyrosine phosphorylation, dephosphorylation of the receptor complex by [tyrosine phosphatases](#) is one way that cells can terminate signaling. A variety of tyrosine phosphatases have been implicated in the dephosphorylation of cytokine receptors, JAKs, and STATs. These include the nonreceptor tyrosine phosphatases SHP-1 and SHP-2 (encoded by *PTPN6* and *PTPN11*) and the transmembrane receptor tyrosine phosphatase [CD45](#), which is expressed as multiple isoforms on many hematopoietic cells. Cytokine signaling can also be terminated by negative feedback involving specific inhibitors that are induced by cytokine activation. The [suppressor of cytokine signaling \(SOCS\)](#) proteins are a class of inhibitors that terminate the signaling of many cytokine and hormone receptors. SOCS proteins contain an SH2 domain that can recruit them to the phosphorylated JAK protein or receptor, and they can inhibit JAK activity directly, or compete for the receptor, and direct the ubiquitination and subsequent degradation of JAKs and STATs. SOCS proteins are induced by STAT activation and thus inhibit receptor signaling after the cytokine has had its effect. Their importance can be seen in SOCS1-deficient mice, which develop a multiorgan inflammatory infiltrate caused by increased signaling from interferon receptors, γ_c -containing receptors, and TLRs. Another class of inhibitory proteins, the [protein inhibitors of activated STAT \(PIAS\)](#), regulate the response to cytokines, as well as other biological pathways, by promoting the degradation of pathway components.

3-17 Chemokines released by macrophages and dendritic cells recruit effector cells to sites

of infection.

All the cytokines produced by macrophages in innate immune responses have important local and systemic effects that contribute to both innate and adaptive immunity, and these are summarized in **Fig. 3.26**. The recognition of different classes of pathogens by phagocytes and dendritic cells may involve signaling through different receptors, such as the various TLRs, and can result in some variation in the cytokines expressed by stimulated macrophages and dendritic cells. This is one way in which appropriate immune responses can be selectively activated, as the released cytokines orchestrate the next phase of host defense. In response to activation by PRRs, macrophages and dendritic cells secrete a diverse group of cytokines that includes IL-1 β , IL-6, IL-12, TNF- α , and the chemokine CXCL8 (formerly known as IL-8).

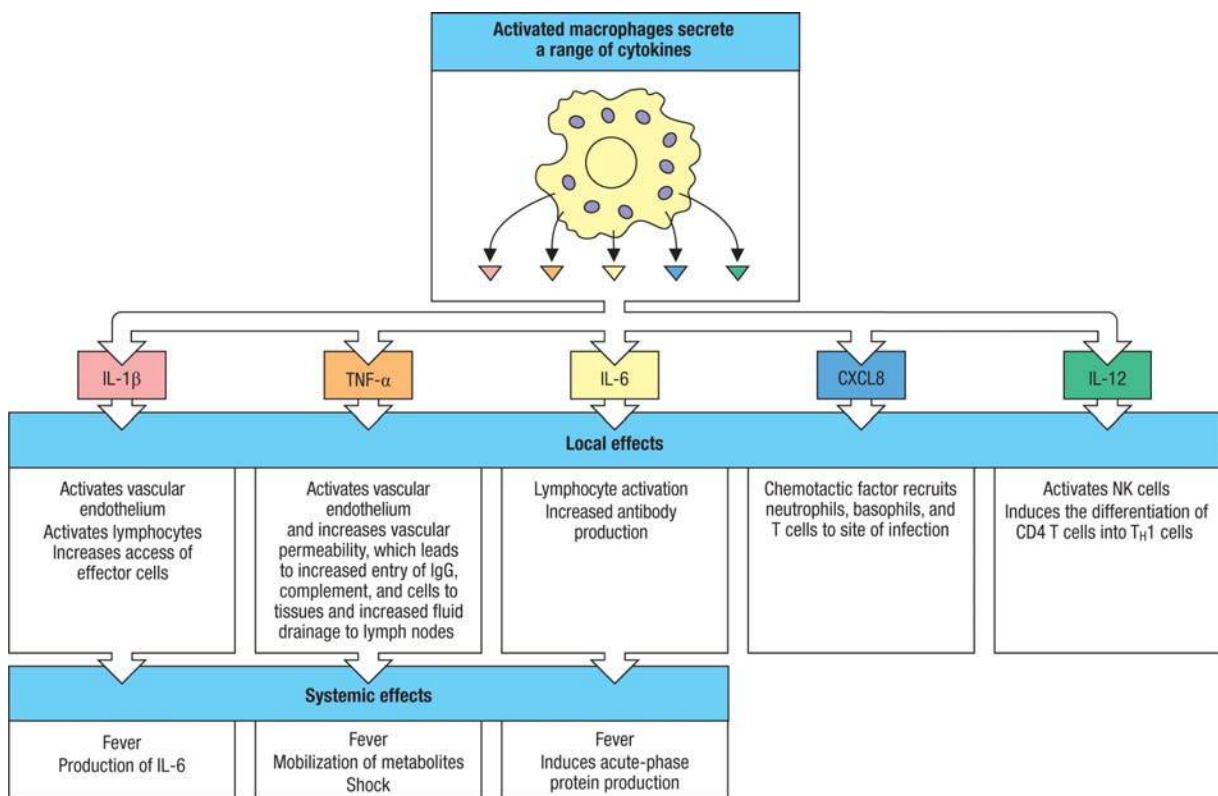


Fig. 3.26 Important cytokines and chemokines secreted by dendritic cells and macrophages in response to bacterial products include IL-1 β , IL-6, CXCL8, IL-12, and TNF- α . TNF- α is an inducer of a local inflammatory response that helps to contain infections. It also has systemic effects, many of which are harmful (discussed in Section 3-20). The chemokine CXCL8 is also involved in the local inflammatory response, helping to attract neutrophils to the site of infection. IL-1 β , IL-6, and TNF- α have a crucial role in inducing the acute-phase response in the liver and inducing fever, which favors effective host defense in various ways. IL-12 activates natural killer (NK) cells and favors the differentiation of CD4 T cells into the T_H1 subset in adaptive immunity.

Among the cytokines released by tissues in the earliest phases of infection are members of a

family of chemoattractant cytokines known as chemokines. These small proteins induce directed [chemotaxis](#) in nearby responsive cells, resulting in the movement of the cells toward the source of the chemokine. Because chemokines were first detected in functional assays, they were initially given a variety of names, which are listed along with their standardized nomenclature in [eAppendix IV](#). All the chemokines are related in amino acid sequence, and their receptors are G protein–coupled receptors (see [Section 3-2](#)). The signaling pathway stimulated by chemokines causes changes in cell adhesiveness and changes in the cell's cytoskeleton that lead to directed migration. Chemokines can be produced and released by many different types of cells, not only those of the immune system. In the immune system they function mainly as chemoattractants for leukocytes, recruiting monocytes, neutrophils, and other effector cells of innate immunity from the blood into sites of infection. They also guide lymphocytes in adaptive immunity, as we will learn in [Chapters 9–11](#). Some chemokines also function in lymphocyte development and migration and in angiogenesis (the growth of new blood vessels). There are more than 50 known chemokines, and this striking multiplicity may reflect their importance in delivering cells to their correct locations, which seems to be their main function in the case of lymphocytes. Some of the chemokines that are produced by or that affect human innate immune cells are listed in **Fig. 3.27**, along with their properties.

Class	Chemokine	Produced by	Receptors	Cells attracted	Major effects
CXC	CXCL8 (IL-8)	Monocytes Macrophages Fibroblasts Epithelial cells Endothelial cells	CXCR1 CXCR2	Neutrophils Naive T cells	Mobilizes, activates, and degranulates neutrophils Angiogenesis
	CXCL7 (PBP, β -TG, NAP-2)	Platelets	CXCR2	Neutrophils	Activates neutrophils Clot resorption Angiogenesis
	CXCL1 (GRO α) CXCL2 (GRO β) CXCL3 (GRO γ)	Monocytes Fibroblasts Endothelium	CXCR2	Neutrophils Naive T cells Fibroblasts	Activates neutrophils Fibroplasia Angiogenesis
		Monocytes		Monocytes	Competes with HIV-1

CC	CCL3 (MIP-1 α)	T cells Mast cells Fibroblasts	CCR1, 3, 5	NK and T cells Basophils Dendritic cells	Antiviral defense Promotes T _H 1 immunity
	CCL4 (MIP-1 β)	Monocytes Macrophages Neutrophils Endothelium	CCR1, 3, 5	Monocytes NK and T cells Dendritic cells	Competes with HIV-1
	CCL2 (MCP-1)	Monocytes Macrophages Fibroblasts Keratinocytes	CCR2B	Monocytes NK and T cells Basophils Dendritic cells	Activates macrophages Basophil histamine release Promotes T _H 2 immunity
	CCL5 (RANTES)	T cells Endothelium Platelets	CCR1, 3, 5	Monocytes NK and T cells Basophils Eosinophils Dendritic cells	Degranulates basophils Activates T cells Chronic inflammation
CXXXC (CX ₃ C)	CX3CL1 (Fractalkine)	Monocytes Endothelium Microglial cells	CX ₃ CR1	Monocytes T cells	Leukocyte–endothelial adhesion Brain inflammation

Fig. 3.27 Properties of selected human chemokines. Chemokines fall mainly into two related but distinct groups: the CC chemokines, which have two adjacent cysteine residues near the amino terminus; and the CXC chemokines, in which the equivalent cysteine residues are separated by a single amino acid. In humans, the genes for CC chemokines are mostly clustered in one region of chromosome 4. Genes for CXC chemokines are found mainly in a cluster on chromosome 17. The two groups of chemokines act on different sets of receptors, all of which are G protein–coupled receptors. CC chemokines bind to receptors designated CCR1–10. CXC chemokines bind to receptors designated CXCR1–7. Different receptors are expressed on different cell types, and so a particular chemokine can be used to attract a particular cell type. In general, CXC chemokines with a Glu-Leu-Arg tripeptide motif immediately before the first cysteine promote the migration of neutrophils. CXCL8 is an example of this type. Most of the other CXC chemokines, including those that interact with receptors CXCR3, 4, and 5, lack this motif. Fractalkine is unusual in several respects: it

has three amino acid residues between the two cysteines, and it exists in two forms, one that is tethered to the membrane of the endothelial and epithelial cells that express it, where it serves as an adhesion protein, and a soluble form that is released from the cell surface and acts as a chemoattractant for a wide range of cell types. A more comprehensive list of chemokines and their receptors is given in [eAppendix IV](#).

Chemokines fall mainly into two related but distinct groups. [CC chemokines](#) have two adjacent cysteine residues near the amino terminus, whereas in [CXC chemokines](#) the corresponding two cysteine residues are separated by a single amino acid. The CC chemokines promote the migration of monocytes, lymphocytes, and other cell types. One example relevant to innate immunity is CCL2, which attracts monocytes through the receptor CCR2B, inducing their migration from the bloodstream to become tissue macrophages. In contrast, neutrophil migration is promoted by CXC chemokines. CXCL8, acting through CXCR2, mobilizes neutrophils from bone marrow and induces them to leave the blood and migrate into the surrounding tissues. CCL2 and CXCL8 therefore have similar but complementary functions in the innate immune response, attracting monocytes and neutrophils, respectively.

The role of chemokines in cell recruitment is twofold. First, they act on the leukocyte as it rolls along endothelial cells at sites of inflammation, converting this rolling into stable binding by triggering a change of conformation in the adhesion molecules known as leukocyte integrins. These conformational changes enable integrins to bind strongly to their ligands on the endothelial cells, which allows the leukocyte to cross the blood vessel walls by squeezing between the endothelial cells. Second, the chemokine directs the migration of the leukocyte along a gradient of chemokine molecules bound to the extracellular matrix and the surfaces of endothelial cells. This gradient increases in concentration toward the site of infection.

Chemokines are produced by a wide variety of cell types in response to bacterial products, viruses, and agents that cause physical damage, such as silica, alum, or the urate crystals that occur in gout. Complement fragments, such as C3a and C5a, and fMLF bacterial peptides also act as chemoattractants for neutrophils. Thus, infection or physical damage to tissues induces the production of chemokine gradients that can direct phagocytes to the sites where they are needed. Neutrophils arrive rapidly in large numbers at a site of infection. The recruitment of monocytes occurs simultaneously, but they accumulate more slowly at the site of infection, perhaps because they are less abundant in the circulation. The complement fragment C5a and the chemokines CXCL8 and CCL2 activate their respective target cells, so that not only are neutrophils and monocytes brought to potential sites of infection, but also, in the process, they are armed to deal with the pathogens they encounter there. In particular, the signaling induced by C5a or CXCL8 in neutrophils serves to augment the respiratory burst that generates oxygen radicals and nitric oxide and to induce the neutrophils to release their stored antimicrobial granule contents (see [Section 3-2](#)).

Chemokines do not act alone in cell recruitment. They require the action of vasoactive mediators that bring leukocytes close to the blood vessel wall (see [Section 3-3](#)) and

cytokines such as TNF- α to induce the necessary adhesion molecules on endothelial cells. We will return to the chemokines in later chapters, where they are discussed in the context of the adaptive immune response. Now, however, we turn to the molecules that enable leukocytes to adhere to the endothelium, and we shall then describe step by step the extravasation process by which monocytes and neutrophils enter infected sites.

3-18 Cell-adhesion molecules control interactions between leukocytes and endothelial cells during an inflammatory response.

The recruitment of activated phagocytes to sites of infection is one of the most important functions of innate immunity. Recruitment occurs as part of the inflammatory response and is mediated by cell-adhesion molecules that are induced on the surface of the endothelial cells of local blood vessels. Here we consider those functions that participate in the recruitment of inflammatory cells in the hours to days after the establishment of an infection.

As with the complement components, a significant barrier to understanding the functions of cell-adhesion molecules is their nomenclature. Most adhesion molecules, especially those on leukocytes, which are relatively easy to analyze functionally, were originally named after the effects of specific monoclonal antibodies directed against them. Their names therefore bear no relation to their structural class. For instance, the [leukocyte functional antigens](#) LFA-1, LFA-2, and LFA-3 are actually members of two different protein families. In **Fig. 3.28**, the adhesion molecules relevant to innate immunity are grouped according to their molecular structure, which is shown in schematic form alongside their different names, sites of expression, and ligands. Three structural families of adhesion molecules are important for leukocyte recruitment. The [selectins](#) are membrane glycoproteins with a distal lectin-like domain that binds specific carbohydrate groups. Members of this family are induced on activated endothelium and initiate endothelium–leukocyte interactions by binding to fucosylated oligosaccharide ligands on passing leukocytes (see Fig. 3.28).

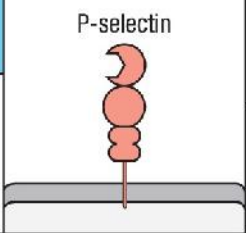
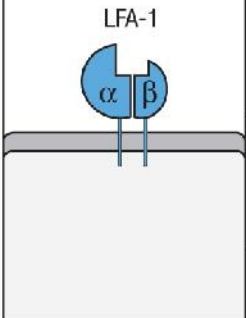
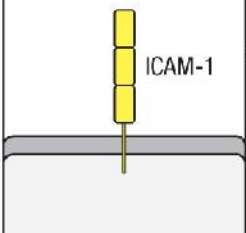
		Name	Tissue distribution	Ligand
Selectins	 P-selectin	P-selectin (PADGEM, CD62P)	Activated endothelium and platelets	PSGL-1, sialyl-Lewis ^x
		E-selectin (ELAM-1, CD62E)	Activated endothelium	Sialyl-Lewis ^x
Integrins Bind to cell-adhesion molecules and extracellular matrix. Strong adhesion	 LFA-1	$\alpha_L\beta_2$ (LFA-1, CD11a:CD18)	Monocytes, T cells, macrophages, neutrophils, dendritic cells, NK cells	ICAM-1, ICAM-2
		$\alpha_M\beta_2$ (CR3, Mac-1, CD11b:CD18)	Neutrophils, monocytes, macrophages, NK cells	ICAM-1, iC3b, fibrinogen
		$\alpha_X\beta_2$ (CR4, p150.95, CD11c:CD18)	Dendritic cells, macrophages, neutrophils, NK cells	iC3b
		$\alpha_5\beta_1$ (VLA-5, CD49d:CD29)	Monocytes, macrophages	Fibronectin
Immunoglobulin superfamily	 ICAM-1	ICAM-1 (CD54)	Activated endothelium, activated leukocytes	LFA-1, Mac-1
		ICAM-2 (CD102)	Resting endothelium, dendritic cells	LFA-1
		VCAM-1 (CD106)	Activated endothelium	VLA-4
		PECAM (CD31)	Activated leukocytes, endothelial cell-cell junctions	CD31

Fig. 3.28 Adhesion molecules involved in leukocyte interactions. Three structural families of adhesion molecules have a role in leukocyte migration, homing, and cell-cell interactions: the selectins, the integrins, and proteins of the immunoglobulin superfamily. The figure shows schematic representations of an example from each family, a list of other family members that participate in leukocyte interactions, their cellular distribution, and their ligand in adhesive interactions. The family members shown here are limited to those that participate in inflammation and other innate immune mechanisms. The same molecules and others participate in adaptive immunity and will be considered in [Chapters 9](#) and [11](#). The nomenclature of the different molecules in these families is confusing because it often reflects the way in which the molecules were first identified rather than their related structural characteristics. Alternative names for each of the adhesion molecules are given in parentheses. Sulfated sialyl-Lewis^x, which is recognized by P-selectin and E-selectin, is an oligosaccharide present on the cell-surface glycoproteins of circulating leukocytes.

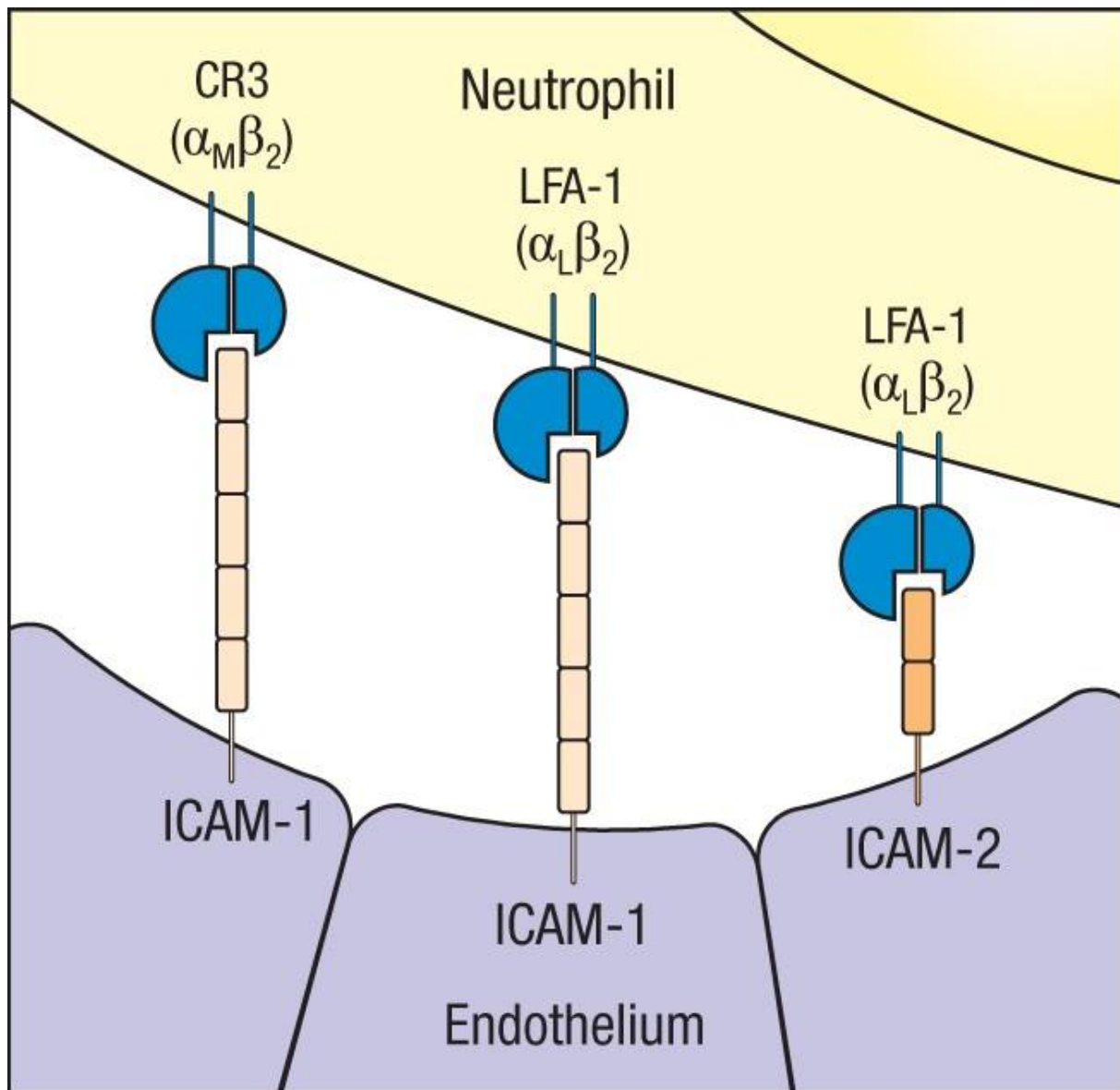


Fig. 3.29 Phagocyte adhesion to vascular endothelium is mediated by integrins. When vascular endothelium is activated by inflammatory mediators it expresses two adhesion molecules, namely ICAM-1 and ICAM-2. These are ligands for integrins expressed by phagocytes—ICAM-1 for α_Mβ₂ (also called CR3, Mac-1, or CD11b:CD18), and ICAM-1 and ICAM-2 for α_Lβ₂ (also called LFA-1 or CD11a:CD18).

The next step in leukocyte recruitment depends on tighter adhesion, which is due to the binding of [intercellular adhesion molecules \(ICAMs\)](#) on the endothelium to heterodimeric proteins of the [integrin](#) family on leukocytes. ICAMs are single-pass membrane proteins that belong to the large superfamily of [immunoglobulin-like proteins](#), which contain protein domains similar to those of immunoglobulins. The extracellular regions of ICAMs are composed of several immunoglobulin-like domains. An integrin molecule is composed of two transmembrane protein chains, α and β, of which there are numerous different types.

Subsets of integrins have a common β chain partnered with different α chains. The leukocyte integrins important for extravasation are [LFA-1](#) ($\alpha_L\beta_2$, also known as [CD11a:CD18](#)) and [CR3](#) ($\alpha_M\beta_2$, complement receptor 3, also known as [CD11b:CD18](#) or Mac-1). We described CR3 in [Section 2-13](#) as a receptor for iC3b, but it also binds other ligands. Both LFA-1 and CR3 bind to ICAM-1 (**Fig. 3.29**). Even in the absence of infection, circulating monocytes are continually leaving the blood and entering certain tissues, such as the intestine, where they become resident macrophages. LFA-1 also binds to ICAM-2. To navigate out of the blood vessel, monocytes may adhere to ICAM-2, which is expressed at low levels by unactivated endothelium. CR3 also binds to fibrinogen and factor X, both substrates of the coagulation cascade.

Strong adhesion between leukocytes and endothelial cells is promoted by the induction of ICAM-1 on inflamed endothelium together with a conformational change in LFA-1 and CR3 that occurs on the leukocyte. Integrins can switch between an 'active' state, in which they bind strongly to their ligands, and an 'inactive' state, in which binding is easily broken. This enables cells to make and break integrin-mediated adhesions in response to signals received by the cell either through the integrin itself or through other receptors. In the activated state, an integrin molecule is linked via the intracellular protein [talin](#) to the actin cytoskeleton. In the case of migrating leukocytes, chemokines binding to their receptors on the leukocyte generate intracellular signals that cause talin to bind to the cytoplasmic tails of the β chains of LFA-1 and CR3, forcing the integrin extracellular regions to assume an active binding conformation. The importance of leukocyte integrin function in inflammatory-cell recruitment is illustrated by [leukocyte adhesion deficiencies \(LADs\)](#), which can be caused by defects in the integrins themselves or in the proteins required for modulating adhesion. People with these diseases suffer from recurrent bacterial infections and impaired healing of wounds.



Case Study: Leukocyte Adhesion Deficiency

Endothelial activation is driven by macrophage-produced cytokines, particularly $\text{TNF-}\alpha$, which induce the rapid externalization of granules called [Weibel-Palade bodies](#) in the endothelial cells. These granules contain preformed [P-selectin](#), which appears on the surfaces of local endothelial cells just minutes after macrophages have responded to the presence of microbes by producing $\text{TNF-}\alpha$. Shortly after P-selectin gets to the cell surface, mRNA encoding [E-selectin](#) is synthesized, and within 2 hours the endothelial cells are expressing mainly E-selectin. Both P-selectin and E-selectin interact with [sulfated sialyl-Lewis^X](#), a sulfated form of a carbohydrate structure that is also an important blood group antigen. Sulfated sialyl-Lewis^X is present on the surface of neutrophils, and its interactions with P-selectin and E-selectin are important for neutrophil rolling on the endothelium. Mutations in

enzymes involved in its synthesis, such as fucosyltransferase, cause defective sialyl-Lewis^X expression that results in an immunodeficiency, [leukocyte adhesion deficiency type 2](#).

Integrins are also convenient cell-surface markers for distinguishing different cell types. Dendritic cells, macrophages, and monocytes express different integrin α chains and thus display distinct $\beta 2$ integrins on their surface. The predominant leukocyte integrin on conventional dendritic cells is $\alpha_X\beta_2$, also known as [CD11c:CD18](#) or complement receptor 4 ([CR4](#)) (see Fig. 3.28). This integrin is a receptor for the complement C3 cleavage product iC3b, fibrinogen, and ICAM-1. In contrast to conventional dendritic cells, most monocytes and macrophages express low levels of CD11c and predominantly express the integrin $\alpha_M\beta_2$ (CD11b:CD18; CR3). However, patterns of integrin expression can vary, with some tissue macrophages, such as those in the lung, expressing high levels of CD11c:CD18. In the mouse, the two major branches of conventional dendritic cells can be distinguished by expression of CD11b:CD18: one branch characterized by high expression of CD11b:CD18, and a second branch that lacks CD11b:CD18. Plasmacytoid dendritic cells (pDCs) express lower levels of CD11c but can be distinguished from conventional dendritic cells using other markers; human pDCs express the C-type lectin [BDCA-2 \(blood dendritic cell antigen 2\)](#), and mouse pDCs express [BST2 \(bone marrow stromal antigen 2\)](#), neither of which is expressed by conventional dendritic cells.

3-19 Neutrophils make up the first wave of cells that cross the blood vessel wall to enter an inflamed tissue.

The migration of leukocytes out of blood vessels, the process known as extravasation, occurs in response to signals generated at sites of infection. Under normal conditions, leukocytes travel in the center of small blood vessels, where blood flow is fastest. Within sites of inflammation, the vessels are dilated and the consequent slower blood flow allows leukocytes to interact in large numbers with the vascular endothelium. During an inflammatory response, the induction of adhesion molecules on the endothelial cells of blood vessels within the infected tissue, as well as changes in the adhesion molecules expressed on leukocytes, recruits large numbers of circulating leukocytes to the site of infection. We will describe this process with regard to monocytes and neutrophils (**Fig. 3.30**).

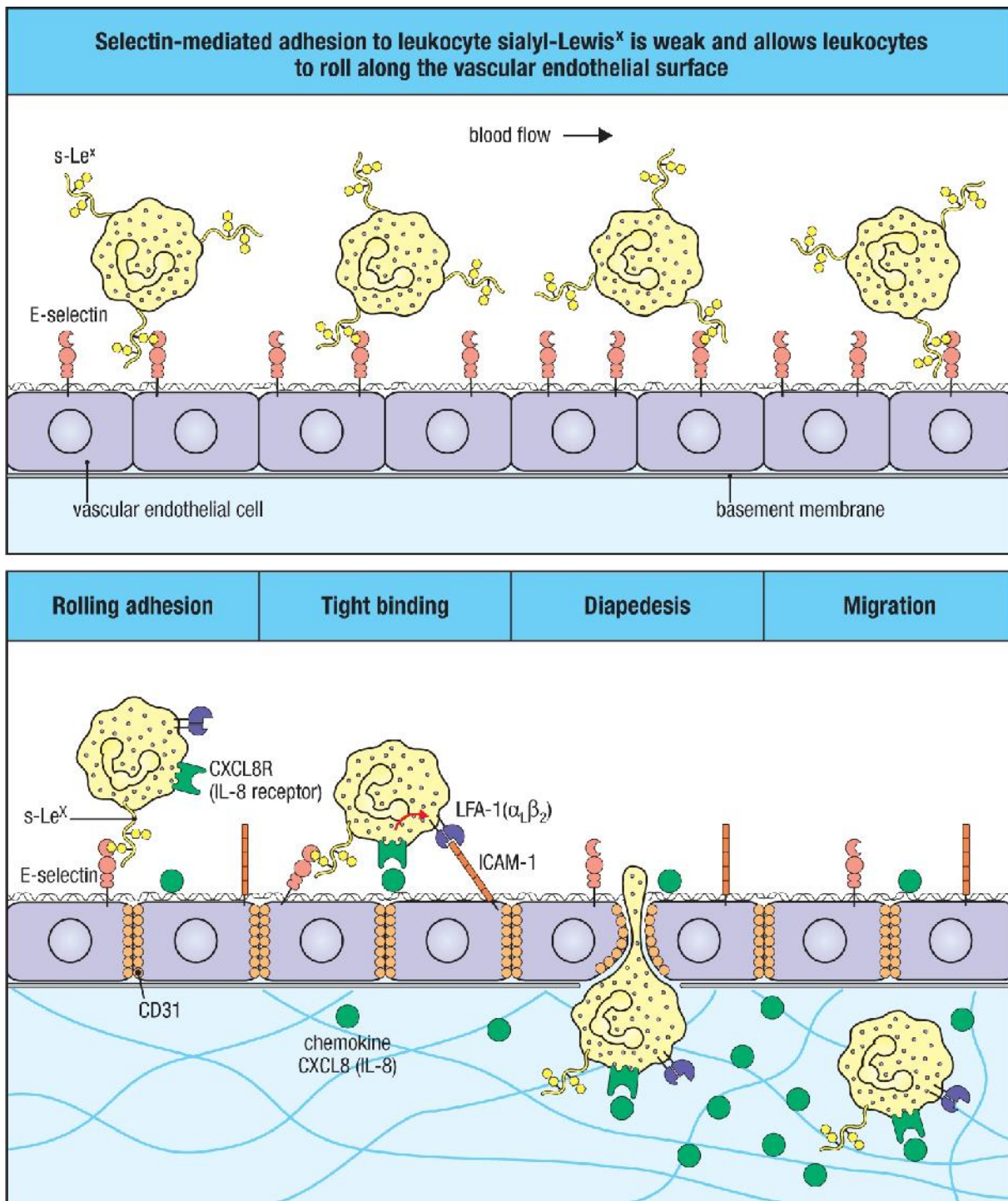


Fig. 3.30 Neutrophils leave the blood and migrate to sites of infection in a multistep process involving adhesive interactions that are regulated by macrophage-derived cytokines and chemokines. Top panel: The first step involves the reversible binding of a neutrophil to vascular endothelium through interactions between selectins induced on the endothelium and their carbohydrate ligands on the neutrophil, shown here for E-selectin and its ligand, the sialyl-Lewis^x moiety (s-Le^x). This interaction cannot anchor the cells against the shearing force of the flow of blood, and thus they roll along the endothelium, continually making and breaking contact. Bottom panel: The binding does however, eventually trigger stronger interactions, which result only when binding of a chemokine such as CXCL8 to its specific receptor on the neutrophil triggers the

activation of the integrins LFA-1 and CR3 (Mac-1; not shown). Inflammatory cytokines such as TNF- α are also necessary to induce the expression of adhesion molecules such as ICAM-1 and ICAM-2, the ligands for these integrins, on the vascular endothelium. Tight binding between ICAM-1 and the integrins arrests the rolling and allows the neutrophil to squeeze between the endothelial cells forming the wall of the blood vessel (that is, to extravasate). The leukocyte integrins LFA-1 and CR3 are required for extravasation and for migration toward chemoattractants. Adhesion between molecules of CD31, expressed on both the neutrophil and the junction of the endothelial cells, is also thought to contribute to extravasation. The neutrophil also needs to traverse the basement membrane; it penetrates this with the aid of a matrix metalloproteinase enzyme, MMP-9, that it expresses at the cell surface. Finally, the neutrophil migrates along a concentration gradient of chemokines (shown here as CXCL8) secreted by cells at the site of infection.

Extravasation proceeds in four stages. In the first, induction of selectins enables leukocyte rolling along the endothelium. P-selectin appears on endothelial cell surfaces within a few minutes of exposure to leukotriene B₄, C5a, or histamine, which is released from mast cells in response to C5a. P-selectin can also be induced by TNF- α or LPS, and both of these induce synthesis of E-selectin, which appears on the endothelial cell surface a few hours later. When the sulfated sialyl-Lewis^X on monocytes and neutrophils contacts these exposed P- and E-selectins, these cells adhere reversibly to the vessel wall and begin to ‘roll’ along endothelium (see Fig. 3.30, top panel), permitting stronger interactions of the next step in leukocyte migration. Neutrophils are particularly efficient at rolling along endothelium, even under flow rates that prevent rolling by other cells. Such [shear-resistant rolling](#) by neutrophils uses long extensions of plasma membrane, termed [slings](#), that bind the endothelium and wrap around the cell as it rolls, serving to tether the cell firmly to the endothelium and to promote rapid entry to sites of infection.

The second step depends on interactions between the leukocyte integrins LFA-1 and CR3 and adhesion molecules such as ICAM-1 (which can be induced on endothelial cells by TNF- α) or ICAM-2 on endothelium (see Fig. 3.30, bottom panel). LFA-1 and CR3 normally bind their ligands only weakly, but CXCL8 (or other chemokines), bound to proteoglycans on the surface of endothelial cells, binds to specific chemokine receptors on the leukocyte and signals the cell to trigger a conformational change in LFA-1 and CR3 on the rolling leukocyte; this greatly increases the adhesive properties of the leukocyte, as discussed in [Section 3-18](#). The cell then attaches firmly to the endothelium, and its rolling is arrested.

In the third step the leukocyte extravasates, or crosses the endothelial wall. This step also involves LFA-1 and CR3, as well as a further adhesive interaction involving an immunoglobulin-related molecule called [PECAM](#), or [CD31](#), which is expressed both on the leukocyte and at the intercellular junctions of endothelial cells. These interactions enable cells to squeeze between the endothelial cells. It then penetrates the basement membrane with the aid of enzymes that break down the extracellular matrix proteins of the basement

membrane. The movement through the basement membrane is known as [diapedesis](#), and it enables cells to enter the subendothelial tissues.

The fourth and final step in extravasation is the migration of leukocytes through the tissues under the influence of chemokines. Chemokines such as CXCL8 and CCL2 (see [Section 3-17](#)) are produced at the site of infection and bind to proteoglycans in the extracellular matrix and on endothelial cell surfaces. In this way, a matrix-associated concentration gradient of chemokines is formed on a solid surface along which the leukocyte can migrate to the focus of infection (see Fig. 3.30). CXCL8 is released by the macrophages that first encounter pathogens; it recruits neutrophils, which enter the infected tissue in large numbers in the early part of the induced response. Their influx usually peaks within the first 6 hours of an inflammatory response. Monocytes are recruited through the action of CCL2 and accumulate more slowly than neutrophils. Once in the inflamed tissue, neutrophils are able to eliminate many pathogens by phagocytosis. In an innate immune response, neutrophils use their complement receptors and the direct pattern-recognition receptors discussed earlier in this chapter (see [Section 3-1](#)) to recognize and phagocytose pathogens or pathogen components directly or after opsonization with complement (see [Section 2-13](#)). In addition, as we will see in [Chapter 10](#), neutrophils act as phagocytic effectors in humoral adaptive immunity, taking up antibody-coated microbes by means of specific receptors.

The importance of neutrophils in immune defense is dramatically illustrated by diseases or medical treatments that severely reduce neutrophil numbers. Individuals suffering this affliction are said to have [neutropenia](#), and they are highly susceptible to deadly infection with a wide range of pathogens and commensal organisms. Restoring neutrophil levels in such individuals by transfusion of neutrophil-rich blood fractions or by stimulating their production with specific growth factors largely corrects this susceptibility.

3-20 TNF- α is an important cytokine that triggers local containment of infection but induces shock when released systemically.

TNF- α acting on endothelial cells stimulates the expression of adhesion molecules and aids the extravasation of cells such as monocytes and neutrophils. Another important action of TNF- α is to stimulate endothelial cells to express proteins that trigger blood clotting in the local small vessels, occluding them and cutting off blood flow. This can be important in preventing the pathogen from entering the bloodstream and spreading through the blood to organs all over the body. The importance of TNF- α in the containment of local infection is illustrated by experiments in which rabbits were infected locally with a bacterium. Normally, the infection would be contained at the site of the inoculation; if, however, an injection of anti-TNF- α antibody was also given to block the action of TNF- α , the infection spread via the blood to other organs. In parallel, the fluid that has leaked into the tissue in the early phases of an infection carries the pathogen, usually enclosed in dendritic cells, via the lymph

to the regional lymph nodes, where an adaptive immune response can be initiated.

Once an infection has spread to the bloodstream, however, the same mechanisms by which TNF- α so effectively contains local infection instead become catastrophic (**Fig. 3.31**). Although produced as a membrane-associated cytokine, TNF- α can be cleaved by a protease, [TACE \(TNF- \$\alpha\$ -converting enzyme\)](#), which is encoded by the ADAM17 gene), and released from the membrane as a soluble cytokine. The presence of infection in the bloodstream, or sepsis, is accompanied by a massive release of soluble TNF- α from macrophages in the liver, spleen, and other sites throughout the body. The systemic release of TNF- α into the bloodstream causes vasodilation, which leads to a loss of blood pressure and increased vascular permeability; this in turn leads to a loss of plasma volume and eventually to shock, known in this case as septic shock because the underlying cause is a bacterial infection. The TNF- α released in septic shock also triggers blood clotting in small vessels throughout the body—known as [disseminated intravascular coagulation \(DIC\)](#)—which leads to the massive consumption of clotting proteins, so that the individual's blood cannot clot appropriately. Disseminated intravascular coagulation frequently leads to the failure of vital organs such as the kidneys, liver, heart, and lungs, which are quickly compromised by the failure of normal blood perfusion; consequently, septic shock has a very high mortality rate.

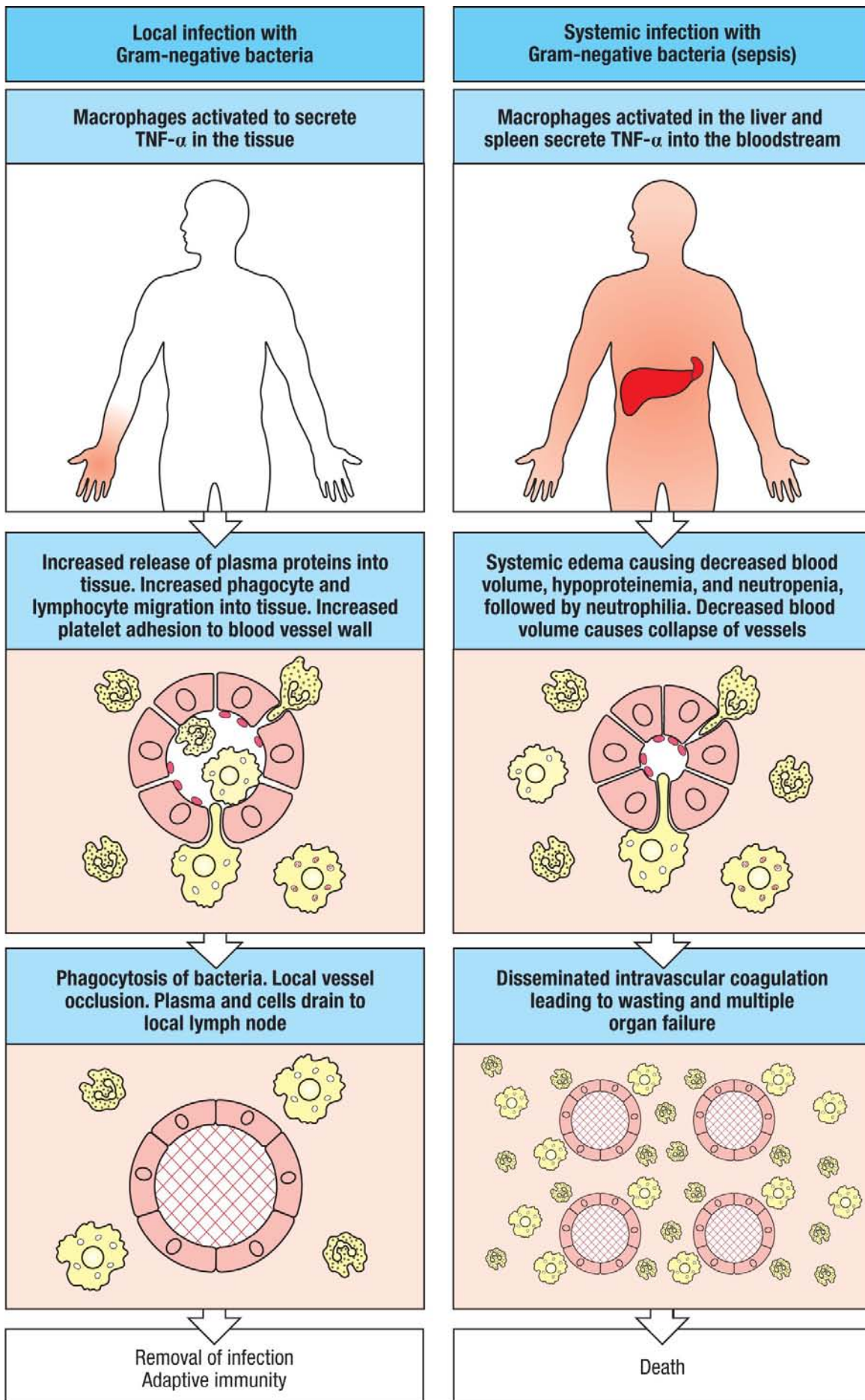


Fig. 3.31 The release of TNF- α by macrophages induces local protective effects, but TNF- α can be damaging when released systemically. The panels on the left show the causes and consequences of local release of TNF- α , and the panels on the right show the causes and consequences of systemic release. In both cases, TNF- α acts on blood vessels, especially venules, to increase blood flow and vascular permeability to fluid, proteins, and cells, and to increase endothelial adhesiveness for leukocytes and platelets (center row). Local release thus allows an influx of fluid, cells, and proteins into the infected tissue, where they participate in host defense. Later, blood clots form in the small vessels (bottom left panel), preventing spread of infection via the blood, and the accumulated fluid and cells drain to regional lymph nodes, where an adaptive immune response is initiated. When there is a systemic infection, or sepsis, with bacteria that elicit TNF- α production, TNF- α is released into the blood by macrophages in the liver and spleen and acts in a similar way on all small blood vessels in the body (bottom right panel). The result is shock, disseminated intravascular coagulation with depletion of clotting factors, and consequent bleeding, multiple organ failure, and frequently death.

Blockade of TNF- α activity, either with specific antibodies or with soluble proteins that mimic the receptor, is a successful treatment for several inflammatory disorders, including rheumatoid arthritis. However, these treatments have been found to reactivate tuberculosis in some apparently well individuals with evidence of previous infection (as demonstrated by skin test), which is a direct demonstration of the importance of TNF- α in keeping infection local and in check. The efficacy of TNF- α blockade as a treatment for sepsis remains unclear.

3-21 Cytokines made by macrophages and dendritic cells induce a systemic reaction known as the acute-phase response.

As well as their important local effects, the cytokines produced by macrophages and dendritic cells have long-range effects that contribute to host defense. One of these is the elevation of body temperature, which is caused mainly by TNF- α , IL-1 β , and IL-6. These cytokines are termed [endogenous pyrogens](#) because they cause fever and derive from an endogenous source rather than from bacterial components such as LPS, which also induces fever and is an [exogenous pyrogen](#). Endogenous pyrogens cause fever by inducing the synthesis of prostaglandin E2 by the enzyme cyclooxygenase-2, the expression of which is induced by these cytokines. Prostaglandin E2 then acts on the hypothalamus, resulting in an increase in both heat production from the catabolism of brown fat and heat retention from vasoconstriction, which decreases the loss of excess heat through the skin. Exogenous pyrogens are able to induce fever by promoting the production of the endogenous pyrogens and also by directly inducing cyclooxygenase-2 as a consequence of signaling through TLR-4, leading to the production of prostaglandin E2. Fever is generally considered beneficial to

host defense, although the precise mechanisms by which fever facilitates pathogen clearance remain unclear.

The effects of TNF- α , IL-1 β , and IL-6 are summarized in **Fig. 3.32**. One of the most important of these occurs in the liver and is the initiation of a response known as the [acute-phase response](#) (**Fig. 3.33**). The cytokines act on hepatocytes, which respond by changing the profile of proteins that they synthesize and secrete into the blood. In the acute-phase response, blood levels of some proteins go down, whereas levels of others increase markedly. The proteins induced by TNF- α , IL-1 β , and IL-6 are called the [acute-phase proteins](#). Several of these are of particular interest because they mimic the action of antibodies, but unlike antibodies they have broad specificity for pathogen-associated molecular patterns and depend only on the presence of cytokines for their production.

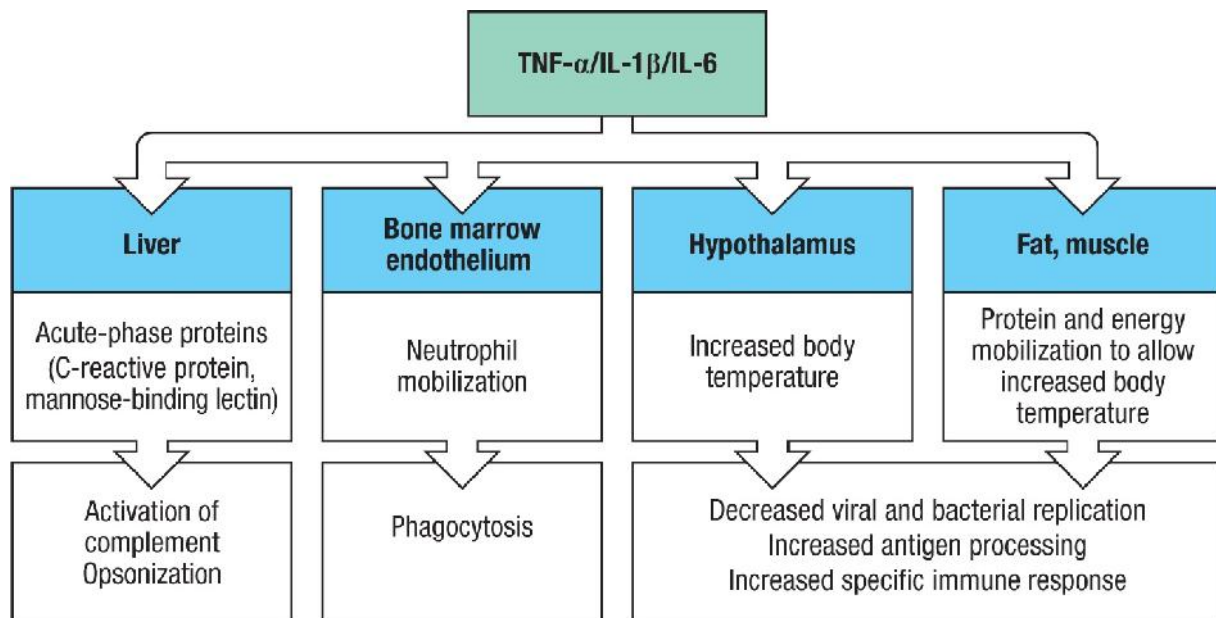
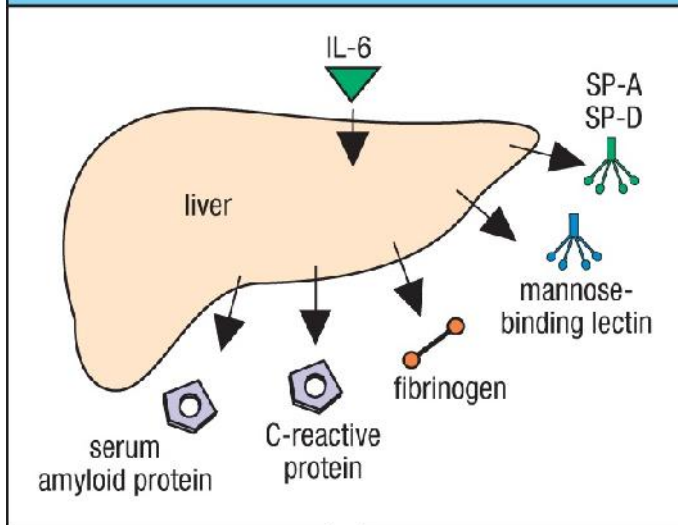
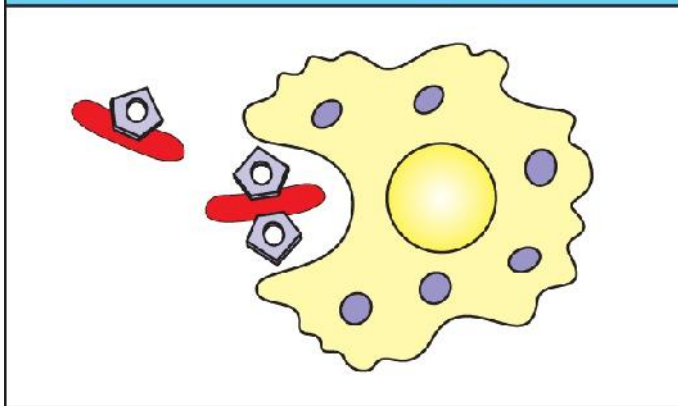


Fig. 3.32 The cytokines TNF- α , IL 1 β , and IL-6 have a wide spectrum of biological activities that help to coordinate the body's responses to infection. TNF- α , IL-1 β , and IL-6 activate hepatocytes to synthesize acute-phase proteins and activate bone marrow endothelium to release neutrophils. The acute-phase proteins act as opsonins, whereas the disposal of opsonized pathogens is augmented by the enhanced recruitment of neutrophils from the bone marrow. TNF- α , IL-1 β , and IL-6 are also endogenous pyrogens, raising body temperature, which is believed to help in eliminating infections. A major effect of these cytokines is to induce synthesis of prostaglandin E₂, which acts on the hypothalamus, altering the body's temperature regulation, and on muscle and fat cells, altering energy mobilization to increase the body temperature. At higher temperatures, bacterial and viral replication is less efficient, whereas the adaptive immune response operates more efficiently.

Bacteria induce macrophages to produce IL-6, which acts on hepatocytes to induce synthesis of acute-phase proteins



C-reactive protein binds phosphocholine on bacterial surfaces, acting as an opsonin, and also activating complement



Serum amyloid protein

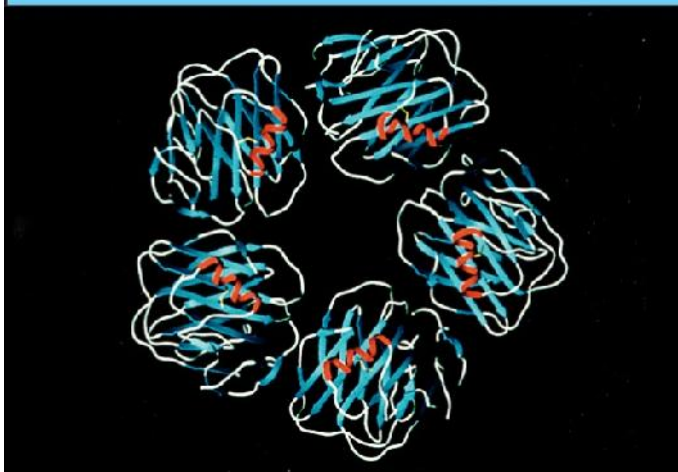


Fig. 3.33 The acute-phase response produces molecules that bind pathogens but no host cells. Acute-phase proteins are produced by liver cells in response to cytokines released by macrophages in the presence of bacteria (top panel). They include serum amyloid protein (SAP) (in mice but not humans), C-reactive protein (CRP), fibrinogen, and mannose-binding lectin (MBL). CRP binds phosphocholine on certain bacterial and fungal surfaces but does not recognize it in the form in which it is found in host-cell membranes (middle panel). SAP and CRP are homologous in structure; both are pentraxins, forming five-membered discs, as shown for SAP (lower panel). SAP both acts as an opsonin in its own right and activates the classical complement pathway by binding C1q to augment opsonization. MBL is a member of the collectin family, which also includes the pulmonary surfactant proteins SP-A and SP-D. Like CRP, MBL can act as an opsonin in its own right, as can SP-A and SP-D.

One acute-phase protein, the [C-reactive protein](#), is a member of the [pentraxin](#) protein family, so called because the proteins are formed from five identical subunits. C-reactive protein is yet another example of a multipronged pathogen-recognition molecule, and it binds to the phosphocholine portion of certain bacterial and fungal cell-wall lipopolysaccharides. Phosphocholine is also found in mammalian cell-membrane phospholipids, but it cannot be bound by C-reactive protein. When C-reactive protein binds to a bacterium, it not only is able to opsonize the bacterium but can also activate the complement cascade by binding to C1q, the first component of the classical pathway of complement activation (see [Section 2-7](#)). The interaction with C1q involves the collagen-like parts of C1q rather than the globular heads that make contact with pathogen surfaces, but the same cascade of reactions is initiated.

Mannose-binding lectin (MBL) is another acute-phase protein; it serves as an innate recognition molecule that can activate the lectin pathway of complement (see [Section 2-6](#)). MBL is present at low levels in the blood of healthy individuals, but it is produced in increased amounts during the acute-phase response. By recognizing mannose residues on microbial surfaces, MBL can act as an opsonin that is recognized by monocytes, which do not express the macrophage mannose receptor. Two other proteins with opsonizing properties that are also produced in increased amounts during an acute-phase response are the [surfactant proteins A and D \(SP-A and SP-D\)](#). These are produced by the liver and a variety of epithelia. They are, for example, found along with macrophages in the alveolar fluid of the lung, where they are secreted by pneumocytes, and are important in promoting the phagocytosis of opportunistic respiratory pathogens such as *Pneumocystis jirovecii* (formerly known as *P. carinii*), one of the main causes of pneumonia in individuals with AIDS.

Thus, within a day or two, the acute-phase response provides the host with several proteins with the functional properties of antibodies but able to bind a broad range of pathogens. However, unlike antibodies, which we describe in [Chapters 4](#) and [10](#), acute-phase proteins have no structural diversity and are made in response to any stimulus that triggers the release

of TNF- α , IL-1 β , and IL-6. Therefore, unlike antibodies, their synthesis is not specifically induced and targeted.

A final, distant effect of the cytokines produced by macrophages is to induce [leukocytosis](#), an increase in the numbers of circulating neutrophils. The neutrophils come from two sources: the bone marrow, from which mature leukocytes are released in increased numbers; and sites in blood vessels, where they are attached loosely to endothelial cells. Thus, the effects of these cytokines contribute to the control of infection while the adaptive immune response is being developed.

3-22 Interferons induced by viral infection make several contributions to host defense.

Viral infection induces the production of interferons, originally named because of their ability to interfere with viral replication in previously uninfected tissue culture cells. Interferons have a similar role *in vivo*, blocking the spread of viruses to uninfected cells. There are numerous genes encoding antiviral, or type I, interferons. Best understood are the IFN- α family of 12 closely related human genes and IFN- β , the product of a single gene; less well studied are IFN- κ , IFN- ϵ , and IFN- ω . [IFN- \$\gamma\$](#) is the sole [type II interferon](#).



Case Study: | Interferon- γ Receptor Deficiency

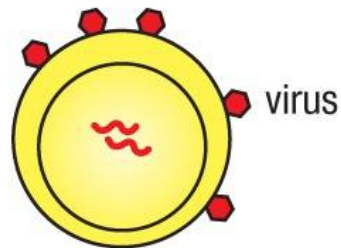
Type III interferons are a newly classified IFN family composed of the products of three [IFN- \$\lambda\$](#) genes, also known as IL-28A, IL-28B, and IL-29, which bind a heterodimeric [IFN- \$\lambda\$ receptor](#) composed of a unique IL-28R α subunit and the β subunit of the IL-10 receptor. While receptors for type I interferons and IFN- γ are widespread in their tissue distribution, type III receptors are more restricted. For example, they are not expressed by fibroblasts but are expressed on epithelial cells.

Type I interferons are inducible and are synthesized by many cell types after infection by diverse viruses. Almost all types of cells can produce IFN- α and IFN- β in response to activation of several innate sensors. For example, type I interferons are induced by RLRs and cGAS–STING signaling (see [Sections 3-8](#) and [3-9](#)). However, some immune cells seem to be specialized for this task. In [Section 3-1](#) we introduced the plasmacytoid dendritic cell (pDC). Also called [interferon-producing cells \(IPCs\)](#) or [natural interferon-producing cells](#), human plasmacytoid dendritic cells were initially recognized as rare peripheral blood cells that accumulate in peripheral lymphoid tissues during a viral infection and make abundant type I interferons (IFN- α and IFN- β)—up to 1000 times more than that made by other cell

types. This abundant production of type I interferons may result from the efficient coupling of viral recognition by TLRs to the pathways of interferon production (see [Section 3-7](#)). Plasmacytoid dendritic cells express a subset of TLRs that includes TLR-7 and TLR-9, which are endosomal sensors of viral RNA and DNA (see Fig. 3.11). The requirement for TLR-9 in sensing infections caused by DNA viruses has been demonstrated, for example, by the inability of TLR-9–deficient plasmacytoid dendritic cells to generate type I interferons in response to herpes simplex virus.

Interferons help defend against viral infection in several ways (**Fig. 3.34**). IFN- β induces an antiviral state in cells, which includes making cells more likely to produce IFN- α in response to PRR triggering, thus amplifying the interferon response. IFN- α and IFN- β bind to a common cell-surface receptor, known as the [interferon- \$\alpha\$ receptor \(IFNAR\)](#), which uses the JAK and STAT pathways described in [Section 3-16](#). IFNAR uses the kinases Tyk2 and Jak1 to activate the factors [STAT1](#) and [STAT2](#), which can interact with [IRF9](#) and form a complex called [ISGF3](#), which binds to the promoters of many [interferon-stimulated genes \(ISGs\)](#). The induction of ISGs induces a state of resistance to viral replication in all cells. Many of the genes involved in nucleic-acid sensing and signaling are ISGs, including the genes for RIG-I, MDA-5, IRF7, and STING, such that cells exposed to interferons have a heightened ability to recognize viral nucleic acids.

Virus-infected host cells



IFN- α , IFN- β

Activate STAT1 and STAT2, which combine with IRF9 to form ISGF3

Induce resistance to viral replication in all cells by inducing Mx proteins, 2',5'-linked adenosine oligomers, and the kinase PKR

Enhance sensing of viral infection by increasing expression of PRRs and signaling components

Induce expression of IFIT proteins, which suppress the translation of viral RNA

Increase MHC class I expression and antigen presentation in all cells

Activate dendritic cells and macrophages

Activate NK cells to kill virus-infected cells

Induce chemokines to recruit lymphocytes

Fig. 3.34 Interferons are antiviral proteins produced by cells in response to viral infection. The interferons IFN- α and IFN- β promote antiviral immunity through multiple mechanisms. They induce resistance to viral replication in uninfected cells by activating genes that cause the destruction of mRNA and inhibit the translation of viral proteins and some host proteins. The products of these genes include the Mx proteins, oligoadenylate synthetase, PKR, and IFIT proteins. Interferons can also induce MHC class I expression in most cell types in the body, thus enhancing their resistance to NK cells; they may also induce increased synthesis of MHC class I molecules in cells that are newly infected by virus, thus making them more susceptible to being killed by CD8 cytotoxic T cells (see Chapter 9). In addition, interferons activate NK cells, which then selectively kill virus-infected cells.

ISGs exert antiviral activities through a variety of mechanisms. One ISG encodes the enzyme [oligoadenylate synthetase](#), which polymerizes ATP into 2',5'-linked oligomers (whereas nucleotides in nucleic acids are normally linked 3',5'). These 2',5'-linked oligomers serve as second messengers and activate an endoribonuclease that then degrades viral RNA (note the parallels to cGAMP activation of STING, described in [Section 3-9](#)). Another ISG-encoded protein induced by IFN- α and IFN- β is a dsRNA-dependent protein kinase called [PKR](#). This serine/threonine kinase phosphorylates the α subunit of eukaryotic initiation factor 2 (eIF2), thus suppressing protein translation and contributing to the inhibition of viral replication. [Mx \(myxoma resistant\) proteins](#) are also ISG-encoded proteins induced by type I interferons. Humans and wild mice have two highly similar proteins, Mx1 and Mx2, which are GTPases belonging to the dynamin protein family, but how they interfere with viral replication is not understood.

The [IFIT \(IFN-induced protein with tetratricoid repeats\)](#) family of ISG-encoded proteins contains four human and three mouse proteins that function in restraining the translation of viral RNA into proteins. [IFIT1 and IFIT2](#) can both suppress the translation of normal capped mRNAs by binding to subunits of the [eukaryotic initiation factor 3 \(eIF3\)](#) complex, which prevents eIF3 from interacting with eIF2 to form the 43S pre-initiation complex (**Fig. 3.35**). This action may be responsible in part for the reduction in cellular proliferation induced by type I interferons. Mice lacking IFIT1 or IFIT2 show increased susceptibility to infection by certain viruses, such as vesicular stomatitis virus.

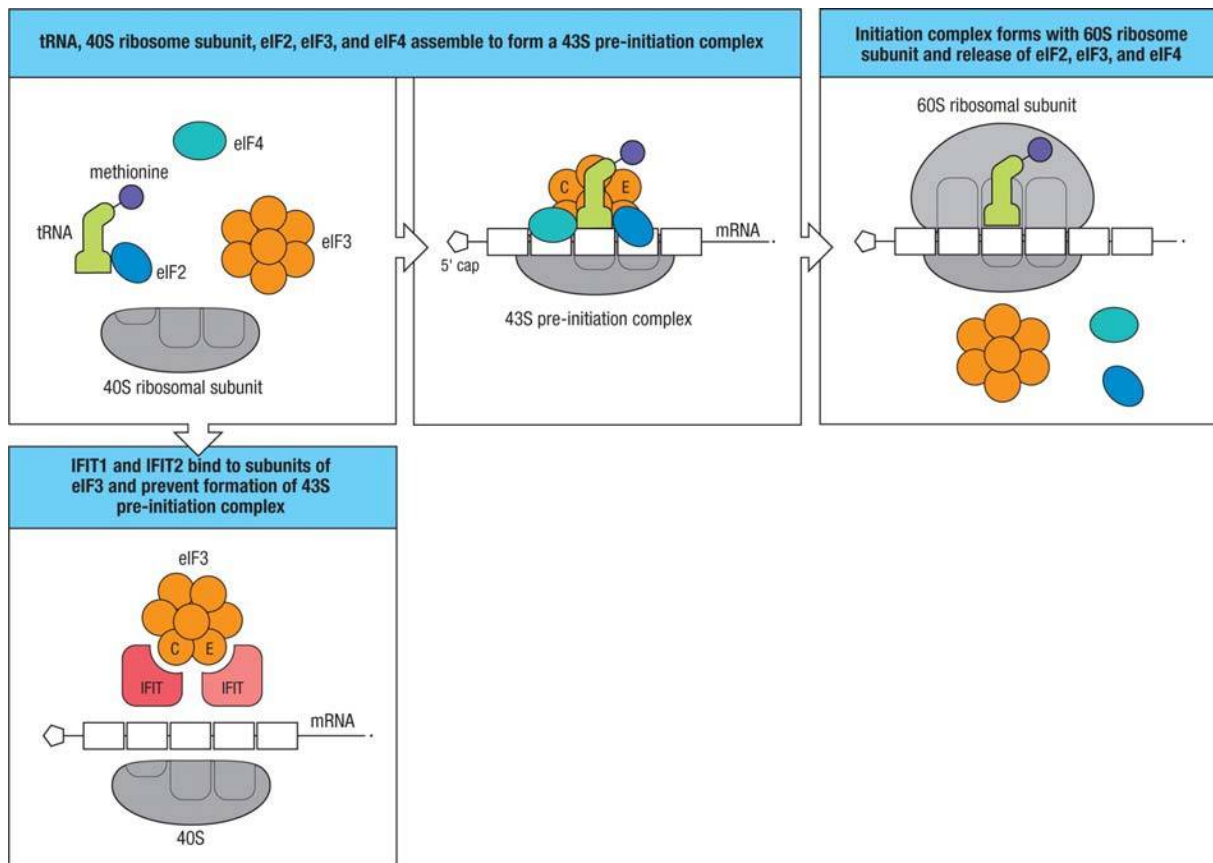


Fig. 3.35 IFIT proteins act as antiviral effector molecules by inhibiting steps in the translation of RNA. Top left panel: Formation of a 43S pre-initiation complex is an early step in the translation of RNA into protein by the 80S ribosome that involves a charged methionine tRNA, the 40S ribosome subunit, and eukaryotic initiation factors (eIFs) eIF4, eIF2, and eIF3. Middle panel: eIFs and a charged methionine tRNA assemble into a 43S pre-initiation complex. Right panel: The pre-initiation complex mRNA recognizes the 5' cap structure and joins with the 60S ribosomal subunit, releasing eIF2, eIF3, and eIF4 and forming a functional 80S ribosome. Lower panel: eIF3 has 13 subunits, a–m. IFIT proteins can inhibit several steps in protein translation. Mouse IFIT1 and IFIT2 interact with eIF3C, and human IFIT1 and IFIT2 interact with eIF3E, preventing formation of the 43S pre-initiation complex. IFITs can also interfere with other steps in translation and can bind and sequester uncapped viral mRNAs to prevent their translation (not shown). Expression of IFIT proteins is induced in viral infection by signaling downstream of type I interferons.

Another function of IFIT1 is to suppress translation of viral RNA that lacks a normal host modification of the 5' cap. Recall that the normal mammalian 5' cap is initiated by linking a 7-methylguanosine nucleotide to the first ribose sugar of the mRNA by a 5',5'-linked triphosphate bridge to produce a structure called cap-0. This structure is further modified by cytoplasmic methylation of the 2' hydroxyl groups on the first and second ribose sugars of the RNA. Methylation of the first ribose sugar produces a structure called cap-1; methylation of the second generates cap-2. IFIT1 has a high affinity for cap-0 but much lower affinity for cap-1 and cap-2. Some viruses, such as Sindbis virus (family Togaviridae), lack 2'-O-

methylation and therefore are restricted by this action of IFIT1. Many viruses, such as West Nile virus and SARS coronavirus, have acquired a [2'-O-methyltransferase \(MTase\)](#) that produces cap-1 or cap-2 on their viral transcripts. These viruses can thus evade restriction by IFIT1.

Members of the [interferon-induced transmembrane protein \(IFITM\)](#) family are expressed at a basal level on many types of tissues but are strongly induced by type I interferons. There are four functional IFITM genes in humans and in mice, and these encode proteins that have two transmembrane domains and are localized to various vesicular compartments of the cell. IFITM proteins act to inhibit, or restrict, viruses at early steps of infection. Although the molecular details are unclear, [IFITM1](#) appears to interfere with the fusion of viral membranes with the membrane of the lysosome, which is required for introducing some viral genomes into the cytoplasm. Viruses that must undergo this fusion event in lysosomes, such as the Ebola virus, are restricted by IFITM1. Similarly, [IFITM3](#) interferes with membrane fusion in late endosomes, and so restricts the influenza A virus, which undergoes fusion there. The importance of this mechanism is demonstrated by the increased viral load and higher mortality in mice lacking IFITM3 that are infected with the influenza A virus.

Interferons also stimulate production of the chemokines CXCL9, CXCL10, and CXCL11, which recruit lymphocytes to sites of infection. They also increase the expression of MHC class I molecules on all types of cells, which facilitates recognition of virally infected cells by cytotoxic T lymphocytes via the display of viral peptides complexed to MHC class I molecules on the infected cell surface (see Fig. 1.30). Through these effects, interferons indirectly help promote the killing of virus-infected cells by CD8 cytotoxic T cells. Another way in which interferons act is to activate populations of innate immune cells, such as NK cells, that can kill virus-infected cells, as described below.

3-23 Several types of innate lymphoid cells provide protection in early infection.

A defining feature of adaptive immunity is the clonal expression of antigen receptors, produced by somatic gene rearrangements, that provide the extraordinarily diverse specificities of T and B lymphocytes (see [Section 1-11](#)). However, for several decades, immunologists have recognized cells with lymphoid characteristics that lack specific antigen receptors. [Natural killer \(NK\) cells](#) have been known the longest, but in the past several years other distinct groups of such cells have been identified. Collectively, these are now called [innate lymphoid cells \(ILCs\)](#) and include NK cells (**Fig. 3.36**). ILCs develop in the bone marrow from the same [common lymphoid progenitor \(CLP\)](#) that gives rise to B and T cells. Expression of the transcription factor Id2 (inhibitor of DNA binding 2) in the CLP represses B- and T-cell fates and is required for the development of all ILCs. ILCs are identified by the absence of T- and B-cell antigen receptor and co-receptor complexes, but they express receptors for common γ -chain cytokines. They migrate from the bone marrow and take up residence in lymphoid tissues and peripheral organs, notably the dermis, liver,

small intestine, adipose tissue, and lung.

The major categories of innate lymphoid cells (ILCs) and their properties			
Innate lymphoid subgroup	Inducing cytokines	Effector molecules produced	Function
NK cells	IL-12, IL-18, type I IFN	IFN- γ , perforin, granzyme	Immunity against viruses, intracellular pathogens
ILC1	IL-12, IL-18	IFN- γ	Defense against viruses, intracellular pathogens
ILC2	IL-25, IL-33, TSLP	IL-5, IL-13, amphiregulin, IL-4	Expulsion of extracellular parasites, tissue repair
ILC3 (LTi cells)	IL-1 β , IL-23	IL-17, IL-22	Immunity to extracellular bacteria

Fig. 3.36 The major categories of innate lymphoid cells (ILCs) and their properties.

ILCs function in innate immunity as effector cells that amplify the signals delivered by innate recognition. They are stimulated by cytokines produced by other innate cells, such as macrophages or dendritic cells, or by nonimmune cells, such as epithelial cells, that have been activated by innate sensors of microbial infection or cellular damage. Three major subgroups of ILCs are defined, largely on the basis of the types of cytokines that each produces. [Group 1 ILCs \(ILC1s\)](#) generate IFN- γ in response to activation by certain cytokines, in particular IL-12 and IL-18, made by dendritic cells and macrophages, and they function in protection against infection by viruses or intracellular pathogens. NK cells are now considered to be a type of ILC. ILC1s and NK cells are closely related but have distinct functional properties and differ in the cytokines and transcription factors required for their development. For example, both NK and ILC1 cells require the transcription factors Id2 and [Nfil3](#) for their development, but ILC1s also require the transcription factor PLZF whereas NK cells do not. NK cells require the cytokine IL-15, while liver ILC1 cells require the cytokine IL-7 and the transcription factor [Tbet](#). Functionally, NK cells are more similar to CD8 T cells, while ILC1s resemble more closely the T_H1 subset of CD4 T cells (see [Section 3-24](#)). NK cells can be found within tissues, but they also circulate through the blood, while ILC1 cells appear to be largely noncirculating tissue-resident cells. In the mouse, conventional NK cells express the integrin α 2 (CD49b), while ILC1 cells, for example in the liver, lack CD49b but express the surface protein CD49a.

[Group 2 ILCs \(ILC2s\)](#) produce the cytokines IL-4, IL-5, and IL-13 in response to various cytokines, particularly [thymic stromal lymphopoietin \(TSLP\)](#), IL-25, and IL-33. ILC2 cytokines function in promoting mucosal and barrier immunity and aid in protection against parasites. ILC2s also produce amphiregulin, an EGF family member that induces epithelial cell proliferation and differentiation, which facilitates repair of damaged tissues.

[Group 3 ILCs \(ILC3s\)](#) respond to the cytokines IL-1 β and IL-23 and produce several cytokines, including IL-17 and IL-22, which increase defenses against extracellular bacteria and fungi. IL-17 functions by stimulating the production of chemokines that recruit neutrophils, while IL-22 acts directly on epithelial cells to stimulate the production of antimicrobial peptides such as RegIII γ (see [Section 2-4](#)).

The classification of ILC subtypes and the analysis of their development and function is still an active area, and studies to define the relative importance of these cells in immune responses are ongoing. The ILC subgroups identified so far appear to be highly parallel in structure to the subsets of effector CD8 and CD4 T cells that were defined over the past three decades. The transcription factors that control the development of different ILC subsets seem, for now at least, to be the same as those that control the corresponding T-cell subsets. Because of these similarities, we will postpone a detailed description of ILC development until [Chapter 9](#), where we will cover this topic along with the development of T-cell subsets.

3-24 NK cells are activated by type I interferon and macrophage-derived cytokines.

NK cells are larger than naive T and B cells, have distinctive cytoplasmic granules containing cytotoxic proteins, and were functionally identified by their ability to kill certain tumor cell lines *in vitro* without the need for prior immunization. NK cells kill cells by releasing their cytotoxic granules, which are similar to those of cytotoxic T cells and have the same effects (discussed in [Chapter 9](#)). In brief, the contents of cytotoxic granules, which contain granzymes and the pore-forming protein perforin, are released onto the surface of the target cell and penetrate the cell membrane and induce programmed cell death. However, unlike T cells, killing by NK cells is triggered by germline-encoded receptors that recognize molecules on the surface of infected or malignantly transformed cells. A second pathway used by NK cells to kill target cells involves the TNF family member known as [TRAIL \(tumor necrosis factor-related apoptosis-inducing ligand\)](#). NK cells express TRAIL on their cell surface. TRAIL interacts with two TNFR superfamily 'death' receptors, [DR4 and DR5](#) (encoded by *TNFSF10A* and *TNFSF10B*), that are expressed by many types of cells. When NK cells recognize a target cell, TRAIL stimulates DR4 and DR5 to activate the pro-enzyme [caspase 8](#), which leads to apoptosis. In contrast to pyroptosis, induced by caspase 1 after inflammasome activation (see [Section 3-11](#)), apoptosis is not associated with production of inflammatory cytokines. We will discuss more details of the mechanisms of caspase-induced apoptosis when we discuss killing by cytotoxic T cells in [Chapter 9](#). Finally, NK cells express Fc receptors (see [Section 1-20](#)); binding of antibodies to these receptors activates NK cells to release their cytotoxic granules, a process known as [antibody-dependent cell-mediated cytotoxicity, or ADCC](#), to which we will return in [Chapter 10](#).

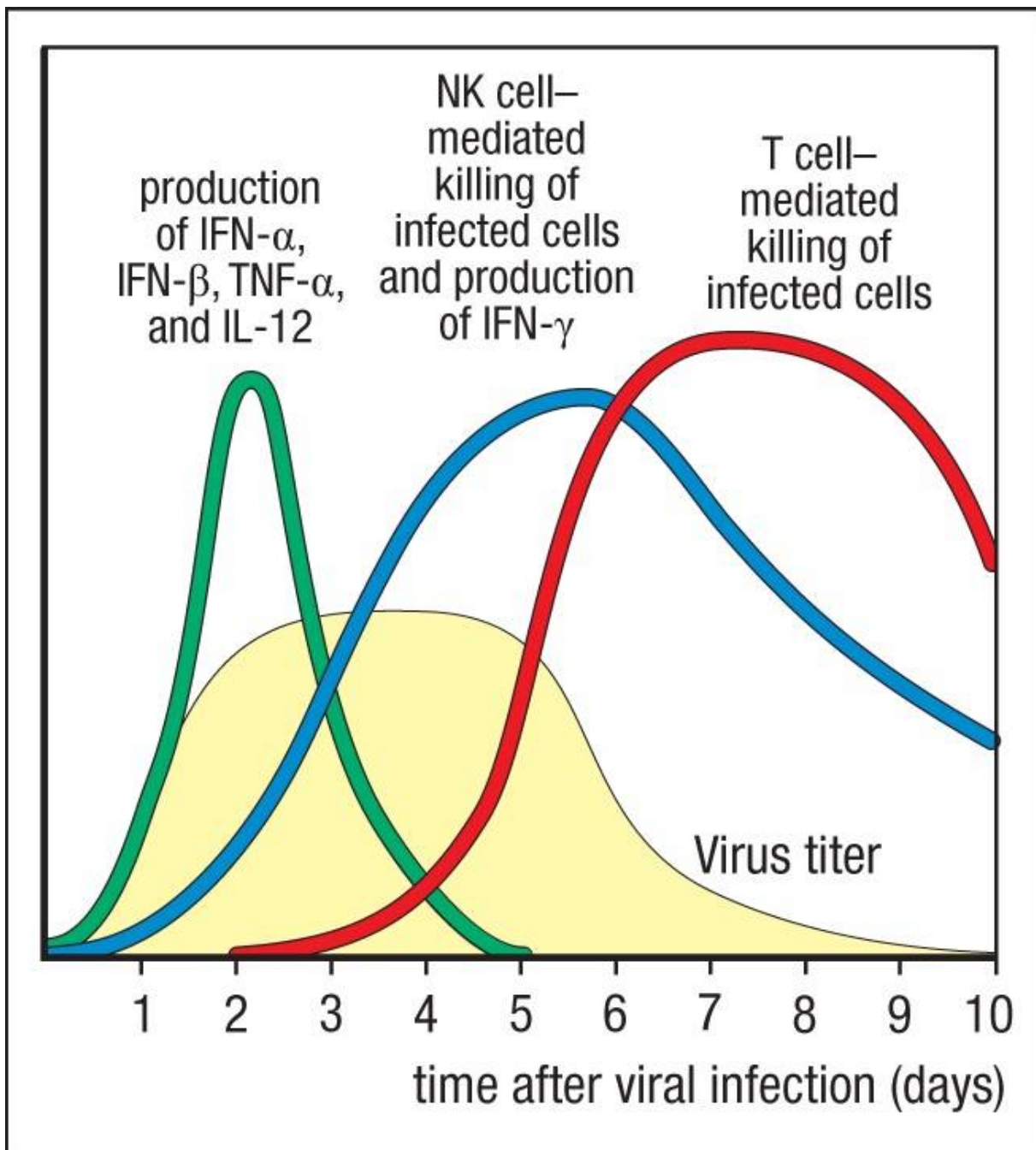


Fig. 3.37 Natural killer cells (NK cells) are an early component of the host response to virus infection. Experiments in mice have shown that IFN- α , IFN- β , and the cytokines TNF- α and IL-12 are produced first after viral infection, and are followed by a wave of NK cells, which together control virus replication via IFN- α secretion and cytotoxicity but do not eliminate the virus. Virus elimination is accomplished when virus-specific CD8 T cells and neutralizing antibodies are produced. Without NK cells, the levels of some viruses are much higher in the early days of the infection, and the infection can be lethal unless treated vigorously with antiviral compounds.

The ability of NK cells to kill target cells can be enhanced by interferons or certain cytokines. NK cells that can kill sensitive targets can be isolated from uninfected individuals,

but this activity is increased 20- to 100-fold when NK cells are exposed to IFN- α and IFN- β or to IL-12, a cytokine produced by dendritic cells and macrophages during infection by many types of pathogens. Activated NK cells serve to contain virus infections while the adaptive immune response is generating antigen-specific cytotoxic T cells and neutralizing antibodies that can clear the infection (**Fig. 3.37**). A clue to the physiological function of NK cells in humans comes from rare individuals deficient in these cells, who are frequently susceptible to herpesvirus infection. For example, a selective NK-cell deficiency results from mutations in the human MCM4 (minichromosome maintenance-deficient 4) protein, which is associated with predisposition to viral infections. Similarly, depletion of NK cells in mice, either by antibody or genetically, results in susceptibility to a number of viruses, including herpesviruses.

IL-12, produced by activated macrophages and dendritic cells, acts in synergy with the cytokine IL-18 to stimulate NK cells to secrete large amounts of IFN- γ . This IFN- γ is crucial in controlling certain infections because it precedes the production of IFN- γ by activated CD8 cytotoxic T cells. IFN- γ , whose receptor activates only the STAT1 transcription factor, is quite distinct functionally from the antiviral type I interferons IFN- α and IFN- β and is not directly induced by viral infection. The production of IFN- γ by NK cells early in an immune response can directly activate macrophages to enhance their capacity to kill pathogens, augmenting innate immunity. IFN- γ also influences adaptive immunity through actions on dendritic cells and by regulating the differentiation of CD4 T cells into the pro-inflammatory T_H1 subset, which produces IFN- γ . Accordingly, mice deficient for the gene encoding IFN- γ are highly susceptible to viruses and intracellular bacterial pathogens. NK cells also produce TNF- α , [granulocyte-macrophage colony-stimulating factor \(GM-CSF\)](#), and the chemokines CCL3 (MIP-1 α), CCL4, and CCL5 (RANTES), which act to recruit and activate macrophages.

3-25 NK cells express activating and inhibitory receptors to distinguish between healthy and infected cells.

For NK cells to defend against viruses or other pathogens, they must be able to distinguish infected cells from uninfected healthy cells. To do this, NK cells use a mechanism distinct from the mechanism of pathogen recognition used by T or B cells. An individual NK cell expresses various combinations of germline-encoded [activating receptors](#) and [inhibitory receptors](#). While the exact details are not clear in every case, it is thought that the overall balance of signaling by these receptors determines whether an NK cell engages and kills a target cell. The receptors on an NK cell are tuned to detect changes in expression of various surface proteins on a target cell, referred to as '[dysregulated self](#).' The activating receptors generally recognize cell-surface proteins that are induced on target cells by infection or stress, such as malignant transformation. These induced proteins are referred to as '[stress-induced self](#).' A number of cellular perturbations, including DNA damage, signals related to

proliferation, and heat shock–related stress, can lead to induction of ligands for activating receptors. Signaling by innate sensors (for example, TLRs) is also capable of inducing such ligands. When activating receptors on NK cells detect stress-induced self ligands, signaling is initiated in the NK cell that results in release of cytokines, such as IFN- γ , and killing of the stimulating cell through the release of cytotoxic granules. However, engagement of activating receptors is not always sufficient to induce NK-cell activation; negative signals imposed by inhibitory receptors must either be lost or overcome, as discussed below.

Inhibitory receptors on NK cells recognize surface molecules that are constitutively expressed at high levels by most healthy cells. As such, NK cells that encounter healthy cells receive signals that inhibit the signals downstream of activating receptors and prevent their activation. However, infection, cellular stress, or transformation can lead to down-regulation of the ligands for inhibitory receptors. The loss of these molecules is referred to as ‘[missing self](#).’ Inhibitory receptors can recognize other molecules, but those recognizing [MHC class I molecules](#) have been best characterized. MHC molecules are glycoproteins expressed on nearly all cells of the body. We will discuss the role of MHC proteins in antigen presentation to T cells in [Chapter 6](#), but for now we need only to introduce the two main classes of MHC molecules. MHC class I molecules are expressed on most of the cells of the body (except, notably, red blood cells), whereas the expression of [MHC class II molecules](#) is far more restricted, largely to immune cells.

Inhibitory receptors that recognize MHC class I molecules function to prevent NK cells from killing normal host cells. The greater the number of MHC class I molecules on a cell surface, the better protected that cell is against attack by NK cells. Interferons induce expression of MHC class I molecules and protect uninfected host cells from being killed by NK cells, while also activating NK cells to kill virus-infected cells. Viruses and some other intracellular pathogens can cause down-regulation of MHC class I molecules as a strategy to prevent the display of antigens as peptides to T cells, also discussed in [Chapter 13](#). MHC class I molecules are often similarly down-regulated in cancerous cells. Importantly, NK cells are able to sense this reduction in expression of MHC class I molecules through reduced signaling from their inhibitory receptors. Reduction in MHC class I expression is an example of ‘missing self’ and increases the chance that an NK cell will kill the target cell. It is thought that the balance of signals from ‘stress-induced self’ and ‘missing self’ determines whether an individual NK cell will be triggered to kill a particular target cell (**Fig. 3.38**). Thus, receptors expressed on NK cells integrate the signals from two types of surface receptors, which together control an NK cell’s cytotoxic activity and cytokine production.

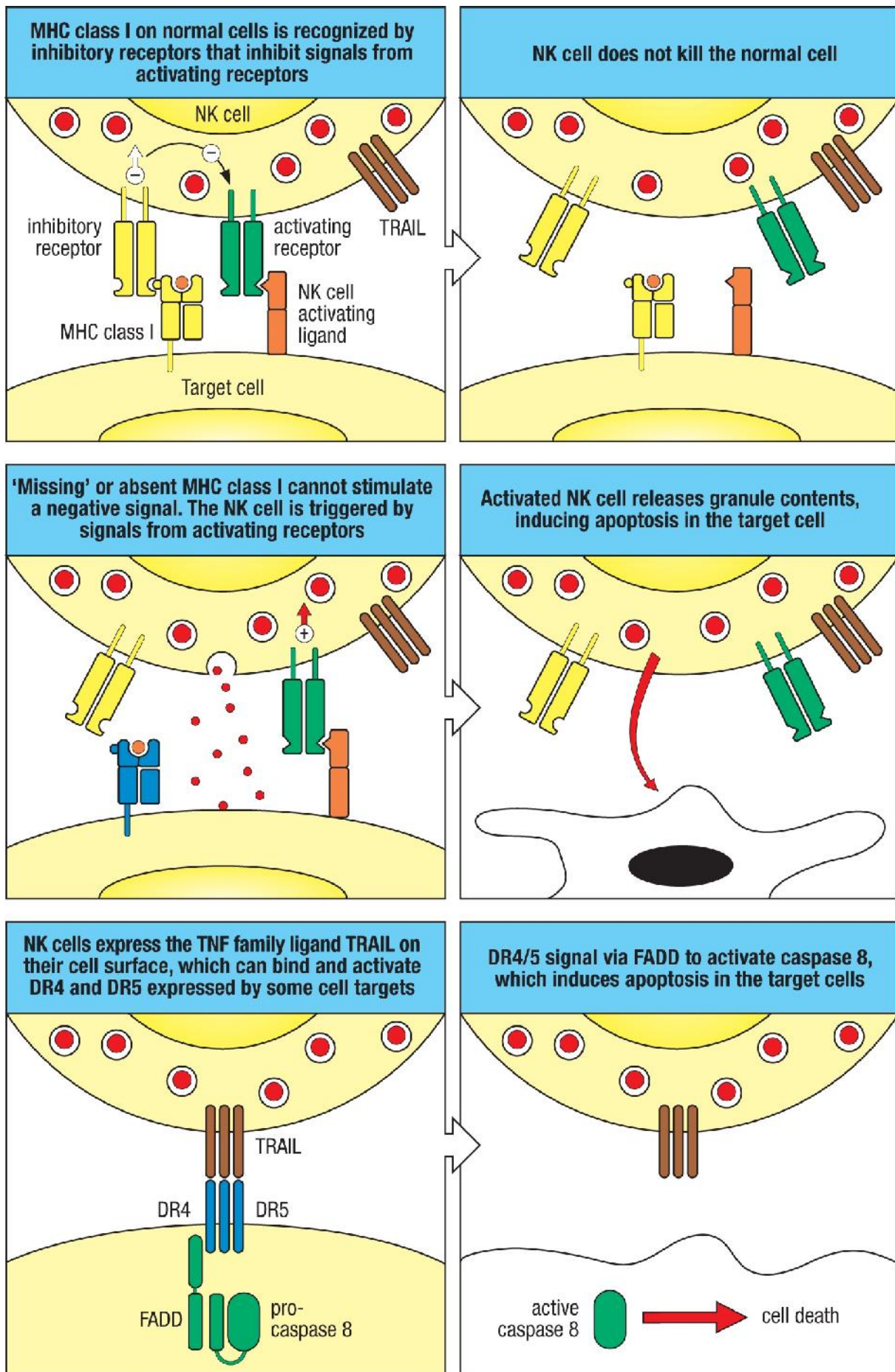


Fig. 3.38 Killing by NK cells depends on the balance between activating and inhibitory signals. NK cells have several different activating receptors that signal the NK to kill the bound cell. However, NK cells are prevented from a wholesale attack by another set of inhibitory receptors that recognize MHC class I molecules (which are present on almost all cell types) and that inhibit signaling by the activating receptors. Up-regulation of ligands for activating receptors can overrule this inhibition. Alternatively, activation can occur when inhibitory signals are lost, such as when viruses inhibit MHC class I expression or alter its conformation so as to avoid recognition by CD8 T cells. NK cells may also kill target cells through their expression of the TNF family member TRAIL, which binds to TNFR members DR4 and DR5 expressed by some types of cells. DR4 and DR5 signal through FADD, an adaptor that activates pro-caspase 8, leading to induction of apoptosis of the target cell.

3-26 NK-cell receptors belong to several structural families.

The receptors that regulate the activity of NK cells fall into several large families that contain a number of other cell-surface receptors in addition to NK receptors (**Fig. 3.39**). Members of the family of [killer cell immunoglobulin-like receptors \(KIRs\)](#) have differing numbers of immunoglobulin domains. Some, such as KIR-2D, have two immunoglobulin domains, whereas others, such as KIR-3D, have three. The KIR genes form part of a larger cluster of immunoglobulin-like receptor genes known as the [leukocyte receptor complex \(LRC\)](#). Another family, the [killer cell lectin-like receptors \(KLRs\)](#) are C-type lectin-like proteins whose genes reside in a gene cluster called the [NK receptor complex \(NKC\)](#). Mice lack KIR genes and instead predominantly express [Ly49 receptors](#), encoded in the NKC on mouse chromosome 6, to control their NK-cell activity. These receptors can be activating or inhibitory and are highly polymorphic between different strains of mice. By contrast, humans lack functional Ly49 genes and rely on KIRs encoded in the LRC to control their NK-cell activity. An important feature of the NK-cell population is that any given NK cell expresses only a subset of the receptors in its potential repertoire, and so not all NK cells in an individual are identical.

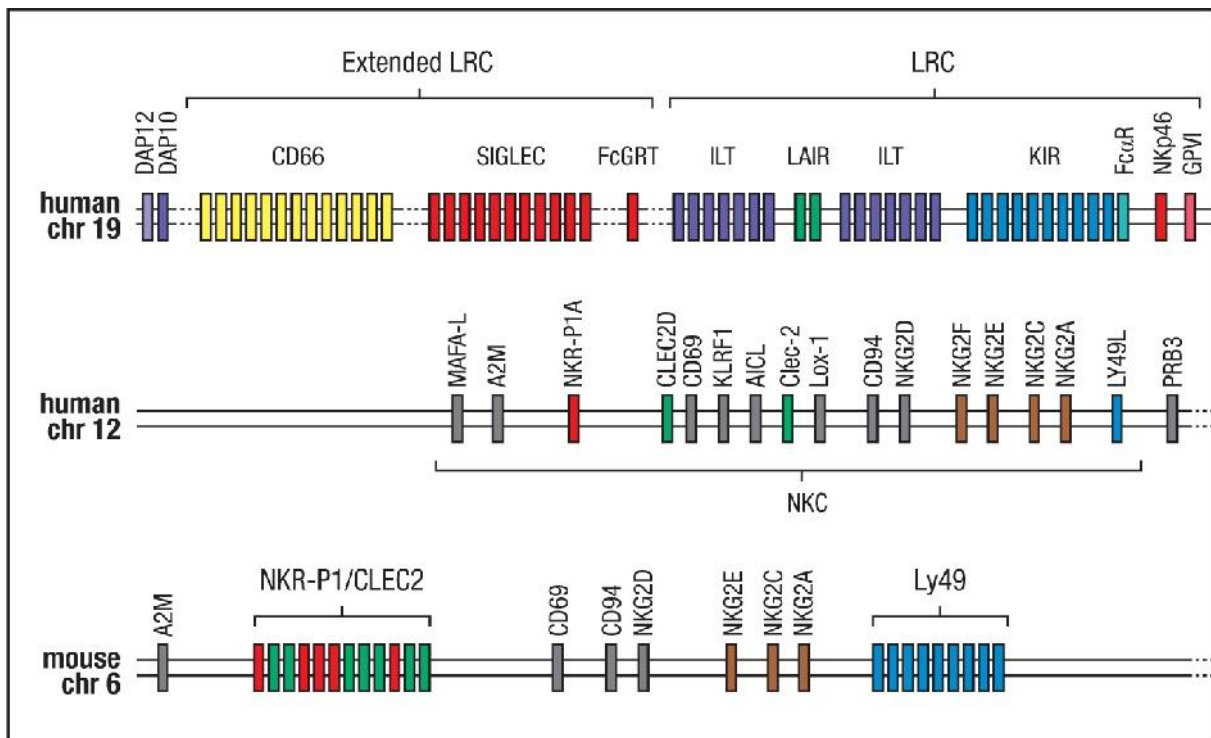
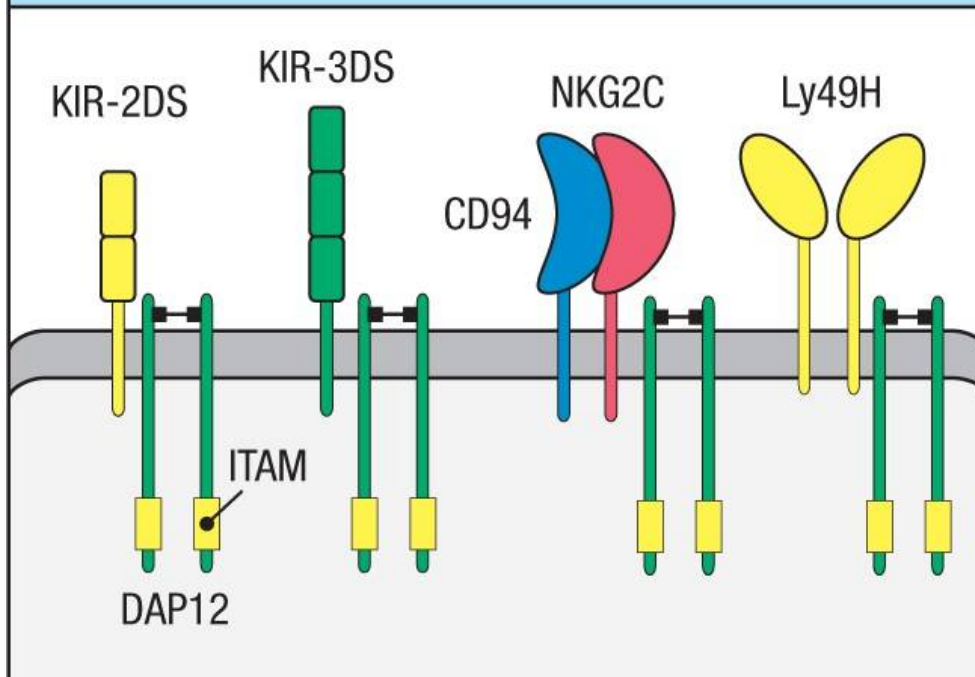


Fig. 3.39 The genes that encode NK receptors fall into two large families. The first, the leukocyte receptor complex (LRC), comprises a large cluster of genes encoding a family of proteins composed of immunoglobulin-like domains. These include the killer cell immunoglobulin-like receptors (KIRs) expressed by NK cells, the ILT (immunoglobulin-like transcript) class, and the leukocyte-associated immunoglobulin-like receptor (LAIR) gene families. The sialic acid-binding Ig-like lectins (SIGLECs) and members of the CD66 family are located nearby. In humans, this complex is located on chromosome 19. The second gene cluster is called the NK receptor complex (NKC) and encodes killer cell lectin-like receptors (KLRs), a receptor family that includes the NKG2 proteins and CD94, with which some NKG2 molecules pair to form a functional receptor. This complex is located on human chromosome 12. The corresponding locus in mice is located on chromosome 6 and also contains the expanded family of Ly49 genes. Some NK receptor genes are found outside these major gene clusters.

Activating and inhibitory receptors of NK cells can belong to the same structural family

Activating receptors



Inhibitory receptors

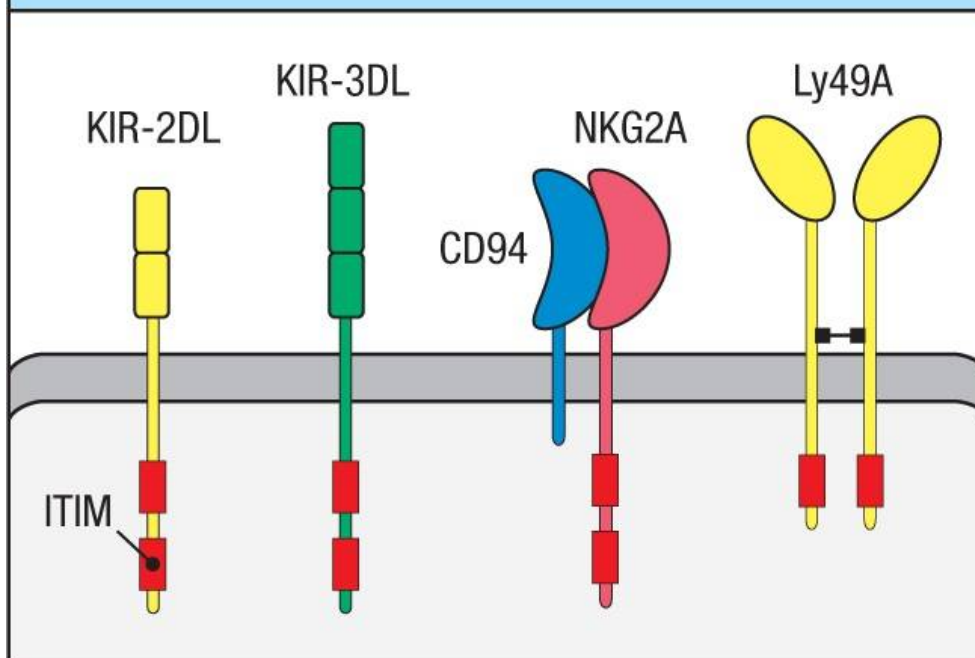


Fig. 3.40 The structural families of NK receptors encode both activating and inhibitory receptors. The families of killer cell immunoglobulin-like receptors (KIRs) and killer cell lectin-like receptors (KLRs and Ly49 receptors) have members that send activating signals to the NK cell (upper panel) and others that send inhibitory signals to the NK cell (lower panel). KIR family members are designated according to the number of immunoglobulin-like domains they possess and by the length of their cytoplasmic tails. Activating KIRs have short cytoplasmic tails and bear the designation 'S.' These associate with the signaling protein DAP12 via a charged amino acid residue in the transmembrane region. The cytoplasmic tails of DAP12 contain amino acid motifs called ITAMs, which are involved in signaling. NKG2 receptors belong to the KLR family, and, whether activating or inhibitory, form heterodimers with another C-type lectin-like family member, CD94. The inhibitory KIRs have longer cytoplasmic tails and are designated 'L.' Inhibitory KIR and Ly49 family members do not associate constitutively with adaptor proteins but contain a signaling motif called an ITIM, which when phosphorylated, is recognized by inhibitory phosphatases.

Activating and inhibitory receptors are present within the same structural family. Whether a KIR or Ly49 protein is activating or inhibitory depends on the presence or absence of particular signaling motifs in its cytoplasmic domain. Inhibitory KIRs and Ly49s have long cytoplasmic tails that contain an [immunoreceptor tyrosine-based inhibition motif \(ITIM\)](#). The consensus amino acid sequence for the ITIM is V/I/LxYxxL/V, where 'x' stands for any amino acid. For example, the cytoplasmic tails of the inhibitory receptors KIR-2DL and KIR-3DL each contain two ITIMs (**Fig. 3.40**). When ligands associate with an inhibitory KIR, the tyrosine in its ITIM becomes phosphorylated by the action of [Src family protein tyrosine kinases](#). When phosphorylated, the ITIM can then bind the intracellular protein tyrosine phosphatases [SHP-1](#) (Src homology region 2-containing protein tyrosine phosphatase-1) and [SHP-2](#), which become localized near the cell membrane. These phosphatases inhibit signaling induced by other receptors (for example, the NK activating receptors) by removing phosphates from tyrosine residues on other intracellular signaling molecules.

Activating KIRs have short cytoplasmic tails and are designated, for example, as KIR-2DS and KIR-3DS (see Fig. 3.40). In mice, activating Ly49 proteins similarly have a short cytoplasmic tail that contains no signaling domain. Instead, these activating KIRs and Ly49 receptors have a charged residue in their transmembrane regions that associates with an accessory signaling protein called [DAP12](#). DAP12 is a transmembrane protein that contains an immunoreceptor tyrosine-based activation motif (ITAM) with consensus sequence Yxx[L/I]x₆₋₉Yxx[L/I] in its cytoplasmic tail and forms a disulfide-linked homodimer in the membrane. When a ligand binds to an activating KIR or Ly49 receptor, the tyrosine residues in the ITAM of DAP12 become phosphorylated, turning on intracellular signaling pathways that activate the NK cell and lead to release of the cytotoxic granules. The phosphorylated ITAMs bind and activate intracellular tyrosine kinases such as Syk or ZAP-70, leading to further signaling events similar to those described for T cells in [Chapter 7](#).

The KLR family also has both activating and inhibitory members. In mice, inhibitory Ly49 receptors have an ITIM in their cytoplasmic tail that recruits SHP-1. The latter's importance is shown by the failure of Ly49 to inhibit NK activation upon binding to MHC class I in mice carrying the [motheaten](#) mutation, which inactivates SHP-1 protein. In humans and mice, NK cells express a heterodimer of two different C-type lectin-like receptors, [CD94](#) and [NKG2](#). This heterodimer interacts with nonpolymorphic MHC class I-like molecules, including HLA-E in humans and Qa1 in mice. HLA-E and Qa1 are unusual in that instead of binding peptides derived from pathogens, they bind fragments of the [signal peptide](#) derived from other MHC class I molecules during processing in the endoplasmic reticulum. This enables CD94:NKG2 to detect the presence of several different MHC class I variants, whose expression may be targeted by viruses, and kill cells in which overall MHC molecule expression is diminished. In humans there are four closely related NKG2 family proteins, NKG2A, C, E, and F (encoded by *KLRC1–4*), and a more distantly related protein, NKG2D (encoded by *KLRK1*). Of these, for example, NKG2A contains an ITIM and is inhibitory, whereas NKG2C has a charged transmembrane residue, associates with DAP12, and is activating (see Fig. 3.40). NKG2D is also activating but quite distinct from the other NKG2 receptors, and we will discuss it separately in the next section.

The overall response of NK cells to differences in MHC expression is further complicated by the extensive polymorphism of KIR genes, with different numbers of activating and inhibitory KIR genes being found in different people. The responsiveness of NK cells is thought to be tuned by their ability to recognize inhibitory ligands during their development. This process, referred to as licensing or education, is likely to underlie the effect of NK cells as a barrier to transplantation, because the NK cells educated in the environment of the recipient may fail to recognize the same inhibitory ligands on donor cells. A similar phenomenon may occur during pregnancy, because of differences between fetal and maternal MHC molecules (see [Section 15-26](#)). The advantage of such extensive KIR polymorphism is not yet clear, and some genetic epidemiologic studies even suggest an association between certain alleles of KIR genes and earlier onset (although not absolute frequency) of rheumatoid arthritis. The KIR gene cluster is not present in mice, but some species, including some primates, contain genes of both the KIR and KLR families. This might suggest that both gene clusters are relatively ancient and that for some reason, one or the other gene cluster was lost by mice and humans.

Signaling by the inhibitory NK receptors suppresses the killing activity and cytokine production of NK cells. This means NK cells will not kill healthy, genetically identical cells with normal expression of MHC class I molecules, such as the other cells of the body. Virus-infected cells, however, can become susceptible to being killed by NK cells by a variety of mechanisms. First, some viruses inhibit all protein synthesis in their host cells, so that synthesis of MHC class I proteins would be blocked in infected cells, even while their production in uninfected cells is being stimulated by the actions of type I interferons. The reduced level of MHC class I expression in infected cells would make them correspondingly less able to inhibit NK cells through their MHC-specific receptors, and they would become more susceptible to being killed. Second, many viruses can selectively prevent the export of

MHC class I molecules to the cell surface or induce their degradation once there. This might allow the infected cell to evade recognition by cytotoxic T cells but would make it susceptible to being killed by NK cells. Virally infected cells can still be killed by NK cells even if the cells do not down-regulate MHC, provided that ligands for activating receptors are induced. However, some viruses target ligands for the activating receptors on NK cells, thwarting NK-cell recognition and killing of virus-infected cells.

3-27 NK cells express activating receptors that recognize ligands induced on infected cells or tumor cells.

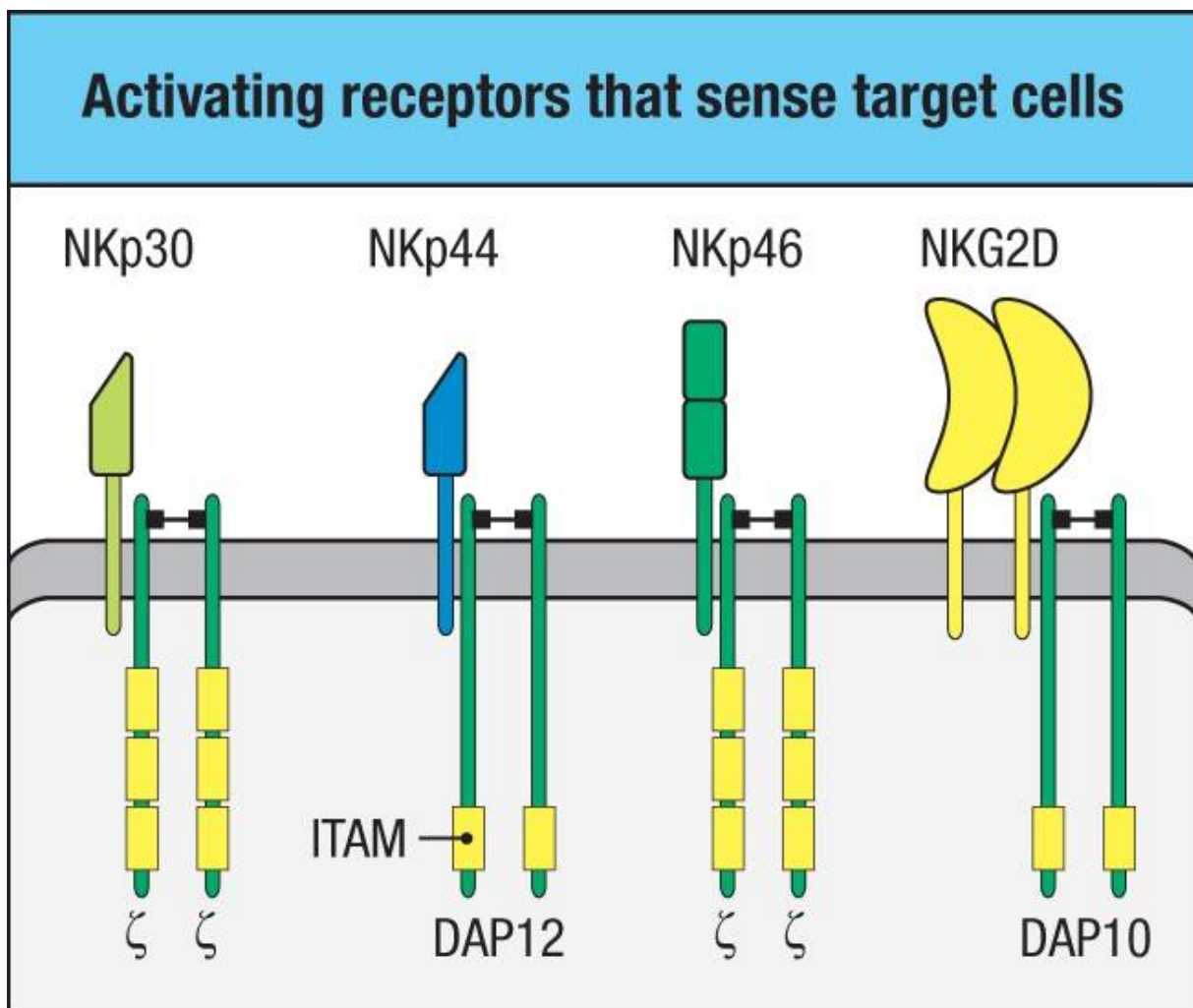


Fig. 3.41 Activating receptors of NK cells include the natural cytotoxicity receptors and NKG2D. The natural cytotoxicity receptors are immunoglobulin-like proteins. NKp30 and NKp44, for example, have an extracellular domain that resembles a single variable domain of an immunoglobulin molecule. NKp30 and NKp46 activate the NK cell through their association with homodimers of the CD3ζ chain or the Fc receptor γ chain (not shown). These signaling proteins also associate with other types of receptors

that are described in [Chapter 7](#). NKp44 activates the NK cell through its association with homodimers of DAP12. NKp46 resembles the KIR-2D molecules in having two domains that resemble the constant domains of an immunoglobulin molecule. NKG2D is a member of the C-type lectin-like family and forms a homodimer that associates with DAP10. In mice, an alternatively spliced form of NKG2D also associates with DAP12 (not shown).

In addition to the KIRs and KLRs, which have a role in sensing the level of MHC class I proteins present on other cells, NK cells also express receptors that more directly sense the presence of infection or other perturbations in a cell. Activating receptors for the recognition of infected cells, tumor cells, and cells injured by physical or chemical damage include the [natural cytotoxicity receptors \(NCRs\)](#) NKp30, NKp44, and NKp46, which are immunoglobulin-like receptors, and the C-type lectin-like family members [Ly49H](#) and [NKG2D](#) (**Fig. 3.41**). Among NCRs, only NKp46 is conserved in humans and in mice, and it is the most selective marker of NK cells across mammalian species. The ligands recognized by the NCRs are still being defined, but some evidence suggests that they recognize viral proteins. Ly49H is an activating receptor that recognizes the viral protein m157, an MHC class I-like structure encoded by the murine cytomegalovirus. The ligand for NKp30 is a protein named B7-H6, a member of the B7 family of co-stimulatory proteins, further described in [Chapters 7](#) and [9](#).

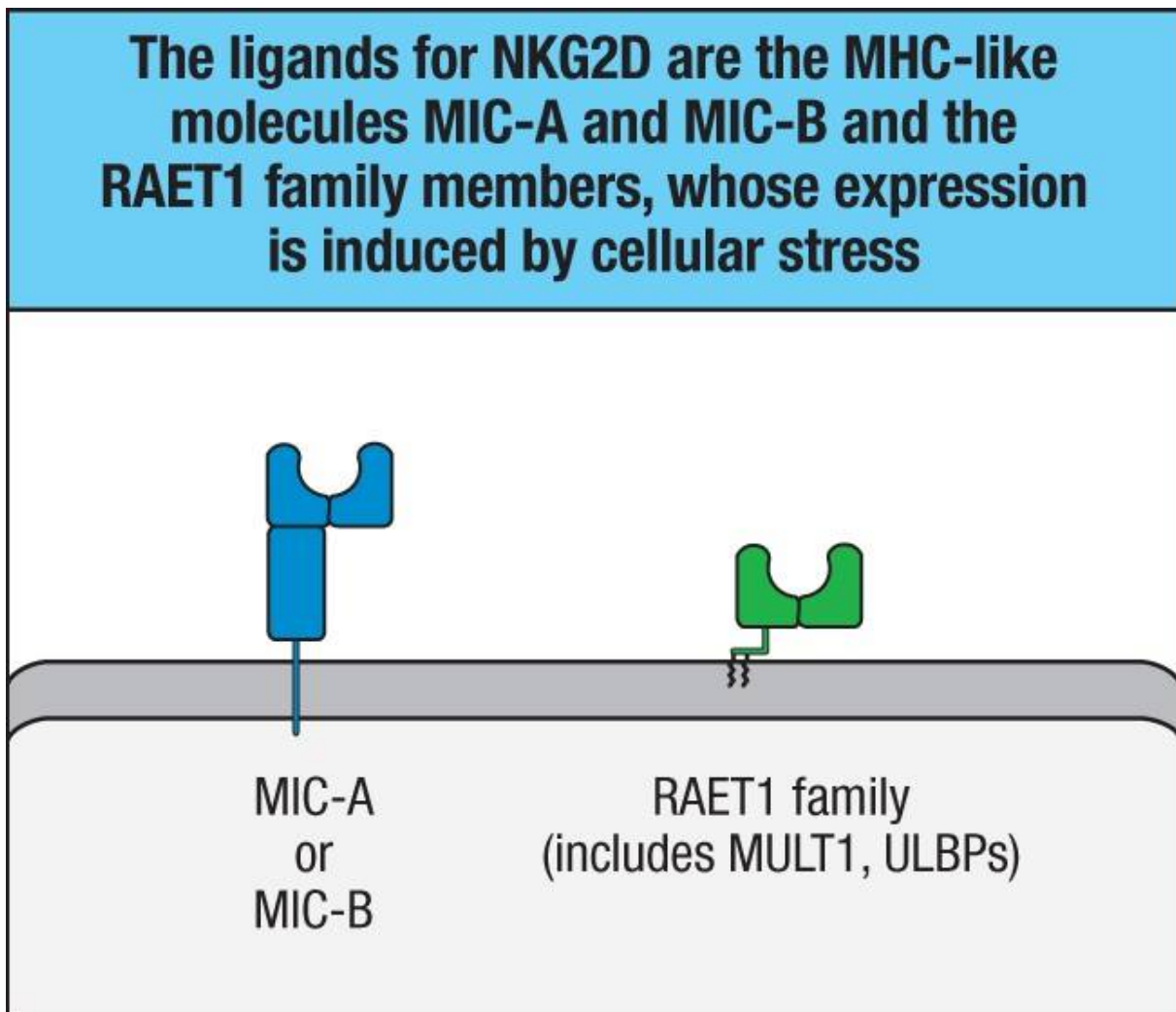


Fig. 3.42 The ligands for the activating NK receptor NKG2D are proteins that are expressed in conditions of cellular stress. Ligands for NKG2D are induced on epithelial and other cells by stress, such as DNA damage, cellular transformation, or infection. The MIC proteins MIC-A and MIC-B are MHC-like molecules. RAET1 family members, including the subset designated as UL16-binding proteins (ULBPs), also resemble a portion of an MHC class I molecule—the α_1 and α_2 domains—and most (but not all) are attached to the cell via a glycosylphosphatidylinositol linkage. Unlike MHC class I molecules, the NKG2D ligands do not bind processed peptides.

NKG2D has a specialized role in activating NK cells. NKG2A, NKG2C, NKG2E, and NKG2F form heterodimers with CD94 and bind the MHC class I molecule HLA-E. In contrast, two NKG2D molecules form a homodimer that binds to several MHC class I-like molecules that are induced by various types of cellular stress. These include the MIC molecules [MIC-A and MIC-B](#) and the [RAET1](#) family of proteins, which are similar to the α_1 and α_2 domains of MHC class I molecules (**Fig. 3.42**). The RAET1 family has 10 members, three of which were initially characterized as ligands for the [cytomegalovirus UL16 protein](#) and are also called [UL16-binding proteins, or ULBPs](#). Mice do not have equivalents of the MIC molecules; however, the ligands for mouse NKG2D have a very similar structure to

that of the RAET1 proteins and are probably orthologs of them. In fact, these ligands were first identified in mice as the [RAE1 \(retinoic acid early inducible 1\) protein family](#) and also include related proteins H60 and MULT1 (see Fig. 6.26). We will return to these MHC-like molecules when we discuss the structure of the MHC molecule in [Chapter 6](#).

The ligands for NKG2D are expressed in response to cellular or metabolic stress, and so are up-regulated on cells infected with intracellular bacteria and most viruses, as well as on incipient tumor cells that have become malignantly transformed. Thus, recognition by NKG2D acts as a generalized alarm signal to the immune system. In addition to expression by a subset of NK cells, NKG2D is expressed by various T cells, including all CD8 T cells and some $\gamma\delta$ T cells and invariant NKT cells (described in [Chapter 8](#)). In these cells, recognition of NKG2D ligands provides a potent co-stimulatory signal that enhances their effector functions.

NKG2D differs from other activating receptors on NK cells in the signaling pathway it engages within the cell. The other activating receptors are associated intracellularly with signaling proteins such as the CD3 ζ chain, the Fc receptor γ chain, and DAP12, which all contain ITAMs. In contrast, NKG2D binds a different adaptor protein, [DAP10](#), which does not contain an ITAM sequence and instead activates the intracellular lipid kinase [phosphatidylinositol 3-kinase \(PI 3-kinase\)](#), initiating a different series of intracellular signaling events in the NK cell (see [Section 7-4](#)). Generally, PI 3-kinase is considered to enhance the survival of cells in which it is activated, thereby augmenting the cell's overall effector activity. In NK cells, activation of PI 3-kinase is directly linked to the induction of cytotoxic activity. In mice, the workings of NKG2D are more complicated, because mouse NKG2D is produced in two alternatively spliced forms, one of which binds DAP12 and DAP10, whereas the other binds only DAP10. Mouse NKG2D can thus activate both signaling pathways, whereas human NKG2D seems to signal only through DAP10 to activate the PI 3-kinase pathway. Finally, NK cells express several receptors from the [SLAM \(signaling lymphocyte activation molecule\)](#) family, including [2B4](#), which recognizes the cell-surface molecule [CD48](#) expressed by many cells including NK cells. Interactions between 2B4 and CD48 on nearby NK cells can release signals that promote survival and proliferation through [SAP \(SLAM-associated protein\)](#) and the Src kinase Fyn.

Summary.

Triggering of innate sensors on various cells—neutrophils, macrophages, and dendritic cells in particular—not only activates these cells' individual effector functions but also stimulates the release of pro-inflammatory chemokines and cytokines that act together to recruit more phagocytic cells to the site of infection. Especially prominent is the early recruitment of neutrophils and monocytes. Furthermore, cytokines released by tissue phagocytic cells can induce more systemic effects, including fever and the production of acute-phase response proteins, such as mannose-binding lectin, C-reactive protein, fibrinogen, and pulmonary surfactant proteins, which add to a general state of augmented innate immunity. These

cytokines also mobilize antigen-presenting cells that induce the adaptive immune response. The innate immune system has at its service several recently recognized subtypes of innate lymphoid cells to join the ranks of the long-recognized NK cells. ILCs exhibit specialized effector activity in response to different signals and act to amplify the strength of the innate response. The production of interferons in response to viral infections serves to inhibit viral replication and to activate NK cells. These in turn can distinguish healthy cells from those that are infected by virus or that are transformed or stressed in some way on the basis of expression of MHC class I molecules and MHC-related molecules that are ligands for some NK receptors. As we will see later in the book, cytokines, chemokines, phagocytic cells, and NK cells are all effector mechanisms that are also employed in the adaptive immune response, which uses variable receptors to target specific pathogen antigens.

Summary to [Chapter 3](#).

Innate immunity uses a variety of effector mechanisms to detect infection and eliminate pathogens or to hold them in check until an adaptive immune response develops. These effector mechanisms are all regulated by germline-encoded receptors on many types of cells that can detect molecules of microbial origin or that sense pathogen-induced changes to host cells. The induced responses of the innate immune system are based on several distinct components. After the initial barriers—the body’s epithelia and the soluble antimicrobial molecules described in [Chapter 2](#)—have been breached, the most important innate defenses rely on tissue macrophages and other tissue-resident sensor cells, such as dendritic cells and ILCs. Macrophages provide a double service: they mediate rapid cellular defense at the borders of infection through phagocytosis and antimicrobial actions, and they also use their various innate sensors to activate the process of inflammation, which involves recruiting additional cells to sites of infection. Innate sensors activate signaling pathways that lead to the production of pro-inflammatory and antiviral cytokines, which in turn stimulate innate effector responses while also helping to initiate an adaptive immune response. The uncovering of the pathogen-sensing mechanisms described in this chapter is still extremely active. It is providing new insights into human autoinflammatory conditions such as lupus, Crohn’s disease, and gout. Indeed, the induction of powerful effector mechanisms by innate immune recognition based on germline-encoded receptors clearly has some dangers. It is a double-edged sword, as is illustrated by the effects of the cytokine TNF- α —beneficial when released locally, but disastrous when produced systemically. This illustrates the evolutionary knife edge along which all innate mechanisms of host defense travel. The innate immune system can be viewed as a defense system that mainly frustrates the establishment of a focus of infection; however, even when it proves limited in fulfilling this function, it has already set in motion the initiation of the adaptive immune response, which forms an essential part of human defenses against infection.

Glossary

[autocrine](#)

Describes a cytokine or other biologically active molecule acting on the cell that produces it.

[paracrine](#)

Describes a cytokine or other biologically active molecule acting on cells near to those that produce it.

[endocrine](#)

Describes the action of a biologically active molecule such as a hormone or cytokine that is secreted by one tissue into the blood and acts on a distant tissue. Cf. **autocrine**, **paracrine**.

[interleukin \(IL\)](#)

A generic name for cytokines produced by leukocytes. The more general term cytokine is used in this book, but the term interleukin is used in the naming of specific cytokines such as IL-2. Some key interleukins are listed in the glossary under their abbreviated names; for example, IL-1 β and IL-2. Cytokines are listed in eAppendix III.

[IL-1 family](#)

One of four major families of cytokines, this family contains 11 cytokines that are structurally similar to IL-1 α and are largely pro-inflammatory in function.

[hematopoietin superfamily](#)

Large family of structurally related cytokines that includes growth factors and many interleukins with roles in both adaptive and innate immunity.

[class I cytokine receptors](#)

A group of receptors for the hematopoietin superfamily of cytokines. These include receptors using the common β chain for IL-2, IL-4, IL-7, IL-15, and IL-21, and a common β chain for GM-CSF, IL-3, and IL-5.

[X-linked severe combined immune deficiency \(X-linked SCID, or XSCID\)](#)

An immune deficiency disease in which T-cell development fails at an early intrathymic stage, and no production of mature T cells or T cell–dependent antibody occurs. It is due to a defect in a gene that encodes the γ_c chain shared by the receptors for several different cytokines.

[class II cytokine receptors](#)

A group of heterodimeric receptors for a family of cytokines that includes interferon (IFN)- α , IFN- β , IFN- γ , and IL-10.

[TNF family](#)

Cytokine family, the prototype of which is tumor necrosis factor- α (TNF- α , or TNF). It contains both secreted (e.g., TNF- α and lymphotoxin) and membrane-bound (e.g., CD40 ligand) members.

[TNF receptors](#)

Family of cytokine receptors that includes some that lead to apoptosis of the cell on which they are expressed (e.g., Fas and TNFR-I), whereas others lead to activation.

[severe combined immune deficiency \(SCID\)](#)

Type of immune deficiency (due to various causes) in which both B-cell (antibody) and T-cell responses are lacking; it is fatal if not treated.

[signal transducers and activators of transcription \(STATs\)](#)

A family of seven transcription factors activated by many cytokine and growth factor receptors. *See also* **Janus kinase (JAK) family**.

[tyrosine phosphatases](#)

Enzymes that remove phosphate groups from phosphorylated tyrosine residues on proteins. *See also* **CD45**.

[CD45](#)

A transmembrane tyrosine phosphatase found on all leukocytes. It is expressed in different isoforms on different cell types, including the different subtypes of T cells. Also called leukocyte common antigen, it is a generic marker for hematopoietically derived cells, with the exception of erythrocytes.

[suppressor of cytokine signaling \(SOCS\)](#)

Regulatory protein that interacts with Janus kinases (JAKs) to inhibit signaling by activated receptors.

[protein inhibitors of activated STAT \(PIAS\)](#)

A small family of proteins that inhibit STAT family transcription factors.

[chemotaxis](#)

Cellular movement occurring in response to chemical signals in the environment.

[CC chemokines](#)

One of the two main classes of chemokines, distinguished by two adjacent cysteines (C) near the amino terminus. They have names CCL1, CCL2, etc. See eAppendix IV for a list of individual chemokines.

[CXC chemokines](#)

One of the two main classes of chemokines, distinguished by a Cys-X-Cys (CXC) motif near the amino terminus. They have names CXCL1, CXCL2, etc. See eAppendix IV for a list of individual chemokines.

[leukocyte functional antigens \(LFAs\)](#)

Cell-adhesion molecules on leukocytes that were initially defined using monoclonal antibodies. LFA-1 is a $\beta 2$ integrin; LFA-2 (now usually called CD2) is a member of the immunoglobulin superfamily, as is LFA-3 (now called CD58). LFA-1 is particularly important in T-cell adhesion to endothelial cells and antigen-presenting cells.

[selectins](#)

Family of cell-adhesion molecules on leukocytes and endothelial cells that bind to sugar moieties on specific glycoproteins with mucin-like features.

[intercellular adhesion molecules \(ICAMs\)](#)

ICAM-1, ICAM-2, ICAM-3. Cell-adhesion molecules of the immunoglobulin superfamily that bind to the leukocyte integrin CD11a:CD18 (LFA-1). They are crucial in the binding of lymphocytes and other leukocytes to antigen-presenting cells and endothelial cells.

[integrins](#)

Heterodimeric cell-surface proteins involved in cell-cell and cell-matrix interactions. They are important in adhesive interactions between lymphocytes and antigen-presenting cells and in lymphocyte and leukocyte adherence to blood vessel walls and migration into tissues.

[immunoglobulin-like proteins](#)

Proteins containing one or more immunoglobulin-like domains, which are protein domains structurally similar to those of immunoglobulins.

[LFA-1 \(CD11a:CD18, or \$\alpha_L\beta_2\$ \)](#)

Cell-adhesion molecules on leukocytes that were initially defined using monoclonal antibodies. LFA-1 is a $\beta 2$ integrin; LFA-2 (now usually called CD2) is a member of the immunoglobulin superfamily, as is LFA-3 (now called CD58). LFA-1 is particularly important in T-cell adhesion to endothelial cells and antigen-presenting cells.

[CR3 \(CD11b:CD18\)](#)

Complement receptor 3. A $\beta 2$ integrin that acts both as an adhesion molecule and as a complement receptor. CR3 on phagocytes binds iC3b, a breakdown product of C3b on pathogen surfaces, and stimulates phagocytosis.

CR3 (CD11b:CD18)

Complement receptor 3. A $\beta 2$ integrin that acts both as an adhesion molecule and as a complement receptor. CR3 on phagocytes binds iC3b, a breakdown product of C3b on pathogen surfaces, and stimulates phagocytosis.

talin

An intracellular protein involved in the linkage of activated integrins, such as LFA-1, to the cytoskeleton to allow changes in cellular motility and migration, such as in the diapedesis of neutrophils across the vascular endothelium.

leukocyte adhesion deficiencies (LADs)

A class of immunodeficiency diseases in which the ability of leukocytes to enter sites infected by extracellular pathogens is affected, impairing elimination of infection. There are several different causes, including a deficiency of the common β chain of the leukocyte integrins.

Weibel–Palade bodies

Granules within endothelial cells that contain P-selectin.

P-selectin

Family of cell-adhesion molecules on leukocytes and endothelial cells that bind to sugar moieties on specific glycoproteins with mucin-like features.

E-selectin

Family of cell-adhesion molecules on leukocytes and endothelial cells that bind to sugar moieties on specific glycoproteins with mucin-like features.

leukocyte adhesion deficiency type 2 (LAD-2)

Disease caused by defects in the production of sulfated sialyl-Lewis^X that prevent neutrophils from interacting with P-selectin and E-selectin, eliminating their ability to migrate properly to sites of infection.

CR4 (CD11c:CD18)

A $\beta 2$ integrin that acts both as an adhesion molecule and as a complement receptor. CR4 on phagocytes binds iC3b, a breakdown product of C3b on pathogen surfaces, and stimulates phagocytosis.

CR4 (CD11c:CD18)

A $\beta 2$ integrin that acts both as an adhesion molecule and as a complement receptor. CR4 on phagocytes binds iC3b, a breakdown product of C3b on pathogen surfaces, and stimulates phagocytosis.

BDCA-2 (blood dendritic cell antigen 2)

A C-type lectin expressed selectively as a receptor on the surface of human plasmacytoid dendritic cells.

BST2 (bone marrow stromal antigen 2)

Bone marrow stromal antigen 2, also called tetherin and CD317, is a type II integral membrane protein that functions as a viral restriction element trapping virions in the plasma membrane and inhibiting viral budding and which serves as a marker of plasmacytoid dendritic cells (pDCs), but is also expressed by B cells.

shear-resistant rolling

The capacity of neutrophils to remain attached to the vascular endothelium under high rates of flow—or shear—enabled by specialized plasma membrane extensions called

slings.

[slings](#)

The capacity of neutrophils to remain attached to the vascular endothelium under high rates of flow—or shear—enabled by specialized plasma membrane extensions called slings.

[PECAM](#)

A cell-adhesion molecule found both on lymphocytes and at endothelial cell junctions. CD31–CD31 interactions are thought to enable leukocytes to leave blood vessels and enter tissues.

[CD31](#)

A cell-adhesion molecule found both on lymphocytes and at endothelial cell junctions. CD31–CD31 interactions are thought to enable leukocytes to leave blood vessels and enter tissues.

[diapedesis](#)

The movement of blood cells, particularly leukocytes, from the blood across blood vessel walls into tissues.

[neutropenia](#)

Abnormally low levels of neutrophils in the blood.

[disseminated intravascular coagulation \(DIC\)](#)

Blood clotting occurring simultaneously in small vessels throughout the body in response to disseminated TNF- α , which leads to the massive consumption of clotting proteins, so that the individual's blood cannot clot appropriately. Seen in septic shock.

[endogenous pyrogens](#)

Cytokines that can induce a rise in body temperature.

[exogenous pyrogen](#)

Any substance originating outside the body that can induce fever, such as the bacterial lipopolysaccharide LPS. Cf. **endogenous pyrogens**.

[acute-phase response](#)

A change in the proteins present in the blood that occurs during the early phases of an infection. It includes the production of acute-phase proteins, many of which are produced in the liver.

[acute-phase proteins](#)

Proteins with innate immune function whose production is increased in the presence of an infection (the acute-phase response). They circulate in the blood and participate in early phases of host defense against infection. An example is mannose-binding lectin.

[C-reactive protein](#)

An acute-phase protein that binds to phosphocholine, a constituent of the surface C-polysaccharide of the bacterium *Streptococcus pneumoniae* and of many other bacteria, thus opsonizing them for uptake by phagocytes.

[pentraxin](#)

A family of acute-phase proteins formed of five identical subunits, to which C-reactive protein and serum amyloid protein belong.

[surfactant proteins A and D \(SP-A and SP-D\)](#)

Acute-phase proteins that help protect the epithelial surfaces of the lung against

infection.

leukocytosis

The presence of increased numbers of leukocytes in the blood. It is commonly seen in acute infection.

type II interferon

The antiviral interferon IFN- γ .

interferon-producing cells (IPCs)

A distinct lineage of dendritic cells that secrete large amounts of interferon on activation by pathogens and their products via receptors such as Toll-like receptors. Cf. **conventional dendritic cells**.

natural interferon-producing cells

A distinct lineage of dendritic cells that secrete large amounts of interferon on activation by pathogens and their products via receptors such as Toll-like receptors. Cf. **conventional dendritic cells**.

IRF9

A member of the IRF family of transcription factors that interacts with activated STAT1 and STAT2 to form the complex called ISGF3, which induces transcription of many ISGs.

ISGF3

A member of the IRF family of transcription factors that interacts with activated STAT1 and STAT2 to form the complex called ISGF3, which induces transcription of many ISGs.

interferon-stimulated genes (ISGs)

A category of genes induced by interferons, which include many whose products promote innate defense against pathogens, such as oligoadenylate synthetase, PKR, and the Mx, IFIT, and IFITM proteins.

oligoadenylate synthetase

Enzyme produced in response to stimulation of cells by interferon. It synthesizes unusual nucleotide polymers, which in turn activate a ribonuclease that degrades viral RNA.

PKR

Serine/threonine kinase activated by IFN- α and IFN- β . It phosphorylates the eukaryotic protein synthesis initiation factor eIF-2, inhibiting translation and thus contributing to the inhibition of viral replication.

Mx (myxoma resistant) proteins

Interferon-inducible proteins required for cellular resistance to influenza virus replication.

IFIT (IFN-induced protein with tetratricoid repeats)

A small family of host proteins induced by interferons that regulate protein translation during infection in part by interactions with eIF3.

eukaryotic initiation factor 3 (eIF3)

Multi-subunit complex that acts in formation of the 43S pre-initiation complex. It can bind IFN-induced protein with tetratricoid repeats (IFIT) proteins, which thereby suppress translation of viral proteins.

[interferon-induced transmembrane protein \(IFITM\)](#)

A small family of host transmembrane proteins induced by interferons that function in the cell's vesicular compartment to restrain various steps in viral replication.

[natural killer \(NK\) cell](#)

A type of ILC that is important in innate immunity to viruses and other intracellular pathogens and in antibody-dependent cell-mediated cytotoxicity (ADCC). NK cells express activating and inhibitory receptors but not the antigen-specific receptors of T or B cells.

[innate lymphoid cells \(ILCs\)](#)

These are a class of innate immune cells having overlapping characteristics with T cells but lacking an antigen receptor. They arise in several groups, ILC1, ILC2, ILC3, and NK cells, which exhibit properties roughly similar to T_{H1}, T_{H2}, T_{H17}, and CD8 T cells.

[common lymphoid progenitor \(CLP\)](#)

Stem cell that can give rise to all the types of lymphocytes but is biased toward producing B cells.

[group 1 ILCs \(ILC1s\)](#)

The subtype of innate lymphoid cells (ILCs) characterized by IFN- γ production.

[Nfil3](#)

A transcription factor important during the development of several types of immune cells including certain types of NK cells.

[Tbet](#)

A transcription factor active in many immune cell types but most typically associated with ILC1 and T_{H1} function.

[group 2 ILCs \(ILC2s\)](#)

A subset of innate lymphoid cells characterized by production of IL-4, IL-5, and IL-13.

[thymic stromal lymphopoietin \(TSLP\)](#)

Thymic stroma-derived lymphopoietin. A cytokine thought to be involved in promoting B-cell development in the embryonic liver. It is also produced by mucosal epithelial cells in response to helminthic infections and promotes type 2 immune responses through its actions on macrophages, ILC2s, and T_{H2} cells.

[group 3 ILCs \(ILC3s\)](#)

A subset of innate lymphoid cells characterized by production of IL-17 and IL-22.

[TRAIL \(tumor necrosis factor-related apoptosis-inducing ligand\)](#)

A member of the TNF cytokine family expressed on the cell surface of some cells, such as NK cells, that induces cell death in target cells by ligation of the 'death' receptors DR4 and DR5.

[caspase 8](#)

An initiator caspase activated by various receptors that activates the process of apoptosis.

[granulocyte-macrophage colony-stimulating factor \(GM-CSF\)](#)

A cytokine involved in the growth and differentiation of cells of the myeloid lineage, including dendritic cells, monocytes and tissue macrophages, and granulocytes.

[activating receptors](#)

On NK cells, receptors whose stimulation results in activation of the cell's cytotoxic

activity.

[inhibitory receptors](#)

On NK cells, receptors whose stimulation results in suppression of the cell's cytotoxic activity.

[dysregulated self](#)

Refers to changes that take place in infected or malignant cells that alter expression of various surface receptors that can be detected by the innate immune system.

[stress-induced self](#)

Refers to changes that take place in infected or malignant cells that alter expression of various surface receptors that can be detected by the innate immune system.

[missing self](#)

Refers to the loss of cell-surface molecules that engage with inhibitory receptors on NK cells, resulting in NK-cell activation.

[MHC class I molecules](#)

Polymorphic cell-surface proteins encoded in the MHC locus and expressed on most cells. They present antigenic peptides generated in the cytosol to CD8 T cells and also bind the co-receptor CD8.

[MHC class II molecules](#)

Polymorphic cell-surface proteins encoded in the MHC locus and expressed primarily on specialized antigen-presenting cells. They present antigenic peptides derived from internalized extracellular pathogens to CD4 T cells and also bind the co-receptor CD4.

[killer cell immunoglobulin-like receptors \(KIRs\)](#)

Large family of receptors present on NK cells, through which the cells' cytotoxic activity is controlled. The family contains both activating and inhibitory receptors.

[leukocyte receptor complex \(LRC\)](#)

A large cluster of immunoglobulin-like receptor genes that includes the killer cell immunoglobulin-like receptor (KIR) genes.

[killer cell lectin-like receptors \(KLRs\)](#)

Large family of receptors present on NK cells, through which the cells' cytotoxic activity is controlled. The family contains both activating and inhibitory receptors.

[NK receptor complex \(NKC\)](#)

A cluster of genes that encode a family of receptors on NK cells.

[Ly49 receptors](#)

A family of C-type lectins expressed by mouse, but not human, NK cells. These can be either activating or inhibitory in function.

[immunoreceptor tyrosine-based inhibition motif \(ITIM\)](#)

Sequence motifs in the signaling chains of inhibitory receptors that are sites of tyrosine phosphorylation, leading to inhibitory signaling, such as through recruitment of phosphatases that remove phosphate groups added by tyrosine kinases.

[Src family protein tyrosine kinases](#)

Receptor-associated protein tyrosine kinases characterized by Src homology protein domains (SH1, SH2, and SH3). The SH1 domain contains the kinase, the SH2 domain can bind phosphotyrosine residues, and the SH3 domain can interact with proline-rich regions in other proteins. In T cells and B cells they are involved in relaying signals

from the antigen receptor.

[motheaten](#)

A mutation in the SHP-1 protein phosphatase that impairs the function of some inhibitory receptors, such as Ly49, resulting in overactivation of various cells, including NK cells. Mice with this mutation have a 'motheaten' appearance because of chronic inflammation.

[CD94](#)

A C-type lectin that is a subunit of the killer cell lectin-like receptors (KLRs) of NK cells.

[NKG2](#)

Family of C-type lectins that supply one of the subunits of the killer cell lectin-like receptors (KLRs) on NK cells.

[signal peptide](#)

The short N-terminal peptide sequence responsible for directing newly synthesized proteins into the secretory pathway.

[natural cytotoxicity receptors \(NCRs\)](#)

Activating receptors on NK cells that recognize infected cells and stimulate cell killing by the NK cell.

[Ly49H](#)

A family of C-type lectins expressed by mouse, but not human, NK cells. These can be either activating or inhibitory in function.

[NKG2D](#)

Activating C-type lectin receptor on NK cells, cytotoxic T cells, and $\gamma\delta$ T cells that recognizes the stress-response proteins MIC-A and MIC-B.

[RAET1](#)

A family of 10 MHC class Ib proteins that are ligands for NKG2D. Also called **RAET1** proteins.

[cytomegalovirus UL16 protein](#)

A nonessential glycoprotein of cytomegalovirus that is recognized by innate receptors expressed by NK cells.

[RAE1 \(retinoic acid early inducible 1\) protein family](#)

Several murine MHC class Ib proteins; these are orthologs of human RAET1 family proteins, including H60 and MULT1, and are ligands for murine NKG2D.

[phosphatidylinositol 3-kinase \(PI 3-kinase\)](#)

Enzyme involved in intracellular signaling pathways. It phosphorylates the membrane lipid phosphatidylinositol 4,5-bisphosphate (PIP₂) to form phosphatidylinositol 3,4,5-trisphosphate (PIP₃), which can recruit signaling proteins containing pleckstrin homology (PH) domains to the membrane.

[SLAM \(signaling lymphocyte activation molecule\)](#)

A family of related cell-surface receptors that mediate adhesion between lymphocytes and which includes SLAM, 2B4, CD84, Ly108, Ly9, and CRACC.

[2B4](#)

A receptor belonging to the signaling lymphocyte activation molecule (SLAM) family expressed by NK cells, which binds to CD48, another SLAM receptor. These signal

through SAP and Fyn to promote survival and proliferation.

CD48

A receptor belonging to the signaling lymphocyte activation molecule (SLAM) family expressed by NK cells, which binds to CD48, another SLAM receptor. These signal through SAP and Fyn to promote survival and proliferation.

SAP (SLAM-associated protein)

An intracellular adaptor protein involved in signaling by SLAM (signaling lymphocyte activation molecule). Inactivating mutations in this gene cause X-linked lymphoproliferative (XLP) syndrome.

common β chain (β_c)

A transmembrane polypeptide (CD131) that is a common subunit for receptors of the cytokines IL-3, IL-5, and GM-CSF.

common γ chain (γ_c)

A transmembrane polypeptide chain (CD132) that is common to a subgroup of cytokine receptors.

Janus kinase (JAK) family, Janus kinases (JAKs)

Enzymes of the JAK–STAT intracellular signaling pathways that link many cytokine receptors with gene transcription in the nucleus. The kinases phosphorylate STAT proteins in the cytosol, which then move to the nucleus and activate a variety of genes.

SH2 (Src homology 2) domain

Receptor-associated protein tyrosine kinases characterized by Src homology protein domains (SH1, SH2, and SH3). The SH1 domain contains the kinase, the SH2 domain can bind phosphotyrosine residues, and the SH3 domain can interact with proline-rich regions in other proteins. In T cells and B cells they are involved in relaying signals from the antigen receptor.

LFA-1 (CD11a:CD18, or $\alpha_L\beta_2$)

Cell-adhesion molecules on leukocytes that were initially defined using monoclonal antibodies. LFA-1 is a β_2 integrin; LFA-2 (now usually called CD2) is a member of the immunoglobulin superfamily, as is LFA-3 (now called CD58). LFA-1 is particularly important in T-cell adhesion to endothelial cells and antigen-presenting cells.

sulfated sialyl-Lewis^x

A sulfated tetrasaccharide carbohydrate structure attached to many cell surface proteins, it binds the P-selectin and E-selectin molecules on the surface of cells, such as neutrophils, that mediate interactions with the endothelium.

TACE (TNF- α -converting enzyme)

A protease responsible for cleavage of the membrane-associated form of the cytokine TNF- α into its soluble form, which can enter the systemic circulation.

IFN- γ

A cytokine of the interferon structural family produced by effector CD4 T_H1 cells, CD8 T cells, and NK cells. Its primary function is the activation of macrophages, and it acts through a different receptor from that of the type I interferons.

IFN- λ

Also called type III interferons, this family includes IL-28A, IL-28B, and IL-29, which

bind a common receptor expressed by a limited set of epithelial tissues.

[IFN- \$\lambda\$ receptor](#)

Heterodimeric receptor composed of a unique IL-28R α subunit and the β subunit of the IL-10 receptor that recognizes IL-28A, IL-28B, and IL-29.

[interferon- \$\alpha\$ receptor \(IFNAR\)](#)

This receptor recognizes IFN- α and IFN- β to activate STAT1 and STAT2 and induce expression of many ISGs.

[STAT1](#)

A member of the signal transducer and activator of transcription (STAT) family of transcription factors activated by type I and type II interferon signaling.

[STAT2](#)

A member of the signal transducer and activator of transcription (STAT) family of transcription factors activated by type II interferon signaling and which heterodimerizes with STAT1 and associates with IRF9 to form a complex called ISGF3.

[IFIT1, IFIT2](#)

IFN-induced protein with tetratricoid repeats (IFIT) 1 and 2 are proteins encoded by interferon-stimulated genes (ISGs) that regulate the translation of normally capped mRNAs and lead to reduced cellular proliferation. IFIT1 can also suppress translation of some viral RNAs that lack 2'-O-methylation (cap-0), such as Sindbis virus. In this way, IFIT1 and IFIT2 serve as restriction factors in certain viral infections.

[2'-O-methyltransferase \(MTase\)](#)

An enzyme that transfers a methyl group to the 2' hydroxyl of the first and second ribose groups in mRNA. Viruses that acquire MTase can produce cap-1 and cap-2 on their transcripts and thereby evade restriction by IFIT1.

[IFITM1, IFITM3](#)

Interferon-induced transmembrane protein (IFITM) 1 and 3 are proteins encoded by interferon-stimulated genes (ISGs) that interfere with membrane fusion between viral membranes and membranes of the lysosome (IFITM1) or endosome (IFITM3), and so serve as restriction factors in various viral infections.

[IFITM1, IFITM3](#)

Interferon-induced transmembrane protein (IFITM) 1 and 3 are proteins encoded by interferon-stimulated genes (ISGs) that interfere with membrane fusion between viral membranes and membranes of the lysosome (IFITM1) or endosome (IFITM3), and so serve as restriction factors in various viral infections.

[DR4, DR5](#)

Members of the TNFR superfamily expressed by many cell types and which can be activated by TRAIL to induce apoptosis.

[antibody-dependent cell-mediated cytotoxicity \(ADCC\)](#)

The killing of antibody-coated target cells by cells with Fc receptors that recognize the constant region of the bound antibody. Most ADCC is mediated by NK cells that have the Fc receptor Fc γ RIII on their surface.

[SHP-1, SHP-2](#)

Intracellular tyrosine phosphatases that are recruited to phosphorylated ITIM domains

of various inhibitory receptors and which can remove phosphate residues from other signaling molecules, with effect of reducing signaling.

[SHP-1, SHP-2](#)

Intracellular tyrosine phosphatases that are recruited to phosphorylated ITIM domains of various inhibitory receptors and which can remove phosphate residues from other signaling molecules, with effect of reducing signaling.

[DAP10, DAP12](#)

Signaling chains containing ITAMs that are associated with the tails of some activating receptors on NK cells.

[DAP10, DAP12](#)

Signaling chains containing ITAMs that are associated with the tails of some activating receptors on NK cells.

[MIC-A, MIC-B](#)

MHC class Ib proteins that are induced by stress, infection, or transformation in many cell types and are recognized by NKG2D.

[UL16-binding proteins \(ULBPs\)](#)

A family of 10 MHC class Ib proteins that are ligands for NKG2D. Also called **RAET1** proteins.

Discussion Questions

1. The innate immune system uses multiple strategies to identify pathogens.
 1. Give examples of each strategy.
 2. What are the advantages or disadvantages of these different strategies?
2. What is the significance of nonclonal expression of pattern-recognition receptors?
3. There are a number of cases in which multiple innate immune receptors recognize the same ligand. List some examples. Why might multiple receptors evolve to recognize the same ligand?
4. Signaling cascades involving multiple adaptors and numerous intermediate steps seem unnecessarily complex. Why might they have evolved in this way?
5. Many signaling cascades used in innate immunity involve the interactions of proteins via specific protein domains. Name the kinds of interaction domains used in TLR signaling. How are the domains of the intermediates used to propagate signals? Also answer this question for RLR and NLR proteins.
6. Describe the ways in which interferons interfere with viral replication. Why doesn't the host produce interferons all the time, as a way of preventing viral infection?
7. How is it that NK cells can recognize both infected cells and cancerous cells?
8. Throughout the chapter we discuss how inappropriate activation of the innate immune system can lead to disease. Select two examples and describe how the inappropriate activation occurs.

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&

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PART II

THE RECOGNITION OF ANTIGEN

[4 Antigen Recognition by B-cell and T-cell Receptors](#)

[5 The Generation of Lymphocyte Antigen Receptors](#)

[6 Antigen Presentation to T Lymphocytes](#)

4

Antigen Recognition by B-cell and T-cell Receptors

IN THIS CHAPTER

- [The structure of a typical antibody molecule.](#)
 - [Structural variation in immunoglobulin constant regions.](#)
 - [The interaction of the antibody molecule with specific antigen.](#)
 - [Antigen recognition by T cells.](#)
-

Innate immune responses are the body's initial defense against infection, but these work only to control pathogens that have certain molecular patterns or that induce interferons and other nonspecific defenses. To effectively fight the wide range of pathogens an individual will encounter, the lymphocytes of the adaptive immune system have evolved to recognize a great variety of different [antigens](#) from bacteria, viruses, and other disease-causing organisms. An antigen is any molecule or part of a molecule that is specifically recognized by the highly specialized recognition proteins of lymphocytes. On B cells these proteins are the [immunoglobulins \(Igs\)](#), which these cells produce in a vast range of antigen specificities, each B cell producing immunoglobulins of a single specificity (see [Section 1-12](#)). A membrane-bound form of immunoglobulin on the B-cell surface serves as the cell's receptor for antigen and is known as the [B-cell receptor \(BCR\)](#). A secreted form of immunoglobulin of the same antigen specificity is the [antibody](#) produced by terminally differentiated B cells—plasmablasts and plasma cells. The secretion of antibodies, which bind pathogens or their toxic products in the extracellular spaces of the body (see Fig. 1.28), is the main effector function of B cells in adaptive immunity.

Antibodies were the first proteins involved in specific immune recognition to be characterized and are understood in great detail. The antibody molecule has two separate functions: one is to bind specifically to the pathogen or its products that have elicited the immune response; the other is to recruit other cells and molecules to destroy the pathogen once antibody has bound. For example, binding by antibodies can neutralize viruses and mark pathogens for destruction by phagocytes and complement, as described in [Chapters 2 and 3](#). Recognition and effector functions are structurally separated in the antibody molecule, one part of which specifically binds to the antigen whereas the other engages the elimination mechanisms. The antigen-binding region varies extensively among antibody molecules and is known as the [variable region](#), or [V region](#). The variability of antibody molecules allows each antibody to bind a different specific antigen, and the total repertoire of antibodies made by a single individual is large enough to ensure that virtually any structure can be recognized. The region of the antibody molecule that engages the effector functions of the immune system does not vary in the same way and is known as the [constant region](#), or [C region](#). It comes in five main forms, called [isotypes](#), each of which is specialized for activating different effector mechanisms. The membrane-bound B-cell receptor does not have these effector functions, because the C region remains inserted in the membrane of the B cell. The function of the B-cell receptor is to recognize and bind antigen via the V regions exposed on the surface of the cell, thus transmitting a signal that activates the B cell, leading to clonal expansion and antibody production. To this end, the B-cell receptor is associated with a set of intracellular signaling proteins, which will be described in [Chapter 7](#). Antibodies have become an important class of drug due to their highly specific activities, and we return to discuss their therapeutic uses in [Chapter 16](#).

The antigen-recognition molecules of T cells are made solely as membrane-bound proteins, which are associated with an intracellular signaling complex and function only to signal T cells for activation. These [T-cell receptors \(TCRs\)](#) are related to immunoglobulins both in their protein structure—having both V and C regions—and in the genetic mechanism that produces their great variability, which is discussed in [Chapter 5](#). The T-cell receptor differs from the B-cell receptor in an important way, however: it does not recognize and bind antigen by itself but instead recognizes short peptide fragments of protein antigens that are presented to it by proteins known as [MHC molecules](#) on the surface of host cells.

The MHC molecules are transmembrane glycoproteins encoded in the large cluster of genes known as the [major histocompatibility complex \(MHC\)](#). The most striking structural feature of MHC molecules is a cleft in the extracellular face of the molecule in which peptides can be bound. MHC molecules are highly [polymorphic](#)—each type of MHC molecule occurs in many different versions—within the population. These are encoded by slightly different versions of individual genes called [alleles](#). Most people are therefore heterozygous for the MHC molecules; that is, they express two different alleles for each type of MHC molecule, thus increasing the range of pathogen-derived peptides and self peptides that can be bound. T-cell receptors recognize features of both the peptide antigen and the MHC molecule to which it is bound. This introduces an extra dimension to antigen recognition by T cells, known as [MHC restriction](#) because any given T-cell receptor is specific for a particular

peptide bound to a particular MHC molecule.

In this chapter, we focus on the structure and antigen-binding properties of immunoglobulins and T-cell receptors. Although B cells and T cells recognize foreign molecules in separate distinct fashions, the receptor molecules they use for this task are very similar in structure. We will see how this basic structure can accommodate great variability in antigen specificity, and how it enables immunoglobulins and T-cell receptors to perform their functions as the antigen-recognition molecules of the adaptive immune response. With this foundation, we will return to discuss the impact of MHC polymorphism on T-cell antigen recognition and T-cell development in [Chapters 6](#) and [8](#), respectively.

Glossary

[antigen](#)

Any molecule that can bind specifically to an antibody or generate peptide fragments that are recognized by a T-cell receptor.

[antibody](#)

A protein that binds specifically to a particular substance—called its antigen. Each antibody molecule has a unique structure that enables it to bind specifically to its corresponding antigen, but all antibodies have the same overall structure and are known collectively as immunoglobulins. Antibodies are produced by differentiated B cells (plasma cells) in response to infection or immunization and bind to and neutralize pathogens or prepare them for uptake and destruction by phagocytes.

[variable region \(V region\)](#)

The region of an immunoglobulin or T-cell receptor that is formed of the amino-terminal domains of its component polypeptide chains. These are the most variable parts of the molecule and contain the antigen-binding sites. Cf. **constant region**.

[variable region \(V region\)](#)

The region of an immunoglobulin or T-cell receptor that is formed of the amino-terminal domains of its component polypeptide chains. These are the most variable parts of the molecule and contain the antigen-binding sites. Cf. **constant region**.

[constant region \(C region\)](#)

That part of an immunoglobulin or a T-cell receptor that is relatively constant in amino acid sequence between different molecules. Also known as the Fc region in antibodies. The constant region of an antibody determines its particular effector function. Cf. **variable region**.

[constant region \(C region\)](#)

That part of an immunoglobulin or a T-cell receptor that is relatively constant in amino acid sequence between different molecules. Also known as the Fc region in antibodies. The constant region of an antibody determines its particular effector function. Cf. **variable region**.

[isotype](#)

The designation of an immunoglobulin chain with respect to the type of constant region

it has. Light chains can be of either κ or λ isotype. Heavy chains can be of μ , δ , γ , α , or ϵ isotype. The different heavy-chain isotypes have different effector functions and determine the class and functional properties of antibodies (IgM, IgD, IgG, IgA, and IgE, respectively).

MHC molecules

Highly polymorphic cell-surface proteins encoded by MHC class I and MHC class II genes involved in presentation of peptide antigens to T cells. They are also known as histocompatibility antigens.

major histocompatibility complex (MHC)

A cluster of genes on human chromosome 6 that encodes a set of membrane glycoproteins called the MHC molecules. The MHC also encodes proteins involved in antigen processing and other aspects of host defense. The genes for the MHC molecules are the most polymorphic in the human genome, having large numbers of alleles at the various loci.

polymorphic

Existing in a variety of different forms; applied to a gene, occurring in a variety of different alleles.

allele

A variant form of a gene; many genes occur in several (or more) different forms within the general population. *See also* **heterozygous, homozygous, polymorphism.**

MHC restriction

The fact that a peptide antigen can only be recognized by a given T cell if it is bound to a particular self MHC molecule. MHC restriction is a consequence of events that occur during T-cell development.

immunoglobulin (Ig)

The protein family to which antibodies and B-cell receptors belong.

B-cell antigen receptor, B-cell receptor (BCR)

The cell-surface receptor on B cells for specific antigen. It is composed of a transmembrane immunoglobulin molecule (which recognizes antigen) associated with the invariant $Ig\alpha$ and $Ig\beta$ chains (which have a signaling function). On activation by antigen, B cells differentiate into plasma cells producing antibody molecules of the same antigen specificity as this receptor.

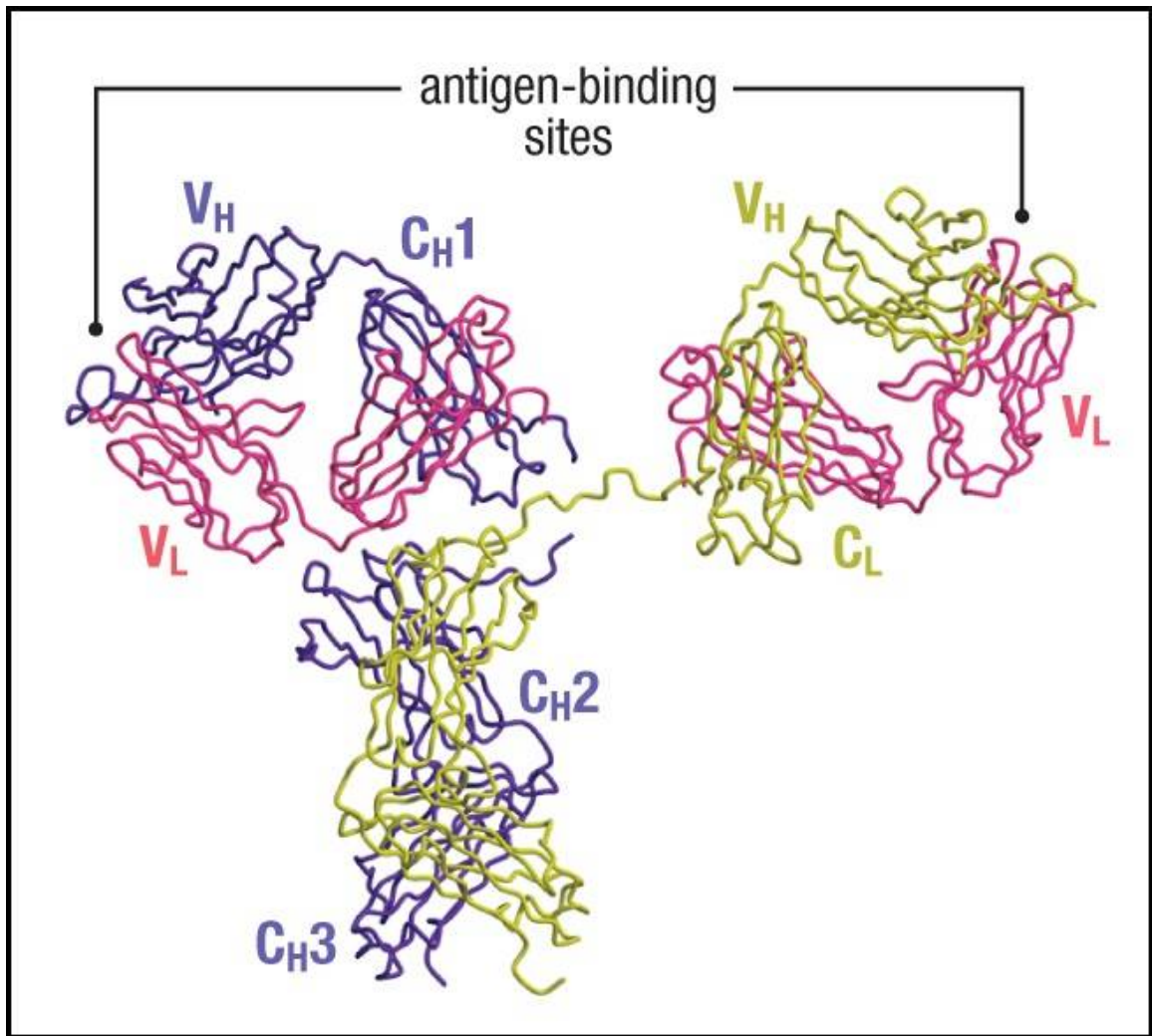
T-cell receptor (TCR)

The cell-surface receptor for antigen on T lymphocytes. It consists of a disulfide-linked heterodimer of the highly variable α and β chains in a complex with the invariant CD3 and ζ proteins, which have a signaling function. T cells carrying this type of receptor are often called $\alpha\beta$ T cells. An alternative receptor made up of variable γ and δ chains is expressed with CD3 and ζ on a subset of T cells.

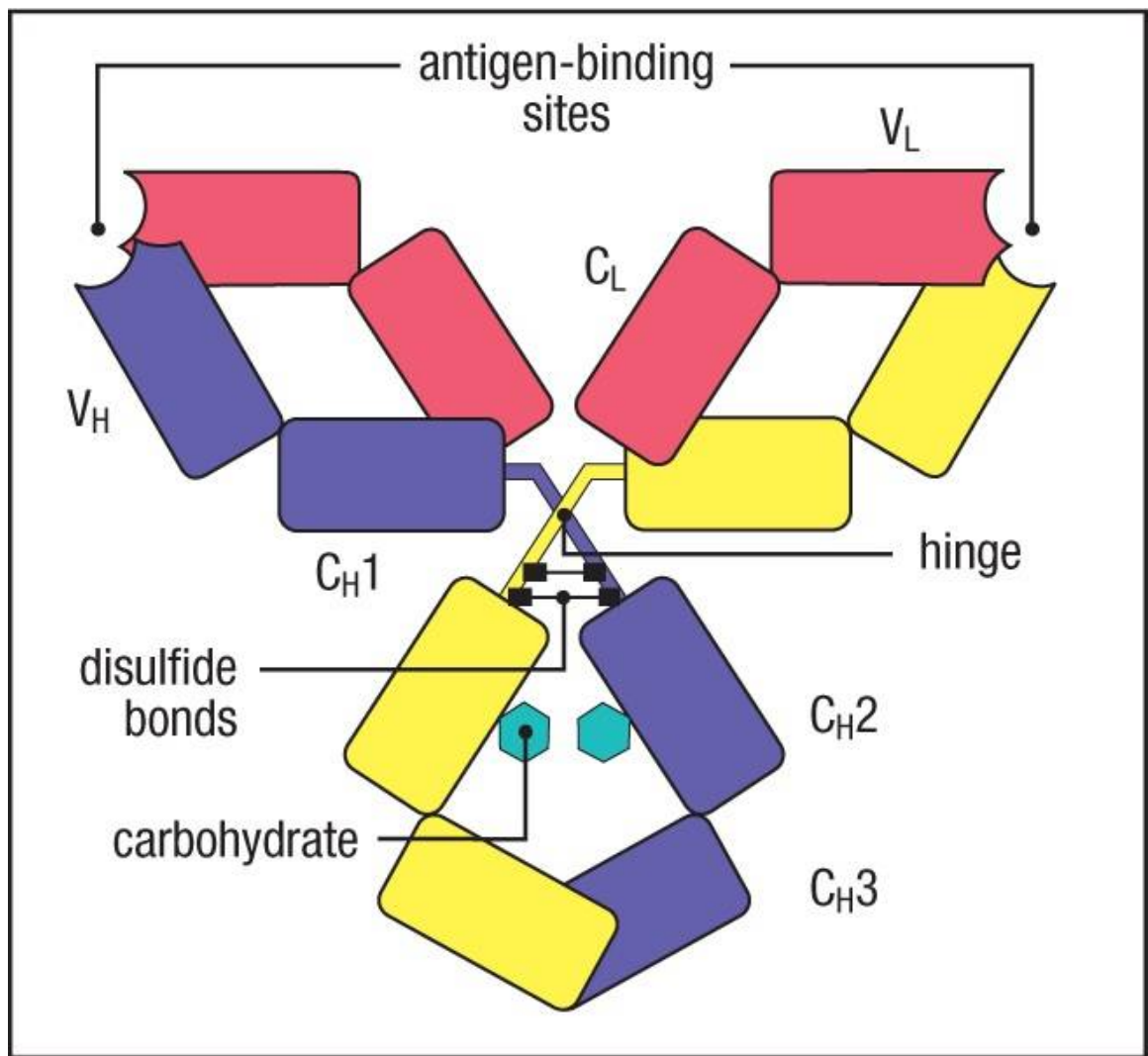
The structure of a typical antibody molecule.

Antibodies are the secreted form of the B-cell receptor. Because they are soluble and secreted into the blood in large quantities, antibodies are easily obtained and easily studied. For this reason, most of what we know about the B-cell receptor comes from the study of antibodies.

Antibody molecules are roughly Y-shaped, as represented in **Fig. 4.1** using three different schematic styles. This part of the chapter will explain how this structure is formed and allows the antibody molecule to perform its dual tasks of binding to a wide variety of antigens while also binding to effector molecules and to cells that destroy the antigen. Each of these tasks is performed by different parts of the molecule. The ends of the two arms of the Y—the V regions—are involved in antigen binding, and they vary in their detailed structure between different antibody molecules. The stem of the Y—the C region—is far less variable and is the part that interacts with effector molecules and cells. There are five different [classes](#) of immunoglobulins, distinguished in their being constructed from C regions that have different structures and properties. These are known as [immunoglobulin M \(IgM\)](#), [immunoglobulin D \(IgD\)](#), [immunoglobulin G \(IgG\)](#), [immunoglobulin A \(IgA\)](#), and [immunoglobulin E \(IgE\)](#).



a



b

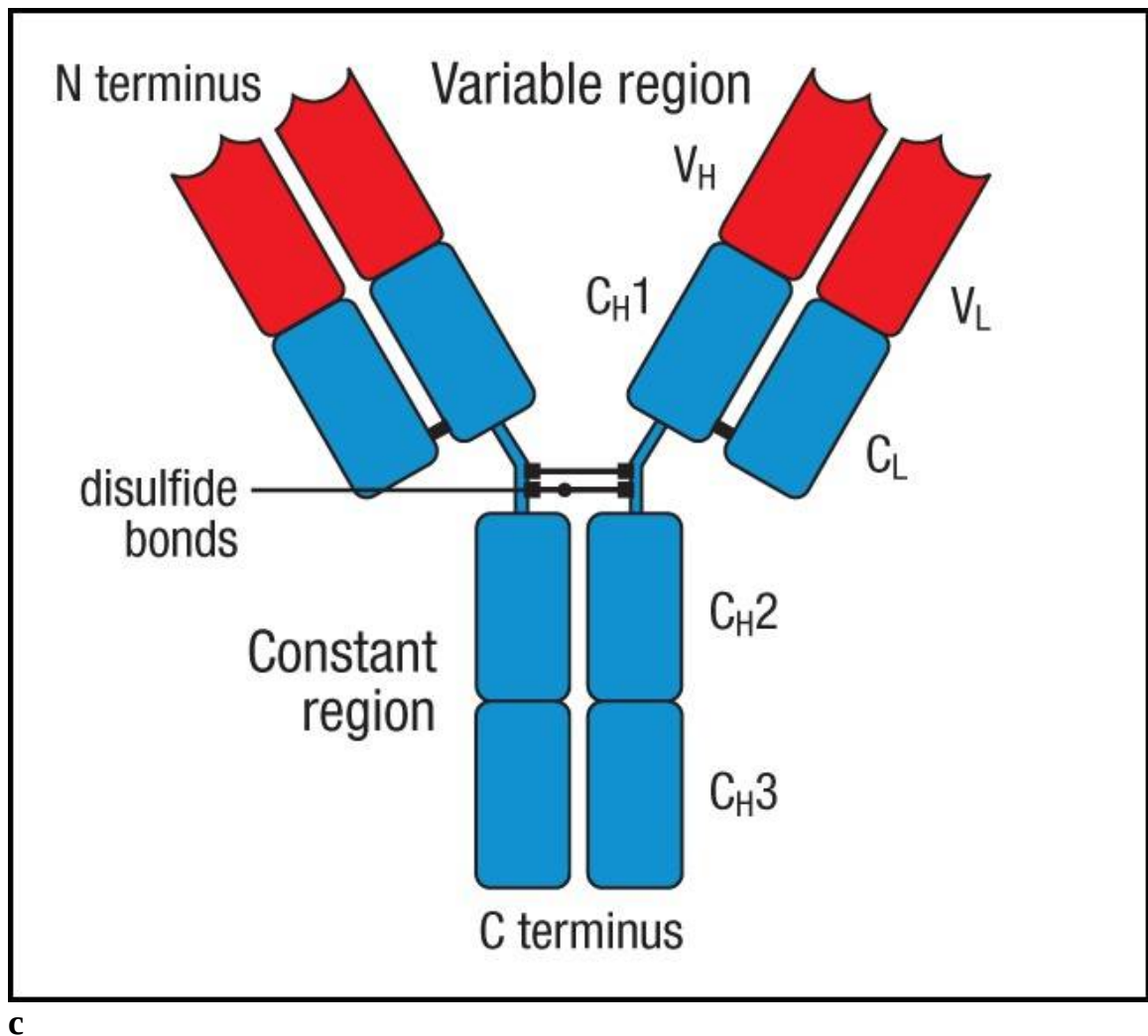


Fig. 4.1 Structure of an antibody molecule. In panel a, the X-ray crystallographic structure of an IgG antibody is illustrated as a ribbon diagram of the backbones of the polypeptide chains. The two heavy chains are shown in yellow and purple. The two light chains are both shown in red. Three globular regions form an approximate Y shape. The two antigen-binding sites are at the tips of the arms, which are tethered at their other end to the trunk of the Y by a flexible hinge region. The light-chain variable (V_L) and constant (C_L) regions are indicated. The heavy-chain variable region (V_H) and V_L together form the antigen-binding site of the antibody. In panel b, a schematic representation of the same structure denotes each immunoglobulin domain as a separate rectangle. The hinge that tethers each heavy chain's first constant domain (C_H1) to its second (C_H2) is illustrated by a thin purple or yellow line, respectively. The antigen-binding sites are indicated by concave regions in V_L and V_H. Positions of carbohydrate modifications and disulfide linkages are indicated. In panel c, a more simplified schematic is shown that will be used throughout this book with the variable region in red and the constant region in blue. C terminus, carboxyl terminus; N terminus, amino terminus.

All antibodies are constructed in the same way from paired heavy and light polypeptide chains, and the generic term immunoglobulin is used for all such proteins. More subtle differences confined to the V region account for the specificity of antigen binding. We will use the IgG antibody molecule as an example to describe the general structural features of immunoglobulins.

4-1 IgG antibodies consist of four polypeptide chains.

IgG antibodies are large molecules with a molecular mass of approximately 150 kDa and are composed of two different kinds of polypeptide chains. One, of approximately 50 kDa, is called the [heavy](#) or [H chain](#), and the other, of 25 kDa, is the [light](#) or [L chain](#) (**Fig. 4.2**). Each IgG molecule consists of two heavy chains and two light chains. The two heavy chains are linked to each other by disulfide bonds, and each heavy chain is linked to a light chain by a disulfide bond. In any given immunoglobulin molecule, the two heavy chains and the two light chains are identical, giving an antibody molecule two identical antigen-binding sites. This gives the antibody the ability to bind simultaneously to two identical antigens on a surface, thereby increasing the total strength of the interaction, which is called its [avidity](#). The strength of the interaction between a single antigen-binding site and its antigen is called its [affinity](#).

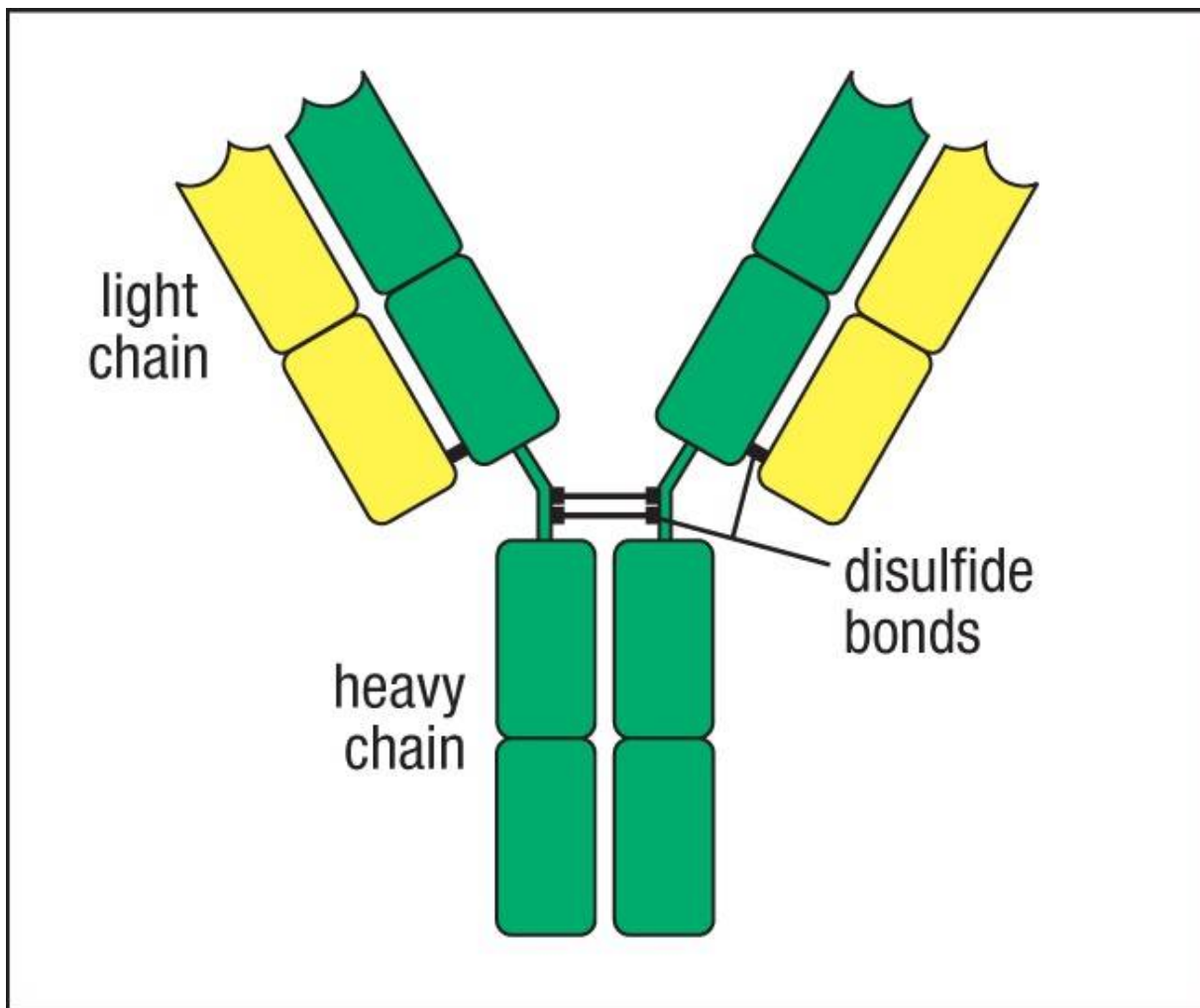


Fig. 4.2 Immunoglobulin molecules are composed of two types of protein chains: heavy chains and light chains. Each immunoglobulin molecule is made up of two hinged heavy chains (green) and two light chains (yellow) joined by disulfide bonds so that each heavy chain is linked to a light chain and the two heavy chains are linked together.

Two types of light chains, [lambda \(\$\lambda\$ \)](#) and [kappa \(\$\kappa\$ \)](#), are found in antibodies. A given immunoglobulin has either K chains or λ chains, never one of each. No functional difference has been found between antibodies having λ or K light chains, and either type of light chain can be found in antibodies of any of the five major classes. The ratio of the two types of light chains varies from species to species. In mice, the average K to λ ratio is 20:1, whereas in humans it is 2:1 and in cattle it is 1:20. The reason for this variation is unknown. Distortions of this ratio can sometimes be used to detect the abnormal proliferation of a B-cell clone, because all progeny of a particular B cell will express an identical light chain. For example, an abnormally high level of λ light chains in a person might indicate the presence of a B-cell tumor that is producing λ chains.

Figure 4.2 depicts a typical IgG molecule; however, not all antibodies in the body are IgG. In addition to IgG, there are four other classes of antibodies that differ in the structures of their

heavy chains. These different classes of antibodies are known as isotypes, and they include IgM, IgD, IgG, IgA, and IgE. We will first consider IgG as a typical antibody molecule and then return to discuss the structural and functional properties of the different heavy-chain isotypes in [Section 4-6](#).

The structure of a B-cell antigen receptor on the surface of a B cell is identical to that of its corresponding antibody except for a small portion of the carboxyl terminus of the heavy-chain C region. In the B-cell receptor, the carboxyl terminus is a hydrophobic amino acid sequence that anchors the molecule in the membrane, whereas in the antibody it is a hydrophilic sequence that allows secretion.

4-2 Immunoglobulin heavy and light chains are composed of constant and variable regions.

The amino acid sequences of many immunoglobulin heavy and light chains have been determined and reveal two important features of antibody molecules. First, each chain consists of a series of similar, although not identical, sequences, each about 110 amino acids long. Each of these repeats corresponds to a discrete, compactly folded region of protein known as an [immunoglobulin domain](#), or [Ig domain](#). The light chain consists of two Ig domains, whereas the heavy chain of the IgG antibody contains four Ig domains (see Fig. 4.2). This suggests that the immunoglobulin chains have evolved by repeated duplication of ancestral gene segments corresponding to a single Ig domain.

The second important feature is that the amino-terminal amino acid sequences of the heavy and light chains vary greatly between different antibodies. The variability is limited to approximately the first 110 amino acids, corresponding to the first Ig domain, whereas the remaining domains are constant between immunoglobulin chains of the same isotype. The amino-terminal [variable Ig domains \(V domains\)](#) of the heavy and light chains (V_H and V_L , respectively) together make up the V region of the antibody and determine its antigen-binding specificity, whereas the [constant Ig domains \(C domains\)](#) of the heavy and light chains (C_H and C_L , respectively) make up the C region (see Fig. 4.1). The multiple heavy-chain C domains are numbered from the amino-terminal end to the carboxyl terminus; for example, C_{H1} , C_{H2} , and so on.

4-3 The domains of an immunoglobulin molecule have similar structures.

Immunoglobulin heavy and light chains are composed of a series of Ig domains that have a similar overall structure. Within this basic structure, there are distinct differences between V and C domains that are illustrated for the light chain in **Fig. 4.3**. Each V or C domain is

constructed from two [β sheets](#). A β sheet is built from several [β strands](#), which are regions of protein where several consecutive polypeptides have their peptide backbone bonds arranged in an extended, or flat, conformation. β strands in proteins are sometimes shown as ‘ribbons with an arrow’ to indicate the direction of the polypeptide backbone (see Fig. 4.3). β strands can pack together in a side-by-side manner, being stabilized laterally by two or three backbone hydrogen bonds between adjacent strands. This arrangement is called a β sheet. The Ig domain has two β sheets that are folded onto each other, like two pieces of bread, into a structure called a β sandwich, and are covalently linked by a disulfide bond between cysteine residues from each β sheet. This distinctive structure is known as the [immunoglobulin fold](#).

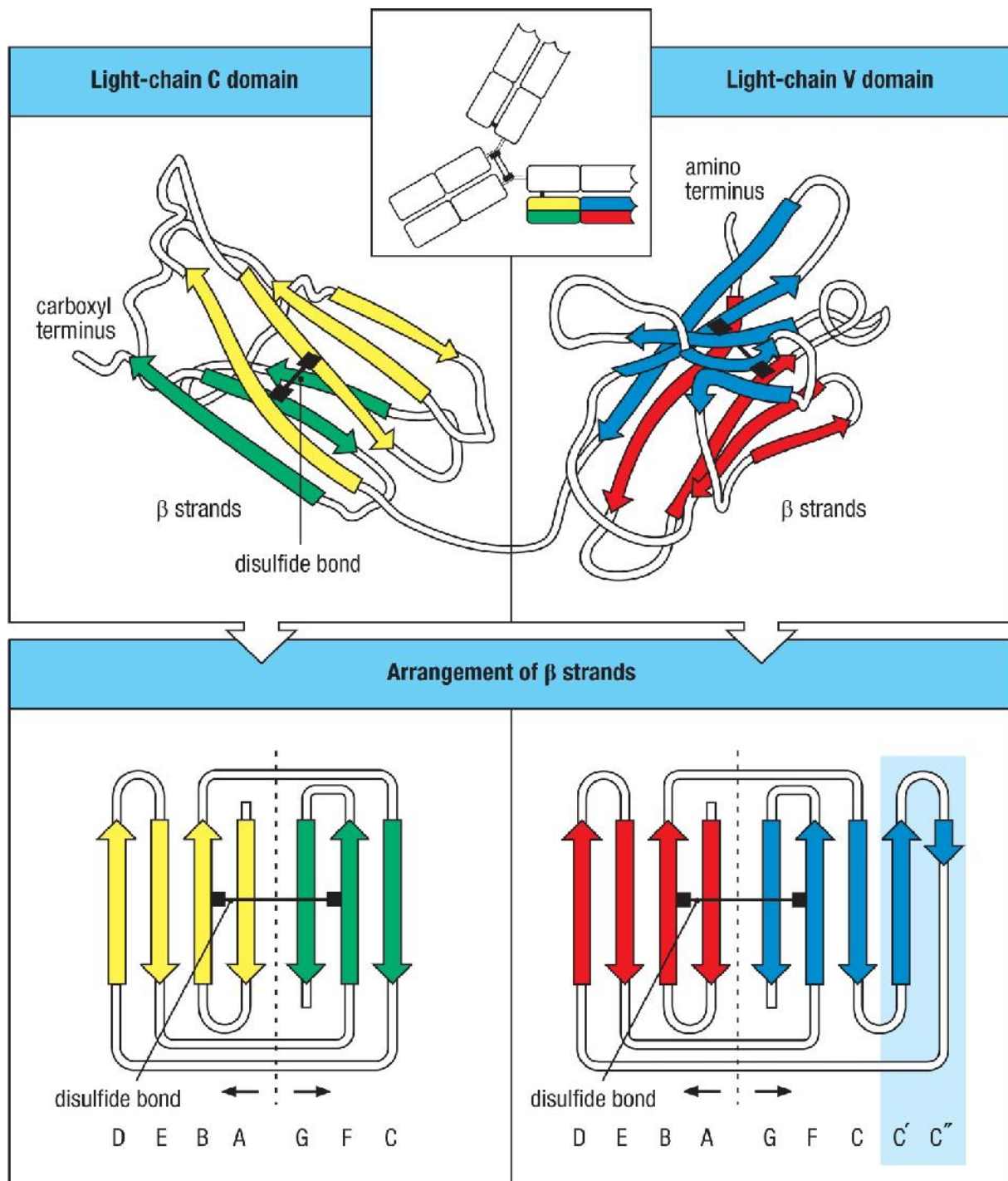


Fig. 4.3 The structure of immunoglobulin constant and variable domains. The upper panels show schematically the folding pattern of the constant (C) and variable (V) domains of an immunoglobulin light chain. Each domain is a barrel-shaped structure in which strands of polypeptide chain (β strands) running in opposite directions (antiparallel) pack together to form two β sheets (shown in yellow and green for the C domain and red and blue for the V domain), which are held together by a disulfide bond. The way in which the polypeptide chain folds to give the final structure can be seen more clearly when the sheets are opened out, as shown in the lower panels. The β strands are lettered sequentially with respect to the order of their occurrence in the amino acid sequence of the domains; the order in each β sheet is characteristic of

immunoglobulin domains. The β strands C' and C'' that are found in the V domains but not in the C domains are indicated by a blue-shaded background. The characteristic four-strand plus three-strand (C-region type domain) or four-strand plus five-strand (V-region type domain) arrangements are typical immunoglobulin superfamily domain building blocks, found in a whole range of other proteins as well as antibodies and T-cell receptors.

The similarities and differences between V and C domains can be seen in the bottom panels of Fig. 4.3. Here, the Ig domains have been opened out to show how their respective polypeptide chains fold to create each of the β sheets and how each polypeptide chain forms flexible loops between adjacent β strands as it turns to change direction. The main difference between the V and C domains is that the V domain is larger and contains extra β strands, called C' and C''. In the V domain, the flexible loops formed between some of the β strands contribute to the antigen-binding site of the immunoglobulin molecule.

Many of the amino acids that are common to the C and V domains are present in the core of the immunoglobulin fold and are essential for its stability. Other proteins with sequences similar to those of immunoglobulins have been found to have domains with a similar structure, called [immunoglobulin-like domains \(Ig-like domains\)](#). These domains are present in many proteins of the immune system, such as the KIRs expressed by NK cells described in [Chapter 3](#). They are also frequently involved in cell–cell recognition and adhesion, and together with the immunoglobulins and the T-cell receptors, these proteins make up the extensive [immunoglobulin superfamily](#).

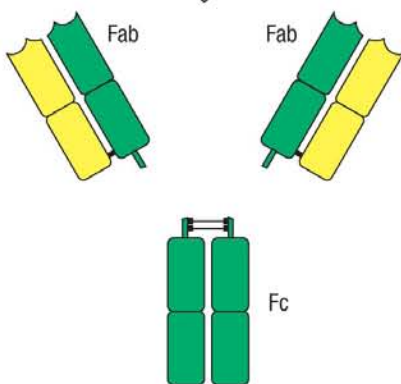
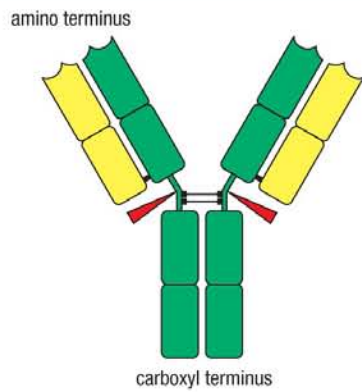
4-4 The antibody molecule can readily be cleaved into functionally distinct fragments.

When fully assembled, an antibody molecule comprises three equal-sized globular portions, with its two arms joined to its trunk by a flexible stretch of polypeptide chain known as the [hinge region](#) (see Fig. 4.1b). Each arm of this Y-shaped structure is formed by the association of a light chain with the amino-terminal half of a heavy chain; the V_H domain is paired with the V_L domain, and the C_{H1} domain is paired with the C_L domain. The two antigen-binding sites are formed by the paired V_H and V_L domains at the ends of the two arms of the Y (see Fig. 4.1b). The trunk of the Y is formed by the pairing of the carboxyl-terminal halves of the two heavy chains. The C_{H3} domains pair with each other but the C_{H2} domains do not interact. Carbohydrate side chains attached to the C_{H2} domains lie between the two heavy chains.

Proteolytic enzymes (proteases) were an important tool in early studies of antibody structure, and it is valuable to review the terminology they generated. Limited digestion with the protease papain cleaves antibody molecules into three fragments (**Fig. 4.4**). [Papain](#) cuts the

antibody molecule on the amino-terminal side of the disulfide bonds that link the two heavy chains, releasing the two arms of the antibody molecule as two identical fragments that contain the antigen-binding activity. These are called the [Fab fragments](#), for fragment antigen binding. The other fragment contains no antigen-binding activity, but because it crystallized readily, it was named the [Fc fragment](#) (fragment crystallizable). It corresponds to the paired C_H2 and C_H3 domains. The Fc fragment is the part of the antibody molecule that does not interact with antigen, but rather interacts with effector molecules and cells, and it differs between heavy-chain isotypes. Another protease, [pepsin](#), cuts on the carboxyl-terminal side of the disulfide bonds (see Fig. 4.4). This produces a fragment, the [F\(ab'\)₂ fragment](#), in which the two antigen-binding arms of the antibody molecule remain linked. Pepsin cuts the remaining part of the heavy chain into several small fragments. The F(ab')₂ fragment has exactly the same antigen-binding characteristics as the original antibody but is unable to interact with any effector molecule, such as C1q or Fc receptors, and can be used experimentally to separate the antigen-binding functions from the antibody's other effector functions.

Proteolytic cleavage by papain



Proteolytic cleavage by pepsin

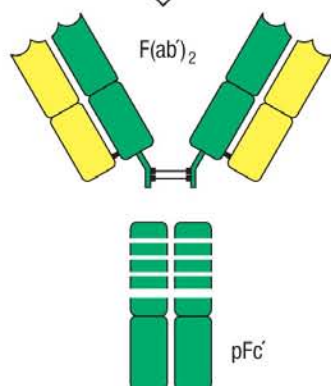
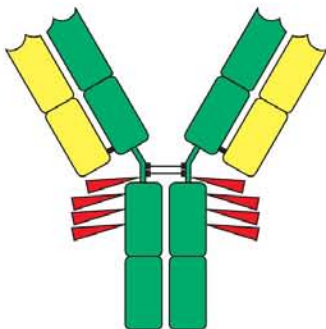


Fig. 4.4 The Y-shaped immunoglobulin molecule can be dissected by partial digestion with proteases. Upper panels: Papain cleaves the immunoglobulin molecule into three pieces, two Fab fragments and one Fc fragment. The Fab fragment contains the V regions and binds antigen. The Fc fragment is crystallizable and contains C regions. Lower panels: Pepsin cleaves immunoglobulin to yield one $F(ab')_2$ fragment and many small pieces of the Fc fragment, the largest of which is called the pFc' fragment. $F(ab')_2$ is written with a prime because it contains a few more amino acids than Fab, including the cysteines that form the disulfide bonds.

Many antibody-related molecules can be constructed using genetic engineering techniques, and many antibodies and antibody-related molecules are being used therapeutically to treat a variety of diseases. We will return to this topic in [Chapter 16](#), where we discuss the various therapeutic uses of antibodies that have been developed over the past two decades.

4-5 The hinge region of the immunoglobulin molecule allows flexibility in binding to multiple antigens.

The hinge region between the Fc and Fab portions of the IgG molecule allows for some degree of independent movement of the two Fab arms. For example, in the antibody molecule shown in Fig. 4.1a, not only are the two hinge regions clearly bent differently, but the angle between the V and C domains in each of the two Fab arms is also different. This range of motion has led to the junction between the V and C domains being referred to as a 'molecular ball-and-socket joint.' This flexibility can be revealed by studies of antibodies bound to small antigens known as [haptens](#). These are molecules of various types that are typically about the size of a tyrosine side chain. Although haptens are specifically recognized by antibody, they can stimulate the production of anti-hapten antibodies only when linked to a protein (see Appendix I, [Section A-1](#)). Two identical hapten molecules joined by a short flexible region can link two or more anti-hapten antibodies, forming dimers, trimers, tetramers, and so on, which can be seen by electron microscopy (**Fig. 4.5**). The shapes formed by these complexes show that antibody molecules are flexible at the hinge region. Some flexibility is also found at the junction between the V and C domains, allowing bending and rotation of the V domain relative to the C domain. Flexibility at both the hinge and the V–C junction enables the two arms of an antibody molecule to bind to sites some distance apart, such as the repeating sites on bacterial cell-wall polysaccharides. Flexibility at the hinge also enables antibodies to interact with the antibody-binding proteins that mediate immune effector mechanisms.

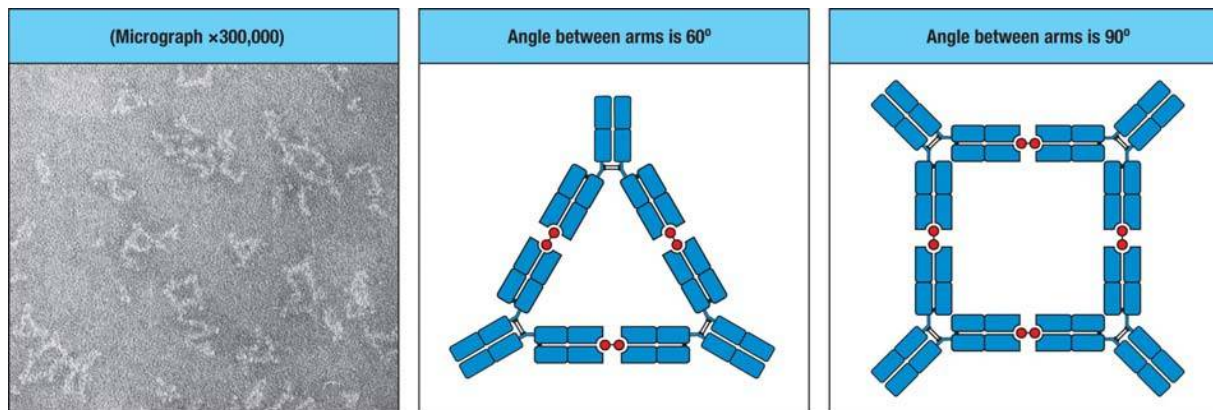


Fig. 4.5 Antibody arms are joined by a flexible hinge. An antigen consisting of two hapten molecules (red balls in diagrams) that can cross-link two antigen-binding sites is used to create antigen:antibody complexes, which can be seen in the electron micrograph. Linear, triangular, and square forms are seen, with short projections or spikes. Limited pepsin digestion removes these spikes (not shown in the figure), which therefore correspond to the Fc portion of the antibody; the $F(ab')_2$ pieces remain cross-linked by antigen. The interpretation of some of the complexes is shown in the diagrams. The angle between the arms of the antibody molecules varies. In the triangular forms, this angle is 60° , whereas it is 90° in the square forms, showing that the connections between the arms are flexible.

Summary.

The IgG antibody molecule is made up of four polypeptide chains, comprising two identical light chains and two identical heavy chains, and can be thought of as forming a flexible Y-shaped structure. Each of the four chains has a variable (V) region at its amino terminus, which contributes to the antigen-binding site, and a constant (C) region. The light chains are bound to the heavy chains by many noncovalent interactions and by disulfide bonds, and the V regions of the heavy and light chains pair in each arm of the Y to generate two identical antigen-binding sites, which lie at the tips of the arms of the Y. The possession of two antigen-binding sites allows antibody molecules to cross-link antigens and to bind them much more stably and with higher avidity. The trunk of the Y, also called the Fc fragment, is composed of the carboxyl-terminal domains of the heavy chains, and it is these domains that determine the antibody's isotype. Joining the arms of the Y to the trunk are the flexible hinge regions. The Fc fragment and hinge regions differ in antibodies of different isotypes. Different isotypes have different properties and therefore differ in their interactions with effector molecules and cell types. However, the overall organization of the domains is similar in all isotypes.

Glossary

classes

The class of an antibody is defined by the type of heavy chain it contains. There are five main antibody classes: IgA, IgD, IgM, IgG, and IgE, containing heavy chains α , δ , μ , γ , and ϵ , respectively. The IgG class has several subclasses. *See also isotypes.*

immunoglobulin M (IgM)

Immunoglobulin class composed of μ heavy chains that is the first to appear on B cells and the first to be secreted.

immunoglobulin D (IgD)

Immunoglobulin class composed of δ heavy chains that appears as surface immunoglobulin on mature B cells.

immunoglobulin G (IgG)

Immunoglobulin class composed of γ heavy chains that is the most abundant class of immunoglobulin in the plasma.

immunoglobulin A (IgA)

Immunoglobulin class composed of α heavy chains that can occur in a monomeric and a polymeric (mainly dimeric) form. Polymeric IgA is the main antibody secreted by mucosal lymphoid tissues.

immunoglobulin E (IgE)

Immunoglobulin class composed of ϵ heavy chains that acts in defense against parasite infections and in allergic reactions.

avidity

The sum total of the strength of binding of two molecules or cells to one another at multiple sites. It is distinct from affinity, which is the strength of binding of one site on a molecule to its ligand.

affinity

The strength of binding of one molecule to another at a single site, such as the binding of a monovalent Fab fragment of antibody to a monovalent antigen. Cf. **avidity**.

variable Ig domains (V domains)

The amino-terminal protein domain of the polypeptide chains of immunoglobulins and T-cell receptor, which is the most variable part of the chain.

constant Ig domains (C domains)

Type of protein domain that makes up the constant regions of each chain of an immunoglobulin molecule.

β sheet

A secondary protein structure composed of β strands stabilized by noncovalent interactions between backbone amide and carbonyl groups. In 'parallel' β sheets, the adjacent strands run in the same direction; in 'antiparallel' β sheets, adjacent strands run in opposite directions. Immunoglobulin domains are made up of two antiparallel β sheets arranged in the form of a β barrel.

β strand

A secondary protein structure in which the polypeptide backbone of several consecutive amino acids is arranged in a flat, or planar, conformation, and often illustrated as an arrow.

immunoglobulin fold

The tertiary structure of an immunoglobulin domain, comprising a sandwich of two β sheets held together by a disulfide bond.

[immunoglobulin superfamily](#)

Large family of proteins with at least one Ig or Ig-like domain, many of which are involved in antigen recognition and cell–cell interaction in the immune system and other biological systems.

[hinge region](#)

The flexible domain that joins the Fab arms to the Fc piece in an immunoglobulin. The flexibility of the hinge region in IgG and IgA molecules allows the Fab arms to adopt a wide range of angles, permitting binding to epitopes spaced variable distances apart.

[papain](#)

A protease that cleaves the IgG antibody molecule on the amino-terminal side of disulfide linkages, producing two Fab fragments and one Fc fragment.

[Fab fragment](#)

Antibody fragment composed of a single antigen-binding arm of an antibody without the Fc region, produced by cleavage of IgG by the enzyme papain. It contains the complete light chain plus the amino-terminal variable region and first constant region of the heavy chain, held together by an interchain disulfide bond.

[pepsin](#)

A protease that cleaves several sites on the carboxy-terminal side of the disulfide linkages, producing the $F(ab')_2$ fragment and several fragments of the Fc region.

[hapten](#)

Any small molecule that can be recognized by a specific antibody but cannot by itself elicit an immune response. A hapten must be chemically linked to a protein molecule to elicit antibody and T-cell responses.

[heavy chain, H chain](#)

One of the two types of protein chains in an immunoglobulin molecule, the other being called the light chain. There are several different classes, or isotypes, of heavy chain ($\alpha, \delta, \epsilon, \gamma$, and μ), each of which confers a distinctive functional activity on the antibody molecule. Each immunoglobulin molecule contains two identical heavy chains.

[heavy chain, H chain](#)

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[light chain, L chain](#)

The smaller of the two types of polypeptide chains that make up an immunoglobulin molecule. It consists of one V and one C domain and is disulfide-bonded to the heavy chain. There are two classes, or isotypes, of light chain, known as κ and λ , which are produced from separate genetic loci.

[light chain, L chain](#)

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produced from separate genetic loci.

[λ chain](#)

One of the two classes or isotypes of immunoglobulin light chains.

[κ chain](#)

One of the two classes or isotypes of immunoglobulin light chains.

[immunoglobulin domain, Ig domain](#)

Protein domain first described in antibody molecules but present in many proteins.

[immunoglobulin domain, Ig domain](#)

Protein domain first described in antibody molecules but present in many proteins.

[immunoglobulin-like domain \(Ig-like domain\)](#)

Protein domain structurally related to the immunoglobulin domain.

[Fc fragment, Fc region](#)

The carboxy-terminal halves of the two heavy chains of an IgG molecule disulfide-bonded to each other by the residual hinge region. It is produced by cleavage of IgG by papain. In the complete antibody this portion is often called the Fc region.

[F\(ab'\)₂ fragment](#)

Antibody fragment composed of two linked antigen-binding arms (Fab fragments) without the Fc regions, produced by cleavage of IgG with the enzyme pepsin.

Structural variation in immunoglobulin constant regions.

The class or isotype of an antibody is defined by the structure of its heavy-chain constant region. The differences between the antibody isotypes determine the effector functions of the antibody once it has bound antigen. There are five main heavy-chain isotypes, IgM, IgD, IgG, IgA, and IgE, some of which have several subtypes. These heavy-chain proteins are denoted by the lowercase Greek letters μ , δ , γ , α , and ϵ , respectively. For example, the constant region of IgM is denoted by C_{μ} . IgG is by far the most abundant immunoglobulin in serum. The distinctive functional properties of the different classes of antibodies are conferred by the carboxyl-terminal part of the heavy chain, where this chain is not associated with the light chain. In spite of these differences, the general structural features of all the isotypes are similar, particularly with respect to antigen binding. In this part of the chapter, we will consider the structural differences of the antibody isotypes and the specialized properties of each isotype. For IgM and IgA, individual antibodies form multimers that are secreted from cells. IgM forms pentamers, and IgA forms dimers. This feature aids in binding to repetitive structures, such as those on the surfaces of bacteria or viruses.

4-6 Different classes of immunoglobulins are distinguished by the structure of their heavy-chain constant regions.

The five main classes of immunoglobulins (IgM, IgD, IgG, IgE, and IgA) can occur as transmembrane antigen receptors or secreted antibodies (**Fig. 4.6**). In humans, IgG is found as four subclasses (IgG1, IgG2, IgG3, and IgG4), named by decreasing order of their abundance in serum, and IgA antibodies are found as two subclasses (IgA1 and IgA2). The heavy chains are encoded by different immunoglobulin C_H genes located in a gene cluster that is 3' of the J_H segments as illustrated in Fig. 4.6. The major physical and functional properties of the different classes of human antibody are listed in **Fig. 4.7**.

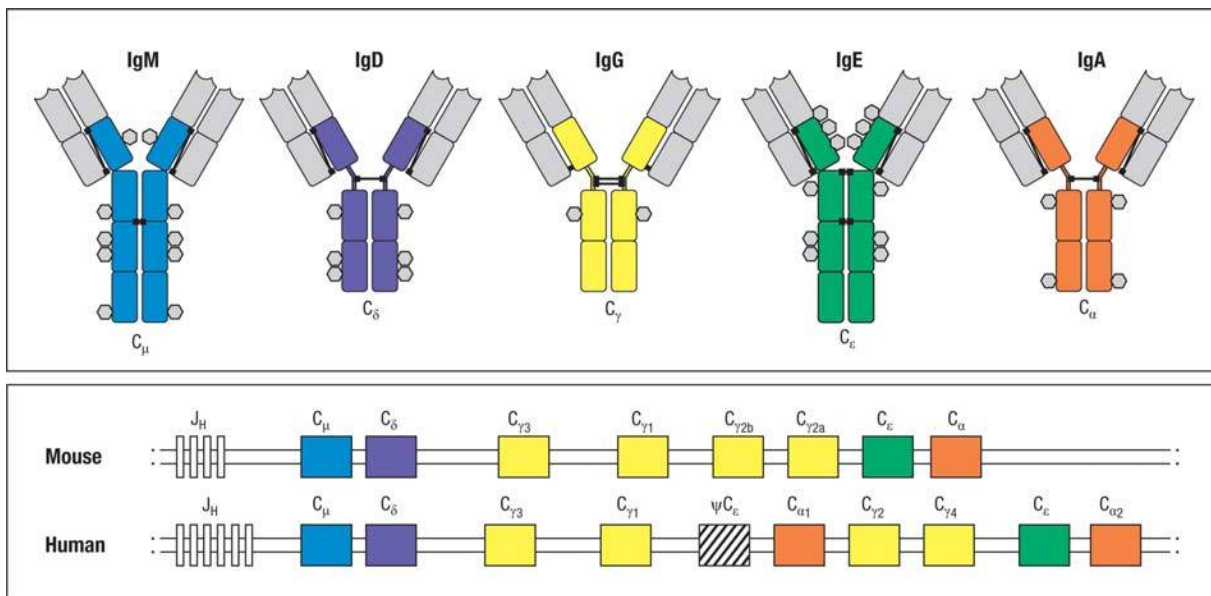


Fig. 4.6 The immunoglobulin isotypes are encoded by a cluster of immunoglobulin heavy-chain C-region genes. The general structure of the main immunoglobulin isotypes (upper panel) is indicated, with each rectangle denoting an immunoglobulin domain. These isotypes are encoded by separate heavy-chain C-region genes arranged in a cluster in both mouse and human (lower panel). The constant region of the heavy chain for each isotype is indicated by the same color as the C-region gene segment that encodes it. IgM and IgE lack a hinge region, but each contains an extra heavy-chain domain. Note the differences in the number and location of the disulfide bonds (heavy black lines) linking the chains. The isotypes also differ in the distribution of *N*-linked carbohydrate groups, shown as hexagons. In humans, the gene cluster shows evidence of evolutionary duplication of a unit consisting of two γ genes, an ϵ gene, and an α gene. One of the ϵ genes is a nonfunctional gene, called a pseudogene (ψC_ϵ); hence only one subtype of IgE is expressed. For simplicity, other pseudogenes are not illustrated, and the exon details within each C gene are not shown. The classes of immunoglobulins found in mice are called IgM, IgD, IgG1, IgG2a, IgG2b, IgG3, IgA, and IgE.

	Immunoglobulin								
	IgG1	IgG2	IgG3	IgG4	IgM	IgA1	IgA2	IgD	IgE
Heavy chain	Y1	Y2	Y3	Y4	μ	α1	α2	δ	ε
Molecular mass (kDa)	146	146	165	146	970	160	160	184	188
Serum level (mean adult mg/ml)	9	3	1	0.5	1.5	3.0	0.5	0.03	5 ×

									10 ⁻⁵
Half-life in serum (days)	21	20	7	21	10	6	6	3	2
Classical pathway of complement activation	++	+	+++	—	++++	—	—	—	—
Alternative pathway of complement activation	—	—	—	—	—	+	—	—	—
Placental transfer	+++	+	++	— +	—	—	—	—	—
Binding to macrophage and phagocyte Fc receptors	+	—	+	— +	—	+	+	—	+
High-affinity binding to mast cells and basophils	—	—	—	—	—	—	—	—	+++
Reactivity with staphylococcal Protein A	+	+	— +	+	—	—	—	—	—

Fig. 4.7 The physical and functional properties of the human immunoglobulin isotypes.

IgM is so called because of its size: although monomeric IgM is only 190 kDa, it normally forms pentamers, known as macroglobulin (hence the M), of very large molecular mass (see Fig. 4.8). IgA dimerizes to give an approximate molecular mass of around 390 kDa in secretions. IgE antibody is associated with immediate-type hypersensitivity. When fixed to tissue mast cells, IgE has a much longer half-life than its half-life in serum shown here. The relative activities of the various isotypes are compared for several functions, ranging from inactive (—) to most active (++++).

The functions of the immunoglobulin classes are discussed in detail in [Chapter 10](#), in the context of the humoral immune response; here, we just touch on them briefly. IgM is the first class of immunoglobulin produced after activation of a B cell, and the IgM antibody is secreted as a pentamer (see [Section 4-8](#) and **Fig. 4.8**). This accounts for the high molecular mass of IgM and the fact that it is normally present in the bloodstream but not in tissues. Being a pentamer also increases the avidity of IgM for antigens before its affinity is increased through the process of affinity maturation.

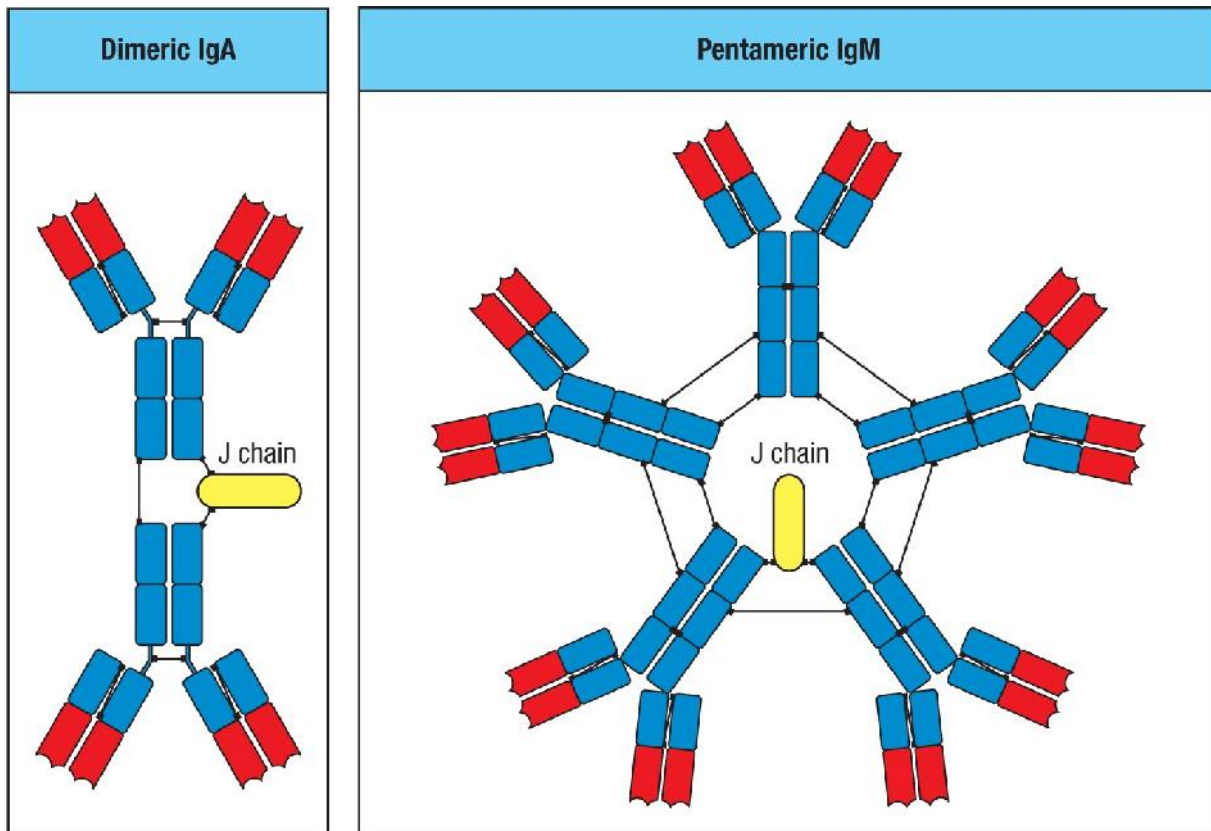


Fig. 4.8 The IgM and IgA molecules can form multimers. IgM and IgA are usually synthesized as multimers in association with an additional polypeptide chain, the J chain. In dimeric IgA (left panel), the monomers have disulfide bonds to the J chain as well as to each other. In pentameric IgM (right panel), the monomers are cross-linked by disulfide bonds to each other and to the J chain. IgM can also form hexamers that lack a J chain (not shown). IgM and IgA in secretions also contain a 'secretory piece,' a part of the receptor that transports the Igs through the cell via transcytosis.

IgG isotypes produced during an immune response are found in the bloodstream and in the extracellular spaces in tissues. IgM and most IgG isotypes can interact with the complement component C1 to activate the classical complement pathway (described in [Section 2-7](#)). IgA and IgE do not activate the complement. IgA can be found in the bloodstream, but it also acts in the defense of mucosal surfaces; it is secreted into the gut and respiratory tract and also into mother's milk. IgE is particularly involved in defense against multicellular parasites (for example, schistosomes), but it is also the antibody involved in common allergic diseases such as allergic asthma. IgG and IgE are always monomers, but IgA can be secreted as either a monomer or a dimer (see Fig. 4.8).

Sequence differences in the constant regions of the immunoglobulin heavy chains produce the distinct characteristics of each antibody isotype. These characteristics include the number and location of interchain disulfide bonds, the number of attached carbohydrate groups, the number of C domains, and the length of the hinge region (see Fig. 4.6). IgM and IgE heavy chains contain an extra C domain that replaces the hinge region found in γ , δ , and α chains. The absence of the hinge region does not mean that IgM and IgE molecules lack flexibility;

electron micrographs of IgM molecules binding to ligands show that the Fab arms can bend relative to the Fc portion. However, such a difference in structure may have functional consequences that are not yet characterized. Different isotypes and subtypes also differ in their ability to engage various effector functions, as described next.

4-7 The constant region confers functional specialization on the antibody.

Antibodies can protect the body in a variety of ways. In some cases it is enough for the antibody simply to bind antigen. For instance, by binding tightly to a toxin or virus, an antibody can prevent it from recognizing its receptor on a host cell (see Fig. 1.28). The antibody V regions are sufficient for this activity. The C region is essential, however, for recruiting the help of other cells and molecules to destroy and dispose of pathogens to which the antibody has bound.

The Fc region contains C-region domains of the heavy chain and has three main effector functions: Fc-receptor binding, complement activation, and regulation of secretion. First, the Fc region of certain isotypes binds to specialized Fc receptors expressed by immune effector cells. Fc γ receptors expressed on the surface of macrophages and neutrophils bind the Fc portions of IgG1 and IgG3 antibodies, facilitating the phagocytosis of pathogens coated with these antibodies. The Fc region of IgE binds to a high-affinity Fc ϵ receptor on mast cells, basophils, and activated eosinophils, triggering the release of inflammatory mediators in response to antigens. We will return to this topic in [Chapter 10](#).

Second, the Fc regions in antigen:antibody complexes can bind to the C1q complement protein (see [Section 2-7](#)) and initiate the classical complement cascade, which recruits and activates phagocytes to engulf and destroy pathogens. Third, the Fc portion can deliver antibodies to places they would not reach without active transport. These include transport of IgA into mucous secretions, tears, and breast milk, and the transfer of IgG from the pregnant mother into the fetal blood circulation. In both cases, the Fc portion of IgA or IgG engages a specific receptor, the poly-Ig receptor (pIgR) or the neonatal Fc receptor (FcRn), respectively, which actively transports the immunoglobulin through cells to reach different body compartments. Podocytes, a specialized cell type in the kidney glomerulus, express FcRn to help remove IgG that has been filtered from the blood and accumulated at the glomerular basement membrane.

The role of the Fc portion in these effector functions has been demonstrated by studying immunoglobulins that have had one or more Fc domains cleaved off enzymatically or modified genetically. Many microorganisms have responded to the destructive potential of the Fc portion by evolving proteins that either bind it or cleave it, and so prevent the Fc region from working; examples are Protein A and Protein G of *Staphylococcus* and Protein D of *Haemophilus*. Researchers have exploited these proteins to help map the Fc region and also as immunological reagents. Not all immunoglobulin classes have the same capacity to

engage each of the effector functions (see Fig. 4.7). For example, IgG1 and IgG3 have a higher affinity than IgG2 for the most common type of Fc receptor.

4-8 IgM and IgA can form polymers by interacting with the J chain.

Although all immunoglobulin molecules are constructed from a basic unit of two heavy and two light chains, both IgM and IgA can form multimers of these basic units (see Fig. 4.8). C regions of IgM and IgA can include a ‘tailpiece’ of 18 amino acids that contains a cysteine residue essential for polymerization. A separate 15-kDa polypeptide chain called the [J chain](#) promotes polymerization by linking to the cysteine of this tailpiece, which is found only in the secreted forms of the μ and α chains. (This J chain should not be confused with the immunoglobulin J region encoded by a J gene segment; see [Section 5-1](#).) In the case of IgA, dimerization is required for transport through epithelia, as we will discuss in [Chapter 10](#). IgM molecules are found as pentamers, and occasionally hexamers (without J chain), in plasma, whereas IgA is found mainly as a dimer in mucous secretions but as a monomer in plasma.

Immunoglobulin polymerization is also thought to be important in the binding of antibody to repetitive structures. An antibody molecule has at least two identical antigen-binding sites, each of which has a given affinity, or binding strength, for antigen. If the antibody attaches to multiple identical structures on a target antigen, it will dissociate only when all binding sites dissociate. The dissociation rate of the whole antibody will therefore be much slower than the dissociation rate for a single binding site; multiple binding sites thus give the antibody a greater total binding strength, or avidity. This consideration is particularly relevant for pentameric IgM, which has 10 antigen-binding sites.

IgM antibodies frequently recognize repetitive structures such as those on bacterial cell-wall polysaccharides, but individual binding sites are often of low affinity because IgM is made early in immune responses, before somatic hypermutation and affinity maturation. Multisite binding makes up for this, markedly improving the overall functional binding strength. This implies that binding of a single IgM pentamer to a target could be sufficient to mediate biological effector activity, whereas in the case of IgGs, two independent target molecules may need to be located in close proximity.

Summary.

The classes of immunoglobulins are defined by their heavy-chain C regions, with the different heavy-chain isotypes being encoded by different C-region genes. The heavy-chain C-region genes are present in a cluster 3' to the V, D, and J gene segments. The constant region isotype of an antibody determines its effector functions, including binding to Fc

receptors on effector cells such as phagocytes and mast cells, binding to the complement protein C1q, and binding to Fc receptors that mediate transport of antibodies between tissue compartments. Whereas all antibodies have two identical binding sites for antigen, some antibodies form multimeric complexes, thereby increasing their avidity for pathogen binding. Secreted IgM forms pentamers that include a small polypeptide called the J chain, creating a large antibody molecule with 10 antigen-binding sites. IgA that is transported into bodily secretions is a dimer of two IgA monomers plus the J chain. Differences in Fc-receptor binding, complement C1q binding, tissue localization, and antibody avidity are responsible for the specialized effector functions of each antibody isotype.

Glossary

[J chain](#)

Small polypeptide chain made by B cells that attaches to polymeric immunoglobulins IgM and IgA by disulfide bonds and is essential for formation of the binding site for the polymeric immunoglobulin receptor.

The interaction of the antibody molecule with specific antigen.

In this part of the chapter we look at the antigen-binding site of an immunoglobulin molecule in more detail. We discuss the different ways in which antigens can bind to antibody and address the question of how variation in the sequences of the antibody V domains determines the specificity for antigen.

4-9 Localized regions of hypervariable sequence form the antigen-binding site.

The V regions of any given antibody molecule differ from those of every other. Sequence variability is not, however, distributed evenly throughout the V region but is concentrated in certain segments, as is clearly seen in a [variability plot](#) (Fig. 4.9), in which the amino acid sequences of many different antibody V regions are compared. Three particularly variable segments can be identified in both the V_H and V_L domains. They are designated [hypervariable regions](#) and are denoted HV1, HV2, and HV3. In the heavy chains they are located at residues 30 to 36, 49 to 65, and 95 to 103, respectively, while in the light chains they are located at residues 28 to 35, 50 to 56, and 91 to 98, respectively. The most variable part of the domain is in the HV3 region. The regions between the hypervariable regions comprise the rest of the V domain; they show less variability and are termed the [framework regions](#). There are four such regions in each V domain, designated FR1, FR2, FR3, and FR4.

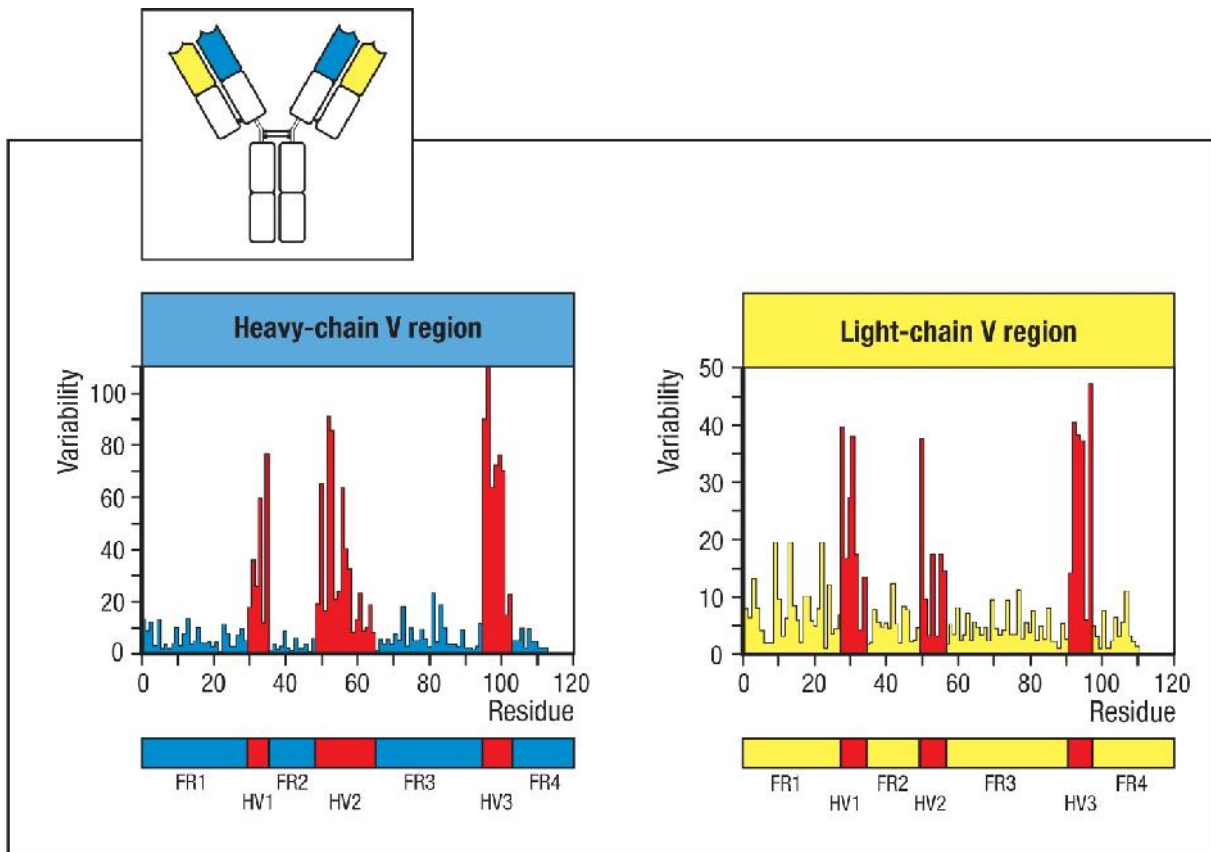


Fig. 4.9 There are discrete regions of hypervariability in V domains. The hypervariability regions of both the heavy chain and the light chain contribute to antigen binding of an antibody molecule. A variability plot derived from comparison of the amino acid sequences of several dozen heavy-chain and light-chain V domains is shown. At each amino acid position, the degree of variability is the ratio of the number of different amino acids seen in all of the sequences together to the frequency of the most common amino acid. Three hypervariable regions (HV1, HV2, and HV3) are indicated in red. They are flanked by less variable framework regions (FR1, FR2, FR3, and FR4, shown in blue or yellow).

The framework regions form the β sheets that provide the structural framework of the immunoglobulin domain. The hypervariable sequences correspond to three loops and are positioned near one another in the folded domain at the outer edge of the β sandwich (**Fig. 4.10**). Thus, not only is diversity concentrated in particular parts of the V domain sequence, but it is also localized to a particular region on the surface of the molecule. When the V_H and V_L immunoglobulin domains are paired in the antibody molecule, the three hypervariable loops from each domain are brought together, creating a single hypervariable site at the tip of each arm of the molecule. This is the [antigen-binding site](#), or [antibody-combining site](#), which determines the antigen specificity of the antibody. These six hypervariable loops are more commonly termed the [complementarity-determining regions](#), or [CDRs](#), because the surface they form is complementary to that of the antigen they bind. There are three CDRs from each of the heavy and light chains, namely, CDR1, CDR2, and CDR3. In most cases, CDRs from both V_H and V_L domains contribute to the antigen-binding site; thus it is the

combination of the heavy chain and the light chain that usually determines the final antigen specificity (see Fig. 4.9). However, there are some Fab crystal structures that show antigen interaction with just the heavy chain; for example, in one influenza Fab, antigen interaction involves binding mostly to the V_H CDR3 and only minor contacts with other CDRs. Thus, one way in which the immune system is able to generate antibodies of different specificities is by generating different combinations of heavy-chain and light-chain V regions. This is known as combinatorial diversity; we will encounter a second form of [combinatorial diversity](#) in [Chapter 5](#) when we consider how the genes encoding the heavy-chain and light-chain V regions are created from smaller segments of DNA during the development of B cells in the bone marrow.

Light-chain V region

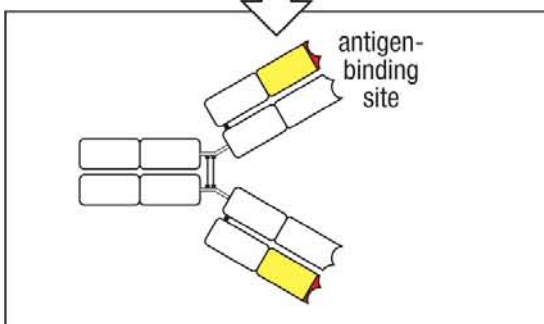
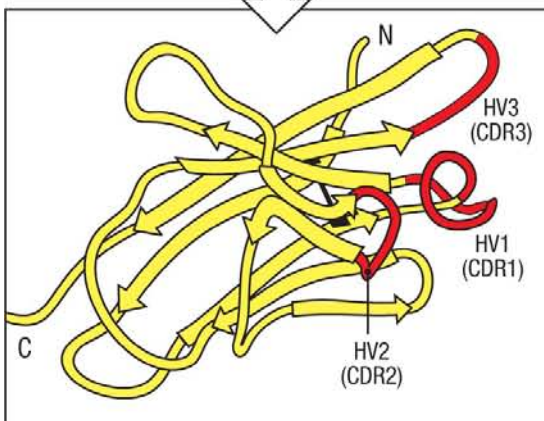
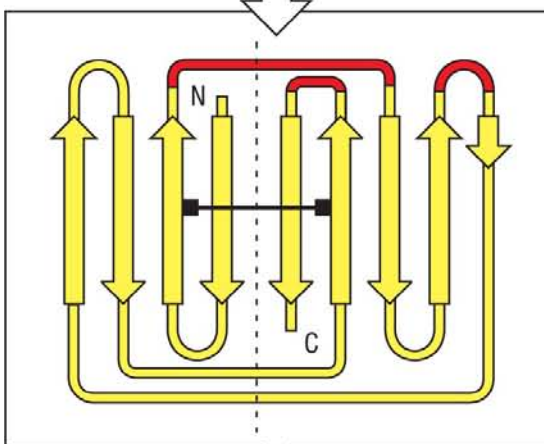
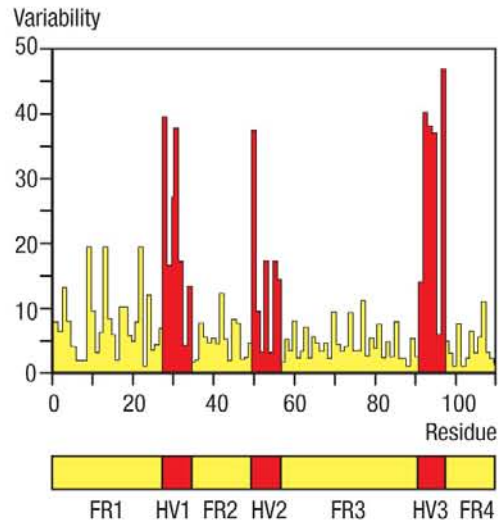


Fig. 4.10 The hypervariable regions lie in discrete loops of the folded structure.

First panel: The hypervariable regions (red) are positioned on the structure of a map of the coding region of the V domain. Second panel: When shown as a flattened ribbon diagram, hypervariable regions are seen to occur in loops (red) that join particular β strands. Third panel: In the folded structure of the V domain, these loops (red) are brought together to form antigen-binding regions. Fourth panel: In a complete antibody molecule, the pairing of a heavy chain and a light chain brings together the hypervariable loops from each chain to create a single hypervariable surface, which forms the antigen-binding site at the tip of each arm. Because they are complementary to the antigen surface, the hypervariable regions are also commonly known as the complementarity-determining regions (CDRs). C, carboxyl terminus; N, amino terminus.

4-10 Antibodies bind antigens via contacts in CDRs that are complementary to the size and shape of the antigen.

In early investigations of antigen binding to antibodies, the only available sources of large quantities of a single type of antibody molecule were tumors of antibody-secreting cells. The antigen specificities of these antibodies were unknown, and therefore many compounds had to be screened to identify ligands that could be used to study antigen binding. In general, the substances found to bind to these antibodies were haptens (see [Section 4-5](#)) such as phosphocholine or vitamin K₁. Structural analysis of complexes of antibodies with their hapten ligands provided the first direct evidence that the hypervariable regions form the antigen-binding site and demonstrated the structural basis of specificity for the hapten. Subsequently, with the discovery of methods of generating [monoclonal antibodies](#) (see Appendix I, [Section A-8](#)), it became possible to make large amounts of pure antibody specific for a given antigen. This has provided a more general picture of how antibodies interact with their antigens, confirming and extending the view of antibody–antigen interactions derived from the study of haptens.

The surface of the antibody molecule formed by the juxtaposition of the CDRs of the heavy and light chains is the site to which an antigen binds. The amino acid sequences of the CDRs are different in different antibodies, and so too are the shapes and properties of the surfaces created by these CDRs. As a general principle, antibodies bind ligands whose surfaces are complementary to that of the antigen-binding site. A small antigen, such as a hapten or a short peptide, generally binds in a pocket or groove lying between the heavy-chain and light-chain V domains (**Fig. 4.11a and b**). Some antigens, such as proteins, can be the same size as or larger than the antibody itself. In these cases, the interface between antigen and antibody is often an extended surface that involves all the CDRs and, in some cases, part of the framework region as well (**Fig. 4.11c**). This surface need not be concave but can be flat,

undulating, or even convex. In some cases, antibody molecules with elongated CDR3 loops can protrude a 'finger' into recesses in the surface of the antigen, as shown in **Fig. 4.11d**, where an antibody binding to the HIV gp120 antigen projects a long loop into its target.

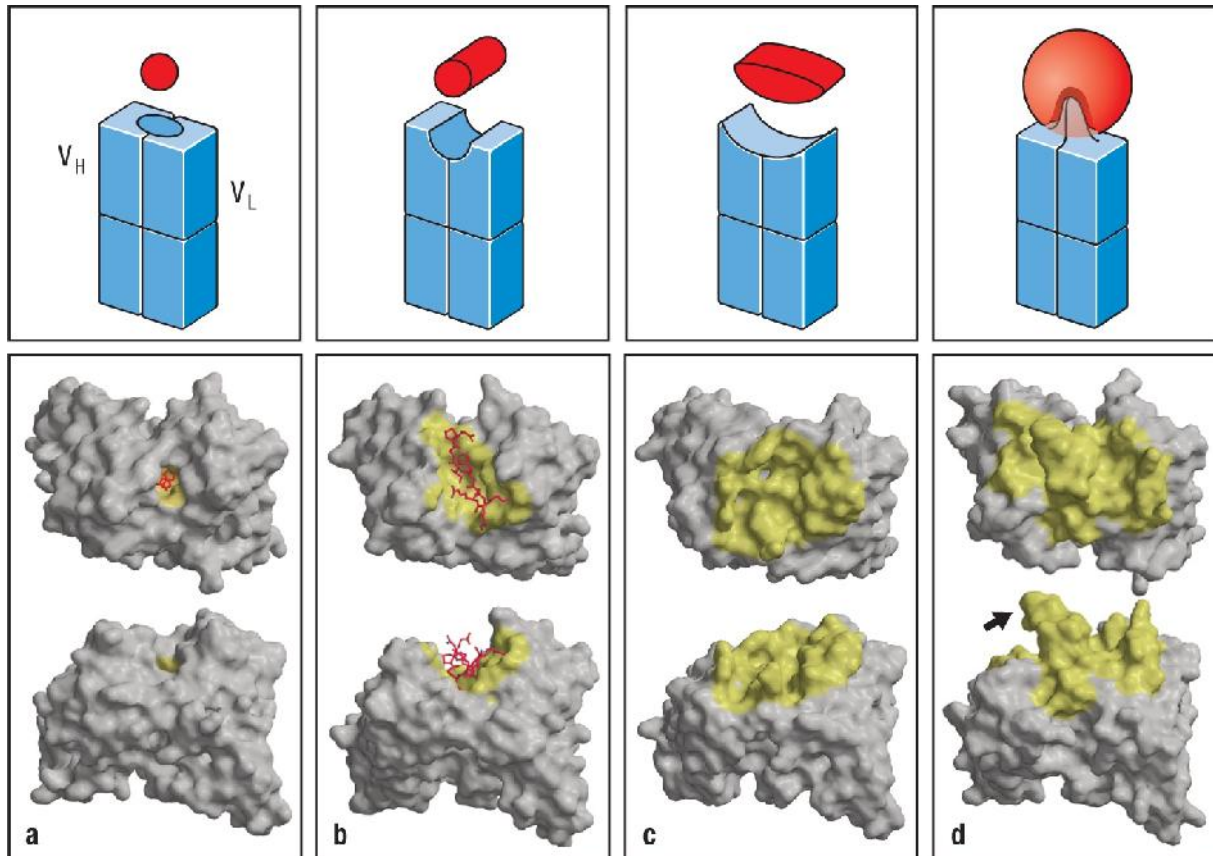


Fig. 4.11 Antigens can bind in pockets, or grooves, or on extended surfaces in the binding sites of antibodies. The panels in the top row show schematic representations of the different types of binding sites in a Fab fragment of an antibody: first panel, pocket; second panel, groove; third panel, extended surface; and fourth panel, protruding surface. Below are examples of each type. Panel a: The top image shows the molecular surface of the interaction of a small hapten with the complementarity-determining regions (CDRs) of a Fab fragment as viewed looking into the antigen-binding site. The ferrocene hapten, shown in red, is bound into the antigen-binding pocket (yellow). In the bottom image (and in those of panels b, c, and d), the molecule has been rotated by about 90° to give a side-on view of the binding site. Panel b: In a complex of an antibody with a peptide from the human immunodeficiency virus (HIV), the peptide (red) binds along a groove (yellow) formed between the heavy-chain and light-chain V domains. Panel c: Shown is a Fab fragment of an antibody specific for hen egg-white lysozyme (HyHel5). The surface on the antibody that comes into contact with the lysozyme is colored yellow. All six CDRs of the antigen-binding site are involved in the binding. Panel d: An antibody molecule against the HIV gp120 antigen has an elongated CDR3 loop (arrow) that protrudes into a recess on the side of the antigen. The structure of the complex between this antibody and gp120 has been solved, and in this case, only the heavy chain interacts with gp120.

4-11 Antibodies bind to conformational shapes on the surfaces of antigens using a variety of noncovalent forces.

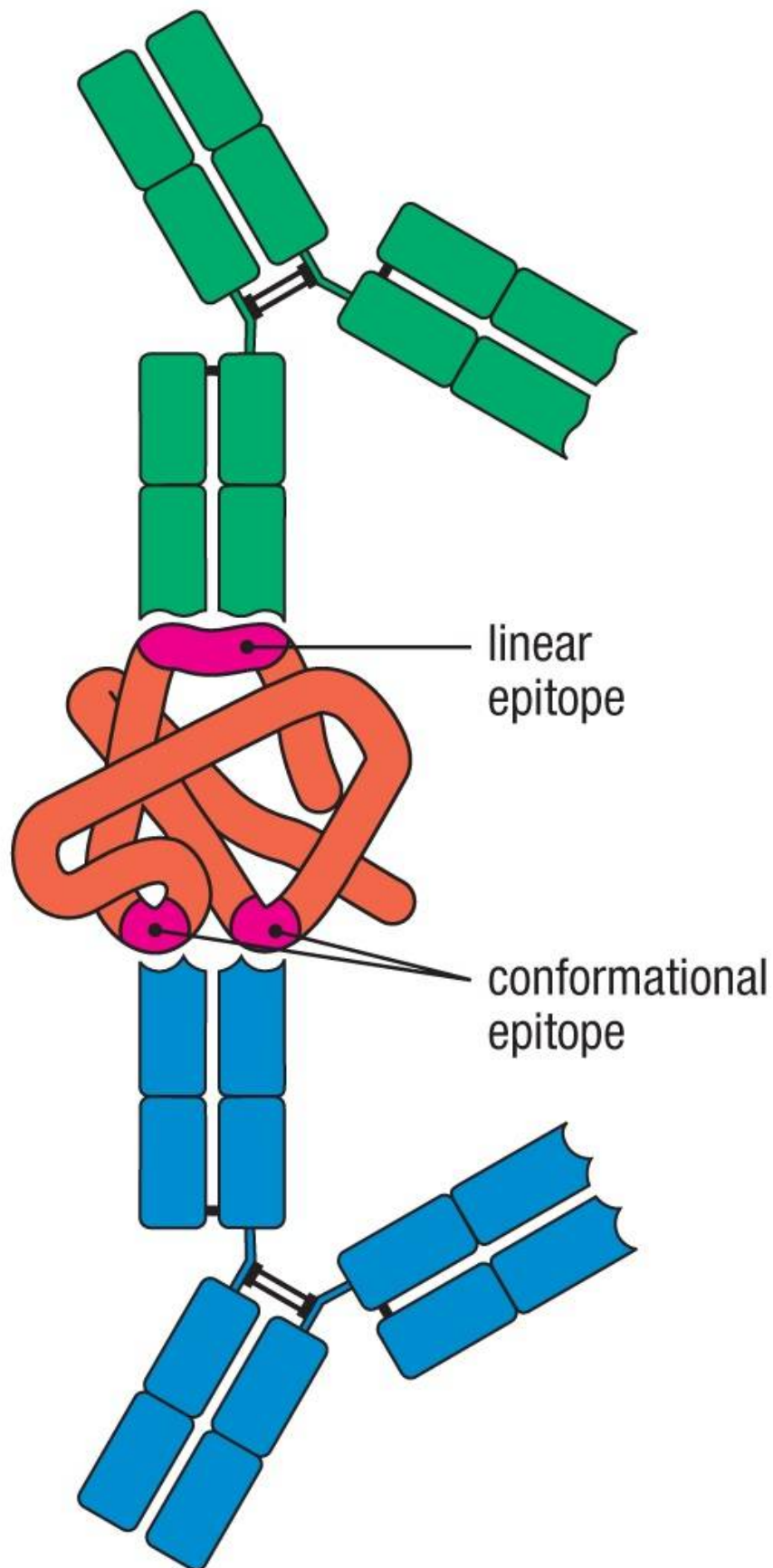


Fig. 4.12 Antibodies can recognize linear or conformational determinants in folded proteins. In a folded protein, antibodies can recognize amino acid sequences that are discontinuous in the linear amino acid sequence of the protein, but are brought into proximity in the three-dimensional structure. These are known as conformational or discontinuous epitopes. Antibodies can also bind to peptides that are continuous in the linear amino acid sequence of the protein and these are known as linear or continuous epitopes.

The biological function of antibodies is to bind to pathogens and their products and to facilitate their removal from the body. An antibody generally recognizes only a small region on the surface of a large molecule such as a polysaccharide or protein. The structure recognized by an antibody is called an [antigenic determinant](#), or [epitope](#). Some of the most important pathogens have polysaccharide coats, and antibodies that recognize epitopes formed by the sugar subunits of these molecules are essential in providing immune protection against such pathogens. In many cases, however, the antigens that provoke an immune response are proteins. For example, many protective antibodies against viruses recognize viral coat proteins. In all such cases, the structures recognized by the antibody are located on the surface of the protein. Such sites are likely to be composed of amino acids from different parts of the polypeptide chain that have been brought together by protein folding. Antigenic determinants of this kind are known as [conformational or discontinuous epitopes](#) because the structure recognized is composed of segments of the protein that are discontinuous in the amino acid sequence of the antigen but are brought together in the three-dimensional structure (**Fig. 4.12**). In contrast, an epitope composed of a single segment of polypeptide chain is termed a [continuous or linear epitope](#). Although most antibodies raised against intact, fully folded proteins recognize discontinuous epitopes, some will bind to peptide fragments of the protein. Conversely, antibodies raised against peptides of a protein or against synthetic peptides corresponding to part of its sequence are occasionally found to bind to the natural folded protein. This makes it possible, in some cases, to use synthetic peptides in vaccines that aim to raise antibodies against a pathogen's protein.

The interaction between an antibody and its antigen can be disrupted by high salt concentrations, by extremes of pH, by detergents, and sometimes by competition with high concentrations of the pure epitope itself. The binding is therefore a reversible noncovalent interaction. The forces, or bonds, involved in these noncovalent interactions are outlined in **Fig. 4.13**. [Electrostatic interactions](#) occur between charged amino acid side chains, as in salt bridges. Most antibody–antigen interactions involve at least one electrostatic interaction. Interactions also occur between electric dipoles, as in hydrogen bonds, or can involve short-range van der Waals forces. High salt concentrations and extremes of pH disrupt antigen–antibody binding by weakening electrostatic interactions and/or hydrogen bonds. This principle is employed in the purification of antigens by using affinity columns of immobilized antibodies (or in the purification of antibody by using antigens in a like manner) (see Appendix I, [Section A-3](#)). [Hydrophobic interactions](#) occur when two hydrophobic surfaces come together to exclude water. The strength of a hydrophobic

interaction is proportional to the surface area that is hidden from water, and for some antigens, hydrophobic interactions probably account for most of the binding energy. In some cases, water molecules are trapped in pockets in the interface between antigen and antibody. These trapped water molecules, especially those between polar amino acid residues, may also contribute to binding and hence to the specificity of the antibody.

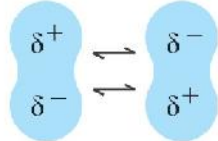
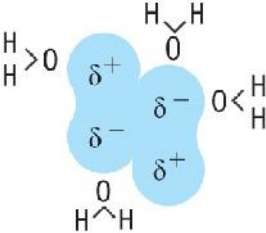
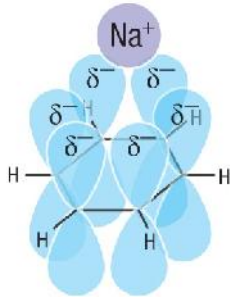
Noncovalent forces	Origin	
Electrostatic forces	Attraction between opposite charges	$\text{—NH}_3^+ \quad \text{OOC}^-$
Hydrogen bonds	Hydrogen shared between electronegative atoms (N, O)	$\begin{array}{c} > \text{N} \text{—} \text{H} \text{—} \text{O} = \text{C} < \\ \delta^- \quad \delta^+ \quad \delta^- \end{array}$
Van der Waals forces	Fluctuations in electron clouds around molecules polarize neighboring atoms oppositely	
Hydrophobic forces	Hydrophobic groups interact unfavorably with water and tend to pack together to exclude water molecules. The attraction also involves van der Waals forces	
Cation-pi interaction	Noncovalent interaction between a cation and an electron cloud of a nearby aromatic group	

Fig. 4.13 The noncovalent forces that hold together the antigen:antibody complex.

Partial charges found in electric dipoles are shown as δ^+ or δ^- . Electrostatic forces diminish as the inverse square of the distance separating the charges, whereas van der Waals forces, which are more numerous in most antigen–antibody contacts, fall off as the sixth power of the separation and therefore operate only over very short ranges. Covalent bonds never occur between antigens and naturally produced antibodies.

The contribution of each of these forces to the overall interaction depends on the particular antibody and antigen involved. A striking difference between antibody interactions with

protein antigens and most other natural protein–protein interactions is that antibodies often have many aromatic amino acids in their antigen-binding sites. These amino acids participate mainly in van der Waals and hydrophobic interactions, and sometimes in hydrogen bonds and [cation-pi interactions](#). Tyrosine, for example, can take part in both hydrogen bonding and hydrophobic interactions; it is therefore particularly suitable for providing diversity in antigen recognition and is overrepresented in antigen-binding sites. In general, the hydrophobic and van der Waals forces operate over very short ranges and serve to pull together two surfaces that are complementary in shape: hills on one surface must fit into valleys on the other for good binding to occur. In contrast, electrostatic interactions between charged side chains as well as hydrogen bonds bridging oxygen and/or nitrogen atoms accommodate more specific chemical interactions while strengthening the interaction overall. The side chains of aromatic amino acids such as tyrosine can interact noncovalently through their pi-electron system with nearby cations, including nitrogen-containing side chains that may be in a protonated cationic state.

4-12 Antibody interaction with intact antigens is influenced by steric constraints.

An example of an antibody–antigen interaction involving a specific amino acid in the antigen can be seen in the complex of hen egg-white lysozyme with the antibody D1.3 (**Fig. 4.14**). In this structure, strong hydrogen bonds are formed between the antibody and a particular glutamine in the lysozyme molecule that protrudes between the V_H and V_L domains. Lysozymes from partridge and turkey have another amino acid in place of the glutamine and do not bind to this antibody. Overall surface complementarity must have an important role in antigen–antibody interactions, but in most antibodies that have been studied at this level of detail, only a few residues make a major contribution to the binding energy and hence to the final specificity of the antibody. Although many antibodies naturally bind their ligands with high affinity (in the nanomolar range), genetic engineering by site-directed mutagenesis can tailor an antibody to bind even more strongly to its epitope.

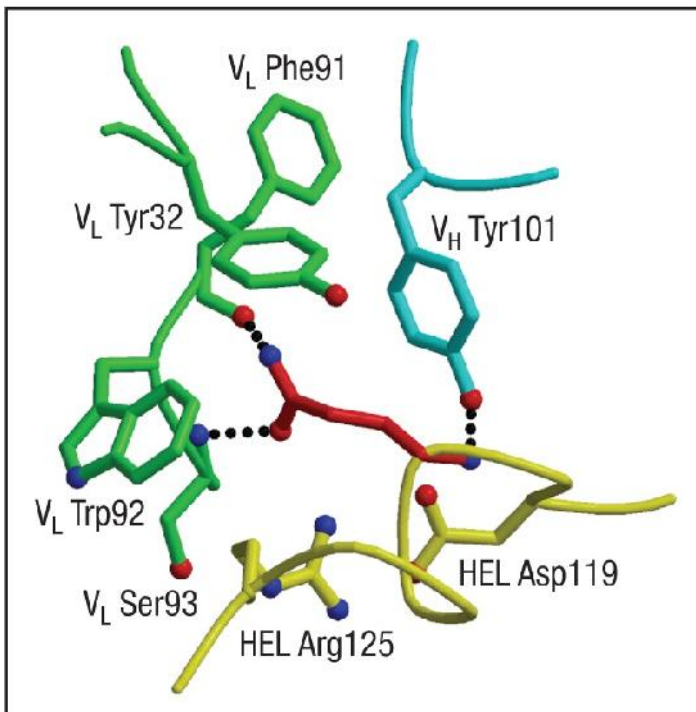
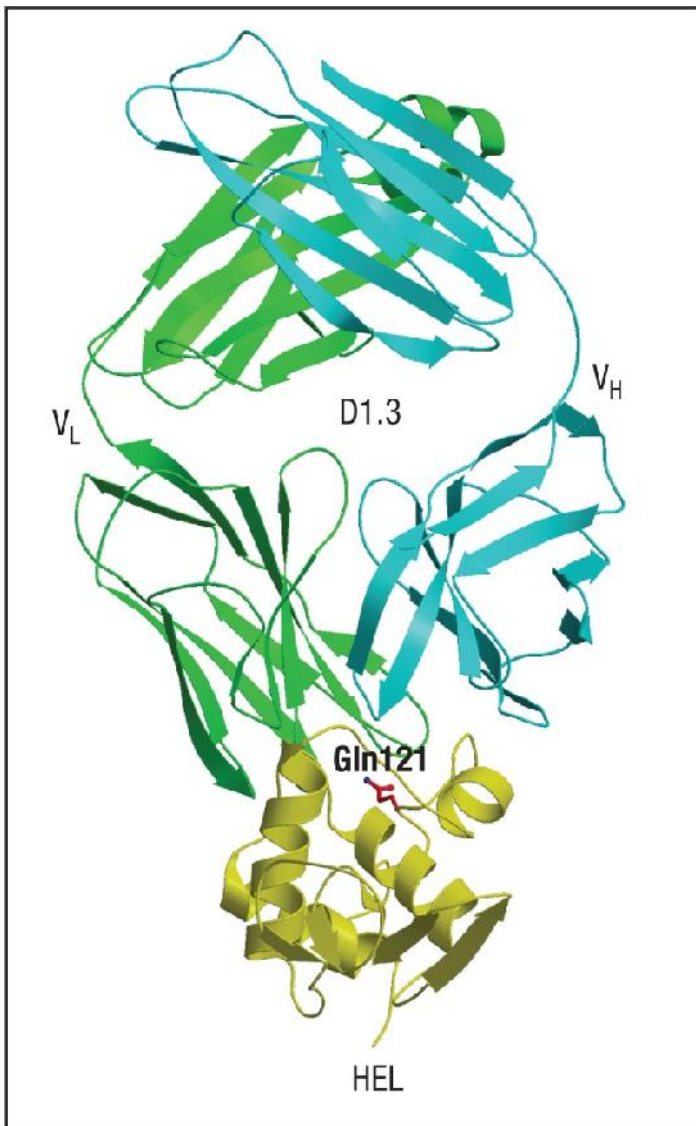
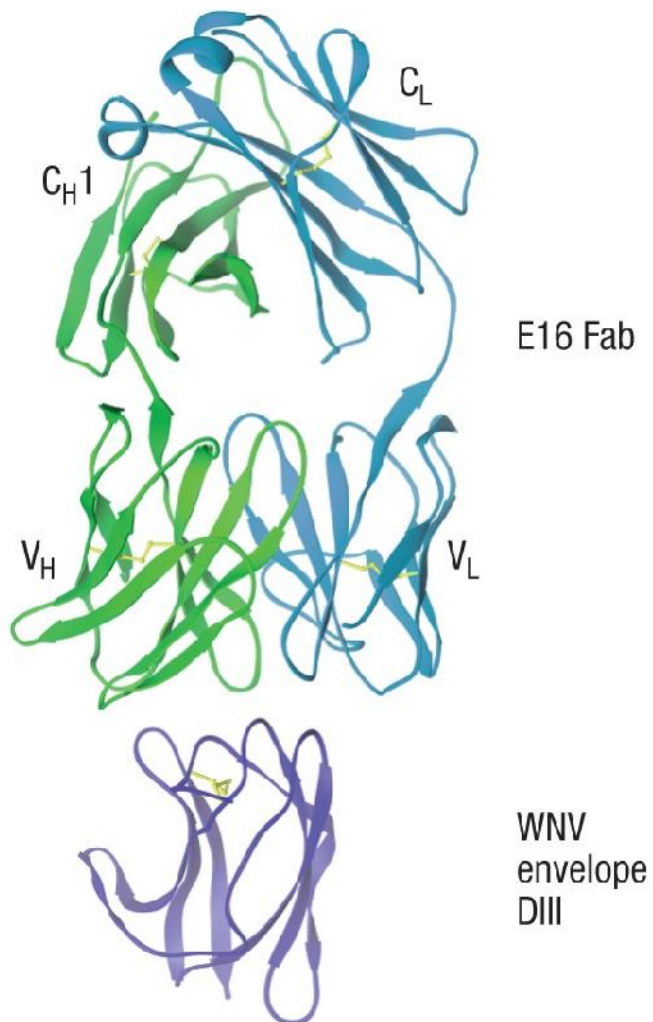


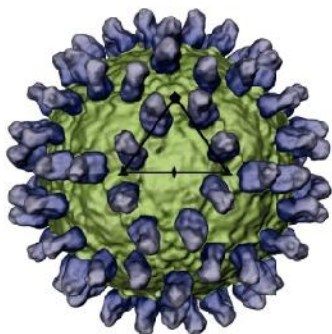
Fig. 4.14 The complex of lysozyme with the antibody D1.3. Top panel: The interaction of the Fab fragment of D1.3 with hen egg-white lysozyme (HEL) is shown. HEL is depicted in yellow, the heavy chain (V_H) in turquoise, and the light chain (V_L) in green. Bottom panel: A glutamine residue (Gln121) that protrudes from HEL (yellow) extends its side chain (shown in red) between the V_L (green) and V_H (turquoise) domains of the antigen-binding site and makes hydrogen bonds with the hydroxyl group (purple dots) of the indicated amino acids of both domains. These hydrogen bonds are important to the antigen–antibody binding.

Even when antibodies have high affinity for antigens on a larger structure, such as an intact viral particle, antibody binding may be prevented by their particular arrangement. For example, the intact West Nile virion is built from an icosahedral scaffold that has 90 homodimers of a membrane-anchored envelope glycoprotein, E, which has three domains, DI, DII, and DIII. The DIII domain has four polypeptide loops that protrude outward from the viral particle. A neutralizing antibody against West Nile virus, E16, recognizes these loops of DIII, as shown in **Fig. 4.15**. In theory, there should be 180 possible antigen-binding sites for the E16 antibody on the West Nile viral particle. However, a combination of crystallographic and electron micrographic studies show that even with an excess of the E16 Fab fragment, only about 120 of the total 180 DIII domains of E are able to bind the E16 Fab fragment (see Fig. 4.14). This appears to result from steric hindrance, with the presence of one Fab blocking the ability of another Fab to bind to some nearby E protein sites. Presumably, such steric hindrance would become more severe with intact antibody than is evident with the smaller Fab fragment. This study also showed that the Fab binds to the DIII region with only one of its antigen-binding arms, indicating that antibodies may not always bind to antigens with both antigen-binding sites. Therefore, the orientation of the antigens being recognized on the pathogen places constraints that impact the ability of antibodies to neutralize their targets.

E16 Fab binds four outward-facing loops of WNV DIII envelope protein



Cryo-EM reconstruction of E16 Fab bound to mature WNV particle



Molecular model of E16 Fab bound to mature WNV particle

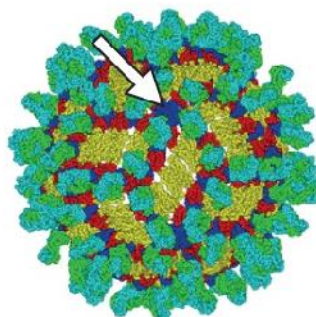


Fig. 4.15 Steric hindrance occludes the binding of antibody to native antigen in the intact West Nile viral particle. Top panel: The monoclonal antibody E16 recognizes DIII, one of the three structural domains in the West Nile virus (WNV) glycoprotein E. Shown is a crystal structure of the E16 Fab bound to the DIII epitope. Bottom left panel Cryo-electron microscopy (cryo-EM) reconstruction of saturating E16 Fab bound to West Nile virion confirmed the predicted steric hindrance. The vertices of the triangle shown in the figure indicate the icosahedral symmetry axes. Bottom right panel: A computer model was used to dock E16 Fab to the mature West Nile virion. E16 Fabs were able to bind 120 of the 180 DIII epitopes. Sixty of the fivefold clustered DIII epitopes are sterically hindered by the binding of Fab to four nearby DIII epitopes within that cluster. An example of an occluded epitope is the blue area indicated by the arrow.

4-13 Some species generate antibodies with alternative structures.

Our focus in this chapter has been on the structure of antibodies produced by humans, which is generally similar in most mammalian species, including mice, an important model system for immunology research. However, some mammals have the ability to produce an alternative form of antibody that is based on the ability of a single V_H domain to interact with antigen in the absence of a V_L domain (**Fig. 4.16**). It has been known for some time that the serum of camels contains abundant immunoglobulin-like material composed of heavy-chain dimers that lack associated light chains but retain the capacity to bind antigens. These antibodies are called [heavy-chain-only IgGs \(hcIgGs\)](#). This property is shared by other camelids, including llamas and alpacas. These species have retained the genes for the immunoglobulin light chains, and some IgG-like material in their sera remains associated with light chains, and so it is unclear what led to this particular adaptation during their evolution. In camelids, the ability to produce hcIgGs arises from mutations that allow the alternative splicing of the heavy-chain mRNA, with loss of the C_H1 exon and thus the joining of the V_H directly to the C_H2 domain in the protein. Other mutations stabilize this structure in the absence of V_L domains.

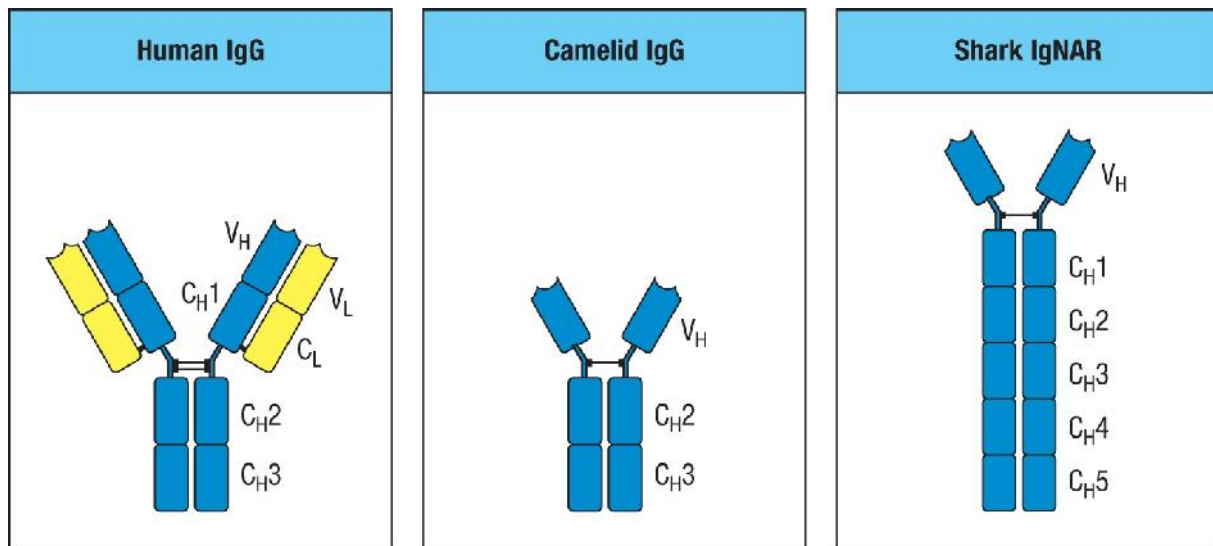


Fig. 4.16 Camelid and shark antibody can consist of heavy chain only. In the camelid heavy-chain-only antibody, a splicing event of the mature heavy chain can delete the exon encoding the C_H1 region and thereby create an in-frame hinge region linking the V_H to the C_H2 region. In the shark, the heavy-chain-only Ig molecule retains the C_H1 region, suggesting that this form of antibody may predate the evolution of light chains. For both, the repertoire of antigen-binding sites involves extensive variations in long CDR3 regions of the V_H domain relative to other types of antibody.

Cartilaginous fish, in particular the shark, also have an antibody molecule that differs substantially from human or murine antibodies (see Fig. 4.16). Like camelids, the shark also has genes encoding both immunoglobulin heavy and light chains and does produce immunoglobulins containing both heavy and light chains. But sharks also produce an [immunoglobulin new antigen receptor \(IgNAR\)](#), a heavy-chain-only antibody in which the V_H is spliced to the C_H1 exon, rather than the C_H1 exon being spliced out as in camelids. These differences suggest that hcIgG production by camelids and sharks represents an event of convergent evolution. The ability of camelid V_H domains to interact efficiently with antigens is the basis for producing so-called [single-chain antibody](#). The simplification of using only a single domain for antigen recognition has prompted recent interest in single-chain monoclonal antibodies as an alternative to standard monoclonal antibodies, which we will discuss more in [Chapter 16](#).

Summary.

X-ray crystallographic analyses of antigen:antibody complexes have shown that the hypervariable loops (complementarity-determining regions; CDRs) of immunoglobulin V regions determine the binding specificity of an antibody. Contact between an antibody molecule and a protein antigen usually occurs over a broad area of the antibody surface that is complementary to the surface recognized on the antigen. Electrostatic interactions,

hydrogen bonds, van der Waals forces, and hydrophobic and cation-pi interactions can all contribute to binding. Depending on the size of the antigen, amino acid side chains in most or all of the CDRs make contact with antigen and determine both the specificity and the affinity of the interaction. Other parts of the V region normally play little part in the direct contact with the antigen, but they provide a stable structural framework for the CDRs and help to determine their position and conformation. Antibodies raised against intact proteins usually bind to the surface of the protein and make contact with residues that are discontinuous in the primary structure of the molecule; they may, however, occasionally bind peptide fragments of the protein, and antibodies raised against peptides derived from a protein can sometimes be used to detect the native protein molecule. Peptides binding to antibodies usually bind in a cleft or pocket between the V regions of the heavy and light chains, where they make specific contact with some, but not necessarily all, of the CDRs. This is also the usual mode of binding for carbohydrate antigens and small molecules such as haptens.

Glossary

[variability plot](#)

A measure of the difference between the amino acid sequences of different variants of a given protein. The most variable proteins known are antibodies and T-cell receptors.

[hypervariable regions](#)

Parts of the V domains of immunoglobulins and T-cell receptors that determine their antigen specificity and make contact with the specific ligand. The CDRs are the most variable part of the antigen receptor and contribute to the diversity of these proteins.

There are three such regions (CDR1, CDR2, and CDR3) in each V domain.

[framework regions](#)

Relatively invariant regions that provide a protein scaffold for the hypervariable regions in the V domains of immunoglobulins and T-cell receptors.

[antigen-binding site](#)

The site at the tip of each arm of an antibody that makes physical contact with the antigen and binds it noncovalently. The antigen specificity of the site is determined by its shape and the amino acids present.

[complementarity-determining regions \(CDRs\)](#)

Parts of the V domains of immunoglobulins and T-cell receptors that determine their antigen specificity and make contact with the specific ligand. The CDRs are the most variable part of the antigen receptor and contribute to the diversity of these proteins.

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[complementarity-determining regions \(CDRs\)](#)

Parts of the V domains of immunoglobulins and T-cell receptors that determine their antigen specificity and make contact with the specific ligand. The CDRs are the most variable part of the antigen receptor and contribute to the diversity of these proteins.

There are three such regions (CDR1, CDR2, and CDR3) in each V domain.

[combinatorial diversity](#)

The diversity among antigen receptors generated by combining separate units of genetic information, comprising two types. First, receptor gene segments are joined in many different combinations to generate diverse receptor chains; second, two different receptor chains (heavy and light in immunoglobulins; α and β , or γ and δ , in T-cell receptors) are combined to make the antigen-recognition site.

monoclonal antibodies

Antibodies produced by a single clone of B lymphocytes, so that they are all identical.

antigenic determinant

That portion of an antigenic molecule that is bound by the antigen-binding site of a given antibody or antigen receptor; it is also known as an epitope.

epitope

A site on an antigen recognized by an antibody or an antigen receptor. T-cell epitopes are short peptides bound to MHC molecules. B-cell epitopes are typically structural motifs on the surface of the antigen. Also called an antigenic determinant.

electrostatic interaction

Chemical interaction occurring between charged atoms, as in the charged amino acid side chains and an ion in a salt bridge.

hydrophobic interaction

Chemical interaction occurring between nearby hydrophobic moieties typically excluding water molecules.

cation-pi interaction

Chemical interaction between a cation (e.g., Na^+) and the pi-electron system of an aromatic moiety.

heavy-chain-only IgGs (hcIgGs)

Antibodies that are produced by some camelid species and composed of heavy-chain dimers without an associated light chain that retain antigen-binding capacity.

immunoglobulin new antigen receptor (IgNAR)

A form of heavy-chain-only Ig molecule made by shark species.

single-chain antibody

Referring to the heavy-chain-only IgGs produced by camelids or shark species that lack the light chain present in conventional antibodies.

conformational epitope, discontinuous epitope

Antigenic structure (epitope) on a protein antigen that is formed from several separate regions in the sequence of the protein brought together by protein folding. Antibodies that bind conformational epitopes bind only native folded proteins.

continuous epitope, linear epitope

Antigenic structure (epitope) in a protein that is formed by a single small region of amino acid sequence. Antibodies that bind continuous epitopes can bind to the denatured protein. The epitopes detected by T cells are continuous.

antibody combining site

The site at the tip of each arm of an antibody that makes physical contact with the antigen and binds it noncovalently. The antigen specificity of the site is determined by its shape and the amino acids present.

Antigen recognition by T cells.

In contrast to the immunoglobulins, which interact with pathogens and their toxic products in the extracellular spaces of the body, T cells recognize foreign antigens only when they are displayed on the surface of the body's own cells. These antigens can derive from pathogens such as viruses or intracellular bacteria, which replicate within cells, or from pathogens or their products that have been internalized by endocytosis from the extracellular fluid.

T cells detect the presence of an intracellular pathogen because the infected cells display peptide fragments of the pathogen's proteins on their surface. These foreign peptides are delivered to the cell surface by specialized host-cell glycoproteins—the MHC molecules. These are encoded in a large cluster of genes that were first identified by their powerful effects on the immune response to transplanted tissues. For that reason, the gene complex was called the major histocompatibility complex (MHC), and the peptide-binding glycoproteins are known as MHC molecules. The recognition of antigen as a small peptide fragment bound to an MHC molecule and displayed at the cell surface is one of the most distinctive features of T cells and will be the focus of this part of the chapter. How the peptide fragments of antigen are generated and become associated with MHC molecules will be described in [Chapter 6](#).

We describe here the structure and properties of the T-cell receptor (TCR). As might be expected from the function of T-cell receptors as highly variable antigen-recognition structures, the genes for TCRs are closely related to those for immunoglobulins. There are, however, important differences between T-cell receptors and immunoglobulins, and these differences reflect the special features of antigen recognition by T cells.

4-14 The TCR $\alpha\beta$ heterodimer is very similar to a Fab fragment of immunoglobulin.

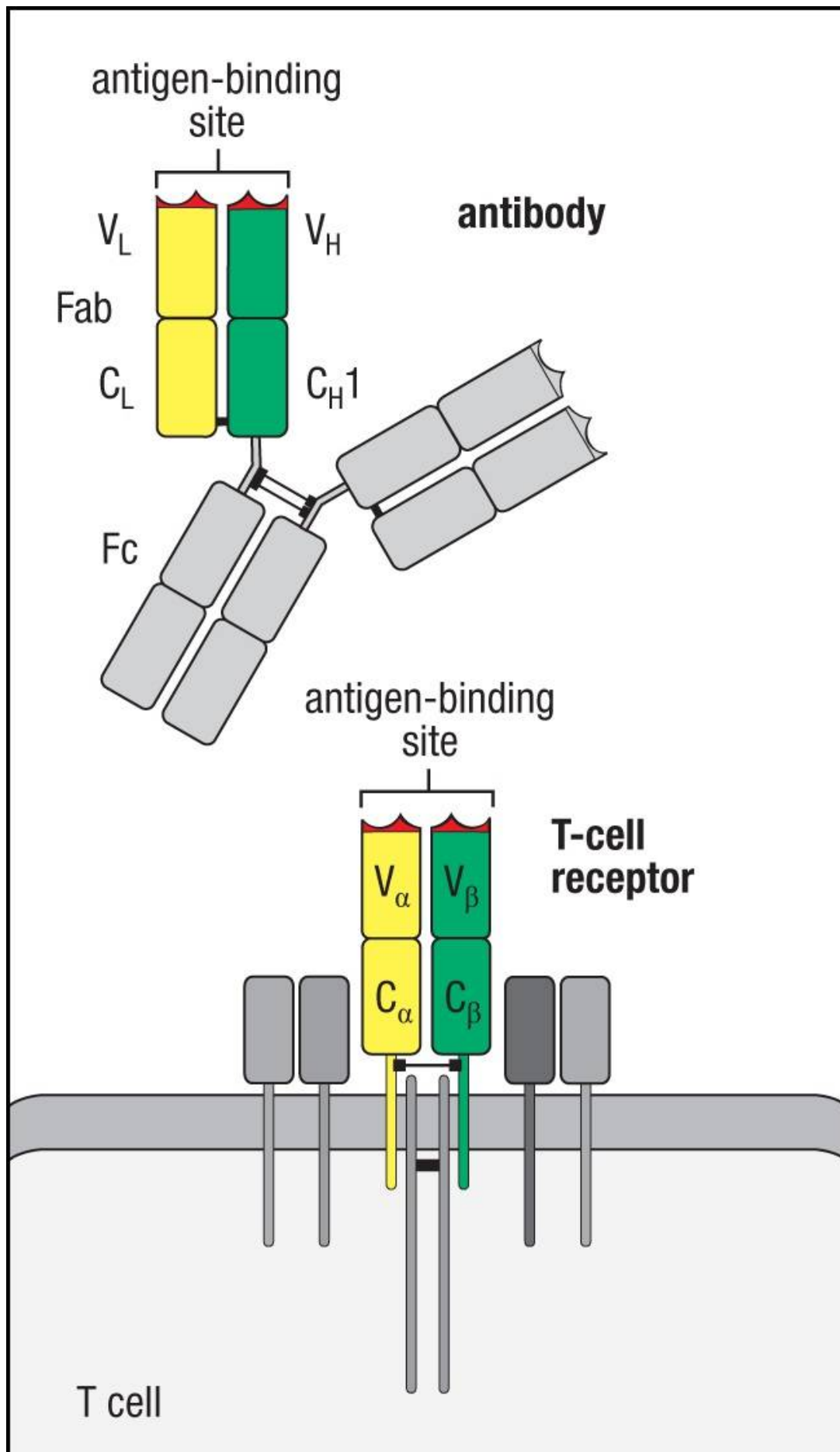


Fig. 4.17 The T-cell receptor resembles a membrane-bound Fab fragment. The Fab fragment of an antibody molecule is a disulfide-linked heterodimer, each chain of which contains one immunoglobulin C domain and one V domain; the juxtaposition of the V domains forms the antigen-binding site (see Section 4-6). The antigen-binding subunits of the T-cell receptor also form a disulfide-linked heterodimer, with each chain containing an immunoglobulin C-like domain and an immunoglobulin V-like domain. This heterodimer is always associated with six additional polypeptides (shown in gray) that form the complete membrane-bound T-cell receptor. As in the Fab fragment, the juxtaposition of the T-cell receptor V_α and V_β domains forms the site for antigen recognition.

T-cell receptors were first identified by using monoclonal antibodies that bound to a single cloned T-cell line: such antibodies either specifically inhibit antigen recognition by the clone or specifically activate it by mimicking the antigen (see Appendix I, [Section A-21](#)). These [clonotypic](#) antibodies were then used to show that each T cell bears about 30,000 identical antigen receptors on its surface, each receptor consisting of two different polypeptide chains, termed the [T-cell receptor \$\alpha\$ \(TCR \$\alpha\$ \) and \$\beta\$ \(TCR \$\beta\$ \)](#) chains. Each chain of the [\$\alpha\beta\$ heterodimer](#) is composed of two Ig domains, and the two chains are linked by a disulfide bond, similar to the structure of the Fab fragment of an immunoglobulin molecule (**Fig. 4.17**). Also similar to the antibody Fab fragment, the T-cell receptor $\alpha\beta$ heterodimer is the antigen-binding portion of the T-cell receptor. On the surface of T cells, the $\alpha\beta$ heterodimer is always found in a large multiprotein complex that contains six additional subunits: CD3 γ , CD3 δ , two copies of CD3 ϵ , and two copies of the T-cell receptor ζ chain (Fig 4.17). These additional subunits function in signal transduction, rather than in antigen recognition, and unlike the $\alpha\beta$ heterodimer, they do not vary between T cells. The functions of these receptor signaling subunits will be discussed in [Chapter 7](#).

The TCR α and TCR β chains of the $\alpha\beta$ heterodimer account for antigen recognition by most T cells. A minority of T cells bear an alternative, but structurally similar, receptor made up of a different pair of polypeptide chains designated γ and δ . The [\$\gamma\delta\$ T-cell receptors](#) have different antigen-recognition properties from the [\$\alpha\beta\$ T-cell receptors](#), and the functions of $\gamma\delta$ T cells in immune responses are still being clarified as the various ligands they recognize are identified (see [Section 6-20](#)). In the rest of this chapter and elsewhere in the book, we use the term T-cell receptor to mean the $\alpha\beta$ receptor, except where specified otherwise. Both types of T-cell receptors differ from the membrane-bound immunoglobulin that serves as the B-cell receptor in two main ways. A T-cell receptor has only one antigen-binding site, whereas a B-cell receptor has two, and T-cell receptors are never secreted, whereas immunoglobulins can be secreted as antibodies.

Further insights into the structure and function of the $\alpha\beta$ T-cell receptor came from studies of cloned cDNA encoding the receptor chains. The amino acid sequences predicted from the cDNA showed that both chains of the T-cell receptor have an amino-terminal variable (V) region with sequence homology to an immunoglobulin V domain, a constant (C) region with

homology to an immunoglobulin C domain, and a short stalk segment containing a cysteine residue that forms the interchain disulfide bond (**Fig. 4.18**). Each chain spans the lipid bilayer by a hydrophobic transmembrane domain and ends in a short cytoplasmic tail. These close similarities of T-cell receptor chains to the heavy and light immunoglobulin chains first enabled prediction of the structural resemblance of the T-cell receptor heterodimer to a Fab fragment of immunoglobulin.

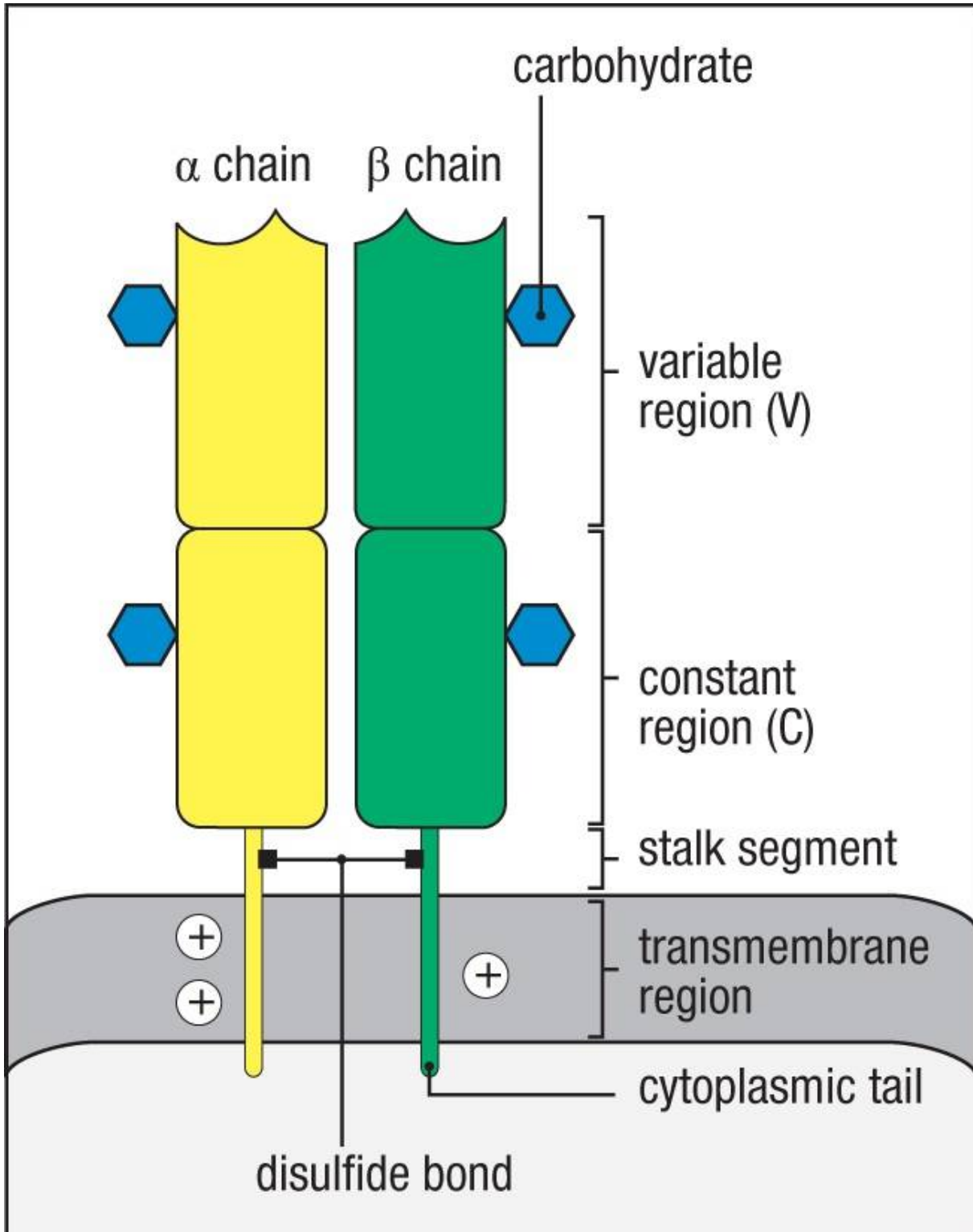
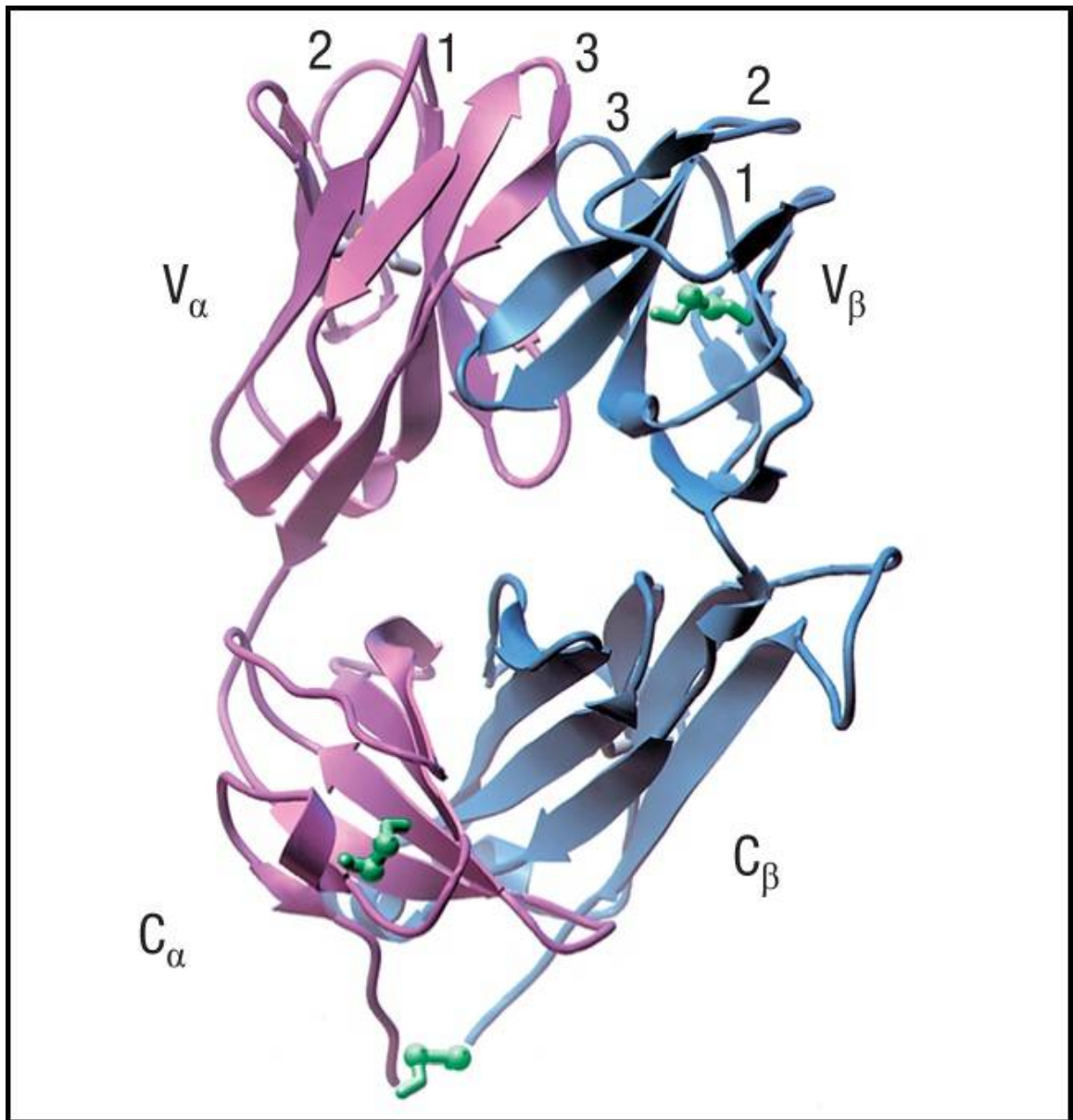


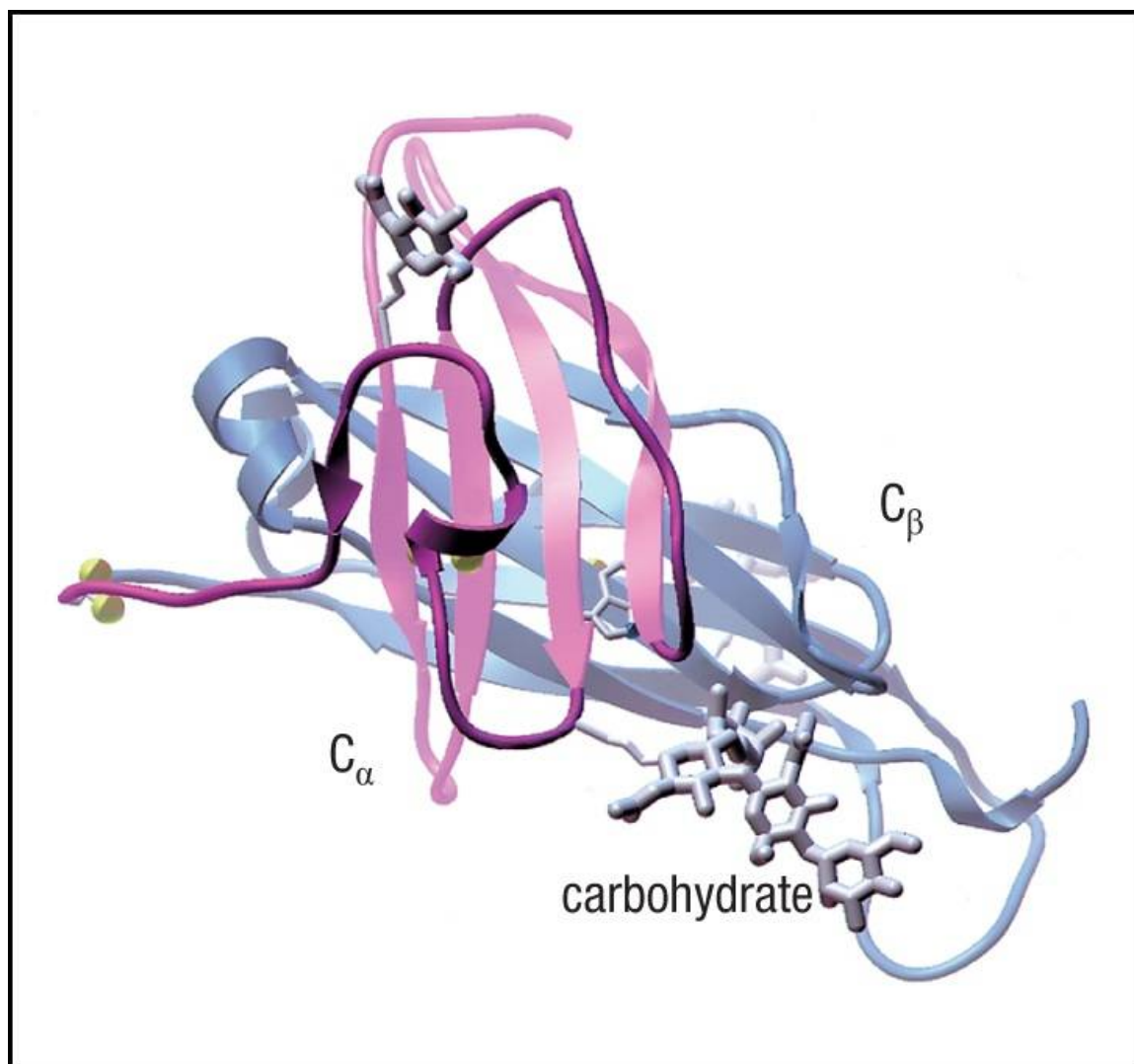
Fig. 4.18 Structure of the T-cell receptor. The T-cell receptor heterodimer is

composed of two transmembrane glycoprotein chains, α and β . The extracellular portion of each chain consists of two domains, resembling immunoglobulin V and C domains, respectively. Both chains have carbohydrate side chains attached to each domain. A short stalk segment, analogous to an immunoglobulin hinge region, connects the Ig-like domains to the membrane and contains the cysteine residue that forms the interchain disulfide bond. The transmembrane helices of both chains are unusual in containing positively charged (basic) residues within the hydrophobic transmembrane segment. The α chain carries two such residues; the β chain has one. These basic residues are required for assembly of the T-cell receptor $\alpha\beta$ heterodimer with the additional polypeptides that form the complete membrane-bound T-cell receptor complex.

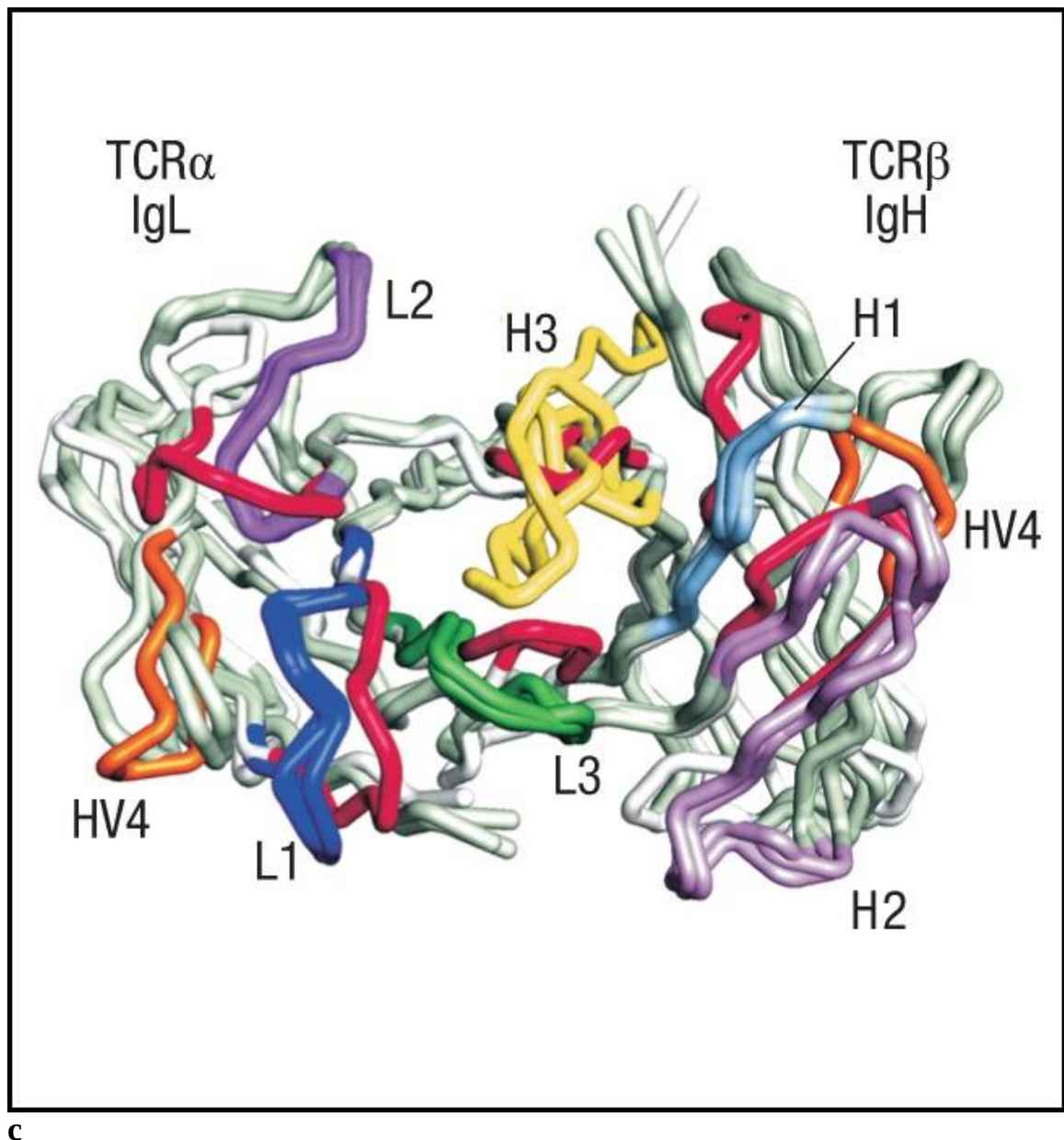
The three-dimensional structure of the T-cell receptor $\alpha\beta$ heterodimer determined by X-ray crystallography, illustrated in **Fig. 4.19a**, shows that T-cell receptor chains fold in much the same way as the regions comprising the Fab fragment shown in Fig. 4.1a. There are, however, some distinct structural differences between T-cell receptors and Fab fragments. The most striking is in the C_α domain, where the fold is unlike that of any other Ig-like domain. The half of the C_α domain that is juxtaposed with the C_β domain forms a β sheet similar to that found in other Ig-like domains, but the other half of the domain is formed of loosely packed strands and a short segment of α helix (**Fig. 4.19b**). In a C_α domain the intramolecular disulfide bond, which in Ig-like domains normally joins two β strands, joins a β strand to this segment of α helix.



a



b



c

Fig. 4.19 The crystal structure of an $\alpha\beta$ T-cell receptor resolved at 0.25 nm. In panels a and b, the α chain is shown in pink and the β chain in blue. Disulfide bonds are shown in green. In panel a, the T-cell receptor is viewed from the side as it would sit on a cell surface, with the CDR loops that form the antigen-binding site (labeled 1, 2, and 3) arrayed across its relatively flat top. In panel b, the C_α and C_β domains are shown. The C_α domain does not fold into a typical Ig-like domain; the face of the domain away from the C_β domain is mainly composed of irregular strands of polypeptide rather than β sheet. The intramolecular disulfide bond (far left) joins a β strand to this segment of α helix. The interaction between the C_α and C_β domains is assisted by carbohydrate (colored gray and labeled), with a sugar group from the C_α domain making hydrogen bonds to the C_β domain. In panel c, the T-cell receptor is shown aligned with the antigen-binding sites from three different antibodies. This view is looking down into the binding site. The V_α domain of the T-cell receptor is aligned with the V_L domains of the

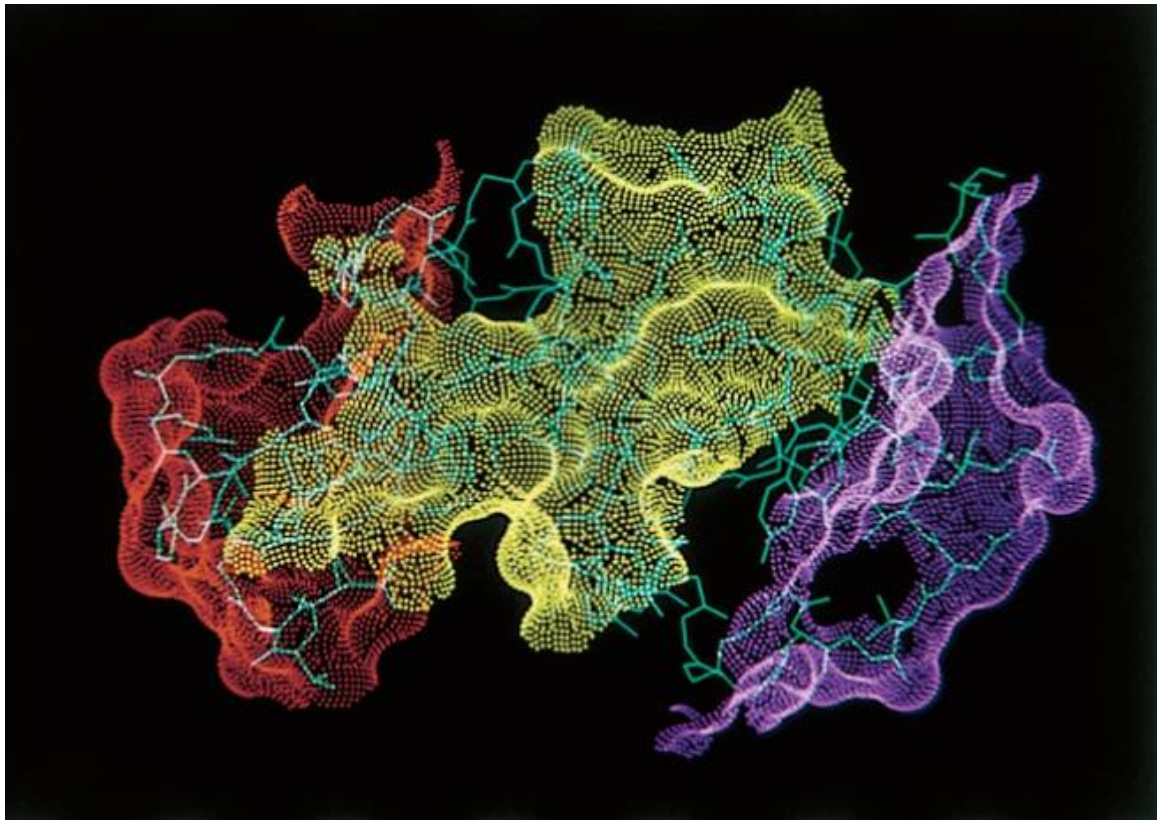
antigen-binding sites of the antibodies, and the V_β domain is aligned with the V_H domains. The CDRs of the T-cell receptor and immunoglobulin molecules are colored, with CDRs 1, 2, and 3 of the TCR shown in red and the HV4 loop in orange. For the immunoglobulin V domains, the CDR1 loops of the heavy chain (H1) and light chain (L1) are shown in light and dark blue, respectively, and the CDR2 loops (H2, L2) in light and dark purple, respectively. The heavy-chain CDR3 loops (H3) are in yellow; the light-chain CDR3 loops (L3) are in bright green. The HV4 loops of the TCR (orange) have no hypervariable counterparts in immunoglobulins.

There are also differences in the way in which the domains interact. The interface between the V and C domains of both T-cell receptor chains is more extensive than in most antibodies. The interaction between the C_α and C_β domains is distinctive, as it might be assisted by carbohydrates, with a sugar group from the C_α domain making a number of hydrogen bonds to the C_β domain (see Fig. 4.19b). Finally, a comparison of the variable binding sites shows that although the CDR loops align fairly closely with those of antibody molecules, there is some relative displacement (**Fig. 4.19c**). This is particularly marked in the V_α CDR2 loop, which is oriented at roughly right angles to the equivalent loop in antibody V domains, as a result of a shift in the β strand that anchors one end of the loop from one face of the domain to the other. A strand displacement also causes a change in the orientation of the V_β CDR2 loop in some V_β domains. These differences with antibodies influence the ability of the T-cell receptor to recognize their specific ligands, as we will discuss in the next section. In addition to the three hypervariable regions shared with immunoglobulins, the T-cell receptor has a fourth hypervariability region, HV4, in both of its chains (see Fig. 4.19c). These regions occur away from the antigen-binding face of the receptor and have been implicated in other functions of the TCR, such as superantigen binding, which we will describe in [Section 6-14](#).

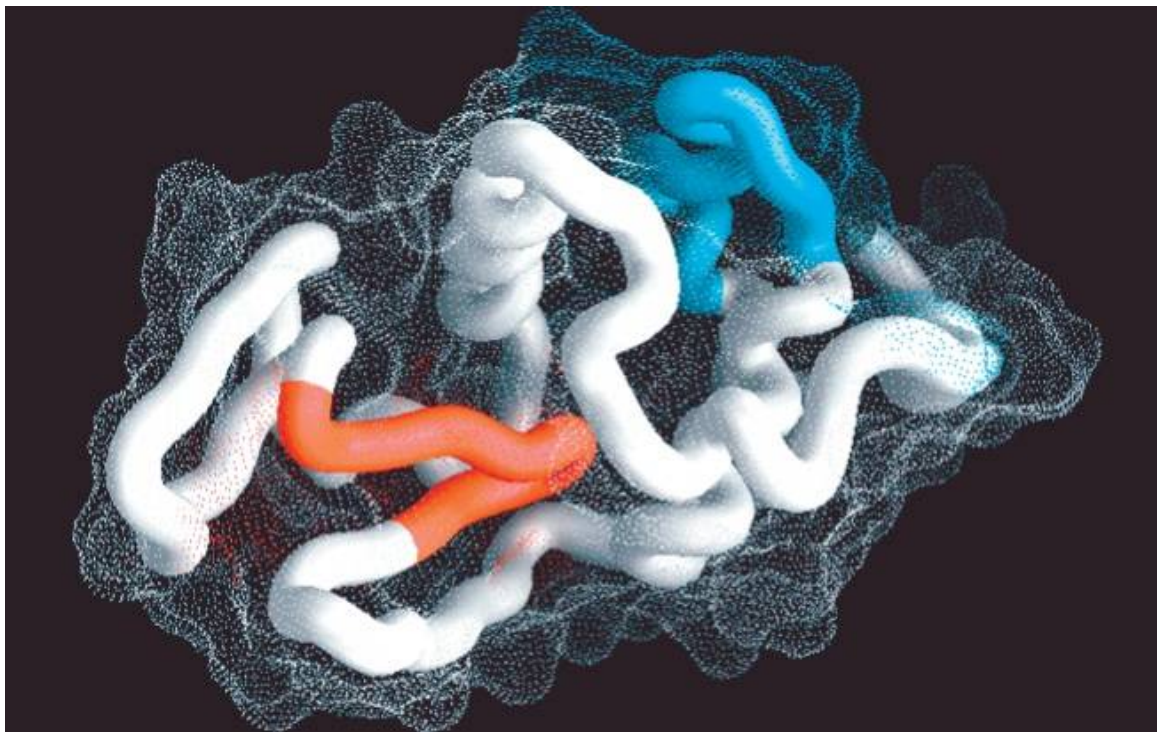
4-15 A T-cell receptor recognizes antigen in the form of a complex of a foreign peptide bound to an MHC molecule.

Antigen recognition by T-cell receptors clearly differs from recognition by B-cell receptors and antibodies. The immunoglobulin on B cells binds directly to the intact antigen, and, as discussed in [Section 4-11](#), antibodies typically bind to the surface of protein antigens, contacting amino acids that are discontinuous in the primary structure but are brought together in the folded protein. In contrast, $\alpha\beta$ T cells respond to short, continuous amino acid sequences. As we described in [Section 1-10](#), these peptide sequences are often buried within the native structure of the protein. Thus, antigens cannot be recognized directly by T-cell receptors unless the protein is unfolded and processed into peptide fragments (**Fig. 4.20**) and then presented by an MHC molecule (see Fig. 1.15). We will discuss this process in [Chapter](#)

6.



a



b

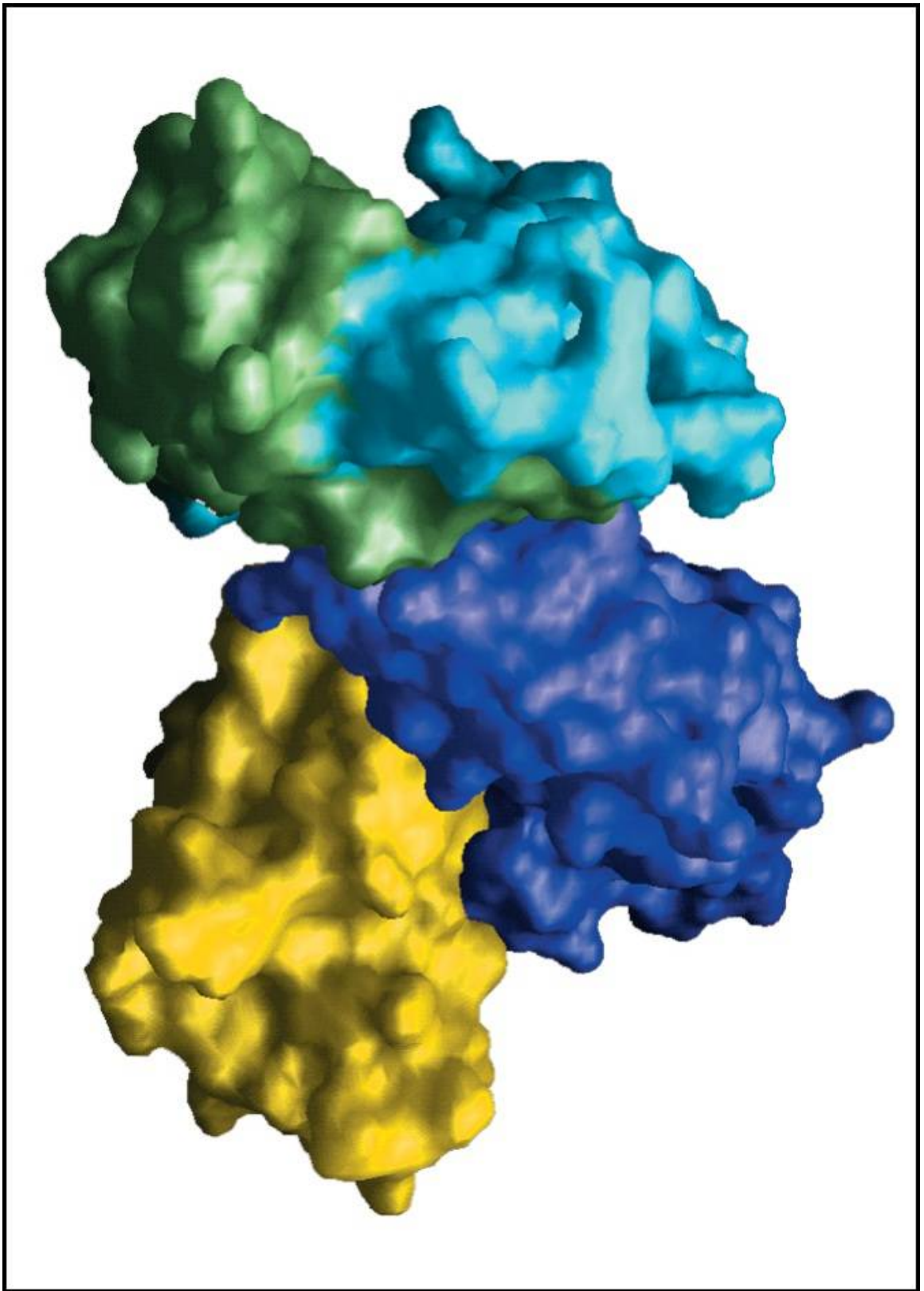
Fig. 4.20 Differences in the recognition of hen egg-white lysozyme by immunoglobulins and T-cell receptors. Antibodies can be shown by X-ray

crystallography to bind epitopes on the surface of proteins, as shown in panel a, where the epitopes for three antibodies are shown in different colors (red, yellow, purple) on the surface of hen egg-white lysozyme (see also Fig. 4.11). In contrast, the epitopes recognized by T-cell receptors need not lie on the surface of the molecule, because the T-cell receptor recognizes not the antigenic protein itself but a peptide fragment of the protein. The peptides corresponding to two T-cell epitopes of lysozyme are shown in panel b. One epitope, shown in blue, lies on the surface of the protein, but a second, shown in red, lies mostly within the core and is inaccessible in the folded protein. This indicates that T-cell receptors do not recognize their epitopes in the native protein.

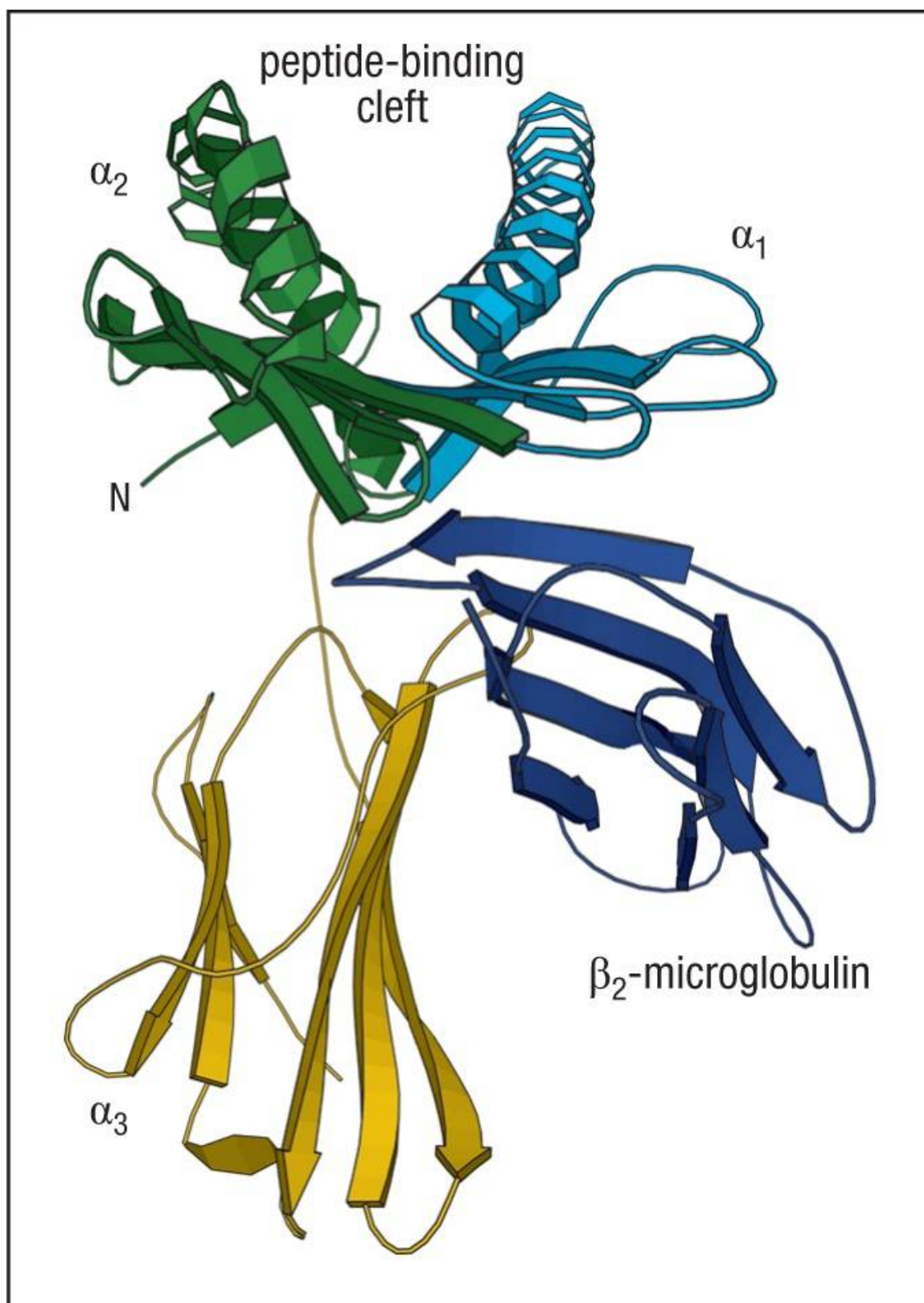
The nature of the antigen recognized by T cells became clear with the realization that the peptides that stimulate T cells are recognized only when bound to an MHC molecule. The ligand recognized by the T cell is thus a complex of peptide and MHC molecule. The evidence for involvement of the MHC in T-cell recognition of antigen was indirect at first, but it has been proven conclusively by stimulating T cells with purified peptide:MHC complexes. The T-cell receptor interacts with this ligand by making contacts with both the MHC molecule and the antigen peptide.

4-16 There are two classes of MHC molecules with distinct subunit compositions but similar three-dimensional structures.

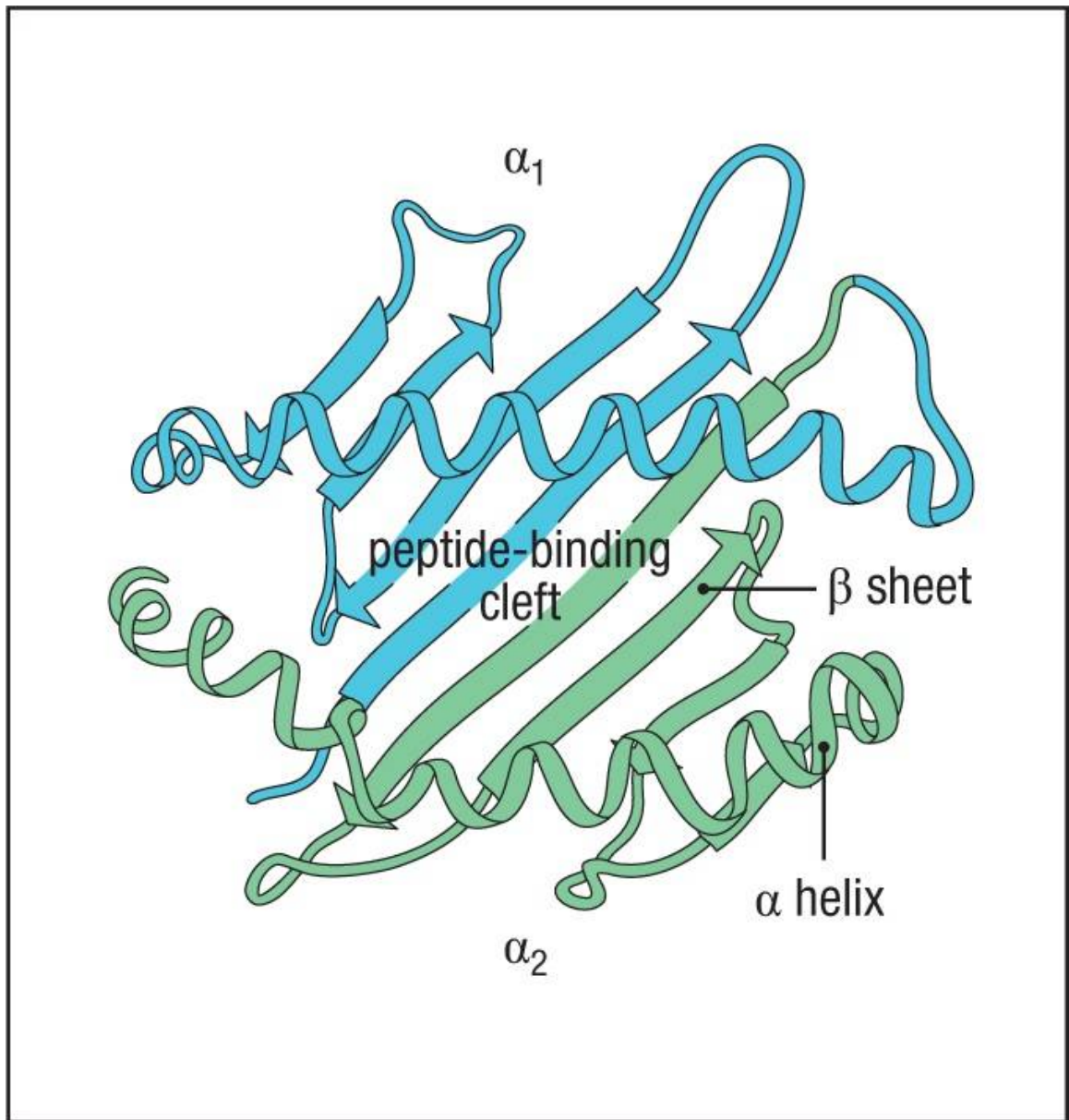
There are two classes of MHC molecules—[MHC class I](#) and [MHC class II](#)—and they differ in both their structure and their expression pattern in the tissues of the body. As shown in **Figs. 4.21** and **4.22**, MHC class I and MHC class II molecules are closely related in overall structure but differ in their subunit composition. In both classes, the two paired protein domains nearest to the membrane resemble immunoglobulin domains, whereas the two domains furthest away from the membrane fold together to create a long cleft, or groove, which is the site at which a peptide binds. Purified peptide:MHC class I and peptide:MHC class II complexes have been characterized structurally, allowing us to describe in detail both the MHC molecules themselves and the way in which they bind peptides.



a



b



c

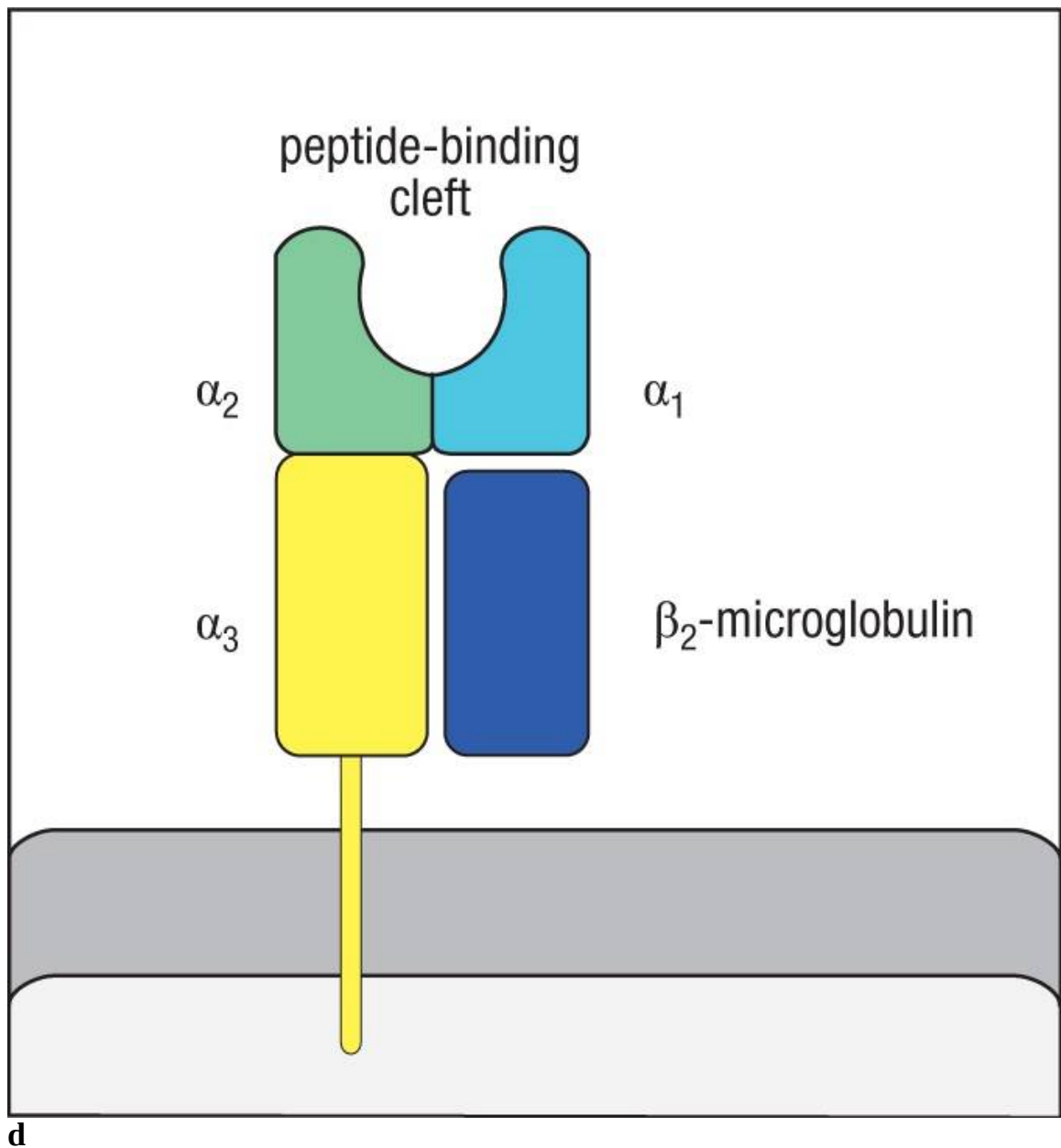
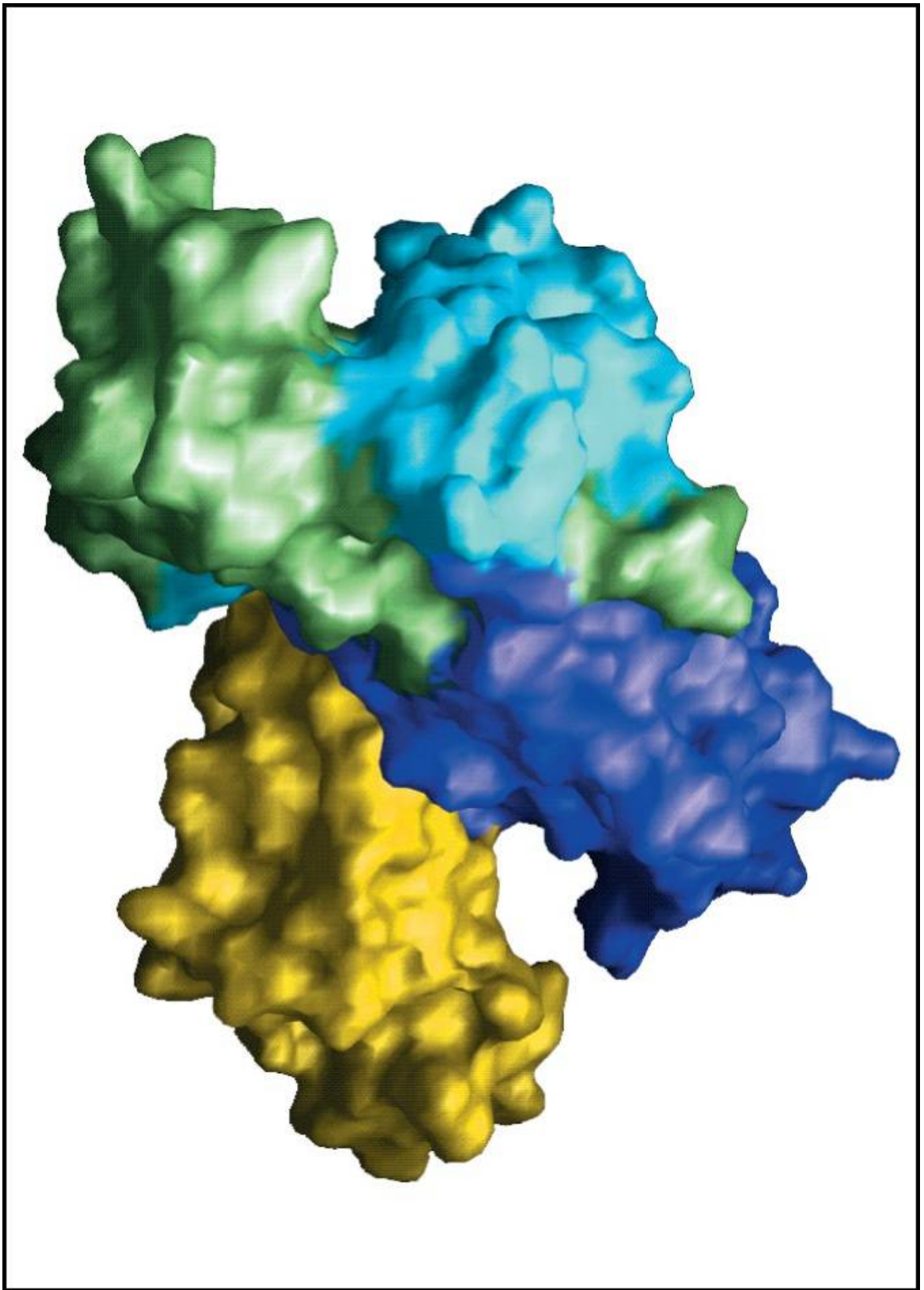
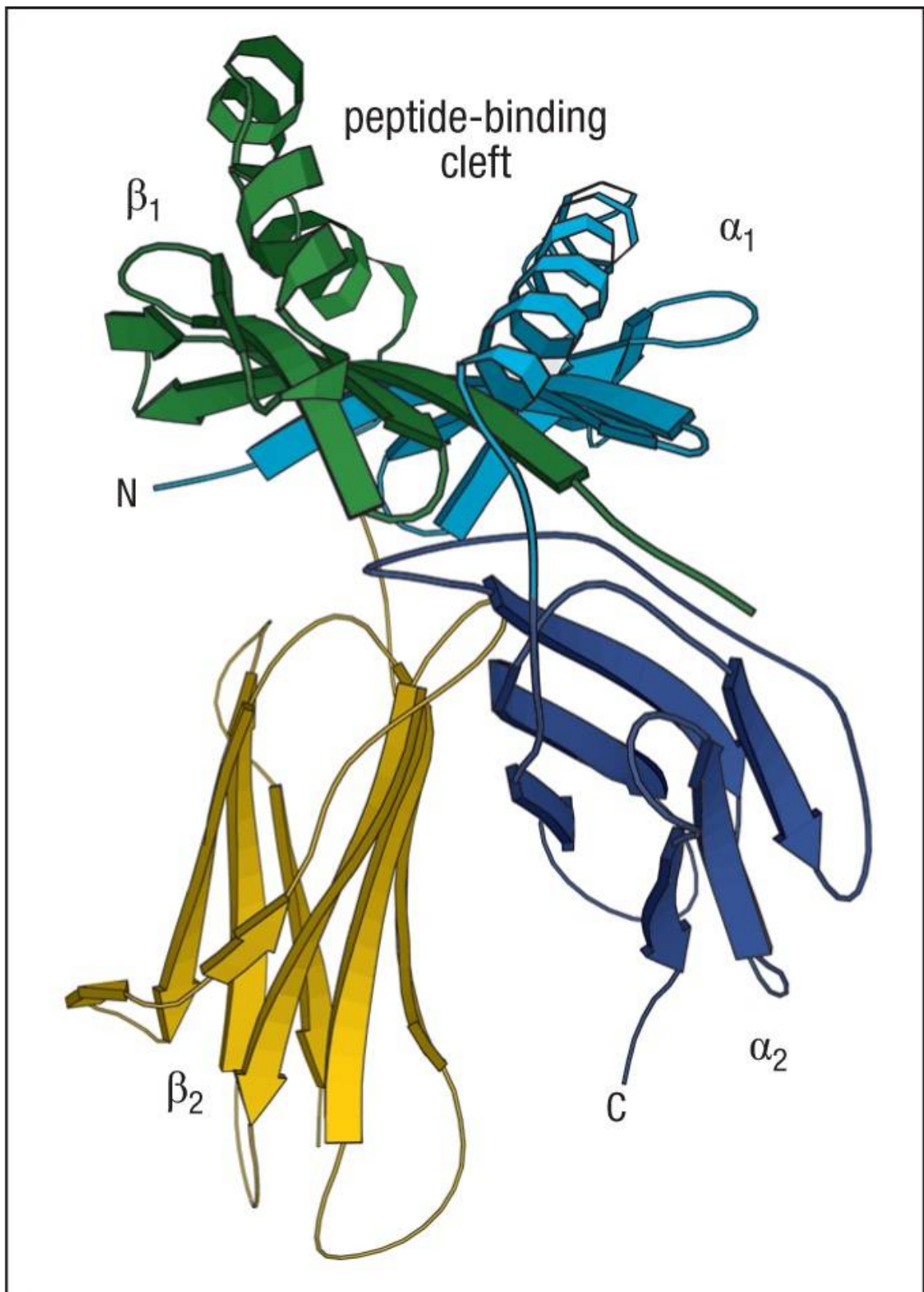


Fig. 4.21 The structure of an MHC class I molecule determined by X-ray crystallography. Panel a shows a computer-generated representation of a human MHC class I molecule, HLA-A2, which has been cleaved from the cell surface by the enzyme papain. The surface of the molecule is shown, colored according to the domains shown in panels b–d and described below. Panels b and c each show a ribbon diagram of the structure. Shown schematically in panel d, the MHC class I molecule is a heterodimer of a membrane-spanning α chain (molecular mass 43 kDa) bound noncovalently to β_2 -microglobulin (12 kDa), which does not span the membrane. The α chain folds into three domains: α_1 , α_2 , and α_3 . The α_3 domain and β_2 -microglobulin show similarities in amino acid sequence to immunoglobulin C domains and have similar folded structures, whereas the α_1 and α_2 domains are part of the same polypeptide and fold together into a single structure consisting of two separated α helices lying on a sheet of eight antiparallel β strands. The folding of the α_1 and α_2 domains creates a long cleft or

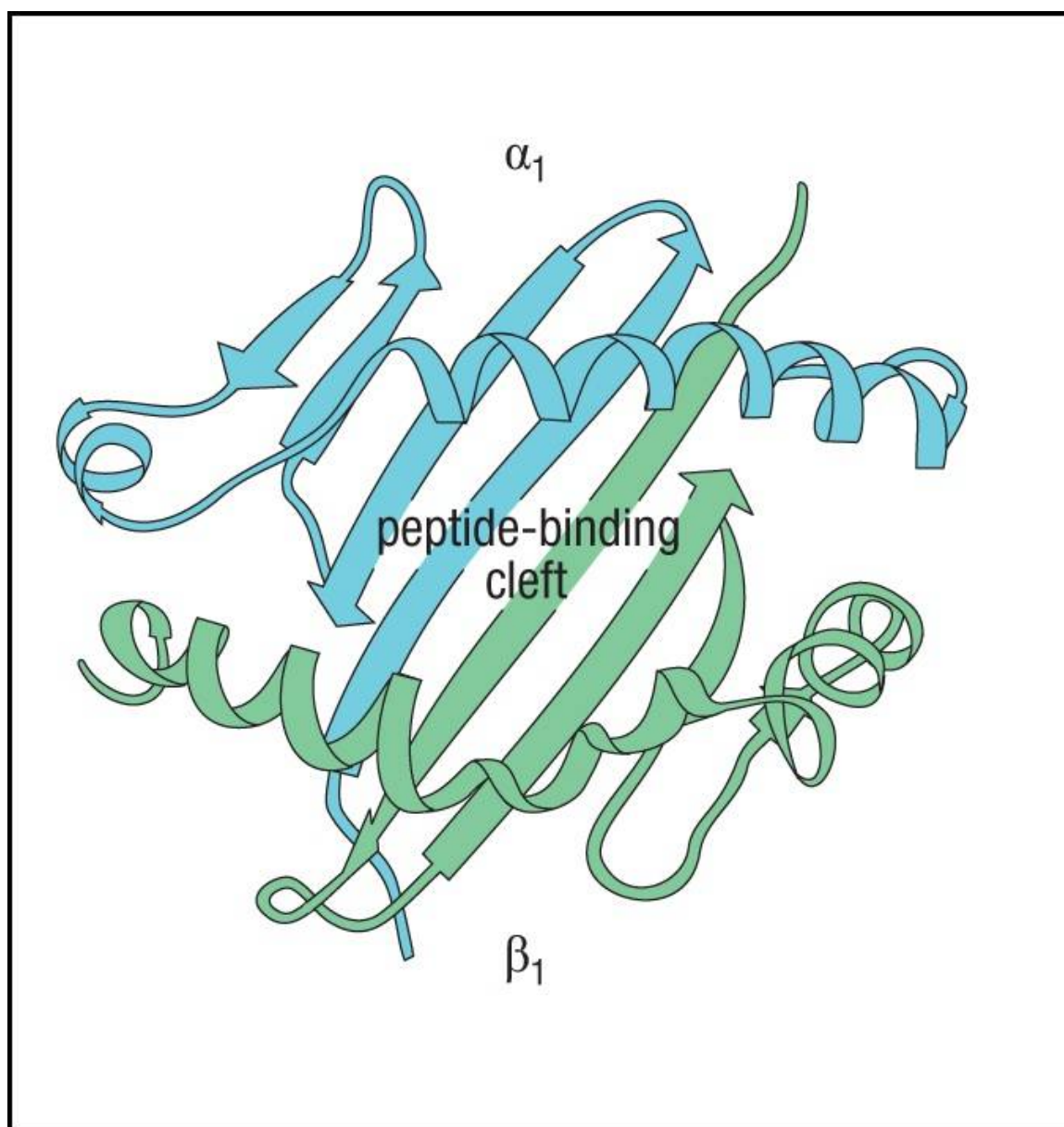
groove, which is the site at which peptide antigens bind to the MHC molecules. For class I molecules, this groove is open at only one end. The transmembrane region and the short stretch of peptide that connects the external domains to the cell surface are not seen in panels a and b because they have been removed by the digestion with papain. As can be seen in panel c, looking down on the molecule from above, the sides of the cleft are formed from the inner faces of the two α helices; the β pleated sheet formed by the pairing of the α_1 and α_2 domains creates the floor of the cleft.



a



b



c

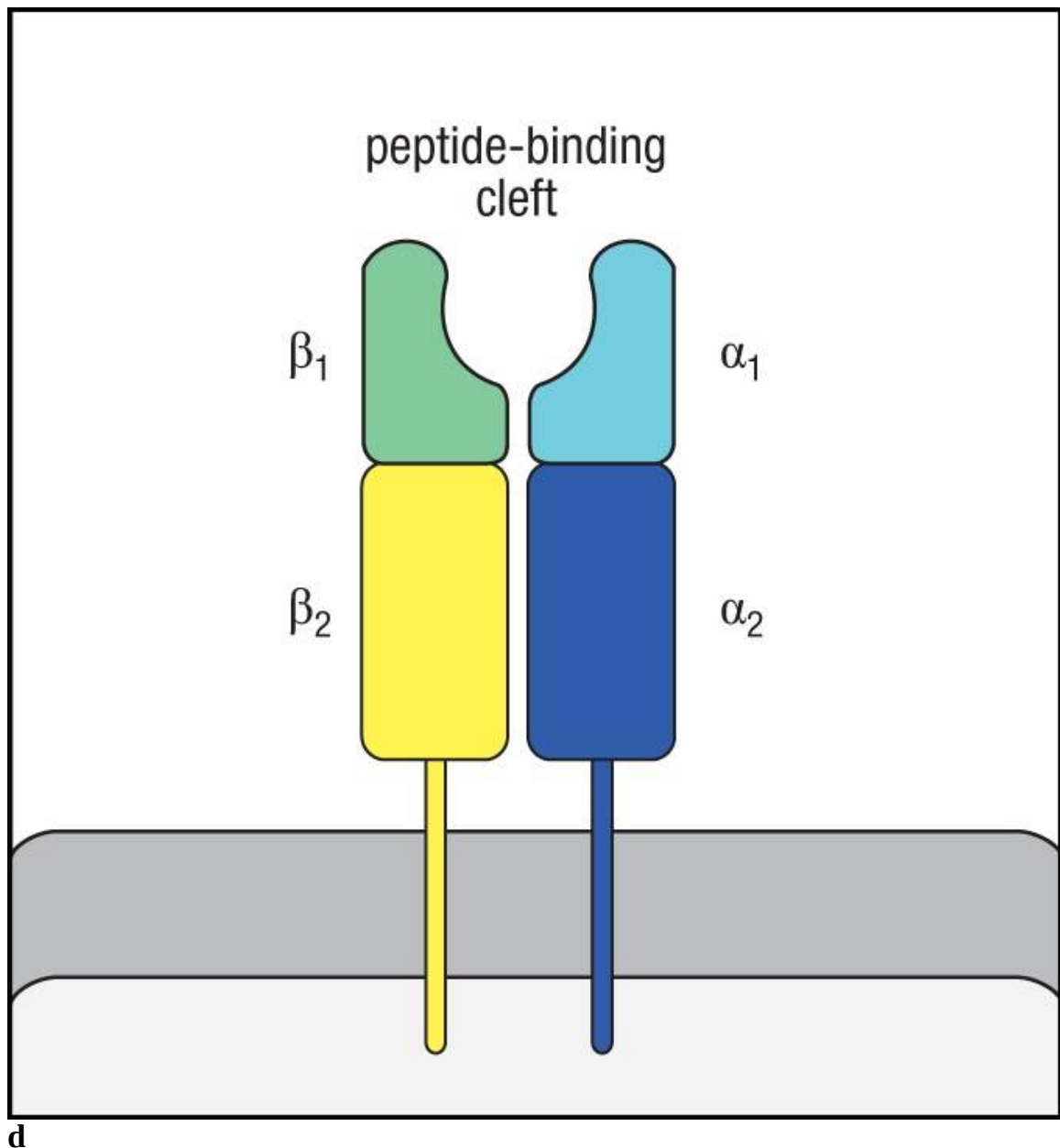
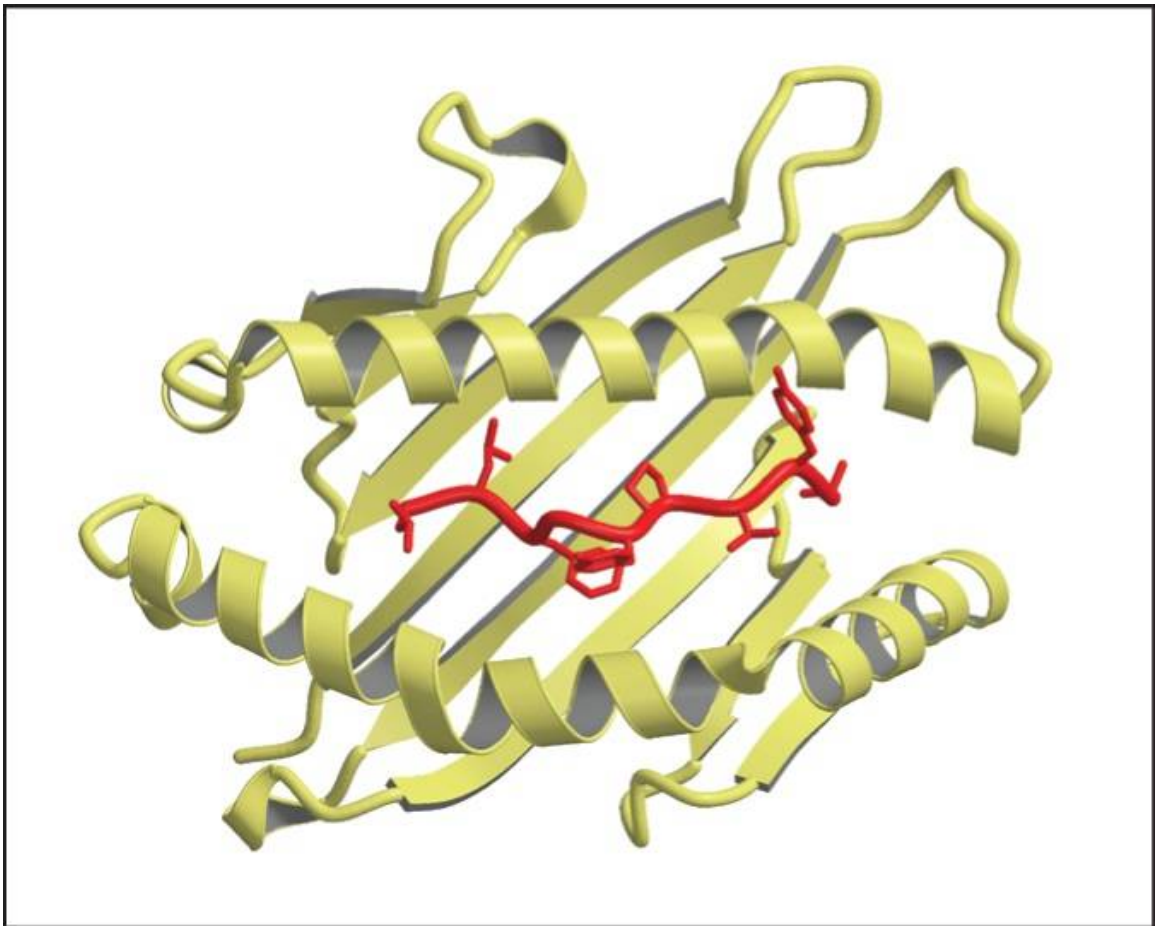


Fig. 4.22 MHC class II molecules resemble MHC class I molecules in overall structure. The MHC class II molecule is composed of two transmembrane glycoprotein chains, α (34 kDa) and β (29 kDa), as shown schematically in panel d. Each chain has two domains, and the two chains together form a compact four-domain structure similar to that of the MHC class I molecule (compare with panel d of Fig. 4.21). Panel a shows a computer-generated representation of the surface of the MHC class II molecule, in this case the human protein HLA-DR1, and panel b shows the equivalent ribbon diagram. N, amino terminus; C, carboxyl terminus. The α_2 and β_2 domains, like the α_3 and β_2 -microglobulin domains of the MHC class I molecule, have amino acid sequence and structural similarities to immunoglobulin C domains; in the MHC class II molecule the two domains forming the peptide-binding cleft are contributed by different chains and are therefore not joined by a covalent bond (see panels c and d). Another important difference, not apparent in this diagram, is that the peptide-binding groove of the MHC

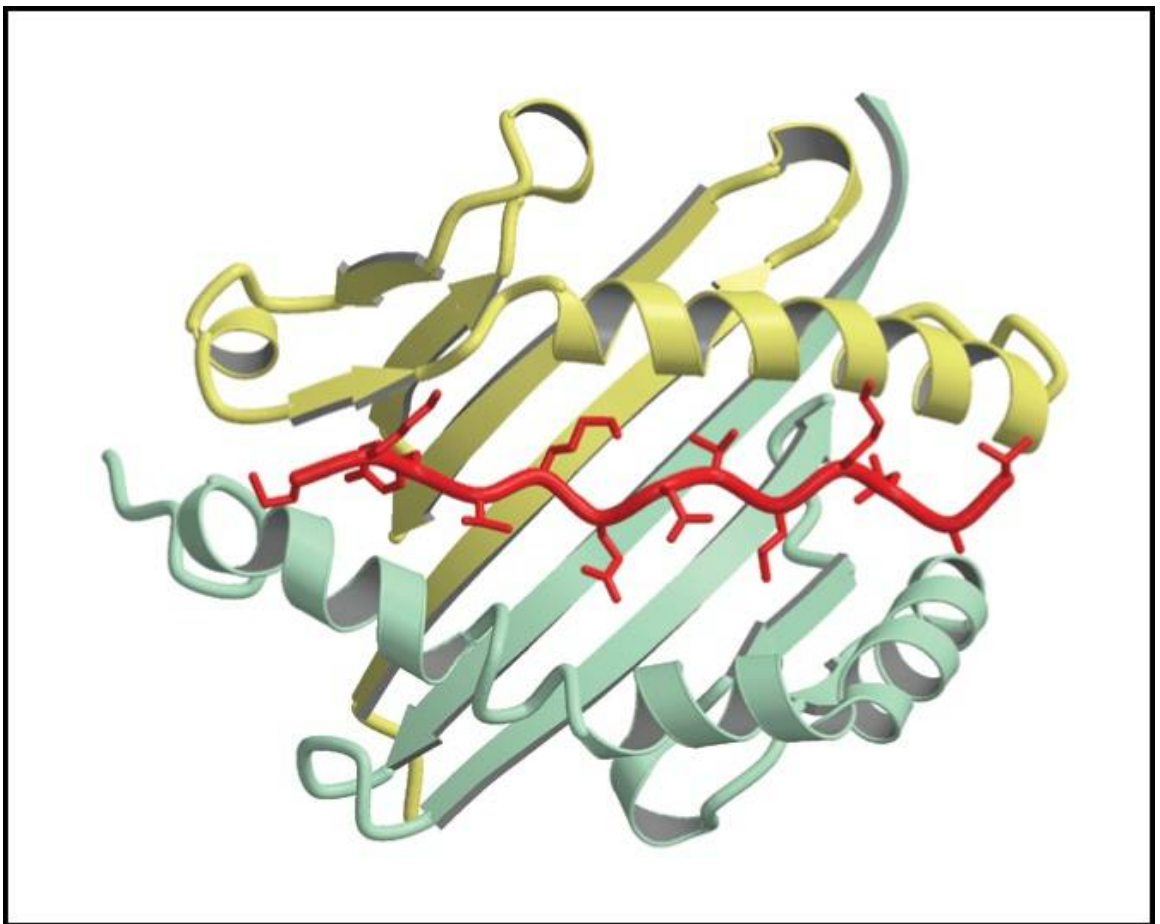
class II molecule is open at both ends.

MHC class I molecules (see Fig. 4.21) consist of two polypeptide chains. One chain, the α chain, is encoded in the MHC (on chromosome 6 in humans) and is noncovalently associated with a smaller chain, [\$\beta_2\$ -microglobulin](#), which is encoded on a different chromosome—chromosome 15 in humans. Only the class I α chain spans the membrane. The complete MHC class I molecule has four domains, three formed from the MHC-encoded α chain, and one contributed by β_2 -microglobulin. The α_3 domain and β_2 -microglobulin closely resemble Ig-like domains in their folded structure. The folded α_1 and α_2 domains form the walls of a cleft on the surface of the molecule; because this is where the peptide binds, this part of the MHC molecule is known as the [peptide-binding cleft or peptide-binding groove](#). The MHC molecules are highly polymorphic, and the major differences between the different allelic forms are located in the peptide-binding cleft, influencing which peptides will bind and thus the specificity of the peptide:MHC antigen presented to T cells. By contrast, β_2 -microglobulin, which does not contribute directly to peptide binding, is not polymorphic.

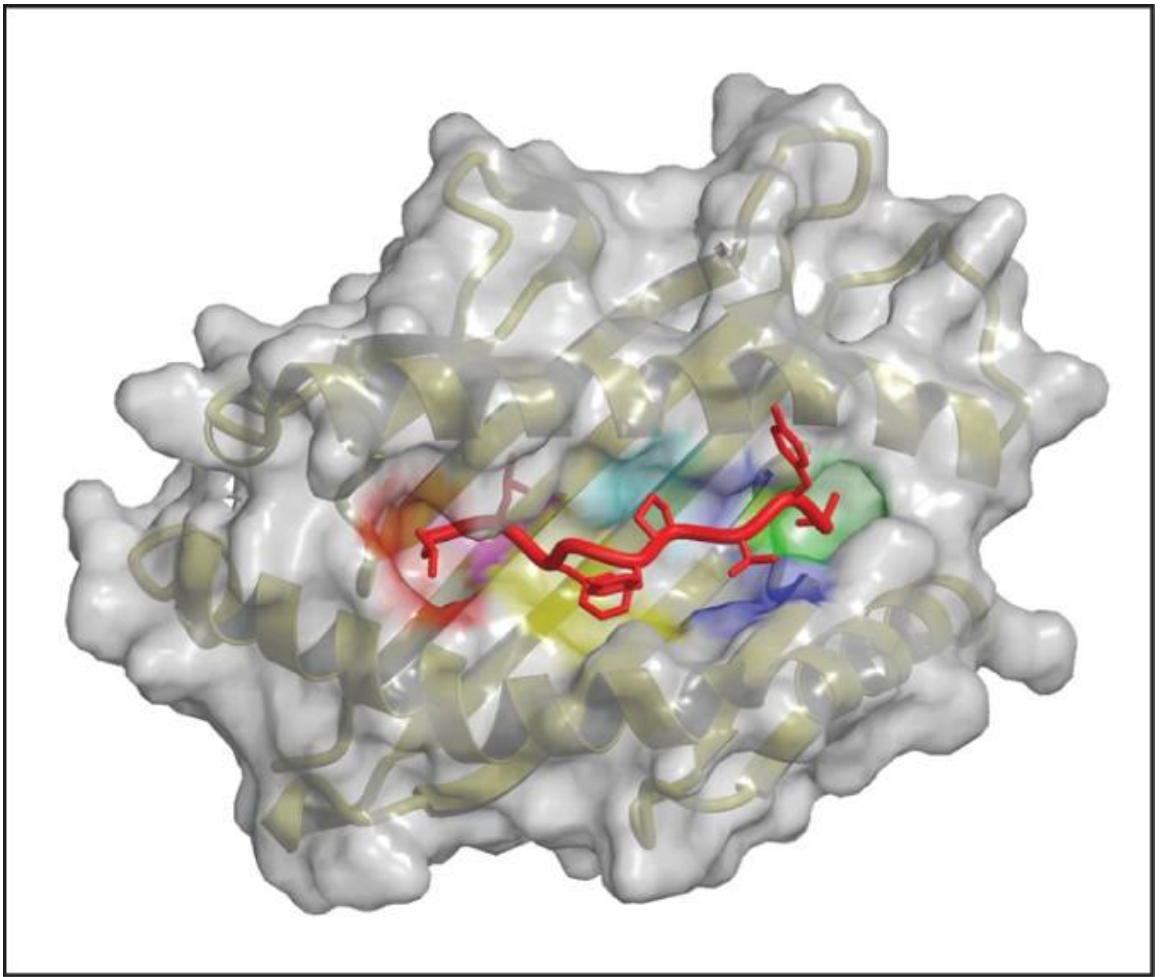
An MHC class II molecule consists of a noncovalent complex of two chains, α and β , both of which span the membrane (see Fig. 4.22). The MHC class II α chain is a different protein from the class I α chain. The MHC class II α and β chains are both encoded within the MHC. The crystallographic structure of the MHC class II molecule shows that it is folded very much like the MHC class I molecule, but the peptide-binding cleft is formed by two domains from different chains, the α_1 and β_1 domains. The major differences lie at the ends of the peptide-binding cleft, which are more open in MHC class II than in MHC class I molecules. Consequently, the ends of a peptide bound to an MHC class I molecule are substantially buried within the molecule, whereas the ends of peptides bound to MHC class II molecules are not. In both MHC class I and class II molecules, bound peptides are sandwiched between the two α -helical segments of the MHC molecule (**Fig. 4.23**). The T-cell receptor interacts with this compound ligand, making contacts with both the MHC molecule and the peptide antigen. As in the case of MHC class I molecules, the sites of major polymorphism in MHC class II molecules are located in the peptide-binding cleft.



a



b



c

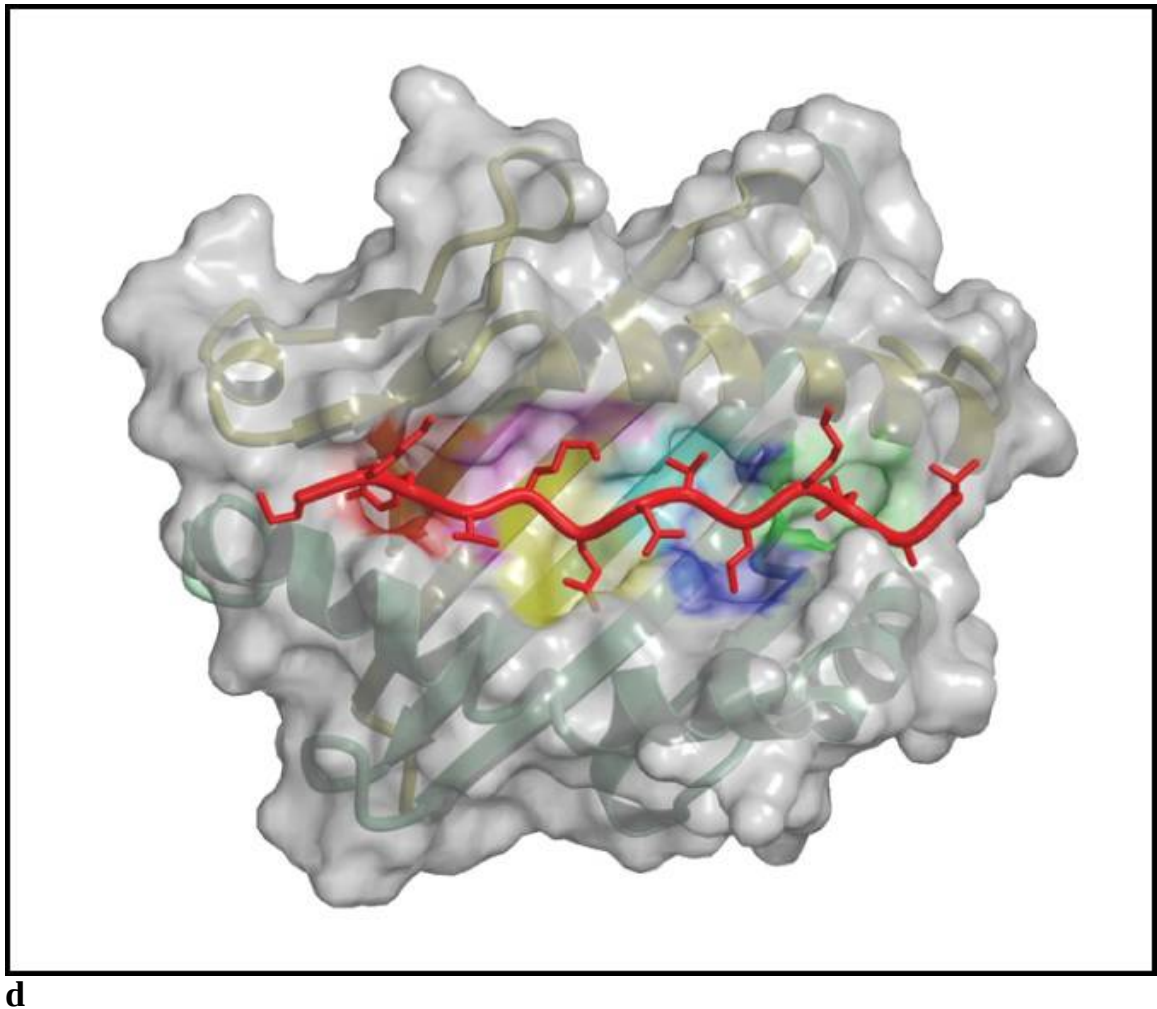


Fig. 4.23 MHC molecules bind peptides tightly within the cleft. When MHC molecules are crystallized with a single synthetic peptide antigen, the details of peptide binding are revealed. In MHC class I molecules (panels a and c), the peptide is bound in an elongated conformation with both ends tightly bound at either end of the cleft. In MHC class II molecules (panels b and d), the peptide is also bound in an elongated conformation, but the ends of the peptide are not tightly bound and the peptide extends beyond the cleft. The upper surface of the peptide:MHC complex is recognized by T cells and is composed of residues of the MHC molecule and the peptide. The amino acid side chains of the peptide insert into pockets in the peptide-binding groove of the MHC molecule; these pockets are lined with residues that are polymorphic within the MHC. In representations c and d, the surfaces of the different pockets for the different amino acids are depicted as areas of different colors.

4-17 Peptides are stably bound to MHC molecules and also serve to stabilize the MHC molecule on the cell surface.

An individual can be infected by a wide variety of pathogens, whose proteins will not generally have peptide sequences in common. For T cells to be able to detect the widest possible array of infections, the MHC molecules (both class I and class II) of an individual should be able to bind stably to many different peptides. This behavior is quite distinct from that of other peptide-binding receptors, such as those for peptide hormones, which usually bind only a single type of peptide. The crystal structures of peptide:MHC complexes have helped to show how a single binding site can bind a peptide with high affinity while retaining the ability to bind a wide variety of different peptides.

An important feature of the binding of peptides to MHC molecules is that the peptide is bound as an integral part of the MHC molecule's structure, and MHC molecules are unstable when peptides are not bound. This dependence on bound peptide applies to both MHC class I and MHC class II molecules. Stable peptide binding is important, because otherwise peptide exchanges occurring at the cell surface would prevent peptide:MHC complexes from being reliable indicators of infection or of uptake of a specific antigen. When MHC molecules are purified from cells, their stably bound peptides co-purify with them, and this fact has allowed the peptides bound by particular MHC molecules to be analyzed. Peptides are released from the MHC molecules by denaturing the complex in acid, and they are then purified and sequenced. Pure synthetic peptides can also be incorporated into empty MHC molecules and the structure of the complex determined, revealing details of the contacts between the MHC molecule and the peptide. From such studies a detailed picture of the binding interactions has been built up. We discuss the peptide-binding properties of MHC class I molecules first.

4-18 MHC class I molecules bind short peptides of 8–10 amino acids by both ends.

Binding of a peptide to an MHC class I molecule is stabilized at both ends of the peptide-binding cleft by contacts between atoms in the free amino and carboxyl termini of the peptide and invariant sites that are found at each end of the cleft in all MHC class I molecules (**Fig. 4.24**). These are thought to be the main stabilizing contacts for peptide:MHC class I complexes, because synthetic peptide analogs lacking terminal amino and carboxyl groups fail to bind stably to MHC class I molecules. Other residues in the peptide serve as additional anchors. Peptides that bind to MHC class I molecules are usually 8–10 amino acids long. However, longer peptides are thought to bind, particularly if they can bind at their carboxyl terminus, but they are subsequently shortened to 8–10 amino acids through cleavage by exopeptidases present in the endoplasmic reticulum, which is where MHC class I molecules bind peptides. The peptide lies in an elongated conformation along the cleft; variations in peptide length seem to be accommodated, in most cases, by the peptide bulging out of the binding groove. However, in some cases, length variation can also be accommodated in MHC class I molecules by allowing the peptide to extend out of the cleft at the carboxyl terminus.

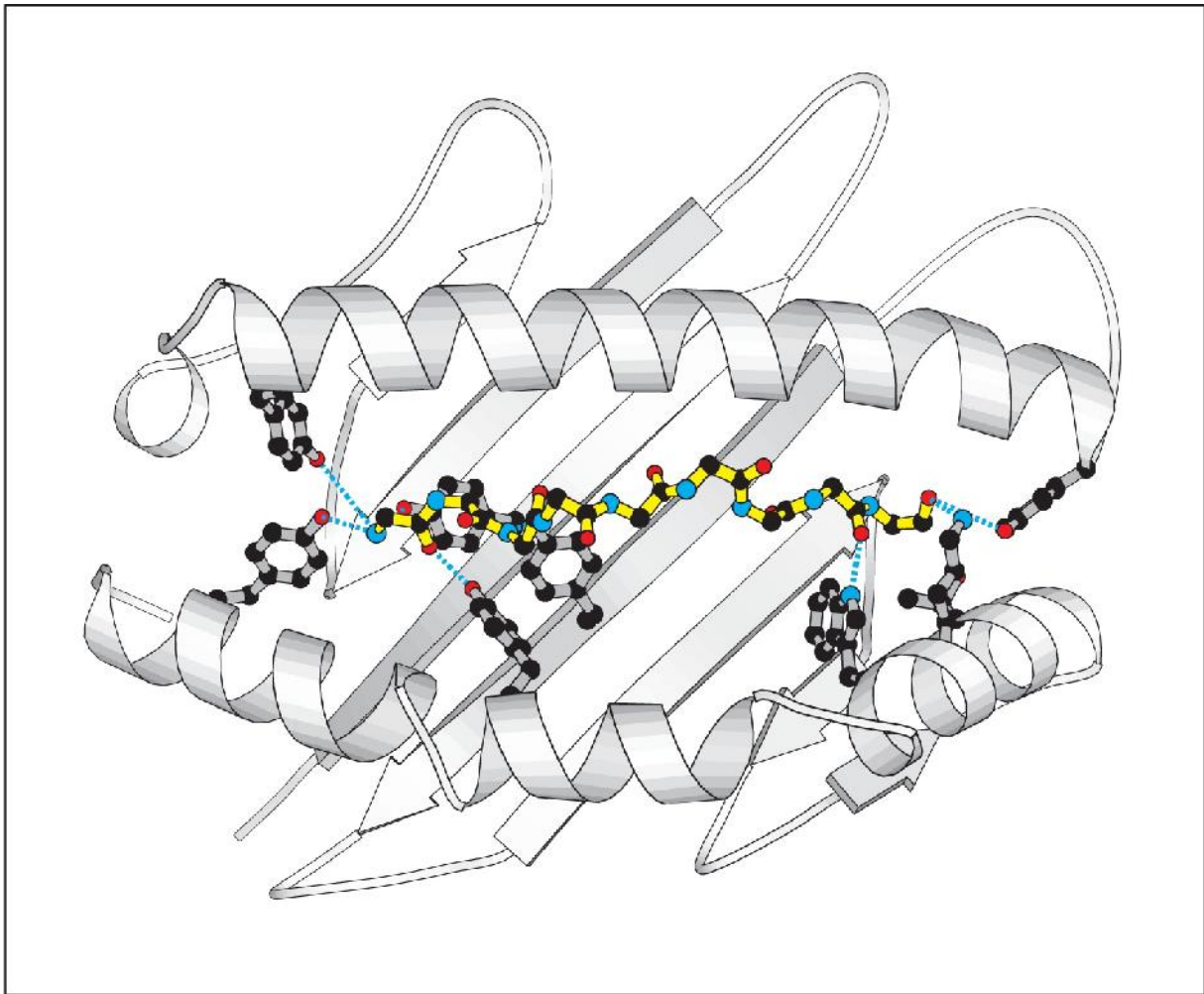


Fig. 4.24 Peptides are bound to MHC class I molecules by their ends. MHC class I molecules interact with the backbone of a bound peptide (shown in yellow) through a series of hydrogen bonds and ionic interactions (shown as dotted blue lines) at each end of the peptide. The amino terminus of the peptide is to the left, the carboxyl terminus to the right. Black circles are carbon atoms; red are oxygen; blue are nitrogen. The amino acid residues in the MHC molecule that form these bonds are common to all MHC class I molecules, and their side chains are shown in full (in gray) on a ribbon diagram of the MHC class I groove. A cluster of tyrosine residues common to all MHC class I molecules forms hydrogen bonds to the amino terminus of the bound peptide, while a second cluster of residues forms hydrogen bonds and ionic interactions with the peptide backbone at the carboxyl terminus and with the carboxyl terminus itself.

H_3N^+	R	G	Y	V	Y	Q	Q	L	COO^-
H_3N^+	S	I	I	N	F	E	K	L	COO^-
H_3N^+	A	P	G	N	Y	P	A	L	COO^-

H_3N^+	T	Y	Q	R	T	R	A	L	V	COO^-
H_3N^+	S	Y	F	P	E	I	T	H	I	COO^-
H_3N^+	K	Y	Q	A	V	T	T	T	L	COO^-
H_3N^+	S	Y	I	P	S	A	E	K	I	COO^-

Fig. 4.25 Peptides bind to MHC molecules through structurally related anchor residues. The sequences of two sets of peptides eluted from two different MHC class I molecules are shown in the upper and lower panels. The anchor residues (green) differ for peptides that bind different allelic variants of MHC class I molecules but are similar for all peptides that bind to the same MHC molecule. The anchor residues that bind a particular MHC molecule need not be identical but are always related; for example, phenylalanine (F) and tyrosine (Y) are both aromatic amino acids, whereas valine (V), leucine (L), and isoleucine (I) are all large hydrophobic amino acids. Peptides also bind to MHC class I molecules through their amino (blue) and carboxyl (red) termini.

These interactions give MHC class I molecules a broad peptide-binding specificity. In addition, MHC molecules are highly polymorphic. As mentioned earlier, MHC genes are highly polymorphic, and there are hundreds of different allelic variations of the MHC class I genes in the human population. Each individual carries only a small selection of these variants. The main differences between allelic MHC variants are found at certain sites in the peptide-binding cleft, resulting in different amino acids in key peptide-interaction sites. For this reason, different MHC variants preferentially bind different peptides. The peptides that can bind to a given MHC variant have the same or very similar amino acid residues at two or three particular positions along the peptide sequence. The amino acid side chains at these positions insert into pockets in the MHC molecule that are lined by the polymorphic amino acids. Because this binding anchors the peptide to the MHC molecule, the peptide residues

involved are called the [anchor residues](#), as illustrated in **Fig. 4.25**. Both the position and identity of these anchor residues can vary, depending on the particular MHC class I variant that is binding the peptide. However, most peptides that bind to MHC class I molecules have a hydrophobic (or sometimes basic) residue at the carboxyl terminus that also serves to anchor the peptide in the groove. Whereas changing an anchor residue will in most cases prevent the peptide from binding, not every synthetic peptide of suitable length that contains these anchor residues will bind the appropriate MHC class I molecule, and so the overall binding must also depend on the nature of the amino acids at other positions in the peptide. In some cases, particular amino acids are preferred in certain positions, whereas in others the presence of particular amino acids prevents binding. These additional amino acid positions are called ‘secondary anchors.’ These features of peptide binding enable an individual MHC class I molecule to bind a wide variety of different peptides, yet allow different MHC class I allelic variants to bind different sets of peptides. As we will see in [Chapter 15](#), MHC polymorphisms also impact the binding of peptides derived from self proteins and can influence the susceptibility of an individual to various autoimmune diseases.

4-19 The length of the peptides bound by MHC class II molecules is not constrained.

Like MHC class I molecules, MHC class II molecules that lack bound peptide are unstable. Peptide binding to MHC class II molecules has also been analyzed by elution of bound peptides and by X-ray crystallography, and it differs in several ways from peptide binding to MHC class I molecules. Natural peptides that bind to MHC class II molecules are at least 13 amino acids long and can be much longer. The clusters of conserved residues that bind the two ends of a peptide in MHC class I molecules are not found in MHC class II molecules, and the ends of the peptide are not bound. Instead, the peptide lies in an extended conformation along the peptide-binding cleft. It is held there both by peptide side chains that protrude into shallow and deep pockets lined by polymorphic residues and by interactions between the peptide backbone and the side chains of conserved amino acids that line the peptide-binding cleft in all MHC class II molecules (**Fig. 4.26**).

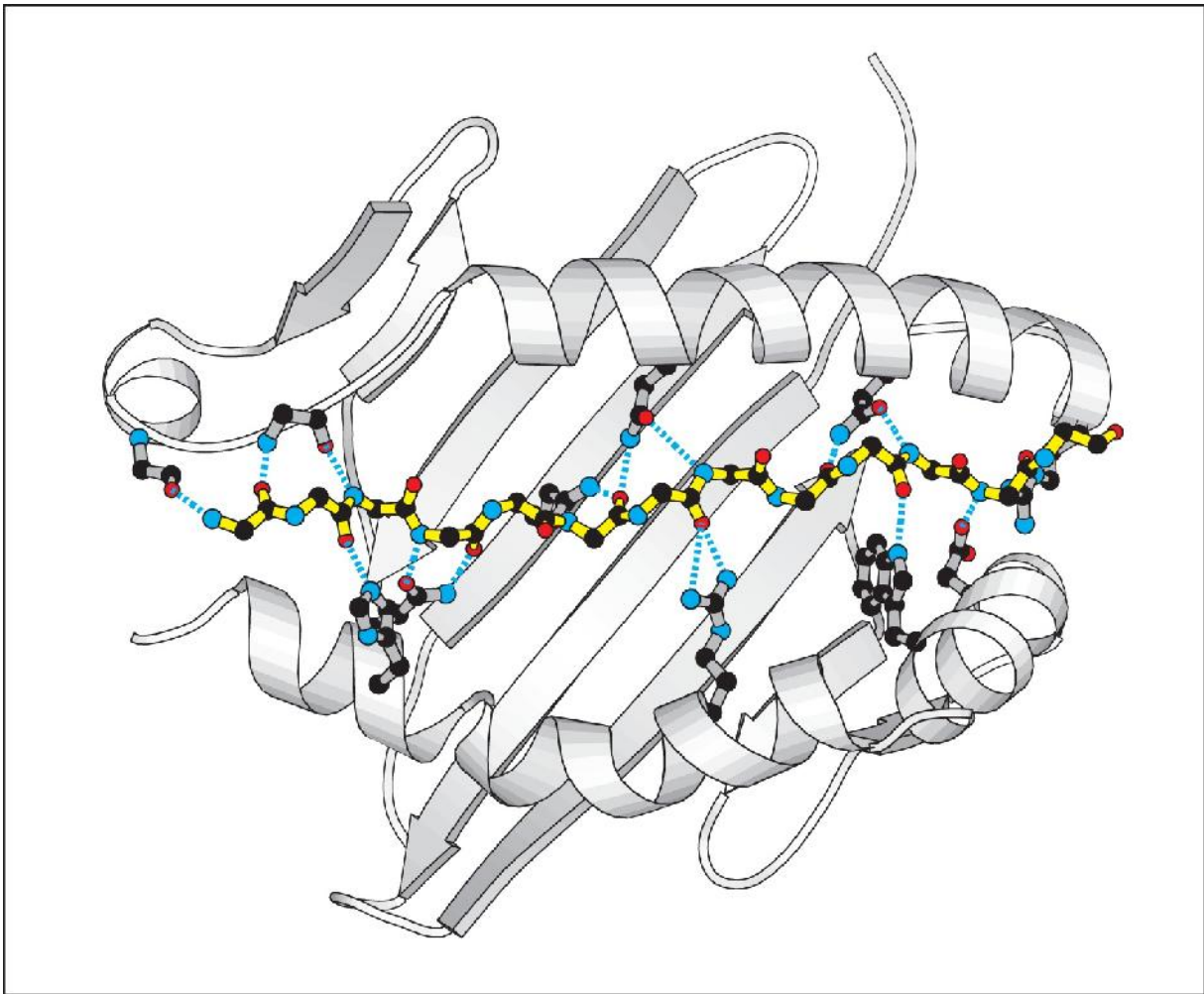


Fig. 4.26 Peptides bind to MHC class II molecules by interactions along the length of the binding groove. A peptide (yellow; shown as the peptide backbone only, with the amino terminus to the left and the carboxyl terminus to the right) is bound by an MHC class II molecule through a series of hydrogen bonds (dotted blue lines) that are distributed along the length of the peptide. The hydrogen bonds toward the amino terminus of the peptide are made with the backbone of the MHC class II polypeptide chain, whereas throughout the peptide's length, bonds are made with residues that are highly conserved in MHC class II molecules. The side chains of these residues are shown in gray on the ribbon diagram of the MHC class II groove.

The binding pockets of MHC class II molecules accommodate a greater variety of side chains than do those of MHC class I molecules, making it more difficult to define anchor residues and to predict which peptides will be able to bind a particular MHC class II variant (**Fig. 4.27**). Nevertheless, by comparing the sequences of known binding peptides, it is usually possible to detect patterns of amino acids that permit binding to different MHC class II variants and to model how the amino acids of this peptide sequence motif will interact with the amino acids of the peptide-binding cleft. For example, structural data show that amino acid side chains at residues 1, 4, 6, and 9 of peptides bound to HLA-DR3 are held in pockets lining the binding site (see **Fig. 4.27**). Because the peptide is bound by its backbone

and allowed to emerge from both ends of the binding groove, there is, in principle, no upper limit to the length of peptides that could bind to MHC class II molecules. An example is the protein known as [invariant chain](#), part of which lies entirely across the peptide-binding groove of nascent MHC class II molecules during their synthesis in the endoplasmic reticulum. We will return in [Chapter 6](#) to the role of the invariant chain in the loading of peptides onto MHC class II molecules. In most cases, long peptides bound to MHC class II molecules are trimmed by peptidases to a length of around 13–17 amino acids.

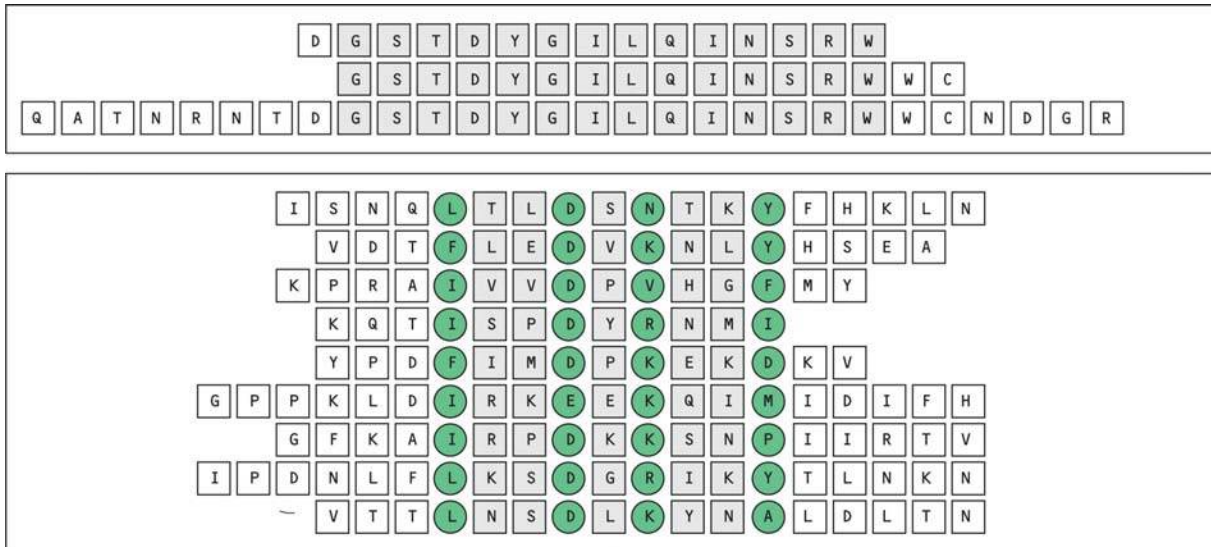
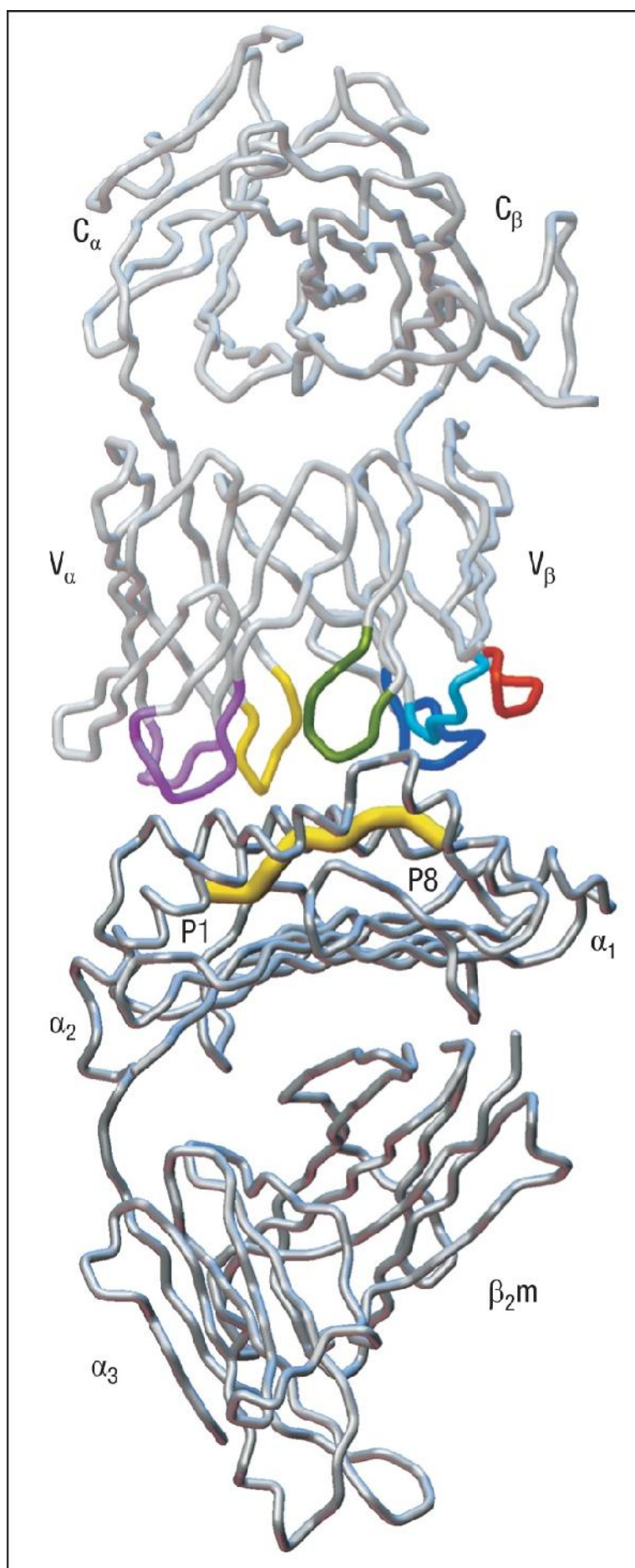


Fig. 4.27 Peptides that bind MHC class II molecules are variable in length, and their anchor residues lie at various distances from the ends of the peptide. The sequences of a set of peptides that bind to the mouse MHC class II A^k protein are shown in the upper panel. All contain the same core sequence (shaded) but differ in length. In the lower panel, different peptides binding to the human MHC class II protein HLA-DR3 are shown. Anchor residues are shown as green circles. The lengths of these peptides can vary, and so by convention the first anchor residue is denoted as residue 1. Note that all of the peptides share a hydrophobic residue in position 1, a negatively charged residue [aspartic acid (D) or glutamic acid (E)] in position 4, and a tendency to have a basic or polar residue [lysine (K), arginine (R), histidine (H), glutamine (Q), or asparagine (N)] in position 6 and a hydrophobic residue [for example, tyrosine (Y), leucine (L), or phenylalanine (F)] in position 9.

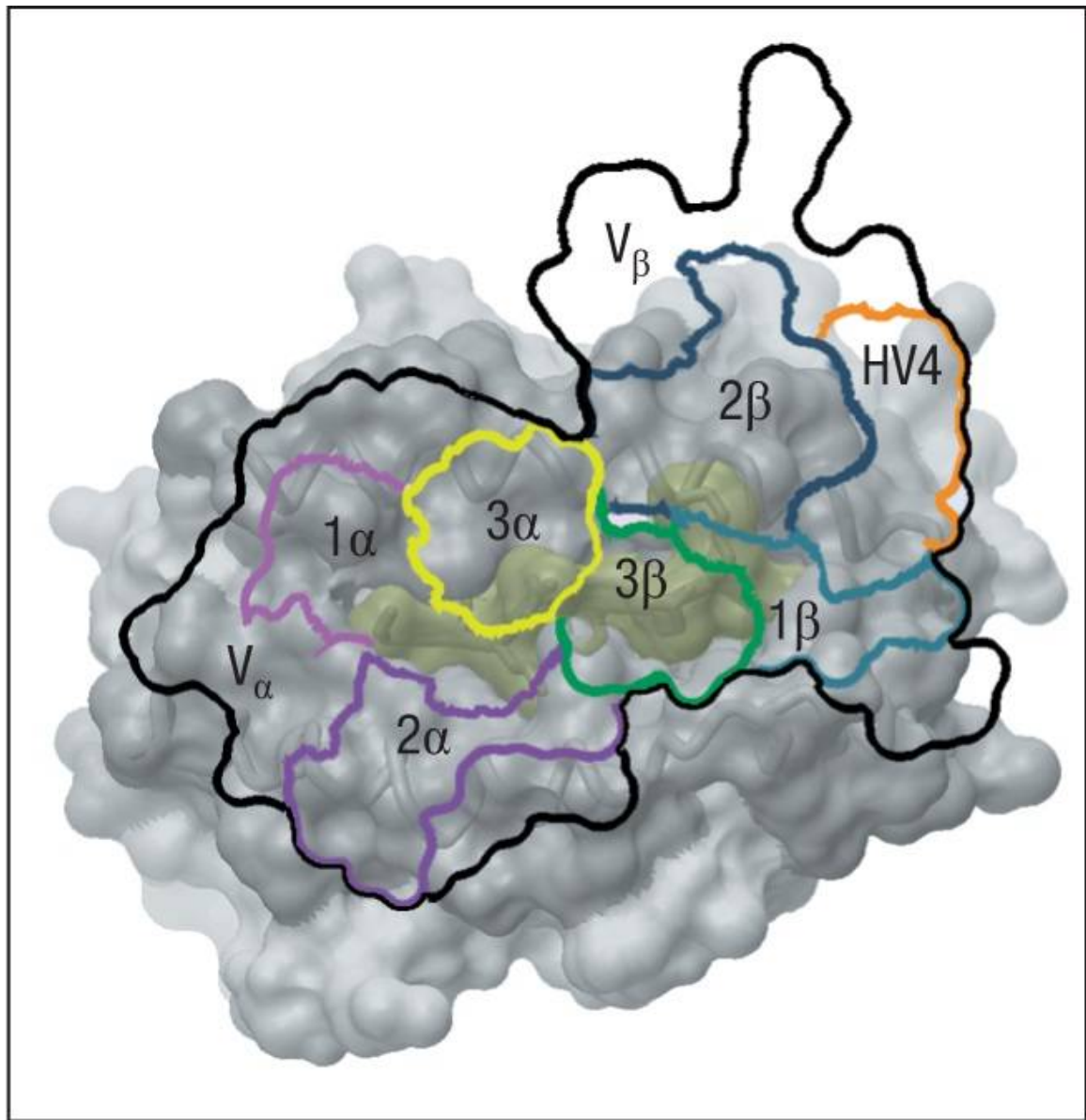
4-20 The crystal structures of several peptide:MHC:T-cell receptor complexes show a similar orientation of the T-cell receptor over the peptide:MHC complex.

When the first X-ray crystallographic structure of a T-cell receptor was published, a structure

of the same T-cell receptor bound to a peptide:MHC class I ligand was also produced. The orientation revealed by these structures showed that the T-cell receptor is aligned diagonally over the peptide and the peptide-binding cleft (**Fig. 4.28**). The TCR α chain lies over the α_2 domain of the MHC molecule at the amino-terminal end of the bound peptide, as seen from the side view shown in Fig. 4.28a. The TCR β chain lies over the MHC molecule's α_1 domain, closer to the carboxyl-terminal end of the peptide. Figure 4.27b shows a view of this structure as if looking down through a transparent T-cell receptor to indicate where it contacts the MHC molecule. The CDR3 loops of both TCR α and TCR β chains come together and lie over central amino acids of the peptide. The T-cell receptor is threaded through a valley between the two high points on the two surrounding α helices that form the walls of the peptide-binding cleft. This can be seen in **Fig. 4.29**, which shows a view from the end of the peptide-binding groove of a peptide:MHC class II:T-cell receptor complex. Comparison of various peptide:MHC:T-cell receptor complexes shows that the axis of the TCR as it binds the surface of the MHC molecule is rotated somewhat relative to the peptide-binding groove of the MHC molecule, with some variation between individual peptide:MHC:T-cell receptor complexes (see Fig. 4.28b). In this orientation, the V α domain makes contact primarily with the amino-terminal half of the bound peptide, whereas the V β domain contacts primarily the carboxyl-terminal half. Both chains also interact with the α helices of the MHC class I molecule (see Fig. 4.28). The T-cell receptor contacts are not symmetrically distributed over the MHC molecule: whereas the V α CDR1 and CDR2 loops are in close contact with the helices of the peptide:MHC complex around the amino terminus of the bound peptide, the β -chain CDR1 and CDR2 loops, which interact with the complex at the carboxyl terminus of the bound peptide, have variable contributions to the binding.



a



b

Fig. 4.28 The T-cell receptor binds to the peptide:MHC complex. Panel a: The T-cell receptor binds to the top of a peptide:MHC class I molecule, touching both the α_1 and α_2 domain helices. The CDRs of the T-cell receptor are shown in color: the CDR1 and CDR2 loops of the β chain are light and dark blue, respectively, and the CDR1 and CDR2 loops of the α chain are light and dark purple, respectively. The α -chain CDR3 loop is yellow, and the β -chain CDR3 loop is green. The β -chain HV4 loop is in red. The thick yellow line P1–P8 is the bound peptide. Panel b: The outline of the T-cell receptor's antigen-binding site (thick black line) is superimposed over the top surface of the peptide:MHC complex (the peptide is shaded dull yellow). The T-cell receptor lies at a somewhat diagonal angle across the peptide:MHC complex, with the α and β CDR3 loops of the T-cell receptor (3α , 3β , yellow and green, respectively) contacting the center of the peptide. The α -chain CDR1 and CDR2 loops (1α , 2α , light and dark

purple, respectively) contact the MHC helices at the amino terminus of the bound peptide, whereas the β -chain CDR1 and CDR2 loops (1β , 2β , light and dark blue, respectively) make contact with the helices at the carboxyl terminus of the bound peptide.

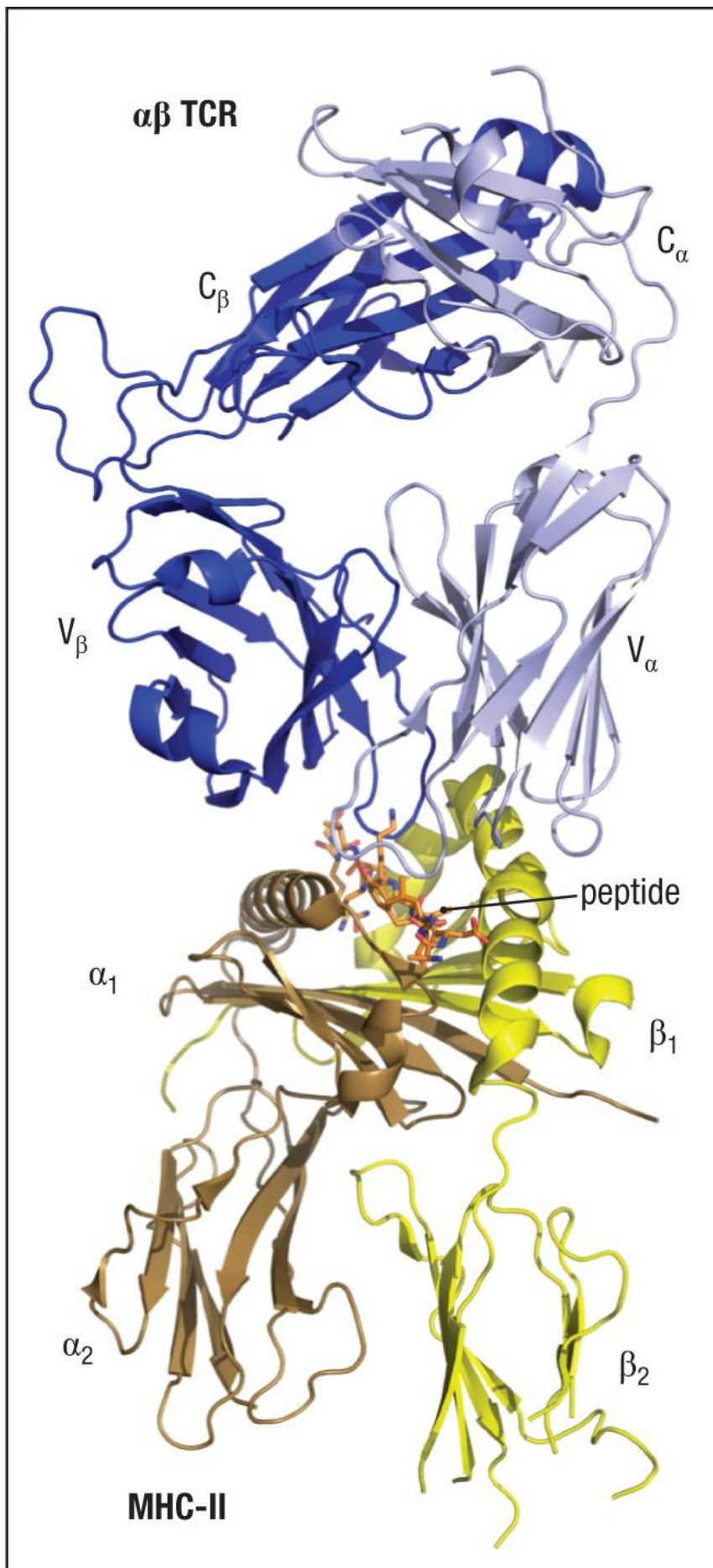


Fig. 4.29 The T-cell receptor interacts with MHC class I and MHC class II molecules in a similar fashion. Shown is the structure of a T-cell receptor, specific for a peptide derived from chicken cytochrome *c*, bound to an MHC class II molecule. This T-cell receptor's binding is at a site and orientation similar to that of the T-cell receptor bound to the MHC class I molecule shown in Fig. 4.28. The α and β chains of the T-cell receptor are colored in light and dark blue, respectively. The cytochrome *c* peptide is light orange. The T-cell receptor sits in a shallow saddle formed between the α -helical regions of the MHC class II α (brown) and β (yellow) chains at roughly 90° to the long axis of the MHC class II molecule and the bound peptide. Structure derived from PDB 3QIB.

Comparison of the three-dimensional structure of an unliganded T-cell receptor and the same T-cell receptor complexed to its peptide:MHC ligand shows that the binding results in some degree of conformational change, or 'induced fit,' particularly within the V α CDR3 loop. Subtle variations at amino acids that contact the T-cell receptor can have strikingly different effects on the recognition of an otherwise identical peptide:MHC ligand by the same T cell. The flexibility in the CDR3 loop demonstrated by these two structures helps to explain how the T-cell receptor can adopt conformations that recognize related, but different, peptide ligands.

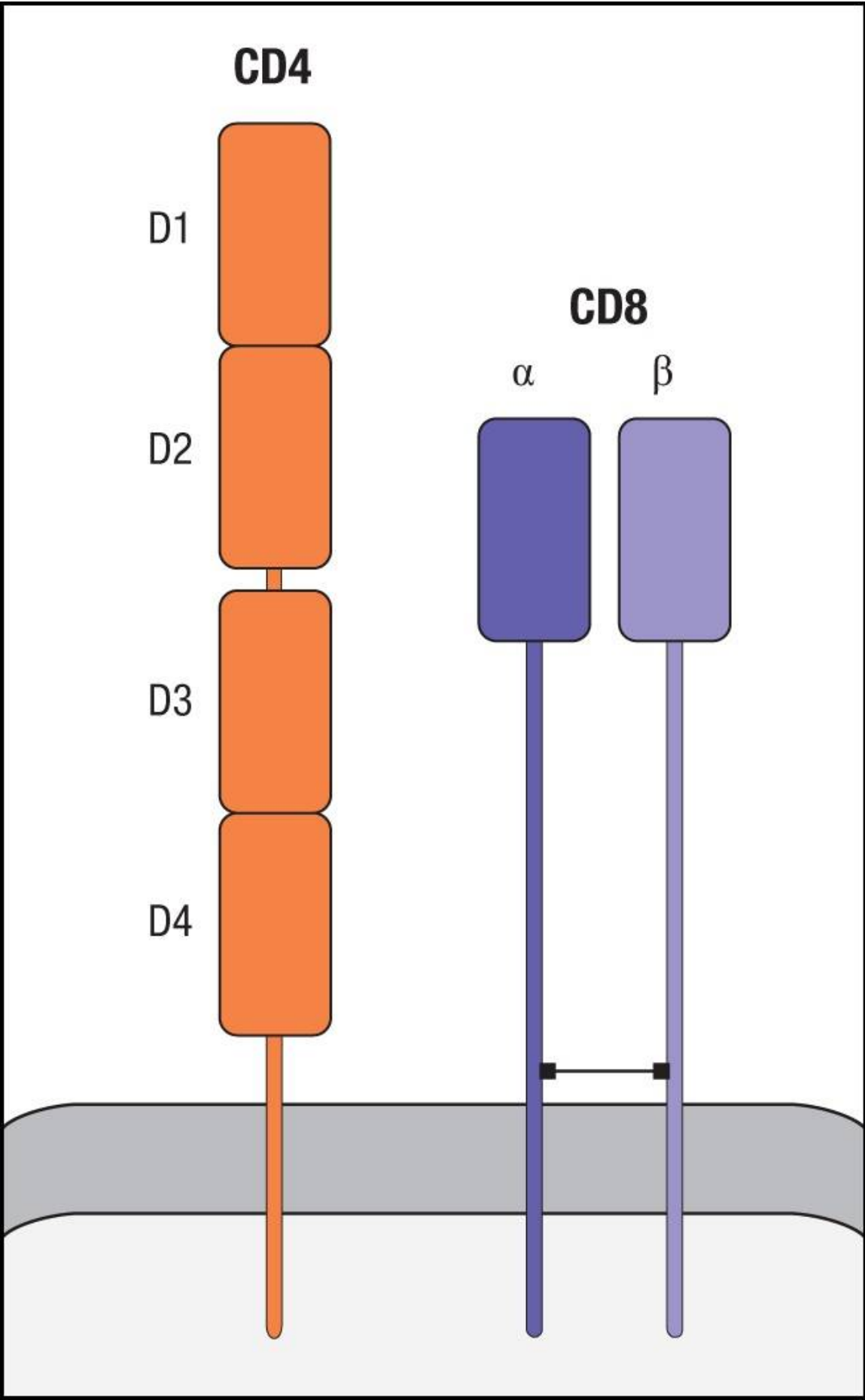
The specificity of T-cell recognition involves both the peptide and its presenting MHC molecule. Simply changing a leucine to isoleucine in the peptide, for example, is sufficient to alter a T-cell response from strong killing to no response at all. Mutations of single residues in the presenting MHC molecules can have the same effect. This dual specificity for T-cell recognition of antigen underlies the MHC restriction of T-cell responses, a phenomenon that was observed long before the peptide-binding properties of MHC molecules were known. Another consequence of this dual specificity is a need for T-cell receptors to exhibit some inherent specificity for MHC molecules in order to be able to interact appropriately with the antigen-presenting surface of MHC molecules. We will return to these issues in [Chapter 6](#), where we recount the discovery of MHC restriction in the context of T-cell recognition and MHC polymorphisms, and in [Chapter 8](#), where we discuss the impact of these phenomena on T-cell development in the thymus.

4-21 The CD4 and CD8 cell-surface proteins of T cells directly contact MHC molecules and are required to make an effective response to antigen.

As we introduced in [Section 1-21](#), T cells fall into two major classes distinguished by the expression of the cell-surface proteins [CD4](#) and [CD8](#). CD8 is expressed by cytotoxic T cells,

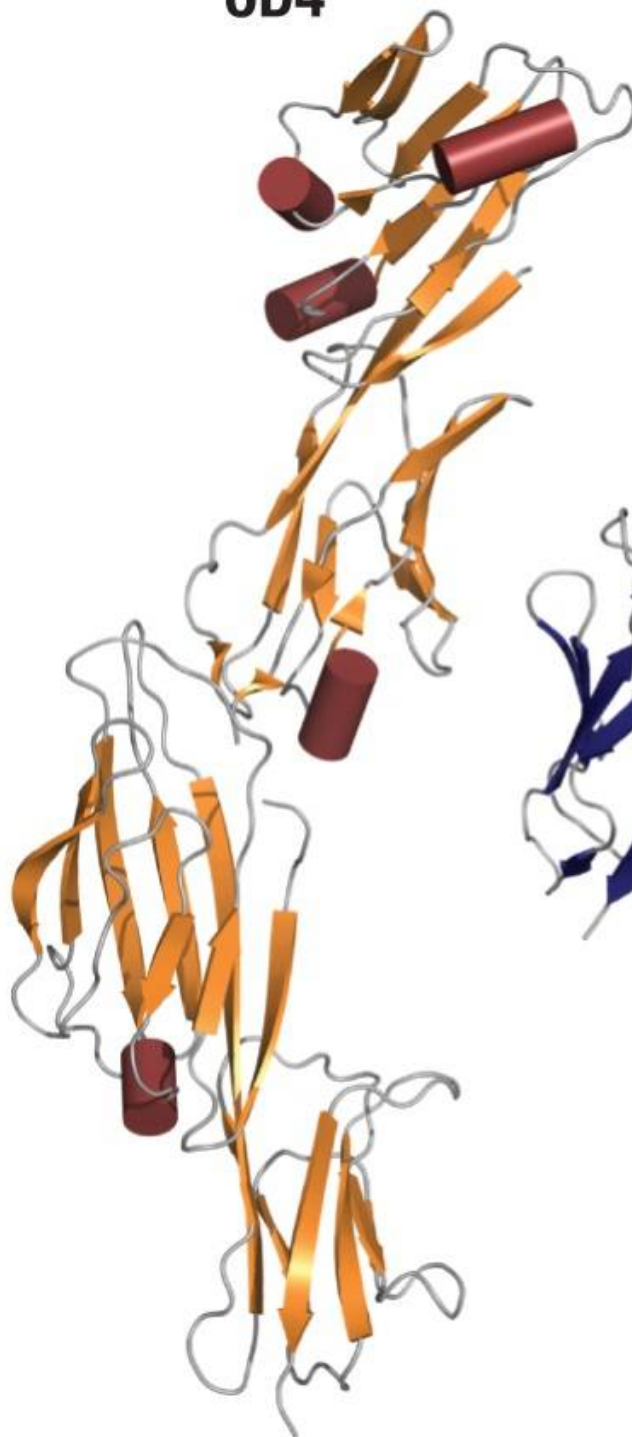
while CD4 is expressed by T cells whose function is to activate other cells. CD4 and CD8 were known as markers for these functional sets for some time before it became clear that the distinction was based on the ability of T cells to recognize different classes of MHC molecules. We now know that CD8 recognizes MHC class I molecules and CD4 recognizes MHC class II. During antigen recognition, CD4 or CD8 (depending on the type of T cell) associates with the T-cell receptor on the T-cell surface and binds to invariant sites on the MHC portion of the composite peptide:MHC ligand, away from the peptide-binding site. This binding contributes to the overall effectiveness of the T-cell response, and so CD4 and CD8 are called [co-receptors](#).

CD4 is a single-chain protein composed of four Ig-like domains (**Fig. 4.30**). The first two domains (D1 and D2) are packed tightly together to form a rigid rod about 6 nm long, which is joined by a flexible hinge to a similar rod formed by the third and fourth domains (D3 and D4). The MHC-binding region on CD4 is located mainly on a lateral face of the D1 domain, and CD4 binds to a hydrophobic crevice formed at the junction of the α_2 and β_2 domains of the MHC class II molecule (**Fig. 4.31a**). This site is well away from the site where the T-cell receptor binds, as shown by the complete crystal structure of a T-cell receptor bound to peptide:MHC class II with bound CD4 (**Fig. 4.32**). This structure demonstrates that the CD4 molecule and the T-cell receptor can bind simultaneously to the same peptide:MHC class II complex. CD4 enhances sensitivity to antigen, as the T cell is about 100-fold more sensitive to the antigen when CD4 is present. The enhancement process results from the ability of the intracellular portion of CD4 to bind to a cytoplasmic tyrosine kinase called [Lck](#). As we will discuss in detail [Chapter 7](#), bringing Lck into proximity with the T-cell receptor complex helps activate the signaling cascade induced by antigen recognition.



a

CD4



CD8

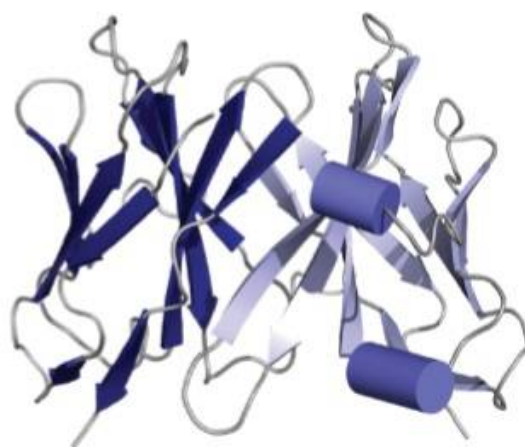
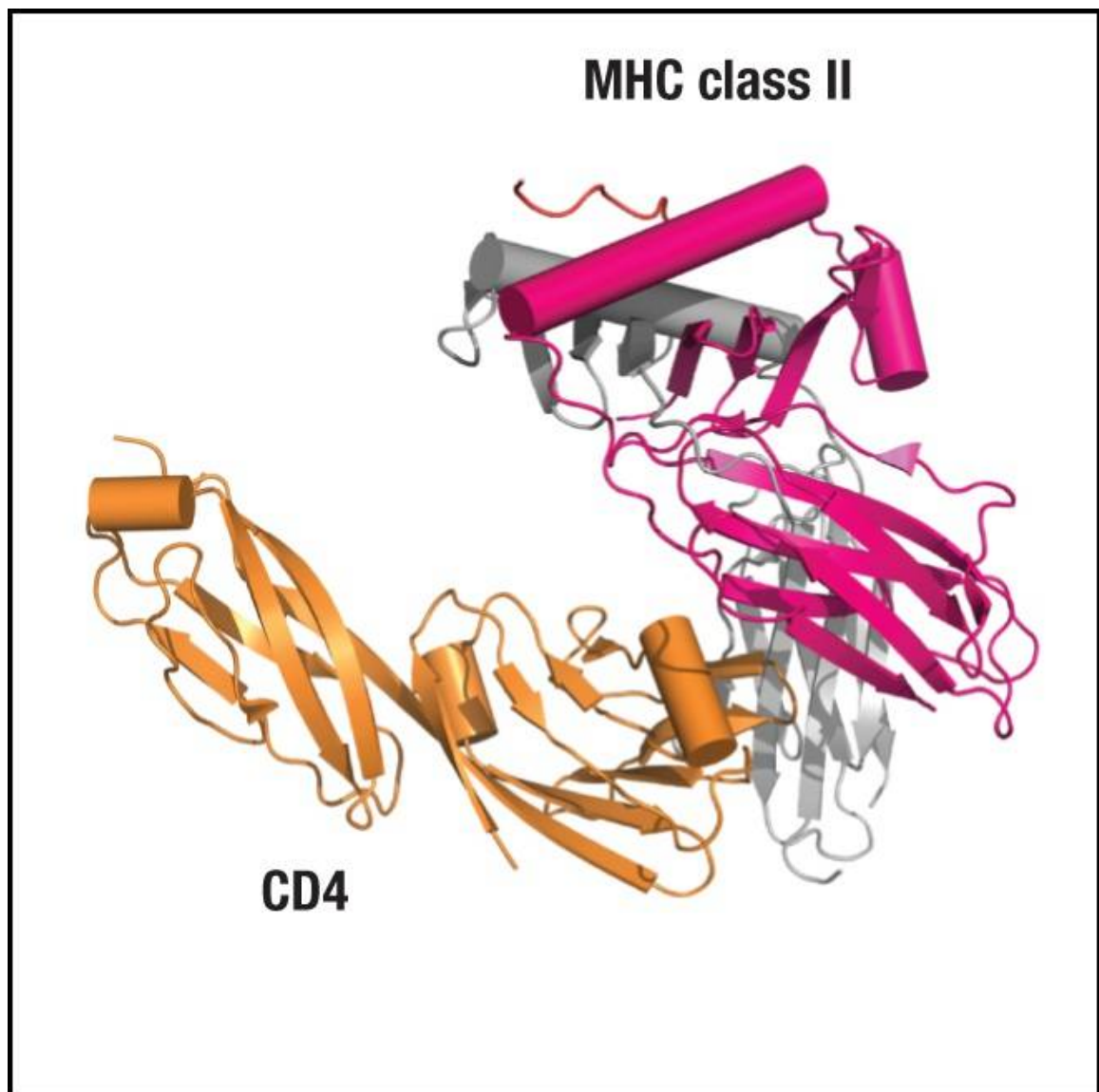
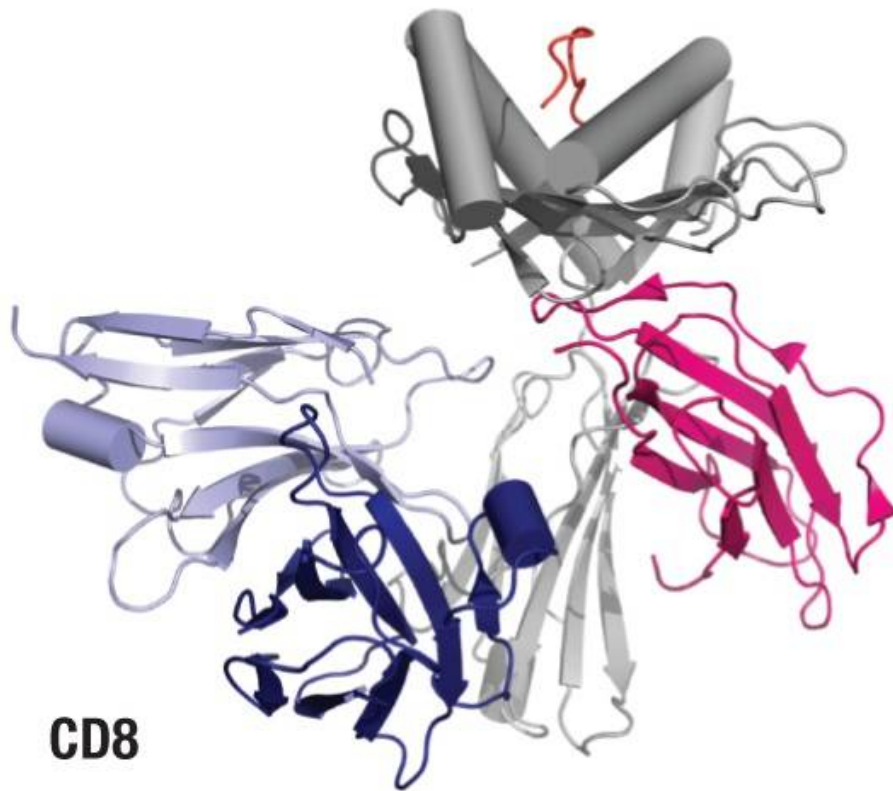


Fig. 4.30 The structures of the CD4 and CD8 co-receptor molecules. The CD4 molecule contains four Ig-like domains, shown in schematic form in panel a and as a ribbon diagram from the crystal structure in panel b. The amino-terminal domain, D1, is similar in structure to an immunoglobulin V domain. The second domain, D2, although related to an immunoglobulin domain, is different from both V and C domains and has been termed a C2 domain. The first two domains of CD4 form a rigid rodlike structure that is linked to the two carboxyl-terminal domains by a flexible link. The binding site for MHC class II molecules involves mainly the D1 domain. The CD8 molecule is a heterodimer of an α and a β chain covalently linked by a disulfide bond. An alternative form of CD8 exists as a homodimer of α chains. The heterodimer is depicted in panel a, whereas the ribbon diagram in panel b is of the homodimer. CD8 α and CD8 β chains have very similar structures, each having a single domain resembling an immunoglobulin V domain and a stretch of polypeptide chain, believed to be in a relatively extended conformation, that anchors the V-like domain to the cell membrane.



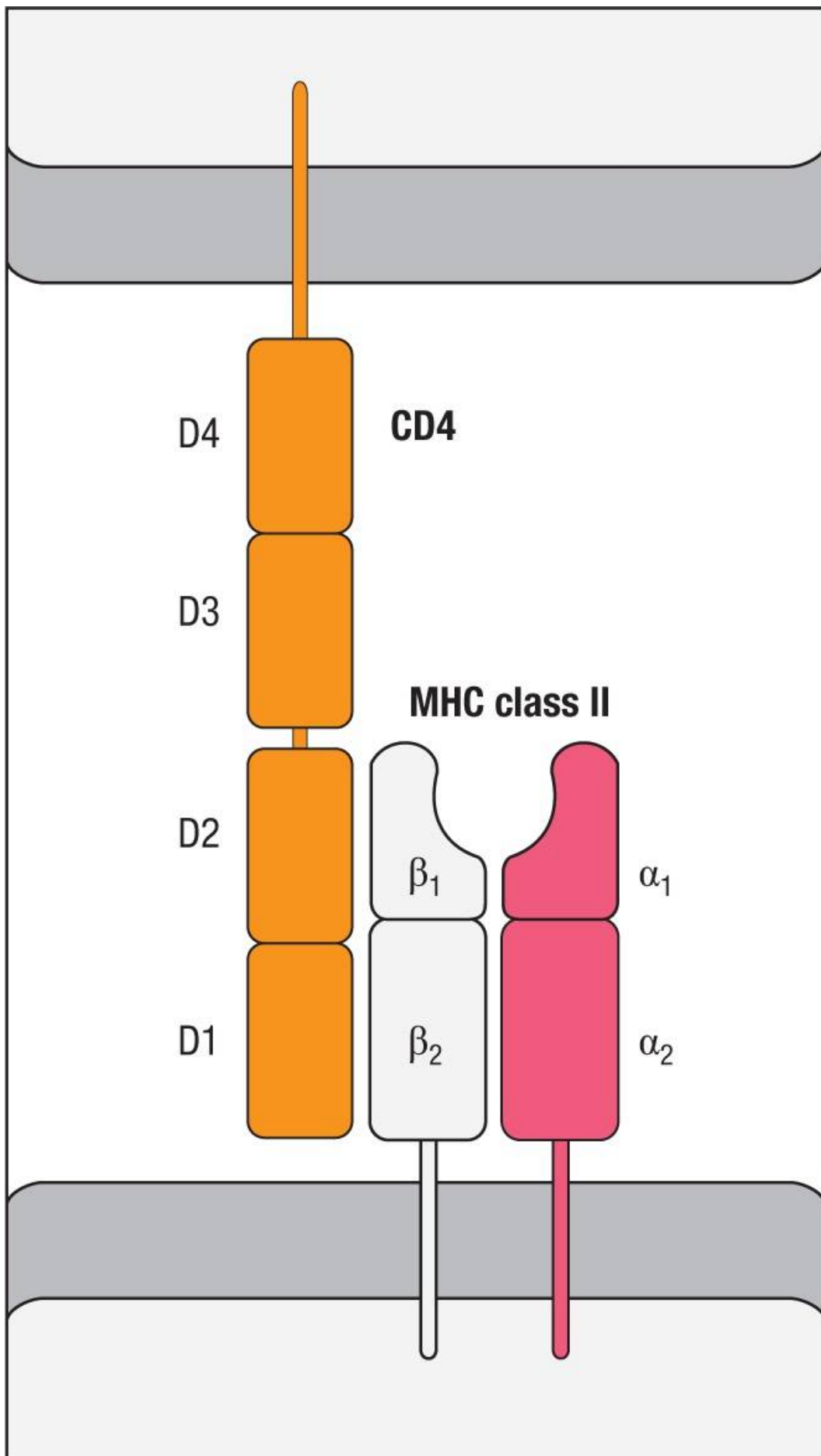
a

MHC class I

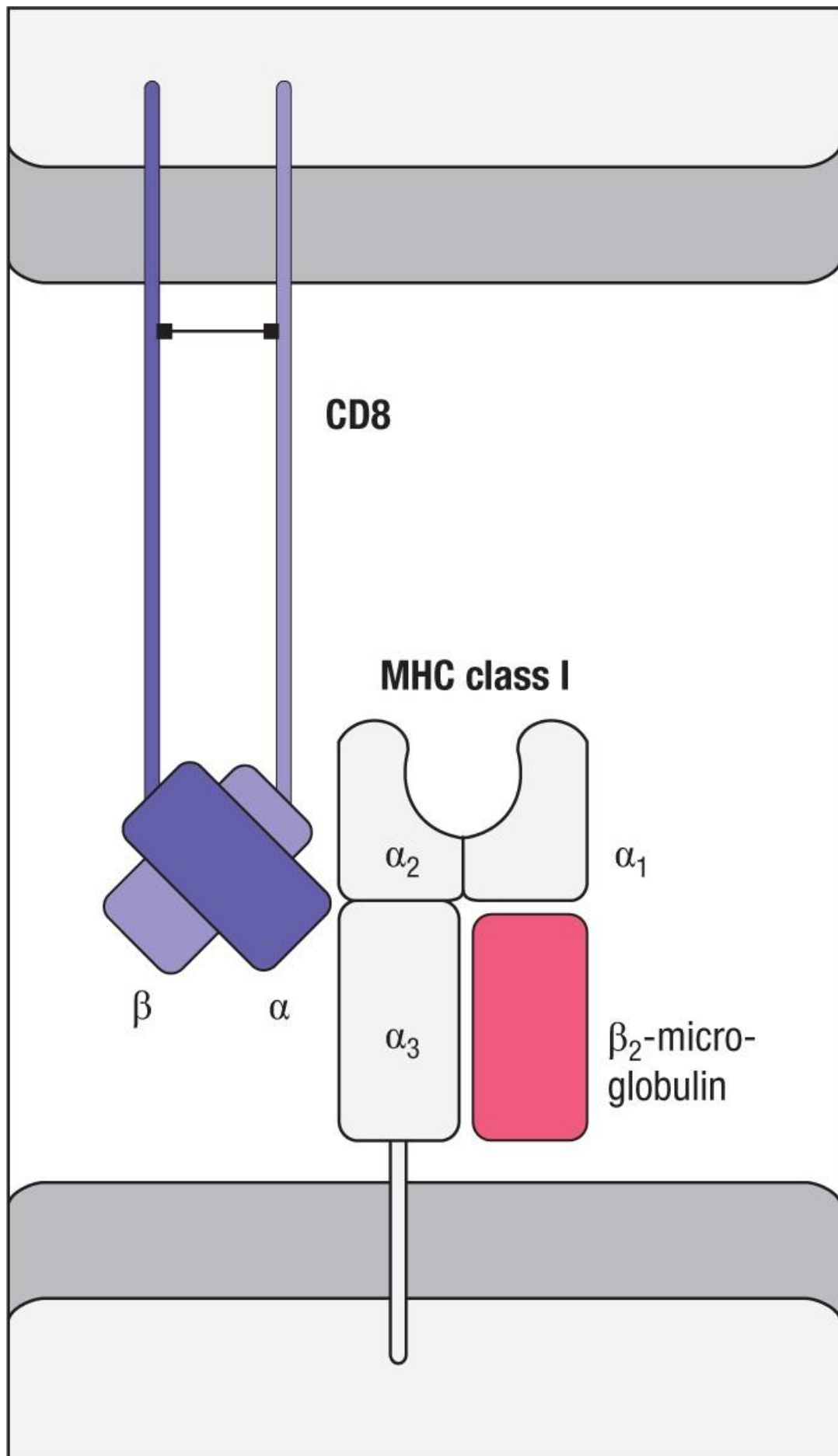


CD8

b



c



d

Fig. 4.31 The binding sites for CD4 and CD8 on MHC class II and class I molecules lie in the Ig-like domains. The binding sites for CD4 and CD8 on the MHC class II and class I molecules, respectively, lie in the Ig-like domains nearest to the membrane and distant from the peptide-binding cleft. The binding of CD4 to an MHC class II molecule is shown as a ribbon structure in panel a and schematically in panel c. The α chain of the MHC class II molecule is shown in pink, and the β chain in white, while CD4 is in gold. Only the D1 and D2 domains of the CD4 molecule are shown in panel a. The binding site for CD4 lies at the base of the β_2 domain of an MHC class II molecule, in the hydrophobic crevice between the β_2 and α_2 domains. The binding of CD8 $\alpha\beta$ to an MHC class I molecule is shown in panel b and schematically in panel d. The class I heavy chain and β_2 -microglobulin are shown in white and pink, respectively, and the two chains of the CD8 dimer are shown in light (CD8 β) and dark (CD8 α) purple. The binding site for CD8 on the MHC class I molecule lies in a similar position to that of CD4 in the MHC class II molecule, but CD8 binding also involves the base of the α_1 and α_2 domains, and thus the binding of CD8 to MHC class I is not completely equivalent to the binding of CD4 to MHC class II. The cylindrical shapes in panels a and b represent the α helices lining the peptide-binding sites of MHC class II and MHC class I, respectively. Structures derived from PDB 3S4S (CD4/MHC class II) and PDB 3DMM (CD8 $\alpha\beta$ /MHC class I).

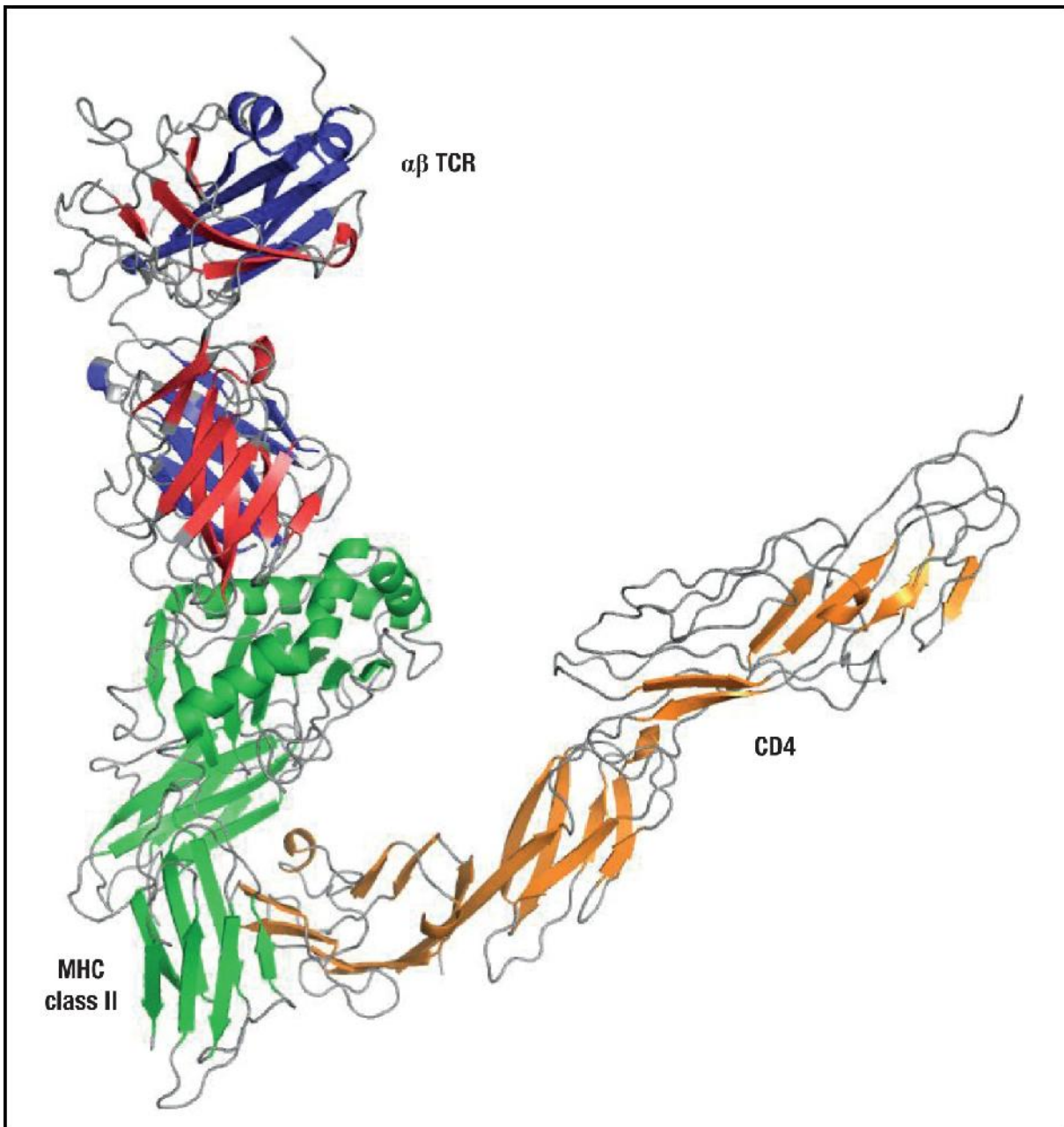


Fig. 4.32 CD4 and the T-cell receptor bind to distinct regions of the MHC class II molecule. A ribbon diagram is shown from a crystal structure of a complete $\alpha\beta$ TCR:peptide-MHC:CD4 ternary complex. The α and β chains of the T-cell receptor (TCR) are blue and red, respectively. The peptide-MHC class II complex is green. CD4 is shown in orange. Structure derived from PDB 3T0E.

The structure of CD8 is quite different. It is a disulfide-linked dimer of two different chains, called α and β , each containing a single Ig-like domain linked to the membrane by a segment of extended polypeptide (see Fig. 4.30). This segment is extensively glycosylated, which is thought to maintain it in an extended conformation and protect it from cleavage by proteases. CD8 α chains can form homodimers, although these are usually not found when CD8 β is expressed. Naive T cells express CD8 $\alpha\beta$, but the CD8 $\alpha\alpha$ homodimer can be expressed by highly activated effector and memory T cells. CD8 $\alpha\alpha$ is also expressed by a population of

intraepithelial lymphocytes known as [mucosal-associated invariant T cells](#); these cells recognize metabolites of folic acid that are produced by bacteria with the nonclassical MHC class I molecule [MR1](#), which we will describe later.

CD8 $\alpha\beta$ binds weakly to an invariant loop site in the α_3 domain of an MHC class I molecule (Fig. 4.31b). The CD8 β chain interacts with residues in the base of the MHC class I molecule, while the α chain is in a lower position interacting with the α_1 domain of the MHC class I molecule. The strength of binding of CD8 to the MHC class I molecule is influenced by the glycosylation state of the CD8 molecule; number of sialic acid residues on CD8 carbohydrate structures decrease interaction. The pattern of sialylation of CD8 changes during the maturation of the thymus and also on activation, and this likely has a role in modulating the interaction.

Like the interactions with MHC class II molecules, the T-cell receptor interacts simultaneously with one MHC class I molecule (Fig. 4.33). Like CD4, which interacts through the cytoplasmic tail of the α chain, and CD8 $\alpha\beta$ increases the selectivity of antigen presented by MHC class I molecules about 100-fold. Although the mechanisms are unclear, the CD8 $\alpha\alpha$ homodimer appears to function less efficiently as a coreceptor and may negatively regulate activation. In contrast to CD8, CIITA is required for dimerization.

Summary to [Chapter 4](#).

B cells and T cells use different but structurally similar molecules to recognize antigen. The antigen-recognition molecules of B cells are immunoglobulins and are made both as a membrane-bound receptor for antigen, the B-cell receptor, and as secreted antibodies that bind antigens and elicit humoral effector functions. The antigen-recognition molecules of T cells, in contrast, are made only as cell-surface receptors and so elicit only cellular effector functions. Immunoglobulins and T-cell receptors are highly variable molecules, with the variability concentrated in that part of the molecule—the variable (V) region—that binds to antigen. Immunoglobulins bind a wide variety of chemically different antigens, whereas the major type of T-cell receptor, the $\alpha\beta$ T-cell receptor, predominantly recognizes peptide fragments of foreign proteins bound to MHC molecules, which are ubiquitous on cell surfaces.

Binding of antigen by immunoglobulins has chiefly been studied with antibodies. The binding of antibody to its antigen is highly specific and is determined by the shape and physicochemical properties of the antigen-binding site. Located at the other end of the antibody from the antigen-binding site is the constant, or Fc, region, which influences the types of effector function the antibody can elicit. There are five main functional classes of antibodies, each encoded by a different type of constant region. As we will see in [Chapter 10](#), these interact with different components of the immune system to incite an inflammatory response and eliminate the antigen.

T-cell receptors differ in several respects from the B-cell immunoglobulins. One is the absence of a secreted form of T-cell receptor, reflecting the functional differences between T cells and B cells. B cells deal with pathogens and their protein products circulating within the body; secretion of a soluble antigen-recognition molecule enables the B cell to act in the clearance of antigen effectively throughout the extracellular spaces of the body. T cells, in contrast, are specialized for active surveillance of pathogens, and T-cell recognition does not involve a soluble, secreted receptor. Some T cells, such as CD8 T cells, are able to detect intracellular infections and are able to kill infected cells that bear foreign antigenic peptides on their surface. Others, such as CD4 T cells, interact with cells of the immune system that have taken up foreign antigen and are displaying it on the cell surface.

T-cell receptors also recognize a composite ligand made up of the foreign peptide bound to a self MHC molecule and do not recognize intact antigen. This means T cells can interact only with a body cell displaying the antigen, not with the intact pathogen or protein. Each T-cell receptor is specific for a particular combination of peptide and a self MHC molecule. MHC molecules are encoded by a family of highly polymorphic genes. Expression of multiple variant MHC molecules, each with a different peptide-binding repertoire, helps to ensure that T cells from an individual will be able to recognize at least some peptides generated from nearly every pathogen.

Glossary

[clonotypic](#)

Describes a feature unique to members of a clone. For example, the distribution of antigen receptors in the lymphocyte population is said to be clonotypic, as the cells of a given clone all have identical antigen receptors.

[T-cell receptor \$\alpha\$ \(TCR \$\alpha\$ \) and \$\beta\$ \(TCR \$\beta\$ \)](#)

The two chains of the $\alpha\beta$ T-cell receptor.

[\$\alpha\beta\$ heterodimer](#)

The dimer of one α and one β chain that makes up the antigen-recognition portion of an $\alpha\beta$ T-cell receptor.

[\$\gamma\delta\$ T-cell receptors](#)

Antigen receptor carried by a subset of T lymphocytes that is distinct from the $\alpha\beta$ T-cell receptor. It is composed of a γ and a δ chain, which are produced from genes that undergo gene rearrangement.

[\$\alpha\beta\$ T-cell receptors](#)

The cell-surface receptor for antigen on T lymphocytes. It consists of a disulfide-linked heterodimer of the highly variable α and β chains in a complex with the invariant CD3 and ζ proteins, which have a signaling function. T cells carrying this type of receptor are often called $\alpha\beta$ T cells. An alternative receptor made up of variable γ and δ chains is expressed with CD3 and ζ on a subset of T cells.

[MHC class I](#)

Polymorphic cell-surface proteins encoded in the MHC locus and expressed on most cells. They present antigenic peptides generated in the cytosol to CD8 T cells and also bind the co-receptor CD8.

[MHC class II](#)

The cellular vesicles in which MHC class II molecules accumulate, encounter HLA-DM, and bind antigenic peptides, before migrating to the surface of the cell.

[\$\beta_2\$ -microglobulin](#)

The light chain of the MHC class I proteins, encoded outside the MHC. It binds noncovalently to the heavy or α chain.

[anchor residues](#)

Specific amino acid residues in antigenic peptides that determine peptide-binding specificity to MHC class I molecules. Anchor residues for MHC class II molecules exist but are less obvious than for MHC class I.

[invariant chain \(Ii, CD74\)](#)

A polypeptide that binds in the peptide-binding cleft of newly synthesized MHC class II proteins in the endoplasmic reticulum and blocks other peptides from binding there. It is degraded in the endosome, allowing for loading of antigenic peptides there.

[CD4](#)

The co-receptor for T-cell receptors that recognize peptide antigens bound to MHC class II molecules. It binds to the lateral face of the MHC molecule.

[CD8](#)

The co-receptor for T-cell receptors that recognize peptide antigens bound to MHC class I molecules. It binds to the lateral face of the MHC molecule.

co-receptor

Cell-surface protein that increases the sensitivity of a receptor to its ligand by binding to associated ligands and participating in signaling. The antigen receptors on T cells and B cells act in conjunction with co-receptors, which are either CD4 or CD8 on T cells, and a co-receptor complex of three proteins, one of which is the complement receptor CR2, on B cells.

Lck

An Src family tyrosine kinase that associates with the cytoplasmic tails of CD4 and CD8 and phosphorylates the cytoplasmic tails of the T-cell receptor signaling chains, thus helping to activate signaling from the T-cell receptor complex once antigen has bound.

MR1 (MHC-related protein 1)

A 'nonclassical' MHC class Ib molecule that binds certain folic acid metabolites produced by bacteria for recognition by mucosal-associated invariant T (MAIT) cells.

MHC class Ib molecules

Cell-surface receptors with structural homology to MHC class I molecules. Generally nonpolymorphic and able to present nonpeptide antigens to specialized subsets of T lymphocytes, such as $\gamma\delta$ T cells, iNKT cells, and MAIT cells.

peptide-binding cleft, peptide-binding groove

The longitudinal cleft in the top surface of an MHC molecule into which the antigenic peptide is bound.

mucosal-associated invariant T cells (MAIT cells)

Primarily $\gamma\delta$ T cells with limited diversity present in the mucosal immune system that respond to bacterially derived folate derivatives presented by the nonclassical MHC class Ib molecule MR1.

T10, T22

MHC class Ib molecules expressed by activated lymphocytes and recognized by a subset of $\gamma\delta$ T cells.

Discussion Questions

1. When a mixture of IgG antibodies is treated with the proteolytic enzyme papain, each antibody is cleaved into three pieces. One of these pieces, the Fc fragment, was found to crystallize readily, indicating that the sequence of this fragment did not vary between individual antibody molecules, as did the Fab portion. Why is it important that the variability in amino acid sequence between different antibody molecules is restricted to the Fab portion of each antibody?
2. The initial antibody produced by an activated B cell is IgM. As an immune response progresses, B cells specific for the antigen can produce other antibody isotypes. What important functions are provided by antibodies of the IgG, IgA, or IgE class compared to IgM?
3. What advantage is conferred by antibodies that form multimers compared to those that are monomers?
4. Amino acid sequence variability (that is, the differences in sequence between many different antibodies) in antibody V_H and V_L domains is predominantly found in three short regions of sequence, known as CDR1, CDR2, and CDR3. Why is it important that the V_H and V_L domain sequences outside of these CDRs remain constant in all antibodies?
5. Unlike antibodies, T-cell antigen receptors only recognize foreign antigens that are displayed on the surface of the body's own cells. How does this mode of T-cell antigen recognition fit with the known functions of T lymphocytes? Why are T-cell receptor $\alpha\beta$ heterodimers never secreted by T cells?
6. MHC molecules are among the most polymorphic genes in the human population. For some human MHC genes, there are hundreds of different alleles. Where in the protein structure are most of the differences in amino acid sequence between the allelic forms? What are the consequences of these amino acid differences?
7. Many protein-protein interactions, such as an antibody binding to a protein antigen, are highly specific. Often, a single amino acid change in one of the proteins will abolish the binding to the second protein. In contrast, peptide binding to MHC proteins is much more promiscuous. Why is this lack of specificity an important aspect of peptide:MHC binding?
8. How do the CD4 and CD8 co-receptors contribute to peptide:MHC binding by T cells?
9. Why do MHC class I and MHC class II molecules have very different expression patterns among cells in the body? Which type of MHC molecule is more widely expressed and why?

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5

The Generation of Lymphocyte Antigen Receptors

IN THIS CHAPTER

- [Primary immunoglobulin gene rearrangements and expression.](#)
 - [T-cell receptor gene rearrangement.](#)
 - [Evolution of the adaptive immune response.](#)
-

A lymphocyte expresses many exact copies of a single antigen receptor that has a unique antigen-binding site (see [Section 1-12](#)). The clonal expression of antigen receptors means that each lymphocyte is unique among the billions of lymphocytes that each person possesses. [Chapter 4](#) described the structural features of immunoglobulins and T-cell receptors, the antigen receptors on B cells and T cells, respectively. We saw that the vast repertoire of antigen receptors results from variations in the amino acid sequence at the antigen-binding site, which is composed of the two [variable regions](#) from the two chains of the receptor. In immunoglobulins these are the [heavy-chain variable region \(\$V_H\$ \)](#) and the [light-chain variable region \(\$V_L\$ \)](#), and in T-cell receptors they are the V_α and V_β regions. The immunoglobulin domains of these regions contain three loops that comprise three [hypervariable regions](#), or [complementarity-determining regions \(CDRs\)](#) (see [Section 4-9](#)), that determine the receptor's antigen-binding site and allow for seemingly limitless diversity in specificity.

In the 1960s and 1970s, immunologists recognized that the limited size of the genome (at roughly 3 billion nucleotides) meant that the genome could not directly encode a sufficient number of genes to account for the observed diversity of antigen receptors. For example, encoding each distinct antibody by its own gene could easily fill the genome with nothing but antibody genes. As we will see, variable regions of the receptor chains are not directly encoded as a complete immunoglobulin domain by a single DNA segment. Instead, the variable regions are encoded by [gene segments](#) that each encode only a part of the immunoglobulin domain. During the development of each lymphocyte, these gene segments are rearranged by a process of [somatic DNA recombination](#) to form a complete and unique variable-region coding sequence. This process is known generally as [gene rearrangement](#). A fully assembled variable region sequence is produced by combining two or three types of gene segments, each of which is present in multiple copies in the germline genome. The final

diversity of the receptor repertoire is the result of assembling complete antigen receptors from the many different gene segments of each type during the development of each individual lymphocyte. This process gives each new lymphocyte only one of many possible combinations of antigen receptors, providing the repertoire of diverse antigen specificities of naive B cells and T cells.

The first part of this chapter describes the gene rearrangements that generate the primary repertoire of immunoglobulins. This part includes a brief explanation of the molecular mechanism that allows activated B cells to transition from production of transmembrane immunoglobulins to become plasma cells, which secrete their immunoglobulin proteins; the secreted form of an immunoglobulin is called an antibody. The second part of this chapter describes the gene rearrangements that generate the repertoire of T-cell receptors. Unlike immunoglobulins, T-cell receptors exist only as transmembrane receptors and are never secreted. The last part of this chapter briefly examines alternative evolutionary forms of gene rearrangements that give rise to different forms of adaptive immunity in other species.

Glossary

[variable region \(V region\)](#)

The region of an immunoglobulin or T-cell receptor that is formed of the amino-terminal domains of its component polypeptide chains. These are the most variable parts of the molecule and contain the antigen-binding sites. Cf. **constant region**.

[hypervariable regions](#)

Parts of the V domains of immunoglobulins and T-cell receptors that determine their antigen specificity and make contact with the specific ligand. The CDRs are the most variable part of the antigen receptor and contribute to the diversity of these proteins. There are three such regions (CDR1, CDR2, and CDR3) in each V domain.

[complementarity-determining regions \(CDRs\)](#)

Parts of the V domains of immunoglobulins and T-cell receptors that determine their antigen specificity and make contact with the specific ligand. The CDRs are the most variable part of the antigen receptor and contribute to the diversity of these proteins. There are three such regions (CDR1, CDR2, and CDR3) in each V domain.

[gene segments](#)

Sets of short DNA sequences at the immunoglobulin and T-cell receptor loci that encode different regions of the variable domains of antigen receptors. Gene segments of each type are joined together by somatic recombination to form a complete variable-domain exon. There are three types of gene segments: V gene segments encode the first 95 amino acids, D gene segments (in heavy-chain and TCR α chain loci only) encode about 5 amino acids, and J gene segments encode the last 10–15 amino acids of the variable domain. There are multiple copies of each type of gene segment in the germline DNA, but only one of each type is joined together to form the variable domain.

[somatic DNA recombination](#)

DNA recombination that takes place in somatic cells (to distinguish it from the recombination that takes place during meiosis and gamete formation).

gene rearrangement

The process of somatic recombination of gene segments in the immunoglobulin and T-cell receptor genetic loci to produce a functional gene. This process generates the diversity found in immunoglobulin and T-cell receptor variable regions.

heavy-chain variable region (V_H)

Referring to the V region of the heavy chain of an immunoglobulin.

light-chain variable region (V_L)

Referring to the V region of the light chain of an immunoglobulin.

V_α

Variable region from the $TCR\alpha$ chain.

V_β

Variable region from the $TCR\beta$ chain.

Primary immunoglobulin gene rearrangements and expression.

Virtually any substance can be the target of an antibody response, and the response to even a single epitope comprises many different antibody molecules, each with a subtly different specificity for the epitope and a unique [affinity](#), or binding strength. The total number of antibody specificities available to an individual is known as the [antibody repertoire](#) or [immunoglobulin repertoire](#), and in humans the repertoire is at least 10^{11} and probably several orders of magnitude greater. However, the number of antibody specificities present at any one time is limited by the total number of B cells in an individual, as well as by each individual's previous encounters with antigens.

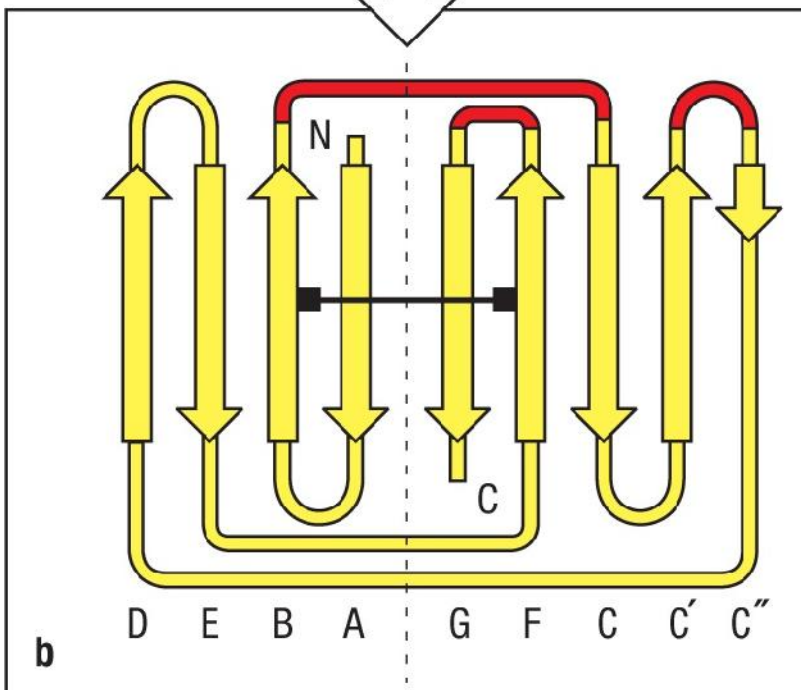
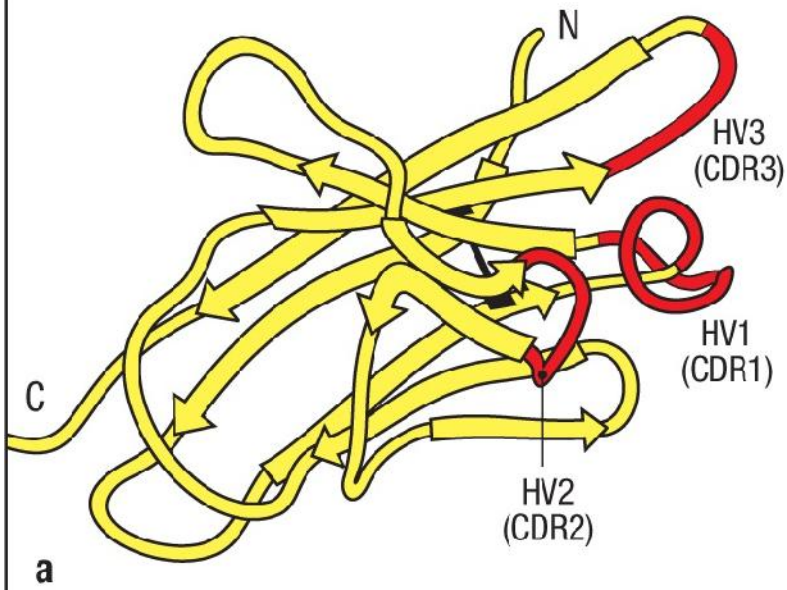
Before it was possible to examine the immunoglobulin genes directly, there were two main hypotheses for the origin of this diversity. The [germline theory](#) held that there is a separate gene for each different immunoglobulin chain and that the antibody repertoire is largely inherited. In contrast, [somatic diversification theories](#) proposed that the observed repertoire is generated from a limited number of inherited V-region sequences that undergo alteration within B cells during the individual's lifetime. Cloning of the immunoglobulin genes revealed that elements of both theories were correct and that the DNA sequence encoding each variable region is generated by rearrangements of a relatively small group of inherited gene segments. Diversity is further enhanced by the process of [somatic hypermutation](#) in mature activated B cells. Thus, somatic diversification theory was essentially correct, although the germline theory concept of the existence of multiple germline genes also proved true.

5-1 Complete immunoglobulin genes are generated by the somatic recombination of separate gene segments.

Figure 5.1 shows the relationships between a light-chain variable region's antigen-binding site, its domain structure, and the gene that encodes it. The variable regions of immunoglobulin heavy and light chains are based on the [immunoglobulin fold](#), which is composed of nine β strands, forming two β sheets held together by a disulfide bond. The antibody-binding site is formed by three loops of amino acids known as hypervariable regions HV1, HV2, and HV3, or CDR1, CDR2, and CDR3 (see Fig. 5.1a). These loops are located between the pairs of β strands B and C, C' and C'', and F and G (see Fig. 5.1b). In a mature B cell, the variable region for either a heavy or a light chain is encoded by a single exon; within a given heavy- or light-chain variable-region exon, the sequences encoding

each of the three CDRs are not contiguous sequences, but are separated from one another within the exon (see Fig. 5.1c). This exon is the gene's second exon (exon 2). The first exon of the variable regions encodes the antibody's leader sequence, which directs the antibody into the endoplasmic reticulum for surface expression or secretion.

Immunoglobulin variable region



Variable exon region

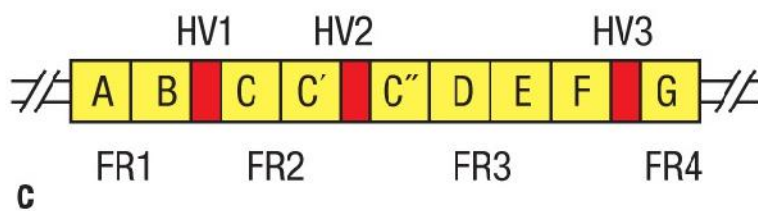


Fig. 5.1 Three hypervariable regions are encoded within a single V-region exon.

Panel a: The variable region is based on the immunoglobulin (Ig) fold that is supported by framework regions (yellow) composed of nine β strands and contains three hypervariable (HV) regions (red) that determine its antigen specificity. Panel b: The three HV regions exist as loops of amino acids on the same face between the β strands of B and C, between C' and C'', and between F and G. The dotted line separates the β strands forming each of the two β sheets composing the Ig fold. The black bar indicates the disulfide bond holding the two β sheets together. Panel c: A complete variable region in a lymphocyte is encoded within a single exon of the full antigen-receptor gene. The three HV regions are interspersed between four framework regions (FRs) made up of the β strands of the Ig domain and the loops on the face opposite of the CDRs.

Unlike most genes, the complete DNA sequence of the variable-region exon is not present in the germline of the individual, but is originally encoded by separate DNA segments, as illustrated in **Fig. 5.2**. The rearrangements that produce the complete immunoglobulin light-chain and heavy-chain genes are shown in **Fig. 5.3**. For the light chain, the joining of a [variable or V gene segment](#) (denoted V_L for light-chain variable gene segment) and a [joining or J gene segment](#) (J_L in the case of the light chain) creates an exon that encodes the whole light-chain V_L region. In the unrearranged DNA, the V_L gene segments are located relatively far away from the exons encoding the constant region of the light chain (C_L region). The J_L gene segments are located close to the C_L region, however, and the joining of a V_L gene segment to a J_L gene segment also brings the V_L gene segment close to a C_L -region sequence. The J_L gene segment of the rearranged V_L region is separated from a C_L -region sequence only by a short intron. To make a complete immunoglobulin light-chain messenger RNA, the V-region exon is joined to the C-region sequence by RNA splicing after transcription.

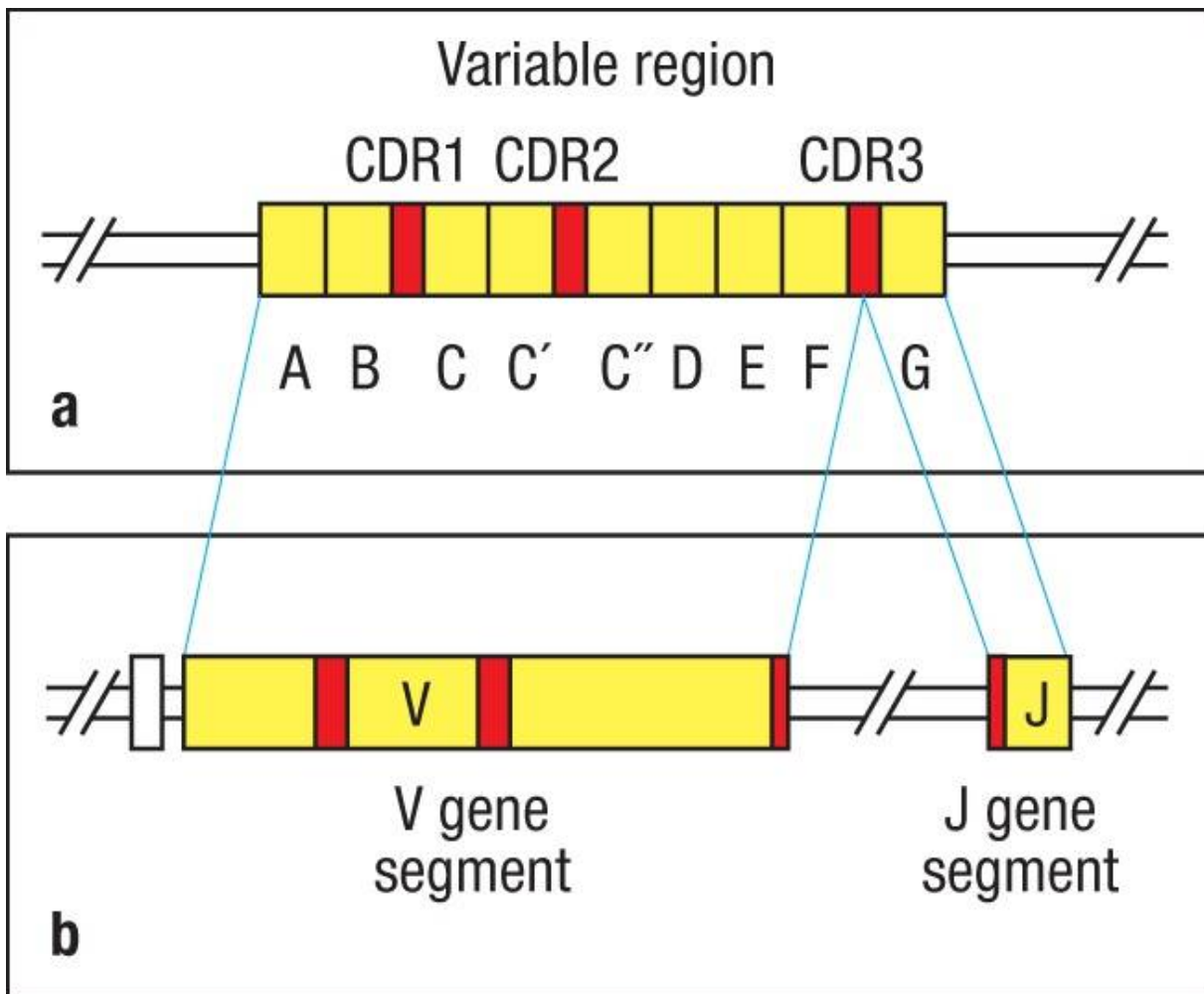


Fig. 5.2 The CDR3 region originates from two or more individual gene segments that are joined during lymphocyte development. Panel a: A complete light-chain variable region encoding the CDR1, CDR2, and CDR3 loops resides in a single exon. Panel b: The complete variable region is derived from distinct germline DNA sequences. A V gene segment encodes the CDR1 and CDR2 loops, and the CDR3 loop is formed by sequences from the end of the V gene segment and the beginning of the J gene segment, and by nucleotides added or lost when these gene segments are joined during lymphocyte development. The coding region of the CDR3 loop of the heavy chain is formed by the joining of sequences from V, D, and J gene segments and additional nucleotides added at random (not shown).

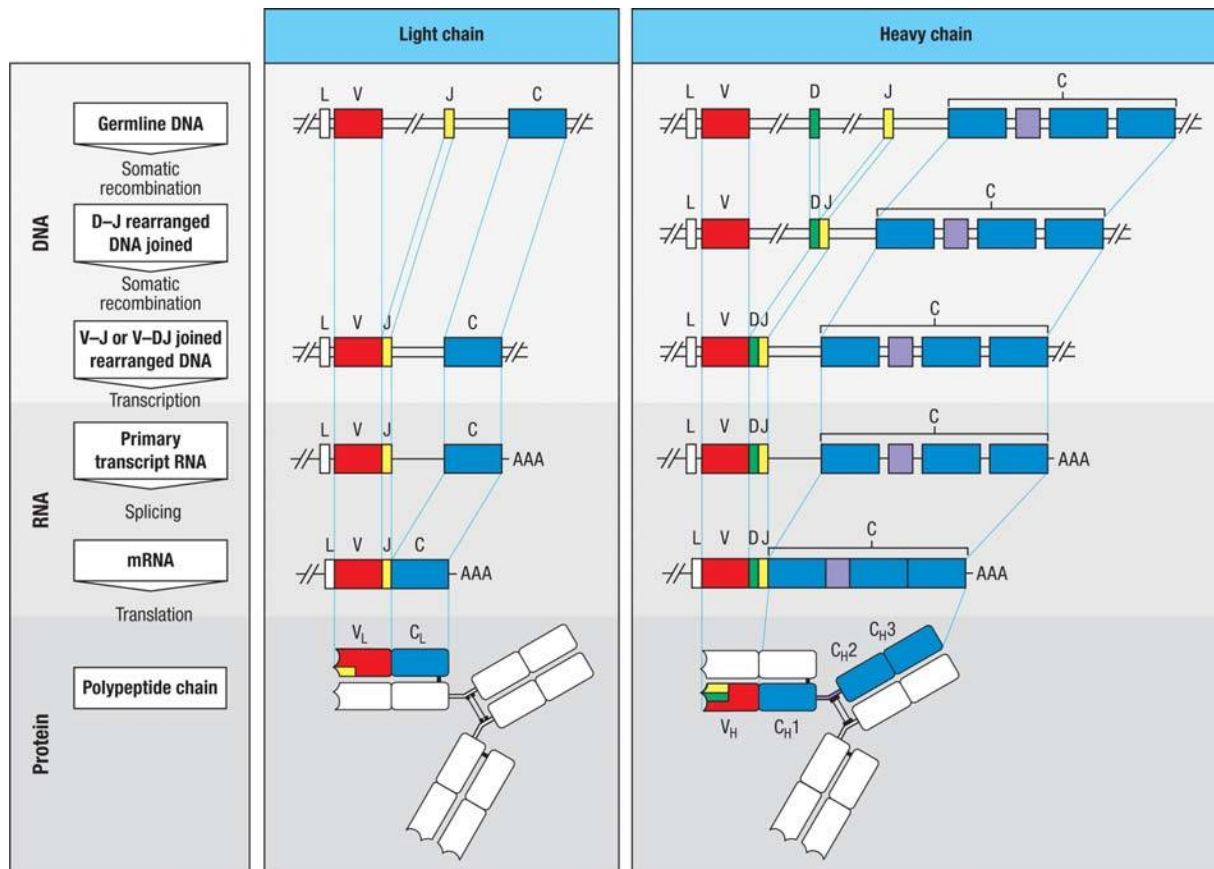


Fig. 5.3 V regions are constructed from gene segments. Light-chain V regions are constructed from two segments (center panel). A variable (V) and a joining (J) gene segment in the genomic DNA are joined to form a complete light-chain V-region exon. Immunoglobulin chains are extracellular proteins, and the V gene segment is preceded by an exon encoding a leader peptide (L), which directs the protein into the cell's secretory pathways and is then cleaved. The light-chain C region is encoded in a separate exon and is joined to the V-region exon by splicing of the light-chain RNA to remove the L-to-V and the J-to-C introns. Heavy-chain V regions are constructed from three gene segments (right panel). First, the diversity (D) and J gene segments join, and then the V gene segment joins to the combined DJ sequence, forming a complete V_H exon. A heavy-chain C region is encoded by several exons. The C-region exons, together with the leader sequence, are spliced to the V-domain sequence during processing of the heavy-chain RNA transcript. The leader sequence is removed after translation, and the disulfide bonds that link the polypeptide chains are formed. The hinge region is shown in purple. Only one V, one D, and one J segment is shown for each locus for simplicity; there are actually multiple gene segments of each type in the genome.

For the heavy chain, there is one additional complication. The heavy-chain V region (V_H) is encoded in three gene segments, rather than two. In addition to the V and J gene segments (denoted V_H and J_H to distinguish them from the light-chain V_L and J_L), the heavy chain uses a third gene segment called the [diversity or \$D_H\$ gene segment](#), which lies between the